

Modern Fractal Geometry

Volume III

Projections, Incidences, and Distance Geometry

Working reader edition — September 2026

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Preface to Volume III

This volume develops projection, incidence, Furstenberg, distance-set, visibility, and Fourier methods in modern fractal geometry. The organizing principle is not historical chronology but proof dependency. We begin with potential-theoretic orthogonal projections and slicing, pass through exceptional-set and entropy mechanisms, then develop nonlinear and radial projections, sharp incidence theory, Furstenberg sets, distance sets, visible parts, and finally Fourier analysis on curves.

The exposition follows a strict distinction between four kinds of statements. Results intended to be proved in the volume are followed by their local arguments. Earlier-volume or external prerequisites are invoked only through explicitly stated interfaces. Guided theorem studios state modern research theorems exactly while marking the research-scale proof blocks that are not reproduced. Terminal questions are presented as questions and have no role in subsequent proofs. An adjacent proof is a statement of proof ownership, not by itself a certificate that every research-scale step has already passed audit; the status of this working edition is recorded below.

The emphasis is on definitions before use, proof architecture, examples and counterexamples, and a clear separation between intuition and formal argument.

Proof status of this revised edition

The projection and CP-chain core through Chapter 3 has local proofs, modulo the declared undergraduate and graduate prerequisites. The rank-one Bourgain theorem and the disjoint regularisation in Chapter 6 are also proved locally. An independent second-reader audit passed the Chapter 9 high-multiplicity bootstrap [III.9.25.4](#) relative to its stated suppliers.

The formerly open sharp-incidence multiscale combination Proposition [III.6.29.12](#) now has a quantified normal-scale loss, a uniformization lemma, separate one-scale estimates for normal, good and bad blocks, and a complete induction with its exponent ledger. The passage from that proposition to the primal and dual sharp incidence theorems is also written out. The earlier conditional status of descendants in Chapters 5–12 that depended only on this missing step has therefore been removed. This is a reader-facing revised manuscript; the present repair has received a targeted mathematical self-audit, not a new independent referee review of every research proof in the volume. Illustrative source theorems and terminal questions are separated from proved claims and are never used as hidden premises.

Architecture and reading routes

The twelve chapters form four overlapping mathematical arcs rather than twelve independent surveys.

Projection foundations.

Chapters 1–2 establish the energy, slicing, and exceptional-set mechanisms for orthogonal projections. Chapter 3 develops the entropy/CP-chain alternative, and Chapter 4 transports generalized-projection technology to radial families. These chapters supply the basic analytic and dynamical currencies used later.

Incidence and Furstenberg theory.

Chapters 5–9 move from radial exceptional sets to Bourgain-type nonlinear projection growth, discretized incidence geometry, the classical Furstenberg problem, and the modern Orponen–Shmerkin advances. Readers interested primarily in incidence geometry may read Chapters 1–2 and then proceed to Chapters 6–9, consulting Chapter 3 when entropy language enters.

Distances, tubes, and visibility.

Chapter 10 combines projection, slicing, and nonlinear incidence mechanisms in the distance-set problem. Chapter 11 develops the related tube-nullity, heavy-line, and visible-part questions. Chapters 4–7 provide the most relevant preparatory material for this route.

Fourier/frontier route.

Chapter 12 uses the incidence and Furstenberg machinery together with Fourier estimates on curves. Its locally proved deductions depend on the earlier chapters exactly where cross-references indicate; the frontier statements and terminal questions are not used in those deductions.

A reader seeking the shortest classical path can read Chapters 1, 2, 4, 5, 8, 10, and 11. A reader interested in the modern discretized machinery should include Chapters 3, 6, 7, and 9. The full order remains the recommended route because later chapters repeatedly compare the energy, entropy, incidence, and Fourier architectures rather than treating them as interchangeable black boxes.

Conventions on prerequisites

Volume I supplies the declared Frostman, energy–dimension, capacity, and standard measure-theoretic interfaces. Earlier volumes also supply the explicitly named entropy/tree-coding inputs used in Chapter 3 and the finite-scale interfaces cited later. Such prerequisites are not reproved merely to make a local proof look self-contained: the exact interface is stated where it is used, and the reader can see which part of the argument is local and which part is imported.

Proof and source conventions

Proved locally.

The argument supplied in this volume is the local proof owner for the stated result, possibly using prerequisites declared at the beginning of the chapter. In the present working edition, this label is a completeness claim only when the declared dependency branch has no outstanding proof module.

Imported interface.

A technical statement proved in an earlier volume or an explicitly identified external source. Its exact statement and hypotheses are recorded before it is used.

Guided theorem studio.

A research theorem quoted with its precise hypotheses and source, while the indicated research-scale proof blocks are left to that source. All deductions made from the theorem inside this volume are nevertheless proved.

Terminal question.

An open or frontier problem. Such a question is never used as proof authority.

LaTeX labels provide cross-references without interrupting the mathematical exposition.

Notation

Unless a chapter explicitly changes convention, the following notation is used throughout.

$\dim_H E$	Hausdorff dimension of a set E .
$\dim_P E$	packing dimension of E .
$\underline{\dim}_H \mu$	lower Hausdorff dimension of a measure μ .
$\mathcal{H}^s$	s -dimensional Hausdorff measure.
$\mathcal{H}_\infty^s$	s -dimensional Hausdorff content.
$\mathcal{L}^m$	m -dimensional Lebesgue measure.
$I_s(\mu)$	Riesz s -energy of a measure.
$\text{spt } \mu$	support of a measure.
P_V	orthogonal projection onto a linear subspace V .
π_e	orthogonal scalar projection in direction e .
π_a	radial projection from the centre a .
$G(n, m)$	Grassmannian of m -planes in $\mathbb{R}^n$.
$N_r(E)$	covering number of E at scale r .
$E^{(r)}$	open or closed r -neighbourhood of E , as specified locally.
$\Delta(E)$	unpinned distance set $\{ x - y : x, y \in E\}$.
$\Delta^\mathcal{V}(E)$	pinned distance set $\{ x - y : x \in E\}$.
$\hat{\mu}$	Fourier transform, with normalization specified where first used.
$A \lesssim B$	$A \leq CB$ for an admissible implicit constant C .
$A \asymp B$	both $A \lesssim B$ and $B \lesssim A$.

Scale-dependent symbols such as dyadic grids, tube families, entropy functionals, CP-chain states, and transversal parameter measures are defined afresh in the chapters where they enter. This is intentional: their normalizations vary between the energy, entropy, incidence, and Fourier settings, and no global convention is imposed when it would conceal a scale or normalization choice.

Part I

Classical Projection Machinery

Chapter III.1

Marstrand Projection and Slicing Theorems

Marstrand's projection theorem [1] is one of the first places where the geometry of a set is controlled by averaging over a parameter space. The guiding principle is that singular behavior may occur in exceptional directions, but the exceptional set is small enough that energy estimates survive for almost every projection. This chapter develops the planar orthogonal-projection and slicing mechanisms in a form that will be reused throughout the volume.

We begin with the angular kernel identity and the projected-energy formula, then pass to an L^2 density theorem at the critical one-energy level. The second half of the chapter develops approximate slice measures and proves the subcritical and exact slicing statements. Frostman and capacitary facts owned by Volume I are used as declared prerequisites rather than reproved here.

Prerequisites and source boundary. The local arguments are self-contained except for the explicitly imported compact Frostman/energy interfaces from Volume I and the analytic compact-approximation theorem. These prerequisites are invoked only at the set-level passage from measure estimates to Hausdorff-dimension statements.

III.1.1 Projection and energy notation

Definition (orthogonal scalar projection). For $e \in S^1$, set $\pi_e(x) = x \cdot e$. We identify the geometric projection onto $\text{span}(e)$ with this scalar coordinate.

Definition (projected measure). For a finite Borel measure μ on $\mathbb{R}^2$, write $\mu_e = (\pi_e)_\# \mu$.

Definition (Riesz potential and energy). For $0 < t < 2$,

$$U_t^\mu(x) = \int |x - y|^{-t} d\mu(y), \quad I_t(\mu) = \int U_t^\mu(x) d\mu(x) \in [0, \infty].$$

For measures on $\mathbb{R}$ we use the same notation in the range $0 < t < 1$.

Definition (direction measure). Throughout the planar projection chapters, σ is normalized arclength probability measure on S^1 , so $d\sigma = d\theta/(2\pi)$ under $e(\theta) = (\cos \theta, \sin \theta)$.

III.1.2 Angular kernel and projected energy

Lemma III.1.1 (Angular kernel identity). *Let $0 < t < 1$. For every $z \in \mathbb{R}^2 \setminus \{0\}$,*

$$\int_{S^1} |e \cdot z|^{-t} d\sigma(e) = c_t |z|^{-t},$$

where

$$c_t = (2\pi)^{-1} \int_0^{2\pi} |\cos \theta|^{-t} d\theta < \infty.$$

Proof. Rotation invariance of σ permits us to assume $z = |z|(1, 0)$. Writing $e = (\cos \theta, \sin \theta)$,

$$|e \cdot z|^{-t} = |z|^{-t} |\cos \theta|^{-t}.$$

Hence the asserted identity follows by integration. The only possible singularities occur at $\theta = \pi/2$ and $3\pi/2$; near either one, $|\cos \theta|$ is comparable to the distance from the singular angle, and r^{-t} is locally integrable exactly when $t < 1$. This proves finiteness of c_t . $\square$

Theorem III.1.2 (Averaged projected-energy identity). *Let $0 < t < 1$. Then*

$$\int_{S^1} I_t(\mu_e) d\sigma(e) = c_t I_t(\mu),$$

with equality in $[0, \infty]$.

In particular, if $I_t(\mu) < \infty$, then $I_t(\mu_e) < \infty$ for σ -almost every e .

Proof. By the pushforward definition,

$$\begin{aligned} I_t(\mu_e) &= \iint |\pi_e(x) - \pi_e(y)|^{-t} d\mu(x) d\mu(y) \\ &= \iint |e \cdot (x - y)|^{-t} d\mu(x) d\mu(y). \end{aligned}$$

The integrand is nonnegative, so Tonelli's theorem allows the order of integration to be changed without an a priori integrability assumption. Applying Lemma III.1.1 for $x \neq y$ gives

$$\int_{S^1} I_t(\mu_e) d\sigma(e) = c_t \iint |x - y|^{-t} d\mu(x) d\mu(y) = c_t I_t(\mu).$$

If $I_t(\mu) < \infty$, the left-hand integral is finite, and therefore $I_t(\mu_e) < \infty$ for almost every e . $\square$

III.1.3 Finite energy forces dimension

Lemma III.1.3 (Bounded-potential subset). *Let ν be a finite nonzero Borel measure on a metric subset of Euclidean space and let $t > 0$. If $I_t(\nu) < \infty$, then there are $N < \infty$ and a Borel set F with $\nu(F) > 0$ such that $U_t^\nu(x) \leq N$ for all $x \in F$.*

Proof. Since $\int U_t^v dv = I_t(v) < \infty$, the potential is finite at ν -almost every point. The sets $F_N = \{x : U_t^v(x) \leq N\}$ increase to a set of full ν -measure. Hence one of them has positive ν -measure. $\square$

Theorem III.1.4 (Finite energy forces a support-dimension lower bound). *Let ν be a finite nonzero Borel measure on $\mathbb{R}^m$. If $I_t(\nu) < \infty$, then*

$$\dim_H(\text{supp } \nu) \geq t.$$

More strongly, there is a Borel set F with $\nu(F) > 0$ and $\mathcal{H}^t(F) > 0$.

Proof. Choose F and N as in Lemma III.1.3 and set $\eta = \nu|_F$. If $x \in F$ and $r > 0$, then

$$\nu(B(x, r))r^{-t} \leq \int_{B(x, r)} |x - y|^{-t} d\nu(y) \leq U_t^v(x) \leq N,$$

so $\eta(B(x, r)) \leq Nr^t$.

Now let $B(a, r)$ be any ball meeting F , and choose $x \in F \cap B(a, r)$. Then $B(a, r) \subset B(x, 2r)$, hence

$$\eta(B(a, r)) \leq N(2r)^t.$$

For any countable ball cover $F \subset \bigcup_i B(a_i, r_i)$,

$$\eta(F) \leq \sum_i \eta(B(a_i, r_i)) \leq N2^t \sum_i r_i^t.$$

Taking the infimum over arbitrarily fine covers gives positive t -dimensional Hausdorff content, hence $\mathcal{H}^t(F) > 0$. Therefore $\dim_H(\text{supp } \nu) \geq \dim_H F \geq t$. $\square$

Corollary (measure-level projection dimension). *If $0 < t < 1$ and $I_t(\mu) < \infty$, then for almost every e ,*

$$\dim_H(\text{supp } \mu_e) \geq t.$$

Proof. Combine Theorem III.1.2 with Theorem III.1.4. $\square$

Interface note. The set-level theorem below uses the Frostman and energy-dimension results established in Volume I; no local re-proof of Frostman/capacitability is required.

III.1.4 A geometric L2 projection theorem

The next proof is geometric and does not use Fourier inversion.

With $\phi_\varepsilon = (2\varepsilon)^{-1}1_{[-\varepsilon, \varepsilon]}$, define the approximate projected density

$$f_{e, \varepsilon} = \phi_\varepsilon * \mu_e.$$

Lemma III.1.5 (Angular strip estimate). *There is an absolute constant C such that for every $z \neq 0$ and $a > 0$,*

$$\sigma(\{e \in S^1 : |e \cdot z| \leq a\}) \leq C \min(1, a/|z|).$$

Proof. By rotation and scaling, it suffices to estimate the normalized length of

$$\{\theta : |\cos \theta| \leq \rho\}, \text{ where } \rho = a/|z|.$$

For $\rho \geq 1$ the bound is trivial. For $0 < \rho < 1$, this set consists of two intervals centered at the zeros $\pi/2$ and $3\pi/2$, each of length comparable to ρ ; for example, the mean value theorem on a fixed neighborhood of each zero gives an upper bound $C\rho$. Normalizing by 2π yields the claim. $\square$

Lemma III.1.6 (Approximate projected-density identity). *Let $\phi_\varepsilon = (2\varepsilon)^{-1}1_{[-\varepsilon, \varepsilon]}$ and $f_{e, \varepsilon} = \phi_\varepsilon * \mu_e$. Then*

$$\|f_{e, \varepsilon}\|_2^2 = \iint K_\varepsilon(e \cdot (x - y)) d\mu(x) d\mu(y),$$

where

$$K_\varepsilon(v) = (2\varepsilon - |v|)_+ / (4\varepsilon^2).$$

In particular,

$$0 \leq K_\varepsilon(v) \leq (2\varepsilon)^{-1}1_{|v| \leq 2\varepsilon}.$$

Proof. Because $f_{e, \varepsilon}$ is the convolution of the finite measure μ_e with ϕ_ε , Tonelli gives

$$\begin{aligned} & \int f_{e, \varepsilon}(u)^2 du \\ &= \iint \int \phi_\varepsilon(u - a) \phi_\varepsilon(u - b) du d\mu_e(a) d\mu_e(b). \end{aligned}$$

The inner integral is $(\phi_\varepsilon * \tilde{\phi}_\varepsilon)(a - b)$, and $\tilde{\phi}_\varepsilon = \phi_\varepsilon$. The convolution of two normalized interval indicators is exactly the displayed triangular kernel. Pulling $a = \pi_e(x)$ and $b = \pi_e(y)$ back through the pushforward proves the identity. $\square$

Theorem III.1.7 (Almost-everywhere L^2 density under finite 1-energy). *Assume μ is finite, compactly supported, and*

$$I_1(\mu) = \iint |x - y|^{-1} d\mu(x) d\mu(y) < \infty.$$

Then for σ -almost every $e \in S^1$, the projected measure μ_e is absolutely continuous with respect to Lebesgue measure and has a density $f_e \in L^2(\mathbb{R})$. Moreover the densities can be chosen jointly measurably so that

$$\int_{S^1} \|f_e\|_2^2 d\sigma(e) \leq CI_1(\mu).$$

Proof. For $\varepsilon > 0$ put $F_\varepsilon(e, u) = f_{e, \varepsilon}(u)$. Lemma III.1.6, Tonelli, and Lemma III.1.5 give

$$\int_{S^1} \int_{\mathbb{R}} |F_\varepsilon(e, u)|^2 du d\sigma(e) \leq CI_1(\mu) \tag{III.1.1}$$

uniformly in ε . Thus $\{F_\varepsilon\}$ is bounded in the Hilbert space $L^2(S^1 \times \mathbb{R}, \sigma \times \mathcal{L}^1)$. Choose $\varepsilon_j \rightarrow 0$ such that F_{ε_j} converges weakly to some F in this Hilbert space. Weak lower semi-continuity yields

$$\|F\|_{L^2(S^1 \times \mathbb{R})}^2 \leq CI_1(\mu). \quad (\text{III.1.2})$$

We identify F fiberwise. Let $a \in C(S^1)$ and $g \in C_c(\mathbb{R})$. By the pushforward definition,

$$\begin{aligned} & \int_{S^1} a(e) \int_{\mathbb{R}} g(u) F_\varepsilon(e, u) du d\sigma(e) \\ &= \int_{S^1} a(e) \int_{\mathbb{R}} (g * \phi_\varepsilon)(v) d\mu_e(v) d\sigma(e). \end{aligned}$$

Since g is uniformly continuous, $g * \phi_\varepsilon \rightarrow g$ uniformly. The total mass of every μ_e equals $\mu(\mathbb{R}^2)$, so the last expression converges to

$$\int_{S^1} a(e) \int_{\mathbb{R}} g(v) d\mu_e(v) d\sigma(e).$$

On the other hand weak convergence gives the limit

$$\int_{S^1} a(e) \int_{\mathbb{R}} g(u) F(e, u) du d\sigma(e).$$

Hence these two quantities agree for every such tensor test function. Choose a countable dense family $\{g_m\}$ in $C_c(\mathbb{R})$ (for the usual inductive family of compact supports). For each m , equality for all continuous a implies

$$\int g_m(u) F(e, u) du = \int g_m d\mu_e$$

for σ -almost every e . Intersecting the countably many full-measure sets and using density gives, for almost every e ,

$$\int g(u) F(e, u) du = \int g d\mu_e \text{ for every } g \in C_c(\mathbb{R}).$$

Therefore $\mu_e = F(e, \cdot) du$ for almost every e . Set $f_e = F(e, \cdot)$ there. Fubini applied to (III.1.2) gives $f_e \in L^2(\mathbb{R})$ for almost every e , and (III.1.2) is exactly the asserted averaged norm bound. $\square$

Theorem III.1.8 (Positive Lebesgue measure of projected support). *Under the assumptions of Theorem III.1.7, if $\mu(\mathbb{R}^2) > 0$, then for almost every e ,*

$$\mathcal{L}^1(\pi_e(\text{supp } \mu)) > 0.$$

Proof. For almost every e , write $\mu_e = f_e du$ with $f_e \in L^2$ and total mass $M = \mu(\mathbb{R}^2) > 0$. Since μ_e is supported on the closed set $\pi_e(\text{supp } \mu)$ when $\text{supp } \mu$ is compact,

$$M = \int_{\pi_e(\text{supp } \mu)} f_e(u) du.$$

Cauchy-Schwarz gives

$$M \leq \mathcal{L}^1(\pi_e(\text{supp } \mu))^{1/2} \|f_e\|_2.$$

Thus $\mathcal{L}^1(\pi_e(\text{supp } \mu)) \geq M^2 / \|f_e\|_2^2 > 0$. $\square$

III.1.5 Arithmetic exact-overlap exceptional direction

Proposition III.1.9 (An arithmetic exact-overlap exceptional direction). *Let*

$$A = \left\{ \sum_{n \geq 1} a_n 4^{-n} : a_n \in \{0, 1\} \right\}$$

and $E = A \times A$. Then $\dim_H(E) = 1$, but for the direction $e = 2^{-1/2}(1, 1)$,

$$\dim_H(\pi_e(E)) = \log 3 / \log 4 < 1.$$

Hence this direction is exceptional for the Marstrand conclusion, and the drop is caused by an arithmetic exact overlap ($0 + 1 = 1 + 0$) in the digit sums.

Proof. The set A is the self-similar attractor of the two similarities

$$x \rightarrow x/4, x \rightarrow (x + 1)/4.$$

Their basic intervals are disjoint, so the open set condition holds and

$$\dim_H(A) = \log 2 / \log 4 = 1/2.$$

The product $E = A \times A$ is the attractor of four similarities of ratio $1/4$ with pairwise disjoint basic rectangles, so

$$\dim_H(E) = \log 4 / \log 4 = 1.$$

Now $\pi_e(x, y) = (x + y)/\sqrt{2}$, and scaling by the nonzero constant $1/\sqrt{2}$ does not change Hausdorff dimension. Because the base is 4 and the largest digit sum is $2 < 4$, addition produces no carries:

$$A + A = \left\{ \sum_{n \geq 1} c_n 4^{-n} : c_n \in \{0, 1, 2\} \right\}.$$

Indeed every c_n is a sum of two digits from $\{0, 1\}$, and conversely each $c_n \in \{0, 1, 2\}$ can be realized digitwise. Thus $A + A$ is the self-similar attractor of the three ratio- $1/4$ similarities corresponding to digits 0, 1, 2. Their basic intervals satisfy the open set condition, so

$$\dim_H(A + A) = \log 3 / \log 4.$$

Therefore $\dim_H(\pi_e(E)) = \log 3 / \log 4 < 1 = \min(\dim_H E, 1)$. □

III.1.6 Slice normalization checks (preparatory, not the slicing theorem)

For $e \in S^1$ and $u \in \mathbb{R}$, put $L(e, u) = \{x : x \cdot e = u\}$ and $S(e, u, \varepsilon) = \{x : |x \cdot e - u| \leq \varepsilon\}$. The approximate slice functional is

$$\mu_{e,u}^\varepsilon(A) = \frac{1}{2\varepsilon} \mu(A \cap S(e, u, \varepsilon)).$$

No limiting slice measure is asserted by this notation.

An admissible slice measure at (e, u) is any weak limit of $\mu_{e,u}^{\varepsilon_j}$ along $\varepsilon_j \downarrow 0$ for which the masses remain bounded. Such a limit is supported on $\text{spt}(\mu) \cap L(e, u)$; uniqueness is not part of the definition.

For $\varepsilon > 0$ define

$$\mu_{e,u}^\varepsilon(A) = (2\varepsilon)^{-1} \mu(A \cap \{|x \cdot e - u| \leq \varepsilon\}).$$

For every nonnegative Borel function h on $\mathbb{R}^2$, Tonelli gives

$$\begin{aligned} & \int_{\mathbb{R}} \left[\int h(x) d\mu_{e,u}^\varepsilon(x) \right] du \\ &= \int h(x) [(2\varepsilon)^{-1} \int_{\mathbb{R}} 1_{|x \cdot e - u| \leq \varepsilon} du] d\mu(x) \\ &= \int h d\mu. \end{aligned}$$

Thus the strip normalization conserves total mass after integration in the slice parameter u . This identity will be the normalization checkpoint when the limiting slice measures are defined. It is **not** yet a claim of existence or uniqueness of such limits.

III.1.7 Energy monotonicity and the planar set theorem

For any directional conclusion $\mathcal{P}(e)$, its exceptional set is $\{e \in S^1 : \mathcal{P}(e) \text{ fails}\}$; here and below, “almost every direction” means with respect to σ .

Lemma III.1.10 (Bounded-support energy monotonicity). *Let $0 < q < t < 2$ and let μ be finite and supported on a compact set of diameter $D > 0$. Then*

$$I_q(\mu) \leq D^{t-q} I_t(\mu).$$

Proof. For $x, y \in \text{supp}(\mu)$ with $x \neq y$,

$$|x - y|^{-q} = |x - y|^{t-q} |x - y|^{-t} \leq D^{t-q} |x - y|^{-t}.$$

Integrate against $\mu \times \mu$. □

Theorem III.1.11 (Marstrand planar projection theorem (analytic-set form)). *Let $E \subset \mathbb{R}^2$ be analytic and write $s = \dim_{\text{H}}(E)$.*

1. If $0 < s \leq 1$, then for σ -almost every e ,

$$\dim_{\text{H}} \pi_e(E) = s.$$

2. If $s > 1$, then for σ -almost every e ,

$$\mathcal{L}^1(\pi_e(E)) > 0.$$

The zero-dimensional and empty cases are immediate.

External/earlier interfaces used. *the analytic compact-approximation theorem supplies compact subsets of analytic E whose dimension approaches $\dim_{\mathbb{H}}(E)$ from below. The compact Frostman/energy measure is then supplied by the Frostman and energy–dimension package from Volume I.*

Proof. First suppose $s \leq 1$. Fix a countable sequence t_j with $0 < t_j < s$ and $t_j \uparrow s$. For each j , choose u_j with $t_j < u_j < s$. By the analytic compact-approximation theorem, choose a compact $K_j \subset E$ with $\dim_{\mathbb{H}}(K_j) > u_j$. The Frostman/energy package from Volume I gives a nonzero compactly supported μ_j on K_j with finite t_j -energy. By [Theorem III.1.2](#), for almost every direction e , $I_{t_j}((\mu_j)_e) < \infty$; by [Theorem III.1.4](#),

$$\dim_{\mathbb{H}} \pi_e(E) \geq \dim_{\mathbb{H}} \text{supp}((\mu_j)_e) \geq t_j.$$

Intersect the countably many full-measure direction sets and let $j \rightarrow \infty$. Then, for almost every e ,

$$\dim_{\mathbb{H}} \pi_e(E) \geq s.$$

Since π_e is 1-Lipschitz, the reverse inequality is automatic. Hence equality holds almost everywhere.

Now suppose $s > 1$. Choose $1 < t < u < s$. By the analytic compact-approximation theorem, choose compact $K \subset E$ with $\dim_{\mathbb{H}}(K) > u$; the Frostman/energy package from Volume I gives μ supported on K with $I_t(\mu) < \infty$. By [Theorem III.1.10](#), $I_1(\mu) < \infty$. [Theorem III.1.8](#) therefore gives

$$\mathcal{L}^1(\pi_e(\text{supp } \mu)) > 0$$

for almost every e ; since $\text{supp } \mu \subset E$, the same holds for $\pi_e(E)$. □

III.1.8 Averaged slice energies

Put $q = t - 1$ for $1 < t < 2$.

Lemma III.1.12 (Approximate slice-energy bound). *Let $1 < t < 2$ and let μ be finite and compactly supported. For the strip measures*

$$\mu_{e,u}^\varepsilon(A) = (2\varepsilon)^{-1} \mu(A \cap \{|x \cdot e - u| \leq \varepsilon\}),$$

one has

$$\int_{S^1} \int_{\mathbb{R}} I_{t-1}(\mu_{e,u}^\varepsilon) du d\sigma(e) \leq C I_t(\mu)$$

for every $\varepsilon > 0$, with an absolute constant C independent of μ and ε .

Proof. Write $q = t - 1$. Expanding the energy and integrating first in u gives

$$\begin{aligned} & \int_{\mathbb{R}} I_q(\mu_{e,u}^\varepsilon) du \\ &= \int \int |x - y|^{-q} K_\varepsilon(e \cdot (x - y)) d\mu(x) d\mu(y), \end{aligned}$$

where the same triangular kernel as in [Theorem III.1.6](#) appears:

$$K_\varepsilon(v) = (2\varepsilon - |v|)_+ / (4\varepsilon^2).$$

Tonelli and the angular strip estimate imply, for $z \neq 0$,

$$\int_{S^1} K_\varepsilon(e \cdot z) d\sigma(e) \leq C|z|^{-1}.$$

Hence

$$\begin{aligned} & \int_{S^1} \int_{\mathbb{R}} I_q(\mu_{e,u}^\varepsilon) du d\sigma(e) \\ & \leq C \int \int |x - y|^{-q-1} d\mu(x) d\mu(y) \\ & = CI_t(\mu). \end{aligned}$$

The diagonal is harmless whenever the right side is finite; otherwise the inequality is interpreted in $[0, \infty]$. $\square$

Lemma III.1.13 (Support and energy of weak slice limits). *Fix $1 < t < 2$, put $q = t - 1$, and let μ be compactly supported. Suppose $\varepsilon_j \rightarrow 0$, the masses $\mu_{e,u}^{\varepsilon_j}(\mathbb{R}^2)$ stay bounded, and*

$$\sup_j I_q(\mu_{e,u}^{\varepsilon_j}) < \infty.$$

Then a subsequence converges weakly to a finite measure ν , and every such limit satisfies

$$\text{supp}(\nu) \subset \text{supp}(\mu) \cap L(e, u)$$

and

$$I_q(\nu) \leq \liminf_j I_q(\mu_{e,u}^{\varepsilon_j}).$$

If in addition the masses converge to $m > 0$, then ν is nonzero and has mass m .

Proof. All measures are supported on the fixed compact set $\text{supp}(\mu)$. Bounded mass therefore gives weak sequential compactness. If an open set O has closure disjoint from $L(e, u)$, then $|x \cdot e - u|$ is bounded below by a positive constant on $O \cap \text{supp}(\mu)$, so $\mu_{e,u}^{\varepsilon_j}(O) = 0$ for all large j ; hence the limit is supported on the fiber.

For $M > 0$ define the bounded continuous kernel

$$k_M(x, y) = \min(M, |x - y|^{-q}),$$

with value M on the diagonal. Weak convergence on the compact support implies convergence of the product measures, hence

$$\begin{aligned} \int \int k_M d\nu d\nu &= \lim_{j \rightarrow \infty} \int \int k_M d\mu_{e,u}^{\varepsilon_j} d\mu_{e,u}^{\varepsilon_j} \\ &\leq \liminf_j I_q(\mu_{e,u}^{\varepsilon_j}). \end{aligned}$$

Let $M \rightarrow \infty$ and use monotone convergence. Convergence of total masses follows by testing against the constant function 1. $\square$

Theorem III.1.14 (Finite-energy slicing theorem). *Let μ be a nonzero finite compactly supported Borel measure on $\mathbb{R}^2$, and let $1 < t < 2$ with $I_t(\mu) < \infty$. Then for σ -almost every direction e , the projected measure μ_e is absolutely continuous and there is a Borel set $U_e \subset \mathbb{R}$ with $\mu_e(\mathbb{R} \setminus U_e) = 0$ and positive Lebesgue measure such that for every $u \in U_e$ there exists a nonzero finite Borel measure $\nu_{e,u}$ satisfying*

$$\text{supp}(\nu_{e,u}) \subset \text{supp}(\mu) \cap L(e, u)$$

and

$$I_{t-1}(\nu_{e,u}) < \infty.$$

Consequently,

$$\dim_H(\text{supp}(\mu) \cap L(e, u)) \geq t - 1$$

for almost every $u \in U_e$.

Proof. By [Theorem III.1.10](#), $I_1(\mu) < \infty$, so [Theorem III.1.7](#) yields for almost every e an L^2 density f_e of μ_e . Fix the dyadic sequence $\varepsilon_n = 2^{-n}$. Lemma [III.1.12](#) and Fatou give

$$\int_{S^1} \int_{\mathbb{R}} \liminf_n I_{t-1}(\mu_{e,u}^{\varepsilon_n}) du d\sigma(e) \leq C I_t(\mu).$$

Thus for almost every e , the displayed liminf is finite for Lebesgue-almost every u .

For such a direction, because $\mu_e = f_e du$,

$$\mu_{e,u}^\varepsilon(\mathbb{R}^2) = (2\varepsilon)^{-1} \mu_e([u - \varepsilon, u + \varepsilon]).$$

At every Lebesgue point of f_e , the Lebesgue differentiation theorem gives convergence of this quantity to $f_e(u)$ as $\varepsilon \rightarrow 0$. Since μ is nonzero, $\int f_e = \mu(\mathbb{R}^2) > 0$, so the set $\{u : f_e(u) > 0\}$ has positive Lebesgue measure. Remove from $\{u : f_e(u) > 0\}$ the null set of non-Lebesgue points and the null set where the slice-energy liminf is infinite; call the remainder U_e . Since $\mu_e = f_e du$, its complement has μ_e -measure zero, and because μ_e is nonzero, U_e has positive Lebesgue measure.

Fix $u \in U_e$. Choose a subsequence ε_{n_j} on which the slice energies are bounded. The masses converge to $f_e(u) > 0$. Lemma [III.1.13](#) produces a nonzero weak limit $\nu_{e,u}$ supported on the fiber, with finite $(t-1)$ -energy. [Theorem III.1.4](#) applied in the fiber ambient space gives the stated dimension lower bound. $\square$

III.1.9 Set-level slicing consequences

Theorem III.1.15 (Subcritical Marstrand slicing). *Let $E \subset \mathbb{R}^2$ be analytic with $\dim_H(E) = s > 1$. For every t with $1 < t < s$, for σ -almost every e there is a set $U_{e,t}$ of positive Lebesgue measure such that*

$$\dim_H(E \cap L(e, u)) \geq t - 1$$

for Lebesgue-almost every $u \in U_{e,t}$.

Proof. Choose u with $t < u < s$. Use the analytic compact-approximation theorem to choose compact $K \subset E$ with $\dim_H(K) > u$; then use the Frostman/energy results from Volume I to choose a nonzero compactly supported μ on K with $I_t(\mu) < \infty$, and apply [Theorem III.1.14](#). $\square$

Lemma III.1.16 (Fixed-direction slicing upper integral). *Let $1 < s < 2$ and let $E \subset \mathbb{R}^2$ be $\mathcal{H}^s$ -measurable with $\mathcal{H}^s(E) < \infty$. For every fixed $e \in S^1$,*

$$\int_{\mathbb{R}} \mathcal{H}^{s-1}(E \cap L(e, u)) du \leq C_s \mathcal{H}^s(E)$$

(up to the multiplicative normalization chosen for Hausdorff measure). In particular,

$$\dim_H(E \cap L(e, u)) \leq s - 1$$

for Lebesgue-almost every u .

Proof. Fix $\delta > 0$ and cover E by balls $B(x_i, r_i)$ with $2r_i < \delta$. For each u , the slice $E \cap L(e, u)$ is covered by the sets $B(x_i, r_i) \cap L(e, u)$ for those indices satisfying $|x_i \cdot e - u| \leq r_i$; each such slice has diameter at most $2r_i$. Hence the $(s - 1)$ -dimensional δ -Hausdorff content of the slice is bounded by

$$\sum_i (2r_i)^{s-1} 1_{|x_i \cdot e - u| \leq r_i}.$$

Integrating in u gives at most

$$\sum_i (2r_i)^{s-1} (2r_i) = \sum_i (2r_i)^s.$$

Take the infimum over δ -covers and then let $\delta \downarrow 0$; monotone convergence on the left yields the asserted inequality. Finiteness of $\mathcal{H}^{s-1}$ implies the dimension upper bound. $\square$

Theorem III.1.17 (Exact slicing on a positive-offset family). *Let $1 < s < 2$ and let $E \subset \mathbb{R}^2$ be Borel with*

$$0 < \mathcal{H}^s(E) < \infty.$$

Use the Frostman/energy interface from Volume I to choose a nonzero compactly supported measure μ on a positive-measure compact piece of E with

$$\mu(B(x, r)) \leq Cr^s \text{ for all } x, r.$$

Then for σ -almost every e there is a Borel set $U_e \subset \mathbb{R}$ with positive Lebesgue measure such that for Lebesgue-almost every $u \in U_e$,

$$\dim_H(E \cap L(e, u)) = s - 1$$

and

$$\mathcal{H}^{s-1}(E \cap L(e, u)) < \infty.$$

Proof. The Frostman bound implies $I_t(\mu) < \infty$ for every $0 < t < s$ by the Frostman-to-energy implication from Volume I. Choose a countable sequence $t_j \uparrow s$, with $1 < t_j < s$. Apply the proof of [Theorem III.1.14](#) to this single measure μ for every t_j . The projection

density f_e is the same for all j ; after intersecting the countably many full-measure direction sets and null offset sets, we obtain for almost every e that for almost every u with $f_e(u) > 0$, $\dim_{\mathbb{H}}(E \cap L(e, u)) \geq t_j - 1$ for every j .

Thus the lower bound is $s - 1$ on the positive-measure set $\{f_e > 0\}$ modulo a null set. Lemma III.1.16 gives the reverse inequality and finiteness of $\mathcal{H}^{s-1}$ for almost every offset. Intersecting these properties yields a positive-measure set U_e on which equality holds almost everywhere. $\square$

Scope note. Mattila, Chapter 10, Theorem 10.11 records a stronger point-through- x formulation. The point-through- x formulation is stronger than the positive-offset formulation proved above and is recorded here as a historical strengthening rather than being inferred from the preceding argument.

Chapter III.2

Kaufman–Mattila Exceptional-Set Estimates

Marstrand’s theorem [1] asserts that bad directions have measure zero. Kaufman–Mattila theory [2, 3] asks the sharper question: what is the Hausdorff dimension of the exceptional set itself? In higher dimensions the parameter space is the Grassmannian, and the geometry of the Schubert locus where a fixed vector collapses governs the answer.

The chapter first derives the singular-direction energy estimate from a Frostman measure on the Grassmannian. It then treats the two natural regimes separately: dimension drop when the source dimension does not exceed the projection dimension, and zero-Lebesgue-measure projections above that threshold. The most general family-of-linear-maps theorem is recorded as a designated external interface rather than used as hidden proof authority.

Prerequisites and source boundary. Frostman and energy inputs come from the preceding chapter and Volume I. The $s > m$ branch is proved locally by a complementary-Schubert Frostman estimate and a Fourier L^2 averaging argument. The stronger Peres–Schlag Sobolev-dimension theorem is included for comparison. The sharp higher-dimensional witness uses the Kaufman–Mattila transformation theorem solely to certify sharpness.

III.2.1 Grassmannian geometry

Definition (Grassmannian). For $1 \leq m < n$, $G(n, m)$ is the compact manifold of m -dimensional linear subspaces of $\mathbb{R}^n$; P_V denotes orthogonal projection onto V .

Definition (Grassmannian metric and invariant measure). We use $d_G(V, W) = \|P_V - P_W\|_{\text{op}}$ and write $\gamma_{n,m}$ for the orthogonally invariant probability measure on $G(n, m)$. Its manifold dimension is $q = m(n - m)$.

Definition (Schubert incidence set). For $z \in S^{n-1}$, $Z_z = \{V \in G(n, m) : P_V z = 0\} = \{V : V \subset z^\perp\}$. Thus $Z_z \cong G(n - 1, m)$ and $\dim Z_z = q - m$.

Definition (Frostman measure on a direction family). A finite Borel measure ν on $G(n, m)$ is β -Frostman if $\nu(B_G(V, r)) \leq C_\nu r^\beta$ for all V and $0 < r \leq 1$.

III.2.2 Geometry of the bad-incidence Schubert set

Lemma III.2.1 (The zero-projection locus). *For every unit vector $z \in S^{n-1}$, the set*

$$Z_z = \{V \in G(n, m) : P_V z = 0\}$$

is naturally isometric, up to the fixed Grassmannian metric equivalence, to $G(n-1, m)$. Hence

$$\dim Z_z = q - m = m(n - m - 1).$$

Moreover there is $C = C(n, m)$ such that, whenever $0 < \rho < 1/2$,

$$\{V : |P_V z| \leq \rho\} \subset N_{C\rho}(Z_z).$$

Proof. The equality $P_V z = 0$ is equivalent to $V \subset z^\perp$, so $Z_z = G(z^\perp, m)$, which is a copy of $G(n-1, m)$. Its dimension is $m((n-1) - m) = q - m$.

Now assume $|P_V z| \leq \rho$. For every unit $v \in V$,

$$|v \cdot z| = |v \cdot P_V z| \leq \rho.$$

Let $Q = P_{z^\perp}|_V$. Then

$$|Qv|^2 = |v|^2 - |v \cdot z|^2 \geq (1 - \rho^2)|v|^2,$$

so Q is injective and $W = Q(V)$ is an m -plane contained in $z^\perp$, hence $W \in Z_z$. Also

$$|v - Qv| = |v \cdot z| \leq \rho|v|.$$

The standard gap-metric estimate between two subspaces now yields

$$\|P_V - P_W\| \leq C\rho.$$

Thus $d_G(V, Z_z) \leq C\rho$. □

Lemma III.2.2 (Frostman transversality near Z_z). *Let ν be a β -Frostman measure on $G(n, m)$ and suppose*

$$\beta > q - m.$$

Put $a = \beta - (q - m) > 0$. Then there is C depending only on n, m, β and the Frostman constant of ν such that for every unit z and $0 < \rho < 1$,

$$\nu(\{V : |P_V z| \leq \rho\}) \leq C\rho^a.$$

Proof. Write

$$k = q - m = m(n - m - 1) = \dim Z_z.$$

We first make the tubular covering estimate explicit and uniform in z . Orthogonal transformations act isometrically on $G(n, m)$ for the metric $d_G(V, W) = \|P_V - P_W\|_{\text{op}}$ and act transitively on S^{n-1} . It is therefore enough to establish the covering estimate for $z = e_n$; the same constants then work for every unit vector z .

Set $H = e_n^\perp$. Thus $Z_{e_n} = G(H, m)$. Fix $E \in G(H, m)$ and put $F = H \cap E^\perp$. On the standard Grassmann chart about E , every plane sufficiently close to E has the form

$$\Gamma(A) = \{u + Au : u \in E\}, \quad A \in \text{Hom}(E, F).$$

The parameter space $\text{Hom}(E, F)$ has dimension $m(n - m - 1) = k$. On a fixed bounded ball in this parameter space, the map $A \mapsto P_{\Gamma(A)}$ is smooth with bounded derivative. After shrinking the chart once, its inverse onto the image is also Lipschitz. Consequently there are constants $c_E, C_E > 0$ such that, for A, A' in that chart,

$$c_E \|A - A'\| \leq d_G(\Gamma(A), \Gamma(A')) \leq C_E \|A - A'\|. \quad (2.1a)$$

Here any fixed Euclidean norm on the finite-dimensional space $\text{Hom}(E, F)$ may be used.

Compactness of $G(H, m)$ gives finitely many such charts, and their parameter domains may be taken bounded. A bounded subset of $\mathbb{R}^k$ is coverable by $O(r^{-k})$ Euclidean balls of radius r for $0 < r < r_0$. Transporting those balls through the finitely many bi-Lipschitz charts yields the uniform metric-entropy estimate

$$N_r(Z_{e_n}; d_G) \leq C_0 r^{-k}, \quad 0 < r < r_0. \quad (2.1b)$$

By orthogonal invariance, (2.1b) holds with the same C_0, r_0 for every Z_z .

Lemma 2.1 supplies a constant $C_1 = C_1(n, m)$ such that

$$\{V : |P_V z| \leq \rho\} \subset N_{C_1 \rho}(Z_z).$$

For $0 < \rho < r_0/(2C_1 + 2)$, cover Z_z by at most $C_0 \rho^{-k}$ balls of radius ρ . Every point of $N_{C_1 \rho}(Z_z)$ then belongs to one of the concentric balls of radius $(C_1 + 1)\rho$. Hence the Frostman bound gives

$$\begin{aligned} \nu\{V : |P_V z| \leq \rho\} &\leq C_0 \rho^{-k} C_\nu (C_1 + 1)^\beta \rho^\beta \\ &\leq C \rho^{\beta-k} = C \rho^{\beta-(q-m)} = C \rho^a. \end{aligned}$$

For the remaining range of ρ , compactness of $G(n, m)$ and the Frostman condition give a uniform bound for $\nu(G(n, m))$; after enlarging C , this is also bounded by $C \rho^a$. The estimate is therefore valid for all $0 < \rho < 1$, uniformly in z . $\square$

Primary-source comparison. *The argument above is deliberately local: it proves the Schubert-neighborhood covering needed by the energy method and does not import a projection theorem. It isolates the Grassmannian codimension- m geometry that underlies the classical exceptional-set framework of Kaufman–Mattila [3] and the later Grassmannian Sobolev estimates of Peres–Schlag [4].*

Lemma III.2.3 (Averaged singular projection kernel). *Under the hypotheses of Lemma 2.2, if*

$$0 < t < a = \beta - (q - m),$$

then for every $x \neq y$ in $\mathbb{R}^n$,

$$\int_{G(n, m)} |P_V(x - y)|^{-t} d\nu(V) \leq C |x - y|^{-t}.$$

Proof. Put $z = (x - y)/|x - y|$ and $X(V) = |P_V z| \in [0, 1]$. The layer-cake identity gives

$$X^{-t} = 1 + t \int_X^1 r^{-t-1} dr.$$

Integrating and using Lemma 2.2,

$$\begin{aligned} & \int X(V)^{-t} d\nu(V) \\ & \leq \nu(G(n, m)) + Ct \int_0^1 r^{a-t-1} dr < \infty \end{aligned}$$

because $a > t$. Rescaling by $|x - y|^{-t}$ gives the claim. $\square$

Lemma III.2.4 (Borel measurability of compact-set exceptional families). *Let $K \subset \mathbb{R}^n$ be compact and $0 < t < m$. Then*

$$E_t(K) = \{V \in G(n, m) : \dim_H P_V(K) < t\}$$

is a Borel subset of $G(n, m)$.

Proof. The map $V \rightarrow P_V(K)$ is continuous from $G(n, m)$ to the hyperspace of nonempty compact subsets of a sufficiently large closed ball, equipped with the Hausdorff metric: indeed

$$d_H(P_V K, P_W K) \leq \|P_V - P_W\| \sup_{x \in K} |x|.$$

Fix rational $r > 0$ and $\delta > 0$. The condition that a compact set F admits a finite cover by balls of radii $< \delta$ with sum of the r th powers of the diameters less than a prescribed rational number is open in the Hausdorff topology after an arbitrarily small enlargement of those balls. Using only rational centers, radii and bounds gives a countable description of the conditions $\mathcal{H}^r(F) = 0$. Hence

$$\{V : \mathcal{H}^r(P_V K) = 0\}$$

is Borel. Finally,

$$E_t(K) = \bigcup_{r \in \mathbb{Q}, 0 < r < t} \{V : \mathcal{H}^r(P_V K) = 0\},$$

because $\dim_H F < t$ iff $\mathcal{H}^r(F) = 0$ for some $r < t$. Thus $E_t(K)$ is Borel. $\square$

III.2.3 Projected energies against a singular family of directions

For a finite Borel measure μ on $\mathbb{R}^n$, set $\mu_V = (P_V)_\# \mu$; for $0 < t < m$, define

$$I_t(\mu_V) = \iint |u - v|^{-t} d\mu_V(u) d\mu_V(v).$$

For $A \subset \mathbb{R}^n$ let

$$E_t(A) = \{V \in G(n, m) : \dim_H P_V(A) < t\}.$$

When $s = \dim_H A \leq m$, we also write $E(A) = \{V : \dim_H P_V(A) < s\}$.

Theorem III.2.5 (Kaufman energy averaging for a Frostman direction measure). *Let μ be a finite compactly supported Borel measure on $\mathbb{R}^n$, let $0 < t < m$, and let ν be a β -Frostman measure on $G(n, m)$ with*

$$\beta > q - m + t.$$

Then

$$\int_{G(n, m)} I_t(\mu_V) d\nu(V) \leq C I_t(\mu).$$

In particular, if $I_t(\mu) < \infty$, then $I_t(\mu_V) < \infty$ for ν -almost every V .

Proof. By the pushforward identity and Tonelli,

$$\begin{aligned} & \int I_t(\mu_V) d\nu(V) \\ &= \int \int \int [|P_V(x - y)|^{-t} d\nu(V)] d\mu(x) d\mu(y). \end{aligned}$$

The exponent assumption is exactly $t < \beta - (q - m)$, so Lemma 2.3 bounds the inner bracket by $C|x - y|^{-t}$. Integration gives the stated estimate. $\square$

Theorem III.2.6 (Parameterized Kaufman–Mattila exceptional-set bound). *Let $A \subset \mathbb{R}^n$ be analytic with $\dim_H A = s$, and let*

$$0 < t < \min\{s, m\}.$$

Then

$$\dim_H E_t(A) \leq q - m + t = m(n - m - 1) + t,$$

where

$$E_t(A) = \{V \in G(n, m) : \dim_H P_V(A) < t\}.$$

Proof. Choose a compact set $K \subset A$ with $\dim_H K > t$; this uses the declared analytic compact-selection interface. Since $P_V K \subset P_V A$,

$$E_t(A) \subset E_t(K).$$

It is therefore enough to bound the compact case.

Assume for contradiction that

$$\dim_{\mathbb{H}} E_t(K) > q - m + t.$$

Choose

$$q - m + t < \beta < \dim_{\mathbb{H}} E_t(K).$$

By the external Frostman interface there is a nonzero finite β -Frostman measure ν supported on a compact subset $F \subset E_t(K)$. Since $\dim_{\mathbb{H}} K > t$, the exact earlier Frostman/energy interface supplies a nonzero finite measure μ supported on K with $I_t(\mu) < \infty$.

Theorem 2.5 implies $I_t(\mu_V) < \infty$ for ν -almost every V . Finite t -energy forces

$$\dim_{\mathbb{H}} \text{supp}(\mu_V) \geq t.$$

But $\text{supp}(\mu_V) \subset P_V(K)$, whereas every $V \in F \subset E_t(K)$ satisfies $\dim_{\mathbb{H}} P_V(K) < t$. This is a contradiction. $\square$

Corollary III.2.7 (Full dimension-drop exceptional set when $s \leq m$). *Let $A \subset \mathbb{R}^n$ be analytic and $0 < s = \dim_{\mathbb{H}} A \leq m$. Then*

$$\begin{aligned} \dim_{\mathbb{H}} \{V \in G(n, m) : \dim_{\mathbb{H}} P_V(A) < s\} \\ \leq s + m(n - m - 1). \end{aligned}$$

In the planar line-projection case $(n, m) = (2, 1)$, this becomes

$$\dim_{\mathbb{H}} \{e : \dim_{\mathbb{H}} \pi_e(A) < s\} \leq s,$$

which is Kaufman's 1968 exceptional-direction estimate.

Proof. Choose $t_j \uparrow s$. If $\dim_{\mathbb{H}} P_V(A) < s$, then $\dim_{\mathbb{H}} P_V(A) < t_j$ for some j . Hence the full exceptional set is contained in $\bigcup_j E_{t_j}(A)$. Theorem 2.6 gives

$$\dim_{\mathbb{H}} E_{t_j}(A) \leq m(n - m - 1) + t_j.$$

Take the supremum over j . $\square$

III.2.4 Positive-measure projections in the supercritical range

For analytic $A \subset \mathbb{R}^n$ with $\dim_{\mathbb{H}} A > m$, the relevant exceptional set is

$$E^0(A) = \{V \in G(n, m) : \mathcal{L}^m(P_V(A)) = 0\}.$$

Definition III.2.8 (Sobolev dimension). *For a finite compactly supported Borel measure λ on $\mathbb{R}^m$, set*

$$\dim_S(\lambda) = \sup \left\{ \sigma : \int |\hat{\lambda}(\xi)|^2 (1 + |\xi|)^{\sigma-m} d\xi < \infty \right\}.$$

Finite a -energy implies $\dim_S(\lambda) \geq a$, and $\dim_S(\lambda) > m$ implies absolute continuity with an L^2 density. The definition is retained because it is useful throughout the volume, although the positive-measure theorem below no longer requires an external Sobolev-dimension estimate.

A stronger comparison theorem. *Peres–Schlag [4], Proposition 6.1, gives the stronger estimate*

$$\dim_H\{V \in G(n, m) : \dim_S(\mu_V) < \sigma\} \leq m(n - m) + \sigma - \dim_S(\mu).$$

This stronger statement is included for comparison, but it is not a premise of the chapter. Theorem 2.10 below is proved locally by the Fourier argument in Proposition 2.9.

Proposition III.2.9 (Local L^2 averaging over a singular Grassmannian family). *Let $q = m(n - m)$ and let $0 < a < n$. Suppose μ is a finite compactly supported Borel measure on $\mathbb{R}^n$ with $I_a(\mu) < \infty$. Let ν be a τ -Frostman measure on $G(n, m)$. If*

$$\tau > q + m - a, \tag{2.9a}$$

then for ν -almost every V the projected measure $\mu_V = (P_V)_\# \mu$ is absolutely continuous with respect to $\mathcal{L}^m \upharpoonright_V$ and has an L^2 density. More quantitatively,

$$\int_{G(n, m)} \left\| \frac{d\mu_V}{d\mathcal{L}^m \upharpoonright_V} \right\|_2^2 d\nu(V) \lesssim \mu(\mathbb{R}^n)^2 + I_a(\mu). \tag{2.9b}$$

The implicit constant depends on n, m, a, τ , the Frostman constant of ν , and a fixed compact neighbourhood of $\text{spt } \mu$.

Proof. Put

$$r_0 = (m - 1)(n - m), \quad \alpha = \tau - r_0.$$

Condition (2.9a) implies $\alpha > n - a > 0$, since

$$q + m - r_0 = m(n - m) + m - (m - 1)(n - m) = n. \tag{2.9c}$$

We first prove the geometric estimate that replaces the imported Sobolev theorem.

Step 1: planes nearly containing a fixed direction. For $e \in S^{n-1}$ let

$$Y_e = \{V \in G(n, m) : e \in V\}.$$

Then $Y_e \cong G(n - 1, m - 1)$ and hence $\dim Y_e = r_0$. We claim that, uniformly in e and $0 < \rho < 1$,

$$\nu\{V : |P_{V^\perp} e| \leq \rho\} \lesssim \rho^{\tau - r_0} = \rho^\alpha. \tag{2.9d}$$

Indeed, if $|P_{V^\perp} e| \leq \rho < 1/2$, then $u = P_V e$ satisfies $|u| \geq (1 - \rho^2)^{1/2}$. With $\hat{u} = u/|u|$, the angle between $\hat{u}$ and e is $O(\rho)$. Choose an orthogonal map R , equal to the identity on the

orthogonal complement of $\text{span}\{e, \hat{u}\}$, which sends $\hat{u}$ to e . Then $\|R - I\| \lesssim \rho$ and $W = RV$ belongs to Y_e . Since $P_W = RP_V R^{-1}$,

$$d_G(V, W) = \|P_V - P_W\|_{op} \lesssim \rho.$$

Thus $\{V : |P_{V^\perp} e| \leq \rho\} \subset N_{C\rho}(Y_e)$. Exactly as in the finite-chart proof of Lemma 2.2, compactness and the smooth r_0 -dimensional manifold structure of Y_e give a uniform cover by $O(\rho^{-r_0})$ Grassmannian balls of radius ρ . The τ -Frostman bound yields (2.9d); the range $1/2 \leq \rho < 1$ is absorbed by enlarging the constant.

Consequently, for $z \neq 0$ and $0 < t \leq |z|$,

$$\nu\{V : d(z, V) \leq t\} = \nu\left\{V : \left|P_{V^\perp} \frac{z}{|z|}\right| \leq \frac{t}{|z|}\right\} \lesssim \left(\frac{t}{|z|}\right)^\alpha. \quad (2.9e)$$

Step 2: a Fourier-incidence kernel. Choose N so large that $M := N - m > \alpha + 1$, and define

$$K(z) = \int_{G(n,m)} \int_V (1 + |w - z|)^{-N} d\mathcal{L}_V^m(w) d\nu(V).$$

For fixed V , writing $z = P_V z + P_{V^\perp} z$ and translating inside V gives

$$\int_V (1 + |w - z|)^{-N} dw \lesssim (1 + d(z, V))^{-M}. \quad (2.9f)$$

If $R = |z| \geq 2$, split the planes into the sets $d(z, V) < 1$ and $2^j \leq d(z, V) < 2^{j+1}$, $j \geq 0$. Using (2.9e),

$$K(z) \lesssim R^{-\alpha} + \sum_{2^j \leq R} 2^{-jM} \left(\frac{2^{j+1}}{R}\right)^\alpha \lesssim R^{-\alpha}, \quad (2.9g)$$

because $M > \alpha$. For $|z| < 2$ the same kernel is uniformly bounded, so

$$K(z) \lesssim (1 + |z|)^{-\alpha}. \quad (2.9h)$$

Step 3: Fourier smoothing and L^2 projection density. Choose $\varphi \in C_c^\infty(\mathbb{R}^n)$ equal to 1 on a neighbourhood of $\text{spt } \mu$. Then $\hat{\mu} = \hat{\varphi} * \hat{\mu}$. Since $\hat{\varphi}$ is Schwartz, Cauchy–Schwarz in the form $|\int f g|^2 \leq (\int |f|)(\int |f| |g|^2)$ gives, after increasing N if needed,

$$|\hat{\mu}(w)|^2 \lesssim \int_{\mathbb{R}^n} (1 + |w - z|)^{-N} |\hat{\mu}(z)|^2 dz. \quad (2.9i)$$

Hence Tonelli and (2.9h) imply

$$\begin{aligned} \int_{G(n,m)} \int_V |\hat{\mu}(w)|^2 dw d\nu(V) &\lesssim \int_{\mathbb{R}^n} |\hat{\mu}(z)|^2 K(z) dz \\ &\lesssim \int |\hat{\mu}(z)|^2 (1 + |z|)^{-\alpha} dz. \end{aligned} \quad (2.9j)$$

Since $\alpha > n - a$, the high-frequency weight satisfies $(1 + |z|)^{-\alpha} \lesssim |z|^{a-n}$ for $|z| \geq 1$. The low-frequency part is bounded by $\mu(\mathbb{R}^n)^2$. The authorized Fourier representation of Riesz energy therefore yields

$$\int_{G(n,m)} \int_V |\hat{\mu}(w)|^2 dw d\nu(V) \lesssim \mu(\mathbb{R}^n)^2 + I_a(\mu) < \infty. \quad (2.9k)$$

For ν -almost every V , the restriction $\hat{\mu}|_V$ is thus in $L^2(V)$. Under an orthonormal identification $V \cong \mathbb{R}^m$ this is the Fourier transform of μ_V . The standard L^2 Fourier criterion (equivalently, apply Plancherel to mollifications and pass weakly) shows that μ_V has an L^2 density, and (2.9b) follows from (2.9k). $\square$

Why this is enough. Proposition 2.9 is a singular-family version of the classical Falconer Fourier argument: the new geometric input is the codimension- $(n - m)$ Schubert incidence $Y_e = \{V : e \in V\}$. The later arguments use this conclusion directly and do not require the stronger Sobolev-dimension theorem.

Theorem III.2.10 (Exceptional set of zero-measure projections when $s > m$). *Let $A \subset \mathbb{R}^n$ be analytic and put $s = \dim_H A > m$. Then*

$$\dim_H \{V \in G(n, m) : \mathcal{L}^m(P_V(A)) = 0\} \leq m(n - m) + m - s.$$

Proof. Fix

$$m < a < \min\{s, n\};$$

when $s = n$, simply take any $m < a < n$. By analytic compact selection and the Frostman/energy interface, choose a compact $K \subset A$ and a nonzero finite measure μ supported on K such that $I_a(\mu) < \infty$.

Set

$$F_K = \{V \in G(n, m) : \mathcal{L}^m(P_V K) = 0\}.$$

This is Borel. Indeed $V \mapsto P_V K$ is continuous in the Hausdorff metric, and Lebesgue measure is upper semicontinuous on compact subsets under Hausdorff convergence; hence $F_K = \bigcap_{j \geq 1} \{V : \mathcal{L}^m(P_V K) < 1/j\}$ is a G_δ set.

Suppose for contradiction that

$$\dim_H F_K > q + m - a, \quad q = m(n - m).$$

Choose

$$q + m - a < \tau < \dim_H F_K$$

and a nonzero τ -Frostman measure ν supported on a compact subset of F_K . Proposition 2.9 gives an L^2 density for μ_V for ν -almost every V . But every such V belongs to F_K , while $\text{spt } \mu_V \subset P_V K$ has m -dimensional Lebesgue measure zero. A nonzero absolutely continuous measure cannot be supported on a null set. This contradiction proves

$$\dim_H F_K \leq q + m - a.$$

Finally,

$$\{V : \mathcal{L}^m(P_V A) = 0\} \subset F_K,$$

so the same bound holds for the exceptional family of A . Letting $a \uparrow s$ proves the theorem. $\square$

The exponent agrees with Falconer's classical Fourier estimate and with the standard higher-dimensional formulation of the positive-measure exceptional projection theorem.

III.2.5 Sharpness and the general-transformation boundary

Theorem III.2.11 (Kaufman–Mattila sharpness example). *For integers $0 < m < n$ and every real $0 < s \leq m$, Kaufman and Mattila construct a compact set $E \subset \mathbb{R}^n$ with $\dim_{\text{H}} E = s$ such that*

$$\begin{aligned} \dim_{\text{H}} \{V \in G(n, m) : \dim_{\text{H}} P_V(E) < s\} \\ = s + m(n - m - 1). \end{aligned}$$

Thus the exponent in Corollary 2.6 is sharp in the full higher-dimensional Grassmannian problem.

Source note. *This sharpness example is imported from Kaufman–Mattila [3], Theorem 5. Its Diophantine construction is not reproduced here; the preceding upper estimate is proved locally.*

Imported theorem (General Kaufman–Mattila transformation theorem). *The chapter’s designated B interface is Proposition 3 of Kaufman–Mattila (1975): for a Borel family $M \subset \text{Hom}(\mathbb{R}^n, \mathbb{R}^k)$ satisfying the paper’s kernel-covariance, Lipschitz perturbation, uniform fiber-Hausdorff-measure, and lower fiber-density hypotheses with exponent v , every Suslin set $E \subset \mathbb{R}^n$ of Hausdorff dimension s satisfies*

$$\dim_{\text{H}} f(E) = s$$

for $\mathcal{H}^{s+v}$ -almost every $f \in M$.

The exact hypotheses are recorded in the source note for this chapter. This general result is not used downstream unless the required specialization is proved locally or supplied through an explicitly stated interface.

III.2.6 What the two regimes show

The two regimes together describe both dimension drop below the projection threshold and zero-measure exceptional projections above it. The Kaufman–Mattila example shows that the subcritical exponent is sharp. The general family-of-linear-transformations proposition is recorded separately because its full hypotheses are broader than what is needed in this volume.

Chapter III.3

Entropy, CP-Chains, and Projection Theorems

Energy methods average over directions. Entropy methods instead analyze the measure itself across many scales. CP-chains package this multiscale information into a stationary process whose typical sceneries encode the local geometry of the original measure. Hochman–Shmerkin’s projection method [5] then converts finite-scale entropy information into robust projection theorems and nonlinear C^1 image results.

This chapter builds the bridge from generated sceneries to the projection function, proves the required semicontinuity and C^1 perturbation statements, and closes with Furstenberg’s dimension-conservation mechanism [8] for ergodic CP shifts. The Hochman–Shmerkin local tree-entropy theorem, finite-scale entropy stability, faithful-map estimates, and Lipschitz-to-tree-morphism machinery are reconstructed in full below. Thus the projection spine is local relative only to the earlier Marstrand measure-projection theorem and standard martingale/ergodic foundations. The triangular Maker average needed in the dimension-conservation branch is also proved locally below from Birkhoff’s theorem and an elementary tail estimate.

Prerequisites and source boundary. All entropy, tree, map, and triangular-average tools are local. Imports: Chapter III.1 for Marstrand measure projection, Volume I for the final mass-distribution corollary, and standard martingale/Birkhoff/disintegration facts. No later-volume theorem is used backwards.

III.3.1 CP-chain state space and dynamics

Definition (boxes, normalized copies, restriction and rescaling). *A box is a Cartesian product of positive-length intervals, with a fixed half-open boundary convention when boxes form a partition. For a box B , let B^* be its normalized translate/dilate and $T_B : B \rightarrow B^*$ the associated orientation-preserving affine similarity. If $\mu(B) > 0$, set $\mu_B = \mu|_B/\mu(B)$ and $\mu^B = (T_B)_\# \mu_B$.*

Definition (ρ -regular filtration and generation). *A refining filtration $F = (F_n)$ is ρ -regular when level- n cells have diameter comparable to ρ^n with uniformly bounded eccentricity. Writing $F_n(x)$ for the cell containing x and $\mu^{x,n} = \mu^{F_n(x)}$, we say that μ generates a distribution*

P along F if $(\mu^{x \cdot n})_{n \geq 0}$ is strongly generic for P for μ -almost every x . Here a sequence (z_n) is strongly generic for P if

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{z_n} \longrightarrow P$$

weakly and, for every integer $q \geq 2$ and every residue $i \in \{0, \dots, q-1\}$, the empirical measures

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{z_{qn+i}}$$

have a weak limit, denoted $Q_{q,i}$. Thus strong genericity records the ordinary limiting distribution and the limiting distribution along every arithmetic progression; no transition map on the state space is implicit in this definition.

Definition (partition operator). A partition operator Δ assigns, equivariantly under translations and scalings, a partition ΔB to each admissible box B . Define $\Delta^0(B) = \{B\}$ and iterate by refining every cell. It is ρ -regular when these iterates form a ρ -regular filtration.

Definition (CP-chain and adapted transition). For a ρ -regular partition operator, a state is a pair (μ, B) with μ a probability measure on B^* . The next cell $C \in \Delta(B^*)$ is chosen with probability $\mu(C)$ and the next state is (μ^C, C) . A CP-chain is a stationary Markov process with this transition; “adapted” refers precisely to this mass-weighted choice.

Definition (stationarity and ergodicity). The state distribution P is stationary when invariant under the CP transition kernel and ergodic when the corresponding stationary Markov shift is ergodic. The measure component of P is its marginal on the measure coordinate.

III.3.2 CP-chain bridge

Once exact dimensionality is established, the common almost-sure dimension of a typical measure in an ergodic CP-chain is called the dimension α of the chain. For a ρ -regular chain it equals the entropy average proved in Lemma III.3.4.

Lemma III.3.1 (Restriction to positive mass preserves generation). Let μ generate a distribution P along a refining filtration $F = (F_n)$, and let A be Borel with $\mu(A) > 0$. Then $\nu = \mu|_A / \mu(A)$ generates the same distribution along the induced filtration at ν -almost every point of A .

Proof. By martingale differentiation, for μ -a.e. $x \in A$,

$$\mu(A \cap F_n(x)) / \mu(F_n(x)) \rightarrow 1.$$

Consequently the normalized restrictions of μ and ν to $F_n(x)$ have total variation distance tending to zero. Rescaling the common cell does not change this. Thus the two scenery sequences are weak-* asymptotic. Strong genericity is invariant under an asymptotic perturbation, so ν generates the same distribution. $\square$

Lemma III.3.2 (Residue-class ergodic averaging). *Let (z_n) be strongly generic for an ergodic shift-invariant distribution P , let $q \geq 2$, and let f be continuous. Then there is a residue $i \in \{0, \dots, q-1\}$ such that*

$$\liminf_N (1/N) \sum_{n=0}^{N-1} f(z_{qn+i}) \geq \int f dP.$$

Proof. Let $Q_{q,i}$ be the limiting distribution on residue class i . Splitting the first qN terms of the ordinary empirical average into residue classes and passing to the limit gives

$$P = \frac{1}{q} \sum_{i=0}^{q-1} Q_{q,i}.$$

Consequently the average of the numbers $\int f dQ_{q,i}$ is $\int f dP$, and at least one of them is not smaller than this average. $\square$

Proposition III.3.3 (Typical measures of an ergodic CP-chain generate its measure component). *Let (μ_n, B_n) be an ergodic CP-chain with stationary distribution P . For P -almost every initial state (μ, B) , μ generates the measure component of P along the filtration $\Delta^n(B^*)$.*

Proof. Condition on (μ_1, B_1) and run the chain forward. Pulling the selected cells back to B_1^* gives nested cells $A_n \in \Delta^{n-1}(B_1^*)$ with

$$\mu_n = (\mu_1)^{A_n}.$$

Regularity forces $\text{diam}(A_n) \rightarrow 0$, hence $\bigcap_n A_n = \{X\}$ almost surely. Adaptedness says that, conditionally on μ_1 , the random point X has distribution μ_1 . Thus the conditional law of the measure path is the law of the scenery sequence $((\mu_1)^{X,n})$ with $X \sim \mu_1$. Birkhoff's theorem, applied also to every power of the shift, shows that almost every stationary path is strongly generic. Disintegration over the initial measure gives generation for P -a.e. μ_1 and μ_1 -a.e. X . $\square$

Lemma III.3.4 (Exact dimensionality and the chain-dimension formula). *For an ergodic ρ -regular CP-chain, P -almost every measure is exact dimensional, with common dimension*

$$\alpha = -(1/\log \rho) \int H(\mu, \Delta(B^*)) dP(\mu, B).$$

Proof. Use the nested cells from Proposition 3.3. At the sampled point X , regularity makes $-\log \text{diam } A_N = N \log(1/\rho) + O(1)$. The conditional-mass chain rule gives

$$-\log \mu_1(A_N) = \sum_{n=1}^{N-1} -\log \mu_n(B_{n+1}).$$

The conditional expectation of the summand given (μ_n, B_n) is

$$H(\mu_n, \Delta(B_n^*)).$$

The information-function ergodic theorem therefore yields the stated almost-sure limit. Regularity converts cell local dimension to Euclidean local dimension. The sampled point has law μ_1 , hence lower and upper local dimensions both equal α at μ_1 -a.e. point. Ergodicity makes α constant. $\square$

III.3.3 Local tree entropy and faithful-map lifting

For a nonzero finite Borel measure ν , its *lower Hausdorff dimension* is

$$\underline{\dim}_H \nu = \inf\{\dim_H A : A \text{ is Borel and } \nu(A) > 0\}.$$

Thus a lower bound for $\underline{\dim}_H \nu$ must hold on every positive- ν -mass set, not merely on $\text{supp } \nu$.

We now prove the two technical tools behind the entropy argument. The proof follows the architecture of Sections 3–5 of [5], but every estimate used below is written out here.

Definition (σ -trees and tree morphisms). *Let Λ be a finite alphabet. A σ -tree, $0 < \sigma < 1$, is a closed $X \subset \Lambda^{\mathbb{N}}$ with the ultrametric*

$$d_\sigma(x, y) = \sigma^{\min\{j : x_j \neq y_j\}}.$$

Its level- n cylinders are denoted $[x_1 \cdots x_n]$. A map $g : X \rightarrow Y$ between σ -trees is a tree morphism if every level- n cylinder of X is mapped into a level- n cylinder of Y .

Lemma (Local entropy averages force mass decay on a tree). *Let θ be a probability measure on a σ -tree. If for θ -almost every x*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} H_{\sigma^{n+1}}(\theta_{[x_1 \cdots x_n]}) \geq a,$$

then

$$\underline{\dim}_H \theta \geq \frac{a}{\log(1/\sigma)}.$$

Proof. Let $\mathcal{F}_n$ be the sigma-algebra generated by level- n cylinders and put

$$I_{n+1}(x) = -\log \theta(x_{n+1} \mid x_1 \cdots x_n).$$

The conditional expectation

$$h_n(x) = \mathbb{E}(I_{n+1} \mid \mathcal{F}_n)(x)$$

is exactly the displayed one-generation entropy. Since the alphabet is finite,

$$\sum_j p_j (\log(1/p_j))^2 \leq C_\Lambda$$

uniformly over probability vectors (p_j) . Hence $D_{n+1} = I_{n+1} - h_n$ is a martingale-difference sequence with uniformly bounded second moments. The martingale

$$\sum_{n \geq 1} \frac{D_n}{n}$$

converges almost surely, because the sum of its L^2 increments is finite. Kronecker's lemma therefore gives

$$\frac{1}{N} \sum_{n=1}^N D_n(x) \longrightarrow 0$$

for θ -almost every x . The conditional-mass chain rule now yields

$$-\frac{1}{N} \log \theta([x_1 \cdots x_N]) = \frac{1}{N} \sum_{n=1}^N I_n(x) = \frac{1}{N} \sum_{n=0}^{N-1} h_n(x) + o(1).$$

Thus, almost everywhere, the lower local dimension is at least

$$\frac{a}{\log(1/\sigma)}.$$

The local-dimension characterization of lower Hausdorff dimension proves the claim. $\square$

Theorem (Tree-morphism entropy-to-dimension theorem). *Let X, Y be σ -trees, let $g : X \rightarrow Y$ be a tree morphism, and let η be a probability measure on X . If for η -almost every branch*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} H_{\sigma^{n+1}}(g\eta_{[x_1 \cdots x_n]}) \geq a,$$

then

$$\underline{\dim}_H(g\eta) \geq \frac{a}{\log(1/\sigma)}.$$

Proof. For a node u of X and an output child symbol j , let

$$F_{u,j} = \{i : g(ui) = g(u)j\}.$$

Whenever $F_{u,j}$ is nonempty, independently choose one representative $i \in F_{u,j}$ with probability

$$\frac{\eta(i \mid u)}{\eta(F_{u,j} \mid u)},$$

and transfer the whole conditional mass $\eta(F_{u,j} \mid u)$ to that chosen child. Doing this at every node defines a random probability measure ϑ on X . It has three properties.

First, whenever $\vartheta[u] > 0$, the vector of nonzero child masses of $\vartheta(\cdot \mid u)$ is exactly the push-forward child-mass vector of $g\eta_{[u]}$. Hence

$$H_{\sigma^{|u|+1}}(\vartheta_{[u]}) = H_{\sigma^{|u|+1}}(g\eta_{[u]}).$$

Second, g is injective on $\text{supp } \vartheta$: if two surviving branches first split at a child of the same node, their output child symbols must differ because only one representative of each fibre $F_{u,j}$ survives. Since g preserves levels, it is therefore an isometry from $\text{supp } \vartheta$ onto its image. Third, for every cylinder $[u]$,

$$\mathbb{E} \vartheta[u] = \eta[u].$$

Indeed, along the path to u the probability that each prescribed child is selected is the product of the ratios $\eta(i \mid v)/\eta(F_{v,j} \mid v)$, while conditioned on survival the cylinder mass is the product of the complementary factors $\eta(F_{v,j} \mid v)$; the products cancel. A monotone-class argument extends this identity to all Borel sets.

Let G be the full η -measure set on which the entropy-average hypothesis holds. Then $\mathbb{E}\vartheta(G) = 1$, so $\vartheta(G) = 1$ almost surely. The preceding one-generation identity and the local entropy-averages lemma give

$$\underline{\dim}_H \vartheta \geq a / \log(1/\sigma)$$

almost surely. The isometry property gives the same bound for $g\vartheta$. Finally $g\eta = \mathbb{E}(g\vartheta)$. If a Borel set has Hausdorff dimension below $a / \log(1/\sigma)$, it has zero $g\vartheta$ -mass almost surely and hence zero $g\eta$ -mass after averaging. Thus the lower Hausdorff dimension of $g\eta$ has the claimed lower bound. $\square$

The preceding argument is the tree-dimensional estimate corresponding to Hochman–Shmerkin [5, Theorem 4.4].

Lemma (Ball entropy and grid entropy differ by a bounded amount). *For a probability measure λ on $\mathbb{R}^k$ and $r > 0$, let $\mathcal{Q}_r$ be the half-open grid of cubes of side $r/(2\sqrt{k})$. Then*

$$|H_r(\lambda) - H(\lambda, \mathcal{Q}_r)| \leq C_k.$$

Proof. Every atom $Q \in \mathcal{Q}_r$ has diameter at most $r/2$, so for $x \in Q$, $\lambda(B_r(x)) \geq \lambda(Q)$ and hence $H_r(\lambda) \leq H(\lambda, \mathcal{Q}_r)$.

Conversely, a ball of radius r meets at most M_k cubes of $\mathcal{Q}_r$, where M_k depends only on k . If $p_Q = \lambda(Q)$ and S_Q is the sum of $p_{Q'}$ over the finitely many grid cubes that can meet an r -ball centred in Q , then $\lambda(B_r(x)) \leq S_Q$ for $x \in Q$. Therefore

$$H_r(\lambda) \geq - \sum_Q p_Q \log S_Q.$$

The neighbourhood relation has degree at most M_k in both variables, and Jensen's inequality gives

$$H(\lambda, \mathcal{Q}_r) - H_r(\lambda) \leq \sum_Q p_Q \log \frac{S_Q}{p_Q} \leq \log \sum_Q S_Q \leq \log M_k.$$

$\square$

Proposition (Finite-scale entropy stability). *Let λ be a probability measure on $\mathbb{R}^k$.*

(i) *For every fixed $C > 0$,*

$$|H_r(\lambda) - H_{Cr}(\lambda)| \leq K_{C,k} \quad (r > 0).$$

(ii) *Translations leave H_r unchanged. If a similarity has ratio in a fixed compact subinterval of $(0, \infty)$, applying it changes H_r by at most $O_{k,I}(1)$, where I is that ratio interval.*

(iii) *If f, g are maps on a probability space (Ω, μ) satisfying $\sup_{\Omega} |f - g| \leq Ar$, then*

$$|H_r(f_{\#}\mu) - H_r(g_{\#}\mu)| \leq K_{A,k}.$$

In particular, on the unit ball, if after a target translation $\sup_x \|D_x g - \pi\| \leq r$ for a linear map π , then $|H_r(g\mu) - H_r(\pi\mu)| = O_k(1)$.

Proof. For (i), compare the two grid entropies supplied by the preceding lemma. A cube at the coarser scale $Cr/(2\sqrt{k})$ meets only $O_{C,k}(1)$ cubes at scale $r/(2\sqrt{k})$, and a fine cube meets only $O_k(1)$ coarse cubes. Passing to the joint refinement therefore changes Shannon entropy by at most the logarithm of these bounded intersection numbers. The grid comparison then gives (i); the case $C < 1$ follows by symmetry. Statement (ii) is immediate from the definition together with (i).

For (iii), if $|f - g| \leq Ar$, then for $x, y \in \Omega$,

$$|f(y) - f(x)| < r \implies |g(y) - g(x)| < (1 + 2A)r.$$

Hence

$$H_r(f_{\#}\mu) \geq H_{(1+2A)r}(g_{\#}\mu) \geq H_r(g_{\#}\mu) - K_{1+2A,k},$$

and the reverse inequality follows by interchanging f and g . For the derivative statement, translate so that $g(0) = \pi(0)$ and integrate the derivative bound along line segments; this gives $\sup |g - \pi| = O(r)$ on the unit ball and reduces to (iii). $\square$

The proposition is the entropy-stability estimate used below; compare Hochman–Shmerkin [5, Lemmas 3.5–3.6].

Definition (Faithful map). *Let X be a ρ -tree and let $F : X \rightarrow \mathbb{R}^k$. We call F C -faithful if, for every level n cylinder $[a]$,*

- (i) *no point of $F[a]$ belongs to more than C of the child images $F[ab]$;*
- (ii) *$F[a]$ contains a Euclidean ball of radius $(C^{-1}\rho)^n$ and is contained in a Euclidean ball of radius $(C\rho)^n$.*

The multiplicity condition iterates: a point belongs to at most C^n images of level- n cylinders. The next proposition is the dimension half of the lifting mechanism.

Proposition (Faithful maps lose only $O(1)/\log(1/\rho)$ dimension). *If $F : X \rightarrow \mathbb{R}^k$ is C -faithful and η is a probability measure on the ρ -tree X , then*

$$\underline{\dim}_H(F\eta) \geq \underline{\dim}_H\eta - \frac{K_{C,k}}{\log(1/\rho)}$$

for a constant $K_{C,k}$ independent of ρ and η .

Proof. Fix $s < \underline{\dim}_H\eta$. By the local-dimension characterization of lower Hausdorff dimension and Egorov’s theorem, for every $\varepsilon > 0$ there is a compact set E_ε with $\eta(E_\varepsilon) > 1 - \varepsilon$ such that, for the normalized restriction $\eta^\varepsilon = \eta|_{E_\varepsilon}/\eta(E_\varepsilon)$, there exists $N = N(\varepsilon, s)$ with

$$\eta^\varepsilon[a] \leq \rho^{sn} \quad (n \geq N)$$

for every level- n cylinder $[a]$ meeting E_ε . It therefore suffices to obtain the asserted estimate uniformly for $F\eta^\varepsilon$.

Fix $y \in \mathbb{R}^k$ and $n \geq N$, and let

$$\Phi_n(y) = \{a : |a| = n, F[a] \cap B_{\rho^n}(y) \neq \emptyset\}.$$

Every $F[a]$, $a \in \Phi_n(y)$, contains a ball of radius $C^{-n}\rho^n$, while the whole union of these images is contained in a ball of radius at most $4C^n\rho^n$ after enlarging the constant harmlessly.

Moreover, by iterating the one-generation multiplicity condition, no point is covered by more than C^n of the level- n images. Comparing Lebesgue volumes gives

$$|\Phi_n(y)| \omega_k(C^{-n} \rho^n)^k \leq C^n \omega_k(4C^n \rho^n)^k,$$

so

$$|\Phi_n(y)| \leq \exp(K_{C,k}n)$$

for a constant depending only on C and k . Hence

$$(F\eta^\varepsilon)(B_{\rho^n}(y)) \leq |\Phi_n(y)| \max_{a \in \Phi_n(y)} \eta^\varepsilon[a] \leq \exp(K_{C,k}n) \rho^{sn} = \rho^{n(s-K_{C,k}/\log(1/\rho))}.$$

Taking the lower local dimension and then using the local-dimension characterization yields

$$\underline{\dim}_H(F\eta^\varepsilon) \geq s - \frac{K_{C,k}}{\log(1/\rho)}.$$

Because the sets E_ε can be chosen with mass tending to one, the same lower bound holds for $F\eta$: if a Borel set had positive $F\eta$ -mass and smaller Hausdorff dimension, then for sufficiently small ε it would also have positive $F\eta^\varepsilon$ -mass. Finally let $s \uparrow \underline{\dim}_H \eta$. $\square$

Proposition (One-generation entropy is stable under a faithful map). *Let $F : X \rightarrow \mathbb{R}^k$ be C -faithful. For every probability η on X and every level- n cylinder $[a]$ with positive mass,*

$$|H_{\rho^{n+1}}(\eta_{[a]}) - H_{\rho^{n+1}}(F\eta_{[a]})| \leq K'_{C,k},$$

where $K'_{C,k}$ is independent of ρ, n, a and η .

Proof. Restrict to $[a]$ and rescale its tree metric by ρ^{-n} , so it is enough to prove

$$|H_\rho(\eta) - H_\rho(F\eta)| = O_{C,k}(1)$$

at the root. For $x \in X$ put

$$B_r^F(x) = F^{-1}(B_r(Fx)).$$

On a ρ -tree, $H_\rho(\eta)$ is the Shannon entropy of the first-symbol partition, equivalently

$$H_\rho(\eta) = - \int \log \eta[x_1] d\eta(x), \quad H_\rho(F\eta) = - \int \log \eta(B_\rho^F(x)) d\eta(x).$$

Because $F[x_1]$ is contained in a ball of radius $C\rho$, its diameter is at most $2C\rho$, and hence $[x_1] \subset B_{2C\rho}^F(x)$. Therefore

$$H_{2C\rho}(F\eta) \leq H_\rho(\eta).$$

The locally proved finite-scale entropy stability proposition gives

$$H_\rho(F\eta) - H_\rho(\eta) \leq O_{C,k}(1).$$

For the reverse inequality, for first-generation symbols u, v define

$$v \in \Lambda(u) \iff \text{dist}(F[u], F[v]) < \rho.$$

A volume comparison using the radius- $C^{-1}\rho$ balls inside the child images, their C -fold overlap, and the fact that all images indexed by $u \in \Lambda(v)$ lie in one $O_C(\rho)$ ball shows that

$$\#\{u : v \in \Lambda(u)\} \leq M_{C,k}$$

for every v . Also, if $x_1 = u$, then

$$B_\rho^F(x) \subset \bigcup_{v \in \Lambda(u)} [v].$$

Using $\log t \leq t$ for $t > 0$,

$$\begin{aligned} H_\rho(\eta) - H_\rho(F\eta) &= \int \log \frac{\eta(B_\rho^F(x))}{\eta[x_1]} d\eta(x) \\ &\leq \sum_u \sum_{v \in \Lambda(u)} \eta[v] \\ &\leq M_{C,k}. \end{aligned}$$

Combining the two inequalities proves the claim. $\square$

Lemma (Nested cube cover). *Let $Q \subset \mathbb{R}^k$ be an axis-parallel cube of side length ℓ and $N \geq 1$ an integer. There is a family of at most $(2N)^k$ closed subcubes of side ℓ/N such that every point of Q belongs to at most 3^k members and every axis-parallel cube $Q' \subset Q$ of side at most $\ell/(2N)$ is contained in one member of the family.*

Proof. After scaling take $Q = [0, 1]^k$. In one dimension use intervals of length $1/N$ whose left endpoints are $j/(2N)$, $0 \leq j \leq 2N - 2$. They cover $[0, 1]$, every point lies in at most three of them, and every interval of length at most $1/(2N)$ is contained in one of them. Taking Cartesian products proves the k -dimensional statement. $\square$

Theorem (Faithful lifting of a Lipschitz map to a tree morphism). *Let X be a ρ -tree and let $f : X \rightarrow \mathbb{R}^k$ be L -Lipschitz. Then there are a ρ -tree Y , a tree morphism $g : X \rightarrow Y$, and a $C_{k,L}$ -faithful map $h : Y \rightarrow \mathbb{R}^k$ such that*

$$f = h \circ g.$$

Moreover, for every probability measure η on X and every level- n cylinder $[a]$ of positive η -mass,

$$|H_{\rho^{n+1}}(f\eta_{[a]}) - H_{\rho^{n+1}}(g\eta_{[a]})| \leq C'_{k,L}.$$

Consequently

$$\underline{\dim}_H(f\eta) \geq \underline{\dim}_H(g\eta) - \frac{C''_{k,L}}{\log(1/\rho)}.$$

Proof. Translate the image and divide the target by $A = \max\{1, L\}$. After replacing the target norm by the equivalent ℓ^∞ norm (and increasing A by a factor depending only on k if needed), the rescaled map is 1-Lipschitz and its image is contained in $[0, 1]^k$. We perform the construction for this map. At the end we undo the fixed affine rescaling; this only changes the faithfulness constant by a factor depending on k and L .

Put $m = \lfloor \rho^{-1} \rfloor$. We first choose integers $N_j \in \{m, m+1\}$ so that, with $P_n = \prod_{j=1}^n N_j$,

$$\frac{1}{2} \leq \rho^n P_n \leq 1 \quad (n \geq 0).$$

This can be done greedily. If $q_n = \rho^n P_n \in [1/2, 1]$, choose $N_{n+1} = m+1$ when $q_n \rho(m+1) \leq 1$, and choose $N_{n+1} = m$ otherwise. In the first case the new value is still at least $1/2$ because $\rho(m+1) > 1$; in the second case $q_n > 1/(\rho(m+1))$, hence $q_n \rho m > m/(m+1) \geq 1/2$. Thus the induction closes.

We now construct Y and h . Start with the cube $Q_\emptyset = [0, 2]^k$. Choose once and for all an alphabet Λ_Y with $|\Lambda_Y| = (2(m+1))^k$. At every vertex b of level n , apply the nested cube cover with $N = N_{n+1}$ to Q_b , label its members by distinct symbols of Λ_Y , and retain only those labelled children in the tree. Thus $Y \subset \Lambda_Y^{\mathbb{N}}$ is a genuine tree with a fixed finite alphabet. Every level- n cube has side length

$$2P_n^{-1},$$

and the preceding choice of P_n gives

$$2\rho^n \leq 2P_n^{-1} \leq 4\rho^n.$$

For a branch $y = (y_j)$, the nested closed cubes $Q_{y_1 \dots y_n}$ have diameters tending to zero, so their intersection is a single point; define $h(y)$ to be this point. Since the child cubes cover the parent cube, every point of Q_b extends recursively to a branch beginning with b , and hence

$$h[b] = Q_b.$$

The overlap bound 3^k supplies the multiplicity part of faithfulness, while the displayed side-length bounds show that each $h[b]$ contains a ball of radius comparable to ρ^n and is contained in a ball of radius comparable to ρ^n . Enlarging one constant gives a C_k -faithful map h .

It remains to construct the morphism. Work with finite words. Set $g(\emptyset) = \emptyset$ and maintain inductively

$$f[a] \subset Q_{g(a)}.$$

Suppose $|a| = n$ and the inclusion is known. For a child $a\alpha$, the 1-Lipschitz property and the diameter ρ^{n+1} of an $(n+1)$ -cylinder imply that $f[a\alpha]$ is contained in an axis-parallel cube of side at most ρ^{n+1} . Since

$$\rho^{n+1} \leq P_{n+1}^{-1} = \frac{2P_n^{-1}}{2N_{n+1}},$$

the containment property of the nested cover provides a child c of $g(a)$ with

$$f[a\alpha] \subset Q_{g(a)c}.$$

Choose one such c and put $g(a\alpha) = g(a)c$. This is a tree morphism. For every infinite branch x , the point $f(x)$ lies in every $Q_{g(x_1 \dots x_n)}$, so the shrinking-intersection definition of h gives $f(x) = h(gx)$.

Finally let $\nu = g\eta_{[a]}$. Since $f = h \circ g$, the preceding one-generation entropy proposition applied to the faithful map h yields

$$|H_{\rho^{n+1}}(f\eta_{[a]}) - H_{\rho^{n+1}}(g\eta_{[a]})| = O_{k,L}(1).$$

The faithful-map dimension proposition applied to h gives the final dimension inequality. This proves every part of the lifting theorem. $\square$

Imported theorem (Interface D: Marstrand measure-projection interface). *If ν is exact dimensional of dimension α , then for Haar-a.e. $\pi \in \Pi_{d,k}$,*

$$\underline{\dim}_H(\pi\nu) = \min(k, \alpha).$$

This is the measure-form consequence of the energy projection theorem proved in Chapter 2; the restriction argument needed to pass from finite energy to lower measure dimension is included next.

Proof of Interface D. If $\min\{k, \alpha\} = 0$, the claimed equality follows at once from nonnegativity of lower Hausdorff dimension and the reverse bound proved in the final paragraph below. We may therefore assume $\min\{k, \alpha\} > 0$. Fix rational $t < \min\{k, \alpha\}$ and choose u with $0 < t < u < \min\{k, \alpha\}$. Exact dimensionality and the local-dimension criterion give bounded Borel sets $E_j \uparrow E$, where $\nu(E) = 1$, such that, after increasing a constant depending on j ,

$$\nu|_{E_j}(B(x, r)) \leq C_j r^u \quad (x \in E_j, 0 < r < 1). \quad (3.D.1)$$

For example, intersect a large bounded ball with the set on which the eventual inequality $\nu(B(x, r)) \leq r^u$ holds uniformly below one fixed radius. The layer-cake formula applied at centers $x \in E_j$ shows that

$$I_t(\nu|_{E_j}) < \infty. \quad (3.D.2)$$

Haar measure on $G(d, k)$ is a full-dimensional Frostman measure. Since $t < k$, Theorem III.2.5 applied to every $\nu|_{E_j}$ gives, for Haar-a.e. π ,

$$I_t(\pi_*(\nu|_{E_j})) < \infty \quad \text{for every } j. \quad (3.D.3)$$

The exceptional set may be chosen simultaneously for the countable family of pairs (t, j) . If A is Borel and $(\pi\nu)(A) > 0$, then $\nu(E_j \cap \pi^{-1}A) > 0$ for some j . The restriction of $\pi_*(\nu|_{E_j})$ to A is therefore nonzero and still has finite t -energy, so $\dim_H A \geq t$. Hence $\underline{\dim}_H(\pi\nu) \geq t$. Letting rational t increase to $\min\{k, \alpha\}$ proves the lower bound.

The ambient bound gives $\underline{\dim}_H(\pi\nu) \leq k$. Exact dimensionality also supplies a full- ν -measure Borel set of Hausdorff dimension at most α ; its Lipschitz image has full $\pi\nu$ -measure and dimension at most α . This proves the reverse bound and the asserted equality. $\square$

III.3.4 The finite-scale projection theorem

For a probability measure λ on a metric space, its r -scale entropy is

$$H_r(\lambda) = - \int \log \lambda(B_r(y)) d\lambda(y).$$

Let $\Pi_{d,k}$ be the compact space of orthogonal projections $\mathbb{R}^d \rightarrow \mathbb{R}^k$. For $q \geq 1$, define

$$e_q(\mu, \pi) = \frac{H_{\rho^q}(\pi\mu)}{q \log(1/\rho)}, \quad E_q(\pi) = \int e_q(\mu, \pi) dP(\mu).$$

The corresponding lower entropy dimension is

$$\dim_e \lambda = \liminf_{r \downarrow 0} \frac{H_r(\lambda)}{\log(1/r)}.$$

Theorem III.3.5 (CP-chain finite-scale projection lower bound). *Let μ generate the measure component of P along the CP filtration. Then, for every $q \geq 1$ and every $\pi \in \Pi_{d,k}$,*

$$\underline{\dim}_H(\pi\mu) \geq E_q(\pi) - C/q.$$

In particular the conclusion holds for P -almost every measure of the chain.

Proof. Fix q and write $c = \log(1/\rho)$.

Step 1: the q -step entropy average. At μ -a.e. x , the scenery sequence is strongly generic. If the q -step process is ergodic, strong genericity gives

$$(1/N) \sum_{n=0}^{N-1} e_q(\mu^{x, qn}, \pi) \rightarrow E_q(\pi).$$

For the general ergodic process, Lemma 3.2 gives, at almost every x , a residue $i(x) \in \{0, \dots, q-1\}$ for which the corresponding $\liminf$ is at least $E_q(\pi)$. We first prove the estimate on a piece where the residue is fixed and then recombine.

Step 2: return from normalized sceneries to the original cells. On a fixed-residue piece replace the filtration by its shifted q -step filtration. If $F_{qn+i}(x)$ is the current cell, linearity of π , ρ -regularity and the local finite-scale entropy stability proposition give

$$\begin{aligned} & H_{\rho^q}(\pi\mu^{x, qn+i}) \\ &= H_{\rho^{qn+i+q}}(\pi\mu_{F_{qn+i}(x)}) + O(1). \end{aligned}$$

Therefore the normalized $\liminf$ of the average of the right-hand entropies is at least

$$E_q(\pi) - C/q.$$

Step 3: code the q -step filtration by a tree. Form the $\sigma = \rho^q$ tree whose level- n vertices are the cells of the shifted filtration at depth $qn + i$. A branch codes a nested chain of cells,

hence a unique point by regularity. Let η be the lift of the normalized restriction of μ to this tree. The coding map $f : X \rightarrow \mathbb{R}^d$ is Lipschitz with a constant depending only on regularity. Apply the faithful lifting theorem proved above to $\pi \circ f$; write

$$\pi \circ f = h \circ g,$$

where g is a tree morphism and h faithful. The local faithful-map entropy comparison transfers Step 2 to

$$\begin{aligned} \liminf_N (1/N) \sum_{n < N} H_{\sigma^{n+1}}(g\eta_{[x_1 \dots x_n]}) \\ \geq qcE_q(\pi) - O(1) \end{aligned}$$

for η -a.e. branch.

Step 4: entropy to dimension. The local tree-morphism entropy-to-dimension theorem yields

$$\underline{\dim}_H g\eta \geq (qcE_q(\pi) - O(1))/(qc) = E_q(\pi) - C/q.$$

The faithful-map dimension proposition gives the same lower bound, after changing C , for

$$h \circ g\eta = \pi(\mu_{\text{piece}}).$$

Step 5: remove the residue partition. The sets on which a given residue is selected form a measurable partition up to a null set. Lemma 3.1 says every positive-mass restriction still generates the required component along the induced filtration, so the argument applies to every positive-mass piece. Since a finite convex combination of measures each having lower Hausdorff dimension at least s again has lower Hausdorff dimension at least s , recombination gives

$$\underline{\dim}_H \pi\mu \geq E_q(\pi) - C/q.$$

Proposition 3.3 supplies the final P -almost-every assertion. □

III.3.5 The projection function

When the limit exists, the projection function is

$$E(\pi) = \lim_{q \rightarrow \infty} E_q(\pi).$$

The theorem below proves existence and lower semicontinuity; this is not a limsup convention. For a rank- k linear map $L : \mathbb{R}^d \rightarrow \mathbb{R}^k$, let π_L be orthogonal projection onto $\ker(L)^\perp$ and define $E(L) = E(\pi_L)$; invertible postcomposition does not change the relevant dimensions.

Theorem III.3.6 (Projection function, semicontinuity, and typical projection dimensions). *For every $\pi \in \Pi_{d,k}$, the limit*

$$E(\pi) = \lim_{q \rightarrow \infty} E_q(\pi)$$

exists. The function E is lower semicontinuous. If the chain dimension is α , then:

1. $E(\pi) = \min(k, \alpha)$ for Haar-a.e. π ; 2. for each fixed π , $\underline{\dim}_H \pi \nu = \underline{\dim}_e \pi \nu = E(\pi)$ for P -a.e. ν ; 3. if μ generates the measure component, then for every π ,

$$\underline{\dim}_e(\pi \mu) \geq \underline{\dim}_H(\pi \mu) \geq E(\pi).$$

Here $\underline{\dim}_e$ is lower entropy dimension.

Proof. Fix π . Let

$$A = \limsup_q E_q(\pi), \quad B = \liminf_q E_q(\pi).$$

For P -a.e. ν , Proposition 3.3 and Theorem 3.5 give

$$\underline{\dim}_H \pi \nu \geq A.$$

On the other hand, by the definition of entropy dimension,

$$\underline{\dim}_e(\pi \nu) = \liminf_q e_q(\nu, \pi).$$

Since $0 \leq e_q \leq k + O(1/q)$, Fatou gives

$$\int \underline{\dim}_e(\pi \nu) dP(\nu) \leq B.$$

Always $\underline{\dim}_H \lambda \leq \underline{\dim}_e \lambda$. Hence

$$A \leq \int \underline{\dim}_H \pi \nu dP \leq \int \underline{\dim}_e \pi \nu dP \leq B \leq A.$$

All inequalities are equalities. Thus $A = B$, proving existence of $E(\pi)$, and the nonnegative difference $\underline{\dim}_e(\pi \nu) - \underline{\dim}_H \pi \nu$ has integral zero while both are bounded below by $E(\pi)$; therefore they equal $E(\pi)$ for P -a.e. ν . This proves (2).

For (1), Lemma 3.4 gives exact dimension α for P -a.e. ν . Interface D says that for each such ν , almost every π satisfies $\underline{\dim}_H(\pi \nu) = \min(k, \alpha)$. Fubini changes the order of the two almost-everywhere quantifiers, and (2) identifies this common value with $E(\pi)$ for Haar-a.e. π .

For lower semicontinuity, fix π and $\varepsilon > 0$. Since $E_q(\pi) \rightarrow E(\pi)$, choose q large enough simultaneously that

$$E_q(\pi) - C/q > E(\pi) - \varepsilon/2$$

and that every $O(1/q)$ perturbation loss below is smaller than $\varepsilon/4$. If π' is sufficiently close to π , specifically $\|\pi' - \pi\| \leq \rho^q$, the local map-perturbation estimate gives uniformly in the component measure

$$E_q(\pi') \geq E_q(\pi) - C_1/q.$$

Applying Theorem 3.5 to a typical generating measure and then using (2) at the fixed projection π' gives

$$E(\pi') \geq E_q(\pi') - C_2/q > E(\pi) - \varepsilon.$$

Thus E is lower semicontinuous.

Finally, if μ generates P , Theorem 3.5 holds for every q ; taking $q \rightarrow \infty$ gives $\underline{\dim}_H \pi \mu \geq E(\pi)$, and $\underline{\dim}_e \geq \underline{\dim}_H$ gives (3). $\square$

Corollary III.3.7 (Dense-open almost-maximal projections). *For every $\varepsilon > 0$, the set*

$$U_\varepsilon = \{\pi : E(\pi) > \min(k, \alpha) - \varepsilon\}$$

is open, dense, and of full Haar measure. Every generating measure μ satisfies

$$\underline{\dim}_H \pi \mu > \min(k, \alpha) - \varepsilon$$

for all $\pi \in U_\varepsilon$.

Proof. Lower semicontinuity makes U_ε open. Theorem 3.6(1) gives full measure, hence density, and Theorem 3.6(3) gives the dimension bound. $\square$

III.3.6 Nonlinear finite-scale stability

Proposition III.3.8 (Uniform finite-scale C^1 perturbation estimate). *Fix $\pi \in \Pi_{d,k}$ and $q \geq 1$. Let $\phi : \text{supp}(\mu) \rightarrow \mathbb{R}^k$ be C^1 , where μ strongly generates the measure component of P . If*

$$\sup_{x \in \text{supp}(\mu)} \|D_x \phi - \pi\| < \rho^q,$$

then

$$\underline{\dim}_H \phi \mu \geq E_q(\pi) - C/q.$$

Proof. Use the same ρ^q -tree coding as in Theorem 3.5, now with the map $\phi \circ f$ in place of $\pi \circ f$. The faithful lifting theorem proved above reduces the dimension estimate to the average cylinder entropies of $\phi \mu_{F_{qn}(x)}$ up to $O(1)$.

Fix a typical x . Rescale the target by ρ^{-qn} . Differentiability and the uniform C^1 modulus imply that on $F_{qn}(x)$ the rescaled map differs, in sup norm tending to zero, from the corresponding rescaled affine map with linear part $D_x \phi$. For all sufficiently large n , this error is at most a fixed multiple of ρ^q , so the local map-perturbation estimate changes the entropy at scale ρ^q by only $O(1)$, uniformly along the tail of the sequence. Target translations do not affect H_{ρ^q} , so only the linear part matters. Since $\|D_x \phi - \pi\| < \rho^q$, the same local estimate again gives, uniformly in the component measure,

$$H_{\rho^q}((D_x \phi)v) \geq H_{\rho^q}(\pi v) - O(1).$$

The rescaling of the source cell differs from the canonical normalization only by a uniformly bounded affine distortion coming from ρ -regularity, producing one more $O(1)$ entropy loss. Consequently the required branch average is bounded below by

$$\begin{aligned} & \int H_{\rho^q}(\pi v) dP(v) - O(1) \\ &= q \log(1/\rho) E_q(\pi) - O(1). \end{aligned}$$

The local tree-morphism theorem and the faithful-map dimension estimate convert this to $\underline{\dim}_H \phi \mu \geq E_q(\pi) - C/q$. Strong generation is exactly what identifies the q -step scenery average with the integral against P . $\square$

Theorem III.3.9 (Uniform C^1 lower semicontinuity at a projection). *Fix $\pi \in \Pi_{d,k}$. For P -almost every μ , the map*

$$g \mapsto \underline{\dim}_H g\mu$$

is lower semicontinuous at π in the C^1 topology. More precisely, for every $\varepsilon > 0$ there is $\delta > 0$, independent of the typical measure μ , such that

$$\|g - \pi\|_{C^1} < \delta \implies \underline{\dim}_H g\mu \geq \underline{\dim}_H \pi\mu - \varepsilon.$$

Proof. By Theorem 3.6(2), for P -a.e. μ , $\underline{\dim}_H \pi\mu = E(\pi)$, and Proposition 3.3 gives strong generation. Choose q with

$$E_q(\pi) - C/q > E(\pi) - \varepsilon.$$

Take $\delta \leq \rho^q$. If $\|g - \pi\|_{C^1} < \delta$, Proposition 3.8 yields

$$\underline{\dim}_H g\mu \geq E_q(\pi) - C/q > E(\pi) - \varepsilon = \underline{\dim}_H \pi\mu - \varepsilon.$$

The same q , hence the same δ , works for every measure in the common full-measure generating set. $\square$

Theorem III.3.10 (Nonsingular C^1 image theorem). *Let μ be a typical measure of the ergodic CP-chain and let $g : \text{supp}(\mu) \rightarrow \mathbb{R}^k$ be C^1 with $\text{rank } D_x g = k$ for every x . Extend E from orthogonal projections to rank- k linear maps by*

$$E(L) = E(\pi_{\ker(L)^\perp}).$$

Then

$$\underline{\dim}_H(g\mu) \geq \text{ess inf}_{x \sim \mu} E(D_x g).$$

Proof. Let $s = \text{ess inf}_{x \sim \mu} E(D_x g)$ and fix $\varepsilon > 0$. Work on the full-measure set on which μ generates P and $E(D_x g) \geq s$. For such a point x , write

$$D_x g = A_x \circ \pi_x,$$

where π_x is the orthogonal projection onto $\ker(D_x g)^\perp$ and A_x is an invertible linear map on $\mathbb{R}^k$. Postcomposition by A_x^{-1} is bi-Lipschitz and therefore does not change lower Hausdorff dimension.

By Theorem 3.6, $E_q(\pi_x) \rightarrow E(\pi_x) = E(D_x g)$. Choose an integer q_x so large that

$$E_{q_x}(\pi_x) - C/q_x > E(\pi_x) - \varepsilon \geq s - \varepsilon.$$

Continuity of Dg gives a neighborhood U_x on which

$$\|A_x^{-1} D_y g - \pi_x\| < \rho^{q_x}.$$

Whenever $\mu(U_x) > 0$, Lemma 3.1 shows that the normalized restriction μ_{U_x} still generates P along the induced filtration. Proposition 3.8, applied to $A_x^{-1} \circ g$ on this restriction, gives

$$\underline{\dim}_H g(\mu_{U_x}) = \underline{\dim}_H (A_x^{-1} \circ g)(\mu_{U_x}) > s - \varepsilon.$$

Now let Y be any Borel set with $g\mu(Y) > 0$, equivalently $\mu(g^{-1}Y) > 0$. Choose a μ -density point x of $g^{-1}Y$ in the full-measure good set above. Shrinking U_x if necessary preserves the derivative estimate and gives $\mu(g^{-1}Y \cap U_x) > 0$. Hence Y has positive mass for $g(\mu_{U_x})$; by the definition of lower Hausdorff dimension,

$$\dim_H Y \geq \underline{\dim}_H g(\mu_{U_x}) > s - \varepsilon.$$

Since Y was arbitrary and $\varepsilon > 0$ was arbitrary, $\underline{\dim}_H g\mu \geq s$. □

III.3.7 Furstenberg dimension conservation

We now turn to Furstenberg's dimension-conservation theorem [8]. The CP statement proved below corresponds to Theorem 3.1 of that paper. We then prove the gallery-selection and homogeneous-fractal culmination directly; neither branch is used in the Hochman–Shmerkin projection arguments above.

For the fixed half-open p -adic partition, let $Q_n(z)$ be the level- n cell containing z . A probability measure θ is p -regular if, for θ -almost every z , the limit

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_p \theta(Q_n(z))$$

exists and is independent of z ; the common value is $\dim_p \theta$.

For a coordinate projection $P : Q' \times Q'' \rightarrow Q'$ and a disintegration $\theta = \int \delta_x \times \theta_x d\bar{\theta}(x)$, we say that θ satisfies p -dimension conservation if $\theta, \bar{\theta}$, and almost every θ_x are p -regular, the fibres have a common dimension δ , and $\dim_p \theta = \dim_p \bar{\theta} + \delta$.

At set level, a Lipschitz map $f : A \rightarrow \mathbb{R}^k$ on compact A is dimension conserving if there is $\delta \geq 0$ such that

$$\delta + \dim_H \{y : \dim_H(A \cap f^{-1}\{y\}) \geq \delta\} \geq \dim_H A.$$

For the final application we use the Furstenberg terminology of Volume I: minisets are subsets of positive homothetic copies in the fixed cube, microsets are Hausdorff limits of minisets, a homogeneous set has every microset as a miniset, and a gallery is a closed family stable under taking minisets.

We specialize here to the base- p CP shift. Write $Q = Q' \times Q''$, with dimensions m_1, m_2 , and let $P(x, y) = x$. For a state $(\theta; x, y)$, write

$$\theta = \int \delta_u \times \theta_u d\bar{\theta}(u)$$

for the disintegration over P .

Imported theorem (Interface E: Triangular ergodic theorem — local proof). *Let (X, T, μ) be probability preserving. Suppose $f_n \in L^1(\mu)$, $f_n \rightarrow f_\infty$ almost everywhere, and*

$$G := \sup_{n \geq 1} |f_n| \in L^1(\mu).$$

Then

$$\frac{1}{N} \sum_{k=0}^{N-1} (f_{N-k} - f_{\infty})(T^k z) \longrightarrow 0 \quad \text{for } \mu\text{-a.e. } z. \quad (3.34)$$

Consequently the triangular averages

$$\frac{1}{N} \sum_{k=0}^{N-1} f_{N-k}(T^k z)$$

have the same almost-everywhere limit as the Birkhoff averages of f_{∞} . This is the exact Maker 1940 statement used by Furstenberg [9]; the label X-MAKER-1940 is retained only for provenance.

Proof. After changing f_{∞} on a null set, $|f_{\infty}| \leq G$. For $m \geq 1$ put

$$h_m = \sup_{n \geq m} |f_n - f_{\infty}|.$$

Thus $0 \leq h_m \leq 2G$ and $h_m \downarrow 0$ a.e., so $h_m \in L^1(\mu)$. Let $\mathcal{J}$ be the invariant sigma-algebra. By Birkhoff, for almost every z ,

$$\frac{1}{N} \sum_{k=0}^{N-1} h_m(T^k z) \longrightarrow \mathbb{E}(h_m \mid \mathcal{J})(z). \quad (3.35)$$

Moreover $\mathbb{E}(h_m \mid \mathcal{J}) \downarrow 0$ almost everywhere by conditional monotone convergence.

We also use the elementary fact that if $H \in L^1(\mu)$, then

$$\frac{H(T^N z)}{N} \longrightarrow 0 \quad \text{for } \mu\text{-a.e. } z. \quad (3.36)$$

Indeed, for every $\varepsilon > 0$, invariance and the tail-sum bound give

$$\sum_{N \geq 1} \mu\{H \circ T^N > \varepsilon N\} = \sum_{N \geq 1} \mu\{H > \varepsilon N\} \leq \varepsilon^{-1} \int H d\mu < \infty,$$

so (3.36) follows from Borel–Cantelli. Apply this to $H = 2G$.

Fix m . For $N > m$, split the triangular error according to whether $N - k \geq m$:

$$\begin{aligned} & \left| \frac{1}{N} \sum_{k=0}^{N-1} (f_{N-k} - f_{\infty})(T^k z) \right| \\ & \leq \frac{1}{N} \sum_{k=0}^{N-m} h_m(T^k z) + \frac{1}{N} \sum_{j=1}^{m-1} 2G(T^{N-j} z). \end{aligned}$$

The second term tends to zero by (3.36), while (3.35) gives

$$\limsup_{N \rightarrow \infty} \left| \frac{1}{N} \sum_{k=0}^{N-1} (f_{N-k} - f_{\infty})(T^k z) \right| \leq \mathbb{E}(h_m \mid \mathcal{J})(z).$$

Letting $m \rightarrow \infty$ proves (3.34). Adding the Birkhoff average of f_{∞} proves the final assertion. $\square$

Lemma III.3.11 (conditional-information maximal bound). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $A \in \mathcal{F}$, and let $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ be finite sub- σ -algebras. Then*

$$\mathbf{1}_A \sup_n [-\log_p \mathbb{P}(A \mid \mathcal{F}_n)]$$

is integrable.

Proof. For an integer $N \geq 0$, inspect, in increasing order of n , the atoms B of $\mathcal{F}_n$ for which $\mathbb{P}(A \mid B) < p^{-N}$, discarding every atom contained in one selected earlier. The remaining selected atoms are disjoint and their union contains

$$\{\omega \in A : \sup_n [-\log_p \mathbb{P}(A \mid \mathcal{F}_n)(\omega)] > N\}.$$

For every selected B , $\mathbb{P}(A \cap B) < p^{-N} \mathbb{P}(B)$. Summing over the disjoint atoms gives

$$\mathbb{P}\left(A \cap \left\{ \sup_n [-\log_p \mathbb{P}(A \mid \mathcal{F}_n)] > N \right\}\right) \leq p^{-N}.$$

The tail-sum criterion now gives integrability because $\sum_N p^{-N} < \infty$. □

Theorem III.3.12 (Furstenberg dimension conservation for ergodic CP shifts). *Let (Φ, T, P) be an ergodic base- p CP-shift system on $Q^{m_1+m_2} = Q' \times Q''$. Then for P -almost every state measure θ , with the Fubini disintegration above,*

1. θ and $\bar{\theta}$ are p -regular; 2. for $\bar{\theta}$ -almost every x , θ_x is p -regular with one common p -dimension δ ; 3. one has the exact identity

$$\dim_p \theta = \dim_p \bar{\theta} + \delta.$$

Thus almost every ergodic CP measure satisfies the preceding measure-level dimension-conservation statement.

Proof. The p -regularity of θ is already contained in Lemma 3.4 specialized to the base- p partition. Let its p -dimension be α .

For a typical state $(\theta; x, y)$, define

$$\begin{aligned} J_n(\theta; x, y) \\ = \theta(Q'_n(x) \times Q''_1(y)) / \theta(Q'_n(x) \times Q''). \end{aligned}$$

By martingale convergence in the base variable,

$$J_n(\theta; x, y) \rightarrow J_\infty(\theta; x, y) = \theta_x(Q''_1(y))$$

for almost every state. Put $K_n = -\log_p J_n$ and

$$K_\infty = -\log_p J_\infty.$$

Step 1: fiber information. Iterating the CP shift and cancelling successive conditional masses gives, for every N ,

$$(1/N) \sum_{k=0}^{N-1} K_\infty(T^k(\theta; x, y))$$

$$= -(1/N) \log_p \theta_x(Q_N''(y)).$$

By Birkhoff's theorem the left side converges almost everywhere to a constant δ because the CP shift is ergodic. The maximal estimate in Lemma 3.11, applied to the finite base-coordinate σ -algebras and then summed over the finitely many first fiber digits, gives integrability of $\sup_n K_n$; hence K_∞ is integrable. Therefore $\delta < \infty$, and the displayed identity proves that almost every conditional measure θ_x is p -regular with common p -dimension δ .

Step 2: triangular information identity. A direct cancellation of the same conditional ratios gives

$$\frac{1}{N} \sum_{k=0}^{N-1} K_{N-k}(T^k(\theta; x, y)) = -\frac{1}{N} \log_p \theta(Q_N'(x) \times Q_N''(y)) + \frac{1}{N} \log_p \bar{\theta}(Q_N'(x)). \quad (\text{III.3.1})$$

The hypotheses of Interface E hold by $K_n \rightarrow K_\infty$ and Lemma 3.11, so the left side of (III.3.1) converges to the same number δ as the Birkhoff averages of

$$K_\infty.$$

The first term on the right of (III.3.1) converges to $\alpha = \dim_p \theta$. Consequently

$$-(1/N) \log_p \bar{\theta}(Q_N'(x)) \rightarrow \alpha - \delta$$

for almost every state. Adaptedness/disintegration transfers this assertion to $\bar{\theta}$ -almost every x , so $\bar{\theta}$ is p -regular with

$$\dim_p \bar{\theta} = \alpha - \delta.$$

Rearranging proves

$$\dim_p \theta = \dim_p \bar{\theta} + \delta.$$

This is exactly Furstenberg's Theorem 3.1, now in the chapter's notation. $\square$

Lemma (p -regular measures control every positive-mass set). *Let λ be a p -regular probability measure on a bounded Euclidean box, of p -dimension a . If A is Borel and $\lambda(A) > 0$, then*

$$\dim_H A \geq a.$$

Proof. The assertion is immediate when $a = 0$. Fix $0 < \varepsilon < a$. By p -regularity and Egorov's theorem there are N and a Borel set $E \subset A$ with $\lambda(E) > 0$ such that

$$\lambda(Q_n(x)) \leq p^{-n(a-\varepsilon)} \quad (x \in E, n \geq N). \quad (3.37)$$

If a level- n half-open p -adic cube Q meets E , then $Q = Q_n(x)$ for some $x \in E \cap Q$, so the same estimate holds for $\lambda(Q)$. Put $\lambda_E = \lambda|_E / \lambda(E)$. A Euclidean ball of radius r with $p^{-(n+1)} < r \leq p^{-n}$ meets at most C_d level- n cubes. Summing (3.37) over those cubes gives

$$\lambda_E(B(y, r)) \leq \frac{C_d}{\lambda(E)} p^{-n(a-\varepsilon)} \leq C_{d,p,E,\varepsilon} r^{a-\varepsilon}.$$

After enlarging the constant this holds at all radii. The mass-distribution principle gives $\dim_H E \geq a - \varepsilon$, hence the same lower bound for A . Let $\varepsilon \downarrow 0$. $\square$

Corollary III.3.13 (support-level dimension conservation for a coordinate projection). *Assume in Theorem 3.12 that, for a typical θ ,*

$$\dim_H \text{supp}(\theta) = \dim_p \theta.$$

Then the coordinate projection P is dimension conserving on $\text{supp}(\theta)$ in the set-level sense described above.

Proof. Write $\alpha = \dim_p \theta$, $\beta = \dim_p \bar{\theta}$ and let δ be the common fiber p-dimension. Theorem 3.12 gives $\alpha = \beta + \delta$.

The preceding lemma gives, for $\bar{\theta}$ -almost every x ,

$$\dim_H \text{supp}(\theta_x) \geq \delta.$$

After changing the disintegration on a null set, θ_x is supported on the vertical section of the closed set $\text{supp}(\theta)$; this follows by applying the disintegration identity to a countable basis of open sets disjoint from that support. Hence $\{x\} \times \text{supp}(\theta_x)$ is contained in the fiber $\text{supp}(\theta) \cap P^{-1}\{x\}$. Choose a Borel set X_0 of full $\bar{\theta}$ -measure on which both this support inclusion and the preceding lower-dimensional conclusion hold. Then

$$X_0 \subset Y_\delta := \{x : \dim_H(\text{supp}(\theta) \cap P^{-1}\{x\}) \geq \delta\}.$$

Applying the preceding lemma to the Borel set X_0 gives $\dim_H X_0 \geq \beta$, and monotonicity gives $\dim_H Y_\delta \geq \beta$ without requiring a separate measurability assertion for Y_δ . Therefore

$$\delta + \dim_H Y_\delta \geq \delta + \beta = \alpha = \dim_H \text{supp}(\theta).$$

This is precisely set-level dimension conservation. □

Theorem III.3.14 (gallery selection and homogeneous-fractal culmination). *Let G be a nonempty gallery of compact subsets of $W = [0, 1]^m$. For the largest number $N_n(G)$ of level- n binary cells meeting a member of G , put $\Delta(G) = \limsup_{n \rightarrow \infty} n^{-1} \log_2 N_n(G)$. There is an ergodic adapted base-2 CP shift, supported on measures θ with $\text{supp} \theta \in G$, for which almost every θ is 2-regular and*

$$\dim_2 \theta = \dim_H \text{supp} \theta = \Delta(G). \quad (3.47)$$

Consequently, for every compact $A \subset \mathbb{R}^m$ and every linear map $L : \mathbb{R}^m \rightarrow \mathbb{R}^k$, some microset A' of A satisfies $\dim_H A' \geq \dim_H A$ and $L|_{A'}$ is dimension conserving. If A is homogeneous, then $L|_A$ is dimension conserving.

Proof. We give the measure-selection step, since it is the part that cannot be deduced merely from the CP dimension-conservation theorem above. The argument follows the finite-entropy construction of Furstenberg [8, Sections 4–6]; all limits and support claims are verified here.

Put $\mathcal{A} = \{0, 1\}^m$, $\Sigma = \mathcal{A}^{\mathbb{N}}$, and $\Pi(\omega) = \sum_{j \geq 1} 2^{-j} \omega_j \in W$. Let $\mathcal{P}(\Sigma)$ have its weak topology. For a word w , write $[w]$ for its cylinder and, when $\sigma([w]) > 0$, put

$$\sigma^w = \text{shift}_{|w|} * (\sigma|_{[w]} / \sigma([w])).$$

Define the compact scenery state space

$$\mathcal{M}_G = \{\sigma \in \mathcal{P}(\Sigma) : \text{supp}(\Pi_*\sigma) \in G\}. \quad (3.48)$$

It is closed: if $\sigma_j \rightarrow \sigma$, pass to a subsequence for which the supports of $\Pi_*\sigma_j$ converge in the Hausdorff metric to some $S \in G$. Weak convergence gives $\text{supp}(\Pi_*\sigma) \subset S$; every nonempty compact subset of S belongs to G , since it is a miniset with magnification 1. The same miniset property implies $\sigma^a \in \mathcal{M}_G$ whenever $\sigma([a]) > 0$: its geometric support is contained in $2 \text{supp}(\Pi_*\sigma) - a$, and it lies in W .

On $\mathcal{M}_G$ use the Markov operator

$$(Pf)(\sigma) = \sum_{a \in \mathcal{A}} \sigma([a])f(\sigma^a). \quad (3.49)$$

It is Feller. Indeed, each summand is continuous where $\sigma([a]) > 0$, and its absolute value is bounded by $\|f\|_\infty \sigma([a])$ where $\sigma([a])$ tends to zero. Set $h(\sigma) = -\sum_a \sigma([a]) \log_2 \sigma([a])$; this is bounded and continuous.

Use here endpoint-corrected binary cells: the last interval in each coordinate is closed at 1. Their occupied-cell count differs from the global half-open dyadic count by at most a factor 2^m , so its exponential growth rate is $\Delta(G)$. Choose $A_n \in G$ attaining $N_n(G)$, one point of A_n in each occupied cell, and for each point choose its canonical binary address (the expansion not eventually all 1 in a nonterminal coordinate). Let σ_n be uniform on those addresses. Then $\sigma_n \in \mathcal{M}_G$, its first- n cylinder entropy is $H_n(\sigma_n) = \log_2 N_n(G)$, and

$$\nu_n = \frac{1}{n} \sum_{j=0}^{n-1} P_*^j \delta_{\sigma_n} \quad (3.50)$$

has a weakly convergent subsequence. Any limit ν is stationary, because $P_*\nu_n - \nu_n = (P_*^n \delta_{\sigma_n} - \delta_{\sigma_n})/n \rightarrow 0$. The finite chain rule gives, for every σ ,

$$H_n(\sigma) = \sum_{j=0}^{n-1} P_*^j h(\sigma), \quad \int h d\nu_n = \frac{\log_2 N_n(G)}{n}. \quad (3.51)$$

Thus $\int h d\nu = \Delta(G)$.

We next show that no ergodic stationary law of the *labelled* chain can have entropy average greater than $\Delta(G)$. Its states are (σ, a) with $\sigma \in \mathcal{M}_G$ and $\sigma([a]) > 0$, where a is the *outgoing* digit. The transition from (σ, a) is to (σ^a, b) with probability $\sigma^a([b])$. Let η be an ergodic stationary law of this chain, and write (σ_j, a_j) for its stationary path. Its state marginal has $\mathbb{P}(a_j = a \mid \sigma_j) = \sigma_j([a])$. Conditionally on σ_0 , the probability of the word $a_0 \cdots a_{n-1}$ is $\sigma_0([a_0 \cdots a_{n-1}])$, by the definition of σ^a . Therefore

$$-\log_2 \sigma_0([a_0 \cdots a_{n-1}]) = \sum_{j=0}^{n-1} -\log_2 \sigma_j([a_j]). \quad (3.52)$$

The integrand on the right is in L^1 , with mean $\int h(\sigma) d\eta(\sigma, a)$. Birkhoff's theorem for this ergodic labelled chain in (3.52) consequently yields, for η -almost every σ_0 and σ_0 -almost

every address,

$$\lim_n -\frac{1}{n} \log_2 \sigma_0([a_0 \cdots a_{n-1}]) = h_\eta := \int h(\sigma) d\eta(\sigma, a). \quad (3.53)$$

This is also the geometric dyadic-cell limit. In each coordinate, ergodicity gives two possibilities: either every selected digit is 1 almost surely, in which case that coordinate of $\Pi(\omega)$ is identically 1 and belongs to the terminal cell at every level; or the frequency of digit 0 is positive almost surely, in which case the address is not eventually 1. In the latter case its binary expansion is unique and the inverse image of its dyadic cell, modulo a σ_0 -null set, is exactly its cylinder. The exceptional event of a noncanonical boundary address has zero probability by the same frequency argument. Hence $\Pi_*\sigma_0$ is 2-regular with dimension h_η . Here is the relevant mass-distribution argument. For each $\epsilon > 0$, on a positive-mass Borel subset of $\Pi_*\sigma_0$ there is N such that every level- n cell containing a point of that subset has mass at most $2^{-n(h_\eta - \epsilon)}$ for all $n \geq N$. A ball of radius 2^{-n} meets at most $O_m(1)$ such cells. The normalized restriction to the subset is therefore an $(h_\eta - \epsilon)$ -Frostman measure, up to a constant depending on N ; letting $\epsilon \downarrow 0$ gives

$$h_\eta \leq \dim_H \text{supp}(\Pi_*\sigma_0) \leq \overline{\dim}_B \text{supp}(\Pi_*\sigma_0) \leq \Delta(G). \quad (3.54)$$

The last bound follows directly from $N_n(\text{supp}(\Pi_*\sigma_0)) \leq N_n(G)$.

Lift ν to the stationary labelled-chain law $d\hat{\nu}(\sigma, a) = d\nu(\sigma)\sigma([a])$, and decompose this law into ergodic stationary components. Equation (3.54) bounds every component's mean entropy by $\Delta(G)$, while their average equals $\Delta(G)$ by (3.51). Therefore almost every component has equality. Choose one. The joint law of $(\Pi_*\sigma_0, \Pi(a_0a_1 \cdots))$ is an adapted ergodic CP shift: the conditional law of the point, given σ_0 , is $\Pi_*\sigma_0$, and magnifying its selected cell sends σ_0 to σ_1 . The endpoint convention checked above makes this identity hold almost surely. Equations (3.53)–(3.54) now prove (3.47).

For completeness, the family of microsets of a compact set is a gallery. It is closed by a diagonal choice of minisets from sequences approximating a convergent sequence of microsets. If B_j are minisets converging to B and $C \subset cB + t$ is a miniset of B , choose points of B_j approaching a finite $1/j$ -net of $c^{-1}(C - t)$. When $c > 1$, replace c by $c(1 - o(1)) \geq 1$ and shift the image an $o(1)$ amount towards the centre of W ; this puts all chosen points in W while preserving their limit C . When $c = 1$, the coordinates with $t_i = 0$ already remain in W ; in the coordinates with $t_i > 0$ (respectively $t_i < 0$), shift t_i slightly down (respectively up). Because $B_j \subset W$, these one-sided shifts place the chosen approximants in W and still converge to C . Each resulting finite set is a miniset of the original compact set by composition of the homotheties. Thus C is a microset, as claimed.

Apply the first part in orthonormal coordinates adapted to $\mathbb{R}^n = (\ker L)^\perp \oplus \ker L$, after a fixed homothety putting the transformed compact set in W . The construction gives a support C in its microset gallery with $\dim_H C = \Delta(G_A) \geq \dim_H A$. The CP dimension-conservation theorem III.3.12 and its support corollary III.3.13 make the coordinate projection onto $(\ker L)^\perp$ dimension conserving on C . Since L factors as this projection followed by an invertible linear map on its range, the same holds for L after undoing the orthogonal change. Here is the window check. Write C as a Hausdorff limit of subsets of $c_j A + t_j$ in the adapted (possibly rotated) cube. If $c_j \rightarrow \infty$ along a subsequence, apply one fixed

homothety that places the entire rotated cube inside W ; its factor $r > 0$ makes $rc_j \geq 1$ eventually. The images of the approximating sets are therefore minisets in the original W , and their limit A' is a microset of A , homothetic to C . If (c_j) has a bounded subsequence, pass to $c_j \rightarrow c > 0$ and $t_j \rightarrow t$; then $C \subset cA + t$. Its inverse homothetic image is a subset $B \subset A$ with dimension at least $\dim_{\mathbb{H}} A$ and on which L is dimension conserving. Hence L is dimension conserving on A itself by fibre monotonicity, and we may take $A' = A$, which is a miniset and thus a microset of itself.

Finally, if A is homogeneous, A' is a miniset of A : $A' \subset cA + t$ for some $c \geq 1$. The inverse homothety sends A' to a subset $B \subset A$ with $\dim_{\mathbb{H}} B = \dim_{\mathbb{H}} A'$. Since $\dim_{\mathbb{H}} B \leq \dim_{\mathbb{H}} A \leq \dim_{\mathbb{H}} A'$, equality holds. Dimension conservation is invariant under this homothety and passes from B to A by monotonicity of all fibre dimensions, proving the last assertion. $\square$

III.3.8 Energy versus entropy architectures

The energy argument in Chapter III.1 averages a singular kernel over the projection parameter and therefore gives a Haar-a.e. statement directly. The entropy argument here instead uses finite-scale information along *sceneries of one measure*. Ergodicity converts those local entropies into the deterministic profile E_q ; tree coding converts the average information into lower dimension; finally lower semicontinuity upgrades an a.e.-maximal projection profile into robust information near every fixed projection.

Furstenberg's dimension-conservation mechanism uses a third architecture: disintegrate an ergodic CP measure over a coordinate projection and telescope conditional information. The triangular Maker average identifies the information lost under the projection with the common information dimension of the fibers. This is logically independent of the HS projection profile $E(\pi)$, even though both are driven by the same magnification dynamics.

These hypotheses are not interchangeable: finite energy is neither assumed nor produced by the CP-chain argument, while generation and ergodicity have no role in the classical energy proof.

Part II

Radial, Nonlinear, and Visible Projections

Chapter III.4

Radial Projections and Visibility

Orthogonal projections are linear; radial projections are singular at the center and nonlinear away from it. Once the center and source are separated, however, the radial family satisfies the transversality conditions introduced by Peres–Schlag [4]. This chapter now proves the required generalized-projection theorem locally, applies it to the radial family, and then removes the artificial separation assumption by a countable localization argument.

The resulting theorems provide the baseline almost-every-center radial dimension statements and the positive-spherical-measure consequences used later in the volume. The final section records the geometric dictionary connecting radial images, rays, line incidence, and visible parts. Sharp exceptional-center estimates are deliberately deferred to the later theorem-studio chapter.

Prerequisites and source boundary. The Peres–Schlag generalized-projection engine is proved locally in this chapter from Littlewood–Paley decomposition, quantitative inverse charts, parameter Frostman estimates, and the multiscale covering argument reconstructed from the primary proof. Radial transversality, singular-center localization, and visibility deductions are also verified locally. The sharper Orponen–Shmerkin exceptional-center theory is not used in this chapter.

III.4.1 Radial maps, measures, and exceptional centres

Definition (radial projection). For $a \in \mathbb{R}^d$, $\pi_a : \mathbb{R}^d \setminus \{a\} \rightarrow S^{d-1}$ is $\pi_a(y) = (y - a)/|y - a|$. If $a \in E$, the notation $\pi_a(E)$ always means $\pi_a(E \setminus \{a\})$.

Definition (pinned direction measure). For a locally finite Borel measure μ , $(\pi_a)_\# \mu$ always means the pushforward of $\mu|_{\mathbb{R}^d \setminus \{a\}}$ when μ has an atom at a ; any normalization is stated separately.

Definition (exceptional centre). Exceptional centres are always tied to a displayed conclusion. For example, $\text{Exc}_t(E) = \{a : \dim_{\mathbb{H}} \pi_a(E) < t\}$.

III.4.2 Elementary radial geometry

For $a \neq y$, $\pi_a(y)$ records the oriented line through a and y ; unoriented-line statements identify antipodal directions. A radial fibre is an open ray, while its unoriented counterpart is a punctured line.

For compact K and $a \notin K$, $\text{Vis}_a(K)$ denotes its visible part. The identity $\pi_a(\text{Vis}_a(K)) = \pi_a(K)$ is an image identity, not a Hausdorff-dimension identity.

Lemma III.4.1 (annular Lipschitz bound). *Let $a, y, z \in \mathbb{R}^d$, with $y, z \neq a$, and suppose*

$$\min\{|y - a|, |z - a|\} \geq r > 0.$$

Then

$$|\pi_a(y) - \pi_a(z)| \leq 2|y - z|/r.$$

Proof. For nonzero vectors u, v ,

$$\begin{aligned} & |u/|u| - v/|v|| \\ & \leq |u - v|/|u| + ||v|^{-1} - |u|^{-1}||v| \\ & \leq 2|u - v|/\min\{|u|, |v|\}. \end{aligned}$$

Apply this with $u = y - a$, $v = z - a$. □

Lemma III.4.2 (radial fibers and line incidence). *For $a \in \mathbb{R}^d$ and $\omega \in S^{d-1}$,*

$$\pi_a^{-1}\{\omega\} = a + (0, \infty)\omega.$$

Consequently the unoriented radial map to projective directions has fibers equal to punctured affine lines through a . In particular, $\omega \in \pi_a(E)$ exactly when the positive ray $a + (0, \infty)\omega$ meets E .

Proof. $\pi_a(y) = \omega$ is equivalent to $y - a = t\omega$ with the positive scalar $t = |y - a|$. Identifying ω and $-\omega$ merges the two rays into the punctured line through a . □

Lemma III.4.3 (differential of the radial map). *Fix $y \neq a$ and put $\omega = \pi_a(y)$. As a function of the centre a ,*

$$D_a\pi_a(y) = -|y - a|^{-1}(I - \omega \otimes \omega).$$

Thus its kernel is $\text{span}(\omega)$, and its restriction to $\omega^\perp$ has all singular values equal to $|y - a|^{-1}$.

Proof. For $N(v) = v/|v|$,

$$D_v N = |v|^{-1}(I - N(v) \otimes N(v)).$$

Use $v = y - a$. □

III.4.3 Radial families are transversal generalized projections

We use the transversality and regularity conditions of Peres–Schlag Definition 7.2. The radial family is verified first; the corresponding generalized-projection theorem is then reconstructed in full before it is applied.

Lemma III.4.4 ($\beta = 0$ transversality and regularity on separated charts). *Let $A, K \subset \mathbb{R}^d$ be compact and suppose $\text{dist}(A, K) > 0$. After covering $A \times K$ by finitely many product patches $U \times V$, and choosing on each patch a smooth coordinate chart $\kappa : W \subset S^{d-1} \rightarrow \mathbb{R}^m$ containing every $\pi_a(y)$ with $(a, y) \in U \times V$, define*

$$\Pi_a(y) = \kappa(\pi_a(y)).$$

On every such compact patch the family $\{\Pi_a\}_{a \in U}$ satisfies Peres–Schlag transversality of order $\beta = 0$ and L, δ -regularity for every finite L and every $0 < \delta < 1$.

Proof. Because the source and parameter patches are positively separated, $(a, y) \mapsto \Pi_a(y)$ is smooth on an open neighbourhood of their product, with all mixed derivatives bounded on the compact patch.

For $y \neq z$ put

$$\Phi(a; y, z) = [\Pi_a(y) - \Pi_a(z)]/|y - z|.$$

Write $z = y + rw$, where $r = |z - y|$ and $w \in S^{d-1}$. By the fundamental theorem of calculus,

$$\Phi(a; y, y + rw) = - \int_0^1 D_y \Pi_a(y + trw) w dt.$$

Hence Φ , together with all derivatives in the parameter a , extends continuously to $r = 0$; at the boundary,

$$\Phi_0(a; y, w) = -D_y \Pi_a(y) w.$$

Let $e = \pi_a(y)$ and $\rho = |y - a|$. Since $D\kappa_e$ is an isomorphism from the tangent space $e^\perp$ to $\mathbb{R}^m$,

$$D_y \Pi_a(y) = D\kappa_e(I - e \otimes e)/\rho,$$

so $\Phi_0(a; y, w) = 0$ if and only if $w = +e$ or $w = -e$.

At such a zero, differentiate with respect to a , keeping the secant direction w fixed. Since $D_a e[h] = -(I - e \otimes e)h/\rho$, differentiation of the identity $D_y \Pi_a(y)e = 0$ gives

$$\begin{aligned} D_a \Phi_0(a; y, +e)[h] \\ = -\rho^{-2} D\kappa_e(I - e \otimes e)h, \end{aligned}$$

up to the irrelevant sign determined by the orientation convention; for $w = -e$ the sign is reversed. In either case the linear map $h \mapsto D_a \Phi_0[h]$ has rank $m = d - 1$. Therefore

$$\det(D_a \Phi_0 D_a \Phi_0^T) > 0$$

on the compact zero set of the extended normalized secant map.

By continuity and compactness there are constants $c, c' > 0$ such that

$$|\Phi(a; y, z)| \leq c \implies \det(D_a \Phi D_a \Phi^T) \geq c'$$

for all distinct y, z in the source patch. This is the $\beta = 0$ transversality condition. The regularity estimates are stronger than required: every parameter derivative of Φ is uniformly bounded because it is a divided difference in the source variable of a smooth function with uniformly bounded mixed derivatives; the top derivative is uniformly Lipschitz in a . Thus Definition 7.2 holds with $\beta = 0$, arbitrary finite L , and any $0 < \delta < 1$. $\square$

III.4.4 The Peres–Schlag generalized-projection theorem

Recall from Chapter 3 that, for a nonzero finite Borel measure ν , its lower Hausdorff dimension is

$$\underline{\dim}_H \nu = \inf \{ \dim_H B : B \text{ Borel, } \nu(B) > 0 \}.$$

The radial argument below uses only the case $\beta = 0$, but the proof mechanism is most transparent when stated in the generality in which it was introduced by Peres–Schlag [4]. We therefore prove the complete analytic engine here. This also fixes the precise meaning of the transversality and finite parameter-regularity hypotheses that appear in Lemma 4.4.

Let (Ω, d) be a compact metric space, let $Q \subset \mathbb{R}^n$ be open, and let

$$\Pi : Q \times \Omega \longrightarrow \mathbb{R}^m, \quad \Pi_\lambda(\omega) = \Pi(\lambda, \omega), \quad n \geq m.$$

For a finite positive Borel measure μ on Ω put $\nu_\lambda = (\Pi_\lambda)_\# \mu$ and

$$\Phi_\lambda(\omega, \tau) := \frac{\Pi_\lambda(\omega) - \Pi_\lambda(\tau)}{d(\omega, \tau)}, \quad \omega \neq \tau.$$

After rescaling d we may, and shall, assume $\text{diam } \Omega \leq 1$. For $\gamma \in \mathbb{R}$ write, as in the primary paper, the homogeneous Sobolev seminorm

$$\|v\|_{2,\gamma}^2 := \int_{\mathbb{R}^m} |\hat{v}(\xi)|^2 |\xi|^{2\gamma} d\xi.$$

In the range used below, $m + 2\gamma > 0$, so its low-frequency part is finite for every finite measure; only the high-frequency part governs Sobolev dimension. We define the Sobolev dimension by

$$\dim_S \nu := \sup \left\{ s \in \mathbb{R} : \int_{\mathbb{R}^m} |\hat{\nu}(\xi)|^2 (1 + |\xi|)^{s-m} d\xi < \infty \right\}.$$

Thus $\dim_S \nu > m$ means that ν has an L^2 density with positive fractional Sobolev smoothness.

Fix a positive integer L and $0 \leq \delta < 1$. The notation $\Pi \in C_\lambda^{L,\delta}(Q)$ means that all parameter derivatives $\partial_\lambda^\eta \Pi_\lambda(\omega)$ with $|\eta| \leq L$ are locally uniformly bounded, uniformly in ω , and the derivatives of order L are locally δ -Hölder in λ . For $\beta \in [0, 1)$ we say that

a compact parameter set $Q_0 \subset Q$ is β -transversal and L, δ -regular if there are constants $c, C > 0$ such that, whenever $\omega \neq \tau$ and $\lambda_1, \lambda_2 \in Q_0$ satisfy

$$|\Phi_{\lambda_1}(\omega, \tau)| + |\Phi_{\lambda_2}(\omega, \tau)| \leq c d(\omega, \tau)^\beta,$$

one has

$$\det(D_\lambda \Phi_{\lambda_1} D_\lambda \Phi_{\lambda_1}^T) \geq c^2 d(\omega, \tau)^{2\beta}, \quad (\text{PS.1})$$

$$|\partial_\lambda^\eta \Phi_{\lambda_1}(\omega, \tau)| \leq C_\eta d(\omega, \tau)^{-\beta|\eta|}, \quad 1 \leq |\eta| \leq L, \quad (\text{PS.2})$$

and

$$\max_{|\eta|=L} |\partial^\eta \Phi_{\lambda_1} - \partial^\eta \Phi_{\lambda_2}| \leq C |\lambda_1 - \lambda_2|^\delta d(\omega, \tau)^{-\beta(L+\delta)}. \quad (\text{PS.3})$$

All constants may depend on Q_0 but not on the pair ω, τ .

Theorem (Peres–Schlag generalized-projection theorem; local proof). *Assume $L + \delta > 1$ and that every compact $Q_0 \subset Q$ is β -transversal and L, δ -regular in the sense above. Let μ be a finite positive measure on Ω with $I_\alpha(\mu) < \infty$ for some $\alpha > 0$. There is a constant $a_0 = a_0(m, n, \delta)$ such that the following hold.*

For every compact $Q' \subset Q$,

$$\int_{Q'} \|v_\lambda\|_{2,Y}^2 d\lambda \lesssim_{Q',Y} I_\alpha(\mu) \quad (\text{PS.4})$$

whenever

$$0 < (m + 2\gamma)(1 + a_0\beta) \leq \alpha, \quad 2\gamma < L + \delta - 1. \quad (\text{PS.5})$$

Moreover, if $0 < \sigma \leq \min\{\alpha, m\}$, then

$$\dim_H\{\lambda \in Q : \dim_S v_\lambda \leq \sigma\} \leq n + \sigma - \min\left\{\frac{\alpha}{1 + a_0\beta}, L + \delta\right\}; \quad (\text{PS.6})$$

if $m < \sigma \leq \alpha$, then

$$\dim_H\{\lambda \in Q : \dim_S v_\lambda \leq \sigma\} \leq n - \frac{S - \sigma}{1 + (\sigma - m)/(L + \delta)}, \quad S := \min\left\{\frac{\alpha}{1 + a_0\beta}, L + \delta\right\}; \quad (\text{PS.7})$$

and, for $0 < \sigma \leq \alpha - a_0\beta$,

$$\dim_H\{\lambda \in Q : \dim_S v_\lambda < \sigma\} \leq n + \sigma - m. \quad (\text{PS.8})$$

Proof. We follow the primary proof but spell out the two steps that are usually hidden in references to “Littlewood–Paley theory” and to a “quantitative inverse function theorem.” The proof is divided into six steps.

Step 1: a dyadic Sobolev decomposition. Choose a radial, radially nonincreasing $\hat{\varphi} \in C_c^\infty(\mathbb{R}^m)$ with $0 \leq \hat{\varphi} \leq 1$, $\hat{\varphi} = 1$ on $B(0, 1)$ and $\hat{\varphi} = 0$ outside $B(0, 2)$, and define

$$\hat{\psi}(\xi) = \hat{\varphi}(\xi/2) - \hat{\varphi}(\xi).$$

Then ψ is Schwartz, $\widehat{\psi} \geq 0$, $\text{spt } \widehat{\psi} \subset \{1 \leq |\xi| \leq 4\}$, and

$$\sum_{j \in \mathbb{Z}} \widehat{\psi}(2^{-j}\xi) = 1, \quad \xi \neq 0. \quad (\text{PS.9})$$

Consequently, for every finite measure ν and every $\gamma \in \mathbb{R}$,

$$\|\nu\|_{2,\gamma}^2 \prec_{\gamma} \sum_{j \in \mathbb{Z}} 2^{2j\gamma} \int (\psi_{2^{-j}} * \nu)(x) d\nu(x), \quad (\text{PS.10})$$

where $\psi_{2^{-j}}(x) = 2^{jm}\psi(2^j x)$. Indeed, Plancherel turns the right-hand side into

$$\int |\widehat{\nu}(\xi)|^2 \sum_j 2^{2j\gamma} \widehat{\psi}(2^{-j}\xi) d\xi,$$

and on the support of the j th summand one has $2^j \asymp |\xi|$. Notice also that all moments of ψ vanish, because $\widehat{\psi}$ vanishes in a neighbourhood of the origin.

We shall use one elementary consequence repeatedly. If $G \in C_c^{M,\delta}(\mathbb{R}^m)$, $R \geq 1$, and M is a nonnegative integer, then

$$\left| \int \psi(Ru)G(u) du \right| \lesssim R^{-m-M-\delta} \|G\|_{C^{M,\delta}}. \quad (\text{PS.11})$$

To prove this, substitute $v = Ru$, subtract the Taylor polynomial of $G(v/R)$ at 0 through order M , use the vanishing moments, and estimate the remainder by $R^{-M-\delta}|v|^{M+\delta}[G]_{C^{M,\delta}}$; the remaining weighted integral of $|\psi(v)|$ is finite. For $\delta = 0$ the same estimate with any exponent $M' < M$ follows by applying the argument to $C^{M-1,\delta'}$ for $0 < \delta' < 1$ and then letting $\delta' \uparrow 1$.

Step 2: the multiscale parameter lemma. We need a parameter-space device converting an averaged dyadic estimate into a Hausdorff-dimension bound. Let $h_j \geq 0$ be uniformly bounded functions on an open subset of $\mathbb{R}^n$ such that

$$\sup_j R^j \int h_j(\lambda) d\lambda < \infty, \quad R > 1, \quad (\text{PS.12})$$

and put $H_j = \sum_{i=0}^j B^{i-j} h_i$, where $B \geq A > 1$. Suppose

$$|\nabla H_j| \leq CA^j H_j. \quad (\text{PS.13})$$

Then, for $1 \leq r < R$ with $r \leq B$,

$$\dim_H \left\{ \lambda : \sum_j r^j h_j(\lambda) = \infty \right\} \leq n - \theta \quad \text{whenever} \quad A^\theta \leq R/r. \quad (\text{PS.14})$$

If $B < r < R$, assume in addition that $h_j \in C^{L,\delta}$ and, locally at $|\lambda - \lambda'| \leq A^{-j}$,

$$\max_{|\eta|=L} |\partial^\eta H_j(\lambda) - \partial^\eta H_j(\lambda')| \leq CA^{j(L+\delta)} |\lambda - \lambda'|^\delta (H_j(\lambda) + H_j(\lambda')). \quad (\text{PS.15})$$

Then

$$\dim_{\mathbf{H}} \left\{ \lambda : \sum_j r^j h_j(\lambda) = \infty \right\} \leq n - \theta \quad (\text{PS.16})$$

provided

$$A^\theta (r/B)^{\theta/(L+\delta)} \leq R/r. \quad (\text{PS.17})$$

For completeness we prove the covering estimate. It is enough to work in a compact cube $Q' \subset Q$. First take $1 < r_0 < r_1 \leq B$, fix $T \geq 1$, and put

$$E_j(T) = \left\{ \lambda \in Q' : h_j(\lambda) > j^{-2} r_1^{-j}, \quad \sum_{i \geq 0} r_0^i h_i(\lambda) \leq T \right\}.$$

Every point where $\sum r_1^j h_j$ diverges but $\sum r_0^j h_j$ converges lies in $\limsup_j E_j(T)$ for some integer T .

Since $r_0 < B$ and all terms are nonnegative,

$$\sum_{k \geq 0} r_0^k H_k(\lambda_0) = \sum_{i \geq 0} r_0^i h_i(\lambda_0) \sum_{k \geq i} (r_0/B)^{k-i} \leq \frac{T}{1 - r_0/B}, \quad (\text{PS.18})$$

for $\lambda_0 \in E_j(T)$. Thus $H_k(\lambda_0) \lesssim_T r_0^{-k}$. From (PS.13), Gronwall's inequality gives

$$|H_k(\lambda) - H_k(\lambda_0)| \lesssim A^k |\lambda - \lambda_0| H_k(\lambda_0), \quad |\lambda - \lambda_0| < cA^{-k}. \quad (\text{PS.19})$$

Using $h_j = B^{-j}(B^j H_j - B^{j-1} H_{j-1})$ and the last two bounds, we obtain

$$|h_j(\lambda) - h_j(\lambda_0)| \lesssim_T |\lambda - \lambda_0| (A/r_0)^j. \quad (\text{PS.20})$$

Choose

$$W_j = c_T j^{-2} (r_0/r_1)^j A^{-j}. \quad (\text{PS.21})$$

For c_T small enough, every cube of side W_j meeting $E_j(T)$ has $h_j \geq \frac{1}{2} j^{-2} r_1^{-j}$ throughout a fixed concentric subcube. Select a disjoint family of such subcubes whose bounded dilates cover $E_j(T)$. If its cardinality is M_j , then (PS.12) yields

$$M_j \lesssim_T j^{2(n+1)} \left(\frac{r_1 A^n}{R} \right)^j \left(\frac{r_1}{r_0} \right)^{jn}. \quad (\text{PS.22})$$

Therefore

$$\sum_j M_j W_j^{n-\theta} < \infty \quad \text{if} \quad A^\theta < \frac{R}{r_1} \left(\frac{r_0}{r_1} \right)^\theta. \quad (\text{PS.23})$$

Hausdorff–Cantelli gives dimension at most $n - \theta$ for the corresponding limsup set. To pass to the endpoint r , choose a finite chain $0 < \rho_1 < 1 < \rho_2 < \dots < \rho_M = r$ with successive ratios arbitrarily close to one. Since the h_j are uniformly bounded, $\sum \rho_1^j h_j$ converges everywhere, and the divergence set for r is the union over $k \geq 2$ of the sets where the ρ_k -series diverges but the ρ_{k-1} -series converges. Apply (PS.23) to each adjacent pair and let the maximal ratio tend to one. This proves (PS.14), including equality in its displayed hypothesis.

Now assume $B < r < R$. On the part of the divergence set where $\sum B^i h_i \leq T$, define

$$E_j(T) = \left\{ \lambda \in Q' : h_j(\lambda) > j^{-2} r^{-j}, \quad \sum_{i \geq 0} B^i h_i(\lambda) \leq T \right\}.$$

For $\lambda_0 \in E_j(T)$ the definition of H_k gives $H_k(\lambda_0) \leq TB^{-k}$. Combining this with (PS.13) shows that, whenever $|\lambda - \lambda_0| \lesssim B^{-k}$,

$$H_k(\lambda) \lesssim_T B^{-k}. \quad (\text{PS.24})$$

Let $N = \lceil L + \delta \rceil$. The standard N th finite-difference identity, together with (PS.15), (PS.24), and $h_j = B^{-j}(B^j H_j - B^{j-1} H_{j-1})$, gives

$$\left| \sum_{i=0}^N (-1)^i \binom{N}{i} h_j(\lambda_0 + iy) \right| \lesssim_T (A^j |y|)^{L+\delta} B^{-j}. \quad (\text{PS.25})$$

Indeed, if $\delta > 0$ this is the integral remainder formula for the N th finite difference; if $\delta = 0$, use the ordinary L th derivative form. Set

$$W_j = c_T j^{-2/(L+\delta)} A^{-j} (B/r)^{j/(L+\delta)}. \quad (\text{PS.26})$$

Then the right-hand side of (PS.25) is at most $\frac{1}{2} j^{-2} r^{-j}$. Integrating the finite-difference identity over $[-W_j, W_j]^n$, using $h_j(\lambda_0) > j^{-2} r^{-j}$ and nonnegativity of h_j , shows

$$\int_{[-NW_j, NW_j]^n} h_j(\lambda_0 + u) du \gtrsim_N j^{-2} r^{-j} W_j^n. \quad (\text{PS.27})$$

Select disjoint cubes of side $2NW_j$ meeting $E_j(T)$ whose bounded dilates cover it. If there are M_j cubes, (PS.12) and (PS.27) imply

$$M_j \lesssim_T j^{2+2n/(L+\delta)} (r/R)^j A^{nj} (r/B)^{jn/(L+\delta)}. \quad (\text{PS.28})$$

Consequently

$$\sum_j M_j W_j^{n-\theta} < \infty$$

whenever the strict form of (PS.17) holds, since its exponential factor is

$$\left[\frac{r}{R} A^\theta (r/B)^{\theta/(L+\delta)} \right]^j.$$

Letting $\theta' \uparrow \theta$ removes strictness. This controls the part where the B -series converges. The remaining part is contained in $\{\sum B^i h_i = \infty\}$ and is controlled by (PS.14), because (PS.17) and $B < r$ imply $A^\theta \leq R/B$. Thus (PS.16) follows, and the multiscale lemma is proved.

Step 3: quantitative inversion where the normalized secant is small. Fix $U \subset Q_0 \subset Q$, distinct ω, τ , and write $r = d(\omega, \tau)$. We first treat $0 < \delta < 1$ and set

$$b_0 = \frac{m+2}{\delta}.$$

There are constants $c_0, c_1, c_2 > 0$ and points $\lambda_1, \dots, \lambda_N$ with

$$N \lesssim r^{-b_0 n \beta} \quad (\text{PS.29})$$

such that

$$\{\lambda \in U : |\Phi_\lambda| \leq c_0 r^{b_0 \beta}\} \subset \bigcup_{v=1}^N B(\lambda_v, c_1 r^{b_0 \beta}) \quad (\text{PS.30})$$

and the doubled balls remain inside the region where $|\Phi_\lambda| \lesssim r^\beta$. On each doubled ball, after a permutation of the parameter coordinates, write $\lambda = (x, y) \in \mathbb{R}^m \times \mathbb{R}^{n-m}$. For every fixed y the map

$$F_y(x) = \Phi_{(x,y)}(\omega, \tau) \quad (\text{PS.31})$$

is a diffeomorphism on the corresponding slice, and

$$|\det DF_y^{-1}| \lesssim r^{-\beta}, \quad \|DF_y^{-1}\| \lesssim r^{-m\beta}. \quad (\text{PS.32})$$

Here is the proof. At every point of the small-secant set, (PS.1) and the Cauchy–Binet formula provide an $m \times m$ coordinate minor of $D_\lambda \Phi$ with determinant $\gtrsim r^\beta$. By (PS.2) its largest singular value is $O(r^{-\beta})$, so its smallest singular value is $\gtrsim r^{m\beta}$. The δ -Hölder control (PS.3), on a ball of radius $r^{b_0 \beta}$, changes this minor by

$$O(r^{b_0 \beta \delta} r^{-\beta(1+\delta)}) = O(r^{(m+1-\delta)\beta}) = o(r^{m\beta}). \quad (\text{PS.33})$$

Thus the same minor stays uniformly invertible on a sufficiently small fixed multiple of that ball. Apply the elementary inverse-function principle to $(DF_y(x_0))^{-1} F_y$: if its derivative differs from the identity by at most $1/2$ on a ball, the map is injective on the concentric one-third ball and its image contains a fixed fraction of the corresponding Euclidean ball. A Vitali subcover of the small-secant set by balls of radius $\asymp r^{b_0 \beta}$ gives (PS.29)–(PS.32).

We shall also need derivatives of the inverse. Differentiating $F_y(H_y(u)) = u$ gives $DH_y = (DF_y \circ H_y)^{-1}$. Repeated differentiation expresses every $D^k H_y$, $2 \leq k \leq L$, as a finite sum of products of copies of $(DF_y \circ H_y)^{-1}$ and derivatives $D^\ell F_y \circ H_y$ with $2 \leq \ell \leq k$. Induction using (PS.2), (PS.32), and (PS.3) therefore yields

$$\|D^k H_y\|_\infty + [D^{L-1} H_y]_{C^{0,\delta}} \lesssim r^{-c_k \beta}, \quad (\text{PS.34})$$

with $c_k \leq C_0(k+1)$ for a constant $C_0 = C_0(m, n, \delta)$; this follows because each differentiation introduces only a bounded number of inverse-Jacobian factors and one derivative of F_y . The same linear-in- k loss holds for derivatives of $|\det DH_y|$. This is the multivariable inverse-derivative formula used in the primary proof, written in the form actually needed below.

Step 4: the transversal Littlewood–Paley kernel estimate. Let $\rho \in C_c^\infty(Q)$ and let ψ be as in Step 1. There exists $a_1 = a_1(m, n, \delta)$ such that, for every distinct ω, τ , every $j \in \mathbb{Z}$, and every

$$0 \leq q < L + \delta + m - 1,$$

one has

$$\left| \int \rho(\lambda) \psi(2^j [\Pi_\lambda(\omega) - \Pi_\lambda(\tau)]) d\lambda \right| \lesssim_q (1 + 2^j r^{1+a_1 \beta})^{-q}. \quad (\text{PS.35})$$

If $2^j r^{1+a_1\beta} \lesssim 1$, the assertion follows from boundedness of ψ and compactness of $\text{spt } \rho$. Otherwise split the integral with a smooth cutoff according as $|\Phi_\lambda| \gtrsim r^{b_0\beta}$ or $|\Phi_\lambda| \lesssim r^{b_0\beta}$. On the first part Schwartz decay gives, for arbitrary N ,

$$O_N((1 + 2^j r^{1+b_0\beta})^{-N}). \quad (\text{PS.36})$$

For the second part use (PS.30) and a partition of unity χ_v subordinate to the doubled inverse-chart balls. It can be chosen with $\|D^k \chi_v\|_\infty \lesssim r^{-b_0 k \beta}$. On one chart, split $\lambda = (x, y)$ as in (PS.31), put $u = F_y(x)$ and change variables. The integral becomes

$$\int_{\mathbb{R}^{n-m}} \int_{\mathbb{R}^m} G_y(u) \psi(2^j r u) du dy, \quad (\text{PS.37})$$

where

$$G_y(u) = \rho(H_y(u), y) \chi_v(H_y(u), y) |\det DH_y(u)|.$$

The cutoff is supported strictly inside the inverse chart, so G_y extends by zero as a compactly supported $C^{L-1, \delta}$ function. By (PS.34), Leibniz, and the derivative bounds for χ_v ,

$$\|G_y\|_{C^{M, \delta}} \lesssim r^{-C_M \beta}, \quad 0 \leq M \leq L-1, \quad (\text{PS.38})$$

where C_M can be chosen uniformly for $M \leq L-1$. Choose M with $q < m + M + \delta$. The vanishing-moment estimate (PS.11), followed by (PS.29), gives

$$|\text{small-secant part}| \lesssim r^{-C\beta} (2^j r)^{-q}. \quad (\text{PS.39})$$

More precisely, the inverse-derivative formula gives $C_M \leq C_0(M+1)$ with $C_0 = C_0(m, n, \delta)$: every differentiation inserts at most a fixed number of inverse-Jacobian factors and one regularity loss, while the chart-radius exponent $b_0 = (m+2)/\delta$ is already fixed by m and δ . For $q \geq q_* := m/2$ choose M minimally with $q < m + M + \delta$; then $(M+1)/q$ is uniformly bounded, independently of L . Choose $a_1 = a_1(m, n, \delta)$ large enough that the derivative loss together with the chart-count loss $N \lesssim r^{-b_0 n \beta}$ is absorbed by replacing r by $r^{1+a_1\beta}$. For $0 < q < q_*$ interpolate between the trivial $q = 0$ bound and the estimate at q_* : $\min(1, X^{-q_*}) \leq X^{-q}$. Thus the same a_1 , independent of q and L , works throughout the stated range. Together with (PS.36) this proves (PS.35). When $\delta = 0$, use the $C^{L-1, \delta'}$ consequence of C^L for any $\delta' < 1$ and let $\delta' \uparrow 1$; the admissible range for q tends to $q < L + m - 1$.

Step 5: averaged Sobolev regularity and the first two exceptional-set bounds. Take $Q' \subset Q$ and choose $\rho \in C_c^\infty(Q)$ with $\rho = 1$ on Q' . Let $s = m + 2\gamma > 0$ and choose q with

$$s < q < L + \delta + m - 1.$$

Using (PS.10), Fubini, and (PS.35),

$$\begin{aligned} \int \|v_\lambda\|_{2, \gamma}^2 \rho(\lambda) d\lambda &\lesssim \iint_{\Omega^2} \sum_{j \in \mathbb{Z}} 2^{js} \left| \int \rho(\lambda) \psi(2^j [\Pi_\lambda(\omega) - \Pi_\lambda(\tau)]) d\lambda \right| d\mu(\omega) d\mu(\tau) \\ &\lesssim \iint_{\Omega^2} \sum_j 2^{js} (1 + 2^j r^{1+a_1\beta})^{-q} d\mu(\omega) d\mu(\tau). \end{aligned} \quad (\text{PS.40})$$

The elementary dyadic estimate

$$\sum_j 2^{js}(1 + 2^j R)^{-q} \lesssim_{s,q} R^{-s}, \quad q > s > 0, \quad (\text{PS.41})$$

gives

$$\int \|v_\lambda\|_{2,Y}^2 \rho(\lambda) d\lambda \lesssim I_{s(1+a_1\beta)}(\mu). \quad (\text{PS.42})$$

Hence (PS.4) follows whenever (PS.5) holds, after increasing the final constant a_0 if necessary.

We now extract exceptional-parameter dimensions from (PS.4). For $j \geq 0$ define

$$h_j(\lambda) = 2^{-jm} \int \widehat{\psi}(2^{-j}\xi) |\widehat{v}_\lambda(\xi)|^2 d\xi, \quad h_0 \text{ enlarged by the bounded low-frequency part.} \quad (\text{PS.43})$$

Then

$$\sum_j 2^{j(m+2Y)} \int_{Q'} h_j \lesssim 1, \quad (\text{PS.44})$$

and

$$\{\dim_S v_\lambda \leq \sigma\} \subset \left\{ \sum_j 2^{j(\sigma+\varepsilon)} h_j(\lambda) = \infty \right\} \quad (\varepsilon > 0). \quad (\text{PS.45})$$

The covering proof in Step 2 uses only two variation estimates for the shell functions. We now verify those estimates directly, rather than hiding them inside a Littlewood–Paley regularity assertion. Choose an integer N so large that every parameter derivative $D^k \psi$, $k \leq L + 1$, is dominated by $K_N(x) = (1 + |x|^2)^{-N}$. The identity

$$K_N(x) = \frac{1}{\Gamma(N)} \int_0^\infty t^{N-1} e^{-t} e^{-t|x|^2} dt$$

shows that K_N is positive definite, hence $\widehat{K}_N \geq 0$; moreover $\widehat{K}_N$ is radial, strictly positive, and rapidly decreasing. Put

$$\mathcal{H}_j(\lambda) := \iint K_N(2^j[\Pi_\lambda(\omega) - \Pi_\lambda(\tau)]) d\mu(\omega) d\mu(\tau). \quad (\text{PS.46})$$

Because $|D^k K_N| \lesssim_k K_N$, the chain rule gives

$$|\nabla \mathcal{H}_j| \lesssim 2^j \mathcal{H}_j. \quad (\text{PS.47})$$

It also gives, for $1 \leq k \leq L$,

$$|D^k h_j(\lambda)| \lesssim 2^{jk} \mathcal{H}_j(\lambda), \quad (\text{PS.48})$$

and, when $|\lambda - \lambda'| \lesssim 2^{-j}$,

$$|D^L h_j(\lambda) - D^L h_j(\lambda')| \lesssim 2^{j(L+\delta)} |\lambda - \lambda'|^\delta (\mathcal{H}_j(\lambda) + \mathcal{H}_j(\lambda')). \quad (\text{PS.49})$$

For (PS.49), expand $D^L h_j$ by the chain rule. Terms in which the top parameter derivative of Π changes use its δ -Hölder bound; in the remaining terms one uses the mean-value

theorem on a derivative of ψ and $2^j|\lambda - \lambda'| \lesssim 1$. The polynomial weight K_N is comparable at arguments a bounded distance apart, so the two endpoint correlations on the right dominate every intermediate term.

There remains one point: on the exceptional level sets used in Step 2, $\mathcal{H}_j$ is controlled by the comparison series. By Fourier inversion,

$$\mathcal{H}_j(\lambda) = c_N 2^{-jm} \int \widehat{K}_N(2^{-j}\xi) |\hat{v}_\lambda(\xi)|^2 d\xi. \quad (\text{PS.50})$$

Insert the dyadic partition (PS.9). On the i th frequency shell the coefficient is

$$c_{i-j} \asymp 2^{m(i-j)} \sup_{|\eta| \asymp 2^{i-j}} \widehat{K}_N(\eta).$$

Hence $c_k \lesssim 2^{mk}$ for $k \leq 0$, while c_k decreases faster than any geometric progression for $k > 0$. After absorbing the bounded low-frequency piece into h_0 ,

$$\mathcal{H}_j(\lambda) \lesssim \sum_{i \geq 0} c_{i-j} h_i(\lambda). \quad (\text{PS.51})$$

If $\sum_i r_0^i h_i(\lambda_0) \leq T$ with $1 < r_0 < 2^m$, then $h_i(\lambda_0) \leq T r_0^{-i}$, and (PS.51) gives

$$\mathcal{H}_j(\lambda_0) \lesssim_T r_0^{-j}. \quad (\text{PS.52})$$

Indeed, for $i \leq j$ the resulting geometric ratio is $r_0/2^m < 1$, and for $i > j$ rapid decay of c_{i-j} makes the tail summable. By (PS.47) and Gronwall, (PS.52) remains true, up to a fixed factor, throughout a ball of radius $O(2^{-j})$ about λ_0 .

Equations (PS.48) and (PS.52) give precisely the variation estimate (PS.20) used to obtain the cube size (PS.21). Equations (PS.49) and (PS.52), with $r_0 = B = 2^m$, give the finite-difference estimate (PS.25) and hence the cube size (PS.26). Therefore the covering conclusions of Step 2 apply to the shell functions (PS.43), with $A = 2$, $B = 2^m$, $R = 2^{m+2\gamma}$ and $r = 2^{\sigma+\varepsilon}$.

If $0 < \sigma < m$, choose $\varepsilon > 0$ so small that $\sigma + \varepsilon \leq m$. Then $r = 2^{\sigma+\varepsilon} \leq B$, so (PS.14) gives, after $\varepsilon \downarrow 0$,

$$\dim_{\text{H}}\{\dim_{\text{S}} v_\lambda \leq \sigma\} \leq n + \sigma - (m + 2\gamma). \quad (\text{PS.55})$$

For the endpoint $\sigma = m$, use instead $\{\dim_{\text{S}} v_\lambda \leq m\} \subset \{\dim_{\text{S}} v_\lambda \leq m + \varepsilon\}$ and apply the second regime below; letting $\varepsilon \downarrow 0$ yields the same bound (PS.55).

If $\sigma > m$ and $m + 2\gamma > \sigma$, use (PS.16)–(PS.17). The largest admissible dimension drop is obtained from

$$\theta \left(1 + \frac{\sigma - m}{L + \delta} \right) \leq m + 2\gamma - \sigma,$$

which gives

$$\dim_{\text{H}}\{\dim_{\text{S}} v_\lambda \leq \sigma\} \leq n - \frac{m + 2\gamma - \sigma}{1 + (\sigma - m)/(L + \delta)}. \quad (\text{PS.56})$$

If $m + 2\gamma \leq \sigma$, the desired estimate at this intermediate value is trivial. Finally let $m + 2\gamma$ increase to $S = \min\{\alpha/(1 + a_0\beta), L + \delta\}$ through values allowed by (PS.5). This proves (PS.6) and (PS.7).

Step 6: the sharp Frostman bound below the target dimension. It remains to prove (PS.8), which is stronger than (PS.6) in its nontrivial range. If $\sigma > m$, the claimed bound $n + \sigma - m \geq n$ is automatic, so assume $0 < \sigma \leq \min\{\alpha, m\}$ and suppose, to the contrary, that for some $Q' \subset Q$ the exceptional set

$$E = \{\lambda \in Q' : \dim_S \nu_\lambda < \sigma\}$$

has Hausdorff dimension strictly larger than $\sigma + n - m$. Choose $t < \sigma$, arbitrarily close to σ , and by Frostman's lemma choose a nonzero measure ϑ supported on E such that

$$\vartheta(B(\lambda, s)) \lesssim s^{\sigma+n-m}. \quad (\text{PS.57})$$

We estimate the t -energy of ν_λ on average with respect to ϑ . For a fixed pair $\omega \neq \tau$ with distance r , Step 3 covers the small-secant parameter set by at most $Cr^{-b_0 n \beta}$ inverse-chart balls. Inside one such ball the change of variables $u = F_y(x)$ shows that $\{|\Phi_\lambda| \leq u\}$ can be covered by

$$\lesssim \left(1 + \frac{r^{b_0 \beta}}{\min\{r^{b_0 \beta}, ur^{-m\beta}\}}\right)^{n-m}$$

balls of diameter $\lesssim \min\{r^{b_0 \beta}, ur^{-m\beta}\}$. Applying (PS.57) and summing over the inverse charts gives, uniformly for $0 < u \lesssim r^\beta$,

$$\vartheta\{\lambda : |\Phi_\lambda| \leq u\} \lesssim r^{-C\beta} u^\sigma. \quad (\text{PS.58})$$

The complementary region contributes at most a further factor $r^{-C\beta}$. The dyadic layer-cake estimate, using $t < \sigma$, therefore yields

$$\int |\Phi_\lambda(\omega, \tau)|^{-t} d\vartheta(\lambda) \lesssim r^{-C\beta}. \quad (\text{PS.59})$$

Consequently

$$\int I_t(\nu_\lambda) d\vartheta(\lambda) \lesssim \iint d(\omega, \tau)^{-t-C\beta} d\mu(\omega) d\mu(\tau). \quad (\text{PS.60})$$

Choose the final constant a_0 larger than the uniform C in this estimate. If $\sigma \leq \alpha - a_0 \beta$ and $t < \sigma$ is sufficiently close to σ , the right-hand side is finite. Hence $I_t(\nu_\lambda) < \infty$ for ϑ -almost every λ , forcing $\dim_S \nu_\lambda \geq t$. Apply this to a countable sequence $t_k \uparrow \sigma$ and intersect the corresponding full- ϑ sets; then $\dim_S \nu_\lambda \geq \sigma$ for ϑ -almost every λ , contradicting the support condition for ϑ . Thus $\dim_H E \leq \sigma + n - m$, proving (PS.8).

Combining Steps 5 and 6 proves all assertions. The endpoint $\delta = 0$ follows, as in Step 4, by replacing $(L, 0)$ by $(L - 1, \delta')$ with $\delta' \uparrow 1$; all displayed bounds converge to the stated ones. This completes the local proof of the Peres–Schlag analytic engine. $\square$

In the radial application we have $n = d$, $m = d - 1$, and $\beta = 0$, so the loss factor $1 + a_0 \beta$ disappears completely.

III.4.5 Returning to the radial family

Lemma III.4.5 (countable localization across the radial singularity). *Let μ be a compactly supported finite Radon measure with $I_\alpha(\mu) < \infty$ for some $\alpha > 0$. Then there is a Lebesgue-null set $N_\mu \subset \mathbb{R}^d$ such that, for every $a \notin N_\mu$:*

1. *for every $0 < t < \min\{\alpha, m\}$, every positive-mass part of μ supported a positive distance from a has radial image measure of finite t -energy after restriction to finitely many sphere charts;*
2. *the pinned measure $\nu_a = (\pi_a)_\# \mu$ has lower Hausdorff dimension at least*

$$\min\{\alpha, m\};$$

3. *if $\alpha \geq m$, then $\nu_a \ll \mathcal{H}^m \upharpoonright_{S^{d-1}}$;*
4. *if in addition $a \notin \text{spt}(\mu)$ and $\alpha \geq m$, then ν_a has an L^2 density in finitely many smooth sphere charts.*

Proof. Choose countable families of rational open balls U, V with compact closures and $\text{dist}(\overline{U}, \overline{V}) > 0$, rich enough that every ordered pair (a, y) , $a \neq y$, belongs to some $U \times V$. For each fixed pair and each finite measure $\mu|_V$, Lemma 4.4 verifies $\beta = 0$ and full regularity after finitely many target charts.

Apply the locally proved Peres–Schlag theorem above. For every rational $0 < t < \min\{\alpha, m\}$, take $\gamma = (t - m)/2$. Then $m + 2\gamma = t \leq \alpha$, and (7.3) gives finite chartwise t -energy for almost every centre in U . If $\alpha \geq m$, take $\gamma = 0$; (7.3) gives an L^2 density for the projected restriction for a.e. centre in U . Take the union of these exceptional sets over the countable family of rational (U, V, t) and the finitely many sphere charts. The resulting set N_μ is Lebesgue null.

Fix $a \notin N_\mu$. Since finite positive energy excludes atoms, $\mu(\{a\}) = 0$. The source $\mathbb{R}^d \setminus \{a\}$ is covered by countably many rational V whose closures stay a positive distance from a . Hence ν_a is a countable sum of projected restrictions having the properties above.

For the lower-dimension claim, let $B \subset S^{d-1}$ be Borel with $\nu_a(B) > 0$. Some separated rational source piece V satisfies

$$\mu(V \cap \pi_a^{-1}(B)) > 0.$$

For every rational $t < \min\{\alpha, m\}$, the projected measure of $\mu|_V$ has finite t -energy, and therefore every Borel set carrying positive mass for it has Hausdorff dimension at least t . Thus $\dim_{\text{H}} B \geq t$; let t increase to $\min\{\alpha, m\}$. This proves item 2.

If $\alpha \geq m$, each separated projected restriction is absolutely continuous, so their countable sum ν_a is absolutely continuous. If $a \notin \text{spt}(\mu)$, the whole compact support is separated from a , so only finitely many source/target patches are needed and the sum of their L^2 chart densities is again L^2 . $\square$

III.4.6 Baseline radial projection theorems

Proposition III.4.6 (pinned measure formulation). *Let μ be a compactly supported finite Radon measure on $\mathbb{R}^d$ with $I_\alpha(\mu) < \infty$, $\alpha > 0$, and let $\nu_a = (\pi_a)_\# \mu$. Then for Lebesgue almost every centre a ,*

$$\underline{\dim}_{\text{H}}(\nu_a) \geq \min\{\alpha, d - 1\}.$$

If $\alpha \geq d - 1$, then for Lebesgue almost every a ,

$$\nu_a \ll \mathcal{H}^{d-1}|_{S^{d-1}}.$$

If a is restricted to $\mathbb{R}^d \setminus \text{spt}(\mu)$, then outside a Lebesgue-null set the latter absolute continuity can be strengthened to chartwise L^2 density.

Proof. This is exactly Lemma 4.5 with $m = d - 1$. $\square$

Theorem III.4.7 (radial Marstrand dimension theorem). *Let $E \subset \mathbb{R}^d$ be analytic and put $s = \dim_{\text{H}} E$. Then for Lebesgue almost every centre $a \in \mathbb{R}^d$,*

$$\dim_{\text{H}} \pi_a(E) = \min\{s, d - 1\}.$$

The assertion uses the convention $\pi_a(E) = \pi_a(E \setminus \{a\})$ when $a \in E$.

Proof. The upper bound holds for every a : by Lemma 4.1, π_a is Lipschitz on each set $\{y : |y - a| \geq 1/j\}$, and

$$E \setminus \{a\} = \bigcup_{j \geq 1} E \cap \{y : |y - a| \geq 1/j\}.$$

Thus $\dim_{\text{H}} \pi_a(E) \leq \min\{s, d - 1\}$.

For the lower bound, fix rational $q < \min\{s, d - 1\}$. By the authorized analytic compact selection theorem choose compact $K \subset E$ with $\dim_{\text{H}} K > q$. The Frostman/energy package from Volume I gives a probability measure μ supported on K with $I_\alpha(\mu) < \infty$ for some $q < \alpha < \dim_{\text{H}} K$. Proposition 4.6 implies

$$\dim_{\text{H}} \pi_a(K) \geq q$$

for Lebesgue-a.e. a . Since $\pi_a(K \setminus \{a\}) \subset \pi_a(E \setminus \{a\})$, the same lower bound holds for E . Intersect the full-measure centre sets over a countable sequence of rational q increasing to $\min\{s, d - 1\}$. $\square$

Theorem III.4.8 (high-dimensional visibility baseline). *Let $E \subset \mathbb{R}^d$ be analytic with $\dim_{\text{H}} E > d - 1$. Then for Lebesgue almost every centre $a \in \mathbb{R}^d$,*

$$\mathcal{H}^{d-1}(\pi_a(E)) > 0.$$

Thus E is visible from Lebesgue almost every centre in the radial-image sense.

Proof. Choose α with $d - 1 < \alpha < \dim_{\text{H}} E$. By analytic compact selection and the I.3 Frostman/energy package, there is a compact $K \subset E$ carrying a nonzero finite measure μ with $I_\alpha(\mu) < \infty$. Proposition 4.6 gives $(\pi_a)_\# \mu \ll \mathcal{H}^{d-1}$ for a.e. a . This projected measure has total mass $\mu(K) > 0$, so its support $\pi_a(K \setminus \{a\})$ cannot have zero $\mathcal{H}^{d-1}$ measure. Since $\pi_a(K \setminus \{a\}) \subset \pi_a(E \setminus \{a\})$, the result follows. $\square$

Proposition III.4.9 (visibility/radial-image dictionary). *Let $K \subset \mathbb{R}^d$ be nonempty and compact and let $a \notin K$. Then*

$$\pi_a(\text{Vis}_a(K)) = \pi_a(K).$$

Consequently

$$\dim_{\text{H}} \pi_a(K) \leq \dim_{\text{H}} \text{Vis}_a(K) \leq \dim_{\text{H}} K.$$

This is an exact image identity, not an identification of the visible subset of K with its direction image on the sphere.

Proof. The image identity is the visibility identity established in Volume I. The first dimension inequality follows because π_a is Lipschitz on the compact set K ; the second is monotonicity under inclusion. Lemma 4.2 supplies the ray/line incidence interpretation. $\square$

III.4.7 Failure and boundary examples

III.4.7.1 Example 4.10 — centres on a line can be exceptional

Let $E \subset L$ where L is an affine line and $\dim_{\text{H}} E > 0$. If $a \in L$, then $\pi_a(E)$ contains at most the two antipodal directions parallel to L , so $\dim_{\text{H}} \pi_a(E) = 0$. Thus exceptional centres can contain a positive-dimensional set, even though Theorem 4.7 says that they have zero d -dimensional Lebesgue measure. This is one reason the next chapter studies exceptional-centre dimension rather than merely almost-everywhere statements.

III.4.7.2 Example 4.11 — visibility is not radial-image dimension

For a compact cylinder $K = A + [0, 1]e$ viewed from the $+e$ direction, the model visibility identity from Volume I gives $\text{Vis}_e(K) = A + e$. The visible set is a subset of the original ambient space, while its projection is the base A . The image identity preserves directions but does not identify the two geometric objects. The same warning applies to point visibility: $\text{Vis}_a(K)$ and $\pi_a(K)$ live in different spaces.

III.4.8 Chapter boundary

The results above are baseline almost-every-centre theorems. They do not estimate the Hausdorff dimension of the exceptional set. Sharp exceptional-centre estimates require additional incidence and projection machinery and are developed in the next chapter.

Chapter III.5

Exceptional Sets for Radial Projections

Chapter III.4 established the almost-every-centre radial projection theorem. The present chapter asks a finer question: how large can the set of exceptional centres be? In the plane, recent incidence methods give sharp or near-sharp estimates in both the low- and high-dimensional regimes. We record the exact research theorems needed later, prove the deductions between their different formulations, and isolate the geometric obstructions that explain the unavoidable one-dimensional exceptional set.

Both the radial Kaufman and radial Falconer estimates are proved below. The Kaufman proof starts with a planar heavy-tube pruning argument and then uses the symmetric thin-tube improvement of Chapter 9. The Falconer proof uses the finite overlap argument of Chapter 6 and the Fu–Ren incidence theorem proved in Chapter 7. The low-dimensional Orponen–Shmerkin exceptional-centre conclusion follows from Kaufman.

Prerequisites and source boundary. Elementary radial notation comes from Chapter III.4. The two proofs below reconstruct the planar Orponen–Shmerkin–Wang estimates [7]: the Kaufman branch uses the locally proved Chapter 9 thin-tube improvement III.9.11; the Falconer branch uses the locally proved overlap lemma III.6.30.3, Katz–Tao decomposition III.7.9, and normalized Fu–Ren theorem III.7.26. These forward references stay within this volume.

III.5.1 Radial exceptional-set notation

Definition (radial projection). For $x \neq y$ in $\mathbb{R}^2$, $\pi_x(y) = (y - x)/|y - x|$. If $x \in Y$, then $\pi_x(Y)$ means $\pi_x(Y \setminus \{x\})$.

Definition (radial exceptional set). For $\sigma \geq 0$, $E_{<\sigma}(Y) = \{x : \dim_{\mathbb{H}} \pi_x(Y \setminus \{x\}) < \sigma\}$, with $E_{\leq\sigma}(Y)$ defined analogously. A superscript “out” indicates that centres in Y are excluded.

III.5.2 Line space and spanned lines

We also use the affine line space $A(2, 1)$. An unoriented line is represented by $(e, p) \in S^1 \times \mathbb{R}$ modulo $(e, p) \sim (-e, -p)$, where the corresponding line is

$$\{z \in \mathbb{R}^2 : z \cdot e = p\}.$$

Hausdorff dimension on $A(2, 1)$ is computed using any of the equivalent smooth quotient metrics. For $X \subset \mathbb{R}^2$, let $\mathcal{L}(X)$ denote the set of affine lines passing through two distinct points of X .

III.5.3 Exceptional-centre theorem studios

Theorem III.5.1 (high-dimensional exceptional centres). *Let $K \subset \mathbb{R}^2$ be Borel with $s = \dim_{\text{H}} K > 1$. For every $\sigma \in [0, 1)$,*

$$\dim_{\text{H}}\{x \in \mathbb{R}^2 \setminus K : \dim_{\text{H}} \pi_x(K) \leq \sigma\} \leq \max\{1 + \sigma - s, 0\}.$$

Proof. The radial Falconer estimate, Theorem III.5.4 below, is proved independently in this chapter. We derive the present exceptional-set form from it, taking care not to assume that the exceptional set itself is Borel.

Write $b = \max\{1 + \sigma - s, 0\}$ and let E denote the set in the statement. If $\dim_{\text{H}} E \leq b$ there is nothing to show. Otherwise choose a and t such that

$$b < a < \dim_{\text{H}} E, \quad 1 < t < s, \quad a + t - 1 > \sigma.$$

Compact inner approximation of the Borel set K gives a compact $K_0 \subset K$ with $\dim_{\text{H}} K_0 > t$. Define

$$F = \{x \in \mathbb{R}^2 \setminus K_0 : \dim_{\text{H}} \pi_x(K_0) \leq \sigma\}.$$

For $x \notin K_0$ the map $x \mapsto \pi_x(K_0)$ is continuous into the Hausdorff hyperspace of nonempty compact subsets of S^1 . The Hausdorff-dimension sublevel sets in that hyperspace are Borel (as recalled in the proof of Theorem III.5.2), so F is Borel. Since $E \subset F$, we have $\dim_{\text{H}} F > a$. Applying Theorem III.5.4 to the nonempty Borel sets F, K_0 gives

$$\sup_{x \in F} \dim_{\text{H}} \pi_x(K_0) \geq \min\{\dim_{\text{H}} F + \dim_{\text{H}} K_0 - 1, 1\} > \sigma,$$

contrary to the definition of F . Hence $\dim_{\text{H}} E \leq b$. □

Theorem III.5.2 (low-dimensional exceptional centres). *Let $K \subset \mathbb{R}^2$ be Borel with $\dim_{\text{H}} K \leq 1$. Then*

$$\dim_{\text{H}}\{x \in \mathbb{R}^2 \setminus K : \dim_{\text{H}} \pi_x(K) < \dim_{\text{H}} K\} \leq 1.$$

Deduction from the radial Kaufman theorem. The radial Kaufman theorem is stated as Theorem III.5.3 immediately below. Put $t = \dim_{\text{H}} K$. If $t = 0$, the set in the conclusion is empty. Otherwise let

$$E = \{x \notin K : \dim_{\text{H}} \pi_x(K) < t\}.$$

If $\dim_{\text{H}} E > 1$, then, by countable stability, for some rational $q < t$ the set $E_q = \{x \in E : \dim_{\text{H}} \pi_x(K) \leq q\}$ has dimension greater than 1. Choose, by inner regularity of Hausdorff dimension for Borel sets, a compact $K_0 \subset K$ with $\dim_{\text{H}} K_0 > q$. Put $F_q = \{x \in \mathbb{R}^2 \setminus K_0 : \dim_{\text{H}} \pi_x(K_0) \leq q\}$. Then $E_q \subset F_q$. The set F_q is Borel: on compact subsets of $\mathbb{R}^2 \setminus K_0$, the map $x \mapsto \pi_x(K_0)$ is continuous in the Hausdorff metric, and the Hausdorff-dimension sublevel sets in the hyperspace of compact subsets of S^1 are Borel (use finite open covers for fixed-scale Hausdorff content, then take countable limits over scales and exponents). Thus $\dim_{\text{H}} F_q > 1$, and F_q cannot lie in an affine line. Theorem III.5.3, applied with $X = F_q$ and $Y = K_0$, then gives

$$\sup_{x \in F_q} \dim_{\text{H}} \pi_x(K_0) \geq \min\{\dim_{\text{H}} F_q, \dim_{\text{H}} K_0, 1\} > q,$$

contrary to the defining upper bound q . This proves the claim. $\square$

III.5.4 Radial Kaufman and Falconer estimates

The low-dimensional theorem needs a first, possibly small, thin-tube exponent before the sharp incidence improvement can begin. The following planar argument proves that seed directly. It is a local version of the spherical-projection mechanism in Shmerkin's Appendix B [16].

Lemma III.5.2.1 (one-scale heavy-tube pruning). *Let $0 < s \leq 1$ and let μ, ν be s -Frostman probability measures in $B(0, 1)$, both with constant at most C . There exists $\eta = \eta(s) > 0$ such that, for all sufficiently small δ (depending also on C), there are a Borel set E_δ and a Borel choice $x \mapsto P_\delta(x)$ of either an empty set or a radius- $\delta^{\eta/2}$ tube through x , with*

$$\mu(E_\delta) \geq 1 - C_1 \delta^{s/2}, \quad (5.1t)$$

and, for every $x \in E_\delta$ and radius- δ tube T through x ,

$$\nu(T \setminus P_\delta(x)) \leq C_1 \delta^\eta. \quad (5.1u)$$

The constants are uniform when s ranges in a compact subset of $(0, 1]$.

Proof. Choose $0 < \eta < s/50$. Use a fixed finite $c\delta$ -net of line parameters in the bounded line space meeting $B(0, 1)$, and enlarge its tubes by a fixed factor. Every radius- δ tube in the unit ball is contained in a member of this family; all overlap and intersection estimates below are invariant under this fixed enlargement. Call a net tube *heavy* if its ν -mass is at least δ^η .

Greedy choose heavy tubes $Y_1, \dots, Y_M$ so that

$$\nu(Y_i \cap Y_j) \leq \frac{1}{2} \delta^{2\eta} \quad (i \neq j), \quad (5.1v)$$

and stop when no heavy tube can be added. This finite procedure is deterministic after fixing an order on the net. If $S = \sum_i \nu(Y_i)$, then Cauchy–Schwarz gives $S^2 \leq S + \frac{1}{2} M^2 \delta^{2\eta}$, whereas $S \geq M \delta^\eta$. Thus $M \leq 2 \delta^{-\eta}$. Indeed, if $M \delta^\eta > 2$, then $S > 2$ and $S^2 - S > S^2/2 \geq$

$M^2\delta^{2\eta}/2$, a contradiction. By maximality, every heavy net tube T satisfies $\nu(T \cap Y_j) > \frac{1}{2}\delta^{2\eta}$ for some j .

If the axes of T and Y_j make angle $a > 0$, their intersection in $B(0, 1)$ lies in a ball of radius $C_2\delta/a$. The s -Frostman bound therefore implies

$$a \leq C_3\delta^{1-2\eta/s}. \quad (5.1w)$$

Since the tubes meet inside $B(0, 1)$, this also places $T \cap B(0, 1)$ inside a tube S_j about the axis of Y_j of width $w = C_4\delta^{1-2\eta/s}$. The heavy tubes are therefore covered by $S_1, \dots, S_M$.

Call a centre x *doubly bad* when it lies in two S_i, S_j whose axes make angle at least δ^η . The intersection of such a pair is contained in a ball of radius $C_5w\delta^{-\eta}$. Consequently

$$\mu\{x : x \text{ doubly bad}\} \leq C_6M^2(w\delta^{-\eta})^s \leq C_7\delta^{s-4\eta-s\eta} \leq C_7\delta^{s/2}. \quad (5.1x)$$

Let E_δ be the complement of this Borel set in $\text{spt } \mu$. For $x \in E_\delta$ which lies in at least one S_j , choose the first such index and let $P_\delta(x)$ be the tube through x parallel to S_j of width $\delta^{\eta/2}$; otherwise put $P_\delta(x) = \emptyset$. This is a Borel choice. If a heavy net tube T contains x , it is contained in some S_i containing x . Because x is not doubly bad, S_i and the selected S_j make angle less than δ^η . Both axes pass within w of x , so inside $B(0, 1)$ $T \subset P_\delta(x)$ for small δ , since $w + \delta^\eta = o(\delta^{\eta/2})$. Thus any radius- δ tube through x that meets $B(0, 1) \setminus P_\delta(x)$ has a light containing net tube. Its remaining ν -mass is at most δ^η ; fixed net enlargement gives (5.1u). All exponent choices can be made uniformly on compact s -intervals. $\square$

Lemma III.5.2.2 (positive thin-tube seed). *Let $0 < s \leq 1$ and let μ, ν be s -Frostman probability measures in a fixed bounded ball. If $\nu(\ell) = 0$ for every affine line ℓ , then for every $\epsilon > 0$ there are $\beta = \beta(s) > 0$ and $K = K(\mu, \nu, \epsilon) < \infty$ and a Borel graph G of $\mu \otimes \nu$ -mass at least $1 - \epsilon$ such that*

$$\nu(G_x \cap T_r(x, e)) \leq Kr^\beta \quad (x, e, 0 < r \leq 1). \quad (5.1y)$$

Proof. Normalize the common support ball to $B(0, 1)$ by a fixed similarity. Line-nullity and compactness of the bounded line space imply

$$\sup_{\ell} \nu(\ell^{(h)}) \longrightarrow 0 \quad (h \downarrow 0). \quad (5.1z)$$

Indeed, otherwise lines ℓ_j and widths $h_j \downarrow 0$ with $\nu(\ell_j^{(h_j)}) \geq c > 0$ have a convergent line subsequence; every fixed neighbourhood of its limit eventually contains those tubes, implying positive mass on the limit line.

Take the exponent η from Lemma III.5.2.1, decreasing it so that $\eta/2 = 2^{-N}$ for an integer $N \geq 1$. This decrease preserves the lemma: enlarge each exceptional tube from width $\delta^{\eta_{\text{old}}/2}$ to width $\delta^{\eta/2}$ and weaken $\delta^{\eta_{\text{old}}}$ to δ^η . Choose $\delta_0 > 0$ so small that the one-scale lemma applies at every $\delta_n = \delta_0^{2^n}$,

$$\sum_{n \geq 0} C_1\delta_n^{s/2} < \epsilon/4, \quad \sum_{n \geq 0} C_1\delta_n^\eta < \epsilon/4, \quad (5.1aa)$$

and, by (5.1z), every tube of width at most $\delta_0^{\eta/2}$ has ν -mass below $\epsilon/(4N)$. At each scale let $E_n, P_n(x)$ be the objects of Lemma III.5.2.1; set $E = \bigcap_{n \geq 0} E_n$ and

$$G = \{(x, y) : x \in E, y \notin \bigcup_{n \geq 0} P_n(x)\}. \quad (5.1ab)$$

The finite-net construction makes G Borel and $\mu(E) > 1 - \epsilon/4$.

For the first N indices, $\sum_{n < N} \nu(P_n(x)) < \epsilon/4$. For $n \geq N$ we have $\delta_n^{\eta/2} = \delta_{n-N}$, so the one-scale estimate at index $n - N$ gives

$$\nu(P_n(x) \setminus P_{n-N}(x)) \leq C_1 \delta_{n-N}^\eta. \quad (5.1ac)$$

It follows from (5.1aa) that $\nu(G_x) > 1 - \epsilon/2$ for every $x \in E$. Hence $(\mu \otimes \nu)(G) > 1 - \epsilon$. If $\delta_n < r \leq \delta_{n-1}$, then any radius- r tube through x is contained in a radius- δ_{n-1} tube and, since G_x avoids $P_{n-1}(x)$, (5.1u) gives

$$\nu(G_x \cap T_r) \leq C_1 \delta_{n-1}^\eta \leq C_1 r^{\eta/2}. \quad (5.1ad)$$

For $r > \delta_0$, enlarge the constant. Thus (5.1y) holds with $\beta = \eta/2$. $\square$

Lemma III.5.2.3 (line-supported centres). *Let $0 < s \leq 1$, let μ, ν be s -Frostman probability measures in a fixed bounded set, and assume $\text{spt } \mu$ lies on a line ℓ while $\text{dist}(\text{spt } \nu, \ell) > 0$. Then for every $0 \leq q < s$ one has $\dim_{\text{H}} \pi_x(\text{spt } \nu) \geq q$ for μ -almost every x .*

Proof. Apply a rigid motion so $\ell = \{(u, 0) : u \in \mathbb{R}\}$. Let $y_i = (a_i, b_i) \in \text{spt } \nu$ with $|b_i| \geq \Delta > 0$, and write $x = (u, 0)$. Since supports are bounded and separated, the implication $|\pi_x(y_1) - \pi_x(y_2)| \leq h|y_1 - y_2|$ forces the following determinant to have absolute value at most $C_0 h|y_1 - y_2|$:

$$|(a_1 - u)b_2 - (a_2 - u)b_1|. \quad (5.1ae)$$

For $h > 0$, the set where this determinant is at most $h|y_1 - y_2|$ has μ -mass at most $C_\Delta h^s$. To see this, if $|b_1 - b_2| \geq c_\Delta |y_1 - y_2|$, the determinant is affine in u with slope of that size, so the permitted u -interval has length $O(h)$. Otherwise $|a_1 - a_2| \asymp |y_1 - y_2|$ and

$$(a_1 - u)b_2 - (a_2 - u)b_1 = (a_1 - a_2)b_2 + (a_2 - u)(b_2 - b_1).$$

Taking c_Δ small in terms of the bounded support diameter makes the second term at most half the first; the sublevel set is then empty for small h , and the bound is trivial for larger h . Layer cake therefore gives, for $q < s$,

$$\int I_q((\pi_x)_\# \nu) d\mu(x) \leq C_{q,s,\Delta} I_q(\nu) < \infty. \quad (5.1af)$$

The last energy is finite by the s -Frostman bound. For μ -almost every x the projected q -energy is finite, and the energy-dimension principle gives the stated lower bound. $\square$

Theorem III.5.3 (radial Kaufman estimate). *Let $X, Y \subset \mathbb{R}^2$ be nonempty Borel sets. If X is not contained in an affine line, then*

$$\sup_{x \in X} \dim_{\text{H}} \pi_x(Y \setminus \{x\}) \geq \min\{\dim_{\text{H}} X, \dim_{\text{H}} Y, 1\}.$$

Proof. Put $d = \min\{\dim_{\text{H}} X, \dim_{\text{H}} Y, 1\}$. There is nothing to prove when $d = 0$. Suppose first that for every $q < d$ some affine line ℓ_q satisfies $\dim_{\text{H}}(Y \cap \ell_q) > q$. Since X is not contained in ℓ_q , choose $x_q \in X \setminus \ell_q$. Radial projection from x_q is bi-Lipschitz on each

compact subsegment of ℓ_q ; hence $\dim_{\mathbb{H}} \pi_{x_q}(Y \cap \ell_q) > q$. Letting $q \uparrow d$ proves the theorem in this case.

Otherwise there are $q_0 < d$ and a number s with $q_0 < s < d$ such that

$$\dim_{\mathbb{H}}(Y \cap \ell) \leq q_0 < s \quad \text{for every affine line } \ell. \quad (5.1ag)$$

By Frostman's lemma choose compactly supported s -Frostman probabilities μ on X and ν on Y . Equation (5.1ag) implies $\nu(\ell) = 0$ for all lines: a positive ν -mass subset of ℓ would have Hausdorff dimension at least s .

If $\mu(\ell) > 0$ for some line ℓ , restrict and renormalize μ to a positive-mass compact part of ℓ . Since $\nu(\ell) = 0$, restrict and renormalize ν to a positive-mass compact part at positive distance from ℓ . The resulting measures retain the s -Frostman exponent. Lemma III.5.2.3 gives $\sup_{x \in X} \dim_{\mathbb{H}} \pi_x(Y) \geq s$.

If instead $\mu(\ell) = 0$ for every line, restrict and renormalize μ, ν to two disjoint positive-mass compact balls and apply a fixed similarity so both supports lie in $B(0, 1)$. They remain s -Frostman, line-null, and separated by $\Delta > 0$. Apply Lemma III.5.2.2 in both orders, with initial error ϵ_0 so small that the intersection of the two Borel seed graphs has mass at least $1 - 2\epsilon_0$. It obeys the $(\beta, K, 1 - 2\epsilon_0)$ -thin-tube bounds in both directions for some $0 < \beta < s$.

The symmetric thin-tube improvement Lemma III.9.11 is proved in Chapter 9 from the Chapter 6 primal sharp incidence theorem. Fix any $\beta < u < s$. Its gain is bounded below as the working exponent ranges over the compact interval $[\beta, u]$; this follows directly from the strict inequalities in its displayed parameter hierarchy, the finite-cover argument in Corollary III.9.12, and compact-uniformity of the Chapter 6 incidence gain. Choose ϵ_0 sufficiently small for the finite number of iterations. Repeated application gives a positive-mass Borel graph G with u -thin tubes for (μ, ν) . Choose $x \in X$ with $\nu(G_x) > 0$. The supports are separated, so the inverse image of any angular arc of radius r lies in a radius- Cr tube through x . The projected measure $(\pi_x)_\#(\nu|_{G_x})$ is therefore u -Frostman. Hence $\dim_{\mathbb{H}} \pi_x(Y) \geq u$. Let $u \uparrow s$ and then $s \uparrow d$ while maintaining (5.1ag). This proves the result. $\square$

A finite projection obstruction

The first ingredient in the Falconer proof converts many rich radial tubes into a regular family in line space. A finite, δ -separated family P in a fixed bounded point or line chart is a *relative* (δ, u, C) -set when $|P \cap B(z, \rho)| \leq C\rho^u|P|$ for $\delta \leq \rho \leq 1$. Tubes through a point are measured by their directions; on a fixed bounded chart this is equivalent to measuring their line parameters. Fixed changes of tube width, bounded charts, and separation constants change only the constants in this definition.

Lemma III.5.3.1 (regular union of radial tubes). *Let $0 \leq s \leq 2$, $0 \leq \sigma < 1$, and $\xi > 0$. There are $\epsilon_0 = \epsilon_0(\sigma, \xi) > 0$ and $\delta_0 = \delta_0(\sigma, \xi) > 0$, independent of s , such that, if $0 < \delta < \delta_0$, $0 < \epsilon < \epsilon_0$, E is a nonempty relative $(\delta, s, \delta^{-\epsilon})$ -set of centres, and each $x \in E$ has a nonempty relative $(\delta, \sigma, \delta^{-\epsilon})$ -family $\mathcal{T}_x$ of δ -tubes through x , then $\bigcup_{x \in E} \mathcal{T}_x$ contains a relative $(C\delta, \gamma, \delta^{-\xi})$ -set of tubes, where*

$$\gamma = \sigma + \min\{s, \sigma\}. \quad (5.1a)$$

The conclusion is valid after a fixed enlargement of all selected tubes; C is absolute for the chosen bounded charts.

Proof. If $\sigma = 0$, a singleton suffices. If $s = 0$, choose one centre; the relative family at that centre already has the required exponent. We may therefore assume $s, \sigma > 0$. Divide point and line space into finitely many bounded duality charts. Pigeonholing retains one chart and one positive fixed fraction of the centres and, for each retained centre, a fixed fraction of its incident tubes. Relative constants change by at most a fixed factor. Move each tube axis by $O(\delta)$ so it passes exactly through its centre. Since the tubes were separated in directions, this changes their separation and regularity only by fixed factors. We absorb all these factors by first replacing ε with 2ε .

Apply the proved Katz–Tao decomposition Lemma III.7.9 to the centres, with loss ε . One colour class E_0 has at least $\delta^{-s+O(\varepsilon)}$ elements and satisfies $|E_0 \cap B(z, \rho)| \leq (\rho/\delta)^s$. Do the same to each $\mathcal{T}_x$ and retain a colour class of at least $\delta^{-\sigma+O(\varepsilon)}$ tubes. By discarding excess elements, we may arrange that all retained fibres have the same integer size $M \geq \delta^{-\sigma+O(\varepsilon)}$; their absolute Katz–Tao bounds persist. All upper cardinality bounds are $O(\delta^{-s})$ or $O(\delta^{-\sigma})$, respectively.

Under the exact chart duality of Lemma III.7.6, the centres become a separated family Ω of dual lines, and each tube family becomes a separated family $P(\ell)$ of dual points incident to its corresponding line. Thus the hypotheses of the locally proved overlap Lemma III.6.30.3 hold with $(\alpha, \beta) = (\sigma, s)$ and $A = \delta^{-O(\varepsilon)}$. It gives

$$\left| \bigcup_{\ell \in \Omega} P(\ell) \right| \geq \delta^{-\gamma+O(\varepsilon)} \quad (5.1b)$$

after weakening the harmless logarithmic factor. This alone does not imply regularity of the union, so we give the required content argument.

Write $U = \bigcup_{\ell \in \Omega} P(\ell)$ and let $N = 1 + \lceil \log_2(1/\delta) \rceil$. The normalized counting measures on Ω and on each $P(\ell)$ satisfy, at radii $\rho \geq \delta$, the Frostman bounds $A\rho^s$ and $A\rho^\sigma$, where $A = \delta^{-O(\varepsilon)}$. We use the following finite form of Frostman’s selection: if a finite set F carries a probability measure m with $m(B(z, \rho)) \leq A\rho^u$ for $r \leq \rho \leq 1$, and $F_0 \subset F$ has $m(F_0) \geq q$, then F_0 contains an r -separated Katz–Tao (r, u, C_u) -set of at least $c_u q A^{-1} r^{-u}$ points. To verify this, run the dyadic capacity recursion in the proof of Lemma III.6.30.1 down to side r . Every cut covering F_0 has capacity at least $c_u q A^{-1} r^{-u}$ by the displayed Frostman bound. The recursion chooses that many leaves, never more than $C_u(R/r)^u$ in a dyadic parent of side R ; a fixed colouring makes their representatives r -separated.

Consider any cover of U by balls whose radii lie in $[\delta, 1]$. Assign every point of U to one covering ball, and divide the balls into the N dyadic radius classes $r \leq \text{rad } B < 2r$. For each $\ell \in \Omega$, some class receives at least $1/N$ of the normalized mass of $P(\ell)$. Pigeonholing again gives one class and a subfamily of lines carrying at least $1/N$ of the normalized mass of Ω . Apply the finite selection just proved to that subfamily and to the assigned fibre points on each selected line, both at scale r . The resulting line and point families meet the hypotheses of Lemma III.6.30.3 with regularity parameter at most CAN , and hence their union occupies at least

$$c(AN)^{-C_0} r^{-\gamma} / (1 + \log(1/r)) \quad (5.1c)$$

r -squares. Choose one dual point in each occupied square and retain one of finitely many residue classes of squares modulo 5 in both coordinates. The resulting points are cr -separated and retain a fixed fraction of the count in (5.1c). Each radius- $2r$ covering ball

contains only $O(1)$ of these points, so the selected class alone has total γ -cost at least $c(AN)^{-C_1}$. The cover was arbitrary: the γ -content of U with covering radii at least δ is bounded below by this quantity.

Apply the same dyadic capacity recursion once more to U at scale $C\delta$. It gives a $C\delta$ -separated Katz–Tao $(C\delta, \gamma, C_\gamma)$ -subset $\mathcal{U}$ with $|\mathcal{U}| \geq c(AN)^{-C_1}\delta^{-\gamma}$. The upper Katz–Tao bound divided by this cardinality is precisely relative nonconcentration with constant $O((AN)^{C_1}) \leq \delta^{-\xi}$, after choosing ε and then δ small enough. These choices are uniform in $s \in [0, 2]$: the doubling and chart constants are fixed, the decomposition proof uses only $4^s \leq 16$, the finite selection capacities use exponents in $[0, 2]$, and the overlap proof bounds every dyadic distance sum by its number of scales rather than by a constant singular as $s \rightarrow \sigma$. Dualize back and enlarge by a fixed factor to restore the original tube supports. $\square$

Lemma III.5.3.2 (discretized radial Falconer obstruction). *Fix $t \in (1, 2]$, $\sigma \in [0, 1)$, and $\xi > 0$. There are $\varepsilon_0, \delta_0 > 0$, uniform in $s \in [0, 2]$, with this property. Suppose $0 < \delta < \delta_0$, $0 < \varepsilon < \varepsilon_0$, P_K is a δ -separated relative $(\delta, t, \delta^{-\varepsilon})$ -set of points, and P_E is a δ -separated relative $(\delta, s, \delta^{-\varepsilon})$ -set of centres. For each $x \in P_E$, suppose a relative $(\delta, \sigma, \delta^{-\varepsilon})$ -family $\mathcal{T}_x$ of δ -tubes passes through x and*

$$|P_K \cap T| \geq \delta^{\sigma+\varepsilon}|P_K| \quad (T \in \mathcal{T}_x). \quad (5.1d)$$

Then $\sigma \geq s + t - 1 - \xi$.

Proof. Put $\gamma = \sigma + \min\{s, \sigma\}$ and choose $\xi > 0$ so small that

$$20\xi < \min\{\xi, 1 - \sigma, t - 1\}. \quad (5.1e)$$

Use Lemma III.5.3.1 with this ξ , taking $\varepsilon \leq \xi$ and then δ sufficiently small. It supplies a relative $(C\delta, \gamma, \delta^{-\xi})$ -family $\mathcal{T}$ drawn from the union of the fibres. Every selected tube has the richness in (5.1d), after a fixed enlargement. Hence

$$|I(P_K, \mathcal{T})| \geq |P_K| |\mathcal{T}| \delta^{\sigma+\varepsilon}. \quad (5.1f)$$

The normalized Fu–Ren theorem III.7.26, proved in Chapter 7, applies at scale $C\delta$ with an arbitrarily small additional loss (absorbed into ξ). Since $t > 1$, it gives

$$|I(P_K, \mathcal{T})| \leq |P_K| |\mathcal{T}| \delta^{\min\{(t+\gamma-1)/2, 1\}-6\xi}. \quad (5.1g)$$

Comparison yields $\sigma + \varepsilon \geq \min\{(t + \gamma - 1)/2, 1\} - 6\xi$. The branch in which the minimum is 1 is impossible by $\sigma < 1 - 6\xi - \varepsilon$. Therefore $2\sigma + 2\varepsilon + 12\xi \geq t + \gamma - 1$. If $s > \sigma$, then $\gamma = 2\sigma$, contradicting $t - 1 > 2\varepsilon + 12\xi$. Thus $s \leq \sigma$ and $\gamma = s + \sigma$, whence $\sigma \geq s + t - 1 - (2\varepsilon + 12\xi)$. Reduce ξ and ε initially so that the parenthesized error is below ξ . $\square$

Lemma III.5.3.3 (one-sided improvement of thin tubes). *Let $0 < s \leq 2$, $1 < t < 2$, $0 \leq \sigma < \min\{s + t - 1, 1\}$, and $0 < \varepsilon < 1/16$. Let μ, ν be probability measures in a fixed bounded ball, at distance at least $\Delta > 0$ from each other, with*

$$\mu(B(z, r)) \leq Cr^s, \quad \nu(B(z, r)) \leq Cr^t \quad (0 < r \leq 1). \quad (5.1h)$$

Suppose there is a Borel graph $G \subset \text{spt } \mu \times \text{spt } \nu$ of product mass at least $1 - \varepsilon$ such that

$$\nu(G_x \cap T_r(x, e)) \leq Kr^\sigma \quad (x \in \text{spt } \mu, e \in S^1, 0 < r \leq 1), \quad (5.1i)$$

where $T_r(x, e)$ is the radius- r tube about the line through x in direction e . Then there are $\eta = \eta(s, t, \sigma) > 0$ and $K' = K'(C, K, \Delta, \epsilon, s, t, \sigma)$ such that the same pair has a Borel graph of mass at least $1 - 4\epsilon$ satisfying

$$\nu(G'_x \cap T_r(x, e)) \leq K' r^{\sigma+\eta} \quad (x, e, r \text{ as above}). \quad (5.1j)$$

The gain can be chosen uniformly positive when (s, t, σ) ranges in a compact subset of the displayed strict parameter region.

Proof. Apply one fixed similarity to the two supports so both lie in $B(0, 1)$. It changes C, K, Δ and the final constant K' by fixed factors, but not any exponent or graph mass. We work in this normalization below. Choose $\zeta > 0$ with $\sigma + 2\zeta < s + t - 1$ and $\sigma + 2\zeta < 1$. Let ϵ_0 be the threshold in Lemma III.5.3.2 for (t, σ, ζ) . Fix $\eta > 0$ so small that

$$\sigma + \eta < 1, \quad 20\eta < \min\{\epsilon_0, \zeta, t - 1\}. \quad (5.1k)$$

All implicit constants below depend on the fixed Frostman data. We show that for sufficiently large K' failure of (5.1j) contradicts the discrete lemma. The graph and tube families will be chosen from fixed finite nets, so all fibre operations are Borel. Let $L = K'/C_*$, where C_* is a fixed constant large enough for the dyadic-scale and angular-net enlargement below. We choose L first, then set $K' = C_*L$.

At every dyadic r , take an r -net of directions and, for each x , the associated $10r$ -tubes through x . Mark a tube when its G_x -mass is at least $Lr^{\sigma+\eta}$. Because the supports are Δ -separated, any $y \in \text{spt } \nu$ lies in at most $C\Delta^{-1}$ of these tubes for fixed x . Consequently the marked family $\mathcal{M}_{x,r}$ has

$$|\mathcal{M}_{x,r}| \leq C\Delta^{-1}L^{-1}r^{-\sigma-\eta} \leq r^{-\sigma-\eta}. \quad (5.1l)$$

Let $H_r \subset G$ consist of pairs (x, y) for which y lies in some marked tube, and put $H = \bigcup_r H_r$. If $(\mu \otimes \nu)(H) < 3\epsilon$, then $G \setminus H$ has mass greater than $1 - 4\epsilon$ and satisfies (5.1j) with constant $K' = C_*L$: any radius- r tube is covered, inside the support ball, by a net tube at the adjacent dyadic scale; if it were heavy on $G \setminus H$, that net tube would have been marked. The choice of C_* accounts for both the scale ratio and this containment. Hence failure implies $(\mu \otimes \nu)(H) \geq 3\epsilon$.

Choose a dyadic $r_0 > 0$ so small that $\sum_{r \leq r_0} 4r^\eta < \epsilon$ and that every constant below is bounded by $r^{-\eta}$ for $r \leq r_0$. Enlarge L until $Lr_0^{\sigma+\eta} > 1$. No tube is then marked at $r > r_0$. Some dyadic $r \leq r_0$ consequently has $(\mu \otimes \nu)(H_r) \geq 4r^\eta$. For $X_r = \{x : \nu((H_r)_x) \geq 2r^\eta\}$, we have $\mu(X_r) \geq 2r^\eta$ after decreasing r_0 . For $x \in X_r$, discard marked tubes carrying less than $r^{\sigma+3\eta}$ of $(H_r)_x$, and let $\mathcal{T}_x$ be the survivors. Their discarded union has mass at most $r^{2\eta}$. Let Y_x be the part of $(H_r)_x$ covered by the survivors. Then

$$\nu(Y_x) \geq r^\eta, \quad r^{\sigma+3\eta} \leq \nu(T \cap Y_x) \leq r^{\sigma-\eta}, \quad r^{-\sigma+2\eta} \leq |\mathcal{T}_x| \leq r^{-\sigma-\eta}. \quad (5.1m)$$

For the upper mass bound, use $Y_x \subset G_x$ and (5.1i), absorbing the fixed tube-width enlargement by $r^{-\eta}$. If the directions of a subfamily lie in an arc of length $\rho \geq r$, then its tubes lie in one $C\rho$ -tube through x . Bounded overlap outside the Δ -ball about x , the lower bound in (5.1m), and the old thin bound show that

$$\#\{T \in \mathcal{T}_x : \angle(T, e) < \rho\} \leq C\Delta^{-1}K\rho^\sigma r^{-\sigma-3\eta} \leq r^{-6\eta}\rho^\sigma |\mathcal{T}_x|. \quad (5.1n)$$

Thus every fibre is a relative $(r, \sigma, r^{-6\eta})$ -set.

The s -Frostman estimate and $\mu(X_r) \geq 2r^\eta$ give a truncated s -content lower bound $cC^{-1}r^\eta$ for X_r . Choose a maximal r -separated net in X_r and assign the restricted measure to nearest net points. The resulting finite measure has the same ball bound up to fixed constants at radii at least r . The finite capacity selection proved in Lemma III.5.3.1 therefore yields an r -separated relative $(r, s, r^{-3\eta})$ -set $P_X \subset X_r$ with

$$cC^{-1}r^{-s+\eta} \leq |P_X| \leq C_s r^{-s}. \quad (5.1o)$$

We next choose one common discretization of ν , rather than separate discretizations for different centres. Put $m_r(y) = \nu(B(y, r))$ and

$$Y_j = \{y \in \text{spt } \nu : 2^{-j-1}Cr^t < m_r(y) \leq 2^{-j}Cr^t\} \quad (j \geq 0). \quad (5.1p)$$

The union of the Y_j has full ν -measure. A maximal r -separated family in Y_j has $O(r^{-2})$ members, and each of its radius- r balls has mass at most $2^{-j}Cr^t$. Hence $\nu(Y_j) \leq 2^{-j}Cr^{t-2}$. Choose $J = O_{C,t,\eta}(\log(1/r))$ so that $\nu(\bigcup_{j>J} Y_j) < r^{3\eta}$. For every $x \in P_X$, some $j(x) \leq J$ has $\nu(Y_x \cap Y_{j(x)}) \geq r^{2\eta}$. Pigeonhole a common j and retain $P'_X \subset P_X$ with $|P'_X| \geq r^\eta |P_X|$. It remains a relative $(r, s, r^{-4\eta})$ -set.

Let P_Y be a maximal $2r$ -separated subset of Y_j . The balls $B(p, r)$, $p \in P_Y$, are disjoint, each with mass between $2^{-j-1}Cr^t$ and $2^{-j}Cr^t$. Covering Y_j locally by radius- r balls centred in Y_j shows that each ball $B(p, 3r)$ meets Y_j in ν -mass at most $C_1 2^{-j}Cr^t$. Therefore

$$c2^j r^{-t+2\eta} \leq |P_Y| \leq C_1 2^j r^{-t}, \quad |P_Y \cap B(z, \rho)| \leq r^{-3\eta} \rho^t |P_Y| \quad (r \leq \rho \leq 1). \quad (5.1q)$$

The lower bound uses $\nu(Y_x \cap Y_j) \geq r^{2\eta}$ for any one $x \in P'_X$; the upper and local bounds use disjointness of the radius- r balls and the t -Frostman estimate. Fixed radii such as $3r$ change only constants.

For each $x \in P'_X$, discard from $\mathcal{T}_x$ the tubes with $\nu(T \cap Y_x \cap Y_j) < r^{\sigma+4\eta}$. By (5.1m), their total mass is at most $r^{3\eta}$, while $\nu(Y_x \cap Y_j) \geq r^{2\eta}$. The retained family $\mathcal{T}'_x$ has at least $r^{-\sigma+3\eta}$ members, and by (5.1n) is a relative $(r, \sigma, r^{-10\eta})$ -set. The local covering estimate for Y_j just used implies

$$|P_Y \cap CT| \geq c2^j r^{-t+\sigma+4\eta} \geq cr^{\sigma+4\eta} |P_Y| \quad (T \in \mathcal{T}'_x). \quad (5.1r)$$

After one fixed rescaling of r and absorption of constants, P'_X , P_Y , and $\mathcal{T}'_x$ satisfy Lemma III.5.3.2 with loss at most $15\eta < \varepsilon_0$. That lemma gives $\sigma \geq s + t - 1 - \zeta$, contradicting the choice of ζ . This proves the improvement.

The choices used only the positive gaps in (5.1k) and the uniform finite-scale thresholds there. On a compact subset of the strict parameter region they can be made uniform, proving the last assertion. $\square$

Theorem III.5.4 (radial Falconer estimate). *Let $X, Y \subset \mathbb{R}^2$ be nonempty Borel sets and assume $\dim_H Y > 1$. Then*

$$\sup_{x \in X} \dim_H \pi_x(Y \setminus \{x\}) \geq \min\{\dim_H X + \dim_H Y - 1, 1\}.$$

Proof. If $\dim_{\text{H}} X = 0$, choose any $x \in X$. For $1 < t < \dim_{\text{H}} Y$ choose a compactly supported t -Frostman measure on $Y \setminus \{x\}$ at positive distance from x . The tube-cover estimate below gives $\dim_{\text{H}} \pi_x(Y) \geq t - 1$; let $t \uparrow \dim_{\text{H}} Y$. Hence assume $\dim_{\text{H}} X > 0$. Let $0 < a < \dim_{\text{H}} X$ and $1 < t < \min\{\dim_{\text{H}} Y, 2\}$, and choose $0 \leq \sigma < \min\{a + t - 1, 1\}$. By Frostman's lemma and compact inner approximation there are probability measures μ, ν on compact subsets of X, Y satisfying the a - and t -Frostman bounds. Restrict to two disjoint positive-mass compact pieces and renormalize; such pieces exist because the t -Frostman measure is nonatomic. Their supports are separated by some $\Delta > 0$ and the same exponents remain valid with a larger constant C .

Every radius- r tube in the support ball is covered by $O(r^{-1})$ radius- Cr balls, so $\nu(T_r) \leq C_0 r^{t-1}$. Thus the whole product is a thin-tube graph at exponent $t - 1$. If $\sigma \leq t - 1$, take G to be the whole product and $K = C_0$. Otherwise, starting with an error $\epsilon_0 > 0$ as small as needed, iterate Lemma III.5.3.3. The triples (a, t, u) with $t - 1 \leq u \leq \sigma$ form a compact subset of its strict parameter region. Its gain therefore has a positive lower bound on this interval. After finitely many steps, choosing $4^N \epsilon_0 < 1/16$ so that every intermediate application is admissible, we obtain a graph G of positive $\mu \otimes \nu$ -mass and a constant K such that

$$\nu(G_x \cap T_r(x, e)) \leq Kr^\sigma \quad (x, e, r). \quad (5.1s)$$

Choose $x \in \text{spt } \mu \subset X$ with $\nu(G_x) > 0$. Since the supports are separated, the preimage under π_x of an arc of radius h lies in a tube $T_{C_1 h}(x, e)$. Therefore the pushforward of $\nu|_{G_x}$ has the σ -Frostman bound. The mass-distribution principle gives $\dim_{\text{H}} \pi_x(Y \setminus \{x\}) \geq \sigma$. Let $\sigma \uparrow \min\{a + t - 1, 1\}$, then $a \uparrow \dim_{\text{H}} X$ and $t \uparrow \dim_{\text{H}} Y$. If the latter dimension is 2, the entire argument uses $t < 2$ and passes to the endpoint only in this final limit. This proves the assertion. $\square$

III.5.5 Complete deductions from the studio theorems

Corollary III.5.5 (low-dimensional exceptional-centre bound). *If $Y \subset \mathbb{R}^2$ is Borel and $t = \dim_{\text{H}} Y \leq 1$, then*

$$\dim_{\text{H}} \{x \in \mathbb{R}^2 \setminus Y : \dim_{\text{H}} \pi_x(Y) < t\} \leq 1.$$

Proof. This is Theorem III.5.2, applied with $K = Y$. $\square$

Corollary III.5.6 (high-dimensional exceptional-centre bound). *Let $Y \subset \mathbb{R}^2$ be Borel with $t = \dim_{\text{H}} Y > 1$. For $0 \leq \sigma \leq 1$,*

$$\dim_{\text{H}} E_{<\sigma}(Y) \leq \max\{1 + \sigma - t, 0\}.$$

Proof. For $\sigma = 0$ the exceptional set is empty. Assume $\sigma > 0$, put $b = \max\{1 + \sigma - t, 0\}$, and suppose that $\dim_{\text{H}} E_{<\sigma}(Y) > b$. Choose $\sigma_j < \sigma$ with $\sigma_j \uparrow \sigma$. Countable stability gives some j for which

$$X_j = \{x : \dim_{\text{H}} \pi_x(Y \setminus \{x\}) \leq \sigma_j\} \quad \text{satisfies} \quad \dim_{\text{H}} X_j > b.$$

Choose a with $b < a < \dim_{\mathbb{H}} X_j$, and then a compact $K \subset Y$ with $t' = \dim_{\mathbb{H}} K > 1 + \sigma_j - a$; this is possible since $a + t - 1 > \sigma_j$. The set

$$F_j = \{x : \dim_{\mathbb{H}} \pi_x(K \setminus \{x\}) \leq \sigma_j\}$$

contains X_j and is Borel. For completeness, write $K_n(x) = \{y \in K : |y - x| \geq 1/n\}$. On each parameter compactum, $x \mapsto \pi_x(K_n(x))$ is an upper-semicontinuous compact-valued map: its graph is closed and the radial map is continuous where $|y - x| \geq 1/n$. For fixed $q > 0$ and scale $h > 0$, the condition that this compact image admit a finite open cover of diameter $< h$ and total q -cost $< c$ is open in x (upper semicontinuity preserves a strict finite cover). Passing through countable choices of n, h, c and rational q shows that $x \mapsto \dim_{\mathbb{H}} \pi_x(K \setminus \{x\})$ is Borel. Thus $\dim_{\mathbb{H}} F_j \geq \dim_{\mathbb{H}} X_j > a$, and Theorem III.5.4, applied to the Borel sets F_j, K , gives

$$\sup_{x \in F_j} \dim_{\mathbb{H}} \pi_x(K \setminus \{x\}) \geq \min\{\dim_{\mathbb{H}} F_j + t' - 1, 1\} > \sigma_j.$$

This contradicts the definition of F_j . □

The hypothesis in the next result is stronger than saying merely that X is not contained in a line: removing any one affine line must preserve its Hausdorff dimension.

We need one special case of the elementary Furstenberg estimate in the *dual* plane. We prove it here rather than import a Furstenberg theorem from a later chapter. The argument is an energy version of the two-line overlap count in [23, Appendix A].

Lemma III.5.7 (the dual (q, q) Furstenberg estimate). *Let $0 < q < 1$, let $F \subset \mathbb{R}^2$ be Borel, and let $\Gamma \subset A(2, 1)$ be a Borel family of affine lines with $\dim_{\mathbb{H}} \Gamma > q$. If $\dim_{\mathbb{H}}(F \cap \ell) > q$ for every $\ell \in \Gamma$, then $\dim_{\mathbb{H}} F \geq 2q$.*

Proof. Choose a q -Frostman probability ν on a compact subfamily of Γ . A countable collection of bounded nonvertical slope charts covers line space, and $F \cap \ell = \bigcup_{R \geq 1} (F \cap \ell \cap \overline{B}(0, R))$. After restricting ν to a positive-mass universally measurable subfamily, one chart and one R work for every retained line: $\dim_{\mathbb{H}}(F \cap \ell \cap \overline{B}(0, R)) > q$. The restriction and renormalization preserve the q -Frostman bound up to a constant. In the chart write $\ell_{a,b} = \{(u, au + b) : u \in \mathbb{R}\}$ and use $d(\ell_{a,b}, \ell_{a',b'}) = |a - a'| + |b - b'|$. For every retained ℓ , Frostman's lemma gives a q -Frostman probability μ_ℓ carried by $F \cap \ell \cap \overline{B}(0, R)$. The standard measurable Frostman selection for Borel sections permits the following uniformization: restrict ν to a positive-mass measurable subfamily and renormalize so that $\ell \mapsto \mu_\ell$ is measurable and

$$\mu_\ell(B(z, \rho)) \leq C\rho^q \quad (z \in \mathbb{R}^2, 0 < \rho \leq 1) \quad (5.22a)$$

with one $C < \infty$. For clarity, this selection follows by considering, for each integer C , the Borel relation of pairs (ℓ, λ) for which λ is a probability carried by $F \cap \ell$ and satisfies the displayed inequalities on a countable dense set of (z, ρ) ; its analytic projection is universally measurable, and one of these projections has positive ν -mass. Restriction multiplies the Frostman constant of ν by at most the reciprocal of that mass.

Set $\mu = \int \mu_\ell d\nu(\ell)$, a probability carried by F . We claim that, for $0 < \rho < 1/2$,

$$\iint \mathbf{1}_{\{|z - z'| < \rho\}} d\mu(z) d\mu(z') \leq C_q \rho^{2q} (1 + |\log \rho|). \quad (5.22b)$$

Indeed, if $d = d(\ell, \ell') > C_0 \rho$, the points of ℓ in the bounded support ball that lie within ρ of ℓ' form an interval of length $O(\rho/d)$, or the set is empty. To verify this geometric assertion, write $|(a - a')u + (b - b')| \leq \rho$ at one such point u . Boundedness of u gives $d \leq |a - a'| + \rho$, hence $|a - a'| \geq d$ in the nonempty case; the asserted interval length follows. By (5.22a), its μ_ℓ -mass is $O((\rho/d)^q)$, while every ρ -ball has $\mu_{\ell'}$ -mass $O(\rho^q)$. If $d \leq C_0 \rho$, use the latter estimate alone. Thus the pair contribution is bounded by

$$C_q \rho^q \min\{1, (\rho/d(\ell, \ell'))^q\}.$$

The q -Frostman bound for ν gives $O(\rho^q)$ mass to $d \leq C_0 \rho$. Each dyadic annulus $2^j \rho < d \leq 2^{j+1} \rho$ contributes at most $C_q \rho^{2q}$ after integration in ℓ' ; there are $O(1 + |\log \rho|)$ annuli. Integrating also in ℓ proves (5.22b).

For every $0 < \nu < 2q$, the layer-cake identity and (5.22b) give

$$I_\nu(\mu) = \nu \int_0^\infty \rho^{-\nu-1} (\mu \otimes \mu) \{ |z - z'| < \rho \} d\rho < \infty.$$

The energy criterion yields $\dim_H F \geq \nu$; let $\nu \uparrow 2q$. □

Corollary III.5.8 (continuum Beck theorem). *Let $X \subset \mathbb{R}^2$ be Borel and suppose that*

$$\dim_H(X \setminus \ell) = \dim_H X \quad \text{for every affine line } \ell.$$

Then

$$\dim_H \mathcal{L}(X) \geq \min\{2 \dim_H X, 2\}.$$

Proof. Put $s = \dim_H X$. The case $s = 0$ is immediate. Fix $0 < q < \min\{s, 1\}$ and then choose u with $q < u < s$. There is a compact $K \subset X$ such that

$$\dim_H(K \setminus \ell) > u \quad \text{for every affine line } \ell. \tag{5.22c}$$

To see this, first choose a compact $K_0 \subset X$ with $\dim_H K_0 > u$. At most one line ℓ_0 can have $\dim_H(K_0 \setminus \ell_0) \leq u$: for two distinct such lines, their complements cover K_0 apart from at most one point, contradicting $\dim_H K_0 > u$. If there is no such line, take $K = K_0$. Otherwise the hypothesis on X supplies a compact $K_1 \subset X \setminus \ell_0$ with $\dim_H K_1 > u$; take $K = K_0 \cup K_1$. This proves (5.22c).

Let

$$B_q = \{x \in K : \dim_H \pi_x(K \setminus \{x\}) < q\}.$$

This set is Borel by the compact-valued truncation argument in the proof of Corollary III.5.6. We claim that $\dim_H(K \setminus B_q) \geq u$. Otherwise $\dim_H B_q = \dim_H K > u$. By (5.22c), B_q cannot be contained in a line. The radial Kaufman theorem III.5.3, applied to B_q and K , would give

$$\sup_{x \in B_q} \dim_H \pi_x(K \setminus \{x\}) \geq \min\{\dim_H B_q, \dim_H K, 1\} > q,$$

contradicting the definition of B_q . The claim follows.

Work in one of finitely many bounded slope charts. For each closed direction arc underlying such a chart, the dimension of its intersection with $\pi_x(K \setminus \{x\})$ is a Borel function

of x : use the same compact truncations and finite-cover test as in Corollary III.5.6. Hence the finite-chart pigeonhole below retains a Borel set. There is a Borel subset $G \subset K \setminus B_q$ with $\dim_{\mathbb{H}} G \geq u$ such that, for every $x \in G$, a set of directions of dimension at least q is represented by lines in this chart spanned by x and K . The point–line duality

$$(a, b) \longleftrightarrow D(a, b) = \{(z, az + b) : z \in \mathbb{R}\}$$

sends the pencil of lines through $x = (x_1, x_2)$ to the dual line $\{(a, b) : b = x_2 - ax_1\}$. On the bounded charts these maps and their inverses are bi-Lipschitz. Hence the set $D^* \mathcal{L}(K)$ of dual points is Borel and meets each dual line indexed by $x \in G$ in a set of dimension at least q . (Here $\mathcal{L}(K)$ is F_σ , since it is the union of compact images of $\{(x, y) \in K^2 : |x - y| \geq 1/n\}$.) Apply Lemma III.5.7 with any $q' < q$. Its line family has dimension at least $u > q'$ and its sections have dimension at least $q > q'$, so

$$\dim_{\mathbb{H}} \mathcal{L}(X) \geq \dim_{\mathbb{H}} \mathcal{L}(K) \geq 2q'.$$

Let $q' \uparrow q$ and then $q \uparrow \min\{s, 1\}$. This gives the claimed $\min\{2s, 2\}$ lower bound. $\square$

III.5.6 Concordance of the two exceptional-set currencies

The preprint-style exceptional-centre formulations and the published Kaufman–Falconer package use slightly different inequalities. The next two propositions show that their numerical conclusions agree where their ranges overlap.

Proposition III.5.8 (high-dimensional concordance). *Let Y be Borel with $t = \dim_{\mathbb{H}} Y > 1$ and $0 \leq \sigma < 1$. Then Corollary III.5.6 implies*

$$\dim_{\mathbb{H}} \{x \in \mathbb{R}^2 \setminus Y : \dim_{\mathbb{H}} \pi_x(Y) \leq \sigma\} \leq \max\{1 + \sigma - t, 0\}.$$

Proof. For every $\tau \in (\sigma, 1]$,

$$\{x \notin Y : \dim_{\mathbb{H}} \pi_x(Y) \leq \sigma\} \subset E_{<\tau}(Y).$$

Corollary III.5.6 bounds the right-hand side by $\max\{1 + \tau - t, 0\}$. Let $\tau \downarrow \sigma$. $\square$

Proposition III.5.9 (low-dimensional concordance). *If $\dim_{\mathbb{H}} Y \leq 1$, Corollary III.5.5 has exactly the same exceptional-centre conclusion as Theorem III.5.2, including the exclusion of centres lying in Y .*

Proof. This follows directly by comparing the two definitions; neither the strict inequality in the projection dimension nor the centre convention changes. $\square$

III.5.7 Boundary tests

Example III.5.10 (line obstruction and sharpness of the bound one). *Let $A \subset \mathbb{R}$ be a nonempty compact set with $0 < \dim_{\mathbb{H}} A \leq 1$, and put $K = A \times \{0\}$. For every centre x on the horizontal line but outside K , the radial image $\pi_x(K)$ has at most two points and hence dimension zero. The set of such centres has Hausdorff dimension one. Thus the universal right-hand side 1 in the low-dimensional theorem cannot be lowered without an additional line-avoidance hypothesis.*

Example III.5.11 (centres on a product set). *Let $Y = A \times B$, where $A, B \subset \mathbb{R}$ are nonempty compact sets, and let $x = (a_0, b_0) \in Y$. Since $\pi_x(x)$ is undefined, all statements use $\pi_x(Y \setminus \{x\})$. Away from the vertical direction the same radial data may be parametrised by the slope set*

$$\left\{ \frac{b - b_0}{a - a_0} : (a, b) \in (A \times B) \setminus \{x\}, a \neq a_0 \right\}.$$

This identity is only a coordinate description. It does not turn a product set into an independent-coordinate projection problem, and none of the preceding theorems assumes such a reduction.

III.5.8 Consequences and scope

The exact exceptional-centre statements and the comparison deductions above will be used later. Their cited proofs contain incidence machinery not reproduced here; the Bourgain, Shmerkin, and sharp incidence developments of the next chapters are therefore established from their own stated inputs.

Chapter III.6

Bourgain Projections and Nonlinear Extensions

This chapter develops a finite-scale projection principle strong enough to survive large-subset restriction and then transports it to smooth nonlinear families. The linear mechanism is Bourgain's: if a set and a family of directions are quantitatively spread across scales, persistent projection collapse forces approximate additive structure that contradicts scalar sum-product expansion. The nonlinear mechanism is Shmerkin's: regularise the source measure into controlled branching pieces, linearise the map on each useful block, and add the resulting entropy gains across scales.

The projection-growth argument uses scalar sum-product expansion and two locally proved rank-one regularity/slicing lemmas. Its multiscale nonlinear part additionally requires disjoint finite-scale measure regularisation, proved in this chapter as X-REG-III.6.

Prerequisites and source boundary. The basic projection-energy estimates of Chapters III.1–III.2 and the finite-scale entropy results of Volume II are available. The disjoint regularisation required later is proved below as Theorem III.6.6. The scalar sum-product mechanism, including the one-dimensional Bourgain dense-ring and filling steps, is reconstructed locally from Bourgain [11], He [14], and the streamlined coefficient-elimination proof of O'Regan–Shmerkin–Wang [13]; the regular bad-direction and codimension-one slicing lemmas are reconstructed locally from the rank-one machinery in He [15]. Bourgain's projection theorem [11] is reconstructed rather than cited as a black box. The multiscale nonlinear architecture follows Shmerkin [16].

III.6.1 Dyadic and nonconcentration conventions

For an integer $m \geq 0$, let $\mathcal{D}_m^d$ be the half-open dyadic cubes of side length 2^{-m} in $\mathbb{R}^d$. For a bounded set A ,

$$N(A, m) = \#\{Q \in \mathcal{D}_m^d : Q \cap A \neq \emptyset\}.$$

Changing from dyadic cubes to balls of radius comparable to 2^{-m} changes N by a factor depending only on d ; all exponential estimates therefore tolerate this change after a harmless adjustment of the slack parameter.

For $\theta \in \mathbb{S}^{d-1}$ write

$$P_\theta(x) = x \cdot \theta.$$

For an affine hyperplane H and $r > 0$, $H^{(r)}$ denotes its open r -neighbourhood. A probability measure ρ on $\mathbb{S}^{d-1}$ has **hyperplane nonconcentration exponent κ at scale m with loss δ** if

$$\rho(H^{(r)} \cap \mathbb{S}^{d-1}) \leq 2^{\delta m} r^\kappa, \quad 2^{-m} \leq r \leq 1. \quad (6.1)$$

A bounded set X has **all-scale $(\alpha, \kappa, \delta, m)$ nonconcentration** if

$$\begin{aligned} N(X, m) &\geq 2^{(\alpha-\delta)m}, \\ N(X \cap B(x, r), m) &\leq 2^{\delta m} r^\kappa N(X, m) \quad (2^{-m} \leq r \leq 1). \end{aligned} \quad (6.2)$$

The exponent in the second line is intentionally written κ : Bourgain's theorem allows the local nonconcentration exponent to be weaker than the global size exponent α .

These conventions distinguish three hypotheses that will never be conflated: Frostman decay, Ahlfors regularity, and finite-scale Bourgain nonconcentration.

Lemma III.6.1 (dyadic/ball covering comparison). *For every d there is $C_d \geq 1$ such that for every bounded $A \subset \mathbb{R}^d$, every m , and every x, r with $r \geq 2^{-m}$,*

$$C_d^{-1} N(A \cap B(x, r), m) \leq N_{2^{-m}}(A \cap B(x, r)) \leq C_d N(A \cap B(x, 2r), m).$$

Here N_ϵ is the minimum number of radius- ϵ balls required to cover a set.

Proof. A radius 2^{-m} ball meets at most $O_d(1)$ dyadic m -cubes, while every such cube is contained in a ball of radius $\sqrt{d}2^{-m}$. Enlarging the ambient ball by a fixed factor absorbs the boundary cubes. $\square$

Lemma III.6.2 (Frostman decay survives finite-scale discretisation). *Suppose μ is a probability measure on $B(0, 1)$ and*

$$\mu(B(x, r)) \leq Cr^s \quad (0 < r \leq 1).$$

Let μ_m be the measure obtained by moving the mass of each dyadic m -cube to an arbitrary point of that cube (equivalently, use the corresponding 2^{-m} -atomic discretisation). Then, for $2^{-m} \leq r \leq 1$,

$$\mu_m(B(x, r)) \leq C_d C(r + 2^{-m})^s \leq C'_{d,C,s} r^s. \quad (6.5)$$

The same statement holds for a restriction μ_X at the cost of multiplying the constant by $\mu(X)^{-1}$.

Proof. If an atom of μ_m lies in $B(x, r)$, its source cube is contained in $B(x, r + C_d 2^{-m})$. Hence

$$\mu_m(B(x, r)) \leq \mu(B(x, r + C_d 2^{-m})) \leq C(r + C_d 2^{-m})^s.$$

Since $r \geq 2^{-m}$, this is at most a constant multiple of r^s . The assertion for μ_X follows by division by $\mu(X)$. $\square$

This lemma is only a discretisation statement. It does **not** assert Ahlfors regularity or near-uniform cube masses.

III.6.2 Robustness and entropy

For a probability measure μ , call μ (a, δ, m) -**robust** if every Borel set A with $\mu(A) > 2^{-\delta m}$ satisfies

$$N(A, m) > 2^{am}. \quad (6.3)$$

If A has positive μ -mass, write $\mu_A = \mu(A)^{-1}\mu|_A$. For a finite partition $\mathcal{P}$, put

$$H(\mu, \mathcal{P}) = \sum_{P \in \mathcal{P}} \mu(P) \log_2 \frac{1}{\mu(P)}.$$

For $\Delta \geq 1$, the Δ -**robust dyadic entropy** is

$$H_m^\Delta(\mu) = \inf\{H(\nu, \mathcal{D}_m) : \nu \in \mathcal{P}(\mathbb{R}^d), \nu \leq \Delta\mu\}. \quad (6.4)$$

Lemma III.6.3 (robustness at all large scales forces dimension). *Let μ be a probability measure. Assume that for every sufficiently large m , μ is (a, δ, m) -robust, with fixed $a, \delta > 0$. Then every Borel set A with $\mu(A) > 0$ has $\dim_{\text{H}} A \geq a$.*

Proof. Fix $\epsilon > 0$. Let E_m be the union of a maximal disjoint family of balls of radius 2^{-m} having μ -mass at least $2^{-(a-\epsilon)m}$. There are at most $2^{(a-\epsilon)m}$ such balls. By the finite-overlap covering lemma, the union of their fixed dilates can be covered by at most $C_d 2^{(a-\epsilon)m}$ balls of radius $O(2^{-m})$. Reindexing the scale by a bounded amount and taking m large, this covering number is $< 2^{am}$. Robustness therefore gives

$$\mu(E_m) \leq 2^{-\delta' m}$$

for some $\delta' > 0$ (after harmlessly decreasing δ to absorb dimensional constants). The series is summable. By Borel–Cantelli and the maximality of the family, for μ -almost every x , all sufficiently small dyadic-scale balls centred at x have mass $< C_d 2^{-(a-\epsilon)m}$. Monotonicity in r gives

$$\mu(B(x, r)) \leq C_x r^{a-2\epsilon}$$

for all sufficiently small r . On a positive-mass subset the constant and starting scale may be chosen uniformly. The mass-distribution principle yields $\dim_{\text{H}} A \geq a - 2\epsilon$ for every positive-mass A ; let $\epsilon \downarrow 0$. $\square$

Lemma III.6.4 (robustness gives robust entropy). *Fix $a, \delta, \epsilon > 0$. For sufficiently large m , if μ is (a, δ, m) -robust, then*

$$H_m^{2^{\delta m/2}}(\mu) \geq (a - \epsilon)m. \quad (6.6)$$

Proof. Let $\nu \leq 2^{\delta m/2} \mu$ be a probability measure. If $\nu(A) > 2^{-\delta m/2}$, then $\mu(A) > 2^{-\delta m}$, so $N(A, m) > 2^{am}$. Hence ν is $(a, \delta/2, m)$ -robust. The union E of dyadic m -cubes with $\nu(Q) > 2^{-am}$ has $\nu(E) \leq 2^{-\delta m/2}$; otherwise E would be supported on at most 2^{am} cubes and violate robustness. Put $F = E^c$. Concavity of entropy gives

$$H_m(\nu) \geq \nu(F)H_m(\nu_F) \geq (1 - 2^{-\delta m/2})(am - 1) \geq (a - \epsilon)m$$

for large m . Take the infimum over ν . □

Lemma III.6.5 (finite mass-level decomposition). *Let μ be supported in $[0, 1]^d$ and be constant on dyadic m -cubes. Fix $\epsilon > 0$. Apart from a set of μ -mass at most*

$$2^{-\epsilon m} + C_d m 2^{-2\epsilon m},$$

the occupied m -cubes split into at most $O_d(m)$ disjoint families X_j such that

$$\mu(X_j) \geq 2^{-2\epsilon m} \tag{6.7a}$$

and the masses of any two occupied cubes in the same X_j differ by at most a factor two. If, in addition, $\mu(B(x, r)) \leq Cr^s$ for $r \geq 2^{-m}$, then every such family satisfies

$$N(X_j \cap B(x, r), m) \leq C'_{d,C} 2^{2\epsilon m} r^s N(X_j, m), \quad 2^{-m} \leq r \leq 1. \tag{6.7b}$$

Proof. Put an occupied cube Q in level j when $2^{-j-1} < \mu(Q) \leq 2^{-j}$. There are at most 2^{dm} occupied cubes. Hence the total mass of cubes with $j > (d + \epsilon)m$ is at most $2^{-\epsilon m}$. Among the remaining $O_d(m)$ levels, discard those whose total mass is $< 2^{-2\epsilon m}$; their combined mass is $O_d(m)2^{-2\epsilon m}$. This proves (6.7a) and the decomposition claim.

Fix a retained level and write $M_j = \mu(X_j)$. If it contains n_j cubes, then M_j is comparable to $n_j 2^{-j}$. A ball $B(x, r)$ meets at most

$$Cr^s 2^{j+1}$$

cubes of this level, because every such cube contributes more than 2^{-j-1} mass and their union lies in $B(x, r + C_d 2^{-m})$. Since $n_j \geq M_j 2^j$ and $M_j \geq 2^{-2\epsilon m}$, division gives (6.7b), after changing the constant. □

The nonlinear arguments below require a measure decomposition stronger than the set-valued tree lemma of Volume II.11.

Definition (terminal-discrete and block-regular measures). *Here and below an m -discrete measure on $[0, 1]^d$ means a probability measure which is a constant multiple of Lebesgue measure on every dyadic cube of side 2^{-m} . In particular, restrictions to unions of terminal cubes remain m -discrete after normalisation. If $m = T\ell$ and $\sigma = (\sigma_1, \dots, \sigma_\ell) \in [0, d]^\ell$, we say that such a measure μ is $(\sigma; T)$ -**regular** when every occupied $Q \in \mathcal{D}_{jT}$, with parent $\hat{Q} \in \mathcal{D}_{(j-1)T}$, satisfies*

$$2^{-T\sigma_j} \mu(\hat{Q})/2 \leq \mu(Q) \leq 2^{-T\sigma_j} \mu(\hat{Q}) \quad (1 \leq j \leq \ell). \tag{6.23}$$

Theorem III.6.6 (Disjoint measure regularisation X-REG-III.6). *For every sufficiently small $\epsilon > 0$ there are $T_0 = T_0(d, \epsilon)$ and $C_d > 0$ such that, for every $T \geq T_0$, all sufficiently large ℓ , and $m = T\ell$, every m -discrete probability μ has pairwise disjoint unions of terminal cubes E_i with*

$$\mu\left([0, 1]^d \setminus \bigcup_i E_i\right) \leq 2^{-\epsilon m}, \quad \mu(E_i) \geq 2^{-C_d \epsilon m}, \quad (6.7d)$$

such that each normalised restriction μ_{E_i} is $(\sigma^{(i)}; T)$ -regular for some $\sigma^{(i)} \in [0, d]^\ell$. One may take $C_d = 2$.

Proof. We first find one regular piece of a definite fraction of an arbitrary m -discrete probability ν . Fix $\lambda = 5/4$, so $\lambda^2 < 2$. Let $\eta = \epsilon/4$. At most 2^{dm} terminal cubes meet $[0, 1]^d$; thus the total ν -mass of cubes Q with

$$0 < \nu(Q) < 2^{-(d+\eta)m}$$

is at most $2^{-\eta m}$. Partition the remaining terminal cubes into geometric mass classes $[a, \lambda a)$. There are at most $K_m \leq C_{d,\epsilon}(m+1)$ classes, so one class carries mass at least $(1 - 2^{-\eta m})/K_m$. Retain precisely its terminal cubes, and call their union F_ℓ . The masses of all its terminal cubes lie in one interval $[a_\ell, \lambda a_\ell)$.

Now work upward. Suppose that F_{j+1} is a union of terminal cubes and that the positive masses $\nu(F_{j+1} \cap Q)$, over occupied level- $(j+1)T$ cubes Q , lie in a common interval $[a_{j+1}, \lambda a_{j+1})$. For each level- jT cube P meeting F_{j+1} , put

$$M(P) = \nu(F_{j+1} \cap P).$$

The cube P contains between one and 2^{dT} occupied level- $(j+1)T$ children. Consequently

$$a_{j+1} \leq M(P) < \lambda 2^{dT} a_{j+1}.$$

Partition these parent cubes into geometric mass classes of ratio λ . Their number is at most $K_T \leq C'_d(T+1)$. Retain the class with largest total $\nu(F_{j+1} \cap P)$, and let F_j be the union of its already selected terminal subtrees. Then

$$\nu(F_j) \geq K_T^{-1} \nu(F_{j+1}),$$

and the positive masses $\nu(F_j \cap P)$ at level jT lie in one interval $[a_j, \lambda a_j)$. Deleting entire subtrees does not change any descendant mass ratio inside a retained subtree. Repeat this step for $j = \ell - 1, \dots, 1$, and put $E = F_1$.

For every j , consider the ratios

$$q(P, Q) = \frac{\nu(E \cap Q)}{\nu(E \cap P)} \quad (Q \in \mathcal{D}_{jT}, P = \hat{Q} \in \mathcal{D}_{(j-1)T}, \nu(E \cap Q) > 0).$$

At $j = 1$ their spread is at most λ , since the denominator is the common root mass. At $j \geq 2$, the selected child masses lie in one ratio- λ interval and the selected parent masses lie in another; their spread is at most $\lambda^2 < 2$. Let q_j be the largest of these ratios. Each

parent has at most 2^{dT} children, whose ratios sum to one, so $2^{-dT} \leq q_j \leq 1$. Define $\sigma_j = -T^{-1} \log_2 q_j \in [0, d]$. The spread bound gives, for every occupied edge,

$$\frac{1}{2} 2^{-T\sigma_j} \leq q(P, Q) \leq 2^{-T\sigma_j}.$$

Normalization by $\nu(E)$ leaves these ratios unchanged; hence ν_E is $(\sigma; T)$ -regular.

The selection retained at least

$$\nu(E) \geq \frac{1 - 2^{-\eta m}}{K_m K_T^{\ell-1}}.$$

Choose T_0 so large that $\log_2 K_T/T < \epsilon/4$ for all $T \geq T_0$. For each such T , take ℓ sufficiently large that $1 - 2^{-\eta m} \geq 1/2$ and $\log_2(2K_m)/m < \epsilon/4$. Then $\nu(E) \geq 2^{-\epsilon m/2}$.

Finally apply this one-piece construction greedily to the normalised restriction of μ to its remaining terminal cubes. As long as the remaining μ -mass exceeds $2^{-\epsilon m}$, extract a regular piece E_i carrying at least a $2^{-\epsilon m/2}$ fraction of that mass. The pieces are disjoint unions of terminal cubes. Each extracted piece has $\mu(E_i) > 2^{-3\epsilon m/2} \geq 2^{-2\epsilon m}$. There are finitely many terminal cubes, so the procedure stops, and its unextracted mass is at most $2^{-\epsilon m}$. This proves (6.7d) with $C_d = 2$. $\square$

III.6.3 Bourgain's discretized projection theorem

Theorem III.6.6 (Bourgain discretized projection theorem). *Fix $0 < \alpha < d$ and $0 < \kappa < 1$. There exist $\delta, \eta > 0$ such that for all sufficiently large m the following holds. Let $X \subset B^d(0, 1)$ and $\rho \in \mathcal{P}(\mathbb{S}^{d-1})$ satisfy*

$$\begin{aligned} N(X, m) &\geq 2^{(\alpha-\delta)m}, \\ N(X \cap B(x, r), m) &\leq 2^{\delta m} r^\kappa N(X, m), \\ \rho(H^{(r)} \cap \mathbb{S}^{d-1}) &\leq 2^{\delta m} r^\kappa, \end{aligned} \tag{6.7}$$

for $2^{-m} \leq r \leq 1$, all x , and all linear hyperplanes H . Then there exists $E \subset \mathbb{S}^{d-1}$ with $\rho(E) \leq 2^{-\delta m}$ such that for every $\theta \notin E$ and every $X' \subset X$ with $N(X', m) \geq 2^{-\delta m} N(X, m)$,

$$\log_2 N(P_\theta X', m) \geq m \left(\frac{\alpha}{d} + \eta \right). \tag{6.8}$$

This is the robust finite-scale form of Bourgain's projection theorem used throughout the rest of the chapter. The proof below is fully local at the research level: both the scalar arithmetic spine and the He regularity/slicing machinery are reconstructed in the chapter.

III.6.3.1 Proof of Bourgain's theorem

We prove the projection-growth conclusion locally by the following rank-one argument. The two regularity/slicing inputs are reconstructed below from the rank-one argument of He. The scalar linear-action expansion is also local through its additive-combinatorial, dense-ring, and approximate-ring layers. Thus no research-level black box is retained in the proof of Theorem 6.6.

Rank-one proof architecture

For this proof only, put

$$h = 2^{-m}.$$

Thus powers $h^{O(\epsilon)}$ encode losses which are exponential in m but whose exponents can be made arbitrarily small before the argument starts. We use projective lines rather than oriented unit vectors whenever convenient; the two conventions differ by a factor two and the hyperplane-neighbourhood hypothesis (6.7) passes between them.

The proof follows the rank-one specialization of the modern detailed proof in Weikun He's *Orthogonal projections of discretized sets*. What is important here is the ownership boundary. We do **not** import He's projection theorem. We reconstruct the two technical regularity/slicing lemmas from that proof in rank one, together with the scalar sum-product action through the Ruzsa–Pluennecke–BSG layer, He's approximate-ring calculus, the probabilistic full-ball expansion, and the local scalar dense-ring theorem. Thus the entire rank-one Bourgain spine is local after the complete rank-one reconstruction.

Local additive-combinatorial calculus at scale h

The scalar argument repeatedly passes between covering numbers and finite additive combinatorics. We record the required calculus here so that no Balog–Szemerédi–Gowers, Ruzsa, or Pluennecke statement remains hidden behind an interface.

For a bounded set $E \subset \mathbb{R}$, let

$$\mathcal{D}_h(E) = \{j \in \mathbb{Z} : E \cap [jh, (j+1)h) \neq \emptyset\}.$$

A ball of radius h meets at most three such cells, while one cell is contained in a ball of radius h . Hence

$$N_h(E) \leq |\mathcal{D}_h(E)| \leq 3N_h(E). \quad (6.A1)$$

Moreover, replacing a sum or difference of cell-index sets by the cell-index set of the corresponding geometric sum or difference changes indices by only an absolute finite set. Thus every finite-set estimate below transfers to covering numbers, at the cost of absolute constants and a fixed change from h to Ch . In one dimension such a fixed scale change changes covering numbers by at most another absolute constant.

Lemma (Ruzsa triangle inequality at scale h). *For bounded $A, B, C \subset \mathbb{R}$ with $N_h(B) > 0$,*

$$N_h(A - C)N_h(B) \lesssim N_h(A - B)N_h(B - C). \quad (6.A2)$$

Proof. It is enough, by (6.A1), to prove the finite-set inequality

$$|A_0 - C_0| |B_0| \leq |A_0 - B_0| |B_0 - C_0|.$$

For each $d \in A_0 - C_0$, fix one representation $d = a_d - c_d$. The map

$$(d, b) \mapsto (a_d - b, b - c_d)$$

from $(A_0 - C_0) \times B_0$ to $(A_0 - B_0) \times (B_0 - C_0)$ is injective: the two coordinates determine d , and then the chosen pair (a_d, c_d) , after which they determine b . This proves the finite inequality and hence (6.A2). $\square$

Lemma (Ruzsa covering at scale h). *If*

$$N_h(A + B) \leq KN_h(A), \quad K \geq 1,$$

then there is a set $T \subset B$ with $|T| \lesssim K$ such that

$$B \subset A - A + T + [-Ch, Ch] \quad (6.A3)$$

for an absolute constant C .

Proof. After cell discretisation, choose a maximal subset $T_0 \subset B_0$ for which the translates $A_0 + t$, $t \in T_0$, are pairwise disjoint. They all lie in $A_0 + B_0$, so $|T_0||A_0| \leq |A_0 + B_0| \lesssim K|A_0|$, and hence $|T_0| \lesssim K$. Maximality says that for every $b \in B_0$ there are $t \in T_0$ and $a, a' \in A_0$ with $a + b = a' + t$. Thus $b \in A_0 - A_0 + T_0$. Returning from cell indices to the original sets gives (6.A3). $\square$

Lemma (Pluenncke–Ruzsa at scale h). *For fixed nonnegative integers p, q , if*

$$N_h(A + A) \leq KN_h(A),$$

then

$$N_h(pA - qA) \lesssim_{p,q} K^{O(p+q)} N_h(A). \quad (6.A4)$$

Proof. Again it suffices to prove the finite statement. We first prove the form of Petridis' minimal-growth lemma that is needed here. Let A_0, B_0 be finite and choose a nonempty $X \subset A_0$ minimizing

$$\lambda = \frac{|X + B_0|}{|X|}.$$

Then for every finite C ,

$$|C + X + B_0| \leq \lambda |C + X|. \quad (6.A5)$$

We prove this by induction on $|C|$. Remove one element c from C , write $C' = C \setminus \{c\}$, and put

$$X_1 = \{x \in X : c + x \in C' + X\}, \quad X_2 = X \setminus X_1.$$

The new part contributed to $C + X$ by the translate $c + X$ has exactly $|X_2|$ elements. The new part contributed to $C + X + B_0$ is contained in

$$c + ((X_2 + B_0) \setminus (X_1 + B_0)),$$

because a representation through $X_1 + B_0$ was already present in $C' + X + B_0$. Its size is therefore at most

$$|X + B_0| - |X_1 + B_0| \leq \lambda |X| - \lambda |X_1| = \lambda |X_2|,$$

where minimality of X gives $|X_1 + B_0| \geq \lambda |X_1|$. Combining this with the induction hypothesis proves (6.A5).

If $|A_0 + B_0| \leq K|A_0|$, then $\lambda \leq K$, and repeated application of (6.A5) gives

$$|X + rB_0| \leq K^r |X| \quad (r \geq 0). \quad (6.A6)$$

Applying the finite Ruzsa triangle inequality with the auxiliary set $-X$ yields

$$|pB_0 - qB_0||X| \leq |pB_0 + X||X + qB_0| \leq K^{p+q}|X|^2 \leq K^{p+q}|A_0|. \quad (6.A7)$$

Take $B_0 = A_0$ and transfer (6.A7) back through (6.A1). The fixed cell errors only enlarge the exponent and the implicit constant by quantities depending on $p + q$, which proves (6.A4). $\square$

Lemma (Balog–Szemerédi–Gowers at scale h). *Let $U, V \subset \mathbb{R}$ be bounded and let $\tilde{U}, \tilde{V}$ be maximal h -separated subsets. Suppose their sizes are comparable to a number M , and the number of quadruples*

$$(u, v, u', v') \in \tilde{U} \times \tilde{V} \times \tilde{U} \times \tilde{V}$$

with

$$|u + v - u' - v'| \leq Ch$$

is at least $K^{-1}M^3$, where $K \geq 2$ and C is fixed. Then there are $U' \subset U, V' \subset V$ with

$$N_h(U'), N_h(V') \geq K^{-O(1)}M \quad (6.A8)$$

and

$$N_h(U' + V') \leq K^{O(1)}M. \quad (6.A9)$$

The same conclusion holds for $U' + aV'$ after a fixed nonzero scalar change whose condition number is absorbed in the displayed powers of K .

Proof. Cell discretisation and pigeonholing over the finitely many possible cell errors reduce the approximate equation to exact additive energy between finite sets A, B , with $|A|, |B| \asymp M$ and

$$E(A, B) = \sum_s r_{A+B}(s)^2 \geq K^{-O(1)}M^3. \quad (6.A10)$$

Put

$$S = \{s \in A + B : r_{A+B}(s) \geq (2K_1)^{-1}M\},$$

where $K_1 = K^{O(1)}$ is the constant in (6.A10). The contribution to the energy from $(A+B) \setminus S$ is at most

$$(2K_1)^{-1}M \sum_s r_{A+B}(s) \leq (2K_1)^{-1}M^3,$$

so at least a fixed fraction of (6.A10) is supported on S . Since $r_{A+B}(s) \leq M$, the graph

$$G = \{(a, b) \in A \times B : a + b \in S\}$$

therefore satisfies

$$|G| \geq K^{-O(1)}M^2, \quad |S| \leq K^{O(1)}M. \quad (6.A11)$$

(The second inequality follows by dividing the total number of pairs by the defining threshold for S .) Thus no logarithmic pigeonholing loss is needed. It remains to pass from the restricted sumset of G to a full sumset.

We prove the required graph form of BSG. Suppose generally that $G \subset A \times B$, $|G| \geq K_0^{-1}|A||B|$, and its restricted sumset

$$S_G = \{a + b : (a, b) \in G\}$$

has size at most $K_0 M$, with $|A|, |B| \leq M$. Remove from A vertices of degree less than $(2K_0)^{-1}|B|$. At least half of the edges remain; call the remaining left vertex set A_0 . Hence

$$|A_0| \geq (2K_0)^{-1}|A|, \quad (6.A12)$$

and the induced graph has density at least $(2K_0)^{-1}$ relative to $A_0 \times B$.

For $a_1, a_2 \in A_0$, call the pair ϑ -bad if

$$|N(a_1) \cap N(a_2)| < \vartheta(2K_0)^{-2}|B|.$$

For $b \in B$, let $Q(b)$ be the number of bad ordered pairs in $N(b)^2$. Every bad pair is counted for fewer than $\vartheta(2K_0)^{-2}|B|$ choices of b , so

$$\frac{1}{|B|} \sum_b Q(b) \leq \vartheta(2K_0)^{-2}|A_0|^2. \quad (6.A13)$$

On the other hand, Cauchy–Schwarz and the edge-density bound give

$$\frac{1}{|B|} \sum_b |N(b)|^2 \geq (2K_0)^{-2}|A_0|^2. \quad (6.A14)$$

Taking $\vartheta \leq 2^{-8}$, consider

$$|N(b)|^2 - \vartheta^{-1/2}Q(b) - \frac{1}{2}(2K_0)^{-2}|A_0|^2.$$

By (6.A13)–(6.A14) its average over $b \in B$ is positive. Hence for some $b_0 \in B$, with $A_1 = N(b_0)$, the displayed quantity is positive. In particular

$$|A_1| \geq 2^{-1/2}(2K_0)^{-1}|A_0|$$

and, using the same positivity once more,

$$Q(b_0) < \vartheta^{1/2}|A_1|^2.$$

Thus, after weakening the absolute constant,

$$|A_1| \geq cK_0^{-1}|A_0|, \quad \#\{\text{bad pairs in } A_1^2\} \leq \vartheta^{1/2}|A_1|^2. \quad (6.A15)$$

Delete from A_1 every vertex having more than $2\vartheta^{1/4}|A_1|$ bad partners. The remaining set A_2 has $|A_2| \geq |A_1|/2$, and every $a \in A_2$ has at most $4\vartheta^{1/4}|A_2|$ bad partners in A_2 .

Put

$$B_2 = \{b \in B : |N(b) \cap A_2| > 8\vartheta^{1/4}|A_2|\}.$$

Because each vertex of $A_2 \subset A_0$ had degree at least $(2K_0)^{-1}|B|$, choosing $\vartheta = (64K_0)^{-4}$ shows that a fixed polynomial fraction of the edges from A_2 land in B_2 . In particular

$$|A_2| \geq K_0^{-O(1)}|A|, \quad |B_2| \geq K_0^{-O(1)}|B|. \quad (6.A16)$$

For every $a \in A_2$ and $b \in B_2$, at least $4\vartheta^{1/4}|A_2|$ vertices $a_1 \in N(b) \cap A_2$ form a non-bad pair with a ; for each such a_1 there are at least $\vartheta(2K_0)^{-2}|B|$ common neighbours b_1 . Consequently the number of length-three paths

$$a - b_1 - a_1 - b$$

is at least $K_0^{-O(1)}|A||B|$.

Each such path gives three restricted sums

$$z_1 = a + b_1, \quad z_2 = a_1 + b_1, \quad z_3 = a_1 + b$$

in S_G , and

$$a + b = z_1 - z_2 + z_3.$$

Choose one representative pair $(a, b) \in A_2 \times B_2$ for each value of $a + b$. For a fixed representative, the triple (z_1, z_2, z_3) determines the path, while triples arising from representatives of two different sum values are disjoint because $a + b = z_1 - z_2 + z_3$. Therefore

$$|A_2 + B_2| K_0^{-O(1)} |A||B| \leq |S_G|^3,$$

which, together with $|S_G| \leq K_0 M$ and comparability of $|A|, |B|$, gives $|A_2 + B_2| \leq K_0^{O(1)} M$. This proves the finite graph BSG theorem. Applying it to the graph from (6.A11) and returning to scale h proves (6.A8)–(6.A9). $\square$

The scalar linear-action expansion reduced to one Bourgain core

We next reconstruct the scalar specialization of He's linear-action argument. This identifies exactly what remains of Bourgain's one-dimensional sum-product theorem.

For $L \geq 1$, define the set of L -good scalars

$$\mathcal{S}_h(X; L) = \{a \in [-L, L] : N_h(X + aX) \leq L N_h(X)\}. \quad (6.A17)$$

Lemma (approximate-ring calculus for good scalars). *Assume $X \subset [-h^{-\eta}, h^{-\eta}]$, $0 < \eta < 1/10$, and $1, a, b \in \mathcal{S}_h(X; h^{-\eta})$. Then, with absolute constants in the exponents,*

$$a + b, a - b, ab \in \mathcal{S}_h(X; h^{-O(\eta)}). \quad (6.A18)$$

If $|a^{-1}| \leq h^{-\eta}$, then also

$$a^{-1} \in \mathcal{S}_h(X; h^{-O(\eta)}). \quad (6.A19)$$

Any scalar within $h^{1-\eta}$ of an $h^{-\eta}$ -good scalar is $h^{-O(\eta)}$ -good. Finally, for fixed s , every scalar obtained from $1, a_1, \dots, a_s$ by at most s additions, subtractions and multiplications is $h^{-O_s(\eta)}$ -good, and goodness propagates from scale h to every scale $r \geq h$ with the same type of loss.

Proof. Since 1 is good,

$$N_h(X + X) \leq h^{-\eta} N_h(X). \quad (6.A20)$$

Ruzsa covering (6.A3) applied to $X + aX$ gives

$$aX \subset X - X + T_a + [-Ch, Ch], \quad |T_a| \leq h^{-O(\eta)}, \quad (6.A21)$$

and similarly for bX . Hence

$$X + (a \pm b)X \subset 3X - 2X + T + [-Ch, Ch], \quad |T| \leq h^{-O(\eta)}.$$

By (6.A4) and (6.A20), the right-hand side has covering number at most $h^{-O(\eta)}N_h(X)$, proving the additive assertions.

For multiplication, (6.A21) for b gives

$$X + abX \subset X + aX - aX + aT_b + [-Ch^{1-\eta}, Ch^{1-\eta}].$$

Ruzsa covering once more for aX , followed by (6.A4), bounds the covering number by $h^{-O(\eta)}N_h(X)$. If a is invertible, then

$$X + a^{-1}X = a^{-1}(X + aX),$$

and the distortion of scale is at most $h^{-\eta}$, which costs only another $h^{-O(\eta)}$ factor. The perturbation assertion follows from

$$X + bX \subset X + aX + [-Ch^{1-2\eta}, Ch^{1-2\eta}].$$

Iterating the already proved operations yields the fixed-word assertion.

For scale propagation, let $r \geq h$ and choose x maximizing $N_h(X \cap B(x, r))$. Then

$$N_h(X) \leq N_r(X)N_h(X \cap B(x, r)). \quad (6.A22)$$

On the other hand, a maximal r -separated subset of $X + aX$, translated by the local piece $X \cap B(x, r)$, produces disjoint-at-fixed-multiple-of- h copies inside $X + X + aX$. Thus

$$N_h(X + X + aX) \gtrsim N_r(X + aX)N_h(X \cap B(x, r)). \quad (6.A23)$$

The fixed-word estimate gives $N_h(X + X + aX) \leq h^{-O(\eta)}N_h(X)$. Combining this with (6.A22)–(6.A23) yields

$$N_r(X + aX) \leq h^{-O(\eta)}N_r(X),$$

as required. $\square$

Lemma (scalar fullness forces expansion). *Fix $\kappa_0 > 0$ and $0 < \sigma < 1$. There is $c = c(\kappa_0, \sigma) > 0$ such that, for sufficiently small h , if*

$$X \subset [-h^{-\eta}, h^{-\eta}], \quad N_r(X) \geq h^\eta r^{-\kappa_0} \quad (h \leq r \leq 1), \quad (6.A24)$$

$$N_h(X) \leq h^{-\sigma-\eta}, \quad (6.A25)$$

and $\eta > 0$ is sufficiently small compared with κ_0, σ , then

$$\max_{|a| \leq 1} N_h(X + aX) \geq h^{-c} N_h(X). \quad (6.A26)$$

Proof. If already $N_h(X + X) \geq h^{-c} N_h(X)$, there is nothing to prove. Otherwise take c temporarily replaced by a small parameter η_1 . From a maximal r -separated subset of X , translating a densest r -ball gives

$$N_h(X + X) \gtrsim N_r(X) \max_w N_h(X \cap B(w, r)).$$

Using (6.A24) and the assumed small doubling yields

$$\max_w N_h(X \cap B(w, r)) \leq h^{-O(\eta_1 + \eta)} r^{\kappa_0} N_h(X). \quad (6.A27)$$

Let $\tilde{X}$ be maximal h -separated and write $N = |\tilde{X}| \asymp N_h(X)$. Average over $a \in [-1, 1]$. The additive energy of the map $(x, y) \mapsto x + ay$ is bounded by counting quadruples $(x, x', y, y') \in \tilde{X}^4$. Split them according to $|y - y'| \geq r$ or $|y - y'| < r$. In the first case the set of a 's for which

$$|x + ay - x' - ay'| \leq Ch$$

has Lebesgue measure $O(h/r)$. In the second case, (6.A27) gives

$$\#\{(y, y') \in \tilde{X}^2 : |y - y'| < r\} \lesssim N \max_w N_h(X \cap B(w, 2r)) \leq h^{-O(\eta_1 + \eta)} r^{\kappa_0} N^2.$$

For fixed a, x, y, y' , the condition $|x + ay - x' - ay'| \leq Ch$ confines the h -separated point x' to an interval of length $O(h)$, so only $O(1)$ choices occur. Consequently

$$\int_{-1}^1 E_h(a) da \leq h^{-O(\eta_1 + \eta)} (hr^{-1} N^4 + r^{\kappa_0} N^3). \quad (6.A28)$$

Since $N \leq h^{-\sigma - \eta}$, choose

$$r = h^{(1 - \sigma)/(1 + \kappa_0)}.$$

Then both terms in parentheses are bounded by

$$h^{\frac{\kappa_0(1 - \sigma)}{1 + \kappa_0} - O(\eta_1 + \eta)} N^3.$$

For sufficiently small η_1, η , some $a \in [-1, 1]$ therefore has

$$E_h(a) \leq h^{c_0} N^3$$

with $c_0 > 0$ depending only on κ_0, σ . Cauchy-Schwarz gives

$$N_h(X + aX) \gtrsim \frac{N^4}{E_h(a)} \geq h^{-c_0} N.$$

Taking $c < c_0/2$ proves (6.A26). This is the one-dimensional probabilistic expansion step in He's Lemma 28/Proposition 29, written here in full. $\square$

Ring-word notation. For a bounded set $A \subset \mathbb{R}$, write $\langle A \rangle_s$ for the set of values of expression trees with at most s leaves from A and internal operations $+$, $-$, $\times$. The precise value of the complexity parameter is immaterial: replacing s by a constant depending only on s allows one to expand, regroup, or distribute such expressions. If moreover $0, 1 \in A$, every bounded-complexity ring word can be written, after increasing s , as an element of

$$sA^{(s)} - sA^{(s)}. \quad (6.A29a)$$

Indeed, distributive expansion gives a signed sum of boundedly many monomials; multiply low-degree monomials by copies of 1, and pad either signed sum by zero monomials.

Lemma (Blichfeldt averaging). *If $V \subset \mathbb{R}^n$ is bounded and measurable, then some $z \in [0, 1]^n$ satisfies*

$$\#(V \cap (z + \mathbb{Z}^n)) \geq |V|. \quad (6.A29b)$$

Proof. By Tonelli and the tiling of $\mathbb{R}^n$ by translates of $[0, 1]^n$,

$$\int_{[0,1]^n} \sum_{k \in \mathbb{Z}^n} \mathbf{1}_V(z+k) dz = \sum_{k \in \mathbb{Z}^n} |V \cap (k + [0, 1]^n)| = |V|.$$

Hence the integrand is at least its average for some z . $\square$

Lemma (one coefficient can be eliminated). *Fix $n \geq 2$. Suppose $A \subset [-M, M]$ is compact, $D = \text{diam } A > 0$, and $v_j \in [1/2, 1]$. If*

$$E = v_1 A + \cdots + v_n A, \quad |E| = m > 0, \quad (6.A29c)$$

then there is an integer $N = N(n, M, D, m)$ such that, with

$$A_1 = NA^{(2)} - NA^{(2)},$$

one has

$$|v_1 A_1 + \cdots + v_{n-1} A_1| \geq Dm. \quad (6.A29d)$$

The bound for N is uniform when M is bounded above and D, m are bounded below.

Proof. Choose $a_+, a_- \in A$ with $a_+ - a_- = D$. Put $\pi(u) = v \cdot u$. For $L \geq 2$, consider the slab

$$V_L = \{u \in [-L, L]^n : |\pi(u)| \leq 1\}.$$

Since every $v_j \in [1/2, 1]$, $|V_L| \geq c_n L^{n-1}$. By Blichfeldt averaging, some translate of the integer lattice contains $\gtrsim_n L^{n-1}$ points of V_L . On a fixed value of the last lattice coordinate there are only $O_n(L^{n-2})$ such points: after the first $n-2$ coordinates are chosen, the slab condition leaves only $O(1)$ choices for the next integer coordinate. Thus, for L sufficiently large, we may select as many points $\tau_1, \dots, \tau_J$ as desired with distinct last lattice coordinates and $|\pi(\tau_j)| \leq 1$.

Take J so large that Jm exceeds the length of a fixed interval containing all sets $E + 2D\pi(\tau_j)$. Two of these translates intersect. Hence for some $x, y \in A^n$ and $\ell = \tau_i - \tau_j \in \mathbb{Z}^n$, with $\ell_n \neq 0$,

$$\sum_{r=1}^n v_r q_r = 0, \quad q_r = x_r - y_r + 2D\ell_r.$$

Moreover $|q_n| \geq 2D - |x_n - y_n| \geq D$. Because $D = a_+ - a_-$ and $|\ell_r| \leq 2L + 1$, every set $q_n A - q_r A$ is contained in $NA^{(2)} - NA^{(2)}$ for a fixed $N = O_n(L)$. Finally,

$$q_n \sum_{r=1}^n v_r a_r = \sum_{r=1}^{n-1} v_r (q_n a_r - q_r a_n), \quad a_r \in A.$$

Thus $q_n E \subset v_1 A_1 + \dots + v_{n-1} A_1$, and taking Lebesgue measure gives (6.A29d). $\square$

Proposition (full-Frostman positive-measure ring expansion). *Fix $0 < \kappa \leq 1$ and $C, M \geq 1$. There are integers $s = s(\kappa, C, M)$ and a constant $c = c(\kappa, C, M) > 0$ with the following property. Let λ be a probability measure supported on a compact $K \subset [-M, M]$, assume $0, 1 \in K$, and suppose*

$$\lambda(I) \leq C|I|^\kappa \quad \text{for every interval } I. \quad (6.A29e)$$

Then

$$|sK^{(s)} - sK^{(s)}| \geq c. \quad (6.A29f)$$

Proof. This is Bourgain's positive-measure ring expansion, but we give a complete one-dimensional proof following the coefficient-elimination architecture of O'Regan–Shmerkin–Wang [13]. Choose the least integer n with $n\kappa > 1$, and put $\nu = \lambda^{\otimes n}$ on K^n . A Euclidean ball of radius r is contained in a product of n intervals of length $2r$, so

$$\nu(B(z, r)) \leq C_n r^{n\kappa}.$$

A dyadic-shell summation therefore gives $I_1(\nu) \leq C_{\kappa, C, n}$. We record the needed higher-dimensional L^2 projection estimate explicitly. For a smooth approximate density $F_{\varepsilon, \theta}$ of the one-dimensional projection $\pi_\theta \nu$, Plancherel and polar coordinates give

$$\int_{S^{n-1}} \|F_{\varepsilon, \theta}\|_2^2 d\sigma(\theta) \leq_n \int_{\mathbb{R}^n} |\hat{\nu}(\xi)|^2 |\xi|^{1-n} d\xi = c_n I_1(\nu).$$

Weak L^2 compactness as $\varepsilon \downarrow 0$ gives the same averaged bound for the projection densities. Restrict the angular integral to a small positive-measure patch around $(1, \dots, 1)/\sqrt{n}$. Some direction in this patch has L^2 -density norm bounded solely in terms of κ, C, n ; by Cauchy–Schwarz its projection support has Lebesgue measure bounded below. After a harmless common rescaling of the coefficients we obtain a vector $v \in [1/2, 1]^n$ such that

$$|v_1 K + \dots + v_n K| \geq c_0(\kappa, C) > 0. \quad (6.A29g)$$

Apply the coefficient-elimination lemma. At each stage the ambient bound stays finite and the projected set has positive measure. If a stage with j remaining coefficients has projection measure m_j , then its underlying set has diameter at least m_j/j ; hence the next lower bound in (6.A29d) is still a positive constant depending only on the preceding constants. After $n - 1$ steps we obtain a compact set $W \subset \langle K \rangle_{s_*}$ with $|W| \geq c_1 > 0$. Since $0, 1 \in K$, the elementary synchronization observation (6.A29a) embeds W in $sK^{(s)} - sK^{(s)}$ for a fixed s , proving (6.A29f). $\square$

Lemma (finite cyclic additive basis). *For $M, m > 0$ there are integers $H = H(M, m)$ and a constant $C_* = C_*(M, m)$ such that the following holds for all sufficiently small $\rho > 0$. If $R \subset [-M, M]$ is compact and*

$$|R + [-\rho, \rho]| \geq m, \quad (6.A29h)$$

then $H(R - R)$ is C_ρ -dense in $[-1, 1]$.*

Proof. We first record the elementary Cauchy–Davenport inequality. If q is prime and $U, V \subset \mathbb{Z}/q\mathbb{Z}$ are nonempty, then

$$|U + V| \geq \min\{q, |U| + |V| - 1\}. \quad (6.A29i)$$

For completeness, in the case $|U| + |V| - 1 \leq q$, if the sumset had at most $|U| + |V| - 2$ elements, enlarge it to a set C of exactly that size and consider $P(X, Y) = \prod_{c \in C} (X + Y - c)$ over $\mathbb{F}_q$. The polynomial vanishes on $U \times V$, while the coefficient of $X^{|U|-1}Y^{|V|-1}$ is $\binom{|U|+|V|-2}{|U|-1} \not\equiv 0 \pmod{q}$. The two-variable Lagrange interpolation formula expresses this coefficient as a weighted sum of the values of P on $U \times V$, a contradiction. The remaining case follows by passing to subsets with cardinalities summing to $q + 1$.

Put $E = R + [-\rho, \rho]$. Since $|E| \geq m$, its diameter is at least m . Choose $r_0, r_1 \in R$ so that

$$p := |r_1 - r_0| \geq m/2 \quad (6.A29j)$$

when ρ is small enough. Work on the circle $\mathbb{R}/p\mathbb{Z}$. The projection of E to this circle has measure at least $m_0 = m_0(M, m) > 0$: indeed the multiplicity of the projection is bounded by $O_{M,m}(1)$, while its integral is $|E|$.

Partition the circle into q equal half-open arcs, where q is a prime to be chosen arbitrarily large, and let $I \subset \mathbb{Z}/q\mathbb{Z}$ be the indices of arcs meeting the projected set. Then $|I| \geq \alpha q$ for some $\alpha = \alpha(M, m) > 0$. By repeated use of (6.A29i), for a fixed $h = O(1/\alpha)$,

$$h(I - I) = \mathbb{Z}/q\mathbb{Z}. \quad (6.A29k)$$

Choose q so large that $(2h + 1)p/q \leq \rho$. Selecting one point of E in each occupied arc and using (6.A29k) shows that $h(E - E)$ is $O(\rho)$ -dense modulo p . Replacing each selected point of E by a point of R changes the sum by at most $2h\rho$, so $h(R - R)$ has the same property.

For $t \in [-1, 1]$, choose $z \in h(R - R)$ and $k \in \mathbb{Z}$ with $|z - t - kp| \leq C\rho$. Because $|z| \leq 2hM$ and $p \geq m/2$, the integer $|k|$ is bounded solely in terms of M, m . Since $p \in R - R$ up to sign, subtracting k copies of p enlarges h only by a fixed amount and gives an $O(\rho)$ -approximation to t in $H(R - R)$. This proves the lemma. $\square$

Proposition (normalized finite-scale Bourgain dense-ring theorem). *Fix $0 < \kappa \leq 1$ and $C \geq 1$. For sufficiently small δ , let μ be a probability measure on $[0, 1]$ satisfying*

$$\mu(I) \leq Cr^\kappa \quad (\delta \leq r \leq 1) \quad (6.A29)$$

for every interval I of length r . Put $S = \text{supp } \mu$ and assume $0, 1 \in S$. Then there is an integer $s_0 = s_0(\kappa, C)$ such that

$$s_0 S^{(s_0)} - s_0 S^{(s_0)} \quad \text{is } C_0\delta\text{-dense in } [-1, 1]. \quad (6.A30)$$

In particular it is $C_0\delta$ -dense in $[0, 1]$.

Proof. Let η_δ be normalized Lebesgue measure on $[-\delta, \delta]$, and set $\lambda = \mu * \eta_\delta$. Its support $K = S + [-\delta, \delta]$ lies in $[-1, 2]$ and contains 0, 1. The truncated estimate (6.A29) becomes a full Frostman estimate. Indeed, for $r \geq \delta$,

$$\lambda(I) \leq \mu(I + [-\delta, \delta]) \leq C'(r + 2\delta)^\kappa \leq C''r^\kappa,$$

whereas for $r < \delta$ the convolution density is at most $C\delta^{\kappa-1}$, hence

$$\lambda(I) \leq Cr\delta^{\kappa-1} = Cr^\kappa(r/\delta)^{1-\kappa} \leq Cr^\kappa. \quad (6.A29l)$$

The full-Frostman expansion therefore gives a fixed N and $c > 0$ with

$$|NK^{(N)} - NK^{(N)}| \geq c. \quad (6.A29m)$$

Every point of K lies within δ of S . Since all factors are uniformly bounded, telescoping a product shows that replacing the N factors of each monomial by nearby points of S changes it by $O_N(\delta)$. Consequently, with

$$R = NS^{(N)} - NS^{(N)} \subset [-N, N],$$

one has

$$|R + [-C_N\delta, C_N\delta]| \geq c. \quad (6.A29n)$$

Apply the finite cyclic additive-basis lemma to R . For a fixed H , $H(R - R)$ is $O(\delta)$ -dense in $[-1, 1]$. But

$$H(R - R) = 2HNS^{(N)} - 2HNS^{(N)}.$$

Put $s_0 = 2HN$. Since $1 \in S$ and $s_0 \geq N$, each N -fold product is also an s_0 -fold product after multiplication by copies of 1. This proves (6.A30). $\square$

Bourgain–Gamburd [12] record the finite-scale dense-ring conclusion as their Proposition 4 and trace it to Bourgain's 2010 Theorem 6. The proof above instead reconstructs the positive-measure core by the modern self-contained coefficient-elimination argument of O'Regan–Shmerkin–Wang [13], then proves the truncated-scale density upgrade locally.

Proposition (real one-scale segment extraction). *Fix $0 < \sigma \leq 1$. There are constants $\beta_0 = \beta_0(\sigma) > 0$, $C_0 = C_0(\sigma) < \infty$, and an integer $s_1 = s_1(\sigma)$ such that, for all sufficiently small δ , every $A \subset [-2, 2]$ with*

$$N_\delta(A) \geq \delta^{-\sigma} \quad (6.A31)$$

admits some $0 \leq \alpha_0 \leq C_0$ for which

$$[-c\delta^{\alpha_0}, c\delta^{\alpha_0}] \subset \langle A \rangle_{s_1} + [-C\delta^{\alpha_0+\beta_0}, C\delta^{\alpha_0+\beta_0}]. \quad (6.A32)$$

The constants c, C depend only on σ .

Proof. Choose a maximal δ -separated subset $E \subset A$, so $n := |E| \gtrsim \delta^{-\sigma}$. Among dyadic radii $t \in [\delta, 4]$, let t_0 be the smallest for which some interval $B(x_0, t_0)$ contains at least

$$(t_0/4)^{\sigma/2}n \quad (6.A33)$$

points of E . Such a radius exists at $t = 4$. Since a δ -separated subset of an interval of radius t_0 has at most Ct_0/δ points, (6.A33) and $n \gtrsim \delta^{-\sigma}$ give

$$t_0 \gtrsim \delta^{a_\sigma}, \quad a_\sigma := \frac{2(1-\sigma)}{2-\sigma} < 1. \quad (6.A34)$$

Put $E_0 = E \cap B(x_0, t_0)$. By minimality, every interval of radius $u < t_0/4$ contains at most $Cu^{\sigma/2}n$ points of E . Moreover $\text{diam } E_0 \gtrsim t_0$: otherwise E_0 would lie in a strictly smaller dyadic interval while still carrying the mass in (6.A33), contradicting the choice of t_0 .

Let x_- and x_+ be the extreme points of E_0 , put $d = x_+ - x_- \asymp t_0$, and define

$$B = d^{-1}(E_0 - x_-) \subset [0, 1].$$

Thus $0, 1 \in B$. If ν is normalized counting measure on B , then for $\delta/d \leq r \leq 1$, the preceding minimality estimate (and the trivial bound for fixed-size r) gives

$$\nu(I) \leq C_\sigma r^{\sigma/2} \quad (6.A35)$$

for every interval I of length r . Apply the normalized finite-scale Bourgain dense-ring theorem at scale δ/d , with exponent $\sigma/2$. Hence $s_0 B^{(s_0)} - s_0 B^{(s_0)}$ is $C\delta/d$ -dense in $[0, 1]$. Taking a difference of two such approximations makes the conclusion symmetric:

$$[-1, 1] \subset \left(s_0 B^{(s_0)} - s_0 B^{(s_0)} \right) - \left(s_0 B^{(s_0)} - s_0 B^{(s_0)} \right) + [-C\delta/d, C\delta/d]. \quad (6.A36)$$

It remains to clear the normalization without inserting an external scalar. Since $d = x_+ - x_-$ with $x_\pm \in A$, multiplying (6.A36) by d^{s_0} turns every normalized product into a bounded-complexity sum of products of members of $A \cup (-A)$: indeed

$$d^{s_0} \prod_{j=1}^{s_0} \frac{a_j - x_-}{d} = \prod_{j=1}^{s_0} (a_j - x_-),$$

and expansion of the right side uses only boundedly many ring words in A . Consequently

$$[-d^{s_0}, d^{s_0}] \subset \langle A \rangle_{s_1} + [-Cd^{s_0-1}\delta, Cd^{s_0-1}\delta]. \quad (6.A37)$$

If $d \geq c_0$, restrict to a fixed interval and take $\alpha_0 = 0$. Otherwise write $d^{s_0} \asymp \delta^{\alpha_0}$. From (6.A34), $\alpha_0 \leq s_0 a_\sigma + o(1)$, while

$$\frac{d^{s_0-1}\delta}{d^{s_0}} = \frac{\delta}{d} \lesssim \delta^{1-a_\sigma} = \delta^{\sigma/(2-\sigma)}.$$

After decreasing the last exponent by a factor two to absorb constants, (6.A32) holds with, for example, $\beta_0 = \sigma/(2(2-\sigma))$ and a fixed C_0 . $\square$

Proposition (uniform scalar segment exponent). *Fix $0 < \sigma \leq 1$. There exist fixed $\alpha > 1$, $\beta > 0$, and $s_2 \geq 1$, depending only on σ , such that every $A \subset [-2, 2]$ satisfying (6.A31) obeys*

$$[-\delta^\alpha, \delta^\alpha] \subset \langle A \rangle_{s_2} + [-C\delta^{\alpha+\beta}, C\delta^{\alpha+\beta}]. \quad (6.A38)$$

Proof. This is the scalar version of the exponent-uniformisation argument in He's Corollary 25, and we give the short proof. Let $b_0 = \beta_0$ be the error gap in (6.A32), set $b_1 = b_0/4$, and let $a_0 \in [0, C_0]$ be the exponent supplied there. If $a_0 < b_1$, restrict the segment in (6.A32) to radius δ^{b_1} ; otherwise leave it unchanged. Thus with

$$a_1 := \max\{a_0, b_1\}, \quad C_1 := \max\{C_0, b_1\},$$

we have $a_1 \in [b_1, C_1]$, while $a_1 - a_0 \leq b_0/4$. Hence the original error $O(\delta^{a_0+b_0})$ is in particular $O(\delta^{a_1+b_1})$. We may therefore work with a base exponent $a_1 \in [b_1, C_1]$ and a uniform positive error gap b_1 .

For $s \geq 1$, let Ω_s be the set of $\omega \geq 0$ for which there is a scalar ξ with $|\xi| \asymp \delta^\omega$ and

$$[0, 1]\xi \subset \langle A \rangle_s + [-C_s \delta^{\omega+b_1/2}, C_s \delta^{\omega+b_1/2}]. \quad (6.A39)$$

The stronger base error above gives $a_1 \in \Omega_{s_1}$. If $\omega \in \Omega_s$ and $\gamma \in [a_1, a_1 + b_1/2]$, the restricted version of (6.A32) supplies a ring word a with $|a| \asymp \delta^\gamma$ and error $O(\delta^{\gamma+b_1/2})$. Multiplying (6.A39) by a therefore shows

$$\omega + \gamma \in \Omega_{s'}, \quad (6.A40)$$

for an s' depending only on s and s_1 ; all product errors are $O(\delta^{\omega+\gamma+b_1/2})$. Induction gives

$$[(k+1)a_1, (k+1)a_1 + kb_1/2] \subset \Omega_{s_k}. \quad (6.A41)$$

Put

$$N := \lceil 2C_1/b_1 \rceil + 2, \quad \alpha = NC_1, \quad K = \lceil 2\alpha/b_1 \rceil.$$

For $n \geq N$, the consecutive intervals

$$I_n = [na_1, na_1 + (n-1)b_1/2]$$

overlap, since $(n-1)b_1/2 \geq C_1 \geq a_1$. Hence $\bigcup_{n \geq N} I_n = [Na_1, \infty)$. Because $\alpha = NC_1 \geq Na_1$, we have $\alpha \in I_n$ for some n ; moreover one may take $n \leq \lceil \alpha/a_1 \rceil + N \leq K+1$. Thus some $k = n-1 \leq K$ has α in the interval in (6.A41). Enlarging the word-complexity bound to $s_2 = \max_{k \leq K} s_k$ and taking $\beta = b_1/4$ proves (6.A38). The chosen α is, in particular, larger than one. $\square$

Proposition (scalar Bourgain filling core). *Fix $\kappa_0 > 0$ and $\epsilon_0 > 0$. There are $\eta > 0$ and an integer $s \geq 1$ such that, for sufficiently small h , whenever*

$$A \subset [-h^{-\eta}, h^{-\eta}], \quad N_r(A) \geq h^\eta r^{-\kappa_0} \quad (h \leq r \leq 1), \quad (6.A42)$$

then

$$[-h^{\epsilon_0}, h^{\epsilon_0}] \subset \langle A \rangle_s + [-Ch, Ch]. \quad (6.A43)$$

Proof. Set $\sigma_* := \min\{\kappa_0/4, 1/2\}$, and let fixed $\alpha > 1, \beta > 0, s_2$ be given by (6.A38) for σ_* . Put

$$q_k = \frac{1}{\alpha} \left(\frac{\alpha}{\alpha + \beta} \right)^k, \quad \delta_k = h^{q_k} \quad (k \geq 0). \quad (6.A44)$$

Then

$$\delta_0^\alpha = h, \quad \delta_k^{\alpha+\beta} = \delta_{k-1}^\alpha \quad (k \geq 1). \quad (6.A45)$$

Before forming the ladder, decrease β once if necessary so that every fixed multiplicative constant in (6.A38) is absorbed into the exponent slack. Choose r so that

$$\left(\frac{\alpha}{\alpha + \beta} \right)^r \leq \frac{\epsilon_0}{2};$$

hence $\delta_r^\alpha \geq h^{\epsilon_0/2}$. The integer r depends only on the fixed parameters.

We verify that (6.A38) is available at every scale δ_k , $1 \leq k \leq r$. Cover the interval $[-h^{-\eta}, h^{-\eta}]$ by $O(h^{-\eta})$ unit intervals. For each k , one such interval contains a subset $A_k \subset A$ satisfying

$$N_{\delta_k}(A_k) \geq h^\eta N_{\delta_k}(A) \geq h^{2\eta} \delta_k^{-\kappa_0}. \quad (6.A46)$$

Translate A_k by one of its points: if $a_k \in A_k$, put $B_k = A_k - a_k$. This places B_k in $[-2, 2]$, preserves the lower covering number up to constants, and any bounded-complexity ring word in B_k expands into a bounded-complexity ring word in A because $a_k \in A$. Since the finitely many exponents q_k are bounded below by $q_r > 0$, choosing η sufficiently small gives

$$h^{2\eta} \delta_k^{-\kappa_0} = \delta_k^{-\kappa_0 + 2\eta/q_k} \geq \delta_k^{-\sigma_*} \quad (6.A47)$$

for every $k \leq r$. The uniform segment theorem (6.A38) was stated on the fixed interval $[-2, 2]$, so no external scalar normalization is introduced here. Therefore (6.A38) yields, after the preceding absorption of constants, for every $1 \leq k \leq r$,

$$[-\delta_k^\alpha, \delta_k^\alpha] \subset \langle A \rangle_{s_3} + [-\delta_k^{\alpha+\beta}, \delta_k^{\alpha+\beta}]. \quad (6.A48)$$

Using (6.A45), this is

$$[-\delta_k^\alpha, \delta_k^\alpha] \subset \langle A \rangle_{s_3} + [-\delta_{k-1}^\alpha, \delta_{k-1}^\alpha]. \quad (6.A49)$$

Iterating from $k = r$ down to $k = 1$ gives

$$[-c\delta_r^\alpha, c\delta_r^\alpha] \subset \langle A \rangle_{rs_3} + [-Ch, Ch]$$

with fixed $c, C > 0$. Since $\delta_r^\alpha \geq h^{\epsilon_0/2}$, for sufficiently small h the left side contains $[-h^{\epsilon_0}, h^{\epsilon_0}]$. This proves (6.A43). This is exactly He's scale ladder, now with every scalar step exposed. $\square$

Proposition (scalar linear-action expansion). *With the locally proved scalar filling theorem above, for every $\kappa_0 > 0$ and $0 < \sigma < 1$ there is $\epsilon_{sp} > 0$ such that, for all sufficiently small h , sets $A, X \subset \mathbb{R}$ satisfying*

$$A, X \subset [-h^{-\epsilon_{sp}}, h^{-\epsilon_{sp}}], \quad (6.8a)$$

$$N_r(A), N_r(X) \geq h^{\epsilon_{sp}} r^{-\kappa_0} \quad (h \leq r \leq 1), \quad (6.8b)$$

and

$$N_h(X) \leq h^{-\sigma - \epsilon_{sp}} \quad (6.8c)$$

satisfy

$$N_h(X + X) + \max_{a \in A} N_h(X + aX) \geq h^{-\epsilon_{sp}} N_h(X). \quad (6.8d)$$

Proof. Assume the contrary. Then 1 and every $a \in A$ belong to $\mathcal{S}_h(X; h^{-\epsilon_{sp}})$. Fix $\epsilon_0 > 0$ much smaller than the expansion exponent $c(\kappa_0, \sigma)$ in (6.A26), and choose ϵ_{sp} small enough that the scalar filling theorem applies with this ϵ_0 . The approximate-ring lemma implies

$$\langle A \rangle_s \subset \mathcal{S}_h(X; h^{-O_s(\epsilon_{sp})}). \quad (6.A50)$$

The perturbation part of the same lemma and the filling conclusion (6.A43) therefore give

$$[-h^{\epsilon_0}, h^{\epsilon_0}] \subset \mathcal{S}_h(X; h^{-O_s(\epsilon_{sp})}). \quad (6.A51)$$

In particular $q = h^{\epsilon_0}$ is good. Since $|q^{-1}| = h^{-\epsilon_0}$, the inverse assertion makes q^{-1} good after a further $h^{-O(\epsilon_0)}$ loss. For every $|u| \leq 1$, the scalar qu lies in the interval in (6.A51); multiplying the two good scalars qu and q^{-1} gives

$$[-1, 1] \subset \mathcal{S}_h(X; h^{-O_s(\epsilon_{sp}) - O(\epsilon_0)}). \quad (6.A52)$$

Choose first ϵ_0 , and then ϵ_{sp} , so small that the exponent on the right of (6.A52) is strictly smaller than the fixed gain in (6.A26). Equation (6.A52) contradicts the scalar fullness lemma. Hence (6.8d) holds. $\square$

The scalar dense-ring argument closes the arithmetic part of Bourgain's method. The dense-ring theorem, one-scale segment extraction, exponent uniformisation, weak-multiscale filling ladder, and scalar linear-action expansion have all been proved above. We next turn to bad-direction regularisation and codimension-one slicing.

Lemma III.6.6.1 (transverse lines, slabs, and product covers). *Let $L_1, \dots, L_j$ be lines in $\mathbb{R}^d$, let v_i be unit vectors spanning L_i , and write*

$$\Delta(L_1, \dots, L_j) = \|v_1 \wedge \dots \wedge v_j\|.$$

When $j = d$, the following hold.

1. *If $x_i \in L_i$ and $z \in \mathbb{R}^d$ satisfies $\|P_{L_i} z - x_i\| \leq r$ for every i , then all such z lie in one ball of radius at most $C_d r \Delta(L_1, \dots, L_d)^{-1}$.*
2. *For every bounded $F \subset \mathbb{R}^d$,*

$$N_h(F) \leq C_d \Delta(L_1, \dots, L_d)^{-d} \prod_{i=1}^d N_h(P_{L_i} F). \quad (6.R1)$$

3. *Gram-Schmidt gives*

$$\Delta(L_1, \dots, L_j) = \prod_{i=2}^j d_{\angle}(L_i, L_1 + \dots + L_{i-1}). \quad (6.R2)$$

The analogous product-cover estimate on $U = L_1 + \dots + L_j$ holds with the same form and d replaced by $\dim U$.

Proof. For $j = d$, consider

$$T : \mathbb{R}^d \longrightarrow L_1 \oplus \cdots \oplus L_d, \quad Tz = (P_{L_1}z, \dots, P_{L_d}z).$$

In orthonormal coordinates, $|\det T| = \Delta(L_1, \dots, L_d)$. Every singular value of T is bounded above by a constant depending only on d , so its least singular value is at least $c_d \Delta$. Hence $\|T^{-1}\| \leq C_d \Delta^{-1}$. This proves the slab-intersection assertion. Covering each $P_{L_i}F$ by h -balls and pulling the resulting product cover back through T^{-1} covers F by balls of radius $C_d h \Delta^{-1}$. Changing from this radius back to h costs at most $C_d \Delta^{-d}$, proving (6.R1). Finally, Gram–Schmidt says that the volume of the parallelepiped $(v_1, \dots, v_j)$ is multiplied at step i by the length of the component of v_i orthogonal to $L_1 + \cdots + L_{i-1}$, which is precisely the corresponding angle factor. This gives (6.R2). Restriction to the span proves the last assertion. $\square$

Lemma III.6.6.2 (averaged intersection of dense cylinders). *Let $A \subset \mathbb{R}^d$ be measurable, let (D, ρ) be a finite measure space, and for every $V \in D$ let $S_V \subset A$ be measurable with $|S_V| \geq \lambda|A|$. If $V_1, \dots, V_d$ are sampled independently according to the normalized restriction of ρ to D , then*

$$\mathbb{P}\{|S_{V_1} \cap \cdots \cap S_{V_d}| \geq \tfrac{1}{2} \lambda^d |A|\} \geq \tfrac{1}{2} \lambda^d. \quad (6.R3)$$

Equivalently, under independent sampling from ρ , the event that all parameters lie in D and the displayed intersection bound holds has probability at least $\tfrac{1}{2} \lambda^d \rho(D)^d$.

Proof. Fubini followed by Jensen gives

$$\mathbb{E}|S_{V_1} \cap \cdots \cap S_{V_d}| = \int_A \left(\frac{1}{\rho(D)} \int_D 1_{S_V}(x) d\rho(V) \right)^d dx \geq \lambda^d |A|.$$

Since the random intersection is bounded above by $|A|$, the part on which it is smaller than $\lambda^d |A|/2$ cannot account for the whole expectation. The stated probability bound follows. $\square$

Lemma III.6.6.3 (regular bad-direction extraction). *Let $d \geq 2$, $0 < \alpha < d$, $\kappa > 0$, and let $0 < \epsilon \ll_{d,\kappa} 1$. Assume $A \subset B(0, 1)$ and a probability measure ρ on projective lines satisfy*

$$N_h(A) \geq h^{-\alpha+\epsilon}, \quad N_h(A \cap B(x, r)) \leq h^{-\epsilon} r^\kappa N_h(A), \quad (6.R4)$$

for $h \leq r \leq 1$, and

$$\rho\{V : d_\perp(V, H) \leq r\} \leq h^{-\epsilon} r^\kappa \quad (6.R5)$$

for every hyperplane H . Let $E_\epsilon(A)$ be the set of lines V for which there is $A' \subset A$ with

$$N_h(A') \geq h^\epsilon N_h(A), \quad N_h(P_V A') \leq h^{-\alpha/d-\epsilon}.$$

Put $\kappa_1 = \kappa/(2d^2)$. Let $E_\epsilon^{\text{reg}}(A)$ consist of those $V \in E_\epsilon(A)$ for which one can choose a witness A' satisfying

$$N_h(A') \geq h^{2\epsilon} N_h(A)$$

and

$$N_h(P_V A' \cap B(u, r)) \leq r^{\kappa_1} h^{-\alpha/d-10\epsilon}, \quad h \leq r \leq 1. \quad (6.9)$$

Then, for all sufficiently small h ,

$$\rho(E_\epsilon(A) \setminus E_\epsilon^{\text{reg}}(A)) \leq h^{2\epsilon}. \quad (6.R6)$$

Proof. This is He's Lemma 27 in the rank-one case; we give the complete argument because it is purely finite-scale geometry.

First note the elementary thickening consequence of (6.R4): for $h \leq r \leq 1$,

$$|A^{(h)} \cap B(x, r)| \leq C_d h^{-\epsilon} r^\kappa |A^{(h)}|, \quad (6.R7)$$

after enlarging r by a fixed factor. Indeed, maximal separated sets give $|A^{(h)}| \asymp_d h^d N_h(A)$, and the same comparison in a ball reduces (6.R7) to (6.R4).

We first show that every nonregular bad direction creates a dense cylinder. Fix $V \in E_\epsilon(A) \setminus E_\epsilon^{\text{reg}}(A)$, and choose a bad witness $A' \subset A$. Define the heavy part of the projected set by

$$Y = \{y \in V : N_h(A' \cap P_V^{-1}(B(y, h))) \geq h^{\alpha/d+3\epsilon} N_h(A)\}$$

and put $A'' = A' \cap P_V^{-1}(Y^{(h)})$. Since $N_h(P_V A') \leq h^{-\alpha/d-\epsilon}$, the discarded light fibres contain at most

$$h^{-\alpha/d-\epsilon} h^{\alpha/d+3\epsilon} N_h(A) = h^{2\epsilon} N_h(A)$$

covering cells. Thus, for small h ,

$$N_h(A'') \geq h^{2\epsilon} N_h(A). \quad (6.R8)$$

Because V is not regular, (6.9) fails for A'' . Hence for some $u \in V$ and some $r \in [h, 1]$,

$$N_h(P_V A'' \cap B(u, r)) \geq r^{\kappa_1} h^{-\alpha/d-10\epsilon}. \quad (6.R9)$$

Take a maximal $6h$ -separated subset of this projected piece. It has $\gg_d r^{\kappa_1} h^{-\alpha/d-10\epsilon}$ points. Each point lies within h of Y , so its $2h$ -fibre contains at least $h^{\alpha/d+3\epsilon} N_h(A)$ cells of A' . The corresponding fibres are disjoint at scale h . Summing them, and harmlessly weakening a fixed constant to a power of h , yields

$$N_h(A \cap P_V^{-1}(B(u, r+2h))) \geq (r+2h)^{\kappa_1} h^{-6\epsilon} N_h(A). \quad (6.R10)$$

In particular the radius in (6.R10) may be assumed at most $h^{6\epsilon/\kappa_1}$, since the left side is at most $N_h(A)$.

Suppose now, for contradiction, that the left side of (6.R6) has mass at least $h^{2\epsilon}$. Divide the possible radii into dyadic intervals. There are only $O_{\epsilon, \kappa, d}(\log(1/h))$ relevant intervals, so for small h there are a radius $r \in [h, 1]$ and a parameter set D with

$$\rho(D) \geq h^{3\epsilon} \quad (6.R11)$$

such that for every $V \in D$ there is a cylinder $C_V = P_V^{-1}(B(u_V, r))$ satisfying, after weakening (6.R10),

$$N_h(A \cap C_V) \geq r^{2\kappa_1} h^{-5\epsilon} N_h(A).$$

The thickening comparison therefore gives

$$|A^{(h)} \cap C_V^{(O(h))}| \geq c_d r^{2\kappa_1} h^{-4\epsilon} |A^{(h)}|. \quad (6.R12)$$

Apply the preceding averaged-intersection lemma to these thickened cylinders. Under independent sampling $V_1, \dots, V_d$ from ρ , the event

$$|A^{(h)} \cap C_{V_1}^{(O(h))} \cap \dots \cap C_{V_d}^{(O(h))}| \geq c_d r^{2d\kappa_1} h^{-4d\epsilon} |A^{(h)}| \quad (6.R13)$$

has probability at least

$$c_d r^{2d\kappa_1} h^{-d\epsilon}. \quad (6.R14)$$

On the other hand, by (6.R2), if $\Delta(V_1, \dots, V_d) < r^{(d-1)/d}$, then for some $j \geq 2$,

$$d_{\mathbb{Z}}(V_j, V_1 + \dots + V_{j-1}) < r^{1/d}.$$

The preceding span is contained in a hyperplane, so (6.R5) and a union bound imply that this nontransverse event has probability at most

$$(d-1)h^{-\epsilon} r^{\kappa/d}. \quad (6.R15)$$

Since $2d\kappa_1 = \kappa/d$, the lower bound (6.R14) is larger than (6.R15) for small h . Hence some d -tuple satisfies both (6.R13) and

$$\Delta(V_1, \dots, V_d) \geq r^{(d-1)/d}. \quad (6.R16)$$

By the transverse-slab lemma, the intersection of the corresponding cylinders is contained in a ball of radius $C_d r^{1/d}$. Therefore (6.R13) gives a lower bound $\gg r^{\kappa/d} h^{-4d\epsilon} |A^{(h)}|$ for the mass of $A^{(h)}$ in that ball, whereas (6.R7) gives the upper bound $\ll h^{-\epsilon} r^{\kappa/d} |A^{(h)}|$. These are incompatible for small h . This proves (6.R6). $\square$

Lemma III.6.6.4 (overlap selection). *Let (Ω, ρ) be a probability space and let $U_\omega \subset U$, $V_\omega \subset V$ be measurable families of unions of h -cells. Suppose that for all ω in a set D ,*

$$|U_\omega| \geq h^c |U|, \quad |V_\omega| \geq h^c |V|,$$

where $|\cdot|$ denotes Lebesgue measure of the h -neighbourhood. Then there are $\omega_* \in D$ and $D' \subset D$ such that

$$\rho(D') \geq h^{4c} \rho(D)/4 \quad (6.10)$$

and, for every $\omega \in D'$,

$$|U_{\omega_*} \cap U_\omega| |V_{\omega_*} \cap V_\omega| \geq \frac{1}{4} h^{4c} |U| |V|. \quad (6.11)$$

Proof. Write 1_{U_ω} and 1_{V_ω} for the characteristic functions. By two applications of Cauchy–Schwarz,

$$\iint_{D \times D} |U_\omega \cap U_{\omega'}| |V_\omega \cap V_{\omega'}| d\rho(\omega) d\rho(\omega') \geq h^{4c} |U| |V| \rho(D)^2.$$

Indeed, after expanding the intersections by Fubini, the left side is the integral on $U \times V$ of the square of the ρ -mass of the parameters whose two sets contain the point. Averaging first in ω_* gives one parameter for which the inner average is at least the displayed right-hand side divided by $\rho(D)$. Markov's inequality applied to the bounded function $|U_{\omega_*} \cap U_\omega| |V_{\omega_*} \cap V_\omega| / (|U| |V|)$ then gives (6.10)–(6.11), after weakening constants. $\square$

Proposition III.6.6.5 (planar weak Bourgain theorem). *Fix $0 < \alpha < 2$ and $\kappa > 0$. There is $\epsilon_0 > 0$ such that, for h small, under the rank-one hypotheses (6.7) in $\mathbb{R}^2$, there is a subset $A_0 \subset X$ with*

$$N_h(A_0) \geq h^{O(\epsilon_0)} N_h(X)$$

for which

$$\rho(E_{\epsilon_0}(A_0)) \leq h^{\epsilon_0}. \quad (6.12)$$

Here $E_\epsilon(A)$ denotes the directions admitting a subset of A of relative covering number at least h^ϵ whose projection has covering number below $h^{-\alpha/2-\epsilon}$.

Proof. Suppose, to the contrary, that every admissible large subset $A \subset X$ has $\rho(E_\epsilon(A)) > h^\epsilon$, with $\epsilon > 0$ to be chosen much smaller than all fixed parameters.

By the regular bad-direction extraction above, choose a regular bad line V_1 and a witness $A_1 \subset X$ such that

$$N_h(A_1) \geq h^{-\alpha+O(\epsilon)}, \quad N_h(P_{V_1}A_1) \leq h^{-\alpha/2-\epsilon}, \quad (6.13)$$

and the projected set satisfies (6.9) with exponent $\kappa/8$. Put $\epsilon_1 = 3\epsilon/\kappa$. Direction non-concentration gives

$$\rho(E_\epsilon(A_1) \setminus V_1^{(h^{\epsilon_1})}) > 0,$$

where, on the projective line, $V_1^{(r)}$ denotes the r -neighbourhood of the single degenerate direction. Choose V_2 there and a witness $A_2 \subset A_1$. Then

$$N_h(A_2) \geq h^{-\alpha+O(\epsilon)}, \quad N_h(P_{V_j}A_2) \leq h^{-\alpha/2-\epsilon}, \quad j = 1, 2. \quad (6.14)$$

The angle between V_1, V_2 is at least $h^{O(\epsilon)}$. Applying an $h^{-O(\epsilon)}$ -bi-Lipschitz linear change of variables, which only changes all covering estimates by $h^{-O(\epsilon)}$, we may take V_1 and V_2 to be the two coordinate axes. Set

$$X_1 = P_{V_1}A_2, \quad Y_1 = P_{V_2}A_2.$$

Since the coordinate map is bi-Lipschitz and $A_2 \subset X_1 \times Y_1$, (6.14) gives

$$h^{-\alpha/2+O(\epsilon)} \leq N_h(X_1), N_h(Y_1) \leq h^{-\alpha/2-\epsilon}. \quad (6.15)$$

Let D be the bad directions for A_2 which stay at distance at least h^{ϵ_1} from both coordinate axes. Again by direction nonconcentration,

$$\rho(D) \geq h^{O(\epsilon)}. \quad (6.16)$$

For $V \in D$, choose a bad witness $A_V \subset A_2$. Both restrictions $P_V|_{V_1}$ and $P_V|_{V_2}$ are invertible with inverse norm at most $h^{-O(\epsilon)}$. Write

$$\phi_V = (P_V|_{V_1})^{-1} \circ P_V|_{V_2} : V_2 \rightarrow V_1.$$

In dimension one this is multiplication by a nonzero scalar. The smallness of $P_V A_V$, together with (6.15), implies that the scale- h additive energy between $P_V X_1$ and $P_V Y_1$ is at least

$$h^{O(\epsilon)} N_h(P_V X_1)^{3/2} N_h(P_V Y_1)^{3/2}. \quad (6.17)$$

To see this directly, discretise all sets by maximal h -separated subsets. The $h^{-\alpha+O(\epsilon)}$ points of A_V determine the same number of pairs in $X_1 \times Y_1$, while their sums under P_V occupy at most $h^{-\alpha/2-\epsilon}$ cells. Cauchy–Schwarz therefore gives at least $h^{-3\alpha/2+O(\epsilon)}$ equal-at-scale- h sum pairs, which is (6.17).

The local scale- h BSG lemma above now gives subsets $X_V \subset X_1$, $Y_V \subset Y_1$ with

$$N_h(X_V), N_h(Y_V) \geq h^{-\alpha/2+O(\epsilon)} \quad (6.18)$$

and

$$N_h(X_V + \phi_V Y_V) \leq h^{-\alpha/2-O(\epsilon)}. \quad (6.19)$$

Apply Lemma 6.6.1 to the h -neighbourhoods of these families. There are $V_* \in D$ and $D' \subset D$, with $\rho(D') \geq h^{O(\epsilon)}$, such that the intersections of X_V with $X_* := X_{V_*}$ and of Y_V with $Y_* := Y_{V_*}$ have covering number $h^{-\alpha/2+O(\epsilon)}$ for every $V \in D'$. Ruzsa's triangle inequality, first with (6.19) and then with the large common intersections, yields

$$N_h(X_* - X_V), N_h(\phi_V Y_* - \phi_V Y_V) \leq h^{-\alpha/2-O(\epsilon)}. \quad (6.20)$$

Using (6.19) once for V and once for V_* , and applying Ruzsa's triangle inequality successively, gives

$$N_h(X_* - \phi_V \phi_*^{-1} X_*) \leq h^{-\alpha/2-O(\epsilon)} \quad (V \in D'), \quad (6.21)$$

where $\phi_* = \phi_{V_*}$. Pluenncke–Ruzsa similarly gives

$$N_h(X_* + X_*) \leq h^{-\alpha/2-O(\epsilon)}. \quad (6.22)$$

Set

$$\mathcal{A} = \{-\phi_V \phi_*^{-1} : V \in D'\} \subset \mathbb{R}.$$

We now verify the hypotheses of the local scalar linear-action theorem above. First, the separation of all directions from the coordinate axes gives $\mathcal{A}, X_* \subset [-h^{-O(\epsilon)}, h^{-O(\epsilon)}]$. Second, the slope map

$$V \mapsto -\phi_V \phi_*^{-1}$$

is $h^{-O(\epsilon)}$ -bi-Lipschitz on this chart. Hence, if $\mathcal{A}_r$ is a maximal r -separated subset of $\mathcal{A}$, the direction nonconcentration hypothesis and $\rho(D') \geq h^{O(\epsilon)}$ imply

$$N_r(\mathcal{A}) \geq h^{O(\epsilon)} r^{-\kappa}, \quad h \leq r \leq 1. \quad (6.23)$$

Third, regularity (6.9) of the first projection, together with $N_h(X_*) \geq h^{-\alpha/2+O(\epsilon)}$, gives

$$N_h(X_* \cap B(x, r)) \leq h^{-O(\epsilon)} r^{\kappa/8} N_h(X_*).$$

Covering X_* by r -balls and dividing by the preceding maximal local covering number yields

$$N_r(X_*) \geq h^{O(\epsilon)} r^{-\kappa/8}. \quad (6.24)$$

Finally, by (6.18), $N_h(X_*) = h^{-\alpha/2+O(\epsilon)}$. Choose $\alpha/2 < \sigma < 1$ and then choose ϵ so small that (6.8c) holds with this σ . Thus the local scalar linear-action proposition applies to $(\mathcal{A}, X_*)$ with exponent $\kappa/8$. Its now fully local conclusion contradicts (6.21)–(6.22) once ϵ is small compared with the sum-product gain. This contradiction proves (6.12). $\square$

Lemma III.6.6.6 (regular codimension-one slice). *Let $d = q \geq 2$, let $0 < \alpha < d$, and suppose the weak rank-one conclusion under construction fails for A at loss ϵ , in the sense that every subset of A of relative covering number at least $h^{O_d(\epsilon)}$ still has a bad direction family of ρ -mass greater than h^ϵ . Then there are an affine hyperplane $y + W$ and*

$$B \subset A^{(h)} \cap (y + W)$$

such that, with

$$\begin{aligned} \beta &= \frac{d-1}{d} \alpha, \\ N_h(B) &\geq h^{-\beta+O_d(\epsilon)} \end{aligned} \quad (6.25)$$

and

$$N_h(B \cap B_W(x, r)) \leq r^{\kappa/(2d^2)} h^{-\beta-O_d(\epsilon)}, \quad h \leq r \leq 1. \quad (6.26)$$

Proof. This is the rank-one form of He, Lemma 28. We spell out the nested-direction and fibre argument.

Apply the regular bad-direction lemma to obtain a line V_1 and a witness $A_1 \subset A$ with

$$N_h(A_1) \geq h^{-\alpha+O_d(\epsilon)}, \quad N_h(P_{V_1} A_1) \leq h^{-\alpha/d-\epsilon}, \quad (6.S1)$$

and

$$N_h(P_{V_1} A_1 \cap B(u, r)) \leq r^{\kappa_1} h^{-\alpha/d-O_d(\epsilon)}, \quad \kappa_1 = \frac{\kappa}{2d^2}. \quad (6.S2)$$

Put $\epsilon_1 = 3\epsilon/\kappa$. We recursively choose lines $V_2, \dots, V_d$ and nested sets $A_1 \supset A_2 \supset \dots \supset A_d$ so that

$$d_\perp(V_j, V_1 + \dots + V_{j-1}) \geq h^{\epsilon_1}, \quad (6.S3)$$

$$N_h(A_j) \geq h^\epsilon N_h(A_{j-1}), \quad N_h(P_{V_j} A_j) \leq h^{-\alpha/d-\epsilon}. \quad (6.S4)$$

Indeed, by the assumed failure of the weak theorem the bad family for A_{j-1} has mass greater than h^ϵ . A hyperplane containing $V_1 + \dots + V_{j-1}$ has an h^{ϵ_1} -neighbourhood of direction mass at most

$$h^{-\epsilon} h^{\epsilon_1 \kappa} = h^{2\epsilon},$$

so a bad direction remains outside it, and its bad witness gives A_j . The harmless changes in the nonconcentration loss caused by passing to a subset of relative size $h^{O_d(\epsilon)}$ are absorbed into the displayed $O_d(\epsilon)$ terms.

From (6.S1)–(6.S4),

$$N_h(A_d) \geq h^{-\alpha+O_d(\epsilon)}. \quad (6.S5)$$

Cover $P_{V_d}A_d$ by h -balls and pigeonhole a fibre. Since $N_h(P_{V_d}A_d) \leq h^{-\alpha/d-\epsilon}$, there is $y_* \in V_d$ such that

$$N_h(A_d \cap P_{V_d}^{-1}(B(y_*, h))) \geq h^{-\beta+O_d(\epsilon)}. \quad (6.S6)$$

Translate so that $y_* = 0$, put $W = V_d^\perp$, let $B_0 = A_d \cap W^{(h)}$, and set $B = P_W B_0$. Every point of B_0 is within h of its projection to W , so covering numbers of B_0 and B are comparable up to a dimensional constant; (6.S6) gives (6.25).

It remains to prove (6.26). Put $U = V_1 + \dots + V_{d-1}$. By (6.S3),

$$d_\perp(U, W^\perp) = d_\perp(U, V_d) \geq h^{\epsilon_1}. \quad (6.S7)$$

The orthogonal projection $P_{U|W} : W \rightarrow U$ is therefore quantitatively invertible: its least singular value is at least $h^{O_d(\epsilon)}$. Consequently, for every $x \in W$ and $r \geq h$,

$$N_h(B \cap B_W(x, r)) \leq h^{-O_d(\epsilon)} N_h(P_U B \cap B(P_U x, 2r)). \quad (6.S8)$$

Since $B \subset A_d^{(h)}$, the last covering number is at most, after a fixed scale enlargement,

$$h^{-O_d(\epsilon)} N_h(P_U A_d \cap B(P_U x, 3r)). \quad (6.S9)$$

Inside U , the lines $V_1, \dots, V_{d-1}$ are quantitatively transverse by (6.S3). Apply the product-cover estimate (6.R1) to the set in (6.S9). Its V_1 -projection is contained in $P_{V_1}A_1 \cap B(x_1, 3r)$, while for $2 \leq j \leq d-1$ its V_j -projection is contained in $P_{V_j}A_j$. Hence (6.S2) and (6.S4) yield

$$\begin{aligned} N_h(P_U A_d \cap B(P_U x, 3r)) &\leq h^{-O_d(\epsilon)} r^{\kappa_1} h^{-\alpha/d} \prod_{j=2}^{d-1} h^{-\alpha/d} \\ &\leq r^{\kappa_1} h^{-\beta-O_d(\epsilon)}. \end{aligned}$$

Combining this with (6.S8)–(6.S9) proves (6.26). $\square$

Lemma III.6.6.7 (projecting the direction measure to a slice). *Let $W \subset \mathbb{R}^d$ be a $(d-1)$ -dimensional linear subspace. For a line V not orthogonal to W , let*

$$\Psi_W(V) = P_W(V) \in \text{Gr}(W, 1).$$

Assume

$$\rho\{V : d_\perp(V, U) \leq r\} \leq h^{-\epsilon} r^\kappa \quad (6.27)$$

for all hyperplanes U and $h \leq r \leq 1$. Put

$$\epsilon_1 = 3\epsilon/\kappa.$$

Conditional on $d_\perp(V, W^\perp) > h^{\epsilon_1}$, the push-forward $\nu = (\Psi_W)_ \rho$ satisfies*

$$\nu\{L : d_\perp(L, U_W) \leq r\} \leq 2h^{-4\epsilon} r^\kappa \quad (6.28)$$

for every hyperplane U_W of W and $h \leq r \leq 1$ (with the right-hand side understood trivially when it exceeds one).

Proof. The excluded set is contained in the h^{ϵ_1} -neighbourhood of the ambient hyperplane containing $W^\perp$. Hence its mass is at most

$$h^{-\epsilon} h^{\epsilon_1 \kappa} = h^{2\epsilon},$$

so the conditioning event has probability at least $1 - h^{2\epsilon}$ and therefore at least $1/2$ for small h . If $L = \Psi_W(V)$ lies within angle r of U_W and $d_\perp(V, W^\perp) > h^{\epsilon_1}$, elementary exterior-product geometry gives

$$d_\perp(V, U_W + W^\perp) \lesssim h^{-\epsilon_1} r.$$

Applying (6.27) to $U_W + W^\perp$ gives at most $h^{-\epsilon - \epsilon_1 \kappa} r^\kappa = h^{-4\epsilon} r^\kappa$. Division by the conditioning probability proves (6.28), after absorbing a fixed geometric constant into the harmless loss. This is the rank-one content of the measure-transfer step in He, Lemma 31, written here locally. $\square$

Lemma III.6.6.8 (bad-direction transfer to the slice). *Let $B \subset W$, put $\epsilon_1 = 3\epsilon/\kappa$, and let V satisfy $d_\perp(V, W^\perp) > h^{\epsilon_1}$. Put $V' = \Psi_W(V)$. If V is bad for B at exponent α/d with loss ϵ , then V' is bad for B as a subset of W at the same target exponent, with loss $O_\kappa(\epsilon)$.*

Proof. On W , the scalar projection $P_V|_W$ factors as an invertible scalar map after $P_{V'}$. The inverse scalar has norm at most $h^{-O(\epsilon_1)} = h^{-O_\kappa(\epsilon)}$ by the angle assumption. Hence for every $B' \subset B$,

$$N_h(P_V, B') \leq h^{-O_\kappa(\epsilon)} N_h(P_{V'}, B').$$

Relative covering-number lower bounds for B' are unchanged. Thus a bad witness for V is a bad witness for V' after replacing ϵ by $O_\kappa(\epsilon)$. This is the rank-one content of He, Lemma 32, but the short comparison is recorded locally. $\square$

Proposition III.6.6.9 (weak rank-one theorem in every dimension). *For every $d \geq 2$, $0 < \alpha < d$ and $\kappa > 0$, there is $\epsilon_d > 0$ such that the weak conclusion (6.12), with target exponent α/d , holds in $\mathbb{R}^d$.*

Proof. The case $d = 2$ is Proposition 6.6.2. Assume the statement has been proved in dimension $d - 1$, including the robust consequence obtained in Lemma 6.6.6 below, and suppose the weak statement fails in dimension d .

Apply the regular codimension-one slice lemma above. After translation we obtain $B \subset W$ satisfying (6.25)–(6.26), with

$$\beta = \frac{d-1}{d} \alpha.$$

The induction target in W is

$$\frac{\beta}{d-1} = \frac{\alpha}{d}. \quad (6.29)$$

Apply Lemma 6.6.3 to push the direction measure to $\text{Gr}(W, 1)$. The resulting measure has the required hyperplane nonconcentration, with exponent κ and a harmlessly

larger loss; (6.26) supplies the spatial nonconcentration of B , with the positive exponent $\kappa/(2d^2)$. Hence the dimension- $d - 1$ theorem and its robust form give an exceptional induced-direction set of ν -mass at most $h^{\epsilon'}$.

On the other hand, because the weak statement was assumed to fail for the original set, it fails for every large subset produced during the slice construction; in particular the ambient bad-direction family for B has ρ -mass at least h^ϵ . Put $\epsilon_1 = 3\epsilon/\kappa$. Directions within h^{ϵ_1} of $W^\perp$ have mass at most $h^{2\epsilon}$ by (6.27). Lemma 6.6.4 maps every remaining ambient bad direction into the induced bad set in W . For ϵ chosen sufficiently small compared with the induction gain, this contradicts the preceding ν -bound. Thus the weak theorem holds in dimension d . $\square$

Lemma III.6.6.10 (robust self-improvement). *Assume that the weak theorem of Proposition 6.6.5 holds with parameter $\epsilon > 0$. Then, after replacing ϵ by a fixed fraction, there is an exceptional set of directions of ρ -mass at most $h^{\epsilon/2}$ outside which **every** subset $Y \subset X$ satisfying*

$$N_h(Y) \geq h^{\epsilon/3} N_h(X)$$

has

$$N_h(P_V Y) \geq h^{-\alpha/d - \epsilon/3}. \quad (6.30)$$

Proof. Iteratively apply the weak theorem. Construct sets $A_1, A_2, \dots \subset X$ whose h -neighbourhoods are pairwise disjoint and such that

$$\rho(E_\epsilon(A_i)) \leq h^\epsilon.$$

At each step apply the weak theorem to the uncovered part, and stop when that uncovered part has covering number at most $h^{\epsilon/2} N_h(X)$. The spatial nonconcentration hypothesis survives this deletion, after halving the slack. The process terminates because the h -neighbourhoods are disjoint. Put $A_0 = \bigcup_i A_i$ and

$$a_i = \frac{N_h(A_i)}{N_h(A_0)};$$

then $\sum_i a_i = 1$ up to a bounded covering-comparison factor, which we absorb by normalising the weights.

Let V be bad in the robust sense and choose a witness $Y \subset X$ with $N_h(Y) \geq h^{\epsilon/3} N_h(X)$. Define

$$I(V) = \{i : N_h(Y^{(2h)} \cap A_i) \geq h^\epsilon N_h(A_i)\}.$$

The uncovered remainder and the pieces with indices outside $I(V)$ together account for at most $O(h^{\epsilon/2}) N_h(X) + h^\epsilon N_h(A_0)$. Therefore, for h small,

$$\sum_{i \in I(V)} a_i \geq h^{\epsilon/2}. \quad (6.31)$$

Moreover, for every $i \in I(V)$, the set $Y^{(2h)} \cap A_i$ is a valid bad witness for A_i (changing scale from h to a fixed multiple changes covering numbers only by a constant), so $V \in E_\epsilon(A_i)$.

For measurable sets $E_i = E_\epsilon(A_i)$, Fubini gives the weighted Markov bound

$$\rho \left\{ V : \sum_{i: V \in E_i} a_i \geq h^{\epsilon/2} \right\} \leq h^{-\epsilon/2} \sum_i a_i \rho(E_i) \leq h^{\epsilon/2}. \quad (6.32)$$

By (6.31), every robustly bad V belongs to the set on the left. This proves the lemma. $\square$

Completion of the proof

Apply Proposition 6.6.5 in dimension d and then Lemma 6.6.6. Translating from the h -notation back to m , choose the internal $\epsilon > 0$ first, then take the theorem's hypothesis-loss parameter $\delta > 0$ sufficiently smaller than every occurrence of ϵ , and finally put $\eta = \epsilon/4$. Fixed-dimensional constants and the changes between balls, dyadic cubes, oriented vectors and projective lines are absorbed by this hierarchy. The resulting exceptional set has

$$\rho(E) \leq h^\delta = 2^{-\delta m},$$

and every $X' \subset X$ with $N(X', m) \geq 2^{-\delta m} N(X, m)$ satisfies

$$N(P_\theta X', m) \geq h^{-\alpha/d-\eta} = 2^{m(\alpha/d+\eta)}$$

for $\theta \notin E$. This is exactly (6.8), and hence Theorem 6.6 is proved.

This proves Bourgain's projection-growth conclusion from the scalar arithmetic argument, bad-direction extraction, codimension-one slicing, direction transfer, and robust self-improvement developed in this section. No result from Chapters III.5, III.7, or III.9 is used.

Proposition III.6.7 (three-regime robust projection theorem). *Fix $0 < \kappa < 1$. There is $\eta = \eta(\kappa) > 0$ such that, after taking $\delta > 0$ sufficiently small and m sufficiently large, the following holds. Suppose*

$$\mu(B(x, r)) \leq 2^{\delta m} r^\alpha, \quad \rho(H^{(r)} \cap S^{d-1}) \leq 2^{\delta m} r^\kappa \quad (6.9)$$

for $2^{-m} \leq r \leq 1$. Then outside a set of ρ -mass at most $2^{-\delta m}$, every projection $P_\theta \mu$ is $(\gamma(\alpha), \delta, m)$ -robust, where

$$\gamma(\alpha) = \begin{cases} \alpha - 6\delta, & \alpha < \kappa/2, \\ \alpha/d + \eta, & \kappa/2 \leq \alpha \leq d - \kappa/2, \\ 1 - 6\delta, & \alpha > d - \kappa/2. \end{cases} \quad (6.10)$$

Proof. Middle regime. Apply Lemma 6.5 to μ . On every significant mass level X_j , the hypotheses (6.7) follow from (6.9), after replacing δ by a fixed multiple. Theorem 6.6 gives

an exceptional set E_j of exponentially small ρ -mass. There are $O_d(m)$ relevant levels, so, after shrinking the theorem's slack, the union has ρ -mass at most $2^{-\delta m}$. If A has projected mass at least $2^{-\delta m}$, then the negligible levels cannot carry all of its preimage; hence some significant level contributes a fixed exponential fraction. Applying (6.8) to that level gives $N(A, m) \geq 2^{(\alpha/d+\eta)m}$.

Small-alpha regime. Choose $\sigma = \alpha < \kappa$. From (6.9), a dyadic annulus sum gives $I_\alpha(\mu) \lesssim m2^{\delta m}$ at the truncated scale. The angular energy estimate already proved in III.2 gives

$$\int I_\alpha(P_\theta \mu) d\rho(\theta) \lesssim_\kappa m2^{2\delta m}.$$

Markov yields an exceptional set of ρ -mass at most $2^{-\delta m}$ on which the projected energy can be ignored. If A has projected mass at least $2^{-\delta m}$ and $\nu = (P_\theta \mu)_A$, then $I_\alpha(\nu) \lesssim m2^{5\delta m}$. On the other hand, Cauchy–Schwarz on the dyadic m -partition gives

$$I_\alpha(\nu) \gtrsim 2^{\alpha m} \sum_I \nu(I)^2 \geq 2^{\alpha m} N(A, m)^{-1}.$$

Thus $N(A, m) \geq 2^{(\alpha-6\delta)m}$ for large m .

Large-alpha regime. The same annulus summation gives $I_{d-\kappa}(\mu) \lesssim 2^{\delta m}$. The III.2 Falconer/Sobolev projection estimate gives

$$\int \|P_\theta \mu\|_2^2 d\rho(\theta) \lesssim 2^{2\delta m}.$$

Outside another set of ρ -mass $\leq 2^{-\delta m}$, the norm is at most $O(2^{3\delta m})$. Restricting to a projected set of mass $\geq 2^{-\delta m}$ multiplies the squared norm by at most $2^{2\delta m}$. Cauchy–Schwarz then yields $N(A, m) \geq 2^m / \|\nu\|_2^2 \geq 2^{(1-6\delta)m}$. $\square$

This proposition is important: once the Bourgain middle regime closes, the endpoint regimes are already rigorous and do not need sum-product theory.

Theorem III.6.8 (continuous linear Bourgain consequence). *Let $A \subset \mathbb{R}^d$ be Borel with $\dim_{\mathbb{H}} A > \alpha$, and let ρ be a compactly supported probability measure on $\mathbb{S}^{d-1}$ satisfying*

$$\rho(H^{(r)}) \leq Cr^\kappa \quad (0 < r \leq 1) \quad (6.11)$$

for all hyperplanes H . Then there is $\eta = \eta_d(\alpha, \kappa) > 0$ such that for ρ -a.e. θ ,

$$\dim_{\mathbb{H}} P_\theta A \geq \frac{\alpha}{d} + \eta. \quad (6.12)$$

In particular at least one such direction lies in $\text{supp}(\rho)$.

Proof. Choose $s > \alpha$ with $s < \dim_{\mathbb{H}} A$ and an s -Frostman probability μ on a compact subset of A . Discretise at scale m and apply Proposition 6.7 with exponent s ; (6.11) supplies the direction hypothesis after absorbing the fixed constant into $2^{\delta m}$. In the small, middle and large regimes respectively, Proposition 6.7 gives robust exponents

$$s - O(\delta), \quad s/d + \eta_0, \quad 1 - O(\delta).$$

Each is strictly larger than α/d once δ is sufficiently small (in the middle regime use $s > \alpha$; in the endpoint regimes use $\alpha < d$). Thus there is a single $\eta > 0$, depending on d, α, κ and the chosen s , such that outside an exceptional set E_m of ρ -mass $O(2^{-\delta m})$, the projected discretisation is $(\alpha/d + \eta, \delta, m)$ -robust. The exceptional masses are summable, so Borel–Cantelli applies.

A projected dyadic source cube meets only $O_d(1)$ target intervals at comparable scale, hence robustness transfers to the actual pushforward after a bounded scale shift. Lemma 6.3 gives $\dim_{\mathbb{H}} P_\theta A \geq \alpha/d + \eta$ for ρ -almost every θ . $\square$

III.6.3.2 Why the hypotheses are needed

Example III.6.9 (line concentration). *If X lies in a 2^{-m} -neighbourhood of a line, directions nearly perpendicular to that line can collapse $P_\theta X$ to $O(1)$ intervals. This is exactly the configuration excluded by the spatial and direction nonconcentration assumptions.*

Example III.6.10 (resonant product). *If $X = A \times B$ and a slope λ satisfies a strong approximate additive relation between A and λB , then $P_{(1,\lambda)} X = A + \lambda B$ may have unusually small covering number. Bourgain’s theorem does not say that no such direction exists; it says that a hyperplane–Frostman family of directions cannot consist predominantly of such resonances.*

Example III.6.11 (one-scale is not all-scale). *A union of many tiny clusters can have large total covering number while failing (6.2) at intermediate scales. This is why the original theorem and Shmerkin’s single-scale refinement have genuinely different hypotheses.*

III.6.4 Nonlinear direction maps and linearisation

Throughout this nonlinear section, Λ is a standard Borel space, ν is a Borel probability on it, and $F : \Lambda \times U \rightarrow \mathbb{R}$ is jointly Borel, with each $F_\lambda : U \subset \mathbb{R}^d \rightarrow \mathbb{R}$ of class C^2 and having nonvanishing gradient. Its spatial gradient is jointly Borel, being the pointwise limit of difference quotients of F . Define

$$\theta_x(\lambda) = \frac{\nabla F_\lambda(x)}{|\nabla F_\lambda(x)|} \in \mathbb{S}^{d-1}. \quad (6.13)$$

The direction hypothesis used below is

$$(\theta_x)_\# \nu_{\Lambda_x}(H^{(r)}) \leq C_x r^\kappa \quad (H \in G(d, d-1), 0 < r \leq 1). \quad (6.14)$$

Lemma III.6.12 (Borel uniformisation of moving direction fibres). *Let X, Λ be standard Borel spaces, let ν be a Borel probability on Λ , and let $\theta : X \times \Lambda \rightarrow \mathbb{S}^{d-1}$ be Borel. Fix $\kappa, c > 0$, with $c \leq 1$. Suppose that for every $x \in X$ there are a ν -measurable set $B_x \subset \Lambda$ of mass at least c and a finite constant C_x such that*

$$(\theta_x)_\# \nu_{B_x}(H^{(r)}) \leq C_x r^\kappa \quad (H \in G(d, d-1), 0 < r \leq 1). \quad (6.14a)$$

Then $X = \bigcup_{n \geq 1} X_n$ for increasing Borel sets X_n , and for each n there is a jointly Borel $G_n \subset X_n \times \Lambda$ whose sections satisfy

$$\nu((G_n)_x) \geq c/2, \quad (\theta_x)_\# \nu_{(G_n)_x}(H^{(r)}) \leq 4nc^{-2}r^\kappa. \quad (6.14b)$$

The same assertion holds if the direction inequality is required only for $r \in [\delta, 1]$, with the same proof and with δ fixed.

Proof. Equip $\mathcal{X} = \{f \in L^\infty(\nu) : 0 \leq f \leq 1\}$ with its weak-* topology. It is compact metrizable because $L^1(\nu)$ is separable. Choose countable dense hyperplanes and rational radii; testing the inequality in (6.14a) on this family is equivalent to testing it on every H, r , by enlarging a neighbourhood slightly and then letting the enlargement decrease to zero. For $n \geq 1$ let $\mathcal{S}_n(x) \subset \mathcal{X}$ consist of the densities f satisfying

$$\int f \, d\nu \geq c, \quad \int f(\lambda) 1_{\{\theta_x(\lambda) \in H^{(r)}\}} \, d\nu(\lambda) \leq nr^\kappa \int f \, d\nu \quad (6.14c)$$

for every test pair. Each section is closed, hence compact. The graph $\{(x, f) : f \in \mathcal{S}_n(x)\}$ is Borel: for a fixed test pair, $x \mapsto 1_{\{\theta_x(\cdot) \in H^{(r)}\}}$ is an $L^1(\nu)$ -valued Borel map, obtained by bounded simple-function approximation, and pairing with f is jointly Borel.

We prove the special compact-section selection needed here, so no general uniformisation theorem is being left as a proof input. List the countably many constraints (6.14c) as $g_j(x, f) \leq 0$, including $c - \int f \, d\nu \leq 0$. Each g_j is Borel in x and continuous in f . For a fixed compact $C \subset \mathcal{X}$, the set of x for which some $f \in C$ obeys every constraint is Borel. Indeed, by compactness it is

$$\bigcap_{N \geq 1} \left\{ x : \min_{f \in C} \max_{j \leq N} g_j(x, f) \leq 0 \right\}. \quad (6.14d)$$

Each minimum is the infimum over one fixed countable dense subset of C , hence is Borel in x . Taking $C = \mathcal{X}$ proves that $X_n = \{x : \mathcal{S}_n(x) \neq \emptyset\}$ is Borel.

To select f_x , fix a dense sequence (h_i) in a metric compatible with the weak-* topology on $\mathcal{X}$. At stage k , choose the least $i = i_k(x)$ such that $\mathcal{S}_n(x)$ meets the closed ball $\bar{B}(h_i, 2^{-k})$ and all previously chosen closed balls. This is a Borel choice: on each of the countably many cells where the earlier indices are fixed, apply (6.14d) with C equal to the intersection of those fixed balls and the candidate ball. The intersection is nonempty at every stage because the h_i are dense. The resulting nested compact intersections have diameter at most 2^{1-k} , hence a unique common point f_x ; moreover $h_{i_k(x)} \rightarrow f_x$. Thus $x \mapsto f_x$ is weak-* Borel and $f_x \in \mathcal{S}_n(x)$. The sets X_n increase and cover X , since $1_{B_x} \in \mathcal{S}_n(x)$ once $n \geq C_x$.

For completeness, the selected L^∞ -classes admit a jointly Borel representative. Take increasing finite Borel partitions $\mathcal{P}_m$ generating the Borel sigma algebra modulo ν -null sets and write the conditional expectation $f_m(x, \lambda)$ of f_x on $\mathcal{P}_m$ using its cell integrals. Every coefficient is Borel in x . Define the jointly Borel representative $f(x, \lambda) = \limsup_{m \rightarrow \infty} f_m(x, \lambda)$; martingale convergence shows that it equals f_x for ν -almost every λ , for each x . Put

$$G_n = \{(x, \lambda) \in X_n \times \Lambda : f(x, \lambda) \geq c/2\}.$$

Because $0 \leq f \leq 1$ and $\int f \, d\nu \geq c$, this level set has $\nu((G_n)_x) \geq c/2$. On G_n one has $1_{G_n} \leq (2/c)f$; therefore

$$\nu_{(G_n)_x} \{\lambda : \theta_x(\lambda) \in H^{(r)}\} \leq \frac{2}{c \nu((G_n)_x)} \int f 1_{\{\theta_x \in H^{(r)}\}} \, d\nu \leq 4nc^{-2}r^\kappa.$$

This proves the claimed uniformisation. $\square$

Lemma III.6.12 (Taylor linearisation on a cube). *Suppose $\|F\|_{C^2(U)} \leq M$ and $|\nabla F| \geq c > 0$. Let Q be a cube of diameter R centred at x_Q , and let $x \in Q$. Then*

$$F(x) = F(x_Q) + |\nabla F(x_Q)| P_{\theta_{x_Q}}(x - x_Q) + E_Q(x), \quad |E_Q(x)| \leq C_d M R^2. \quad (6.15)$$

Consequently, if the target is observed at resolution r and $M R^2 \leq C r$, the dyadic entropy/covering number of $F(Q \cap X)$ at scale r differs from that of the indicated linear projection by at most an additive $O_{d,C,c,M}(1)$ in logarithmic scale.

Proof. Taylor's formula with integral remainder gives the first assertion. The second follows because two subsets of $\mathbb{R}$ at Hausdorff distance at most $C r$ meet comparable numbers of intervals of length r ; multiplying by the nonzero scalar $|\nabla F(x_Q)|$ changes this number only by a bounded factor. $\square$

Lemma III.6.13 (entropy perturbation). *If $f, g : X \rightarrow \mathbb{R}$ satisfy $|f - g| \leq C 2^{-n}$ pointwise, then for every probability μ on X ,*

$$|H(f\mu, \mathcal{D}_n) - H(g\mu, \mathcal{D}_n)| \leq C'(C), \quad (6.16)$$

and the same bounded-loss statement holds for robust entropy.

Proof. Each dyadic interval at level n for one image can meet at most $O_C(1)$ intervals for the other. Refining one partition by the other increases entropy by at most the logarithm of this overlap multiplicity. Apply the estimate in both directions. If $v \leq \Delta\mu$, then $f v$ and $g v$ satisfy the same comparison; take the infimum. $\square$

Lemma III.6.14 (large almost-linear subinterval). *For every $\epsilon > 0$, every 1-Lipschitz function $f : [a, b] \rightarrow \mathbb{R}$ has a subinterval $[c, d]$ of length at least*

$$\delta(\epsilon)(b - a), \quad \delta(\epsilon) = \epsilon^{\lfloor 1/\epsilon \rfloor},$$

on which

$$|f(x) - L_{f,c,d}(x)| \leq \epsilon(d - c) \quad (c \leq x \leq d). \quad (6.17)$$

Proof. Replace f by $-f$ if necessary so that its slope on $[a, b]$ is nonnegative. If $[a, b]$ is not ϵ -linear, after reflecting the variable if necessary there is an x for which $f(x)$ lies more than $\epsilon(b - a)$ above the chord. Then $[a, x]$ has length at least $\epsilon(b - a)$ and its slope exceeds the original slope by at least ϵ . Iterate. Slopes of a 1-Lipschitz function lie in $[-1, 1]$ and, after the initial sign choice, the iterated slopes lie in $[0, 1]$; hence the process stops after at most $\lfloor 1/\epsilon \rfloor$ slope increments. The surviving interval has the asserted relative length and is ϵ -linear. $\square$

Lemma III.6.15 (almost-linear covering decomposition). *For every $\epsilon > 0$ there is $\tau = \tau(\epsilon) > 0$ such that every 1-Lipschitz $f : [a, b] \rightarrow \mathbb{R}$ admits disjoint intervals $[c_j, d_j]$ satisfying*

$$\begin{aligned}
(f, c_j, d_j) &\text{ is } \epsilon\text{-linear}, \\
d_j - c_j &\geq \tau(b - a), \\
\mathcal{L}([a, b] \setminus \cup_j [c_j, d_j]) &\leq \epsilon(b - a).
\end{aligned} \tag{6.18}$$

Proof. Apply Lemma 6.14 to $[a, b]$, remove the resulting interval, and apply it recursively to every remaining component. At each generation at least a fixed fraction $\delta(\epsilon)$ of the remaining length is removed. After $N = N(\epsilon)$ generations, the uncovered length is at most $\epsilon(b - a)/2$. There are at most $2^N - 1$ selected intervals. Delete those shorter than $\epsilon 2^{-(N+1)}(b - a)$; their total length is at most $\epsilon(b - a)/2$. The remaining intervals satisfy (6.18). $\square$

Corollary III.6.16 (almost-linear pieces away from the initial scale). *For every $\epsilon > 0$ there is $\tau > 0$ such that the intervals in Lemma 6.15 can be chosen so that, after changing ϵ, τ by fixed factors,*

$$\tau(b - a) \leq d_j - c_j \leq c_j - a. \tag{6.19}$$

Proof. Partition the portion of $[a, b]$ away from a into the dyadic annuli

$$[a + 2^k(b - a), a + 2^{k+1}(b - a)]$$

for the finitely many negative k whose union misses only an $O(\epsilon)$ fraction near a . Apply Lemma 6.15 on each annulus and collect the pieces. On each annulus its length is at most its distance from a , while the smallest retained piece is a fixed ϵ -dependent fraction of $b - a$. $\square$

For an interval $[u, v]$ write

$$s_f(u, v) = \frac{f(v) - f(u)}{v - u}.$$

We say (f, u, v) is ϵ -**superlinear** if

$$f(x) \geq L_{f, u, v}(x) - \epsilon(v - u), \quad u \leq x \leq v. \tag{6.20}$$

Lemma III.6.17 (ordered superlinear decomposition). *For every $\epsilon > 0$ there is $\tau = \tau(\epsilon) > 0$ such that every 1-Lipschitz function on $[a, b]$ admits disjoint ordered intervals $[a_k, b_k]$ which cover all but $\epsilon(b - a)$ length, have length at least $\tau(b - a)$, are ϵ -superlinear, and whose slopes $s_f(a_k, b_k)$ are nondecreasing from left to right.*

Proof. Apply Lemma 6.15 with accuracy $\epsilon^2/4$, obtaining almost-linear intervals and an exceptional set E of length at most $\epsilon^2(b - a)/4$. Let C be the finite set consisting of their endpoints and a . For $y > a$, let $P(y)$ be the largest point $x \in C \cap [a, y)$ maximizing $s_f(x, y)$. Starting from $y_0 = b$, define $y_{j+1} = P(y_j)$. The resulting chord slopes are nondecreasing in their left-to-right order.

Discard intervals $[y_{j+1}, y_j]$ for which either (i) the fraction of E is at least $\epsilon/2$, or (ii) the interval is shorter than the τ supplied by Lemma 6.15. The total length discarded for

reason (i) is at most $\epsilon(b-a)/2$; an interval discarded for reason (ii) lies in E , so the total loss from (ii) is also at most $\epsilon(b-a)/2$ after decreasing τ .

It remains to verify superlinearity on a retained interval $[y, y']$. On every almost-linear component $[c_i, d_i]$ inside it, maximality of $y = P(y')$ implies that the chord through $(c_i, f(c_i)), (d_i, f(d_i))$ lies above the chord through $(y, f(y)), (y', f(y'))$. Almost-linearity therefore gives (6.20) there with error $\epsilon^2(b-a)/4$. Every point of the residual set is within $\epsilon(y' - y)/2$ of one of these components because the residual fraction in the retained interval is $< \epsilon/2$; the Lipschitz bound transfers the same lower estimate to the residual points. The total error is $< \epsilon(y' - y)$. $\square$

Proposition III.6.18 (intermediate-slope profile decomposition). *Fix $s, t \in (0, 1)$. There is $\xi = \xi(s, t) > 0$ such that, for all sufficiently small $\epsilon > 0$, there is $\tau = \tau(\epsilon) > 0$ with the following property. If $f : [0, B] \rightarrow \mathbb{R}$ is nondecreasing and 1-Lipschitz and*

$$f(0) = 0, \quad f((1-s)B) \geq tB, \quad f(B) = sB, \quad (6.21)$$

then there are ordered disjoint intervals $[a_j, b_j]$ such that

$$\begin{aligned} (f, a_j, b_j) &\text{ is } \epsilon\text{-superlinear,} \\ \tau B &\leq b_j - a_j \leq a_j, \\ \mathcal{L}([0, B] \setminus \cup_j [a_j, b_j]) &\leq \epsilon B, \\ \sum_{s_f(a_j, b_j) \in [\xi, 1-\xi]} (b_j - a_j) &\geq \xi B. \end{aligned} \quad (6.22)$$

The constants can be chosen uniformly when (s, t) ranges in a compact subset of $(0, 1)^2$.

Proof. Scale to $B = 1$. Choose a small $\sigma = \sigma(s, t) > 0$ and partition $[0, 1]$ into $I_0 = [0, 4\sigma]$ and intervals $I_n = [c_n, c_{n+1}]$, $n \geq 1$, of length between σ and 2σ , arranging $c_{n_0} = 1 - s$. Choose $\epsilon_0 \ll \sigma$, and apply Corollary 6.16 to I_0 and Lemma 6.17 to every I_n , obtaining pieces $[a_{n,k}, b_{n,k}]$ with slopes $s_{n,k}$ and total omitted length at most $\epsilon_0 |I_n|$.

Put $\zeta = \sigma/11$. Classify $n \geq 1$ as follows: class 1 if slope $\leq \zeta$ occupies at least $(1 - \sigma)|I_n|$; class 2 if slope $\geq 1 - \zeta$ occupies at least that much; class 3 if neither occurs but slopes outside $[\zeta, 1 - \zeta]$ occupy at least $(1 - \sigma/2)|I_n|$; and class 4 otherwise.

If class 4 occurs, the pieces in that single interval with slope in $[\zeta, 1 - \zeta]$ already have total length at least $\sigma|I_n|/4 \geq \sigma^2/4$. Together with all the other selected pieces they satisfy (6.22), after decreasing τ .

Assume therefore that class 4 is empty. Two elementary telescoping observations now force a transition from high to low slopes. First, not every index from n_0 to the right can be in class 2: otherwise monotonicity and the near-complete high-slope coverage give

$$f(1) \geq f(1-s) + (1-\sigma)^2 s \geq t + (1-\sigma)^2 s > s$$

when σ is small enough. Second, not every index up to n_0 can be in class 1: otherwise the low-slope pieces and $|I_0| = 4\sigma$ give $f(1-s) \leq 6\sigma < t$. Hence there is an adjacent pair I_{n-1}, I_n for which the left interval contains at least $\sigma|I_{n-1}|/2$ of pieces with slope $\geq 1 - \zeta$, while the right interval contains at least $\sigma|I_n|/2$ with slope $\leq \zeta$.

Let $\tilde{a}$ be the left endpoint of the first high-slope piece in I_{n-1} , and $\tilde{b}$ the right endpoint of the last low-slope piece in I_n . Ordered superlinearity gives, up to an $O(\epsilon_0|I_{n-1}|)$ error, slope at least $1 - \zeta$ from $\tilde{a}$ to c_n ; similarly the right side grows with slope at most ζ up to an $O(\epsilon_0|I_n|)$ error. Since

$$c_n - \tilde{a} \geq \frac{\sigma}{2}|I_{n-1}|, \quad \tilde{b} - c_n \geq \frac{\sigma}{2}|I_n|,$$

and adjacent interval lengths are comparable, direct division yields

$$\zeta \leq s_f(\tilde{a}, \tilde{b}) \leq 1 - \zeta$$

provided $\epsilon_0 \leq \zeta$ and $\zeta = \sigma/11$. The same two one-sided estimates show that $(f, \tilde{a}, \tilde{b})$ is $(4\epsilon_0/\sigma)$ -superlinear. Moreover

$$\tilde{b} - \tilde{a} \geq \sigma^2/2, \quad \tilde{b} - \tilde{a} \leq \tilde{a}.$$

Replace the pieces intersecting this interval by $[\tilde{a}, \tilde{b}]$ and retain all other pieces. The uncovered length remains $O(\epsilon_0)$, and the new intermediate piece alone contributes at least $\sigma^2/2$. Taking $\epsilon_0 = \sigma\epsilon/4$ and $\xi = \min\{\zeta, \sigma^2/2\}$ proves (6.22). All restrictions on σ are strict and continuous in (s, t) , giving the uniformity on compact parameter sets. $\square$

Corollary III.6.19 (lattice-endpoint version). *In Proposition 6.18 the endpoints may be required to lie in $(B/\ell)\mathbb{N}_0$ once ℓ is sufficiently large in terms of ϵ , after decreasing τ, ξ by fixed factors and replacing ϵ by 2ϵ .*

Proof. Move each left endpoint right and each right endpoint left to the nearest lattice point. The displacement is at most B/ℓ . Since every original interval has length at least τB , for large ℓ all length and coverage estimates persist. The 1-Lipschitz property changes every chord slope by $O((\tau\ell)^{-1})$, and the superlinear error by the same order. Choose ℓ so this is at most ϵ . $\square$

Recall that all block-regular measures are understood at terminal resolution 2^{-m} as in the [definition above](#). We have kept the normalisation of X-REG-III.6 explicit because its disjointness and mass bounds are not consequences of the set-valued lemma in Volume II.11.

For such a parent cube Q , μ^Q denotes the restriction to Q , normalized and rescaled affinely back to $[0, 1)^d$.

Lemma III.6.20 (regular-branching estimates). *If μ is $(\sigma; T)$ -regular and $m = T\ell$, then for every occupied $Q \in \mathcal{D}_{JT}$,*

$$2^{-j}2^{-T(\sigma_1+\dots+\sigma_j)} \leq \mu(Q) \leq 2^{-T(\sigma_1+\dots+\sigma_j)}, \quad (6.24)$$

and, writing X for the dyadic support,

$$2^T \sum_i \sigma_i \leq N(X, m) \leq 2^T \sum_i \sigma_i + \ell. \quad (6.25)$$

The normalized Lebesgue measure on the occupied m -cubes and μ are mutually dominated with factor 2^ℓ .

Proof. Iterate (6.23) down the tree to obtain (6.24). At level m all occupied cube masses therefore differ by at most 2^ℓ ; summing their masses to one gives (6.25) and the final mutual domination. $\square$

Theorem III.6.21 (Shmerkin multiscale Frostman-piece decomposition). *For every $u, \epsilon > 0$ there are $\xi = \xi(u) > 0$ and $\tau = \tau(\epsilon) > 0$ with the following property for all sufficiently large T and then sufficiently large ℓ . Let μ be $(\sigma; T)$ -regular on $[0, 1]^d$, let X be its dyadic support, and put $m = \ell T$. If*

$$\mu(B(x, |X|^{1/d})) \leq 2^{-um} \quad \text{for every } x, \quad (6.26)$$

then there are pairwise disjoint integer intervals $[A_j, B_j)$ and $\alpha_j \in [0, d]$, with $m_j = T(B_j - A_j)$, such that

$$\begin{aligned} \tau \ell &\leq B_j - A_j \leq A_j, \\ \mu^Q(B(x, r)) &\leq 2^{\epsilon m_j r^{\alpha_j}} \quad (Q \in \mathcal{D}_{TA_j}, \quad 2^{-m_j} \leq r \leq 1), \\ \sum_j \alpha_j m_j &\geq \log_2 N(X, m) - 2\epsilon m, \\ \sum_{\alpha_j \in [\xi, d-\xi]} m_j &\geq \xi m. \end{aligned} \quad (6.27)$$

Proof. Define the nondecreasing 1-Lipschitz branching profile

$$f(0) = 0, \quad f(j) = \frac{\sigma_1 + \dots + \sigma_j}{d},$$

and interpolate linearly. Set $s = f(\ell)/\ell$. First locate s away from 0, 1. The support is a union of $N(X, m)$ cubes of volume 2^{-dm} . Condition (6.26) implies, up to constants absorbed by taking m large,

$$2^{(u/2-d)m} \leq |X| \leq 2^{-um/2}. \quad (6.28)$$

Indeed, write $r_0 = |X|^{1/d}$. Since X is a union of terminal m -cubes, $r_0 \geq 2^{-m}$. Each terminal cube is contained in $O_d(1)$ balls of radius r_0 , so (6.26) gives terminal-cube mass $\leq C_d 2^{-um}$. Since the masses of the terminal cubes sum to one, $N(X, m) \geq c_d 2^{um}$, and therefore $|X| = N(X, m) 2^{-dm} \geq c_d 2^{(u-d)m}$. Conversely, $[0, 1]^d$ can be covered by $C_d r_0^{-d} = C_d |X|^{-1}$ balls of radius r_0 ; (6.26) and total mass one give $1 \leq C_d |X|^{-1} 2^{-um}$, hence $|X| \leq C_d 2^{-um}$. Weakening the exponents absorbs these dimensional constants and yields (6.28). Combining (6.28) with (6.25) gives, after taking T large,

$$s \in [u/4, 1 - u/2]. \quad (6.29)$$

Next put $j_0 = \lfloor (1 - s)\ell \rfloor$. A $j_0 T$ -cube meeting X is coverable by a bounded number (depending on d, T) of balls of radius comparable to $2^{-(1-s)m} \leq |X|^{1/d}$. Thus (6.26) gives an upper bound $\mu(Q) \leq 2^{(\epsilon-u)m}$ for large ℓ . The lower bound in (6.24) gives

$$\mu(Q) \geq 2^{-\epsilon m} 2^{-T(\sigma_1 + \dots + \sigma_{j_0})}.$$

Comparing the two estimates yields

$$f((1-s)\ell) \geq \frac{u}{2d}\ell \quad (6.30)$$

after reducing ϵ in terms of u . Hence Proposition 6.18 applies with $(s, t) = (s, u/(2d))$. Uniformity on the compact range (6.29), followed by Corollary 6.19, gives integer intervals $[A_j, B_j]$ satisfying the first and fourth claims of (6.27), if we set

$$\alpha_j = d s_f(A_j, B_j). \quad (6.31)$$

We verify the Frostman estimate. Superlinearity means that for $0 \leq k \leq B_j - A_j$,

$$\sigma_{A_j+1} + \cdots + \sigma_{A_j+k} \geq k\alpha_j - d\epsilon(B_j - A_j). \quad (6.32)$$

Applying (6.24) to the conditional tree inside an occupied TA_j -cube gives, for every descendant R at relative depth kT ,

$$\mu^Q(R) \leq 2^{d\epsilon m_j} 2^{-kT\alpha_j}.$$

A ball of radius $r \in [2^{-m_j}, 1]$ is covered by $O_{d,T}(1)$ such cubes at the unique block scale $2^{-kT} \leq r < 2^{-(k-1)T}$. After increasing the initial scale and absorbing the bounded factor and d into the chosen error parameter, this gives the second line of (6.27).

For the third line, complete the selected intervals by the complementary intervals $[C_i, D_i]$. Their total length is at most $\epsilon\ell$, while every slope of f is at most one. Telescoping gives

$$f(\ell)\ell = \sum_j (B_j - A_j)s_f(A_j, B_j) + \sum_i (D_i - C_i)s_f(C_i, D_i).$$

Multiplying by dT and using (6.25) yields

$$\sum_j \alpha_j m_j \geq T \sum_{r=1}^{\ell} \sigma_r - d\epsilon m \geq \log_2 N(X, m) - 2\epsilon m$$

after the usual harmless rescaling of ϵ . This proves all four assertions. $\square$

Proposition III.6.21.1 (linear-profile companion decomposition). *For every $\epsilon > 0$ there is $\tau = \tau(\epsilon) > 0$ with the following property once T and then ℓ are sufficiently large. Let μ be a $(\sigma; T)$ -regular probability measure on $[0, 1]^d$, let X be its dyadic support, and put $m = T\ell$. Then there are pairwise disjoint integer intervals $[A_j, B_j] \subset [0, \ell]$ and numbers $\alpha_j \in [0, d]$, with $m_j = T(B_j - A_j)$, such that*

$$\tau\ell \leq B_j - A_j \leq A_j, \quad (6.27a)$$

for every occupied $Q \in \mathcal{D}_{TA_j}$, every $x \in \text{supp}(\mu^Q)$, and $2^{-m_j} \leq r \leq 1$,

$$2^{-\epsilon m_j} r^{\alpha_j} \leq \mu^Q(B(x, r)) \leq 2^{\epsilon m_j} r^{\alpha_j}, \quad (6.27b)$$

and

$$\sum_j \alpha_j m_j \geq \log_2 N(X, m) - 2\epsilon m. \quad (6.27c)$$

This is the reader-level form of Shmerkin's second multiscale decomposition (Theorem 4.9 of [16]).

Proof. Define the branching profile

$$f(0) = 0, \quad f(j) = \frac{\sigma_1 + \dots + \sigma_j}{d}, \quad 0 \leq j \leq \ell,$$

and interpolate linearly. It is nondecreasing and 1-Lipschitz. Apply Corollary 6.16 to f on $[0, \ell]$, with an error parameter much smaller than the stated ϵ . We obtain disjoint almost-linear intervals which cover all but an $O(\epsilon)$ -fraction of $[0, \ell]$, have length at least $\tau\ell$, and satisfy $B - A \leq A$. Moving their endpoints inward to the nearest integers changes every slope and every relative length by $O((\tau\ell)^{-1})$, so for large ℓ the same conclusions hold with integer endpoints. Put

$$\alpha_j = d s_f(A_j, B_j).$$

Almost-linearity gives, for $0 \leq k \leq B_j - A_j$,

$$\left| \sum_{q=A_j+1}^{A_j+k} \sigma_q - k\alpha_j \right| \leq C_d \epsilon (B_j - A_j). \quad (6.27d)$$

Fix an occupied TA_j -cube Q . Lemma 6.20, applied inside the conditional regular tree, shows that every occupied descendant R at relative depth kT satisfies

$$2^{-C\epsilon m_j} 2^{-kT\alpha_j} \leq \mu^Q(R) \leq 2^{C\epsilon m_j} 2^{-kT\alpha_j}; \quad (6.27e)$$

the factor 2^{-k} in the lower regular-branching estimate is absorbed by choosing $T \gg \epsilon^{-1}$. If $2^{-(k+1)T} < r \leq 2^{-kT}$, a ball of radius r meets only $O_{d,T}(1)$ relative kT -cubes, giving the upper inequality in (6.27b). Conversely, descend a bounded number of additional block generations from the relative kT -cube containing x , stopping at terminal depth if necessary. Before terminal depth the resulting cube has diameter at most r and its mass is controlled from below by (6.27e). At terminal depth the measure has constant density on the terminal cube, and $B(x, r)$ captures a fixed dimensional fraction of that cube. In either case this gives the lower inequality. After increasing T and ℓ , all bounded geometric factors are absorbed into the stated ϵ .

Finally the omitted intervals have total length $O(\epsilon\ell)$, while every slope of f lies in $[0, 1]$. Telescoping therefore gives

$$\sum_j \alpha_j m_j \geq T \sum_{q=1}^{\ell} \sigma_q - O(\epsilon m).$$

Lemma 6.20 identifies the first term with $\log_2 N(X, m)$ up to $O(\ell)$, which is again absorbed for large T . This proves (6.27c). $\square$

Proposition III.6.22 (multiscale entropy linearisation for smooth images). *Let $1 \leq k < d$. Let μ, ν be probability measures on $[0, 1]^d$, and let $[A_i, B_i]$, $1 \leq i \leq q$, be pairwise disjoint subintervals of $(0, m]$ satisfying $B_i \leq 2A_i$. Let $F : U \rightarrow \mathbb{R}^k$ be C^2 on a neighbourhood of $\text{supp}(\mu)$, with $DF(x)$ of rank k there, and put*

$$V(x) = \ker DF(x)^\perp \in G(d, k).$$

If $v \leq \Delta\mu$, then, for arbitrary $x_Q \in Q$, one has

$$H_m(Fv) \geq -C_{F,d,k}q + \sum_{i=1}^q \sum_{Q \in \mathcal{D}_{A_i}} v(Q) H_{B_i-A_i}^{m\Delta}(P_{V(x_Q)}\mu^Q). \quad (6.33)$$

The constant depends only on d, k , an upper bound for $\|F\|_{C^2}$, and a positive lower bound for the k -Jacobian of DF on $\text{supp}(\mu)$; hence it is uniform on a fixed nonsingular C^2 family.

Proof. We first prove the non-robust estimate with v in place of μ on the right. Since the intervals are disjoint, the entropy chain rule and non-negativity of the discarded conditional entropies give

$$H(Fv, \mathcal{D}_m) \geq -O_{F,d,k}(1) + \sum_i H(Fv, \mathcal{D}_{B_i} \mid \mathcal{D}_{A_i}).$$

Condition on $Q \in \mathcal{D}_{A_i}$ and use concavity of conditional entropy. This lowers the last expression by at most $O(q)$ to

$$\sum_i \sum_Q v(Q) H(F(v_Q), \mathcal{D}_{B_i}) - O_{F,d,k}(q). \quad (6.34)$$

Fix such a cube Q and $x_Q \in Q$. Taylor's formula gives, for $z \in Q$,

$$|F(z) - F(x_Q) - DF(x_Q)(z - x_Q)| \leq C_F 2^{-2A_i} \leq C_F 2^{-B_i}.$$

Moreover $DF(x_Q)|_{V(x_Q)}$ is an isomorphism whose condition number is uniformly bounded. Therefore the dyadic B_i -partitions induced on Q by F and by $P_{V(x_Q)}$ have uniformly bounded mutual overlap. Entropy changes by only $O_{F,d,k}(1)$ under such a replacement. Rescaling Q to the unit cube and using linearity of the projection yields

$$H(F(v_Q), \mathcal{D}_{B_i}) \geq H(P_{V(x_Q)}v^Q, \mathcal{D}_{B_i-A_i}) - O_{F,d,k}(1). \quad (6.35)$$

This proves the non-robust estimate.

It remains to replace the conditional v -entropy by robust entropy of μ . For a fixed i , call Q bad when $v(Q) < \mu(Q)/m$. Then

$$\sum_{Q \text{ bad}} v(Q) < 1/m. \quad (6.36)$$

On a good cube,

$$v_Q(S) \leq m\Delta \mu_Q(S),$$

and after rescaling and projecting this gives $P_{V(x_Q)}v^Q \leq m\Delta P_{V(x_Q)}\mu^Q$. Hence its entropy is at least $H_{B_i-A_i}^{m\Delta}(P_{V(x_Q)}\mu^Q)$. The total robust-entropy contribution of bad cubes is at most

$$m^{-1}\{k(B_i - A_i) + O_d(1)\}.$$

Summing over the disjoint intervals costs only $O_{d,k}(1)$ (and in any event is absorbed by $O(q)$). Substitution in (6.34)–(6.35) proves (6.33). $\square$

Proof status. The following nonlinear projection theorem and the all-scales robust-image proposition use the disjoint measure decomposition proved as Theorem III.6.6. The rank-one Bourgain theorem proved earlier does not depend on this decomposition.

Theorem III.6.23 (Shmerkin discretized nonlinear projection theorem). *Fix $d \geq 2$ and $\kappa > 0$. There are $\eta = \eta_d(\kappa) > 0$ and $\epsilon_0 = \epsilon_0(d, \kappa) > 0$ such that the following holds for $0 < \epsilon < \epsilon_0$ and all sufficiently large m . Let $X \subset [0, 1]^d$ be a union of dyadic m -cubes, let U be an open neighbourhood of X , and let (Λ, ν) and F_λ satisfy the joint Borel and C^2 hypotheses of the nonlinear-family convention above. Assume, uniformly in $\lambda \in \Lambda$ and $x \in X$, that $\|F_\lambda\|_{C^2(U)} \leq C_{\mathcal{F}} < \infty$ and $|\nabla F_\lambda(x)| \geq c_{\mathcal{F}} > 0$; the lower threshold for m may depend on these constants. Write $|\cdot|$ for d -dimensional Lebesgue measure. Suppose*

$$|X \cap B(x, |X|^{1/d})| \leq 2^{-\kappa m} |X| \quad (x \in \mathbb{R}^d),$$

and that for each $x \in X$ there is a Borel set $\Lambda_x \subset \Lambda$ with $\nu(\Lambda_x) \geq 2^{-\epsilon m}$ such that

$$(\theta_x)_\# \nu_{\Lambda_x}(H^{(r)}) \leq 2^{\epsilon m} r^\kappa \quad (H \in G(d, d-1), 2^{-m} \leq r \leq 1).$$

Then there are $\lambda \in \text{supp } \nu$ and a measurable $X' \subset X$ with $|X'| \geq 2^{-2\epsilon m} |X|$ such that every measurable $X'' \subset X'$ with $|X''| \geq 2^{-\epsilon m} |X'|$ satisfies

$$N(F_\lambda X'', m) \geq 2^{\eta m} N(X, m)^{1/d}. \quad (6.19)$$

If $\Lambda_x = \Lambda$ for all x , one may take $X' = X$ and the conclusion holds for all λ outside a set of ν -measure at most $2^{-\epsilon m}$.

Proof. Replace X by the union of its occupied terminal m -cubes and let μ give each such cube the same mass, uniformly as Lebesgue measure inside the cube. This is an m -discrete probability in the definition above; its terminal-cube counts and the set covering numbers differ by at most dimensional constants. Apply X-REG-III.6 to pass, with loss $2^{-O(\epsilon m)}$, to a $(\sigma; T)$ -regular restriction μ_Y ; the single natural-scale anti-concentration hypothesis survives this restriction after halving κ . Apply Theorem 6.21 to μ_Y . We obtain disjoint intervals $[A_j, B_j)$ and exponents α_j satisfying (6.27), with at most $1/\tau$ intervals.

Before using Fubini, apply Lemma III.6.12 at this fixed scale with $c = 2^{-\epsilon m}$ and $n = \lceil 2^{\epsilon m} \rceil$. It supplies jointly Borel sections, still denoted Λ_x , with mass at least $2^{-\epsilon m}/2$ and directional bound $O(2^{3\epsilon m})r^\kappa$. The constant is absorbed by using 4ϵ in the intermediate nonconcentration error budget; ϵ was chosen after the fixed projection gain. For each $x \in Y$ put $\rho_x = (\theta_x)_\# \nu_{\Lambda_x}$. The hyperplane-neighbourhood hypothesis implies that ρ_x has the finite-scale nonconcentration required by Proposition 6.7. For each j , Proposition 6.7 applied to the conditional measure in the TA_j -cube containing x gives an exceptional direction set $E_j(x)$ of exponentially small ρ_x -mass such that outside $E_j(x)$ the projected conditional measure has robust exponent $\gamma(\alpha_j)$, where, after fixing the parameters in their dependency order,

$$\gamma(a) \geq a/d - \zeta \quad (0 \leq a \leq d), \quad \gamma(a) \geq a/d + \eta_0 \quad (a \in [\xi, d - \xi]). \quad (6.37)$$

The first inequality uses the Kaufman/Falconer endpoint branches of Proposition 6.7; the second is the Bourgain middle branch. Since there are at most $1/\tau$ selected intervals, choose ζ and then ϵ so that the union of the $E_j(x)$ still has ρ_x -mass less than one half. Since $\nu(\Lambda_x) \geq 2^{-\epsilon m}$, Fubini gives a parameter $\lambda \in \text{supp } \nu$ and a set $Z \subset Y$ of μ_Y -mass at least $2^{-\epsilon m}/4$. The one-piece construction in Theorem III.6.6 can retain $\mu(Y) \geq 2^{-\epsilon m/2}$; hence, for large m , $\mu(Z) \geq 2^{-2\epsilon m}$, equivalently $|Z| \geq 2^{-2\epsilon m}|X|$. For this set,

$$\theta_x(\lambda) \notin E_j(x) \quad \text{for every selected } j \text{ and every } x \in Z. \quad (6.38)$$

For every occupied TA_j -cube meeting Z , choose $x_Q \in Q \cap Z$. Let $Z' \subset Z$ have relative mass at least $2^{-\epsilon m}$. Then $\mu_{Z'} \leq 2^{O(\epsilon m)} \mu_Y$. Proposition 6.22, applied to F_λ , gives

$$H_m(F_\lambda \mu_{Z'}) \geq -O_{\mathcal{F}}(1/\tau) + \sum_j \sum_Q \mu_{Z'}(Q) H_{m_j}^{2^{O(\epsilon m)}}(P_{\theta_{x_Q}(\lambda)} \mu_Y^Q). \quad (6.39)$$

By (6.38), Lemma 6.4 and the conditional Frostman estimates in Theorem 6.21, each inner entropy is at least $(\gamma(\alpha_j) - o(1))m_j$. Averaging over Q and using (6.37) together with the last two conclusions of (6.27) gives

$$\frac{1}{m} H_m(F_\lambda \mu_{Z'}) \geq \frac{1}{dm} \log_2 N(X, m) + c(\kappa, d) - O(\zeta + \epsilon) - o(1). \quad (6.40)$$

Indeed the global a/d part telescopes to the baseline $d^{-1} \log_2 N(X, m)$, while the positive proportion of scales with $\alpha_j \in [\xi, d - \xi]$ contributes a fixed additional amount. Choose the parameters so that the errors are at most half that gain. Entropy is bounded above by the logarithm of the number of terminal dyadic intervals meeting the image, so (6.40) yields

$$N(F_\lambda Z', m) \geq 2^{\eta m} N(X, m)^{1/d}.$$

This holds for every sufficiently large $Z' \subset Z$, giving $X' = Z$. If $\Lambda_x = \Lambda$ for all x , apply disjoint regularisation with auxiliary loss 2ϵ . Its retained union R satisfies $\mu(R^c) \leq 2^{-2\epsilon m}$, and every piece E_i has $\mu(E_i) \geq 2^{-4\epsilon m}$. On each piece construct the bad-direction set $E_i(x)$ as above. After fixing the intermediate gain before ϵ , its ν -mass is at most $2^{-\zeta m}$, uniformly in i and $x \in E_i$, for some $\zeta > 4\epsilon$. The bad incidence set

$$B_\lambda = \{x \in R : \theta_x(\lambda) \in E_i(x) \text{ for the unique piece } E_i \ni x\}$$

is Borel by the same finite-test construction used above, and Fubini gives $\int \mu(B_\lambda) d\nu(\lambda) \leq 2^{-\zeta m}$. Thus, outside a parameter set of ν -mass at most $2^{-(\zeta - 3\epsilon)m} \leq 2^{-\epsilon m}$, one has $\mu(B_\lambda) < 2^{-3\epsilon m}$. Fix such a λ and any measurable $X'' \subset X$ with $\mu(X'') \geq 2^{-\epsilon m}$. Removing $R^c \cup B_\lambda$ leaves at least $\frac{1}{2} 2^{-\epsilon m}$ of the original mass for all large m . Since the pieces are disjoint, one of them satisfies

$$\mu_{E_i}(X'' \cap E_i \setminus B_\lambda) \geq \frac{1}{2} 2^{-\epsilon m}.$$

On this piece all selected source points are good for λ , and the preceding multiscale argument applies. Natural-scale nonconcentration loses at most 4ϵ in its exponent after normalising μ to E_i . Because μ is uniform on the occupied terminal cubes, $N(E_i, m) = \mu(E_i)N(X, m) \geq 2^{-4\epsilon m}N(X, m)$. Taking the fixed intermediate projection gain larger than

the $4\epsilon/d$ comparison loss gives $N(F_\lambda X'', m) \geq 2^{\eta m} N(X, m)^{1/d}$, after decreasing the final η . This proves the uniform-parameter clause. The multiscale and entropy-linearisation ingredients needed for both clauses have been proved above. $\square$

Proposition III.6.23.1 (all-scales robust nonlinear images). *Fix $0 < \beta < d$, $\kappa > 0$, and $c > 0$. There is $\eta = \eta_d(\beta, \kappa) > 0$, continuous in (β, κ) , with the following property. Let μ be a compactly supported β -Frostman probability measure on U , and suppose that for μ -almost every x there is a Borel set $\Lambda_x \subset \Lambda$, with $v(\Lambda_x) \geq c$, such that*

$$(\theta_x)_{\#} v_{\Lambda_x}(H^{(r)}) \leq C_x r^\kappa \quad (H \in G(d, d-1), 0 < r \leq 1). \quad (6.40a)$$

Then there are $c_0 = c_0(c) > 0$ and a Borel set $\Lambda_0 \subset \Lambda$, with $v(\Lambda_0) \geq c_0$, such that for every $\lambda \in \Lambda_0$ there is a Borel set A_λ with $\mu(A_\lambda) \geq c_0$ for which

$$F_\lambda(\mu_{A_\lambda}) \quad \text{is} \quad (\beta/d + \eta, \eta, m)\text{-robust} \quad (6.40b)$$

for every sufficiently large m . The threshold scale may depend on λ , on the Frostman and direction-decay constants, and on the C^2 -norm of F_λ , but the gain η depends only on d, β, κ .

Proof. This is the all-scales strengthening behind Shmerkin's Theorem 5.2 [16]; we give the argument because it is exactly the uniformity missing from a scale-by-scale compactness proof of Theorem 6.24.

Apply one fixed similarity to the source coordinates so that $\text{supp } \mu \subset (0, 1)^d$ before taking dyadic discretisations. Compose each F_λ with its inverse similarity and transform U accordingly. The direction laws are unchanged by this scalar derivative, while the Frostman and C^2 constants change by fixed factors only. Robust covering exponents at all sufficiently large scales are unaffected by a fixed similarity, after a bounded shift of scale.

Apply Lemma III.6.12 to the μ -conull Borel set on which the direction premise holds. Choose one of its Borel levels X_n with $\mu(X_n) > 3/4$, and use its jointly Borel fibres as the new Λ_x on X_n . Their mass is at least $c/2$, and their direction constant is one finite C . Set the selected fibres to be empty outside X_n . The discarded mass is fixed and small; it only changes the final retained-mass constant as a function of c , not the projection gain. On X_n , $x \mapsto v_{\Lambda_x}$ is a Borel probability kernel, so the composed direction law $\rho_x = (\theta_x)_{\#} v_{\Lambda_x}$ is Borel. Choose the parameters in the following order: first the three-regime projection gain in Proposition 6.7, then a small entropy error ζ , then the profile error ϵ , then a block length T , and finally the number of blocks ℓ . All choices up to ℓ depend only on d, β, κ , not on the output scale $m = T\ell$.

For every sufficiently large ℓ , put $m = T\ell$ and first take the terminal dyadic 2^{-m} -discretisation of μ . Moving mass inside an m -cube changes every F_λ -image by $O_\lambda(2^{-m})$, hence changes scale- m entropy and covering numbers only by $O_\lambda(1)$, by the same partition-overlap argument as Lemma 6.13. Apply Theorem III.6.6 to this discretisation. It gives disjoint regular pieces $X_{\ell,i}$, with retained union R_ℓ satisfying $\mu(R_\ell^c) \leq 2^{-\epsilon m}$. Since $\sum_\ell \mu(R_\ell^c) < \infty$, the Borel–Cantelli lemma gives

$$R_\infty := \bigcup_{L \geq 1} \bigcap_{\ell \geq L} R_\ell, \quad \mu(R_\infty) = 1.$$

We henceforth restrict all base points to $R_\infty \cap X_n$. If $x \in X_{\ell,i} \cap X_n$, repeat the bad-direction construction in the proof of Theorem 6.23 for the regular conditional measure on that piece

and call the resulting union of exceptional directions $E(x, \ell)$; set it to be empty outside the retained pieces. Take the exceptional set to be the canonical failure set for the finitely many dyadic robustness tests at this scale. Each test is an inequality for the mass of one finite union of dyadic output cells under a Borel kernel, so $\{(x, \lambda) : \theta_x(\lambda) \in E(x, \ell)\}$ is Borel. All finite regularisation choices use a fixed lexicographic tie-break, and hence $x \mapsto E(x, \ell)$ is measurable as a set-valued kernel. Proposition 6.7 gives, uniformly in x ,

$$\rho_x(E(x, \ell)) \leq C_1 m 2^{-\zeta m}. \quad (6.40c)$$

Here the factor m only counts the bounded number of multiscale blocks and harmless regularisation choices. Consequently, after increasing a starting block number ℓ_1 ,

$$E(x) := \bigcup_{\ell \geq \ell_1} E(x, \ell) \quad \text{satisfies} \quad \rho_x(E(x)) \leq \epsilon \quad (6.40d)$$

for every retained $x \in X_n$. This is the decisive all-scales step: the exponential bad-set bound is summable, so the parameter is selected only once.

Since ρ_x is the normalized restriction of ν to Λ_x , (6.40d) gives

$$\nu\{\lambda \in \Lambda_x : \theta_x(\lambda) \notin E(x)\} \geq (c/2)(1 - \epsilon) \quad (x \in X_n).$$

Fubini therefore yields a set Λ_0 of ν -measure at least $c_0 = c_0(c) > 0$ such that, for every $\lambda \in \Lambda_0$,

$$Z_\lambda := \{x : \lambda \in \Lambda_x, \theta_x(\lambda) \notin E(x)\} \quad \text{satisfies} \quad \mu(Z_\lambda) \geq c_0. \quad (6.40e)$$

Fix such a λ and write $Z = Z_\lambda$.

We prove robustness. Let $m = T\ell$ be sufficiently large and let $B \subset \mathbb{R}$ be Borel with

$$F_\lambda(\mu_Z)(B) > 2^{-\eta m}.$$

Set $Y = Z \cap F_\lambda^{-1}(B)$. Since $\mu(R_\ell^c) \leq 2^{-\epsilon m}$, subtracting the unretained part still leaves $\mu(Y \cap R_\ell) \geq \frac{1}{2}\mu(Y)$ for all large m once $\eta < \epsilon$. If η is chosen much smaller than the regularisation error, the mass form of the regularisation interface supplies one $(\sigma; T)$ -regular piece $X_{\ell,i}$, of μ -mass at least $2^{-O(\epsilon m)}$, for which, with $\omega = \mu_{X_{\ell,i}}$,

$$\omega(Y) \geq 2^{-\epsilon m}. \quad (6.40f)$$

Put

$$\bar{\sigma} = \frac{1}{\ell} \sum_{q=1}^{\ell} \sigma_q, \quad \bar{\beta} = \frac{d + \beta}{2}.$$

There are two branches, exactly as in the primary proof.

Moderate branching: $\bar{\sigma} \leq \bar{\beta}$. The support of the regular piece has $N(X_{\ell,i}, m) = 2^{\bar{\sigma}m + O(\ell)}$. Hence its natural volume scale is

$$r_0 = |X_{\ell,i}|^{1/d} = 2^{-(1-\bar{\sigma}/d)m + O(\ell)}.$$

Because $\mu(X_{\ell,i}) \geq 2^{-O(\epsilon m)}$ and μ is β -Frostman,

$$\omega(B(x, r_0)) \leq 2^{O(\epsilon m)} r_0^\beta \leq 2^{-u_0 m}, \quad u_0 = \frac{\beta(d - \beta)}{4d} > 0, \quad (6.40g)$$

after decreasing ϵ . Thus Theorem 6.21 applies with a scale-independent natural-scale exponent. Since $\theta_x(\lambda) \notin E(x, \ell)$ on Y , the entropy calculation is exactly the fixed-scale calculation in the proof of Theorem 6.23, now with Theorem 6.21 supplying the multiscale blocks: the global a/d contribution gives $d^{-1} \log_2 N(X_{\ell,i}, m)$, and the positive proportion of intermediate exponents gives a fixed additional gain. Moreover the Frostman bound and $\mu(X_{\ell,i}) \geq 2^{-O(\epsilon m)}$ imply

$$\log_2 N(X_{\ell,i}, m) \geq (\beta - O(\epsilon))m.$$

Consequently

$$H_m(F_\lambda \omega_Y) \geq (\beta/d + \eta_1)m \quad (6.40h)$$

for some $\eta_1 = \eta_1(d, \beta, \kappa) > 0$, once the errors are fixed small enough.

High branching: $\bar{\sigma} > \bar{\beta}$. Here the natural-scale hypothesis need not hold. Apply instead Proposition III.6.21.1. On every selected block, (6.27b) supplies the two-sided regularity needed for the endpoint branches of Proposition 6.7, and goodness of $\theta_x(\lambda)$ gives robust projected entropy at least $(\alpha_j/d - \zeta)m_j$. Proposition 6.22 and (6.27c) therefore give

$$\frac{1}{m} H_m(F_\lambda \omega_Y) \geq \frac{\bar{\sigma}}{d} - O(\zeta + \epsilon) > \frac{\beta}{d} + \frac{d - \beta}{2d} - O(\zeta + \epsilon). \quad (6.40i)$$

Thus this branch has an even larger fixed surplus over β/d .

Choose $\eta > 0$ smaller than half the gains in (6.40h) and (6.40i), and also smaller than every regularisation/robust-entropy error fixed above. Since $F_\lambda \omega_Y$ is supported on B , entropy is bounded above by $\log_2 N(B, m) + O(1)$. Hence

$$N(B, m) > 2^{(\beta/d + \eta)m}$$

at every sufficiently large block multiple $m = T\ell$. To cover an arbitrary large integer n , put $m = T\lfloor n/T \rfloor$. After replacing η by a smaller positive number, the mass threshold at level n implies the threshold at level m , while $N(B, n) \geq N(B, m)$ and $n - m < T$ absorb the bounded gap in the exponent. This proves the same robust statement for every sufficiently large n , and hence exactly (6.40b) with $A_\lambda = Z_\lambda$.

All positive gaps used above are strict and depend continuously on (β, κ) . On a compact subrectangle of $(0, d) \times (0, \infty)$ their minimum is positive, which gives the stated continuous choice of η . $\square$

Proof status. The continuous nonlinear theorem uses the regularisation theorem proved above. The jointly measurable choice of positive-mass parameter fibres needed for Fubini is supplied by Lemma III.6.12, rather than assumed silently.

Theorem III.6.24 (Shmerkin continuous nonlinear theorem). *Let $0 < \alpha < d$, $\kappa > 0$. Assume the family F_λ is C^2 without singular points in the jointly Borel nonlinear-family convention above, and that for every x in a Borel set $A \subset U$ of dimension at least α , there is a positive- v parameter subset Λ_x satisfying (6.14). Then for some $\lambda \in \text{supp}(v)$,*

$$\dim_H F_\lambda(A) \geq \frac{\alpha}{d} + \eta_d(\alpha, \kappa). \quad (6.20)$$

Proof. Let $g_d(\beta, \kappa)$ denote the continuous positive gain furnished by Proposition III.6.23.1. Choose $\alpha' < \alpha$ so close to α that

$$\frac{\alpha'}{d} + \frac{1}{2}g_d(\alpha', \kappa) \geq \frac{\alpha}{d} + \frac{1}{4}g_d(\alpha, \kappa). \quad (6.40j)$$

By Frostman compact selection, there is a compact $A' \subset A$ carrying an α' -Frostman probability measure μ . Apply the compact-fibre construction in the proof of Lemma III.6.12 for each $c = 1/j$ and each integer bound n on the direction constant. Its Borel witness domains $X_{j,n}$ cover A' , so one has positive μ -mass. On that domain the lemma's level-set construction supplies jointly Borel fibres with uniform positive mass and uniform direction decay. By inner regularity retain a positive- μ compact subset, still denoted A' , and constants $c, C > 0$ such that

$$v(\Lambda_x) \geq c, \quad (\theta_x)_\# v_{\Lambda_x}(H^{(r)}) \leq Cr^\kappa \quad (6.40k)$$

for μ -almost every $x \in A'$. Renormalizing μ on this subset only changes the Frostman constant.

Apply Proposition III.6.23.1. There is $\lambda \in \text{supp}(v)$ and a set $A_\lambda \subset A'$ of positive μ -mass such that, for every sufficiently large m ,

$$F_\lambda(\mu_{A_\lambda}) \text{ is } (\alpha'/d + g_d(\alpha', \kappa), g_d(\alpha', \kappa), m)\text{-robust.}$$

Lemma III.6.3 therefore gives

$$\dim_H F_\lambda(A) \geq \dim_H F_\lambda(A_\lambda) \geq \frac{\alpha'}{d} + g_d(\alpha', \kappa).$$

Together with (6.40j), this proves the theorem with, for example,

$$\eta_d(\alpha, \kappa) = \frac{1}{4}g_d(\alpha, \kappa) > 0.$$

Because g_d is continuous and positive, so is this choice. Notice that no limiting sequence of parameters is required: one nearby Frostman exponent $\alpha' < \alpha$ already yields the claimed endpoint. $\square$

Local spherical-direction theorem

The pinned theorem will follow once spherical-direction decay has been established in a form that survives the norm Gauss map. We give the quantitative argument from Appendix B of Shmerkin [16], with the exponent choices made explicit because continuity in the two decay parameters is needed at the Hausdorff-dimension endpoint.

Let $1 \leq k < d$, and let $p, q > 0$. After replacing p, q by $\min\{p, 1\}, \min\{q, 1\}$, which only weakens the hypotheses, assume that $0 < p, q \leq 1$. Suppose $\mu, \nu \in \mathcal{P}(B(0, 1))$ satisfy

$$\mu(V^{(r)}) \leq C_\mu r^p, \quad \nu(V^{(r)}) \leq C_\nu r^q \quad (V \in \mathcal{A}(d, k-1), 0 < r \leq 1),$$

and suppose that $\nu(L) = 0$ for every affine k -plane L . Then there is a continuous positive function $q_{d,k}(p, q) > 0$ and a set E with $\mu(E) \geq 1/2$ such that, for every $x \in E$, there is a compact set $K(x)$ with $\nu(K(x)) \geq 1/2$ for which

$$e_x(\nu_{K(x)})(H^{(r)}) \leq r^{q_{d,k}(p,q)}, \quad H \in G(d, k), \quad 0 < r \leq r_0. \quad (6.20b)$$

Here $e_x(y) = (y - x)/|y - x|$, $v_{K(x)}$ is the normalized restriction, and the starting scale r_0 may depend on d, C_μ, C_ν, p, q and on ν , but the power exponent depends only on d, p, q .

Step 1: heavy k -plates cluster. A k -plate of width δ means $L^{(\delta)} \cap B(0, 1)$ with $L \in \mathcal{A}(d, k)$. Fix a small number $\tau > 0$. Assume only the single-scale bound

$$\nu(V^{(\delta)}) \leq C_\nu \delta^q, \quad V \in \mathcal{A}(d, k-1).$$

Construct greedily heavy δ -plates $Y_1, \dots, Y_M$ so that

$$\nu(Y_i) \geq \delta^\tau, \quad \nu(Y_i \cap Y_j) \leq \frac{1}{2} \delta^{2\tau} \quad (i \neq j),$$

and stop when no further plate can be added. If $f = \sum_i \mathbf{1}_{Y_i}$, then Cauchy–Schwarz gives

$$\left(\sum_i \nu(Y_i) \right)^2 \leq \int f^2 d\nu \leq \sum_i \nu(Y_i) + \frac{1}{2} M^2 \delta^{2\tau}.$$

Put $S = \sum_i \nu(Y_i)$. If $M > 2\delta^{-\tau}$, then $S \geq M\delta^\tau > 2$, hence $S^2 - S \geq S^2/2$. The preceding estimate gives $S^2 - S < M^2 \delta^{2\tau}/2$, whereas $S^2/2 \geq M^2 \delta^{2\tau}/2$, a contradiction. Thus $M \leq 2\delta^{-\tau}$.

Now let W be any heavy δ -plate. By maximality, $\nu(W \cap Y_j) > \delta^{2\tau}/2$ for some j . If $\Theta(W, Y_j)$ denotes the largest principal angle between their direction k -planes, elementary two-plane geometry puts $W \cap Y_j$ inside a box with $k-1$ unit directions, one direction of length $O_d(\delta/\Theta)$, and the remaining $d-k$ directions of length $O_d(\delta)$. Partitioning the elongated direction into $O_d(1/\Theta)$ pieces and using the $(k-1)$ -plane bound gives

$$\nu(W \cap Y_j) \lesssim_d C_\nu \frac{\delta^q}{\Theta(W, Y_j)}.$$

Hence

$$\Theta(W, Y_j) \lesssim_d C_\nu \delta^{q-2\tau}.$$

Because the two plates meet, W is therefore contained in a k -plate T_j with the same central direction as Y_j and width

$$R \asymp_d C_\nu \delta^{q-2\tau}.$$

Thus all heavy δ -plates are contained in at most $2\delta^{-\tau}$ enlarged plates of width $O_d(C_\nu \delta^{q-2\tau})$.

Step 2: one-scale pruning. Choose

$$\tau_0(p, q) = \min \left\{ \frac{1}{16}, \frac{q}{16}, \frac{pq}{16(3p+2)} \right\} > 0$$

and fix $0 < \tau \leq \tau_0(p, q)$. Apply Step 1. Let Bad be the set of points x through which there passes a heavy δ -plate. Let BadBad consist of those $x \in \text{Bad}$ which lie in two enlarged plates T_i, T_j satisfying $\Theta(T_i, T_j) \geq \delta^\tau$.

The intersection of two such enlarged plates lies in the $O_d(C_\nu \delta^{q-3\tau})$ -neighbourhood of an affine $(k-1)$ -plane. Since there are $O(\delta^{-2\tau})$ pairs, the decay assumption for μ gives

$$\mu(\text{BadBad}) \lesssim_{d, C_\mu, C_\nu} \delta^{pq-(3p+2)\tau}.$$

Our choice of τ_0 leaves a positive exponent margin; after taking δ smaller to absorb the fixed implied constant,

$$\mu(\text{BadBad}) \leq \frac{1}{2} \delta^{pq/2}.$$

Let $E^0 = \text{supp}(\mu) \setminus \text{BadBad}$. Order the enlarged plates and set

$$E_j^0 = E^0 \cap T_j \setminus \bigcup_{i < j} T_i, \quad E_\emptyset^0 = E^0 \setminus \text{Bad}.$$

By inner regularity choose compact subsets $E_j \subset E_j^0$ and $E_\emptyset \subset E_\emptyset^0$ whose total discarded μ -mass is at most $\delta^{pq/2}/2$, and put $E_\delta = E_\emptyset \cup \bigcup_j E_j$. Then

$$\mu(\mathbb{R}^d \setminus E_\delta) \leq \delta^{pq/2}.$$

For $x \in E_\emptyset$ set $P_\delta(x) = \emptyset$. If $x \in E_j$, let $P_\delta(x)$ be the k -plate of width $\delta^{\tau/2}$ centered on the affine plane defining T_j . This compact partition also makes

$$\{(x, y) : x \in E_\delta, y \in \text{supp}(\nu) \setminus P_\delta(x)\}$$

compact. Our choice of τ_0 ensures $q - 2\tau > \tau/2$, so, for small δ , T_j is much thinner than $P_\delta(x)$. Suppose that a δ -plate W through x meets $\text{supp}(\nu) \setminus P_\delta(x)$ and satisfies $\nu(W) > \delta^\tau$. Step 1 places W in some T_i . If $\Theta(T_i, T_j) < \delta^\tau$, the width and angular errors imply $W \subset P_\delta(x)$, a contradiction. Hence $\Theta(T_i, T_j) \geq \delta^\tau$, which puts $x \in \text{BadBad}$, again a contradiction. Therefore

$$\nu(W) \leq \delta^\tau \tag{6.20b1}$$

for every such W , while

$$\mu(\mathbb{R}^d \setminus E_\delta) \leq \delta^{pq/2}. \tag{6.20b2}$$

Step 3: pass to every small scale. Choose an integer $N \geq 1$ and a dyadic exponent

$$\tau = 2^{1-N} \leq \tau_0(p, q) < 2\tau.$$

Thus $\tau/2 = 2^{-N}$ and $\tau > \tau_0/2$. Since the affine k -planes meeting $B(0, 1)$ form a compact family and ν gives each of them zero mass,

$$\sup_{L \in \mathcal{A}(d, k)} \nu(L^{(s)}) \longrightarrow 0 \quad (s \downarrow 0).$$

Fix $\varepsilon = 1/4$ and choose $\bar{\delta} > 0$ so that the last supremum is at most $\varepsilon/(4N)$ when $s \leq \bar{\delta}$.

Choose $\delta_0 > 0$ so small that Step 2 applies below δ_0 , $\delta_0^{\tau/2} \leq \bar{\delta}$, and, with $\delta_n = \delta_0^{2^n}$,

$$\sum_{n \geq 0} \delta_n^{pq/2} < \varepsilon/2, \quad \sum_{n \geq 0} \delta_n^\tau < \varepsilon/4.$$

For every n apply Step 2 at scale δ_n , producing E_n and $P_n(x)$. Set

$$E = \bigcap_{n \geq 0} E_n, \quad K(x) = \text{supp}(\nu) \setminus \bigcup_{n \geq 0} P_n(x).$$

Then $\mu(E) > 1/2$, and $\{(x, y) : x \in E, y \in K(x)\}$ is compact as an intersection of the compact single-scale admissible sets. For $0 \leq n < N$, the width of $P_n(x)$ is at most $\tilde{\delta}$, so $v(P_n(x)) \leq \varepsilon/(4N)$. If $n \geq N$, then

$$\text{width}(P_n(x)) = \delta_n^{\tau/2} = \delta_{n-N}.$$

By (6.20b1) at scale δ_{n-N} ,

$$v(P_n(x) \setminus P_{n-N}(x)) \leq \delta_{n-N}^{\tau}.$$

Iterating backwards by N shows

$$v\left(\bigcup_{n \geq 0} P_n(x)\right) \leq \frac{\varepsilon}{4} + \sum_{m \geq 0} \delta_m^{\tau} < \frac{\varepsilon}{2},$$

so $v(K(x)) > 1/2$.

Finally, if $H \in G(d, k)$ and $e_x(y) \in H^{(r)}$, then y lies in an $O_d(r)$ -plate around the affine plane $x + H$. For r small choose n so that $\delta_{n+1} < C_d r \leq \delta_n$. Since $K(x)$ avoids $P_n(x)$, (6.20b1) gives

$$v\{y \in K(x) : e_x(y) \in H^{(r)}\} \lesssim_d \delta_n^{\tau} \lesssim_d r^{\tau/2}.$$

After normalizing $v|_{K(x)}$ and decreasing the starting scale, the multiplicative constant is absorbed by lowering the exponent once more. Since $\tau > \tau_0(p, q)/2$, we may take, for example,

$$q_{d,k}(p, q) = \frac{1}{16} \tau_0(p, q),$$

which is positive and continuous in (p, q) . This proves (6.20b) in full.

Why the curved-norm transfer needs care. The published proof of the normed extension observes that, for $\psi(e) = \text{dir}(DN(e))$, positive Gaussian curvature makes ψ locally bi-Lipschitz and then states that the hyperplane-neighbourhood decay above transfers through ψ [16, Section 6.2 and Remark 6.4]. In dimension $d = 2$ this implication is valid for a special reason: a linear hyperplane meets S^1 in one antipodal pair, and the Gauss map of a centrally symmetric norm is odd. In dimensions $d \geq 3$, however, local bi-Lipschitzness alone does not preserve global decay near great hyperplane sections; inverse images of such sections are generally curved hypersurfaces. The counterexample below gives an explicit positively curved norm for which this invariance assertion fails. We therefore separate the valid planar transfer from the unresolved higher-dimensional source claim.

Lemma III.6.24.1 (Planar Gauss-map transfer). *Let $\psi : S^1 \rightarrow S^1$ be an odd bi-Lipschitz homeomorphism. Suppose that a probability measure σ on S^1 satisfies*

$$\sigma(L^{(r)}) \leq Cr^q \quad (L \in G(2, 1), 0 < r \leq r_0).$$

Then

$$\psi_{\#}\sigma(L^{(r)}) \leq C'r^q \quad (L \in G(2, 1), 0 < r \leq r'_0), \quad (6.20b3)$$

where C', r'_0 depend on the bi-Lipschitz constants and on C, r_0 , but the exponent q is unchanged.

Proof. Write $L \cap S^1 = \{u, -u\}$ and let $v = \psi^{-1}(u)$. Oddness gives $\psi^{-1}(-u) = -v$, hence the exact inverse image of the great section is $L' \cap S^1$, where $L' = \text{span}(v)$. If M is a Lipschitz constant for ψ^{-1} , then for all sufficiently small r ,

$$\psi^{-1}(L^{(r)} \cap S^1) \subset (L')^{(C_M r)} \cap S^1.$$

Applying the assumed decay to L' proves (6.20b3). $\square$

Lemma III.6.24.2 (Ellipsoidal Gauss-map transfer). *Let A be symmetric positive definite and $N(x) = (x^T A x)^{1/2}$. If σ on S^{d-1} has hyperplane-neighbourhood decay with exponent q , then so does $\psi_{\#}\sigma$, where $\psi(e) = \text{dir}(DN(e))$; the exponent is unchanged.*

Proof. Here $\psi(e) = \text{dir}(Ae)$. If $H = n^\perp$, then $\psi(e) \in H^{(r)}$ implies $|n \cdot Ae| \lesssim_A r$, because $|Ae|$ is bounded above and below on S^{d-1} . Thus e lies in an $O_A(r)$ -neighbourhood of the hyperplane $(A^T n)^\perp$. The claimed decay follows immediately. $\square$

Lemma III.6.24.3 (Supercritical curved-norm direction decay). *Let N be a C^2 norm whose unit ball has everywhere positive Gaussian curvature, and put $\psi(e) = \text{dir}(DN(e))$. Let $X, Y \subset \mathbb{R}^d$ be bounded compact sets with $\text{dist}(X, Y) > 0$, and let ν be a κ -Frostman probability measure on Y , where $\kappa > d - 1$. Then, uniformly for $x \in X$ and $H \in G(d, d - 1)$,*

$$(\psi \circ e_x)_{\#}\nu(H^{(r)}) \lesssim_{N, X, Y, \nu} r^{\kappa - (d-1)}, \quad 0 < r \leq 1. \quad (6.20b4)$$

The power $\kappa - (d - 1)$ is independent of N ; only the implicit constant may depend on the norm.

Proof. Positive curvature makes the Gauss map of the norm sphere a diffeomorphism. Composing it with radial parametrisation of that sphere shows that $\psi : S^{d-1} \rightarrow S^{d-1}$ is a C^1 diffeomorphism, hence ψ^{-1} is Lipschitz on the compact sphere.

Fix $H \in G(d, d - 1)$. The spherical belt $H^{(r)} \cap S^{d-1}$ is covered by $O_d(r^{-(d-2)})$ spherical balls of radius $O_d(r)$. Pulling the cover back through ψ^{-1} gives $O_d(r^{-(d-2)})$ spherical balls of radius $O_N(r)$. Let one of these balls be centred at ω . Since X and Y are separated and bounded, there are constants $0 < a < R < \infty$ such that $a \leq |y - x| \leq R$ on $X \times Y$. If $e_x(y)$ lies in the chosen spherical ball and $t = |y - x|$, then

$$|y - (x + t\omega)| \lesssim_N r.$$

Thus its inverse image in Y lies in an $O_N(r)$ -neighbourhood of the bounded ray segment $\{x + t\omega : a \leq t \leq R\}$, which is covered by $O_{N, X, Y}(r^{-1})$ Euclidean balls of radius $O_{N, X, Y}(r)$. Altogether, $(\psi \circ e_x)^{-1}(H^{(r)}) \cap Y$ is covered by $O_{N, X, Y}(r^{-(d-1)})$ such balls. Summing the κ -Frostman estimate over this cover proves (6.20b4) for all sufficiently small r . Enlarging the constant handles the remaining compact range of scales up to 1. $\square$

Lemma III.6.24.4 (Restriction of a positively curved norm). *Let N be a C^2 norm on $\mathbb{R}^d$ whose unit ball has everywhere positive Gaussian curvature, and let $V \subset \mathbb{R}^d$ be a linear hyperplane. If $d \geq 3$, then $N|_V$ is a C^2 norm on V whose unit ball has everywhere positive Gaussian curvature in the intrinsic dimension of V . If $d = 2$, the restriction to V is a constant multiple of the absolute value.*

Proof. Let $z \in V \setminus \{0\}$ with $N(z) = 1$. The tangent space to the restricted unit sphere is

$$V \cap \ker DN(z).$$

For $d \geq 3$, positive curvature of the ambient norm sphere says that the quadratic form $D^2N(z)$ is positive definite on the tangent space $\ker DN(z)$. Its restriction to $V \cap \ker DN(z)$ is therefore positive definite. This is exactly positivity of the second fundamental form of the unit sphere of $N|_V$. The one-dimensional claim for $d = 2$ follows from homogeneity and symmetry of a norm on a line. $\square$

Lemma III.6.24.5 (Curved directions from hyperplane-supported centres). *Let $P \subset \mathbb{R}^d$ be an affine hyperplane, let N be as in Lemma III.6.24.4, and put $\psi(e) = \text{dir}(DN(e))$. Let $X \subset \mathbb{R}^d \setminus P$ and $Y \subset P$ be bounded compact sets with $\text{dist}(X, P) > 0$, and let ν be a κ -Frostman probability measure on Y , where $\kappa > d - 2$. Then, uniformly for $x \in X$ and $H \in G(d, d - 1)$,*

$$(\psi \circ e_x)_\# \nu(H^{(r)}) \lesssim_{N, X, Y, \nu} r^{\kappa - (d-2)}, \quad 0 < r \leq 1. \quad (6.20b5)$$

The exponent is independent of the norm.

Proof. On the bounded support of ν , radial projection from any $x \in X$ maps P bi-Lipschitzly onto its spherical image, with constants uniform in x : positive distance from P prevents grazing directions and boundedness controls the radial parameter. Hence $(e_x)_\# \nu$ is κ -Frostman on the sphere.

Positive Gaussian curvature makes $\psi : S^{d-1} \rightarrow S^{d-1}$ a C^1 diffeomorphism; compactness makes ψ and ψ^{-1} Lipschitz. Thus $(\psi \circ e_x)_\# \nu$ remains κ -Frostman with a norm-dependent constant. A belt $H^{(r)} \cap S^{d-1}$ is covered by $O_d(r^{-(d-2)})$ spherical balls of radius $O_d(r)$. Summing the ball-Frostman estimate over this cover proves (6.20b5) for small r , and enlarging the constant covers the remaining scales. Notice that this transfers ball-Frostman control through a bi-Lipschitz map; it does not assert the false general invariance of hyperplane decay exhibited by the counterexample below. $\square$

Lemma III.6.24.6 (Translated norm-cone tangent geometry). *Let N be as above, let $n \in S^{d-1}$, and define*

$$C_n = \{z \neq 0 : n \cdot DN(z) = 0\}.$$

Then C_n is a C^1 hypersurface away from the origin. If $y = x + z$ with $z \in C_n$, the tangent affine hyperplane to $x + C_n$ at y contains the vertex x . Consequently, if two translated cones $x_1 + C_{n_1}$ and $x_2 + C_{n_2}$ are tangent at the same point, their common tangent hyperplane contains both x_1 and x_2 .

Proof. Set $f_n(z) = n \cdot DN(z)$. On C_n , the vector n is tangent to the norm sphere, so positive curvature gives

$$n^T D^2N(z)n > 0.$$

Hence $\nabla f_n(z) = D^2N(z)n \neq 0$, and the implicit-function theorem gives the first claim. Since N is one-homogeneous, differentiating Euler's identity yields $D^2N(z)z = 0$. If $g = D^2N(z)n$ is the normal to C_n at z , then

$$g \cdot z = n \cdot D^2N(z)z = 0.$$

Thus z lies in the tangent direction space of C_n . The tangent affine hyperplane to $x + C_n$ at $y = x + z$ is $y + g^\perp$, and $x = y - z$ belongs to it. At a tangency the two tangent hyperplanes coincide, so both vertices belong to the common one. $\square$

Lemma III.6.24.7 (Uniform null-hyperplane modulus). *Let σ be a compactly supported finite Borel measure on $\mathbb{R}^d$ and assume $\sigma(P) = 0$ for every affine hyperplane P . Then*

$$\omega_\sigma(r) := \sup_{P \in \mathcal{A}(d, d-1)} \sigma(P^{(r)}) \longrightarrow 0 \quad (r \downarrow 0). \quad (6.20b6)$$

The supremum may equivalently be restricted to hyperplanes meeting one fixed bounded neighbourhood of $\text{supp } \sigma$.

Proof. Only hyperplanes whose distance from the compact support is bounded need be considered. Modulo the sign of the unit normal, their normal–offset parameters form a compact set. If the conclusion failed, there would be $r_j \downarrow 0$ and affine hyperplanes P_j with $\sigma(P_j^{(r_j)}) \geq \varepsilon > 0$. Pass to a subsequence for which P_j converges to an affine hyperplane P . Then there are $s_j \downarrow 0$ such that $P_j^{(r_j)} \cap \text{supp } \sigma \subset P^{(s_j)}$. Hence $\sigma(P^{(s_j)}) \geq \varepsilon$. Continuity from above gives $\sigma(P) \geq \varepsilon$, a contradiction. $\square$

Lemma III.6.24.8 (Two-contact curved train-track dichotomy). *Let N be a C^2 norm whose unit ball has everywhere positive Gaussian curvature. Fix $0 < r_- < r_+ < \infty$. For $n \in S^{d-1}$ and $z \in C_n$ define the projective unit normal*

$$\mathbf{g}_n(z) = \frac{D^2 N(z)n}{|D^2 N(z)n|} \in S^{d-1}/\{\pm 1\}.$$

Let $\Sigma_0 = a + C_n$ and $\Sigma_1 = x + C_m$. Suppose that a finite nonempty family of paired points

$$p_j = a + z_j \in \Sigma_0, \quad q_j = x + w_j \in \Sigma_1$$

satisfies

$$r_- \leq |z_j|, |w_j| \leq r_+, \quad |p_j - q_j| \leq \varepsilon, \quad d_\pm(\mathbf{g}_n(z_j), \mathbf{g}_m(w_j)) \leq \vartheta, \quad (6.20b7)$$

where $d_\pm(u, v) = \min\{|u - v|, |u + v|\}$. Then for every $0 < \rho < 1/4$ one of the following alternatives holds, with constants depending only on d, r_+ :

(T) transverse-normal alternative. *There are i, j for which $d_\pm(\mathbf{g}_n(z_i), \mathbf{g}_n(z_j)) \geq \rho$, and x lies within*

$$C \frac{\varepsilon + \vartheta}{\rho} \quad (6.20b8)$$

of an affine $(d - 2)$ -plane through a .

(H) hyperplane-track alternative. *There is an affine hyperplane P through a such that every contact point p_j and the moving vertex x lie within*

$$C(\varepsilon + \vartheta + \rho) \quad (6.20b9)$$

of P .

Proof. Choose signs for the normals at each paired contact so that unit representatives g_j, h_j satisfy $|g_j - h_j| \leq \vartheta$. By Lemma III.6.24.6,

$$g_j \cdot z_j = 0, \quad h_j \cdot w_j = 0.$$

Since $p_j = a + z_j$ and $q_j = x + w_j$,

$$\begin{aligned} |g_j \cdot (x - a)| &\leq |g_j \cdot (x - q_j)| + |g_j \cdot (q_j - p_j)| + |g_j \cdot (p_j - a)| \\ &\leq |(g_j - h_j) \cdot w_j| + \varepsilon \\ &\leq r_+ \vartheta + \varepsilon. \end{aligned} \tag{6.20b10}$$

Thus x lies within $O(\varepsilon + \vartheta)$ of every tangent hyperplane $P_j = a + g_j^\perp$ represented by a contact.

If two projective normals are separated by at least ρ , choose their signs so that the acute angle between them is at least $c\rho$. The two hyperplanes P_i, P_j intersect in an affine $(d-2)$ -plane L through a . Elementary two-hyperplane linear algebra gives

$$\text{dist}(x, L) \lesssim_d \frac{\max\{\text{dist}(x, P_i), \text{dist}(x, P_j)\}}{\sin \angle(g_i, g_j)} \lesssim_{d, r_+} \frac{\varepsilon + \vartheta}{\rho},$$

which is (T).

Otherwise all projective normals g_j lie within ρ of g_1 . Choose their signs so that $|g_j - g_1| < \rho$ and put $P = a + g_1^\perp$. Since $g_j \cdot z_j = 0$,

$$\text{dist}(p_j, P) = |g_1 \cdot z_j| \leq |g_1 - g_j| |z_j| \leq r_+ \rho.$$

Equation (6.20b10) for $j = 1$ also gives $\text{dist}(x, P) \lesssim \varepsilon + \vartheta$. This is (H). No derivative of D^2N is used; the argument is therefore valid at the stated C^2 regularity. $\square$

Lemma III.6.24.9 (Homogeneous Legendre linearization). *Let N be a C^2 norm whose unit ball has everywhere positive Gaussian curvature, and put*

$$Q(z) = \frac{1}{2}N(z)^2, \quad \mathcal{G}(z) = DQ(z) = N(z)DN(z) \quad (z \neq 0), \quad \mathcal{G}(0) = 0.$$

Then $\mathcal{G} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is an odd, positively one-homogeneous global bi-Lipschitz homeomorphism and a C^1 diffeomorphism off the origin. Moreover, for every $n \in S^{d-1}$,

$$\mathcal{G}(C_n) = n^\perp \setminus \{0\}. \tag{6.20b11}$$

Consequently, for every vertex x , the translated map

$$\mathcal{G}_x(y) = x + \mathcal{G}(y - x)$$

sends the entire bad cone $x + C_n$ exactly to the affine hyperplane $x + n^\perp$.

Proof. For $z \neq 0$,

$$D^2Q(z) = DN(z) \otimes DN(z) + N(z)D^2N(z). \tag{6.20b12}$$

Write $v = az + w$, where $DN(z) \cdot w = 0$. Euler homogeneity gives $DN(z) \cdot z = N(z)$ and $D^2N(z)z = 0$, so

$$v^T D^2Q(z)v = a^2 N(z)^2 + N(z) w^T D^2N(z)w.$$

Positive Gaussian curvature makes the second term positive definite on $\ker DN(z)$. Compactness on the unit sphere, followed by the zero-homogeneity of D^2Q , therefore gives constants $0 < c_N \leq C_N < \infty$ such that

$$c_N |v|^2 \leq v^T D^2Q(z)v \leq C_N |v|^2 \quad (z \neq 0).$$

Integrating along line segments, with the possible crossing of the origin occurring at only one parameter value, gives

$$(\mathcal{G}(u) - \mathcal{G}(v)) \cdot (u - v) \geq c_N |u - v|^2, \quad |\mathcal{G}(u) - \mathcal{G}(v)| \leq C_N |u - v|.$$

Hence $\mathcal{G}$ is injective and bi-Lipschitz onto its image. For any $w \in \mathbb{R}^d$, the function $Q(z) - w \cdot z$ is coercive and strictly convex, so it has a unique minimizer; its Euler equation is $\mathcal{G}(z) = w$. Thus $\mathcal{G}$ is onto. Oddness and one-homogeneity follow from evenness and two-homogeneity of Q . Finally,

$$n \cdot \mathcal{G}(z) = N(z) n \cdot DN(z),$$

which proves (6.20b11) and the translated assertion. $\square$

Lemma III.6.24.10 (Same-vertex cone overlap and vertex freezing). *Fix bounded compact sets of base and centre points separated by a positive distance, and let ν be a κ -Frostman probability measure on the centre set with $\kappa > d - 2$. Write*

$$B_{x,n}(r) = \{y : \text{dist}(\text{dir } DN(y - x), n^\perp) \leq r\}.$$

If the projective angle between $n, m \in S^{d-1}$ is at least $\theta > 0$, then, for r sufficiently small,

$$\nu(B_{x,n}(r) \cap B_{x,m}(r)) \lesssim_N \left(\frac{r}{\theta}\right)^{\kappa-(d-2)}. \quad (6.20b13)$$

Also, uniformly in y ,

$$|\mathcal{G}_x(y) - \mathcal{G}_a(y)| \leq (1 + \text{Lip}(\mathcal{G}))|x - a|. \quad (6.20b14)$$

Thus, if $|x - a| \leq R$, the fixed map $\mathcal{G}_a$ sends every r -bad cone through x into an affine hyperplane plate of width $O_N(r + R)$.

Proof. On the separated compact product, $|\mathcal{G}(y - x)|$ is bounded above and below.

Lemma III.6.24.9 therefore gives the following geometric form of the condition defining $B_{x,n}(r)$: in the $\mathcal{G}_x$ coordinates, the point lies within $O_N(r)$ of the affine hyperplane $x + n^\perp$. Two such plates whose normals make angle at least θ are contained in the $O_N(r/\theta)$ -neighbourhood of the affine $(d - 2)$ -plane

$$(x + n^\perp) \cap (x + m^\perp).$$

Inside a fixed bounded region this neighbourhood is coverable by $O_N((r/\theta)^{-(d-2)})$ balls of radius $O_N(r/\theta)$. Pull the balls back by the bi-Lipschitz map $\mathcal{G}_x^{-1}$ and apply the Frostman estimate to obtain (6.20b13). Finally,

$$\begin{aligned} |\mathcal{G}_x(y) - \mathcal{G}_a(y)| &\leq |x - a| + |\mathcal{G}(y - x) - \mathcal{G}(y - a)| \\ &\leq (1 + \text{Lip}(\mathcal{G}))|x - a|, \end{aligned}$$

which is (6.20b14). Combining it with the plate dictionary proves the final claim. $\square$

Lemma III.6.24.11 (Transport stability of the Legendre coordinates). *Let $\sigma_x = (\mathcal{G}_x)_\# \nu$ and put $L_N = 1 + \text{Lip}(\mathcal{G})$. The coupling*

$$\pi_{x,a} = (\mathcal{G}_x, \mathcal{G}_a)_\# \nu$$

of σ_x, σ_a is supported on

$$\{(u, v) : |u - v| \leq L_N |x - a|\}. \quad (6.20b15)$$

In particular, for every Borel set $S \subset \mathbb{R}^d$,

$$\sigma_x(S) \leq \sigma_a(S^{(L_N|x-a|)}), \quad \sigma_a(S) \leq \sigma_x(S^{(L_N|x-a|)}). \quad (6.20b16)$$

Thus freezing the vertex produces only a same-scale width error; it does not create a new cross-scale compatibility condition.

Proof. Couple σ_x and σ_a by sending the same y to $(\mathcal{G}_x(y), \mathcal{G}_a(y))$. Equation (6.20b14) bounds the displacement of this coupling by $L_N|x - a|$, which is (6.20b15). The two set inequalities follow immediately from the same coupling. $\square$

Lemma III.6.24.12 (Same-base superexponential telescope). *Let $X, Y \subset \mathbb{R}^d$ be disjoint compact sets and let μ, ν be probability measures supported on X, Y . For a linear hyperplane H through the origin write*

$$\mathcal{B}_x(H, r) = \{y \in Y : \text{dist}(\mathcal{G}_x(y), x + H) \leq r\}.$$

Fix $\tau = 2^{1-N}$ for an integer $N \geq 1$. Suppose there are constants $A, C, a > 0$ such that, for every sufficiently small δ , there is a compact $E_\delta \subset X$ with $\mu(X \setminus E_\delta) \leq C\delta^a$ and, for each $x \in E_\delta$, a hyperplane $H_\delta(x)$ for which

$$P_\delta(x) = \mathcal{B}_x(H_\delta(x), A\delta^{\tau/2})$$

has the following one-scale pruning property: whenever a belt $W = \mathcal{B}_x(H, A\delta)$ meets $Y \setminus P_\delta(x)$,

$$\nu(W) \leq C\delta^\tau. \quad (6.20b17)$$

Assume also that there is a compact $X_ \subset X$ with $\mu(X_*) > 0$ such that*

$$\nu(\mathcal{B}_x(H, 0)) = 0 \quad (x \in X_*, H \text{ a linear hyperplane}). \quad (6.20b18)$$

Then, after restricting μ to a positive-mass compact subset of X_ , there are a compact set E of positive μ -mass and compact sets $K(x) \subset Y$ with $\nu(K(x)) \geq 1/2$ such that*

$$\nu(K(x) \cap \mathcal{B}_x(H, r)) \leq r^{\tau/2} \quad (6.20b19)$$

uniformly for $x \in E$, all linear hyperplanes H , and all sufficiently small r . After normalization, the same positive-power hyperplane-strip decay holds for $(\text{dir } DN(\cdot - x))_\#(\nu|_{K(x)})$.

Proof. The first point is that (6.20b18) gives a *uniform* small-width modulus on X_* . Indeed, if this failed, there would be $r_j \downarrow 0$, $x_j \in X_*$ and hyperplanes H_j with $\nu(\mathcal{B}_{x_j}(H_j, r_j)) \geq \varepsilon > 0$. Passing to a subsequence gives $x_j \rightarrow x \in X_*$ and $H_j \rightarrow H$. Uniform continuity of $(x, y) \mapsto \mathcal{G}_x(y)$ on $X_* \times Y$ implies that, for every $s > 0$ and all large j , $\mathcal{B}_{x_j}(H_j, r_j) \subset \mathcal{B}_x(H, s)$. Hence $\nu(\mathcal{B}_x(H, s)) \geq \varepsilon$ for every $s > 0$, and continuity from above contradicts (6.20b18). Therefore

$$\sup_{x \in X_*} \sup_H \nu(\mathcal{B}_x(H, r)) \longrightarrow 0 \quad (r \downarrow 0). \quad (6.20b20)$$

Choose $\varepsilon > 0$ smaller than $1/4$. Take $\bar{\delta} > 0$ so that the last supremum is at most $\varepsilon/(2N)$ whenever $r \leq A\bar{\delta}$. Now choose δ_0 so small that the one-scale hypothesis applies below δ_0 , $\delta_0^{\tau/2} \leq \bar{\delta}$, and, for $\delta_n = \delta_0^{2^n}$,

$$C \sum_{n \geq 0} \delta_n^a < \frac{1}{2} \mu(X_*), \quad C \sum_{n \geq 0} \delta_n^\tau < \frac{1}{4}.$$

Put

$$E = X_* \cap \bigcap_{n \geq 0} E_{\delta_n}.$$

After inner regularity we may replace E by a compact positive-mass subset. For $x \in E$ abbreviate $P_n(x) = P_{\delta_n}(x)$ and set

$$\tilde{K}(x) = Y \setminus \bigcup_{n \geq 0} P_n(x).$$

For $0 \leq n < N$, equation (6.20b20) gives $\nu(P_n(x)) \leq \varepsilon/(2N)$. If $n \geq N$, then

$$A\delta_n^{\tau/2} = A\delta_{n-N}.$$

If $P_n(x) \setminus P_{n-N}(x)$ is empty there is nothing to prove. Otherwise $P_n(x)$ is itself an $A\delta_{n-N}$ -belt through the same base point x and it meets $Y \setminus P_{n-N}(x)$. Applying (6.20b17) at the older scale therefore yields

$$\nu(P_n(x) \setminus P_{n-N}(x)) \leq C\delta_{n-N}^\tau. \quad (6.20b21)$$

Iterating by residue classes modulo N gives $\nu(\tilde{K}(x)) > 1/2$ after decreasing ε and δ_0 once if necessary. By inner regularity choose a compact $K(x) \subset \tilde{K}(x)$ with $\nu(K(x)) \geq 1/2$. Notice that no relation between $H_n(x)$ and $H_{n-N}(x)$ was used.

Finally let r be small and choose n with $\delta_{n+1} < r/A \leq \delta_n$ after harmlessly adjusting the fixed constants. Enlarge $\mathcal{B}_x(H, r)$ to an $A\delta_n$ -belt. If it meets $K(x)$, it meets $Y \setminus P_n(x)$, so (6.20b17) gives mass at most $C\delta_n^\tau$. Since $\delta_{n+1} = \delta_n^2$, this is $O(r^{\tau/2})$, proving (6.20b19). On the separated compact product $X \times Y$, the magnitude $|\mathcal{G}_x(y - x)| = N(y - x)|DN(y - x)|$ is bounded above and below. Hence distance of $\mathcal{G}_x(y) - x$ from a linear hyperplane is comparable, up to fixed norm-dependent constants, with distance of $\text{dir } DN(y - x)$ from the corresponding great hyperplane. This transfers the same positive power to the normalized direction push-forward. $\square$

Lemma III.6.24.13 (Atomic remainder for exact Legendre belts). *Let $X, Y \subset \mathbb{R}^d$ be disjoint compact sets and let ν be a κ -Frostman probability measure on Y , where $\kappa > d - 2$. For $x \in X$ write*

$$Z_x(H) = \mathcal{B}_x(H, 0), \quad \mathcal{H}_x = \{H : \nu(Z_x(H)) > 0\}, \quad A_x = \bigcup_{H \in \mathcal{H}_x} Z_x(H).$$

Then $\mathcal{H}_x$ is at most countable and, for distinct $H, H' \in \mathcal{H}_x$,

$$\nu(Z_x(H) \cap Z_x(H')) = 0. \quad (6.20b21a)$$

Consequently, with

$$\Lambda_x^\circ = Y \setminus A_x, \quad m_x = \nu(\Lambda_x^\circ), \quad (6.20b21b)$$

the restriction $\nu|_{\Lambda_x^\circ}$ gives zero mass to every exact belt $Z_x(H)$. If $m_x > 0$, the normalized restriction $\nu_x^\circ = m_x^{-1}\nu|_{\Lambda_x^\circ}$ is again κ -Frostman, with an x -dependent Frostman constant. Moreover

$$\{(x, y) \in X \times Y : y \in A_x\} \quad (6.20b21c)$$

is Borel.

Assume in addition that the one-scale pruning hypothesis of Lemma III.6.24.12 holds with summable base-point loss along its superexponential scale sequence. Then for almost every base point x which is eventually one-scale good and satisfies $m_x > 0$, there is a compact $K_x \subset \Lambda_x^\circ$ of positive ν -mass such that the normalized restriction to K_x has positive-power hyperplane-strip decay in the direction coordinates. Thus, under the hypothetical global one-scale pruning statement (later shown false in this generality), the exact-atom obstruction reduces to the pure configuration

$$m_x = 0, \quad \text{equivalently} \quad \nu(A_x) = 1. \quad (6.20b21d)$$

Proof. If $H \neq H'$, Lemma III.6.24.9 gives

$$\mathcal{G}_x(Z_x(H) \cap Z_x(H')) \subset x + (H \cap H').$$

The right-hand side is an affine $(d-2)$ -plane. Since $\mathcal{G}_x$ is bi-Lipschitz, the preimage of an r -neighbourhood of this plane is coverable, inside the fixed compact set Y , by $O(r^{-(d-2)})$ balls of radius $O(r)$. The Frostman estimate therefore gives mass $O(r^{\kappa-(d-2)})$, which tends to zero. This proves (6.20b21a). A finite measure contains at most countably many pairwise disjoint modulo null sets of positive mass, so $\mathcal{H}_x$ is countable. The definition of A_x then immediately implies that $\nu|_{\Lambda_x^\circ}(Z_x(H)) = 0$ for every H , and normalization merely multiplies the Frostman constant by m_x^{-1} .

For measurability, let linear hyperplanes range over the compact space $G(d, d-1)$, and put

$$f(x, H, y) = \text{dist}(\mathcal{G}_x(y), x + H), \quad M(x, H) = \nu\{y : f(x, H, y) = 0\}.$$

The function f is continuous on the compact product. If $(x_j, H_j) \rightarrow (x, H)$, then, for every $s > 0$ and all large j ,

$$\{f(x_j, H_j, \cdot) = 0\} \subset \{f(x, H, \cdot) \leq s\}.$$

Taking a limsup in measure and then $s \downarrow 0$ shows that M is upper semicontinuous. Hence, for every integer $r \geq 1$, the set

$$\{(x, H, y) : f(x, H, y) = 0, M(x, H) \geq 1/r\}$$

is compact. Its projection to $X \times Y$ is compact, and the union of these projections over r is exactly (6.20b21c); the graph is therefore Borel.

Now suppose $m_x > 0$ and x belongs to all sufficiently late one-scale good sets. Every one-scale estimate for v passes to v_x° with its constant multiplied by m_x^{-1} . At this fixed x , compactness of $G(d, d-1)$ and continuity from above give

$$\sup_H v_x^\circ(\mathcal{B}_x(H, r)) \longrightarrow 0 \quad (r \downarrow 0),$$

because every exact belt has zero v_x° -mass. The proof of Lemma III.6.24.12 can therefore be started at an x -dependent late index and yields the asserted positive-power decay. Summability of the base losses gives eventual one-scale goodness almost everywhere by Borel–Cantelli. Theorem III.6.24 explicitly allows the positive parameter subset and its constants to depend on x , followed by countable uniformization. Hence any positive-mass base set with $m_x > 0$ is already admissible once the global one-scale pruning input is available, leaving only (6.20b21d). $\square$

Lemma III.6.24.14 (Global exclusion of positive exact Legendre-belt atoms). *Let $X, Y \subset \mathbb{R}^d$ be disjoint compact sets. Let μ be an α -Frostman probability measure on X and ν a κ -Frostman probability measure on Y , where*

$$\alpha > d - 2, \quad \kappa > d - 2.$$

Assume that $\nu(P) = 0$ for every affine hyperplane P . Then

$$\mu\{x \in X : \exists H \in G(d, d-1) \text{ with } \nu(\mathcal{B}_x(H, 0)) > 0\} = 0. \quad (6.20b21h)$$

Thus in the zero-hyperplane centre branch, after discarding a μ -null set of vertices, every exact Legendre belt through the vertex has zero ν -mass.

Proof. Suppose the exceptional set has positive μ -measure. By countable uniformization there are $c > 0$ and a compact set $E \subset X$ of positive μ -measure such that for every $x \in E$ some hyperplane H satisfies $\nu(\mathcal{B}_x(H, 0)) \geq c$. As in Lemma III.6.24.13,

$$M(x, H) = \nu(\mathcal{B}_x(H, 0))$$

is upper semicontinuous on the compact product $E \times G(d, d-1)$. Hence the nonempty compact-valued correspondence $\{H : M(x, H) \geq c\}$ has a Borel selector; write the selected hyperplane as $H_x = n_x^\perp$ and put $S_x = \mathcal{B}_x(H_x, 0)$. A Borel choice of the projective normal $[n_x]$ is sufficient below.

The measure algebra of the finite Borel measure ν is separable for the metric $d_\nu(A, B) = \nu(A \Delta B)$. Choose a countable dense family $\{D_j\}$. Since $\nu(S_x) \geq c$, the Borel sets

$$E_j = \{x \in E : d_\nu(S_x, D_j) < c/8\}$$

cover E . For one j the set E_j has positive μ -measure. Normalize μ on a compact positive-mass subset of E_j and call the result μ_0 . Then for all x, a in this subset,

$$\nu(S_x \cap S_a) \geq c - \nu(S_x \triangle S_a) > 3c/4 =: c_0. \quad (6.20b21i)$$

Put $Q = N^2/2$ and $A(z) = D^2Q(z) = D\mathcal{G}(z)$. If $y \in S_x \cap S_a$, projective normals to the two exact hypersurfaces at y are

$$[A(y-x)n_x], \quad [A(y-a)n_a].$$

Whenever these normals are distinct, transversality makes their local intersection a C^1 submanifold of codimension two. Since $\kappa > d-2$, this intersection has ν -measure zero by a ball-covering argument. Taking the countable union over quantitative angle cutoffs shows that for every fixed pair (x, a) , almost every $y \in S_x \cap S_a$ is a tangency point. The corresponding triple-incidence set is Borel, so Tonelli upgrades this to a statement for $\mu_0 \times \mu_0 \times \nu$ -almost every incident triple.

Let

$$s(y) = \mu_0\{x : y \in S_x\}.$$

By (6.20b21i),

$$\int s(y)^2 d\nu(y) = \iint \nu(S_x \cap S_a) d\mu_0(x) d\mu_0(a) \geq c_0.$$

Thus $B = \{y : s(y) > 0\}$ has positive ν -measure. For almost every $y \in B$, normalize μ_0 on the incident set $\{x : y \in S_x\}$. Two independent incident vertices have the same projective tangent normal almost surely. A probability law whose two independent samples agree almost surely is a point mass, so there is one projective normal $[g_y]$ shared by almost every incident vertex.

Lemma III.6.24.6 then implies that the affine hyperplane

$$P_y = y + g_y^\perp$$

contains a positive- μ set of incident vertices. Distinct affine hyperplanes of positive μ -mass intersect in an affine set of dimension at most $d-2$, which has zero μ -mass because $\alpha > d-2$. Therefore there are at most countably many affine hyperplanes of positive μ -mass. Since $y \in P_y$, almost all of B is contained in their countable union. This contradicts the hypothesis $\nu(P) = 0$ for every affine hyperplane. Hence (6.20b21h) holds. $\square$

Lemma III.6.24.15 (Pointwise reduction to two heavy Legendre directions). *Assume $d \geq 3$, $d-2 < \kappa \leq d-1$, and put $q = \kappa - (d-2) > 0$. Fix $\tau > 0$ so small that*

$$q(1 - \tau/2) > 2\tau. \quad (6.20b21j)$$

For a fixed sufficiently large constant A , call $\mathcal{B}_x(H, A\delta)$ heavy when its ν -mass exceeds $C\delta^\tau$. Up to changing the harmless fixed constants, for every vertex x and every sufficiently small δ one of the following alternatives is available:

1. *all heavy $A\delta$ -belts through x are contained in one belt $\mathcal{B}_x(H_0, A'\delta^{\tau/2})$;*
2. *there are two heavy belts with hyperplane directions H_1, H_2 satisfying*

$$\angle(H_1, H_2) \geq c\delta^{\tau/2}. \quad (6.20b21k)$$

Moreover, in the second alternative,

$$v(\mathcal{B}_x(H_1, A\delta) \cap \mathcal{B}_x(H_2, A\delta)) \lesssim \delta^{q(1-\tau/2)} = o(\delta^{2\tau}). \quad (6.20b21l)$$

Consequently, if T_δ denotes the set of vertices satisfying the second alternative, then any estimate

$$\mu(T_\delta) \lesssim \delta^a \quad (a > 0) \quad (6.20b21m)$$

provides the one-scale pruning input required by Lemma III.6.24.12.

Proof. If no heavy belt exists there is nothing to prove. Otherwise fix one heavy direction H_0 . If every other heavy direction makes projective angle at most $c\delta^{\tau/2}$ with H_0 , then, in the bounded exact affine coordinates supplied by Lemma III.6.24.9, every width- $A\delta$ plate is contained in a width- $A'\delta^{\tau/2}$ plate about H_0 . This is the first alternative. If this fails, two heavy directions satisfy (6.20b21k). The same-vertex intersection estimate (6.20b13), with $\theta \asymp \delta^{\tau/2}$ and $r \asymp \delta$, gives (6.20b21l); the exponent choice (6.20b21j) makes this $o(\delta^{2\tau})$.

Outside T_δ , the first alternative supplies exactly one exceptional belt of width $O(\delta^{\tau/2})$, and every $A\delta$ -belt meeting its complement is light. Thus a power bound (6.20b21m) would be sufficient to supply the summable base-point loss in the one-scale hypothesis of Lemma III.6.24.12. Lemma III.6.24.17 below shows, however, that this sufficient estimate is false for general admissible norms; the present lemma is therefore a diagnostic dichotomy rather than a closure of the residual branch. $\square$

Lemma III.6.24.16 (Separated Legendre directions need not give transverse physical tracks). *There is a C^∞ norm on $\mathbb{R}^3$ whose unit sphere has everywhere positive Gaussian curvature, two distinct exact Legendre hyperplane directions, and one contact point on each corresponding bad cone such that the two physical tangent hyperplanes coincide. Hence the implication*

$$\text{separated Legendre directions} \implies \text{transverse physical train tracks}$$

is false even inside the admissible norm class.

Proof. For $z = (u, v, w) \neq 0$ set

$$\Phi(z) = \frac{uw(u^2 - v^2)}{u^2 + v^2 + w^2}, \quad Q_\varepsilon(z) = \frac{1}{2}|z|^2 + \varepsilon\Phi(z).$$

The function Φ is smooth off the origin, even, and homogeneous of degree two. For sufficiently small nonzero ε , the Hessian $A_\varepsilon = D^2Q_\varepsilon$ is uniformly positive definite on the sphere. Hence Q_ε is an even strictly convex two-homogeneous function and $N_\varepsilon = (2Q_\varepsilon)^{1/2}$ is a norm. The second fundamental form of the level set $Q_\varepsilon = 1/2$ is the restriction of A_ε , divided by the nonzero gradient length, so the unit sphere has everywhere positive Gaussian curvature.

Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$ and $g = e_3$. Direct differentiation gives

$$A_\varepsilon(e_1) = \begin{pmatrix} 1 & 0 & \varepsilon \\ 0 & 1 & 0 \\ \varepsilon & 0 & 1 \end{pmatrix}, \quad A_\varepsilon(e_2) = \begin{pmatrix} 1 & 0 & -\varepsilon \\ 0 & 1 & 0 \\ -\varepsilon & 0 & 1 \end{pmatrix}.$$

Put $n_i = A_\varepsilon(e_i)^{-1}g$. Projectively,

$$[n_1] = [(-\varepsilon, 0, 1)], \quad [n_2] = [(\varepsilon, 0, 1)],$$

so the two Legendre hyperplane directions are distinct. Euler homogeneity gives $A_\varepsilon(z)z = DQ_\varepsilon(z)$. Therefore

$$n_i \cdot DQ_\varepsilon(e_i) = g \cdot e_i = 0,$$

so e_i lies on the exact bad cone with Legendre normal n_i . The physical normal to that cone at e_i is $A_\varepsilon(e_i)n_i = g$. Since also $g \cdot e_i = 0$, both tangent hyperplanes are exactly $g^\perp$. This proves the claim. $\square$

Lemma III.6.24.17 (Persistent common tracks obstruct one-exceptional-belt pruning). *Fix $1 < \alpha, \kappa \leq 2$, $\tau > 0$, and positive constants A, C . There are disjoint compact sets $X, Y \subset \mathbb{R}^3$, an α -Frostman probability μ on X , a κ -Frostman probability ν on Y , and a C^∞ norm N off the origin whose unit sphere has everywhere positive Gaussian curvature, with the following properties:*

1. $\nu(P) = 0$ for every affine plane $P \subset \mathbb{R}^3$;
2. *there are two fixed distinct linear hyperplanes H_1, H_2 and constants $c_0, c_1 > 0$ such that, for every sufficiently small δ and every $x \in X$,*

$$\nu(\mathcal{B}_x(H_i, A\delta)) \geq \frac{c_0}{\log(1/\delta)} > C\delta^\tau, \quad i = 1, 2, \quad (6.20b21n)$$

while $\angle(H_1, H_2) \geq c_1$;

3. *for every fixed $A_0 > 0$, every sufficiently small δ , every $x \in X$, and every linear hyperplane H_0 , at least one of the two heavy belts in (6.20b21n) meets*

$$Y \setminus \mathcal{B}_x(H_0, A_0\delta^{\tau/2}). \quad (6.20b21o)$$

Consequently $\mu(T_\delta) = 1$ for all sufficiently small δ , so the sufficient power estimate (6.20b21m) is false in this generality. More strongly, the one-exceptional-belt pruning hypothesis (6.20b17) is not a universal consequence of the low-centre zero-hyperplane hypotheses.

Proof. Write $z = (u, v, w)$ and $P = \{w = 0\}$. Choose disjoint antipodally symmetric conic neighbourhoods of $\{\pm e_1\}$ and $\{\pm e_2\}$, together with smooth even degree-zero cutoffs χ_1, χ_2 having disjoint supports and equal to one on smaller conic plateaux U_1, U_2 . Put

$$Q_\varepsilon(z) = \frac{1}{2}|z|^2 + \varepsilon\chi_1(z/|z|)uw + \varepsilon\chi_2(z/|z|)vw. \quad (6.20b21p)$$

The perturbation is smooth off the origin, even, and two-homogeneous. Its Hessian is homogeneous of degree zero and bounded on the sphere. Hence, for sufficiently small nonzero ε , D^2Q_ε is uniformly positive definite and $Q_\varepsilon > 0$ off the origin. Thus the symmetric sublevel set $\{Q_\varepsilon \leq 1/2\}$ is convex and $N_\varepsilon = (2Q_\varepsilon)^{1/2}$ is its Minkowski functional, hence a norm smooth off the origin. The second fundamental form of the level set $Q_\varepsilon = 1/2$ is the restriction of D^2Q_ε divided by $|DQ_\varepsilon|$, so its Gaussian curvature is positive everywhere.

On U_1 and U_2 , respectively, the Legendre map $\mathcal{G}_\varepsilon = DQ_\varepsilon$ is exactly

$$\mathcal{G}_1(u, v, w) = (u + \varepsilon w, v, w + \varepsilon u), \quad \mathcal{G}_2(u, v, w) = (u, v + \varepsilon w, w + \varepsilon v).$$

For

$$n_1 = (-\varepsilon, 0, 1), \quad n_2 = (0, -\varepsilon, 1), \quad H_i = n_i^\perp,$$

one has

$$n_i \cdot \mathcal{G}_i(u, v, w) = (1 - \varepsilon^2)w, \quad D^2Q_i n_i = (1 - \varepsilon^2)e_3. \quad (6.20b21q)$$

The hyperplanes H_1, H_2 are distinct, but on their two conic plateaux the corresponding bad cones contain open sectors of the same physical plane P and have the same physical tangent normal e_3 .

Choose a small compact α -Frostman set $X \subset P$ near the origin and an α -Frostman probability μ on it. Choose compact planar sets $K_1, K_2 \subset P$, each not contained in a line, centred near Le_1 and Le_2 for L large, and put κ -Frostman probabilities η_i on them. We may choose the supports small and L large so that, for every $x \in X$, $p \in K_i$ and $0 \leq t \leq h/e$, the vector $(p, t) - x$ lies in U_i .

Let λ be the atomless probability on $[0, h/e]$ given by

$$d\lambda(t) = \frac{\mathbf{1}_{(0, h/e]}(t)}{t(\log(h/t))^2} dt, \quad \lambda([0, r]) = \frac{1}{\log(h/r)} \quad (0 < r \leq h/e), \quad (6.20b21r)$$

and set

$$\nu = \frac{1}{2}(\eta_1 \times \lambda) + \frac{1}{2}(\eta_2 \times \lambda).$$

Projection of a three-dimensional r -ball to P lies in a planar r -ball, so $\nu(B(z, r)) \lesssim r^\kappa$. Thus ν is κ -Frostman. If an affine plane has a nonzero w -coefficient, each fibre over P is a singleton and Fubini, together with atomlessness of λ , gives zero mass. If its w -coefficient is zero, its section in P is a line; a planar κ -Frostman measure with $\kappa > 1$ gives every line zero mass. Hence ν charges no affine plane.

For $x \in X$ and (p, t) in the i th component, (6.20b21q) gives

$$\text{dist}(\mathcal{G}_x(p, t), x + H_i) = \frac{1 - \varepsilon^2}{|n_i|} t.$$

Therefore, uniformly in x and i ,

$$\nu(\mathcal{B}_x(H_i, A\delta)) \gtrsim \lambda([0, c\delta]) \asymp \frac{1}{\log(1/\delta)}.$$

This is larger than $C\delta^\tau$ at every sufficiently small scale, while the fixed positive angle between H_1 and H_2 is larger than $c\delta^{\tau/2}$. Hence every $x \in X$ lies in T_δ for all sufficiently small δ , proving $\mu(T_\delta) = 1$. Notice also that the selected exact belts have zero mass because their local equation is $t = 0$ and $\lambda(\{0\}) = 0$.

It remains to prove the stronger pruning failure. For fixed x set

$$F_i(x) = \{\mathcal{G}_\varepsilon((p, 0) - x) : p \in K_i\} \subset H_i.$$

The plateau map is linear and invertible, while K_i is not contained in a line, so $F_i(x)$ affinely spans H_i . Since $H_1 \neq H_2$, no linear hyperplane contains both $F_1(x)$ and $F_2(x)$. Hence

$$\rho(x, H) = \max_{i=1,2} \sup_{z \in F_i(x)} \text{dist}(z, H)$$

is a continuous strictly positive function on the compact product of X with the hyperplane Grassmannian. Its minimum is some $\rho_0 > 0$. Because $0 \in \text{supp } \lambda$, every $(p, 0)$ with $p \in K_i$ belongs to Y . If $A_0 \delta^{\tau/2} < \rho_0$, then for every H_0 one of the two exact footprints has a support point outside the proposed exceptional belt $\mathcal{B}_x(H_0, A_0 \delta^{\tau/2})$. That support point still lies in the corresponding heavy $A\delta$ -belt, so the heavy belt meets the complement of the exceptional belt. Its mass is bounded below by (6.20b21n), contradicting (6.20b17). This proves (6.20b21o) and the lemma. $\square$

Lemma III.6.24.18 (Scale mismatch for naive cellwise Legendre freezing). *Assume $d \geq 3$ and $d - 2 < \kappa \leq d - 1$, and put $q = \kappa - (d - 2) \in (0, 1]$. Freeze the Legendre coordinate at a point a and restrict base points to $|x - a| \leq R$. If an original bad belt has width δ , then Lemma III.6.24.11 places its image under $\mathcal{G}_a$ in an affine hyperplane plate of width $r \asymp_N \delta + R$. The fixed transported measure $\sigma_a = (\mathcal{G}_a)_\# \nu$ satisfies*

$$\sigma_a(V^{(s)}) \lesssim_N s^q \quad (V \in \mathcal{A}(d, d - 2), 0 < s \leq 1). \quad (6.20b21e)$$

If one now copies the Euclidean Appendix-B Step 1 estimate independently in this cell, using the required heaviness threshold δ^τ , the argument only certifies containment in representative hyperplane plates at the working width

$$W_1 = C_N(r + r^q \delta^{-2\tau}), \quad W_2 = C_N(r + r^q \delta^{-3\tau}) \quad (6.20b21f)$$

after the Step 2 transverse-angle cutoff δ^τ . Since $q \leq 1$ and $0 < r \leq 1$, one has $r^q \geq r \gtrsim_N R$, and hence

$$\frac{W_1}{R} \gtrsim_N \delta^{-2\tau}, \quad \frac{W_2}{R} \gtrsim_N \delta^{-3\tau}. \quad (6.20b21g)$$

Thus these frozen Euclidean estimates do not furnish a transverse localization smaller than the base cell. In particular, the strategy “freeze each cell and then sum the Euclidean Step 1/Step 2 bad-base estimates independently” cannot obtain the required global summable base loss from these estimates alone in the residual range.

Proof. Bi-Lipschitzness of $\mathcal{G}_a$ preserves the κ -Frostman ball estimate up to fixed constants. Covering the s -neighbourhood of an affine $(d - 2)$ -plane in a fixed bounded region by $O(s^{-(d-2)})$ balls of radius $O(s)$ gives (6.20b21e).

For affine plates of width r , the greedy proof in Appendix-B Step 1 is unchanged except that the codimension-two bound contributes r^q while the overlap threshold is still $\delta^{2\tau}$. Hence the resulting angular clustering error is $O(r^q \delta^{-2\tau})$, and on a bounded region this gives the first working width in (6.20b21f). Dividing by the transverse cutoff δ^τ in Step 2 gives the second. Finally $q \leq 1$ implies $r^q \geq r$ for $r \leq 1$, while transport stability requires $r \gtrsim_N R$. This proves (6.20b21g).

For the natural choice $R = O(\delta)$ and $0 < \tau < q/3$, the two nontrivial terms are $\delta^{q-2\tau}$ and $\delta^{q-3\tau}$, both much larger than δ as $\delta \downarrow 0$. The conclusion is deliberately about the information supplied by the direct frozen-cell adaptation: it does not rule out a joint varying-vertex incidence argument, which is exactly what remains to be found. $\square$

Lemma III.6.24.19 (Qualitative hyperplane nullity has no power rate). *Let $d \geq 3$ and $d - 2 < \kappa \leq d - 1$. There is a compactly supported κ -Frostman probability measure σ on $\mathbb{R}^d$ such that $\sigma(P) = 0$ for every affine hyperplane P , but for $P_0 = \mathbb{R}^{d-1} \times \{0\}$,*

$$\sigma(P_0^{(r)}) \geq \frac{1}{\log(1/r)} \quad (0 < r \leq e^{-1}). \quad (6.20b22)$$

In particular the hypotheses of Lemma III.6.24.7 do not imply $\sigma(P^{(r)}) \lesssim r^\eta$ for any $\eta > 0$.

Proof. Choose a compactly supported κ -Frostman probability measure μ_0 on $\mathbb{R}^{d-1}$; for $\kappa = d - 1$ one may take normalized Lebesgue measure on a cube, while for $\kappa < d - 1$ this follows from Frostman compact selection in that cube. Since $\kappa > d - 2$, a covering by $O(r^{-(d-2)})$ radius- r balls shows that μ_0 gives zero mass to every affine $(d - 2)$ -plane.

Let λ be the probability measure on $[0, e^{-1}]$ with density

$$\frac{\mathbf{1}_{(0, e^{-1}]}(t)}{t(\log(1/t))^2} dt.$$

It is non-atomic and

$$\lambda([0, r]) = \frac{1}{\log(1/r)} \quad (0 < r \leq e^{-1}).$$

Put $\sigma = \mu_0 \times \lambda$. Projection of any radius- r ball to the first factor gives

$$\sigma(B((u, t), r)) \leq \mu_0(B(u, r)) \lesssim r^\kappa,$$

so σ is κ -Frostman. If an affine hyperplane has a nonzero coefficient in the last coordinate, its fibre over each u is a singleton and Fubini together with atomlessness of λ gives zero mass. If the last-coordinate coefficient is zero, the first-factor section is an affine $(d - 2)$ -plane, which μ_0 does not charge. Thus every affine hyperplane has σ -mass zero. The slab estimate (6.20b22) follows from the displayed formula for $\lambda([0, r])$. Since $1/\log(1/r)$ dominates every positive power of r as $r \downarrow 0$, no uniform positive power can follow from qualitative hyperplane nullity plus a κ -Frostman ball estimate alone. $\square$

Lemma III.6.24.20 (Variable-multiplicity same-base telescope). *Let $X, Y \subset \mathbb{R}^d$ be disjoint compact sets, let μ, ν be probability measures supported on X, Y , and fix $\tau = 2^{1-N}$ with $N \geq 1$. Put $\delta_n = \delta_0^{2^n}$. Suppose that $X_* \subset X$ is compact with $\mu(X_*) > 0$ and that, for every $n \geq 0$, there is a compact set $E_n \subset X_*$ such that*

$$\sum_{n \geq 0} \mu(X_* \setminus E_n) < \frac{1}{2} \mu(X_*). \quad (6.20b23)$$

For $x \in E_n$, suppose there are hyperplanes $H_{n,j}(x)$, $1 \leq j \leq m_n(x) \leq M_n < \infty$, and put

$$P_n(x) = \bigcup_{j=1}^{m_n(x)} \mathcal{B}_x(H_{n,j}(x), A\delta_n^{\tau/2}).$$

Assume that every $A\delta_n$ -belt through x which meets $Y \setminus P_n(x)$ has mass at most $C\delta_n^\tau$. If

$$\sum_{n \geq N} M_n \delta_{n-N}^\tau < \infty \quad (6.20b24)$$

and, with $E = X_* \cap \bigcap_{n \geq 0} E_n$,

$$\liminf_{m \rightarrow \infty} \sum_{r=0}^{N-1} \sup_{x \in E} v(P_{m+r}(x)) = 0, \quad (6.20b25)$$

then, after replacing E by a positive-mass compact subset, there are compact $K(x) \subset Y$ with $v(K(x)) \geq 1/2$ such that

$$v(K(x) \cap \mathcal{B}_x(H, r)) \lesssim r^{\tau/2} \quad (6.20b26)$$

uniformly for $x \in E$, all linear hyperplanes H , and all sufficiently small r .

In particular, if every exact belt has zero mass uniformly on the compact base in the sense of (6.20b20) and $\sup_n M_n < \infty$, then (6.20b25) is automatic. Hence every uniformly bounded exceptional-belt multiplicity is harmless at the telescopic level.

Proof. The base set E has positive μ -mass by (6.20b23). The scale identity used in the same-base telescope lemma remains

$$\delta_n^{\tau/2} = \delta_{n-N}.$$

Thus each component of $P_n(x)$ is itself an $A\delta_{n-N}$ -belt. If such a component meets $Y \setminus P_{n-N}(x)$, the older-scale pruning hypothesis bounds its entire mass by $C\delta_{n-N}^{\tau}$; otherwise it contributes nothing to $P_n(x) \setminus P_{n-N}(x)$. Summing over the at most M_n components gives

$$v(P_n(x) \setminus P_{n-N}(x)) \leq CM_n \delta_{n-N}^{\tau}. \quad (6.20b27)$$

Choose m so large that the seed quantity in (6.20b25) and the tail of the convergent series (6.20b24), multiplied by C , have total smaller than $1/2$. For each residue $0 \leq r < N$,

$$\bigcup_{k \geq 0} P_{m+r+kN}(x) \subset P_{m+r}(x) \cup \bigcup_{k \geq 1} (P_{m+r+kN}(x) \setminus P_{m+r+(k-1)N}(x)).$$

Equation (6.20b27) therefore shows that

$$v\left(Y \setminus \bigcup_{n \geq m} P_n(x)\right) > \frac{1}{2}.$$

Inner regularity supplies a compact subset $K(x)$ of this complement with mass at least $1/2$. The final radius argument is exactly the last paragraph of the proof of Lemma III.6.24.12: an $A\delta_n$ -belt meeting $K(x)$ meets $Y \setminus P_n(x)$ and is therefore light. Since $\delta_{n+1} = \delta_n^2$, this yields (6.20b26).

Finally, if $M_n \leq M$ and exact belts are uniformly atom-free, then

$$\sup_{x \in E} v(P_n(x)) \leq M \sup_{x \in E, H} v(\mathcal{B}_x(H, A\delta_n^{\tau/2})) \longrightarrow 0.$$

Hence the seed condition holds. The transition series is automatically convergent because the synchronized scales are superexponential. $\square$

Lemma III.6.24.21 (No fixed finite multibelt multiplicity is universal). *There are disjoint compact sets $X, Y \subset \mathbb{R}^3$, 2-Frostman probability measures μ, ν supported on them, a compact $E \subset X$ with $\mu(E) > 0$, and a C^∞ norm N off the origin whose unit sphere has everywhere positive Gaussian curvature, such that $\nu(P) = 0$ for every affine plane P , and the following holds. For every integer $M \geq 1$ and every $\tau, A, C, A_0 > 0$, there are $M + 1$ distinct linear hyperplanes $H_1, \dots, H_{M+1}$ and $\delta_M > 0$ such that, for every $0 < \delta < \delta_M$ and every $x \in E$,*

$$\nu(\mathcal{B}_x(H_j, A\delta)) > C\delta^\tau \quad (1 \leq j \leq M + 1), \quad (6.20b28)$$

and, for every choice of linear hyperplanes $K_1, \dots, K_M$, at least one of these $M + 1$ heavy belts meets

$$Y \setminus \bigcup_{\ell=1}^M \mathcal{B}_x(K_\ell, A_0\delta^{\tau/2}). \quad (6.20b29)$$

Consequently no uniformly bounded number of exceptional belts per scale can be a universal replacement for the same-base telescope lemma one-exceptional-belt pruning principle.

Proof. Write $z = (p, w)$ with $p = (u, v)$ and $P = \{w = 0\}$. Let $\theta_j = 2^{-j}$ and $a_j = (\cos \theta_j, \sin \theta_j)$. Choose pairwise disjoint antipodally symmetric spherical caps Ω_j around $\pm(a_j, 0)$, with radii $\rho_j \asymp 2^{-j}$ sufficiently small, and smooth even degree-zero cutoffs χ_j supported in Ω_j and equal to one on smaller caps U_j . They may be chosen with

$$\|D^m \chi_j\| \leq C_m \rho_j^{-m}.$$

Set $\varepsilon_j = \varepsilon_* \rho_j^{j+6}$ and

$$Q(z) = \frac{1}{2}|z|^2 + \sum_{j \geq 1} \varepsilon_j \chi_j(z/|z|)(a_j \cdot p)w. \quad (6.20b30)$$

The supports are disjoint. For every fixed derivative order m , $\varepsilon_j \rho_j^{-m} \rightarrow 0$, so the perturbation and all of its derivatives tend to zero at the only accumulation directions $\pm e_1$. Hence Q is C^∞ off the origin. Moreover the Hessian perturbation is uniformly $O(\varepsilon_*)$; after decreasing ε_* , D^2Q is uniformly positive definite. Thus $N = (2Q)^{1/2}$ is a smooth norm and the level set $Q = 1/2$ has everywhere positive Gaussian curvature.

On U_j one has

$$Q_j(p, w) = \frac{1}{2}(|p|^2 + w^2) + \varepsilon_j(a_j \cdot p)w, \quad \mathcal{G}_j = DQ_j = (p + \varepsilon_j w a_j, w + \varepsilon_j a_j \cdot p).$$

For $n_j = (-\varepsilon_j a_j, 1)$ and $H_j = n_j^\perp$,

$$n_j \cdot \mathcal{G}_j(p, w) = (1 - \varepsilon_j^2)w, \quad D^2Q_j n_j = (1 - \varepsilon_j^2)e_3. \quad (6.20b31)$$

The hyperplanes H_j are pairwise distinct.

Let $D \subset P$ be a small closed disk, let μ be normalized planar area on D , and choose a smaller concentric disk $E \subset D$. Fix L large. Let μ_j be the translate of μ by La_j , choose $h_j \asymp L\rho_j$, and let λ_j be the atomless probability on $[0, h_j/e]$ satisfying

$$d\lambda_j(t) = \frac{\mathbf{1}_{(0, h_j/e]}(t)}{t(\log(h_j/t))^2} dt, \quad \lambda_j([0, r]) = \frac{1}{\log(h_j/r)} \quad (0 < r \leq h_j/e). \quad (6.20b32)$$

With weights $w_j = 2^{-j}$ put

$$v = \sum_{j \geq 1} w_j (\mu_j \times \lambda_j).$$

The support closure is compact and is disjoint from D when L is large. Projection of a spatial radius- r ball to P is contained in a planar radius- r ball, so each component, and hence v , is uniformly 2-Frostman. If an affine plane has a nonzero w coefficient, its vertical fibres are singletons and are killed by the atomlessness of λ_j ; if that coefficient is zero, its planar section is a line and is killed by planar area. Thus v charges no affine plane.

For all large j , every $x \in E$, every x' with $|x' - x| \leq cL\rho_j$, and every $0 \leq t \leq cL\rho_j$, the vector $(x' + La_j, t) - x$ lies in U_j . Therefore, for sufficiently small δ ,

$$v(\mathcal{B}_x(H_j, A\delta)) \gtrsim \frac{w_j \rho_j^2}{\log(h_j/\delta)}. \quad (6.20b33)$$

For every fixed j the coefficient $w_j \rho_j^2$ is positive and the right-hand side dominates $C\delta^\tau$ as $\delta \downarrow 0$.

Now fix M and choose $M + 1$ sufficiently large distinct indices, relabelled $1, \dots, M + 1$. Their hyperplanes have a positive minimum projective separation, so for all sufficiently small δ they are also separated on the $\delta^{\tau/2}$ scale. At height zero define the compact footprints

$$F_j = \{\mathcal{G}_j(La_j + h, 0) : |h| \leq cL\rho_j\} \subset H_j.$$

Each F_j contains a relative open subset of H_j . On the compact product $G(3, 2)^M$ consider

$$R(K_1, \dots, K_M) = \max_{1 \leq j \leq M+1} \sup_{z \in F_j} \min_{1 \leq \ell \leq M} \text{dist}(z, K_\ell). \quad (6.20b34)$$

Its minimum is positive. Otherwise a minimizing M -tuple would cover every F_j by $K_1 \cup \dots \cup K_M$. A relative open subset of the plane H_j cannot be covered by finitely many proper linear sections of H_j ; hence some K_ℓ would have to equal H_j for every j , impossible for $M + 1$ distinct H_j .

Because $0 \in \text{supp } \lambda_j$, the height-zero footprints correspond to points of Y . Once $A_0\delta^{\tau/2}$ is smaller than the positive minimum in (6.20b34), one footprint point lies outside every candidate union of M exceptional belts, while remaining inside its heavy $A\delta$ -belt. Together with (6.20b33), this proves (6.20b28)–(6.20b29). $\square$

Lemma III.6.24.22 (Tracked-birth multibelt telescope). *Retain the hypotheses and synchronized scales of Lemma III.6.24.20. For each $n \geq N$ and x in the common good base set, partition the components of $P_n(x)$ into inherited and born components, and suppose every inherited component is contained in $P_{n-N}(x)$. Let $Q_n(x)$ be the union of the born components and let $B_n(x)$ denote their number. Then*

$$P_n(x) \setminus P_{n-N}(x) \subset Q_n(x) \setminus P_{n-N}(x), \quad (6.20b35)$$

and the old-scale pruning estimate gives

$$v(P_n(x) \setminus P_{n-N}(x)) \leq CB_n(x)\delta_{n-N}^\tau. \quad (6.20b36)$$

More sharply, if

$$a_n := \sup_x v(Q_n(x) \setminus P_{n-N}(x)), \quad (6.20b37)$$

then the residue-class telescope closes whenever $\sum_{n \geq N} a_n < \infty$ and the seed condition (6.20b25) holds. Thus total active multiplicity may be unbounded; only genuinely new exceptional mass needs to be summable.

Proof. The inclusion (6.20b35) is exactly the inheritance hypothesis. Each born component at scale n has width $A\delta_n^{\tau/2} = A\delta_{n-N}$. If it meets $Y \setminus P_{n-N}(x)$, the pruning statement at the older scale δ_{n-N} bounds its whole mass by $C\delta_{n-N}^\tau$; otherwise it contributes nothing to the transition difference. Summing over the born components gives (6.20b36). The estimate by a_n is immediate from (6.20b35), and the remainder of the residue-class argument is the proof of Lemma III.6.24.20 with a_n in place of $CM_n\delta_{n-N}^\tau$. $\square$

Lemma III.6.24.23 (Pointwise positive-mass telescope criterion). *Fix one base point x for which the one-scale pruning statements hold at all sufficiently small synchronized scales. Suppose that for some m ,*

$$\sum_{r=0}^{N-1} v(P_{m+r}(x)) + \sum_{n \geq m+N} v(P_n(x) \setminus P_{n-N}(x)) < 1. \quad (6.20b38)$$

Then there is a Borel set $K(x) \subset Y$ of positive v -mass such that, after normalizing $v|_{K(x)}$,

$$v_{K(x)}(\mathcal{B}_x(H, r)) \leq C_x r^{\tau/2} \quad (6.20b39)$$

for every linear hyperplane H and all sufficiently small r , where C_x may depend on x .

Proof. For each residue class modulo N , telescope the exceptional unions starting at $m+r$. Equation (6.20b38) gives

$$v\left(\bigcup_{n \geq m} P_n(x)\right) < 1.$$

Hence its complement $K(x)$ has positive mass. The final radius comparison is the same as in Lemma III.6.24.12: if $\delta_{n+1} < Cr \leq \delta_n$ and $n \geq m$, then an $O(r)$ -belt meeting $K(x)$ also meets $Y \setminus P_n(x)$, so one-scale pruning bounds its mass by $C\delta_n^\tau \lesssim r^{\tau/2}$. Normalization only changes the constant, which is allowed to depend on x . $\square$

Tracked-birth and pointwise telescope refinement. Lemma III.6.24.22 replaces the multibelt telescope charge on total active multiplicity by a charge on genuinely new exceptional mass. Inherited belts that remain inside the older exceptional union are free, so persistent common-track directions do not pay repeatedly. The strongest form is the mass-Carleson condition $\sum a_n < \infty$ in (6.20b37); the birth-count series is only a sufficient upper bound.

Lemma III.6.24.23 also records the weaker endpoint actually consumed by Shmerkin's continuous nonlinear projection theorem. The source hypothesis asks only for a parameter restriction of positive mass at each base point, with an x -dependent decay constant.

Hence it is enough to find the condition on a Borel base subset of Hausdorff dimension at least α ; in particular, a measurable full- μ good set for an α -Frostman base measure is sufficient after choosing a Borel full-measure subset. On that base set one needs only one starting scale for which the seed-plus-transition bound (6.20b38) is strictly below one. A uniform half-mass seed is stronger than the conclusion requires. Any tracked-family construction must nevertheless also retain the measurability needed to select such a Borel good base set.

The persistent-belt counterexamples do not refute this tracked formulation: they force persistent active directions and unbounded total multiplicity, but do not provide a lower bound on the mass of genuinely new directions born at each synchronized transition. Thus the residual telescopic question is a birth-mass/entropy Carleson problem. The radial-Frostman argument below bypasses this difficulty.

Lemma III.6.24.24 (Pointwise Frostman data need not admit any diffuse restriction). *Fix $d \geq 3$, $d - 2 < \kappa < d - 1$, and $\tau, A, C > 0$. Let $\delta_n = \delta_0^{2^n}$ with $\delta_0 > 0$ sufficiently small. There are a compact set*

$$Y \subset \{y \in \mathbb{R}^d : 1 \leq |y| \leq 2\}$$

and a κ -Frostman probability measure ν on Y such that $\nu(P) = 0$ for every affine hyperplane P , while at the Euclidean base point 0 the following hold.

For all sufficiently large n there is a finite family $\mathcal{H}_n$ of linear hyperplanes satisfying

$$Y \subset \bigcup_{H \in \mathcal{H}_n} H^{(A\delta_n)}, \quad \nu(H^{(A\delta_n)}) > C\delta_n^\tau \quad (H \in \mathcal{H}_n). \quad (6.20b40)$$

Consequently, if a Borel set $P_n \subset Y$ has the one-scale pruning property

$$H^{(A\delta_n)} \cap (Y \setminus P_n) \neq \emptyset \implies \nu(H^{(A\delta_n)}) \leq C\delta_n^\tau \quad (6.20b41)$$

for every linear hyperplane H , then $\nu(P_n) = 1$.

More strongly, if $K \subset Y$ is Borel with $\nu(K) > 0$ and $\zeta = \text{dir}_(\nu_K)$, then for every $s > 0$ and $D < \infty$ there are arbitrarily large n and $H \in G(d, d - 1)$ such that*

$$\zeta(H^{(A\delta_n)}) > D\delta_n^s. \quad (6.20b42)$$

Thus no positive-mass restriction at this prescribed base point has any polynomial hyperplane-neighbourhood decay.

Proof. Put $q = \kappa - (d - 2) \in (0, 1)$. Choose an integer m so large that, with $r = m^{-1/q}$,

$$\frac{(m-1)r}{1-r} < \frac{1}{16}. \quad (6.20b43)$$

On $\Omega = \{0, \dots, m-1\}^{\mathbb{N}}$ take the uniform Bernoulli probability β and set

$$x(\omega) = \sum_{j \geq 1} \omega_j r^j.$$

The first differing digit dominates the remaining tail by (6.20b43), so the push-forward of β by x is a q -Ahlfors regular Cantor measure.

Choose $\varepsilon > 0$ so small that all chart constants below are absorbed by A and put

$$b_{j+1} = \varepsilon \delta_j, \quad f(\omega) = \sum_{j \geq 1} \omega_j b_j. \quad (6.20b44)$$

After decreasing δ_0 , the superexponential relation $\delta_{j+1} = \delta_j^2$ gives

$$(m-1) \sum_{j > n} b_j \leq c_A \delta_n \quad (6.20b45)$$

with c_A as small as needed. Hence each level- n graph cylinder

$$\Gamma_w = \{(x(\omega), f(\omega)) : \omega|_n = w\}$$

lies in a $c_A \delta_n$ -strip about the affine line $y = f_w$, where f_w is the prefix sum. Since $b_j/r^j \rightarrow 0$ superexponentially, projection to the x -coordinate is bi-Lipschitz on the graph Γ , so its Bernoulli graph measure λ_Γ is still q -Ahlfors regular.

The graph measure gives zero mass to every affine line. Vertical lines meet Γ in at most one point. For a nonvertical line $y = sx + t$, the digit coefficients are $c_j = b_j - sr^j$. If $s \neq 0$, then eventually $|c_j| \geq |s|r^j/2$, while (6.20b43) and the superexponential decay of b_j make the remaining digit tail smaller than $|c_j|/2$. Thus two graph points on the same line cannot first differ at a sufficiently large index. Hence, for some J depending on the line, each word of length J contains at most one intersection point, so the line meets Γ in at most m^J points. If $s = 0$, the same finite-intersection conclusion follows directly from the superlacunarity of the b_j .

Let σ be normalized Lebesgue measure on a small cube in $\mathbb{R}^{d-3}$, omitting this factor when $d = 3$, and use the projective chart

$$\Phi(u, x, y) = \frac{(u, x, y, 1)}{|(u, x, y, 1)|} \in S^{d-1}.$$

On the chosen compact box Φ is bi-Lipschitz. Therefore

$$\xi = \Phi_*(\sigma \times \lambda_\Gamma)$$

is $(\kappa - 1)$ -Frostman. Finally let ρ be normalized Lebesgue measure on $[1, 2]$ and define

$$\nu = (s, \theta \mapsto s\theta)_*(\rho \times \xi).$$

The polar map is bi-Lipschitz on this annular patch, so ν is κ -Frostman. If an affine hyperplane does not pass through the origin, each angular fibre meets it in at most one radial point and Fubini with ρ gives zero mass. For a linear hyperplane, the projective chart gives an affine equation $a \cdot u + bx + cy + d = 0$. If $a \neq 0$, Fubini in u gives zero mass; if $a = 0$, the line-nullity of λ_Γ does. Thus every affine hyperplane is ν -null.

For a word w of length n , the affine chart hyperplane $y = f_w$ homogenizes to a linear hyperplane $H_w \subset \mathbb{R}^d$. By (6.20b45), the full radial lift of the cylinder is contained in $H_w^{(A\delta_n)}$. Its ν -mass is exactly m^{-n} . Since $-\log \delta_n \asymp 2^n$,

$$\frac{m^{-n}}{\delta_n^\tau} \rightarrow \infty.$$

This proves (6.20b40). If P_n satisfied (6.20b41) and omitted a point of Y , the cylinder containing that point would supply a heavy strip meeting $Y \setminus P_n$, a contradiction. Hence $v(P_n) = 1$.

It remains to prove the restriction statement. For K of positive mass, the direction measure $\zeta = \text{dir}_*(\nu_K)$ is absolutely continuous with respect to ξ . Pull back by Φ and integrate out u ; this gives a nonzero measure $\beta_K \ll \beta$ on Ω . If $h = d\beta_K/d\beta$, martingale differentiation on symbolic cylinders gives, for β_K -almost every ω with $h(\omega) > 0$,

$$\beta_K([\omega_1 \cdots \omega_n]) \geq c(\omega)m^{-n}$$

for all large n . The corresponding cylinder lies in $H_{\omega|_n}^{(A\delta_n)}$, while $m^{-n}/\delta_n^s \rightarrow \infty$ for every $s > 0$. This is (6.20b42). $\square$

A superflat obstruction and the joint-base pivot. Lemma III.6.24.24 shows that the tracked-birth telescope missing input cannot be manufactured from the transformed parameter measure one base point at a time. Even in Euclidean geometry, throughout the open band $d - 2 < \kappa < d - 1$, ambient κ -Frostman decay and zero affine-hyperplane mass are compatible with full-mass heavy-strip covers at every synchronized scale and with failure of every polynomial hyperplane-decay exponent on every positive restriction. The tracked-birth telescope transition bookkeeping can therefore be perfect while the seed obstruction remains maximal.

This does not contradict the Euclidean spherical-direction theorem above: its proof uses a Frostman family of base points and discards a small base exceptional set before constructing the restrictions. The new conclusion is instead methodological and exact: no further pointwise Carleson refinement can close the residual curved-norm clause from Frostman/hyperplane-null data alone. A future proof must use the base variable jointly, for example through incidence or entropy estimates for the translated Legendre cones

$$\{(x, y) : \text{dist}(D(N^2/2)(y - x), H) \leq \delta\},$$

or bypass the telescope with a different nonlinear projection mechanism. The endpoint $\kappa = d - 1$ is not addressed by this obstruction. The radial-Frostman argument below avoids the pointwise restriction and closes the stated branch.

Lemma III.6.24.25 (Conditioned determinant witnesses for heavy Legendre belts). *Retain the separated compact setup above, assume $\kappa > d - 2$, and put $q = \kappa - (d - 2)$. There is $C_* > 1$ with the following property. For $x \in X$ let*

$$\sigma_x = (y \mapsto \mathcal{G}(y - x))_{\#} \nu.$$

Then, for every affine j -plane V with $0 \leq j \leq d - 2$ and $0 < r \leq 1$,

$$\sigma_x(V^{(r)}) \leq C_* r^q. \quad (6.20b46)$$

Fix $H \in G(d, d - 1)$ and suppose

$$E = \mathcal{B}_x(H, A\delta), \quad v(E) = m > 0.$$

Set

$$\rho = \left(\frac{m}{4C_*} \right)^{1/q}. \quad (6.20b47)$$

For $\mathbf{y} = (y_0, \dots, y_{d-1}) \in E^d$ write $v_i = \mathcal{G}(y_i - x)$. Let $\mathcal{W}_x(H, \delta)$ be the set of tuples satisfying

$$\text{dist}(v_j, \text{aff}(v_0, \dots, v_{j-1})) \geq \rho, \quad 1 \leq j \leq d-1. \quad (6.20b48)$$

Then

$$v^d(\mathcal{W}_x(H, \delta)) \geq m(3m/4)^{d-1} \geq c_d m^d. \quad (6.20b49)$$

Every $\mathbf{y}$ in this set satisfies

$$W_{\mathbf{y}}(x) := |(v_1 - v_0) \wedge \dots \wedge (v_{d-1} - v_0)| \geq \rho^{d-1}, \quad (6.20b50)$$

and, with

$$D_{\mathbf{y}}(x) = \det(v_0, \dots, v_{d-1}),$$

one has

$$|D_{\mathbf{y}}(x)| \leq C_A \delta. \quad (6.20b51)$$

If $H_{\mathbf{y}}(x)$ denotes the direction of the affine span of the v_i , then, provided $A\delta \leq c\rho^{d-1}$,

$$\angle(H_{\mathbf{y}}(x), H) \leq C_A \delta \rho^{-(d-1)}. \quad (6.20b52)$$

In particular, if $m \geq c\delta^\tau$ and τ is decreased, if necessary, so that

$$1 - \frac{(d-1)\tau}{q} > \frac{\tau}{2}, \quad (6.20b53)$$

then $\rho \geq \delta^{\tau/q}$ and the error in (6.20b52) is $o(\delta^{\tau/2})$.

Proof. The maps $y \mapsto \mathcal{G}(y - x)$ are uniformly bi-Lipschitz on the separated compact product. Hence the Frostman ball bound transfers to σ_x with a uniform constant. Inside a fixed bounded region, the r -neighbourhood of an affine j -plane is coverable by $O(r^{-j})$ balls of radius $O(r)$, and therefore has σ_x -mass $O(r^{K-j})$. Since $j \leq d-2$ and $r \leq 1$, this proves (6.20b46).

Fix a prefix $v_0, \dots, v_{j-1}$ satisfying (6.20b48). Its affine span has dimension at most $d-2$; extend it, if necessary, to an affine $(d-2)$ -plane. By (6.20b46) and (6.20b47), the set of possible next points violating (6.20b48) has v -mass at most $m/4$. Since E has mass m , Fubini induction leaves at least $3m/4$ choices at each of the final $d-1$ stages. This gives (6.20b49). Gram–Schmidt’s volume identity gives (6.20b50).

Choose a unit normal n to H and write $v_i = h_i + t_i n$ with $h_i \in H$ and $|t_i| \leq A\delta$. The vectors v_i stay in a fixed bounded annulus. Expanding the determinant along the normal components gives $|D_{\mathbf{y}}(x)| \lesssim_A \delta$, which is (6.20b51). Put

$$w = (v_1 - v_0) \wedge \dots \wedge (v_{d-1} - v_0).$$

Replacing every difference by its orthogonal projection to H changes w by $O_A(\delta)$. Equation (6.20b50) and $A\delta \leq c\rho^{d-1}$ therefore show that the projected wedge is nonzero and that the projective directions of w and n differ by at most $C_A \delta \rho^{-(d-1)}$. This is (6.20b52). Finally $m \geq c\delta^\tau$ gives $\rho \geq \delta^{\tau/q}$, and (6.20b53) makes the last angular error $o(\delta^{\tau/2})$. $\square$

Lemma III.6.24.26 (Two-heavy bases reduce to a product-mass determinant incidence). *Use the notation of Lemma III.6.24.15, and decrease τ once more so that (6.20b53) holds. Let $\mathfrak{F}_\delta$ be the set of all*

$$(x, \mathbf{y}, \mathbf{y}') \in X \times Y^d \times Y^d$$

for which

$$|D_{\mathbf{y}}(x)| + |D_{\mathbf{y}'}(x)| \leq C\delta, \quad (6.20b54)$$

$$W_{\mathbf{y}}(x), W_{\mathbf{y}'}(x) \geq c\delta^{(d-1)\tau/q}, \quad (6.20b55)$$

and the two fitted hyperplane directions satisfy

$$\angle(H_{\mathbf{y}}(x), H_{\mathbf{y}'}(x)) \geq c\delta^{\tau/2}. \quad (6.20b56)$$

After adjusting the fixed constants, $\mathfrak{F}_\delta$ is Borel and

$$(\mu \times \nu^{2d})(\mathfrak{F}_\delta) \gtrsim \delta^{2d\tau} \mu(T_\delta). \quad (6.20b57)$$

Thus the joint-base problem has an exact finite-dimensional determinant-incidence interface with no pointwise witness selection.

Proof. The diagnostic set T_δ is universally measurable: it is the projection to X of the Borel subset of $X \times G(d, d-1)^2$ defined by the two heavy-mass inequalities and the separation condition. We therefore work in the μ -completion. For $x \in T_\delta$, choose the two heavy belts only to estimate the fibre. Each has mass at least $c\delta^\tau$. Lemma III.6.24.25 provides at least $c\delta^{d\tau}$ product mass of conditioned d -tuples in each belt. Their fitted hyperplane directions differ from the original heavy directions by $o(\delta^{\tau/2})$, so the original separation survives after changing the fixed constant. Every resulting tuple pair satisfies (6.20b54)–(6.20b56). Hence the ν^{2d} -measure of the fibre of $\mathfrak{F}_\delta$ over x is at least $c\delta^{2d\tau}$. Integrating in x proves (6.20b57). The defining functions are continuous wherever the wedge lower bounds hold, so the displayed incidence set is Borel. No measurable choice of the original pair of heavy hyperplanes is needed. $\square$

Conditioned witnesses and determinant incidence. Lemmas III.6.24.25 and III.6.24.26 close the tuple-conditioning issue left open in the exact-belt exclusion. A heavy Legendre belt contains a positive ν^d -mass family of d -point witnesses whose transformed images affinely span the belt at the polynomial scale $\rho \asymp \delta^{\tau/(\kappa-(d-2))}$. Consequently the two-heavy branch injects, with the quantitative lower bound (6.20b57), into a finite-dimensional determinant sublevel set. Measurable pointwise witness selection is therefore no longer part of the blocker.

the persistent-belt counterexample prevents one from seeking an unrestricted power upper bound for $\mathfrak{F}_\delta$: such a bound would recover the already refuted estimate on $\mu(T_\delta)$. The exact next split is inside the determinant incidence itself. One must obtain a power saving on the quantitatively transverse determinant-gradient part and organize the determinant-gradient-degenerate part into the persistent physical track geometry already exhibited by the persistent-belt counterexample, or else extract the nonlinear projection conclusion directly on that degenerate branch. At C^2 regularity no quantitative implicit-function modulus for the determinant gradients is inserted here. The witness lemma also applies at the endpoint $\kappa = d - 1$ (where $q = 1$), although the superflat pointwise obstruction fixed-base obstruction was only an open-band result.

Lemma III.6.24.27 (Modulus-aware transverse sublevel cover). *Let μ be an α -Frostman probability measure on a compact $X \subset \mathbb{R}^d$, where*

$$s := \alpha - (d - 2) > 0.$$

Let T be compact and let $F_t = (f_t, g_t), t \in T$, be a family of C^1 maps on a fixed neighbourhood of X such that $D_x F_t$ is jointly continuous in (x, t) . Put

$$J_t(x) = |\nabla f_t(x) \wedge \nabla g_t(x)|$$

and let M bound $\|D_x F_t\|$ on the compact domain. Define the common modulus

$$\omega(r) = \sup_{\substack{t \in T \\ |x-x'| \leq r}} \|D_x F_t(x) - D_x F_t(x')\|. \quad (6.20b58)$$

There are constants $c_0, C > 0$ with the following property. For $0 < j \leq 1$, choose $0 < r_j \leq r_0$ so that

$$\omega(4r_j) \leq c_0 j. \quad (6.20b59)$$

Then, whenever $0 < \delta \leq c_0 j r_j$,

$$\sup_{t \in T} \mu\{x \in X : |F_t(x)|_\infty \leq C\delta, J_t(x) \geq j\} \leq C j^{-\alpha} r_j^{-2} \delta^s. \quad (6.20b60)$$

If, moreover, $D_x F_t$ is uniformly β -Hölder in x , $0 < \beta \leq 1$, then one may take $r_j \gtrsim j^{1/\beta}$ and hence

$$\sup_{t \in T} \mu\{x \in X : |F_t(x)|_\infty \leq C\delta, J_t(x) \geq j\} \lesssim \delta^s j^{-\alpha-2/\beta} \quad (6.20b61)$$

provided $\delta \lesssim j^{1+1/\beta}$.

Proof. At a point x_0 with $J_t(x_0) \geq j$, the Euclidean identity expressing the squared wedge norm as the sum of the squares of the 2×2 coordinate minors shows that one such minor, say in coordinates $p = (x_a, x_b)$, has determinant at least $c_d j$. Since its operator norm is bounded by M , its least singular value is at least $c j$. By decreasing the dimensional constant in (6.20b59), the same p -minor on the $4r_j$ -neighbourhood of x_0 differs from its value at x_0 by at most one quarter of that lower singular value.

Work in a coordinate cube of side comparable with r_j about x_0 , writing the remaining coordinates as $q \in \mathbb{R}^{d-2}$. For fixed q , integration of the p -derivative along line segments gives

$$|F_t(p, q) - F_t(p', q)| \geq c j |p - p'|. \quad (6.20b62)$$

If (p, q) and (p', q') both belong to the displayed sublevel set and $|q - q'| \leq \delta$, then the uniform derivative bound and (6.20b62) give $|p - p'| \leq \delta/j$. Partitioning the q -projection of the cube into $O((r_j/\delta)^{d-2})$ cells therefore covers the sublevel set in this cube by the same number of balls of radius $O(\delta/j)$. The condition $\delta \leq c_0 j r_j$ keeps these balls inside a fixed enlargement of the cube.

A maximal r_j -separated family of points in the sublevel set has cardinality $O(r_j^{-d})$ on the fixed bounded base region, and its $O(r_j)$ -cubes cover that set. Frostman summation now gives

$$r_j^{-d} \left(\frac{r_j}{\delta} \right)^{d-2} \left(\frac{\delta}{j} \right)^\alpha = r_j^{-2} j^{-\alpha} \delta^{\alpha-(d-2)},$$

which is (6.20b60). If the derivatives are uniformly β -Hölder, then (6.20b59) holds with $r_j \asymp j^{1/\beta}$, giving (6.20b61); the stated restriction on δ is exactly $\delta \lesssim jr_j$. $\square$

Lemma III.6.24.28 (Transverse determinant incidence is summable). *Retain the hypotheses and notation of Lemma III.6.24.26, and decrease τ , if necessary, so that*

$$s - 2d\tau > 0, \quad s = \alpha - (d - 2). \quad (6.20b63)$$

For $\mathbf{t} = (\mathbf{y}, \mathbf{y}') \in Y^d \times Y^d$, put

$$F_{\mathbf{t}}(x) = (D_{\mathbf{y}}(x), D_{\mathbf{y}'}(x)), \quad J_{\mathbf{t}}(x) = |\nabla_x D_{\mathbf{y}}(x) \wedge \nabla_x D_{\mathbf{y}'}(x)|.$$

Let r_j be chosen from Lemma III.6.24.27 for this compact determinant family, and define

$$\mathfrak{T}_\delta^{\text{tr}}(j) = \{(x, \mathbf{t}) \in \mathfrak{T}_\delta : J_{\mathbf{t}}(x) \geq j\}.$$

Then, for $\delta \leq cjr_j$,

$$(\mu \times \nu^{2d})(\mathfrak{T}_\delta^{\text{tr}}(j)) \lesssim j^{-\alpha} r_j^{-2} \delta^s. \quad (6.20b64)$$

Moreover the proof of Lemma III.6.24.26 supplies $c_* > 0$ such that every $x \in T_\delta$ satisfies $\nu^{2d}((\mathfrak{T}_\delta)_x) \geq c_* \delta^{2d\tau}$. Hence, with

$$T_\delta^{\text{tr}}(j) = \left\{ x \in T_\delta : \nu^{2d}((\mathfrak{T}_\delta^{\text{tr}}(j))_x) \geq \frac{c_*}{2} \delta^{2d\tau} \right\},$$

one has

$$\mu(T_\delta^{\text{tr}}(j)) \lesssim j^{-\alpha} r_j^{-2} \delta^{s-2d\tau}. \quad (6.20b65)$$

For every $x \in T_\delta \setminus T_\delta^{\text{tr}}(j)$,

$$\nu^{2d}((\mathfrak{T}_\delta \cap \{J_{\mathbf{t}} < j\})_x) \geq \frac{c_*}{2} \delta^{2d\tau}. \quad (6.20b66)$$

For the bare C^2 norm hypothesis there is a deterministic nonincreasing sequence $j_n \downarrow 0$ along the synchronized scales $\delta_n = \delta_0^{2^n}$ such that

$$\sum_n \mu(T_{\delta_n}^{\text{tr}}(j_n)) < \infty. \quad (6.20b67)$$

If instead N is $C^{2,\beta}$ off the origin for some $0 < \beta \leq 1$, then, after choosing $\zeta > 0$ and decreasing τ once more so that

$$\zeta(1 + 1/\beta) < 1, \quad s - 2d\tau - (\alpha + 2/\beta)\zeta > 0, \quad (6.20b68)$$

one may take $j = \delta^\zeta$ and obtains the genuine power estimate

$$\mu(T_\delta^{\text{tr}}(\delta^\zeta)) \lesssim \delta^{s-2d\tau-(\alpha+2/\beta)\zeta}. \quad (6.20b69)$$

Thus the quantitatively transverse determinant branch is closed under $C^{2,\beta}$ regularity; at bare C^2 it is summably removable only at a modulus-adapted absolute gradient floor.

Proof. The family $F_{\mathbf{t}}$ is C^1 in x with jointly continuous derivative on the separated compact product, because $\mathcal{G} = D(N^2/2)$ is C^1 there. Applying Lemma III.6.24.27 fibrewise in $\mathbf{t}$ and integrating against ν^{2d} proves (6.20b64). Equations (6.20b65) and (6.20b66) then follow from the fibre lower bound recorded in the proof of Lemma III.6.24.26 and Fubini. The fibre-measure functions are measurable because the incidence sets are Borel.

For the diagonal bare- C^2 assertion, put $A(j) = j^{-\alpha} r_j^{-2}$. For every fixed k , $A(1/k) < \infty$, and the superexponential sequence δ_n eventually satisfies both $\delta_n \leq ck^{-1} r_{1/k}$ and $A(1/k) \delta_n^{s-2d\tau} \leq 2^{-n}$. Choose increasing indices N_k for which these two inequalities hold for every $n \geq N_k$, and set $j_n = 1/k$ on $N_k \leq n < N_{k+1}$ (modifying finitely many initial terms arbitrarily). Then $j_n \downarrow 0$ and (6.20b67) follows from (6.20b65).

If N is $C^{2,\beta}$, then $\mathcal{G}$ is $C^{1,\beta}$ on the separated annulus, and therefore the determinant-gradient family is uniformly β -Hölder. Equation (6.20b61) with $j = \delta^\zeta$, followed by the same fibre comparison, gives (6.20b69). The two inequalities in (6.20b68) ensure respectively that the local patch condition holds and that the resulting exponent is positive. $\square$

Lemma III.6.24.29 (Normalized determinant gradient as an averaged physical normal). *Let $\mathbf{y} = (y_0, \dots, y_{d-1})$ be a witness tuple with $W_{\mathbf{y}}(x) > 0$. Write $\mathbf{v}_i = \mathcal{G}(y_i - x)$ and choose the oriented vector*

$$\mathbf{w} = *(\mathbf{v}_1 - \mathbf{v}_0) \wedge \dots \wedge (\mathbf{v}_{d-1} - \mathbf{v}_0)$$

so that $D_{\mathbf{y}}(x) = \mathbf{w} \cdot \mathbf{v}_0$. Put $W = |\mathbf{w}|$, $n = \mathbf{w}/W$, and $h = n \cdot \mathbf{v}_i = D_{\mathbf{y}}(x)/W$, which is independent of i . Set $u_i = \mathbf{v}_i - hn \in n^\perp$. There are unique barycentric coefficients λ_i satisfying

$$\sum_{i=0}^{d-1} \lambda_i = 1, \quad \sum_{i=0}^{d-1} \lambda_i u_i = 0. \quad (6.20b70)$$

With

$$A_i = D\mathcal{G}(y_i - x), \quad \mathcal{N}_{\mathbf{y}}(x) = \sum_{i=0}^{d-1} \lambda_i A_i n, \quad (6.20b71)$$

one has, uniformly on the separated compact product,

$$|\lambda_i| \lesssim W^{-1}, \quad \left| \frac{\nabla_x D_{\mathbf{y}}(x)}{W} + \mathcal{N}_{\mathbf{y}}(x) \right| \lesssim \frac{|D_{\mathbf{y}}(x)|}{W^2}. \quad (6.20b72)$$

In particular, at an exact determinant zero the second relation is an identity with zero error.

For a pair $(\mathbf{y}, \mathbf{y}')$ in $\mathfrak{F}_\delta$, put $\gamma = (d-1)\tau/q$. Then

$$\left| \frac{|\nabla D_{\mathbf{y}} \wedge \nabla D_{\mathbf{y}'}|}{WW'} - |\mathcal{N}_{\mathbf{y}} \wedge \mathcal{N}_{\mathbf{y}'}| \right| \lesssim \delta^{1-3\gamma}. \quad (6.20b73)$$

Thus, after decreasing τ so that $3\gamma < 1$, normalized determinant-gradient degeneracy is, up to a vanishing error, exactly degeneracy of two barycentrically averaged physical cone normals.

Proof. Let $\mathbf{c}_i$ be the cofactor vector obtained by differentiating $\det(\mathbf{v}_0, \dots, \mathbf{v}_{d-1})$ with respect to the i th column. Translating every column by the same vector leaves all differences $\mathbf{v}_i - \mathbf{v}_0$ fixed, so multilinearity gives the exact identity

$$\sum_i \mathbf{c}_i = \mathbf{w}.$$

When $h = 0$, all $u_i = v_i$ lie in the linear hyperplane $n^\perp$ and affinely span it. Hence every cofactor is parallel to w ; writing $c_i = \lambda_i w$, the preceding identity gives $\sum_i \lambda_i = 1$, while the adjugate identity gives $\sum_i \lambda_i u_i = 0$. This proves (6.20b70) in the zero-offset configuration.

For general h , the differences are unchanged, so the projected points u_i have the same affine conditioning and the same w . Expanding the cofactor of $v_i = u_i + hn$ multilinearly, all terms containing two copies of n vanish. Therefore

$$c_i = \lambda_i w + e_i, \quad |e_i| \lesssim |h|. \quad (6.20b74)$$

The compact annular bound gives $|c_i(u)| \lesssim 1$, hence $|\lambda_i|W \lesssim 1$. Since $A_i = D\mathcal{G}(y_i - x)$ is symmetric, column differentiation gives

$$\nabla_x D_y = - \sum_i A_i c_i.$$

Using (6.20b74), $|h| = |D_y|/W$, and the uniform bound on A_i proves (6.20b72).

On $\mathfrak{F}_\delta$ one has $|D_y| + |D_{y'}| \lesssim \delta$ and $W, W' \gtrsim \delta^\gamma$. Equation (6.20b72) gives errors $O(\delta/W^2)$ and $O(\delta/(W')^2)$, while $|\mathcal{N}_y| \lesssim W^{-1}$ and $|\mathcal{N}_{y'}| \lesssim (W')^{-1}$ by the barycentric bound. Expanding the wedge of the two normalized gradients therefore gives an error $O(\delta^{1-3\gamma})$, proving (6.20b73). $\square$

Transverse determinant submersion. Lemma III.6.24.27 supplies the first quantitative upper bound for the conditioned-witness reduction joint incidence without importing any external projection theorem. At bare C^2 regularity it keeps the full modulus dependence visible and removes, along the synchronized scales, all witness fibres whose absolute determinant-gradient two-Jacobian stays above a suitably slow floor $j_n \downarrow 0$. Under any $C^{2,\beta}$ improvement the floor can be polynomial, and the transverse-dominated base set has the power loss (6.20b69). Lemma III.6.24.29 then identifies the scale-correct normalization: after dividing by the two witness wedges, the determinant gradients are barycentric averages of the physical cone normals up to an $O(\delta^{1-3(d-1)\tau/q})$ error. Hence, in the $C^{2,\beta}$ subcase, choosing $\eta > 0$ and the polynomial floor $j = \delta^{2(d-1)\tau/q+\eta}$ places every witness pair with $|\mathcal{N}_y \wedge \mathcal{N}_{y'}| \gtrsim \delta^\eta$ in the transverse incidence at that floor. Consequently any base set on which a positive fraction of the witness fibre has this property inherits the power loss (6.20b69), provided τ, η are chosen small enough to satisfy (6.20b68) with $\zeta = 2(d-1)\tau/q + \eta$ and $\eta < 1 - 3(d-1)\tau/q$.

This determinant estimate alone does not close the theorem. One must still understand the product-mass witness configurations in (6.20b66) after normalizing for the scale-dependent witness wedges. Small absolute two-Jacobian alone is not yet synonymous with a physical common track, because the conditioned wedge lower bound itself tends to zero with scale. The next step is therefore a structural normalized-degeneracy lemma, or a direct nonlinear projection argument on that branch. No Hölder modulus for D^2N is silently assumed in the bare- C^2 statement.

Lemma III.6.24.30 (Robust anchor-test normalization). *Retain the separated compact setup, put*

$$q = \kappa - (d-2) > 0, \quad \mathcal{G} = D(N^2/2), \quad A = D\mathcal{G},$$

and let

$$E = \{y : \text{dist}(\mathcal{G}(y - x), H_0) \leq A_0 \delta\}$$

be a same-base Legendre belt of mass $m = \nu(E)$. There is a constant C_* , depending only on the fixed compact setup, such that if

$$\rho = \left(\frac{m}{8(d-1)C_*} \right)^{1/q}, \quad (6.20b75)$$

then a subset of E^{d-1} of ν^{d-1} -mass at least $\frac{7}{8}m^{d-1}$ consists of anchor tuples $\mathbf{a} = (y_1, \dots, y_{d-1})$ for which, writing $a_i = \mathcal{G}(y_i - x)$,

$$\text{dist}(a_i, \text{span}\{a_j : j \neq i\}) \geq \rho \quad (1 \leq i \leq d-1). \quad (6.20b76)$$

For every such tuple the matrix $M_{\mathbf{a}} = [a_1 \cdots a_{d-1}]$ satisfies

$$\|M_{\mathbf{a}}^+\| \lesssim \rho^{-1}, \quad W_{\mathbf{a}} := |\ast(a_1 \wedge \cdots \wedge a_{d-1})| \gtrsim \rho^{d-2}. \quad (6.20b77)$$

If $n_{\mathbf{a}}$ is the unit normal to $H_{\mathbf{a}} = \text{span}(a_1, \dots, a_{d-1})$, then

$$\angle(H_{\mathbf{a}}, H_0) \lesssim \delta/\rho. \quad (6.20b78)$$

Fix now $y \in E$, put $v = \mathcal{G}(y - x)$ and $u = P_{H_{\mathbf{a}}} v$, and let $\mathcal{T}_{\mathbf{a}} : H_{\mathbf{a}} \rightarrow \mathbb{R}^d$ be the linear map determined by

$$\mathcal{T}_{\mathbf{a}}(a_i) = A(y_i - x)n_{\mathbf{a}}.$$

Then $\|\mathcal{T}_{\mathbf{a}}\| \lesssim \rho^{-1}$. Define the affine-normal defect

$$\mathcal{R}_{\mathbf{a}}(v) := A(\mathcal{G}^{-1}(v))n_{\mathbf{a}} - \mathcal{T}_{\mathbf{a}}(P_{H_{\mathbf{a}}} v). \quad (6.20b79)$$

For

$$D_{\mathbf{a},y}(x) = \det(a_1, \dots, a_{d-1}, v)$$

one has

$$|D_{\mathbf{a},y}(x)| \lesssim \delta, \quad \left| -\frac{\nabla_x D_{\mathbf{a},y}(x)}{W_{\mathbf{a}}} - \mathcal{R}_{\mathbf{a}}(v) \right| \lesssim \frac{\delta}{\rho^2}. \quad (6.20b80)$$

At exact anchor-test incidence the second identity has zero error. Consequently, for two δ^τ -heavy belts, the wedge comparison error is $O(\delta^{1-3\tau/q})$.

Proof. For fixed values of the other $d-2$ anchors, the values violating (6.20b76) lie in the ρ -neighbourhood of a linear subspace of dimension at most $d-2$. The conditioned-witness reduction Frostman-plane estimate bounds their ν -mass by $C_* \rho^q = m/(8(d-1))$. Fubini and a union bound prove the $7/8$ assertion. The dual basis of the anchor columns has norm $O(\rho^{-1})$, giving the pseudoinverse bound. Since the transformed annulus is bounded away from the origin, one singular value is bounded below absolutely while the remaining ones are $\geq \rho$, which gives (6.20b77). The fact that all anchors lie within $O(\delta)$ of H_0 , together with the pseudoinverse bound, gives (6.20b78).

Write $v = u + hn_{\mathbf{a}}$, so $h = D_{\mathbf{a},y}/W_{\mathbf{a}}$. Equation (6.20b78) and the original belt condition give $|h| \lesssim \delta/\rho$. If $u = \sum_i b_i a_i$, then $|b_i| \lesssim \rho^{-1}$. At $h = 0$, the test-column cofactor is $W_{\mathbf{a}} n_{\mathbf{a}}$

and the i th anchor cofactor is $-W_{\mathbf{a}}b_i n_{\mathbf{a}}$, so column differentiation gives the exact identity $-\nabla D/W_{\mathbf{a}} = \mathcal{R}_{\mathbf{a}}(v)$. For nonzero h , the tangential part of the i th anchor cofactor is h times the $(d-2)$ -wedge of the other anchors. Robustness gives this wedge size at most $W_{\mathbf{a}}/\rho$. After division by $W_{\mathbf{a}}$ the total error is $O(|h|/\rho)$, proving (6.20b80). Finally $m \gtrsim \delta^\tau$ gives $\rho \gtrsim \delta^{\tau/q}$; expanding the wedge of two normalized gradients gives $O(\delta/\rho^3) = O(\delta^{1-3\tau/q})$. $\square$

Lemma III.6.24.31 (Radial rigidity and modulus-aware thin-branch bound). *Fix a robust anchor tuple from Lemma III.6.24.30. For $v = r\theta \neq 0$, homogeneity gives*

$$\mathcal{R}_{\mathbf{a}}(r\theta) = g_{\mathbf{a}}(\theta) - r\mathcal{T}_{\mathbf{a}}(P_{H_{\mathbf{a}}}\theta), \quad g_{\mathbf{a}}(\theta) = A(\mathcal{G}^{-1}(\theta))n_{\mathbf{a}}. \quad (6.20b81)$$

Fix a projective line $[m]$, let P_m be orthogonal projection onto $m^\perp$, and set

$$a(\theta) = P_m g_{\mathbf{a}}(\theta), \quad b(\theta) = P_m \mathcal{T}_{\mathbf{a}}(P_{H_{\mathbf{a}}}\theta).$$

If $\text{dist}(\mathcal{R}_{\mathbf{a}}(r\theta), \text{span}(m)) \leq \varepsilon$, then $|a(\theta) - rb(\theta)| \leq \varepsilon$. For every fixed $\sigma > 0$ there are two alternatives.

If $|b(\theta)| \leq \sigma$, then

$$\text{dist}(y, x + m^\perp) \lesssim \delta/\rho + \sigma + \varepsilon. \quad (6.20b82)$$

If $|b(\theta)| > \sigma$, then on that projective ray the admissible radii have diameter at most $2\varepsilon/\sigma$. The same one-radius conclusion holds for the small-defect condition $|\mathcal{R}_{\mathbf{a}}(r\theta)| \leq \varepsilon$, with a fixed positive lower bound in place of σ .

Let ω_A be a common modulus of continuity of $\theta \mapsto A(\mathcal{G}^{-1}(\theta))n$, uniform in unit n . Let S be a family of transformed test points satisfying

$$|n_{\mathbf{a}} \cdot v| \leq h, \quad \text{dist}(\mathcal{R}_{\mathbf{a}}(v), \text{span}(m)) \leq \varepsilon, \quad |b(\theta)| \geq \sigma.$$

If $0 < R < 1$ satisfies

$$h \leq c\rho\sigma R, \quad \varepsilon \leq c\sigma R, \quad \omega_A(C\rho\sigma R) \leq c\sigma R, \quad (6.20b83)$$

then, for $\sigma_x = (\mathcal{G}_x)_\# v$,

$$\sigma_x(S) \lesssim (\rho\sigma)^{-(d-2)} R^{\kappa-(d-2)}. \quad (6.20b84)$$

The same estimate holds for the small-defect branch with σ replaced by an absolute constant.

Proof. Equation (6.20b81) follows because $\mathcal{G}^{-1}$ is one-homogeneous and $A = D\mathcal{G}$ is zero-homogeneous. If $|b| \leq \sigma$, then $|P_m g_{\mathbf{a}}(\theta)| \lesssim \sigma + \varepsilon$. Uniform positive definiteness gives $|g_{\mathbf{a}}(\theta)| \gtrsim 1$, so $g_{\mathbf{a}}(\theta)$ lies within $O(\sigma + \varepsilon)$ of $[m]$. Symmetry and Euler homogeneity give

$$g_{\mathbf{a}}(v) \cdot \mathcal{G}^{-1}(v) = n_{\mathbf{a}} \cdot v,$$

and (6.20b82) follows from $|n_{\mathbf{a}} \cdot v| \lesssim \delta/\rho$. If instead $|b| > \sigma$, two admissible radii r, r' satisfy $|(r - r')b| \leq 2\varepsilon$, proving the radial interval bound. The small-defect variant is identical: once the defect is small, the uniformly nonzero vector $g_{\mathbf{a}}$ forces $|\mathcal{T}_{\mathbf{a}}(P_H\theta)| \gtrsim 1$.

For the covering estimate, put $s = c\rho\sigma R$ and cover the great sphere $S(H_{\mathbf{a}})$ by $O(s^{-(d-2)})$ caps of radius s . The first condition in (6.20b83) absorbs the normal angular thickness. On one cap, b varies by $O(s/\rho) = O(\sigma R)$, while a varies by $O(\omega_A(Cs)) = O(\sigma R)$. The radial rigidity and the second condition in (6.20b83) therefore place all admissible transformed points in one $O(R)$ -ball. Since σ_x is uniformly κ -Frostman, summing over the caps proves (6.20b84). $\square$

Lemma III.6.24.32 (Hölder exclusion of the non-track degenerate fibre). *Assume in addition that N is $C^{2,\beta}$ off the origin, restrict to the the exact-belt exclusion full base set on which the transformed measure σ_x charges no affine hyperplane, and put $q = \kappa - (d - 2)$. If*

$$\beta(2\kappa - (d - 2)) > \kappa, \quad (6.20b85)$$

then the free parameters may be chosen so that, for every δ^τ -heavy belt of mass m and every robust anchor tuple, the small-defect and non-track radial test sets have $o(m)$ mass, uniformly as $\delta \downarrow 0$.

More precisely, choose c with

$$\frac{\kappa}{q} < c < \frac{\beta}{1 - \beta}$$

when $\beta < 1$ (and any $c > \kappa/q$ when $\beta = 1$), and put $R = \rho^c$. Then

$$v\{\text{radial non-track tests}\} \lesssim \rho^{cq-(d-2)} = o(\rho^q) = o(m). \quad (6.20b86)$$

After excluding the transverse determinant estimate transverse branch and taking the small-defect threshold $R_0 = \rho^c$, a non-small first defect may be used to define its own projective line $[m]$. If a second defect has wedge at most ξ , then

$$\text{dist}(\mathcal{R}_{\mathbf{a}'}(v'), \text{span}(m)) \leq \xi/R_0. \quad (6.20b87)$$

Choosing $\xi \leq cR_0^2$ therefore gives the track/radial split pairwise, with no projective-cap loss. Fubini and (6.20b86) show that every witness-pair class containing a radial component is $o(m^d)$. Hence any positive fraction of the degenerate witness fibre that survives is asymptotically track-track.

Proof. The $C^{2,\beta}$ hypothesis gives $\omega_A(t) \lesssim t^\beta$. On the stated the exact-belt exclusion base set, compactness of the hyperplane parameter space and $\sigma_x(H) = 0$ for every affine hyperplane imply $\sup_H \sigma_x(H^{(r)}) \rightarrow 0$ as $r \downarrow 0$; hence the masses m of shrinking heavy belts tend to zero. Condition (6.20b85) is exactly the condition allowing a c in the displayed interval. With $R = \rho^c$, the modulus condition in (6.20b83) follows from $c < \beta/(1 - \beta)$, while the normal-thickness condition follows after decreasing τ so that $\delta \ll \rho^{2+c}$. Lemma III.6.24.31 then gives (6.20b86); the inequality $cq > \kappa = (d - 2) + q$ makes the right-hand side $o(\rho^q)$.

For (6.20b87), take $[m]$ to be the line generated by the first non-small defect. The distance of the second defect from this line is its wedge with the first divided by the first defect length. The final statement follows by applying the pointwise $o(m)$ bound to the test variable, uniformly in the anchors and in the pair-dependent line, and then integrating over the robust anchor families. The transverse determinant estimate polynomial transverse threshold is compatible with $\xi \leq cR_0^2$ after decreasing τ and the auxiliary exponent once more. $\square$

Anchor-test radial rigidity. Lemma III.6.24.30 replaces the symmetric the transverse determinant estimate normalization by $d - 1$ robust anchors and one test point. It improves the normalized wedge error to $O(\delta^{1-3\tau/q})$ and identifies the normalized determinant gradient with the affine interpolation defect (6.20b79). Homogeneity then gives the exact radial formula (6.20b81). Every near-parallel pair therefore splits, pairwise and without a

global projective-cap choice, into a physical affine-hyperplane track or a one-radius-per-direction radial branch. Lemma III.6.24.31 gives the exact modulus-aware Frostman bound (6.20b84) for the latter. Under the stronger $C^{2,\beta}$ condition (6.20b85), Lemma III.6.24.32 makes every non-track radial component too small to carry a positive fraction of a heavy witness fibre, so only the genuine track-track branch survives.

Under the original bare C^2 hypothesis the modulus in (6.20b83) can be superflat, so the radial graph branch still has no universal polynomial mass bound. The physical track branch is also not base-small in general, as the persistent-belt counterexample already shows. The next exact task is therefore a persistence/entropy theorem on the track-track branch, or a direct nonlinear projection argument there; no return to one-belt pruning or a universal T_δ power estimate is justified.

Lemma III.6.24.33 (Euclidean-radial preconditioning closes the Hölder two-heavy branch). *Retain the separated compact zero-hyperplane setup in dimension $d \geq 3$, put*

$$q = \kappa - (d - 2) > 0, \quad p_0 = \alpha - (d - 2) > 0,$$

and let $q_0 > 0$. Suppose N is $C^{2,\beta}$ off the origin, with $0 < \beta \leq 1$. Suppose moreover that on a compact base set X_ there is a Borel parameter kernel*

$$dv_x(y) = f(x, y) dv(y), \quad 0 \leq f \leq C_0,$$

such that every v_x is a probability measure,

$$\sup_{x \in X_*} \sup_H v_x(\mathcal{B}_x(H, r)) \longrightarrow 0 \quad (r \downarrow 0), \quad (6.20b88a)$$

and

$$(e_x)_\# v_x(H^{(r)}) \leq C_0 r^{q_0} \quad (H \in G(d, d-1), 0 < r \leq r_0). \quad (6.20b88)$$

Set

$$\Theta = \kappa \left(\frac{1}{q} + \frac{1}{q_0} \right). \quad (6.20b89)$$

Assume either $\beta = 1$, or

$$\frac{\beta}{1 - \beta} > \Theta. \quad (6.20b90)$$

Then one may choose $\tau > 0$ and $a > 0$, depending only on the displayed exponents and the fixed compact setup, so that for every sufficiently small δ there is a Borel set $T_\delta^\Gamma \subset X_$ with*

$$\mu(T_\delta^\Gamma) \lesssim \delta^a \quad (6.20b91)$$

and the following property: if $x \in X_ \setminus T_\delta^\Gamma$, then all v_x -heavy $A\delta$ Legendre belts through x are contained in one $A'\delta^{\tau/2}$ Legendre belt. Here “heavy” means v_x -mass at least $C\delta^\tau$. After increasing A' by a harmless fixed factor, the exceptional hyperplane may be chosen Borel as a function of x on $X_* \setminus T_\delta^\Gamma$.*

Proof. Choose exponents $s, c > 0$ such that

$$sq_0 > q, \quad cq > \kappa + (d-2)s, \quad s + c < \frac{\beta}{1-\beta} \quad (6.20b92)$$

when $\beta < 1$; for $\beta = 1$ omit the last inequality. Such a choice exists precisely under (6.20b90), because the infimum of $s + c$ subject to the first two strict inequalities is $\kappa/q_0 + \kappa/q = \Theta$.

By (6.20b88a), the supremum of all shrinking heavy-belt masses tends to zero uniformly on X_* . Hence every power comparison below which is written as $o(m)$ is uniform in the base point, hyperplane direction, and heavy belt once δ is sufficiently small.

Fix two $A\delta$ -belts which are heavy for ν_x , with masses m_1, m_2 and robust anchor scales $\rho_i \asymp m_i^{1/q}$ from Lemma III.6.24.30. Order them so that $\rho_1 \leq \rho_2$ and put

$$\sigma_i = \rho_i^s, \quad R_i = \rho_i^c, \quad \xi = \rho_1^{s+2c}. \quad (6.20b93)$$

Decrease the already free τ so that all the conditioned-witness reduction-70 small- τ requirements hold and, in addition,

$$\tau(3 + s + 2c) < q. \quad (6.20b94)$$

Then $\delta/\rho_i \ll \rho_i^s$ and the normalized-gradient error $O(\delta/\rho_1^3)$ is $o(\xi)$.

The small-defect part of the i th witness fibre is estimated by Lemma III.6.24.31 with covering radius R_i and the fixed coefficient lower bound from its small-defect clause. Hence its test mass is

$$\lesssim \rho_i^{cq-(d-2)} = o(\rho_i^q) = o(m_i). \quad (6.20b95)$$

For a non-track radial branch use $\sigma_i = \rho_i^s$ and $R_i = \rho_i^c$. The $C^{2,\beta}$ modulus condition is exactly $\beta(1 + s + c) > s + c$, while the second inequality in (6.20b92) gives

$$\nu_x\{\text{radial non-track tests}\} \lesssim \rho_i^{cq-(d-2)(1+s)} = o(\rho_i^q) = o(m_i). \quad (6.20b96)$$

It remains to control the physical-track alternative. Equation (6.20b82) and the choice $\xi/R_1 \leq \rho_2^{s+c}$ place every track test from the i th belt in an ordinary physical slab through x of width $O(\rho_i^s)$. Since X_* and Y are separated, this slab maps under e_x into an $O(\rho_i^s)$ great-hyperplane strip. Therefore (6.20b88) gives

$$\nu_x\{\text{physical-track tests}\} \lesssim \rho_i^{sq_0} = o(\rho_i^q) = o(m_i). \quad (6.20b97)$$

The first inequality in (6.20b92) is exactly the last comparison.

For each heavy belt, Lemma III.6.24.30 supplies a robust anchor family of mass $\gtrsim m_i^{d-1}$, and the unrestricted test variable has mass m_i . Thus the two belts supply an anchor-test witness-pair fibre of mass $\gtrsim m_1^d m_2^d$. If the first defect is smaller than R_1 , or the second is smaller than R_2 , then (6.20b95) and Fubini make that witness-pair class $o(m_1^d m_2^d)$. Otherwise use the first defect line as in (6.20b87). If the normalized defect wedge is at most ξ , then

$$\xi/R_1 \leq \rho_2^{s+c} = \sigma_2 R_2,$$

so, after the first witness is fixed, the second witness falls into the robust-anchor decomposition track/radial dichotomy relative to the fixed line generated by the first defect. The first witness is used only to define this line and is not itself charged to a moving-line exceptional set. Uniformly in that first witness and in both anchor tuples, (6.20b96) and (6.20b97) make the second-test fibre $o(m_2)$. Fubini therefore makes the corresponding witness-pair class $o(m_1^d m_2^d)$. Hence a fixed positive fraction of the full anchor-test witness-pair fibre must satisfy normalized defect wedge at least ξ .

The absolute determinant two-Jacobian on this remaining fibre is at least

$$j_\delta \geq \rho_1^{d-2} \rho_2^{d-2} \xi \geq \delta^\zeta, \quad \zeta = \frac{\tau}{q}(2(d-2) + s + 2c). \quad (6.20b98)$$

Decrease τ once more so that

$$\zeta(1 + 1/\beta) < 1, \quad p_0 - 2d\tau - (\alpha + 2/\beta)\zeta > 0. \quad (6.20b99)$$

Let $\mathfrak{F}_\delta^\Gamma$ be the Borel set of pairs of anchor-test tuples for which both determinant sublevels are $O(\delta)$ and the absolute two-Jacobian is at least the right-hand side of (6.20b98). Let T_δ^Γ be the Borel level set on which the $\nu_x^d \times \nu_x^d$ fibre mass of $\mathfrak{F}_\delta^\Gamma$ is at least a fixed small multiple of $\delta^{2d\tau}$. The preceding witness argument shows that, for all sufficiently small δ , every base carrying two separated heavy belts belongs to T_δ^Γ .

Apply Lemma III.6.24.27 to the pair of anchor-test determinants. The $C^{2,\beta}$ regularity gives exactly the transverse determinant estimate Hölder radius $r_j \geq j^{1/\beta}$. Moreover the product kernel satisfies

$$\mu(dx) \nu_x^d(dy) \nu_x^d(dy') \leq C_0^{2d} \mu(dx) \nu^d(dy) \nu^d(dy')$$

as Borel measures. Hence Fubini, the transverse sublevel bound, and (6.20b99) give the same power base loss as (6.20b69); this proves (6.20b91).

Outside T_δ^Γ , two separated heavy belts are therefore impossible. The geometric first alternative of Lemma III.6.24.15 gives the one exceptional $A'\delta^{\tau/2}$ belt. For the measurability assertion, fix a finite $O(\delta)$ -net (H_j) in the Grassmannian and use belts with a harmlessly enlarged $A\delta$ width. For each j the map $x \mapsto \nu_x(\mathcal{B}_x(H_j, 2A\delta))$ is Borel, since the kernel and belt incidence relation are Borel. Select the first net direction whose enlarged belt is heavy, with one fixed default direction if none is heavy. Every genuinely heavy belt is contained in one such enlarged net belt, and outside T_δ^Γ all selected heavy net directions lie within $O(\delta^{\tau/2})$ of one another. Increasing A' once more therefore gives a Borel exceptional hyperplane with the claimed pruning property. $\square$

Lemma III.6.24.34 (Local closure for sufficiently Hölder curved norms). *In the residual zero-hyperplane branch of Theorem III.6.25, let $d \geq 3$ and $d-2 < \kappa \leq d-1$. Put*

$$q = \kappa - (d-2), \quad q_0 = q_{d,d-1}(\alpha - (d-2), q),$$

where $q_{d,d-1}$ is the continuous positive exponent from the locally proved spherical-direction theorem, and define

$$\beta_*(\alpha, \kappa) = \frac{\kappa(q^{-1} + q_0^{-1})}{1 + \kappa(q^{-1} + q_0^{-1})} < 1. \quad (6.20b100)$$

If N is $C^{2,\beta}$ off the origin with everywhere positive Gaussian curvature and

$$\beta > \beta_*(\alpha, \kappa), \quad (6.20b101)$$

then, for this fixed β , there is a positive pinned gain $c = c(d, \alpha, \kappa, \beta)$, independent of the particular norm, in this exponent pair. In particular, at $\beta = 1$ the gain depends only on d, α, κ ; hence the theorem-form normed conclusion is local for every $C^{2,1}$ positively curved norm throughout the full residual band $d - 2 < \kappa \leq d - 1$.

Proof. In the strict dual argument choose the Frostman measures μ and ν of exponents α and κ as in the existing proof. The affine $(d - 2)$ -plane Frostman estimates give the exponents $p_0 = \alpha - (d - 2)$ and $q = \kappa - (d - 2)$. Apply the locally proved spherical-direction theorem with $k = d - 1$. Use the standard Borel replacement and countable uniformisation from the proof of Theorem III.6.24. After restricting to a positive- μ compact base set, obtain a Borel kernel ν_x with uniform positive original ν -mass and the radial decay (6.20b88) at exponent q_0 .

Intersect this base with the exact-belt exclusion full exact-belt-null set and use compactness to get the uniform exact-belt atom-free modulus. The density bound $\nu_x \leq C\nu$ preserves all Frostman and codimension-two estimates. Since (6.20b101) is exactly (6.20b90), Lemma III.6.24.33 supplies a one-scale pruning statement with power-summable base loss for the kernels ν_x , together with a Borel exceptional direction at every retained scale.

Choose the synchronized superexponential scales of Lemma III.6.24.12. After discarding the summable base losses, its residue-class argument applies pointwise to each fixed kernel ν_x . Because the scale-by-scale exceptional directions are Borel, the complement of their countable union is a Borel parameter fibre of uniformly positive ν_x -mass. In the present application ν_x is the normalized restriction of the original ν to the Borel Euclidean-radial fibre Λ_x , whose original ν -mass is uniformly positive. The final fibre therefore has uniformly positive original ν -mass, and normalizing directly from ν gives the same measure as restricting ν_x in two stages. Its normed direction measure has a positive hyperplane-decay power, exactly in the interface of Proposition III.6.23.1. That proposition then yields the desired pinned image.

For the endpoint $\dim_{\mathbb{H}} A \geq \alpha$, choose $\alpha' < \alpha$ sufficiently close to α . Continuity of $q_{d,d-1}$ makes $\beta_*(\alpha', \kappa) \rightarrow \beta_*(\alpha, \kappa)$, so the strict inequality in (6.20b101) persists; the existing continuity argument for the pinned gain then gives the endpoint exactly as before. $\square$

Euclidean-radial preconditioning and Hölder closure. Lemma III.6.24.33 uses the Euclidean radial restriction already produced by the locally proved spherical-direction theorem before any norm Gauss-map transfer is attempted. On that positive-mass parameter kernel, ordinary radial great-hyperplane strips have power decay (6.20b88). Since the physical track in (6.20b82) is exactly such a strip, the track component can be made $o(m)$ simultaneously with the robust-anchor decomposition small-defect and radial non-track components. The required exponent balance is (6.20b92); its sharp feasibility threshold is (6.20b90). Consequently two separated heavy Legendre belts can occur only in the transverse determinant estimate transverse branch, which has the power-small base loss (6.20b91). The same-base telescope lemma telescope then closes the one-scale pruning on the preconditioned kernel.

Lemma III.6.24.34 therefore gives, for every fixed $\beta > \beta_*(\alpha, \kappa)$ from (6.20b100), a positive norm-independent pinned gain which may depend on β . At $\beta = 1$ this is a gain depending only on d, α, κ , so the theorem-form conclusion is local for every $C^{2,1}$ positively curved norm throughout $d-2 < \kappa \leq d-1$. The original bare- C^2 clause is still not promoted: after radial preconditioning the remaining obstruction is precisely the superflat-modulus failure of the robust-anchor decomposition radial one-radius estimate.

Lemma III.6.24.35 (Critical little-Hölder closure of the preconditioned two-heavy branch). *Retain the separated compact zero-hyperplane setup and the Euclidean-radial preconditioned kernel from Lemma III.6.24.33. Put*

$$\begin{aligned} q &= \kappa - (d-2), & q_0 &= q_{d,d-1}(\alpha - (d-2), q), \\ s_0 &= \frac{q}{q_0}, & c_0 &= \frac{\kappa + (d-2)s_0}{q}, & \Theta &= s_0 + c_0 = \kappa \left(\frac{1}{q} + \frac{1}{q_0} \right), \end{aligned}$$

and

$$\beta_* = \frac{\Theta}{1 + \Theta}. \quad (6.20b102)$$

Let $A = D\mathcal{G}$, with $\mathcal{G} = D(N^2/2)$, and let ω_A be a common modulus of continuity of A on the separated annulus. Assume the critical little-Hölder condition

$$\eta(r) := \sup_{0 < t \leq r} \frac{\omega_A(t)}{t^{\beta_*}} \longrightarrow 0 \quad (r \downarrow 0). \quad (6.20b103)$$

Then one may choose $\tau > 0$ and $a > 0$ from d, α, κ alone (the constants and starting scale may depend on the fixed compact setup and the norm), so that the conclusion of Lemma III.6.24.33 holds: outside a Borel set T_δ^{crit} with

$$\mu(T_\delta^{\text{crit}}) \lesssim \delta^a, \quad (6.20b104)$$

all v_x -heavy $A_0\delta$ Legendre belts through x are contained in one $A_1\delta^{\tau/2}$ Legendre belt, with a Borel exceptional direction. In particular the same-base telescope closes on the preconditioned kernel.

Proof. The critical identities are

$$s_0 q_0 = q, \quad c_0 q = \kappa + (d-2)s_0, \quad \beta_*(1 + \Theta) = \Theta. \quad (6.20b105)$$

Choose

$$a_0 = 1, \quad b_0 = 1 + \frac{d-2}{q}, \quad K = a_0 + b_0.$$

After enlarging the harmless constant in the argument of η , define, for small ρ ,

$$L(\rho) = \min \left\{ |\log \rho|, \eta(C\rho^{1+\Theta})^{-1/[2K(1-\beta_*)]} \right\}, \quad (6.20b106)$$

with the second entry interpreted as $+\infty$ when the numerator vanishes. Then $L(\rho) \rightarrow \infty$ and $L(\rho) = \rho^{-o(1)}$. For a heavy belt of mass m and robust-anchor scale $\rho \asymp m^{1/q}$, set

$$\sigma(\rho) = \rho^{s_0} L(\rho)^{-a_0}, \quad R(\rho) = \rho^{c_0} L(\rho)^{-b_0}. \quad (6.20b107)$$

The Euclidean radial preconditioner gives

$$v_x\{\text{physical-track tests}\} \lesssim \sigma^{q_0} = \rho^q L^{-q_0} = o(m). \quad (6.20b108)$$

The robust-anchor decomposition radial estimate gives

$$v_x\{\text{radial non-track tests}\} \lesssim (\rho\sigma)^{-(d-2)} R^q = \rho^q L^{a_0(d-2)-b_0q} = \rho^q L^{-q} = o(m). \quad (6.20b109)$$

The small-defect version is even smaller, since

$$\rho^{-(d-2)} R^q = \rho^{q+(d-2)s_0} L^{-b_0q} = o(\rho^q). \quad (6.20b110)$$

For this small-defect cover the coefficient lower bound is absolute, so its modulus condition is $\omega_A(C\rho R) \lesssim R$. At the critical exponent,

$$\beta_*(1 + c_0) - c_0 = \frac{s_0}{1 + \Theta} > 0,$$

and therefore $\omega_A(C\rho R)/R \rightarrow 0$ by (6.20b103); this branch has a strict power margin.

It remains to verify the harder radial non-track modulus condition in (6.20b83). Equations (6.20b105)–(6.20b107) give

$$\rho\sigma R = \rho^{1+\Theta} L^{-K}, \quad \sigma R = \rho^\Theta L^{-K}.$$

Hence

$$\frac{\omega_A(C\rho\sigma R)}{\sigma R} \lesssim \eta(C\rho^{1+\Theta}) L^{K(1-\beta_*)} \rightarrow 0 \quad (6.20b111)$$

by the definition of L . Since $L = \rho^{-\alpha(1)}$ and every heavy belt satisfies $\rho \gtrsim \delta^{\tau/q}$, decreasing the free τ also gives

$$\delta/\rho = o(\rho\sigma R), \quad \delta/\rho^3 = o(\rho^{s_0+2c_0} L^{-a_0-2b_0}). \quad (6.20b111a)$$

Hence the normal-thickness condition in (6.20b83) and the robust-anchor decomposition normalized-gradient comparison error are both negligible. Thus all three degenerate second-test fibres are $o(m)$, uniformly on the retained compact base.

For a pair of heavy belts, let ρ_i, σ_i, R_i be the corresponding scales. To avoid any monotonicity requirement on the slowly varying factor, set

$$\xi = c \min(R_1, R_2) \min(\sigma_1 R_1, \sigma_2 R_2). \quad (6.20b112)$$

If the first defect is non-small, use the belt with the smaller R_i as the first witness. Then a normalized defect wedge at most ξ puts the second defect within its own the robust-anchor decomposition track/radial tolerance relative to the fixed first-defect line. The first witness is used only to define that line. Fubini together with (6.20b108)–(6.20b110) therefore makes all low-wedge or small-defect witness-pair classes $o(m_1^d m_2^d)$.

Put

$$E_0 = 2(d-2) + s_0 + 2c_0.$$

Because $L(\rho) \leq |\log \rho|$, robust conditioning and (6.20b112) give, on the surviving witness fraction,

$$J \gtrsim \rho_{\min}^{E_0} |\log \rho_{\min}|^{-C_0} \gtrsim \delta^\xi, \quad \zeta = \frac{2\tau E_0}{q}, \quad (6.20b112a)$$

for all sufficiently small δ ; the factor two absorbs every logarithmic loss. The little-Hölder assumption also implies an ordinary β_* -Hölder bound below a norm-dependent starting scale. Decrease τ once more so that

$$\zeta(1 + 1/\beta_*) < 1, \quad \alpha - (d - 2) - 2d\tau - (\alpha + 2/\beta_*)\zeta > 0. \quad (6.20b112b)$$

The transverse determinant estimate Hölder sublevel estimate therefore applies with $\beta = \beta_*$ and gives a positive power base loss, proving (6.20b104). The finite Grassmannian-net Borel selector and Lemma III.6.24.12 then apply exactly as in the Euclidean-radial preconditioning. $\square$

Lemma III.6.24.36 (Sharp modulus boundary of the robust-anchor decomposition–71 one-scale balance). *Retain the Euclidean-radial preconditioning preconditioned setup and write $m = \rho^q$. Suppose arbitrary positive functions $\sigma(\rho), R(\rho)$ are chosen so that the two estimates used to discard the physical-track and radial non-track pieces satisfy*

$$\sigma(\rho)^{q_0} = o(\rho^q), \quad (\rho\sigma(\rho))^{-(d-2)}R(\rho)^q = o(\rho^q). \quad (6.20b113)$$

Then necessarily

$$\sigma = o(\rho^{q/q_0}), \quad R = o(\rho^{\kappa/q} \sigma^{(d-2)/q}), \quad u := \sigma R = o(\rho^\Theta). \quad (6.20b114)$$

Consequently any proof which also invokes the robust-anchor decomposition cover condition must find $u = o(\rho^\Theta)$ for which

$$\omega_A(Cpu) \lesssim u. \quad (6.20b115)$$

This compatibility need not hold for a general continuous modulus. In particular, if

$$\omega_A(t) \gtrsim \frac{1}{\log(e/t)} \quad (6.20b116)$$

for all sufficiently small t , then (6.20b115) is impossible for all sufficiently small ρ .

Logarithmically slow moduli of the form (6.20b116) are compatible with C^2 positive curvature. Thus, once a heavy belt of scale ρ is present, retuning σ , R , or the ambient δ -scale does not alter the scalar incompatibility above. A universal bare- C^2 proof within this architecture would therefore need a separate mechanism excluding precisely those incompatible heavy-belt regimes.

Proof. The first relation in (6.20b113) gives $\sigma = o(\rho^{q/q_0})$. The second gives

$$R^q = o(\rho^\kappa \sigma^{d-2}),$$

which is the second assertion in (6.20b114). Multiplying and using $q + (d - 2) = \kappa$ yields

$$\sigma R = o(\rho^{\kappa/q} \sigma^{\kappa/q}) = o(\rho^{\kappa/q + \kappa/q_0}) = o(\rho^\Theta).$$

The robust-anchor decomposition modulus requirement is precisely (6.20b115). Under (6.20b116), if (6.20b115) held then

$$1 \lesssim u \log \frac{e}{Cpu}.$$

But $u = o(\rho^\Theta)$ implies

$$u \log(1/u) \longrightarrow 0, \quad u \log(1/\rho) \longrightarrow 0,$$

so the right-hand side tends to zero. This contradiction proves incompatibility.

For the final assertion, choose in a coordinate chart of the sphere an even C^2 angular function whose second derivative contains a cutoff of $t \mapsto 1/\log(e/|t|)$, with value zero at $t = 0$, and symmetrize at the antipodal chart. Its two-homogeneous extension Q is C^2 off the origin and has Hessian modulus bounded below by a constant multiple of (6.20b116). For sufficiently small $\varepsilon > 0$,

$$F(z) = \frac{1}{2}|z|^2 + \varepsilon Q(z)$$

has positive-definite Hessian on the unit sphere. Then $N(z) = \sqrt{2F(z)}$ is an even strongly convex C^2 norm with positive Gaussian curvature, while $D^2F = D\mathcal{G}$ retains the logarithmically slow modulus. $\square$

Lemma III.6.24.37 (Local closure at the critical little-Hölder modulus). *In the residual zero-hyperplane branch of Theorem III.6.25, let $d \geq 3$ and $d - 2 < \kappa \leq d - 1$, and define $\beta_*(\alpha, \kappa)$ by (6.20b100). If*

$$\sup_{0 < t \leq r} \frac{\omega_{D\mathcal{G}}(t)}{t^{\beta_*(\alpha, \kappa)}} \longrightarrow 0 \quad (r \downarrow 0), \quad (6.20b117)$$

then there is a positive pinned gain $c = c(d, \alpha, \kappa) > 0$, independent of the particular norm, in this exponent pair. In particular this includes critical moduli $O(r^{\beta_}/\log(e/r)^M)$ for every $M > 0$.*

Proof. Apply the locally proved spherical-direction theorem first and uniformize exactly as in the proof of Lemma III.6.24.34. The resulting Borel kernel satisfies the hypotheses of Lemma III.6.24.35. Its power-small two-heavy base loss gives Borel one-scale pruning, and the same-base telescope furnishes a positive-mass original- ν restriction with positive-power Legendre hyperplane decay. Apply Theorem III.6.24. The exponent choices use only d, α, κ ; norm-dependent constants change only the starting scale. $\square$

Critical modulus balance. Lemma III.6.24.35 shows that the strict inequality $\beta > \beta_*(\alpha, \kappa)$ in the Euclidean-radial preconditioning is not the exact endpoint of the local method. At the critical exponent, a slowly varying loss in the robust-anchor decomposition parameters still makes physical tracks, radial non-track fibres, and small defects all $o(m)$ whenever the Hessian modulus is little-Hölder of order β_* . Lemma III.6.24.37 therefore adds the critical little-Hölder class to the locally proved normed theorem, with a gain depending only on d, α, κ .

Lemma III.6.24.36 also identifies the exact limitation of this one-scale architecture. Any simultaneous $o(m)$ elimination of the Euclidean-radial preconditioning physical track and the robust-anchor decomposition radial graph forces $u = \sigma R = o(\rho^\Theta)$, while the radial cover requires $\omega_A(C\rho u) \lesssim u$. Bare C^2 allows logarithmically slow Hessian moduli for which these conditions are incompatible. This is an obstruction to the robust-anchor decomposition–71 balancing mechanism, not a counterexample to the source-stated curved-norm pinned-distance theorem. The remaining arbitrary- C^2 branch

therefore requires a genuinely multiscale entropy/dimension-drop argument or a joint base-parameter treatment of the radial graph before one imposes a uniform cap scale.

Lemma (Antipodal zero lemma). *If $N \geq J \geq 1$ and $F : \mathbb{S}^N \rightarrow \mathbb{R}^J$ is continuous and odd, then $F(x) = 0$ for some $x \in \mathbb{S}^N$.*

Proof. Adding zero coordinates reduces to $J = N$. We prove the equivalent non-extension assertion: there is no continuous map $g : B^N \rightarrow \mathbb{S}^{N-1}$ whose restriction to ∂B^N is odd. Indeed, a nowhere-zero odd map on $\mathbb{S}^N$, restricted to its closed upper hemisphere and normalized, would give such a g .

For completeness, here is the finite parity argument behind that assertion, following the Tucker-lemma proof in Matoušek [36, §2.3]. Let $C_N = \{x \in \mathbb{R}^N : \sum_i |x_i| \leq 1\}$, which is radially homeomorphic to B^N by a map odd on the boundary. A *triangulation* of C_N means a finite collection of simplices (convex hulls of affinely independent vertices) covering C_N and intersecting face-to-face. Start with the cones from 0 over the faces of ∂C_N , then take repeated barycentric subdivisions. Here barycentric subdivision replaces each simplex by the convex hulls of increasing chains of its faces' barycentres. The resulting triangulations have arbitrarily small mesh (maximum simplex diameter), retain 0 as a vertex, are antipodally symmetric on the boundary, and refine each coordinate-sign orthant.

We claim that if the vertices of any such triangulation are labelled by $\{\pm 1, \dots, \pm N\}$ with opposite labels at antipodal boundary vertices, some edge has opposite labels at its endpoints. Suppose otherwise. For a simplex σ , let $S(\sigma)$ contain $+i$ or $-i$ according as the i th coordinate is positive or negative in the relative interior of σ ; a zero coordinate contributes nothing. Refinement of the sign orthants makes $S(\sigma)$ well-defined. Write $L(\sigma)$ for the set of its vertex labels. Call σ *happy* if $S(\sigma) \subset L(\sigma)$, and put $q = |S(\sigma)|$. It lies in a q -dimensional coordinate subspace and has at least q vertices with distinct labels. Thus a happy simplex has dimension $q - 1$ (*tight*) or q (*loose*). Every happy simplex on ∂C_N is tight.

Make a finite undirected graph whose vertices are the happy simplices. Join a boundary happy simplex to its antipode. Also join a happy facet σ to a happy simplex τ containing it whenever $L(\sigma) = S(\tau)$. The vertex $\{0\}$ has degree one: precisely the first edge from 0 in the signed coordinate direction given by its label is adjacent to it. Every other graph vertex has degree two, as the following exhaustive count shows.

If σ is tight, its q labels are exactly $S(\sigma)$. Within the q -dimensional coordinate crosspolytope, σ is a facet of exactly two q -simplices if it is interior, or exactly one if it is on the boundary. In the boundary case its antipode is the second graph neighbour. If σ is loose and $L(\sigma) = S(\sigma)$, exactly one label is repeated; deleting either of the two vertices with that label gives its two graph neighbours. In the remaining loose case $L(\sigma)$ has exactly one extra signed label $a \notin S(\sigma)$. Its opposite $-a$ is not in $S(\sigma)$, by the assumption that no edge has opposite labels. Deleting the vertex with label a gives one neighbour. On the a -side of the relevant $(q + 1)$ -dimensional coordinate crosspolytope, σ is a boundary facet of precisely one $(q + 1)$ -simplex with sign set $S(\sigma) \cup \{a\}$; this is its other neighbour. This completes the degree count. It contradicts the elementary fact that a finite graph has an even number of odd-degree vertices. The claimed opposite-labelled edge therefore exists.

If an odd-boundary map $g : C_N \rightarrow \mathbb{S}^{N-1}$ existed, choose a mesh small enough that $\|g(u) - g(v)\|_\infty < 2/\sqrt{N}$ on every edge. At each vertex v choose the least index i with

$|g_i(v)| \geq 1/\sqrt{N}$, and label v by $+i$ or $-i$ according to the sign of $g_i(v)$. The boundary labels are antipodal. Opposite labels on an edge force one coordinate difference at least $2/\sqrt{N}$, a contradiction. Hence no such g exists and the antipodal zero assertion follows. $\square$

Theorem (endpoint multilinear Kakeya). *Let $d \geq 3$ and $2 \leq r \leq d$. Here a unit tube means the radius-one neighbourhood of a finite line segment of positive length, with no upper bound on that length; the constant below is independent of all segment lengths. For $1 \leq i \leq r$, let $\mathcal{V}_i$ be finite families of such tubes in $\mathbb{R}^d$. If $e(V)$ is the unit direction of V , then*

$$\int_{\mathbb{R}^d} \left(\sum_{V_1 \in \mathcal{V}_1} \cdots \sum_{V_r \in \mathcal{V}_r} |e(V_1) \wedge \cdots \wedge e(V_r)| \mathbf{1}_{V_1 \cap \cdots \cap V_r}(x) \right)^{1/(r-1)} dx \lesssim_{r,d} \left(\prod_{i=1}^r |\mathcal{V}_i| \right)^{1/(r-1)}. \quad (6.20b116a)$$

The same estimate holds after a common dilation, with the corresponding scaling factor. This is the wedge-weighted form used by Ren [53, Theorem 2.28]. We give the polynomial-visibility proof of Carbery–Valdimarsson [35], including the case $r < d$. In addition to standard area-formula and finite-dimensional analysis facts, the non-elementary analytic inputs are the finite-dimensional John ellipsoid theorem in the form $E \subset K \subset \sqrt{d} E$ for a centrally symmetric convex body K , and the Euclidean isoperimetric inequality $\mathcal{H}^{d-1}(\partial U) \geq d \omega_d^{1/d} |U|^{(d-1)/d}$ for bounded domains with piecewise smooth boundary, where $\omega_d = |B(0, 1)|$. The antipodal zero lemma needed for the polynomial construction has just been proved.

Proof of the bilinear case. For two unit tubes V, W of angle θ , project onto the plane spanned by their directions. The two unit-width strips have intersection area at most $C/|\sin \theta|$; in each orthogonal fibre the intersection has $(d-2)$ -volume at most C_d . Thus

$$|V \cap W| \lesssim_d |e(V) \wedge e(W)|^{-1}$$

when the wedge is nonzero. If it vanishes, the weighted summand is zero. Tonelli's theorem now gives

$$\int \sum_{V \in \mathcal{V}_1, W \in \mathcal{V}_2} |e(V) \wedge e(W)| \mathbf{1}_{V \cap W} \lesssim_d |\mathcal{V}_1| |\mathcal{V}_2|,$$

which is (6.20b116a) for $r = 2$.

Proof of the cases $3 \leq r \leq d$. Write $n = d$ for the ambient dimension and $m = r$ for the number of families. We first work with finite collections of finite tubes; monotone convergence gives the corresponding assertion for unbounded tubes. The proof has two parts: construct a polynomial hypersurface whose directional area is large in every occupied unit cube, and use its degree to factor the wedge weight. We give both parts, since the factorisation is precisely where the exponent $1/(m-1)$ arises.

For a polynomial p of degree at most k , let $Z_p = \{p = 0\}$. If $U \subset \mathbb{R}^n$ is bounded and open, define its directional surface area by

$$A_e(p, U) = \int_{Z_p \cap U} |e \cdot \nu(z)| d\mathcal{H}^{n-1}(z), \quad e \in \mathbb{S}^{n-1},$$

where ν is the ordinary unit normal on the regular part of Z_p . Replacing p by its square-free part shows that the omitted singular part is $\mathcal{H}^{n-1}$ -null. The area formula on lines parallel to e and the elementary fact that a degree- k polynomial has at most k distinct roots on a line give

$$A_e(p, U) \leq k |\pi_{e^\perp} U|_{n-1}, \quad A_{e(T)}(p, T) \leq C_n k \quad (6.20b116a.1)$$

for every unit tube T of arbitrary length. It also holds for a fixed dilation of T and for unit cubes. If p is not square-free, the first inequality follows by applying the argument to its square-free part.

Let $\mathcal{Q}$ be the unit-cube lattice. Put

$$K(p, Q) = \left\{ \nu \in B(0, 1) : \int_{Z_p \cap Q} |\nu \cdot \nu(z)| d\mathcal{H}^{n-1}(z) \leq 1 \right\}, \quad \text{vis}(p, Q) = |K(p, Q)|^{-1/n}.$$

The body $K(p, Q)$ is centrally symmetric and convex. A *John ellipsoid* of a symmetric convex body K is an ellipsoid of maximal volume among those contained in K . To make its dependence on p continuous, fix k and equip the unit sphere $\mathcal{P}_k^*$ of the finite-dimensional space of degree- $\leq k$ polynomials with its round probability measure. For $\varepsilon > 0$ replace $A_e(p, U)$ by its average $A_{e,\varepsilon}(p, U)$ over the ε -ball about p in $\mathcal{P}_k^*$; define $K_\varepsilon(p, Q)$ and $\text{vis}_\varepsilon(p, Q)$ by the same formulas. For fixed ε , boundedness in (6.20b116a.1) and continuity in measure of a moving spherical ball imply that $p \mapsto A_{e,\varepsilon}(p, Q)$ is continuous, uniformly for $e \in \mathbb{S}^{n-1}$. Hence $p \mapsto K_\varepsilon(p, Q)$ is continuous in the Hausdorff metric. All cylinder estimates remain true with $A_{e,\varepsilon}$.

We need two geometric estimates. Suppose $S_e = A_{e,\varepsilon}(p, Q)$, $1 \leq S_e \leq D$ for every unit e , and set $V = \text{vis}_\varepsilon(p, Q)$. For unit $v_1, \dots, v_m$ the convex body contains, up to a dimensional factor, $\pm v_j / S_{v_j}$. Complete these vectors to a basis by orthonormal vectors in their orthogonal complement, and compare the volume of the resulting cross-polytope with $|K_\varepsilon(p, Q)|$. This proves

$$|v_1 \wedge \dots \wedge v_m|^{1/n} V \leq C_n D^{(n-m)/n} \prod_{j=1}^m S_{v_j}^{1/n}. \quad (6.20b116a.2)$$

The inequality is immediate if the wedge vanishes. For the second estimate, let E be a John ellipsoid of $K_\varepsilon(p, Q)$, with semiaxes $\ell_1 \geq \dots \geq \ell_n$. If $S_e \leq 1$ for some e , then $\ell_1 \geq c_n$. Since $\ell_1 \dots \ell_n \asymp_n V^{-n}$, it follows that $\ell_n \leq C_n V^{-n/(n-1)}$. The inclusion $K_\varepsilon(p, Q) \subset C_n E$ then gives $S_{e_n} \geq c_n / \ell_n$, where e_n is the shortest-axis direction. By (6.20b116a.1), $S_{e_n} \leq C_n k$. Thus

$$(\exists e \in \mathbb{S}^{n-1} : S_e \leq 1) \implies V^{n/(n-1)} \leq C_n k. \quad (6.20b116a.3)$$

The same argument with any fixed small threshold in place of 1 changes only the dimensional constant.

Here is the polynomial-visibility assertion, proved below: for every finitely supported $W : \mathcal{Q} \rightarrow [0, \infty)$ there are $k \leq C_n(1 + \sum_Q W(Q)^n)^{1/n}$, a nonzero $p \in \mathcal{P}_k^*$, and $\varepsilon > 0$ such that

$$\text{vis}_\varepsilon(p, Q) \geq c_n W(Q) \quad \text{for all } Q. \quad (6.20b116a.4)$$

We explain the Borsuk–Ulam construction rather than taking this assertion as an additional input.

We use the following elementary form of the antipodal covering principle. If $A_i \subset S^N$, $1 \leq i \leq J \leq N$, and $A_i \cap -A_i = \emptyset$, then the $2J$ sets $A_i, -A_i$ cannot cover S^N . Indeed,

$$x \mapsto (\text{dist}(x, -A_i) - \text{dist}(x, A_i))_{i=1}^J$$

is continuous and odd. A zero supplied by the preceding antipodal zero lemma belongs to none of the sets, because the displayed separation makes its i th coordinate strictly positive on A_i and strictly negative on $-A_i$.

To apply it, discard the cubes with $W(Q)$ below a fixed dimensional constant: their desired visibility bound follows from $K_\varepsilon(p, Q) \subset B(0, 1)$ for every p . Choose k with $\dim \mathcal{P}_k - 1 \geq C_n \sum_{Q: W(Q) \geq_n 1} W(Q)^n$; increasing C_n gives the degree bound in (6.20b116a.4). A John ellipsoid of a centrally symmetric body is comparable to that body by dimensional constants. We use a coloured net $\mathcal{E}$ of centred ellipsoids with the following properties: every such body is comparable to an ellipsoid in $\mathcal{E}$ by a fixed dimensional factor a_n ; the net has only C_n colours; and, within one colour, at most one ellipsoid is a_n -comparable to any given body. Moreover, if bodies $K_j \rightarrow K$ and ellipsoids E_j, E of one colour are respectively a_n -comparable to K_j, K , then eventually $E_j = E$. To construct the net, use the metric

$$d(E, F) = \inf\{\log a : a^{-1}E \subset F \subset aE\}$$

on positive-definite quadratic forms. Unit-radius balls in this finite-dimensional homogeneous metric have a uniform packing bound (conjugate a centre to the identity and compare eigenvalues). A maximal separated net therefore covers, and its bounded-degree proximity graph can be coloured with C_n colours. Choose its same-colour separation larger than $2 \log a_n + 1$. These observations prove all three claimed properties.

Fix a small constant c_n in the definition of a bad polynomial, $\text{vis}_\varepsilon(p, Q) < c_n W(Q)$. Since visibility is bounded below by a dimensional constant, only cubes with $W(Q) \geq_n 1$ matter. Divide bad polynomials into dyadic classes

$$\text{vis}_\varepsilon(p, Q) \asymp 2^{-j} W(Q) =: L_{Q,j}, \quad 1 \lesssim L_{Q,j} \lesssim W(Q).$$

There are finitely many such classes. In each class choose a colour Θ of an ellipsoid $E(p) \in \mathcal{E}$ comparable to $K_\varepsilon(p, Q)$. Restriction to one colour makes $E(p)$ unique. Its volume is $\asymp L_{Q,j}^{-n}$ and, because $K_\varepsilon(p, Q) \subset B(0, 1)$, all its semiaxes are bounded by C_n . For a fixed sufficiently small $\eta = \eta_n > 0$, one can place at least $c_n \eta^{-n} L_{Q,j}^n$ disjoint translates of $\eta E(p)$ inside Q : use a rectangular lattice along the principal axes in a small ball about the centre of Q . Label the translates by $1, \dots, N_{Q,j}$ with $N_{Q,j} \asymp_n L_{Q,j}^n$. The labels can be chosen consistently for every ellipsoid of this colour and volume class. There are only finitely many such ellipsoids: each is contained in a fixed ball and has volume bounded below, so the separated net is finite in this compact region. Thus all candidate ellipsoids and labelled translates, for the finite list of relevant Q, j, Θ , are fixed from W, k, η before the mollification radius ε is chosen.

At most $C_n \eta^{-(n-1)} L_{Q,j}^n$ of the available translates can be bisected by Z_p . Here is the count. If a polynomial zero set bisects a translate $\eta E + z$, both sign regions occupy half its volume. More generally, when both signs occupy at least 40% of a unit ball B , write their relative volumes as $a, b \in [0.4, 0.6]$. The isoperimetric inequality applied to the two sign

regions gives

$$2\mathcal{H}^{n-1}(Z_p \cap B) \geq \mathcal{H}^{n-1}(\partial B)(a^{(n-1)/n} + b^{(n-1)/n} - 1) \geq c_n > 0.$$

Indeed, the total perimeter of the two regions is at most the perimeter of B plus twice their common interior interface. The affine change of variables determined by the principal axes e_i and semiaxes ℓ_i of E now gives

$$\sum_{i=1}^n \ell_i A_{e_i}(p, \eta E + z) \geq c_n \eta^{n-1} |E|. \quad (6.20b116a.5)$$

The same inequality holds if each sign occupies at least 40% of the translate. Only finitely many translates occur in all the relevant Q, j, Θ, E . Since a nonzero polynomial has a Lebesgue-null zero set, compactness of $\mathcal{P}_k^*$ lets us choose one $\varepsilon > 0$ such that every polynomial in its ε -ball approximately bisects each translate bisected by the original polynomial. Average (6.20b116a.5), and sum it over disjoint bisected translates. Comparability of E with $K_\varepsilon(p, Q)$ gives $\ell_i A_{e_i, \varepsilon}(p, Q) \leq C_n$, whereas $|E| \asymp L_{Q,j}^{-n}$. Consequently, the number of bisected translates is at most $C_n \eta^{-(n-1)} L_{Q,j}^n$. Choose η so small that this is strictly smaller than $N_{Q,j}$. Each bad polynomial therefore fails to bisect at least one labelled translate.

For fixed Q, j, Θ and label a , split the polynomials failing to bisect the a th translate into $A_{Q,j,\Theta,a}^+$ and $A_{Q,j,\Theta,a}^-$ according as the positive or negative sign occupies more than half of that translate. Negation interchanges these sets. They satisfy

$$A_{Q,j,\Theta,a}^+ \cap \overline{A_{Q,j,\Theta,a}^-} = \emptyset. \quad (6.20b116a.6)$$

Indeed, if $p_h \in A^-$ tends to $p \in A^+$, continuity of K_ε and the same-colour net property imply $E(p_h) = E(p)$ eventually. Thus the labelled translated ellipsoid is eventually fixed. Lebesgue-nullity of $\{p = 0\}$ implies convergence of its positive-sign volume under $p_h \rightarrow p$, which contradicts the two strict sign inequalities. Notice that mere disjointness of A^+ and A^- would not suffice for the covering principle; the closure separation is essential.

The number of antipodal pairs is bounded by

$$C_n \sum_Q \sum_{j \geq 0} 2^{-jn} W(Q)^n \leq C_n \sum_Q W(Q)^n.$$

By the choice of k this does not exceed $\dim \mathcal{P}_k - 1$. The antipodal covering principle produces a polynomial outside every bad class, proving (6.20b116a.4). The harmless cases $W(Q) \leq 1$ are covered by the universal lower bound on visibility, after the fixed constant c_n is chosen small enough.

We now extract the Kakeya estimate. For a finitely supported $M : \mathcal{Q} \rightarrow [0, \infty)$ with $\sum_Q M(Q)^n = 1$, choose $\lambda \gg_n 1$ so large that $\lambda M(Q)^n \gg_n 1$ for every Q with $M(Q) > 0$. Apply (6.20b116a.4) to $W = \lambda M$. We obtain

$$k \leq C_n \lambda, \quad V_Q := \text{vis}_\varepsilon(p, Q) \geq c_n \lambda M(Q).$$

Increasing λ further if necessary makes $k < c_n V_Q^{n/(n-1)}$ for every occupied Q . By the contrapositive of (6.20b116a.3), $A_{e,\varepsilon}(p, Q) > 1$ for all unit e on such Q . The cylinder estimate also gives $A_{e,\varepsilon}(p, Q) \leq C_n \lambda$. For a tube T and a unit cube Q meeting it, set

$$S(Q, T) = \lambda^{-1} A_{e(T), \varepsilon}(p, Q).$$

If a cube meets T , it lies in a fixed dilation CT ; hence

$$\sum_{Q: Q \cap T \neq \emptyset} S(Q, T) \leq C_n \lambda^{-1} A_{e(T), \varepsilon}(p, CT) \leq C_n. \quad (6.20b116a.7)$$

Applying (6.20b116a.2) in each occupied Q to any m tubes T_i meeting Q , with $D = C_n \lambda$, gives

$$|e(T_1) \wedge \cdots \wedge e(T_m)| M(Q)^n \leq C_n \prod_{i=1}^m S(Q, T_i). \quad (6.20b116a.8)$$

Finally put $F(Q) = \sum_{T_i \cap Q \neq \emptyset} |e(T_1) \wedge \cdots \wedge e(T_m)|$. The original integral is at most $\sum_Q F(Q)^{1/(m-1)}$. Set $H = \sum_Q F(Q)^{1/(m-1)}$ and, if $H > 0$, take

$$M(Q)^n = H^{-1} F(Q)^{1/(m-1)}.$$

Then $\sum_Q M(Q)^n = 1$, and $F(Q)^{1/(m-1)} = F(Q)^{1/m} M(Q)^{n/m} H^{1/m}$. Use (6.20b116a.8), followed by Hölder in the cube variable and (6.20b116a.7), to obtain

$$H \leq C_n H^{1/m} \sum_Q \prod_{i=1}^m \left(\sum_{T_i \in \mathcal{T}_i: T_i \cap Q \neq \emptyset} S(Q, T_i) \right)^{1/m} \leq C_n H^{1/m} \prod_{i=1}^m |\mathcal{T}_i|^{1/m}.$$

Rearrangement yields $H \leq C_n (\prod_i |\mathcal{T}_i|)^{1/(m-1)}$. If $H = 0$ the conclusion is immediate. A common dilation changes both sides by its n th power, completing the proof. $\square$

Lemma III.6.24.38 (Furstenberg gain from endpoint multilinear Kakeya). *Fix an integer $0 \leq k < d - 1$, an exponent $0 \leq u < k + 1$, and $\kappa_0 \in (0, 1]$. There is $c_F = c_F(u, k, \kappa_0, d) > 0$ such that the following holds for all sufficiently small dyadic ϱ . Let $\mathcal{V}$ be a $(\varrho, \kappa_0, \varrho^{-c_F}, k)$ -set of ϱ -tubes with $|\mathcal{V}| \geq \varrho^{-2u+c_F}$. If every $V \in \mathcal{V}$ meets a family P_V of at least ϱ^{-1+c_F} pairwise essentially distinct ϱ -balls, then*

$$\left| \bigcup_{V \in \mathcal{V}} P_V \right| \geq \varrho^{-(u+1)-c_F}. \quad (6.20b116b)$$

Proof. Write $P = \bigcup_V P_V$ and argue by contradiction. Throughout the proof $\varrho^{O(c_F)}$ losses are harmless; at the end c_F is decreased once. Assume $|P| \leq \varrho^{-(u+1)+O(c_F)}$ and write $\mathcal{V}(p)$ for the tubes through p . Removing the ϱ^{2c_F} -ball about p from each tube and using bounded overlap of separated tubes gives the bush bound

$$|\mathcal{V}(p)| \leq \varrho^{-u-O(c_F)}. \quad (6.20b116c)$$

On the other hand

$$I(P, \mathcal{V}) = \sum_{p \in P} |\mathcal{V}(p)| = \sum_{V \in \mathcal{V}} |P_V| \geq \varrho^{-2u-1+O(c_F)}.$$

Together with (6.20b116c) this forces, after dyadic pigeonholing, a subfamily $P' \subset P$ with

$$|P'| = \varrho^{-(u+1)+O(c_F)}, \quad |\mathcal{V}(p)| = \varrho^{-u+O(c_F)} \quad (p \in P'). \quad (6.20b116d)$$

We next transfer the k -plane nonconcentration of the tubes to $(k+1)$ -plane nonconcentration of P' . If H is an $(r, k+1)$ -plate, decompose tubes meeting $P' \cap H$ according to the dyadic scale $r' \in [r, 1]$ at which they leave the concentric dilate of H . A tube in the r' shell meets at most $O(r/(r'\varrho))$ essentially distinct ϱ -balls of H , while the $(\varrho, \kappa_0, \varrho^{-c_F}, k)$ hypothesis gives $|\mathcal{V} \cap H_{r'}| \lesssim |\mathcal{V}| \varrho^{-c_F} (r')^{\kappa_0}$. Summing the geometric series and using (6.20b116d) yields

$$|P' \cap H| \lesssim \varrho^{-O(c_F)} |P'| r^{\kappa_0}. \quad (6.20b116e)$$

This is exactly the point-set transfer in Ren's proof of Theorem 3.5.

Refine the incidence graph twice. First discard tubes carrying fewer than a $\varrho^{O(c_F)}$ fraction of the average number of points of P' ; then discard points whose degree into the retained tube family is smaller than a $\varrho^{O(c_F)}$ fraction of the average. We obtain $\mathcal{V}' \subset \mathcal{V}$ and $\tilde{P} \subset P'$ such that

$$|\mathcal{V}'| = \varrho^{-2u+O(c_F)}, \quad |\tilde{P}| = \varrho^{-(u+1)+O(c_F)},$$

each $V \in \mathcal{V}'$ contains $\varrho^{-1+O(c_F)}$ points of P' , and each $p \in \tilde{P}$ lies in $\varrho^{-u+O(c_F)}$ tubes of $\mathcal{V}'$. After removing from each bush the $\varrho^{B c_F}$ -ball about its root, with B fixed large, the bush estimate (6.20b116c) implies that for every $p_0 \in \tilde{P}$ the union of the remaining points of P' reached by tubes in $\mathcal{V}'(p_0)$ has cardinality $\varrho^{-(u+1)+O(c_F)}$.

Put $\rho = \varrho^{A c_F}$ with $A \gg B$. Starting from $p_0 \in \tilde{P}$, choose successively p_i from that reached set, always together with a tube $V_i \in \mathcal{V}'$ satisfying $p_0, p_i \in V_i$, for $1 \leq i \leq k+2$. At stage i exclude the ρ -neighbourhood of the affine span of $p_0, \dots, p_{i-1}$. This span has dimension at most $k+1$ and hence is contained in an affine $(k+1)$ -plane. By (6.20b116e), the excluded points form at most a $\rho^{\kappa_0} \varrho^{-O(c_F)}$ fraction of the reached set. Thus, for A large and c_F small, there are

$$\varrho^{-(u+1)(k+3)+O(c_F)}$$

incidence-connected tuples $((p_0, \dots, p_{k+2}; V_1, \dots, V_{k+2}))$ for which

$$|e(V_1) \wedge \dots \wedge e(V_{k+2})| \geq \varrho^{O(c_F)}. \quad (6.20b116f)$$

The points are not chosen arbitrarily: every p_i is connected to the root p_0 by its recorded tube V_i . Pigeonholing $p_1, \dots, p_{k+2}$ leaves at least $\varrho^{-(u+1)+O(c_F)}$ admissible roots p_0 .

For $1 \leq i \leq k+2$, let $\mathcal{V}_i$ be the tubes through this fixed p_i which occur in the retained tuples. Then $|\mathcal{V}_i| \leq \varrho^{-u+O(c_F)}$. Dilate the ϱ -tubes to unit thickness and apply (6.20b116a) with $r = k+2$. Every admissible p_0 contributes one ϱ -ball on which one tube from each family overlaps, and the wedge loss in (6.20b116f) is absorbed into $\varrho^{-O(c_F)}$. The rescaled Kakeya inequality therefore bounds the number of admissible p_0 by

$$\varrho^{-u(k+2)/(k+1)-O(c_F)}.$$

Because $u < k+1$, one has $u(k+2)/(k+1) < u+1$. Taking c_F below a fixed fraction of this strict gap contradicts the preceding lower bound. This proves (6.20b116b). $\square$

Lemma III.6.24.39 (robust slicing from the local scalar filling spine). *Fix an integer $0 \leq k < d-1$, an exponent $0 \leq u < k+1$, and $\kappa_0 \in (0, 1]$. There is $c_S = c_S(u, k, \kappa_0, d) > 0$ with the following property. Let $\mathcal{V}$ be a $(\varrho, \kappa_0, \varrho^{-c_S}, k)$ -set of ϱ -tubes, all making a fixed positive*

angle with the horizontal hyperplane, with $|\mathcal{V}| \geq \varrho^{-2u+c_s}$. Let λ be a probability measure on the vertical coordinate satisfying

$$\lambda(B(z, r)) \leq \varrho^{-c_s} r^{k_0}, \quad \varrho \leq r \leq 1. \quad (6.20b116g)$$

Then there is $D \subset \mathbb{R}$ with $\lambda(D) \geq 1 - \varrho^{c_s}$ such that for every $z \in D$ and every $\mathcal{V}' \subset \mathcal{V}$ with $|\mathcal{V}'| \geq \varrho^{c_s} |\mathcal{V}|$,

$$N_\varrho\left(\bigcup \mathcal{V}' \cap \{x_d = z\}\right) \geq \varrho^{-u-c_s}. \quad (6.20b116h)$$

Proof. We expose the two ingredients because this is the only place where Ren's higher-dimensional slicing chain enters the Section 5 chain.

Weak slice. Suppose first, to the contrary, that every $z \in \text{spt } \lambda$ has slice covering number $\varrho^{-u+O(c_s)}$. Let z_1, z_2 be the extreme points of the support. From (6.20b116g), after deleting two endpoint intervals of length ϱ^{2c_s/k_0} , the remaining set has positive λ -mass and $|z_1 - z_2| \geq \varrho^{O(c_s)}$. Put X and Y for the two endpoint slices. Each tube is specified, up to $O(\varrho)$, by its intersections with these two hyperplanes, so

$$N_\varrho(X), N_\varrho(Y) = \varrho^{-u+O(c_s)}, \quad |\mathcal{V}| = \varrho^{-2u+O(c_s)}. \quad (6.20b116i)$$

For an intermediate height z , affine interpolation writes the slice as $X + f(z)Y$ after a harmless scalar normalization, where f is bi-Lipschitz on the trimmed height set. Hence $N_\varrho(X + f(z)Y) \leq \varrho^{-u+O(c_s)}$ there.

Fix one height z_* . The scale- ϱ Ruzsa triangle and Pluennecke–Ruzsa inequalities proved in the scalar spine above imply

$$N_\varrho(X - Xa) \leq \varrho^{-u+O(c_s)} \quad \text{for every } a = f(z)/f(z_*), \quad (6.20b116j)$$

and also $N_\varrho(X + X) \leq \varrho^{-u+O(c_s)}$. The pushforward of λ by $a = f(z)/f(z_*)$ retains the Frostman estimate (6.20b116g). Apply the locally proved scalar filling core (6.A42)–(6.A49) and approximate-ring closure (6.A50)–(6.A52). For any fixed small $\epsilon_0 > 0$, all $w \in [1 - \varrho^{\epsilon_0}, 1 + \varrho^{\epsilon_0}]$ are therefore good scalars:

$$N_\varrho(X + wf(z_*)Y) \leq \varrho^{-u+O(c_s)}. \quad (6.20b116k)$$

The corresponding vertical interval has length $\varrho^{\epsilon_0+O(c_s)}$. Take on every $V \in \mathcal{V}$ the ϱ -balls whose heights lie in this interval. There are at least $\varrho^{-1+\epsilon_0+O(c_s)}$ such balls per tube, while (6.20b116k) makes their union at most $\varrho^{-(u+1)+\epsilon_0+O(c_s)}$. For ϵ_0 and then c_s sufficiently small this contradicts Lemma III.6.24.38. Thus at least one good slice exists.

Robustification. For a tube family $\mathcal{A}$, write $E(\mathcal{A})$ for the heights at which some subfamily of size at least $\varrho^{c_s} |\mathcal{A}|$ has slice covering number below ϱ^{-u-c_s} . Suppose the intermediate assertion fails. Then one may choose z_1 and $\mathcal{V}^{(3)} \subset \mathcal{V}$ with $|\mathcal{V}^{(3)}| \geq \varrho^{c_s} |\mathcal{V}|$ and a small slice at z_1 , while $\lambda(E(\mathcal{V}^{(3)})) \geq \varrho^{c_s}$. The Frostman estimate (6.20b116g) leaves a point $z_2 \in E(\mathcal{V}^{(3)})$ outside the ϱ^{3c_s/k_0} -neighbourhood of z_1 . Choose $\mathcal{V}^{(2)} \subset \mathcal{V}^{(3)}$ witnessing small slices at both heights and prune further to $\mathcal{V}'$ so that a pair of endpoint ϱ -balls (x, y) determines at most one retained tube. If X, Y denote the two endpoint slices, incidence counting gives

$$N_\varrho(X), N_\varrho(Y) = \varrho^{-u+O(c_s)}, \quad |\mathcal{V}'| = \varrho^{-2u+O(c_s)}.$$

Let $D = E(\mathcal{V}') \setminus (B(z_1, \varrho^{3c_s/\kappa_0}) \cup B(z_2, \varrho^{3c_s/\kappa_0}))$; then $\lambda(D) \geq \varrho^{O(c_s)}$.

Write $z = az_1 + (1-a)z_2$ for $z \in D$, so $a, 1-a = \varrho^{O(c_s)}$. A bad witness at z gives a dense bipartite relation $P_z \subset X \times Y$ whose restricted affine sum $aX +_{P_z} (1-a)Y$ has covering number $\varrho^{-u+O(c_s)}$. The locally proved Balog–Szemerédi–Gowers theorem therefore furnishes $X_z \subset X$, $Y_z \subset Y$, and $\mathcal{V}_z \subset \mathcal{V}'$ with

$$N_\varrho(X_z), N_\varrho(Y_z) = \varrho^{-u+O(c_s)}, \quad |\mathcal{V}_z| = \varrho^{-2u+O(c_s)}, \quad N_\varrho\left(X_z + \frac{1-a}{a}Y_z\right) = \varrho^{-u+O(c_s)}.$$

We use only the elementary overlap lemma behind Ren’s Lemma 2.22: if a probability-indexed family of measurable sets all has measure at least K^{-1} , then integration of pairwise intersections and Fubini yield one member which meets a set of indices of positive measure in intersection measure at least $K^{-O(1)}$. Apply this to the thickened products $X_z^{(\varrho)} \times Y_z^{(\varrho)}$. We obtain $z_* \in D$, sets $X_* = X_{z_*}$ and $Y_* = Y_{z_*}$, and $D' \subset D$ with $\lambda(D') = \varrho^{O(c_s)}$ such that, for every $z \in D'$,

$$N_\varrho(X_*^{(\varrho)} \cap X_z^{(\varrho)}), N_\varrho(Y_*^{(\varrho)} \cap Y_z^{(\varrho)}) = \varrho^{-u+O(c_s)}.$$

Three applications of the scale- ϱ Ruzsa triangle inequality transfer the small-sum bound from (X_z, Y_z) to the fixed pair (X_*, Y_*) :

$$N_\varrho\left(X_* + \frac{1-a}{a}Y_*\right) = \varrho^{-u+O(c_s)} \quad (z \in D').$$

The tube family $\mathcal{V}_{z_*}$ is still a $(\varrho, \kappa_0, \varrho^{-O(c_s)}, k)$ -set of size $\varrho^{-2u+O(c_s)}$, whereas the preceding display gives small slices on the positive-mass height set D' . This contradicts the weak-slice argument. Hence the intermediate robust statement holds.

Finally perform the formal exhaustion. Repeatedly remove an intermediate-good subfamily until the remainder has size at most $\varrho^{c_s/2}|\mathcal{V}|$. Let the disjoint removed families be $\mathcal{V}_i$ and put $a_i = |\mathcal{V}_i|/|\bigcup_i \mathcal{V}_i|$. If a subfamily of the original configuration has relative size at least $\varrho^{c_s/3}$, then the indices on which it occupies at least a ϱ^{c_s} fraction of $\mathcal{V}_i$ have total weight $\sum a_i \geq \varrho^{c_s/2}$. At a height where its slice is too small, every one of those indices lies in the exceptional set of $\mathcal{V}_i$. Integrating the weighted indicator and using $\lambda(E(\mathcal{V}_i)) \leq \varrho^{c_s}$ gives an exceptional height mass at most $\varrho^{c_s/2}$. Decrease c_s by a fixed factor to absorb the displayed fractions. This proves (6.20b116h) and reproduces Ren’s weak-slice/intermediate-slice/ formal-exhaustion chain without a hidden slicing theorem. $\square$

Proposition III.6.24.40 (local quasi-product incidence gain). *Fix an integer $0 \leq k < d-1$, an exponent $0 \leq u < k+1$, and $\tau, \kappa_0 > 0$. There is $c_Q = c_Q(u, k, \tau, \kappa_0, d) > 0$ such that the following holds. Let $Y \subset \varrho\mathbb{Z} \cap [0, 1]$ be a $(\varrho, \tau, \varrho^{-c_Q})$ -set. For each $y \in Y$, let $X_y \subset (\varrho\mathbb{Z})^{d-1} \cap [0, 1]^{d-1}$ satisfy*

$$|X_y| \geq \varrho^{-u+c_Q}, \quad |X_y \cap H^{(r)}| \leq \varrho^{-c_Q}|X_y|r^{\kappa_0} \quad (6.20b116l)$$

for all affine k -planes H and $\varrho \leq r \leq 1$. Put $Z = \bigcup_{y \in Y} X_y \times \{y\}$. If every $z \in Z$ carries at least ϱ^{-u+c_Q} incident nonhorizontal ϱ -tubes, then their union $\mathcal{V}$ satisfies

$$|\mathcal{V}| \geq \varrho^{-2u-c_Q}. \quad (6.20b116m)$$

Proof. Assume the contrary and, by discarding tubes inside each point fibre if necessary, make every point fibre have cardinality $\varrho^{-u+O(c_Q)}$. For fixed y , let $\mathcal{V}(y) = \bigcup_{x \in X_y} \mathcal{V}(x, y)$. A nonhorizontal tube meets only $O(1)$ ϱ -balls in the level $x_d = y$, so

$$|\mathcal{V}(y)| \gtrsim \varrho^{-u+O(c_Q)} |X_y|.$$

The counter-assumption $|\mathcal{V}| \leq \varrho^{-2u-c_Q}$, together with the lower bound on $|X_y|$, forces $|X_y| = \varrho^{-u+O(c_Q)}$ and $|\mathcal{V}(y)| = \varrho^{-2u+O(c_Q)}$ uniformly in y .

We first transfer the horizontal nonconcentration to each tube family. If W is an $(r, k+1)$ -plate and $\mathcal{V}(y) \cap W \neq \emptyset$, then, because all tubes make a fixed positive angle with the horizontal hyperplane, the slice $W \cap \{x_d = y\}$ is contained in an $O(r)$ -neighbourhood of an affine k -plane. Hence

$$|\mathcal{V}(y) \cap W| \leq \sum_{x \in X_y \cap W^{(O(r))}} |\mathcal{V}(x, y)| \leq \varrho^{-O(c_Q)} r^{\kappa_0} |\mathcal{V}(y)|.$$

Thus every $\mathcal{V}(y)$ is a $(\varrho, \kappa_0, \varrho^{-O(c_Q)}, k)$ -set.

Since $|\mathcal{V}(y)| = \varrho^{-2u+O(c_Q)}$ and the assumed global family has the same order of magnitude, popularity in the incidence relation $\{(y, V) : V \in \mathcal{V}(y)\}$ yields $\overline{\mathcal{V}} \subset \mathcal{V}$ with $|\overline{\mathcal{V}}| = \varrho^{-2u+O(c_Q)}$ such that every retained tube belongs to $\varrho^{O(c_Q)} |Y|$ of the families $\mathcal{V}(y)$. Averaging the preceding plate bound then gives, for every $(r, k+1)$ -plate W ,

$$|\overline{\mathcal{V}} \cap W| \leq \varrho^{-O(c_Q)} r^{\kappa_0} |\overline{\mathcal{V}}|,$$

so $\overline{\mathcal{V}}$ itself has the required k -plane nonconcentration. A second popularity refinement gives $\overline{Y} \subset Y$, still a $(\varrho, \tau, \varrho^{-O(c_Q)})$ -set, such that

$$|\overline{\mathcal{V}} \cap \mathcal{V}(y)| = \varrho^{-2u+O(c_Q)} \quad (y \in \overline{Y}).$$

Normalize counting measure on $\overline{Y}$ and apply Lemma III.6.24.39 with this k to $\overline{\mathcal{V}}$. Outside a set of small height mass, every such large subfamily must have slice covering number at least ϱ^{-u-c} for the slicing gain $c > 0$. But for each $y \in \overline{Y}$ its slice lies in $X_y^{(O(\varrho))}$, whose covering number is $\varrho^{-u+O(c_Q)}$. Taking $c_Q \ll c$ is a contradiction. This is exactly the two-stage popularity argument in Ren's Proposition 3.1 and proves (6.20b116m). $\square$

Lemma III.6.24.41 (angular-to-ball transfer). *Fix an integer $0 \leq k < d-1$ and exponents $0 \leq u < t \leq d$ with $k < t$. Let P be a (Δ, t, Δ^{-a}) -set in a unit cube. Let $\Gamma \subset S^{d-1}$ be both a $(\Delta, u, \Delta^{-a}, 0)$ -set and a $(\Delta, \kappa_0, \Delta^{-a}, k)$ -set. For every $b > 0$, after discarding at most half of Γ , each remaining direction σ has the following robust property: every $P' \subset P$ with $|P'| \geq \Delta^b |P|$ has $\pi_\sigma(P')$ containing a subset of cardinality $\Delta^{-u+O(a+b)}$ which is a $(\Delta, \kappa', \Delta^{-O(a+b)}, k)$ -set, where*

$$\kappa' = \frac{1}{k+1} \min \left\{ \frac{t-k}{2}, \kappa_0 \right\} > 0. \quad (6.20b116n)$$

Proof. Let μ be normalized Lebesgue measure on the union of the Δ -cubes of P and put $\beta = \min\{(t-k)/2, \kappa_0\}$. The point Frostman hypothesis and the wedge-energy calculation used in Lemma 2.30(a) of Ren give

$$\int (|(x_0 - x_1) \wedge \cdots \wedge (x_0 - x_{k+1})| + \Delta)^{-\beta} d\mu^{k+2} \lesssim \Delta^{-O(a)}. \quad (6.20b116o)$$

Indeed, condition successively on $x_0, \dots, x_j$: the distance of x_{j+1} from their affine span is controlled by the t -Frostman bound, and $\beta < t - k$ makes the final dyadic sum convergent.

Average the corresponding projected $(k + 1)$ -fold wedge energy over $\sigma \in \Gamma$. Multilinearity of the wedge gives

$$|\pi_\sigma(x_0 - x_1) \wedge \dots \wedge \pi_\sigma(x_0 - x_{k+1})| = |(x_0 - x_1) \wedge \dots \wedge (x_0 - x_{k+1})| \operatorname{dist}(\sigma, L),$$

where L is the $(k + 1)$ -plane spanned by the differences. The $(\Delta, \kappa_0, \Delta^{-a}, k)$ condition and $\beta \leq \kappa_0$ therefore give, after a dyadic angular sum,

$$\frac{1}{|\Gamma|} \sum_{\sigma \in \Gamma} I_{\beta, k+1}^\Delta(\pi_\sigma \mu) \lesssim \Delta^{-O(a)}.$$

Simultaneously, the ordinary $(\Delta, u, \Delta^{-a}, 0)$ condition and the usual Kaufman Fubini calculation give the projected one-point energy at any exponent $u' = \max\{0, u - O(a + b)\} < u$:

$$\frac{1}{|\Gamma|} \sum_{\sigma \in \Gamma} I_{u'}^\Delta(\pi_\sigma \mu) \lesssim \Delta^{-O(a)}.$$

Markov leaves at least half the directions with both bounds after decreasing the retained fraction once. Replacing μ by the normalized restriction to any P' of relative size at least Δ^b costs only $\Delta^{-O(b)}$ in both estimates.

Apply the local energy-to-Frostman pruning simultaneously to these two energies. The β -energy produces a $(\Delta, \beta/(k + 1), \Delta^{-O(a+b)}, k)$ -set because a $(k + 1)$ -fold wedge is $O(r^{k+1})$ when all points lie in an r -neighbourhood of a k -plane. Since $\beta/(k + 1) = \kappa'$, this is the claimed plate nonconcentration. The ordinary energy forces at least $\Delta^{-u+O(a+b)}$ occupied Δ -balls (trivially so when $u = 0$). This is Ren's Proposition 4.3 with its energy argument written locally. $\square$

Lemma III.6.24.42 (higher-dimensional elementary tube-incidence baseline). *Let $0 \leq u < v \leq d - 1$. Let P be a (ρ, v, C_P) -set of ρ -cubes, and for every $p \in P$ let $\mathcal{U}(p)$ be a $(\rho, u, C_T, 0)$ -set of exactly M incident ρ -tubes. Put $\mathcal{U} = \bigcup_p \mathcal{U}(p)$. Up to a fixed power of $\log(1/\rho)$,*

$$|\mathcal{U}| \gtrsim (C_P C_T)^{-1} M \rho^{-u} (M \rho^u)^{(v-u)/(d-1-u)}.$$

The same conclusion holds for a t -set with $t > d - 1$ after replacing v by $d - 1$.

Proof. Let $I = \{(p, U) : U \in \mathcal{U}(p)\}$, so $|I| = M|P|$. Cauchy-Schwarz gives

$$|I|^2 \leq |\mathcal{U}| \sum_{p, p' \in P} |\mathcal{U}(p) \cap \mathcal{U}(p')|.$$

For $r = |p - p'| + \rho$, a tube meeting both cubes has direction in a spherical cap of diameter $O(\rho/r)$. The fibre nonconcentration and the packing bound for ρ -separated directions therefore give

$$|\mathcal{U}(p) \cap \mathcal{U}(p')| \lesssim \min\{C_T M(\rho/r)^u, r^{-(d-1)}\}.$$

Put $\theta = (d - 1 - v)/(d - 1 - u)$. Since $v = u\theta + (d - 1)(1 - \theta)$, interpolation of the two terms yields

$$|\mathcal{U}(p) \cap \mathcal{U}(p')| \lesssim (C_T M \rho^u)^\theta r^{-v}.$$

A dyadic-annulus sum and the (ρ, v, C_P) condition give $\sum_{p'} (|p - p'| + \rho)^{-v} \lesssim_{\log} C_P |P|$. Substitution into Cauchy–Schwarz and cancellation of $|P|^2$ proves the displayed estimate. If the original point exponent is $t > d - 1$, it is also a $(\rho, d - 1, C_P)$ -set, so the final sentence follows. This is the full elementary estimate used in Ren’s Corollary 2.10, written here in the exact form needed below. $\square$

Lemma III.6.24.43 (local two-scale refinement for the regular configuration). *Fix an integer $0 \leq k < d - 1$. Let $\rho = \Delta^2$, and suppose that P and the incident tube fibres $\mathcal{U}(p)$ satisfy the hypotheses of Proposition III.6.24.44, with all loss constants bounded by ρ^{-a} . Assume in addition, as in the regular-set application below, that*

$$|P| = \rho^{-t+O(a)}, \quad N_\Delta(P) = \Delta^{-t+O(a)}.$$

After passing to subfamilies which retain a $\rho^{O(a)}$ fraction of the points and incidences, one may arrange the following simultaneously.

1. The occupied Δ -cubes form a family $\mathcal{Q}$, the number of fine points in every $Q \in \mathcal{Q}$ is dyadically constant, and the normalized descendants in each Q inherit the same point Frostman exponent.
2. For every $Q \in \mathcal{Q}$ there is a coarse fibre $\mathcal{U}_\Delta(Q)$ of exactly M_Δ incident Δ -tubes; these fibres are simultaneously $(\Delta, u, \rho^{-O(a)}, 0)$ -sets and $(\Delta, \kappa, \rho^{-O(a)}, k)$ -sets.
3. Inside every retained coarse incidence (Q, U) , the fine incidences have dyadically constant multiplicity. After normalizing Q to unit scale one obtains a fine-scale configuration $(P_Q, \mathcal{U}_Q)$ at scale Δ with fibre size M_Q , with the same two nonconcentration properties, and

$$\frac{|\mathcal{U}|}{M} \gtrsim \rho^{O(a)} \frac{|\mathcal{U}_\Delta|}{M_\Delta} \frac{|\mathcal{U}_Q|}{M_Q}.$$

The local ratio may be chosen uniformly over the retained Q after one more dyadic pigeonhole. Moreover there is a dyadically constant packet size N_Δ such that, for every retained coarse tube U ,

$$|\{T \in \mathcal{U} : A_\Delta(T) = U\}| \asymp N_\Delta, \quad |\mathcal{U}| \gtrsim \rho^{O(a)} N_\Delta |\mathcal{U}_\Delta|.$$

4. Every retained coarse tube $U \in \mathcal{U}_\Delta(Q)$ supports

$$\rho^{O(a)} \frac{M|P \cap Q|}{M_\Delta}$$

fine incidences (p, T) with $p \in P \cap Q$, $T \in \mathcal{U}(p)$, and $T \subset U$.

Proof. We give the packet bookkeeping explicitly, because neither coarse nor fine nonconcentration is inherited from arbitrary pruning. First dyadically pigeonhole the occupancies $|P \cap Q|$, retaining a class which carries a $\rho^{O(a)}$ fraction of P . The assumptions $|P| = \rho^{-t+O(a)}$ and $N_\Delta(P) = \Delta^{-t+O(a)}$ imply, after renaming the retained set,

$$|\mathcal{Q}| = \Delta^{-t+O(a)}, \quad |P \cap Q| = \Delta^{-t+O(a)} \quad (Q \in \mathcal{Q}). \quad (6.20b116m1)$$

If B is a ball of normalized radius $r \in [\Delta, 1]$, its pullback to Q lies in a ball of radius $O(\Delta r)$. The original t -set estimate together with the occupancy bound in (6.20b116m1) therefore gives, for the normalized descendant P_Q ,

$$|P_Q \cap B| \leq \rho^{-O(a)} |P_Q| r^t. \quad (6.20b116m2)$$

Thus the point Frostman exponent survives the scale change.

We next construct the coarse tube fibre through a fixed Q . Choose once and for all a dyadic Δ -ancestor $A_\Delta(T)$ for every fine ρ -tube T ; bounded ambiguity is absorbed into dimensional constants. For $p \in P \cap Q$ and a coarse ancestor U put

$$n_p(U) = |\{T \in \mathcal{U}(p) : A_\Delta(T) = U\}|.$$

Since $\sum_U n_p(U) = M$, a dyadic pigeonhole gives, for every p , a family $\mathcal{A}_p$ of ancestors and a dyadic number $m(p)$ such that $n_p(U) \asymp m(p)$ on $\mathcal{A}_p$ and

$$m(p) |\mathcal{A}_p| \gtrsim_{\log} M.$$

Pigeonholing $m(p)$ over $p \in P \cap Q$ leaves a subset P_Q^0 of relative size $\rho^{O(a)}$ and a common value m_Q^{anc} with the same lower bound. For a coarse ancestor U define its point degree

$$d_Q(U) = |\{p \in P_Q^0 : U \in \mathcal{A}_p\}|.$$

The selected fine-incidence mass satisfies

$$\sum_U m_Q^{\text{anc}} d_Q(U) \gtrsim \rho^{O(a)} M |P \cap Q|. \quad (6.20b116m3)$$

Pigeonhole the nonzero degrees $d_Q(U)$ by their contribution to this sum. We obtain a family $\overline{\mathcal{U}}_\Delta(Q)$, a dyadic degree d_Q , and $H_Q := m_Q^{\text{anc}} d_Q$ such that every $U \in \overline{\mathcal{U}}_\Delta(Q)$ supports at least cH_Q fine incidences (p, T) with $p \in P_Q^0$ and $A_\Delta(T) = U$, while

$$H_Q |\overline{\mathcal{U}}_\Delta(Q)| \gtrsim \rho^{O(a)} M |P \cap Q|. \quad (6.20b116m4)$$

This is the incidence-weight lower bound that replaces the invalid shortcut “pruning pre-serves nonconcentration.”

Let H be an r -tube with $\Delta \leq r \leq 1$. If $U \in \overline{\mathcal{U}}_\Delta(Q)$ lies in H , all fine tubes counted in its H_Q incidences lie in a fixed $O_d(r)$ -enlargement H^+ . Hence

$$H_Q |\{U \in \overline{\mathcal{U}}_\Delta(Q) : U \subset H\}| \lesssim \sum_{p \in P_Q^0} |\mathcal{U}(p) \cap H^+| \leq \rho^{-O(a)} |P \cap Q| M r^u.$$

Using (6.20b116m4),

$$|\{U \in \overline{\mathcal{U}}_\Delta(Q) : U \subset H\}| \leq \rho^{-O(a)} |\overline{\mathcal{U}}_\Delta(Q)| r^u. \quad (6.20b116m5)$$

The identical double count with an $(r, k+1)$ -plate gives the exponent κ . Thus the provisional coarse fibres have both required nonconcentration properties.

We now synchronize these fibres over Q . Dyadically pigeonhole $|\overline{\mathcal{U}}_\Delta(Q)|$ and retain a $\rho^{O(a)}$ fraction of the coarse cubes; trim within a factor two so that every retained fibre has the same cardinality M_Δ . The preceding estimates survive with a $\rho^{-O(a)}$ loss. Next let

$$N(U) = |\{T \in \mathcal{U} : A_\Delta(T) = U\}|, \quad U \in \mathcal{U}_\Delta := \bigcup_Q \mathcal{U}_\Delta(Q).$$

Pigeonhole $N(U)$ on the coarse-incidence set $\{(Q, U) : U \in \mathcal{U}_\Delta(Q)\}$. Retain a class on which $N(U) \asymp N_\Delta$, then retain those Q for which a $\rho^{O(a)}$ fraction of their coarse fibre lies in this class and trim once more to a common M_Δ . Since every retained coarse tube already obeyed (6.20b116m4), item (4) now follows in the form

$$|\{(p, T) : p \in P \cap Q, T \in \mathcal{U}(p), A_\Delta(T) = U\}| \gtrsim \rho^{O(a)} \frac{M|P \cap Q|}{M_\Delta}. \quad (6.20b116m6)$$

The coarse nonconcentration estimates remain valid because every subsequent fibre is a $\rho^{O(a)}$ -large subfamily of the provisional one.

It remains to build the fine normalized configuration and prove the product inequality. For a retained Q , let $\mathcal{U}(Q)$ be the union of retained fine tubes whose Δ -ancestor belongs to $\mathcal{U}_\Delta(Q)$. Summing (6.20b116m4) over the retained coarse fibre shows that the fine incidence mass of $\mathcal{U}(Q)$ is $\rho^{O(a)} M|P \cap Q|$. A point-popularity pruning therefore leaves $P_Q^1 \subset P \cap Q$ of relative size $\rho^{O(a)}$ on which each point carries $\rho^{O(a)} M$ such fine tubes. Because P_Q^1 is a $\rho^{O(a)}$ -large subset of the normalized descendant from (6.20b116m2), its normalized t -Frostman estimate worsens only by a factor $\rho^{-O(a)}$. Intersect these tubes with Q , rescale Q to unit size, and identify coincident tubelets. For each $p \in P_Q^1$, dyadically pigeonhole the number of original tubes represented by one tubelet; then pigeonhole this multiplicity over p . We obtain a common multiplicity m_Q and a normalized fibre size M_Q such that

$$m_Q M_Q = \rho^{O(a)} M. \quad (6.20b116m7)$$

For a fixed normalized point fibre, if a normalized r -tube, or an $(r, k+1)$ -plate, contains L retained tubelets, then each tubelet represents $\asymp m_Q$ distinct original tubes and all of those original tubes lie in an $O_d(\Delta r)$ -tube, respectively an $O_d(\Delta r, k+1)$ -plate. Since $\Delta r \in [\rho, \Delta]$, the original fine-fibre bounds and (6.20b116m7) give

$$L \leq \rho^{-O(a)} M_Q r^u, \quad L \leq \rho^{-O(a)} M_Q r^k. \quad (6.20b116m8)$$

Dividing the original fibre bounds by the common multiplicity m_Q proves the two nonconcentration statements for the normalized descendant.

Finally, unique ancestry and $N(U) \asymp N_\Delta$ give the global packet count

$$|\mathcal{U}| \gtrsim \rho^{O(a)} N_\Delta |\mathcal{U}_\Delta|. \quad (6.20b116m9)$$

For a fixed retained Q , its M_Δ coarse ancestors contain at most $\rho^{-O(a)} M_\Delta N_\Delta$ fine tubes. On the other hand, the tubelet normalization gives

$$|\mathcal{U}(Q)| \gtrsim \rho^{O(a)} m_Q |\mathcal{U}_Q| \gtrsim \rho^{O(a)} \frac{M}{M_Q} |\mathcal{U}_Q|.$$

Consequently

$$M_\Delta N_\Delta \gtrsim \rho^{O(a)} \frac{M}{M_Q} |\mathcal{U}_Q|. \quad (6.20b116m10)$$

Combining (6.20b116m9) and (6.20b116m10) yields

$$\frac{|\mathcal{U}|}{M} \gtrsim \rho^{O(a)} \frac{|\mathcal{U}_\Delta|}{M_\Delta} \frac{|\mathcal{U}_Q|}{M_Q}.$$

One final dyadic pigeonhole makes the local ratio uniform over the retained Q . This proves all four items. The three cardinality comparisons above are the explicit packet bookkeeping behind Ren's Proposition 2.13 and the proof of its confined version, Proposition 6.2; no incidence gain is used. $\square$

Proposition III.6.24.44 (local regular-set M -sensitive gain). *Fix an integer $0 \leq k < d - 1$, $0 \leq u < k + 1$, $t > \max\{u, k\}$ and $\kappa > 0$. There is $c_R = c_R(u, t, \kappa, k, d) > 0$ such that, for sufficiently small dyadic ϱ , the following holds. Let P be a $(\varrho, t, \varrho^{-c_R})$ -set with*

$$N_{\varrho^{1/2}}(P) \leq \varrho^{-t/2-c_R}, \quad (6.20b116p)$$

and let every $p \in P$ carry exactly M incident ϱ -tubes forming both a $(\varrho, u, \varrho^{-c_R}, 0)$ -set and a $(\varrho, \kappa, \varrho^{-c_R}, k)$ -set. Then the union $\mathcal{U}$ of all tube fibres obeys

$$|\mathcal{U}| \geq M \varrho^{-u-c_R}. \quad (6.20b116q)$$

Proof. Set $\rho = \varrho$ and $\Delta = \rho^{1/2}$ and suppose, for contradiction, that $|\mathcal{U}| \leq M \rho^{-u-c_R}$. Put $v = \min\{t, d - 1\} > u$. Applying Lemma III.6.24.42 to the whole configuration gives

$$M \rho^{-u} (M \rho^u)^{(v-u)/(d-1-u)} \lesssim \rho^{-O(c_R)} |\mathcal{U}| \leq M \rho^{-u-O(c_R)}.$$

Hence $M \rho^u \leq \rho^{-O(c_R)}$. Conversely, the fibre $(\rho, u, \rho^{-c_R}, 0)$ condition, applied at the smallest occupied direction cell, gives $M \rho^u \geq \rho^{O(c_R)}$. Thus

$$M = \rho^{-u+O(c_R)}, \quad |\mathcal{U}| \leq \rho^{-2u-O(c_R)}. \quad (6.20b116r)$$

A standard dyadic Frostman pruning (choose at most the admissible number of descendants in every dyadic cube, from the top scale downward) retains a $\rho^{O(c_R)}$ fraction of P with $|P| = \rho^{-t+O(c_R)}$ and preserves both the t -set condition and (6.20b116p). We work with this subset. The t -set upper bound in one Δ -cube and the displayed size of P give $N_\Delta(P) \geq \Delta^{-t+O(c_R)}$, while (6.20b116p) gives the reverse inequality up to the same power loss. Hence $N_\Delta(P) = \Delta^{-t+O(c_R)}$, so the extra size hypotheses of the next lemma are satisfied.

Apply Lemma III.6.24.43. The refinement gives

$$|\mathcal{Q}| = \Delta^{-t+O(c_R)}, \quad |P \cap Q| = \Delta^{-t+O(c_R)} \quad (Q \in \mathcal{Q}).$$

Apply the elementary incidence baseline to the coarse configuration and to every normalized fine configuration. Since their ratio product is bounded above by $|\mathcal{U}|/M \leq \Delta^{-2u-O(c_R)}$,

while each ratio is at least its baseline $\Delta^{-u+O(c_R)}$, the positive factors in the elementary estimate cannot be large. Thus, uniformly after the preceding pruning,

$$M_\Delta = \Delta^{-u+O(c_R)}, \quad |\mathcal{U}_\Delta| = \Delta^{-2u+O(c_R)}, \quad M_Q = \Delta^{-u+O(c_R)}, \quad |\mathcal{U}_Q| = \Delta^{-2u+O(c_R)}.$$

This is the balanced square-root-scale reduction that in Ren's proof is recorded as (G1)–(G5).

Apply Lemma III.6.24.41 in every Q to the directions of $\mathcal{U}_\Delta(Q)$, and discard the bad half. The retained coarse fibres are still u -sets of cardinality $\Delta^{-u+O(c_R)}$, and every large subset of the normalized fine point descendants has a projected subset of cardinality $\Delta^{-u+O(c_R)}$ with k -plane exponent $\kappa' > 0$ from (6.20b116n).

We next choose the special coarse tube rather than assume it. For two occupied coarse cubes Q, Q' at distance r , the common retained coarse tubes form a directional cap of diameter $O(\Delta/r)$; hence their number is at most $\Delta^{-O(c_R)}r^{-u}$. Therefore, with

$$E(U) = \sum_{\substack{Q \neq Q' \\ U \in \mathcal{U}_\Delta(Q) \cap \mathcal{U}_\Delta(Q')}} |Q - Q'|^{-(t-u)},$$

we have

$$\sum_{U \in \mathcal{U}_\Delta} E(U) \lesssim \Delta^{-O(c_R)} \sum_{Q \neq Q'} |Q - Q'|^{-t} \lesssim \Delta^{-2t-O(c_R)}. \quad (6.20b116r1)$$

Here the last estimate is the dyadic-annulus energy bound for the retained $(\Delta, t, \Delta^{-O(c_R)})$ family $\mathcal{Q}$.

Choose a numerical constant A much larger than every implicit loss constant already used and call U *energy-bad* when

$$E(U) > \Delta^{-2(t-u)-Ac_R}. \quad (6.20b116r2)$$

By (6.20b116r1) and $|\mathcal{U}_\Delta| = \Delta^{-2u+O(c_R)}$, the bad family has cardinality

$$|\mathcal{B}| \leq \Delta^{(A-O(1))c_R} |\mathcal{U}_\Delta|. \quad (6.20b116r3)$$

We now justify that these few bad tubes cannot dominate many point fibres. Fix a large constant B_0 depending only on the preceding loss constants. Suppose instead that a subfamily $\mathcal{Q}_{\text{bad}} \subset \mathcal{Q}$ of size at least $\Delta^{B_0 c_R} |\mathcal{Q}|$ had at least $\Delta^{B_0 c_R} M_\Delta$ tubes from $\mathcal{B}$ in each of its fibres. Passing to those bad parts worsens the point and fibre nonconcentration constants by at most $\Delta^{-O(c_R)}$. Lemma III.6.24.42, applied to this subconfiguration and using $M_\Delta = \Delta^{-u+O(c_R)}$, would then give $|\mathcal{B}| \geq \Delta^{-2u-O(c_R)}$. This contradicts (6.20b116r3) once A was chosen large enough. Hence fewer than $\Delta^{B_0 c_R} |\mathcal{Q}|$ coarse cubes can have that property. Discard those cubes and, in every remaining fibre, discard its bad tubes. Every retained fibre still has $\Delta^{-u+O(c_R)}$ good tubes.

The number of remaining coarse incidences is therefore $\Delta^{-(t+u)+O(c_R)}$, while the union of the good tubes has cardinality at most $\Delta^{-2u+O(c_R)}$. Averaging gives a good tube U_0 meeting

$$L := |\mathcal{Q}(U_0)| \geq \Delta^{-(t-u)+O(c_R)} \quad (6.20b116r4)$$

coarse cubes. Since U_0 is good, $E(U_0) \leq \Delta^{-2(t-u)-Ac_R}$. Put

$$e(Q) = \sum_{Q' \in \mathcal{Q}(U_0) \setminus \{Q\}} |Q - Q'|^{-(t-u)}.$$

A second Markov pruning, now at threshold $\Delta^{-(t-u)-2Ac_R}$, removes at most $\Delta^{c_1 c_R} L$ cubes for some $c_1 = c_1(A) > 0$ (after choosing A large enough to dominate the previous losses). On the remaining family $\mathcal{Q}_0(U_0)$ we have, for every $r \in [\Delta, 1]$,

$$|\{Q' \in \mathcal{Q}_0(U_0) : |Q - Q'| \leq r\}| \leq \Delta^{-(t-u)-2Ac_R} r^{t-u} \leq \Delta^{-O(c_R)} |\mathcal{Q}_0(U_0)| r^{t-u}. \quad (6.20b116r5)$$

Together with (6.20b116r4), this is precisely the $(\Delta, t-u, \Delta^{-O(c_R)})$ subfamily required below. We rename it $\mathcal{Q}(U_0)$.

Item (4) of the two-scale refinement and the upper bound $|\mathcal{U}(p) \cap U_0| \leq \rho^{-O(c_R)} M \Delta^u = \Delta^{-u-O(c_R)}$ imply, for every $Q \in \mathcal{Q}(U_0)$ after another popularity pruning, a subset $P_Q \subset P \cap Q$ of size $\Delta^{-t+O(c_R)}$ such that every $p \in P_Q$ is incident to $\Delta^{-u+O(c_R)}$ fine tubes contained in U_0 . The choice of the good direction of U_0 and the angular-to-ball lemma give inside each Q a projected fibre X_Q of cardinality $\Delta^{-u+O(c_R)}$ satisfying (6.20b116l) with exponent κ' .

Rotate U_0 to the vertical axis and dilate its transverse directions by Δ^{-1} . The family $\mathcal{Q}(U_0)$ becomes a height set Y of dimension $t-u > 0$; the sets X_Q become the horizontal fibres X_y ; and the fine tubes contained in U_0 become Δ -tubes making a fixed positive angle with the horizontal hyperplane. Every product point carries $\Delta^{-u+O(c_R)}$ such tubes, while their total number is $\Delta^{-2u-O(c_R)}$: indeed, (6.20b116m9), $|\mathcal{U}| \leq \Delta^{-4u-O(c_R)}$, and $|\mathcal{U}_\Delta| = \Delta^{-2u+O(c_R)}$ give $N_\Delta \leq \Delta^{-2u-O(c_R)}$, and U_0 contains only one such coarse packet. Proposition III.6.24.40, with the same k , gives the opposite strict power inequality when c_R is small relative to its gain. This contradiction proves (6.20b116q). The parameter choices above use t only through positive lower bounds for $t-u$ and $t-k$; hence c_R may be chosen uniformly when t ranges over a compact subinterval of $(\max\{u, k\}, d]$. $\square$

Lemma III.6.24.45 (general-set branching assembly). *Fix an integer $0 \leq k < d-1$ and exponents $k \leq u < k+1$ and $u < t \leq d$. For every sufficiently small $\epsilon > 0$, every $(\varrho, t, \varrho^{-\epsilon})$ point set has a $\varrho^{O(\epsilon)}$ -large uniform subset and scales $1 = \Delta_0 > \Delta_1 > \dots > \Delta_N = \varrho$ which split into structured scales S and bad scales B so that:*

1. $\prod_{j \in B} (\Delta_j / \Delta_{j+1}) \leq \varrho^{-\epsilon}$;
2. on every $j \in S$, every normalized descendant is either a $(\rho_j, t_j, \rho_j^{-O(\epsilon)}, \rho_j^{-O(\epsilon)})$ -regular set with $t_j > u$, or a $(\rho_j, u, \rho_j^{-O(\epsilon)})$ -set, where $\rho_j = \Delta_{j+1} / \Delta_j$;
3. if $S_1 = \{j \in S : t_j \geq (u+t)/2\}$, then

$$\prod_{j \in S_1} (\Delta_j / \Delta_{j+1})^{t_j} \geq \varrho^{-(t-u)/2 + O_{u,t,d}(\epsilon)}. \quad (6.20b116s0)$$

At the same time the tube fibres admit compatible multiscale refinements for which

$$\frac{|\mathcal{U}|}{M} \gtrsim \varrho^{O(\epsilon)} \prod_{j=0}^{N-1} \frac{|\mathcal{U}_j|}{M_j}. \quad (6.20b116s)$$

Proof. Uniformize the point set on powers 2^{-mT} with T so large that the logarithmic pigeonholing loss is at most ϱ^ϵ . Let f be the logarithmic branching function. The Frostman bound is the lower barrier $f(x) \geq tx - O(\epsilon m)$. Apply the ordered superlinear decomposition proved earlier in Lemma III.6.17, with the reference slope u , and merge neighbouring pieces whose slopes differ by $O(\epsilon)$. The result is Ren's Section 5 dichotomy: apart from blocks of total logarithmic length $O(\epsilon m)$, a normalized descendant is either regular with an exponent $t_j > u$, or obeys the u -set bound. This proves (1)–(2).

Let $S_1 = \{j \in S : t_j \geq (u+t)/2\}$ and $S_2 = S \setminus S_1$. The product form of the branching-number identity gives

$$\prod_{j \in S} (\Delta_j / \Delta_{j+1})^{t_j} \geq |P| \varrho^{O_{u,t,d}(\epsilon)}.$$

On S_2 one has $t_j < (u+t)/2$. Since $|P| \geq \varrho^{-t+O(\epsilon)}$ and $\prod_{j \in S_2} (\Delta_j / \Delta_{j+1}) \leq \varrho^{-1}$, division by the S_2 factors gives (6.20b116s0). Since $t_j \leq d$, this also implies

$$\prod_{j \in S_1} (\Delta_j / \Delta_{j+1}) \geq \varrho^{-(t-u)/(2d) + O_{u,t,d}(\epsilon)}.$$

Thus the gain-bearing regular scales occupy a fixed positive logarithmic proportion. Notice that every $j \in S_1$ satisfies $t_j \geq (u+t)/2 > u \geq k$, so the regular-set theorem is available on every scale on which the strict gain will be used.

Finally we record the tube telescope rather than hide it in the phrase “multiply the identities.” Merge adjacent bad intervals first. Because every structured block has logarithmic length at least the τ supplied by the superlinear decomposition, the resulting number of boundaries is $O_{u,t,d,\epsilon}(1)$. At boundary j put $\rho_j = \Delta_{j+1}/\Delta_j$. Let $\mathcal{U}^{(j)}$ be the tube union entering that boundary, let F_j be its common point-fibre size, and set $R_j = |\mathcal{U}^{(j)}|/F_j$. Perform the dyadic tube-packet decomposition from the proof of Lemma III.6.24.43. Only its packet-count and tube-fibre nonconcentration steps are used here; those steps do not require the regular point hypothesis. Write $\mathcal{U}_j$ for its coarse tube union, M_j for the coarse fibre size, and N_j for the dyadically constant number of current tubes inside a retained coarse ancestor. Then the global packet count gives

$$|\mathcal{U}^{(j)}| \gtrsim \rho_j^{O(\epsilon)} N_j |\mathcal{U}_j|. \quad (6.20b116s1)$$

Choose a retained descendant cube Q_j whose normalized union-to-fibre ratio is maximal, and denote its normalized configuration by $(\mathcal{U}^{(j+1)}, F_{j+1})$. The M_j coarse ancestors through Q_j contain at most $\rho_j^{-O(\epsilon)} M_j N_j$ current tubes. On the other hand, the tubelet normalization in that descendant says that each normalized tubelet represents $\rho_j^{O(\epsilon)} F_j / F_{j+1}$ current tubes. Consequently

$$M_j N_j \gtrsim \rho_j^{O(\epsilon)} \frac{F_j}{F_{j+1}} |\mathcal{U}^{(j+1)}|. \quad (6.20b116s2)$$

Divide (6.20b116s1) by F_j and use (6.20b116s2); this gives the one-step telescope

$$R_j \gtrsim \rho_j^{O(\epsilon)} \frac{|\mathcal{U}_j|}{M_j} R_{j+1}. \quad (6.20b116s3)$$

These are precisely the three cardinality comparisons behind the single-scale product inequality: global coarse-packet counting, the local upper packet count, and the lower count coming from normalized tubelet multiplicity. In particular, no intermediate multiplicity is left uncanceled.

Iterating (6.20b116s3) down the compatible descendant chain and using only the terminal bound $R_N = |\mathcal{U}^{(N)}|/F_N \geq 1$ gives

$$\frac{|\mathcal{U}|}{M} = R_0 \gtrsim \left(\prod_{j=0}^{N-1} \rho_j^{O(\epsilon)} \right) \prod_{j=0}^{N-1} \frac{|\mathcal{U}_j|}{M_j}.$$

Since $\prod_j \rho_j = \varrho$, the first product is $\varrho^{O(\epsilon)}$. Each boundary uses only $O(\log(1/\rho_j))$ dyadic classes; because the number of boundaries is bounded in terms of the fixed decomposition parameters, their total polylogarithmic loss is also $\varrho^{-O(\epsilon)}$ once ϱ is sufficiently small. This proves (6.20b116s). It is the explicit iteration of the single-scale product inequality used in Ren's Corollary 2.14 and Section 5. $\square$

Proposition III.6.24.46 (Unconfined co-rank-one M -sensitive gain — local relative to X -HA). *Fix $d \geq 3$, $0 \leq u < d - 1$, $\max\{u, d - 2\} < t \leq d$, and $\kappa > 0$. There are numbers*

$$\gamma_* = \gamma_*(u, t, \kappa, d) > 0, \quad \eta_* = \eta_*(u, t, \kappa, d) > 0, \quad (6.20b117a)$$

such that, for all sufficiently small dyadic ϱ , the following holds. Let $\mathcal{Q}$ be a $(\varrho, t, \varrho^{-\eta_})$ -set of ϱ -cubes in a fixed bounded region. For each $q \in \mathcal{Q}$ let $\mathcal{U}(q)$ consist of exactly M dyadic ϱ -tubes meeting q , and assume that $\mathcal{U}(q)$ is simultaneously a*

$$(\varrho, u, \varrho^{-\eta_*}, 0) \quad \text{and} \quad (\varrho, \kappa, \varrho^{-\eta_*}, d - 2) \quad (6.20b117b)$$

set of tubes. Then

$$\left| \bigcup_{q \in \mathcal{Q}} \mathcal{U}(q) \right| \geq M \varrho^{-u-\gamma_*}. \quad (6.20b117c)$$

Proof. The point that matters for the bootstrap is that u may be smaller than $d - 2$. We therefore make explicit Ren's preliminary reduction rather than silently applying the co-rank endpoint at the wrong plate index. Put

$$k_u = \min\{d - 2, \lfloor u \rfloor\}.$$

Then $0 \leq k_u < d - 1$ and $k_u \leq u < k_u + 1$. Moreover every $(r, d - 1)$ -plate contains every $(r, k_u + 1)$ -plate after enlarging the implicit geometric constant. Hence the second condition in (6.20b117b) implies that every fibre is also a $(\varrho, \kappa, \varrho^{-O(\eta_*)}, k_u)$ -set. This is precisely Ren's observation that the plate index can be lowered before Sections 3–5.

Set $k = k_u$ and $\kappa_0 = \min\{\kappa, 1\}$. Choose the regular-set gain from Proposition III.6.24.44, uniformly for regular exponents in the compact interval

$$\left[\frac{u+t}{2}, d \right].$$

This interval lies strictly above $u \geq k$. Apply Lemma III.6.24.45 with a bookkeeping exponent ϵ much smaller than that regular-set gain. On each scale in S_1 , apply Proposition III.6.24.44 with the effective plate index k ; this contributes the baseline factor ρ_j^{-u} together with a fixed extra negative power. On the remaining structured scales use Lemma III.6.24.42, which gives the baseline $\rho_j^{-u-O(\epsilon)}$. Bad scales cost only $\varrho^{-O(\epsilon)}$.

Multiply the scale estimates in (6.20b116s). The quantitative length bound (6.20b116s0) ensures that the scales in S_1 occupy a fixed positive logarithmic proportion depending only on u, t ; hence their strict regular-set gains multiply to $\varrho^{-\gamma_*}$ for some $\gamma_* = \gamma_*(u, t, \kappa, d) > 0$. Choose first ϵ and then the original fibre loss exponent η_* sufficiently small to absorb uniformization, plate-index lowering, rescaling, and dyadic-pigeonholing losses. This proves (6.20b117c).

For provenance, the local chain is the full specialization required by the confined bootstrap. Lemma III.6.24.38 proves Ren's Theorem 3.5 for every effective k . Lemma III.6.24.39 reconstructs Theorems 3.6–3.7 and the formal exhaustion. Proposition III.6.24.40 is Proposition 3.1. Lemma III.6.24.41 and Proposition III.6.24.44 reconstruct Section 4, while Lemma III.6.24.45 is the Section 5 assembly. The only external theorem in this reconstruction is the harmonic-analysis estimate (6.20b116a). $\square$

Lemma III.6.24.47 (Anisotropic normalization of a confined co-rank-one configuration). *Let H be an $(r_0, d-1)$ -plate and let S_H be an affine map which is an isometry in the $d-1$ long directions of H and expands its short direction by r_0^{-1} . Put $\theta = \delta/r_0$. Under the hypotheses of Proposition III.6.24.51, with a generic loss exponent $a > 0$ in place of ϵ_* , one has, for $\theta \leq R \leq 1$,*

$$\begin{aligned} |\mathcal{P} \cap S_H^{-1}(Q)| &\lesssim \theta^{-a} |\mathcal{P}| R^t, & Q \in \mathcal{D}_R^d, \\ |\mathcal{T}(p) \cap S_H^{-1}(U)| &\lesssim \theta^{-a} |\mathcal{T}(p)| R^u, & U \text{ an } R\text{-tube}, \\ |\mathcal{T}(p) \cap S_H^{-1}(W)| &\lesssim \theta^{-a} |\mathcal{T}(p)| R^\kappa, & W \text{ an } (R, d-1)\text{-plate}. \end{aligned} \quad (6.20b117d)$$

The implicit constants depend only on d and may be absorbed by decreasing the final power gain.

Proof. The first estimate follows because $S_H^{-1}(Q)$ is contained in a ball of radius $O_d(R)$. For the second, $S_H^{-1}(U)$ is contained in a rectangular box with $d-2$ sides of length $O_d(R)$ and two sides of length $O_d(Rr_0)$. Cover it by $O_d(r_0^{-(d-2)})$ balls of radius $O_d(Rr_0)$. The ordinary tube nonconcentration hypothesis gives, in each ball,

$$\theta^{-a} r_0^{d-2-u} |\mathcal{T}(p)| (Rr_0)^u,$$

and the covering multiplicity cancels the factor r_0^{d-2} . Finally, $S_H^{-1}(W)$ lies in an $O_d(Rr_0)$ -neighbourhood of an affine hyperplane, so the co-rank-one nonconcentration hypothesis gives $\theta^{-a} r_0^{-\kappa} (Rr_0)^\kappa |\mathcal{T}(p)|$. This is precisely the cancellation behind Ren's normalized statistics (6.15)–(6.17). $\square$

Lemma III.6.24.48 (Good intermediate branch scale). *Fix $t_0 < t$ and $0 < \beta < 1$ with*

$$t_0 < \frac{t - d\beta}{1 - \beta}. \quad (6.20b117e)$$

For every bookkeeping number $\zeta > 0$ there is a dyadic $\Delta = \Delta(t, t_0, \beta, d, \zeta) \in (0, 1/2)$ such that the following holds for all sufficiently small $\theta = \delta/r_0$. After replacing θ by a comparable power of Δ and retaining at least a θ^ζ fraction of $\mathcal{P}$, there is an intermediate scale

$$\theta \leq R \leq \theta^\beta \quad (6.20b117f)$$

with these properties.

1. The occupied R -cubes of $S_H(\mathcal{P})$ form an $(R, t_0, \theta^{-O(a+\zeta)})$ -set if the original point-set constant is θ^{-a} .
2. For every retained occupied R -cube Q , the descendant point set, rescaled from Q to unit size and pulled back by S_H^{-1} , obeys the two point-counting bounds required in Lemma III.6.24.49: exponent t_0 above the new confinement threshold and exponent t_0 below it, with constant at most $\theta^{-O(\zeta+a)}$ if the original point-set constant is θ^{-a} .

Proof. Write first $\theta = \Delta^m$; rounding to adjacent dyadic scales changes all counts by dimensional constants. Successively pigeonhole the numbers of descendants in each occupied Δ^j -cube, $j = 1, \dots, m$. At one level there are $O_d(\log(1/\Delta))$ dyadic cardinality classes. Hence the retained proportion is at least

$$(C_d \log(1/\Delta))^{-m} = \theta^{c(\Delta)}, \quad c(\Delta) = \frac{\log(C_d \log(1/\Delta))}{\log(1/\Delta)}.$$

Choose Δ so small that $c(\Delta) < \zeta/4$ and absorb the initial cardinality pigeonhole into the remaining $3\zeta/4$. We may therefore assume that the retained set is uniform at the scales $1, \Delta, \dots, \Delta^m$.

Let N_j denote the common number of finest descendants inside an occupied Δ^j -cube. Since the original set is a (δ, t, θ^{-a}) -set, its branching function lies above the affine line of slope t , up to the additive $o_\Delta(m) + O(am)$ loss just produced. Choose the largest index j_* , with $\beta m \leq j_* \leq m$, for which

$$N_{j_*} \geq |\mathcal{P}'| \Delta^{d\beta m + (j_* - \beta m)t_0}.$$

The index $\lceil \beta m \rceil$ is admissible by the trivial bound on the number of occupied cubes. Maximality gives, for $j_* \leq j \leq m$,

$$N_j \leq N_{j_*} \Delta^{(j-j_*)t_0}. \quad (6.20b117g)$$

To verify the small-scale estimate explicitly, the retained proportion gives $|\mathcal{P}| \leq \theta^{-\zeta} |\mathcal{P}'|$ (after enlarging the harmless constant), while the defining lower bound for N_{j_*} gives

$$N_{j_*} \geq |\mathcal{P}'| \theta^{\beta d} R^{t_0} \theta^{-\beta t_0}.$$

Hence, for a physical ball of radius $r \leq \theta$, the original t -Frostman bound is dominated by the desired descendant estimate provided

$$r^{t-t_0} \leq \theta^{\beta(d-t_0)}.$$

The worst case is $r = \theta$, and this is precisely $t_0 \leq (t - d\beta)/(1 - \beta)$; the strict inequality (6.20b117e) absorbs the $a + \zeta$ bookkeeping losses. At scales above R , every occupied R -cube contains the same N_{j_*} descendants, so the original t -bound, divided by N_{j_*} , gives an

$(R, t_0, \theta^{-O(a+\zeta)})$ bound for the coarse occupied cubes. Finally (6.20b117g) gives the relative t_0 -bound inside every retained R -cube. Taking $R = \Delta^{j_*}$ proves (6.20b117f) and both asserted spacing statements. This is the branching-number proof of Ren's Lemma 6.4, with the logarithmic uniformization loss written as the arbitrarily small power θ^ζ . $\square$

Lemma III.6.24.49 (Elementary confined incidence bound). *Put $k = d - 2$ and assume $t_0 > \max\{u, k\}$. Let $\mathcal{Q}$ be a set of δ_1 -cubes contained in an $(r_0, d - 1)$ -plate H such that, for a constant $C_P \geq 1$,*

$$|\mathcal{Q} \cap S_H^{-1}(B_R)| \leq C_P |\mathcal{Q}| R^{t_0} \quad (\delta_1/r_0 \leq R \leq 1),$$

and

$$|\mathcal{Q} \cap B_r| \leq C_P |\mathcal{Q}| r^{t_0} \quad (\delta_1 \leq r \leq \delta_1/r_0).$$

Suppose each $q \in \mathcal{Q}$ carries exactly M_1 incident δ_1 -tubes and

$$|\mathcal{U}(q) \cap U_r| \leq C_T M_1 r_0^{k-u} r^u, \quad \delta_1 \leq r \leq r_0, \quad (6.20b117h)$$

for every r -tube U_r . Then

$$\left| \bigcup_{q \in \mathcal{Q}} \mathcal{U}(q) \right| \gtrsim_{d,u,t_0} (C_P C_T)^{-1} M_1 r_0^{u-k} \delta_1^{-u}. \quad (6.20b117i)$$

Proof. For fixed $p \in \mathcal{Q}$, let J_p count pairs (q, T) with $q \in \mathcal{Q}$ and $T \in \mathcal{U}(p) \cap \mathcal{U}(q)$. Split according to the dyadic distance $r \sim |p - q|$.

If $r > \delta_1/r_0$, then any tube through p, q which remains in H forces $S_H(p), S_H(q)$ to be $O_d(r)$ apart. Thus there are $O(C_P |\mathcal{Q}| r^{t_0})$ choices for q . All tubes through both cubes lie in one $O(\delta_1/r)$ -tube, so (6.20b117h) gives at most

$$O(C_T M_1 r_0^{k-u} (\delta_1/r)^u)$$

choices for T . Since $t_0 > u$, summing these dyadic shells contributes $O(C_P C_T |\mathcal{Q}| M_1 r_0^{k-u} \delta_1^u)$.

If $r < \delta_1/r_0$, use the physical ball bound for q . The $O(\delta_1/r)$ -tube through p, q , intersected with H , is covered by $O_d((\delta_1/(r r_0))^k)$ many r_0 -tubes. Applying (6.20b117h) at scale r_0 leaves $O(C_T M_1 (\delta_1/r)^k)$ tubes. The shell contribution is therefore $O(C_P C_T |\mathcal{Q}| M_1 \delta_1^k r^{t_0-k})$. Because $t_0 > \max\{k, u\}$, the dyadic sum is again $O(C_P C_T |\mathcal{Q}| M_1 r_0^{k-u} \delta_1^u)$. Consequently

$$J_p \lesssim C_P C_T |\mathcal{Q}| M_1 r_0^{k-u} \delta_1^u.$$

If $\mathcal{U} = \bigcup_q \mathcal{U}(q)$, Cauchy-Schwarz gives

$$(M_1 |\mathcal{Q}|)^2 \leq |\mathcal{U}| \sum_{p \in \mathcal{Q}} J_p,$$

which rearranges to (6.20b117i). This is the full co-rank-one form of Ren's Lemma 6.6 and Corollary 6.5. $\square$

Lemma III.6.24.50 (Two-scale factorization of tube-to-fibre ratios). *In the normalized setting of Lemma III.6.24.47, fix an intermediate dyadic scale $R \in (\theta, 1)$. For every $\zeta > 0$, after retaining θ^ζ fractions and worsening nonconcentration constants by at most $\theta^{-O(\zeta)}$, one can construct a coarse scale- R incidence configuration $(\mathcal{P}_R, \mathcal{T}_R)$ and, for every retained R -cube Q , a rescaled fine configuration $(\mathcal{P}_Q, \mathcal{T}_Q)$ such that their fibre cardinalities M_R, M_Q are constant and*

$$\frac{|\mathcal{T}|}{M} \gtrsim \theta^\zeta \frac{|\mathcal{T}_R|}{M_R} \max_Q \frac{|\mathcal{T}_Q|}{M_Q}. \quad (6.20b117j)$$

Both the ordinary u -tube statistic and the co-rank-one κ -plate statistic pass to the coarse and fine fibres with the stated power loss.

Proof. Associate to every fine tube one of the $O_d(1)$ dyadic R -tube ancestors containing its image under S_H . For a fine base cube p , dyadically pigeonhole first the number of such ancestors meeting $\mathcal{T}(p)$ and then the number of fine descendants of each retained ancestor. Repeat the same pigeonhole over base cubes inside each occupied R -cube and finally over the occupied R -cubes themselves. There are only $O(\log(1/\theta))$ possible dyadic classes at each of these finitely many stages, so, for sufficiently small θ , their total loss is at most θ^ζ .

After this pruning, write M_R for the common number of retained coarse ancestors per coarse base cube and M_Q for the common number of fine descendants per relevant coarse ancestor in Q . Up to the same $\theta^{\pm\zeta}$ factor, $M \asymp M_R M_Q$. Pigeonhole once more so that every retained coarse tube contains a comparable number K of retained fine tubes. Then $|\mathcal{T}| \gtrsim \theta^\zeta K |\mathcal{T}_R|$, while for a maximizing Q the rescaled fine family has $|\mathcal{T}_Q| \lesssim K$ up to the same loss. Dividing by $M \asymp M_R M_Q$ gives (6.20b117j).

For nonconcentration, a coarse r -tube or $(r, d-1)$ -plate collects exactly the ancestors of fine tubes lying in a fixed $O_d(r)$ enlargement, so the normalized bounds (6.20b117d) pass to $\mathcal{T}_R$. Inside a fixed Q , affine rescaling sends the corresponding fine containers to tubes or hyperplates with comparable relative width; subset selection and the dyadic uniformizations cost only the already recorded factor $\theta^{-O(\zeta)}$. This is the finite combinatorial content of Ren's Proposition 6.2 specialized to $k = d-2$. $\square$

Proposition III.6.24.51 (Confined co-rank-one Furstenberg kernel from the unconfined gain). *Let $d \geq 3$, $0 \leq u < d-1$, $\max\{u, d-2\} < t \leq d$, $\kappa > 0$, and $0 < r_0 \leq 1$. There is $\epsilon_* = \epsilon_*(u, t, \kappa, d) > 0$ such that, for all sufficiently small dyadic δ/r_0 , the following holds. Let H be an $(r_0, d-1)$ -plate, let $\mathcal{P}$ be a $(\delta, t, (\delta/r_0)^{-\epsilon_*})$ -set of δ -cubes contained in H , and for each $p \in \mathcal{P}$ let $\mathcal{T}(p)$ be a family of δ -tubes meeting p . Assume that $\mathcal{T}(p)$ has simultaneously*

$$(\delta, u, (\delta/r_0)^{-\epsilon_*} r_0^{d-2-u}, 0) \quad \text{and} \quad (\delta, \kappa, (\delta/r_0)^{-\epsilon_*} r_0^{-\kappa}, d-2)$$

nonconcentration in the sense of Ren's Definition 1.7, the first condition being required from scale r_0 down to δ . Then

$$\left| \bigcup_{p \in \mathcal{P}} \mathcal{T}(p) \right| \geq (\delta/r_0)^{-\epsilon_*} \delta^{-2u} r_0^{2(u-(d-2))}. \quad (6.20b118)$$

Moreover ϵ_ may be chosen uniformly when (u, t) ranges in a compact subset of $\{(u, t) : 0 \leq u < d-1, \max\{u, d-2\} < t \leq d\}$.*

Proof. Set $k = d - 2$ and $\theta = \delta/r_0$. Choose $\beta > 0$ so small that

$$\frac{t - d\beta}{1 - \beta} > \max\{u, k\},$$

and choose

$$\max\{u, k\} < t_0 < \min\left\{d - 1, \frac{t - d\beta}{1 - \beta}\right\}.$$

(The interval is nonempty because $u < d - 1$ and $k = d - 2$.) Apply Proposition III.6.24.46 with parameters (u, t_0, κ, d) , and denote its gain and admissible-loss exponents by γ_*, η_* . Fix $\zeta > 0$ and then the dyadic Δ from Lemma III.6.24.48, with

$$0 < \zeta \ll \beta \min\{\gamma_*, \eta_*\}.$$

Finally choose $\epsilon_* > 0$ sufficiently small in terms of all these quantities. All choices depend only on u, t, κ, d .

First pigeonhole the fibre cardinalities. The ordinary tube condition at the finest scale gives, for every nonempty fibre,

$$|\mathcal{T}(p)| \geq \theta^{\epsilon_*} r_0^{u-k} \delta^{-u}. \quad (6.20b117k)$$

There are only logarithmically many dyadic cardinality classes, so after retaining a θ^ζ fraction of the base cubes and deleting tubes we may assume $|\mathcal{T}(p)| = M$ for all retained p , with the right side of (6.20b117k) weakened only by another factor θ^ζ . Subsets inherit all upper nonconcentration estimates after the same harmless power loss.

Lemma III.6.24.47 converts the confined hypotheses into the scale-free statistics (6.20b117d). Apply Lemma III.6.24.48 and obtain $R \in [\theta, \theta^\beta]$. Then apply the two-scale factorization Lemma III.6.24.50 at R .

The coarse configuration has point exponent t_0 and both required fibre nonconcentration estimates. Since $R \leq \theta^\beta$, the accumulated coefficient $\theta^{-O(\epsilon_* + \zeta)}$ is at most $R^{-\eta_*}$ once ϵ_*, ζ are small enough. Proposition III.6.24.46 therefore yields

$$\frac{|\mathcal{T}_R|}{M_R} \geq R^{-u-\gamma_*}. \quad (6.20b117l)$$

For every retained R -cube Q , the fine descendant set has the two t_0 -spacing bounds supplied by Lemma III.6.24.48. Apply Lemma III.6.24.49 at fine physical scale δ/R . Its constants are at most $\theta^{-O(\epsilon_* + \zeta)}$, and hence

$$\frac{|\mathcal{T}_Q|}{M_Q} \geq \theta^{O(\epsilon_* + \zeta)} r_0^{u-k} (\delta/R)^{-u}. \quad (6.20b117m)$$

Combining (6.20b117j), (6.20b117l), and (6.20b117m) gives

$$\frac{|\mathcal{T}|}{M} \geq \theta^{O(\epsilon_* + \zeta)} r_0^{u-k} \delta^{-u} R^{-\gamma_*}.$$

Using (6.20b117k) once more,

$$|\mathcal{T}| \geq \theta^{O(\epsilon_* + \zeta)} R^{-\gamma_*} r_0^{2(u-k)} \delta^{-2u}.$$

Because $R \leq \theta^\beta$, the gain is at least $\theta^{-\beta\gamma_*}$. The initial choice $\epsilon_*, \zeta \ll \beta\gamma_*$ absorbs every displayed bookkeeping loss and the fixed Δ -constants, leaving exactly (6.20b118) after possibly decreasing ϵ_* .

The compact-uniform assertion follows by the same finite-cover argument as Ren's Remark 1.12: after halving the admissible exponents, the strict inequalities in $u < t$, $t > k$, and (6.20b117e) persist in a neighbourhood of each parameter pair; a finite subcover of a compact parameter set supplies common choices. Thus the confined Theorem 1.11 specialization is local relative only to the unconfined M -sensitive gain Proposition III.6.24.46. $\square$

Lemma III.6.24.52 (Dense hyperplates cluster). *Let λ be a finite measure of mass at most one and suppose that, for some $\kappa > 0$,*

$$\lambda(W) \leq C_\lambda \rho^\kappa \quad (6.20b119)$$

for every $(\rho, d-2)$ -plate W , $r < \rho < 1$. If an $(r, d-1)$ -plate is called a -heavy when its λ -mass is at least a , then all a -heavy plates from a fixed bounded-overlap discretization are contained in at most $2a^{-1}$ plates of width

$$R \asymp_d r(C_\lambda/a^2)^{1/\kappa}. \quad (6.20b120)$$

Consequently the number of discretized a -heavy r -plates is at most $(C_\lambda/a)^N$ for some $N = N(d, \kappa)$.

Proof. Choose a maximal family $Y_1, \dots, Y_m$ of a -heavy r -plates with $\lambda(Y_i \cap Y_j) \leq a^2/2$ for $i \neq j$. If $f = \sum_j \mathbf{1}_{Y_j}$ and $S = \int f d\lambda$, then

$$S^2 \leq \int f^2 d\lambda \leq S + \frac{1}{2}m^2a^2.$$

If $m > 2a^{-1}$, then $S \geq ma > 2$, hence $S^2 - S > S^2/2 \geq m^2a^2/2$, a contradiction. Thus $m \leq 2a^{-1}$.

For any other heavy plate Y , maximality gives $\lambda(Y \cap Y_j) > a^2/2$ for some j . If θ is the largest principal angle between their central hyperplanes, then $Y \cap Y_j$ is contained in an $O_d(r/\theta)$ -neighbourhood of an affine $(d-2)$ -plane. By (6.20b119), $a^2 \leq C_\lambda(r/\theta)^\kappa$, whence $\theta \leq r(C_\lambda/a^2)^{1/\kappa}$. Since the two plates meet in the fixed bounded ambient region, Y lies in the enlargement (6.20b120) of Y_j . The polynomial cardinality bound follows because a width- R hyperplate contains only polynomially many elements of the fixed r -net of hyperplates. This is Ren's Lemmas 7.5–7.6 in the present co-rank-one case, with the counting exponent left as the harmless integer $N(d, \kappa)$. $\square$

Lemma III.6.24.53 (Local power decay around hyperplates). *Let $s > d-2$, and let μ, ν be finite measures of mass at most one which satisfy (6.20b119) with the same exponent $\kappa > 0$ (possibly different constants). Fix $r_0 \leq 1$ and $K \geq K_0(d, \kappa)$. Let $\mathcal{A}$ be the set of pairs which lie in a common K^{-1} -concentrated $(r_0, d-1)$ -plate. Then there are $\eta_0 = \eta_0(d, \kappa) > 0$ and a Borel set $\mathcal{B}_0$ with*

$$(\mu \times \nu)(\mathcal{B}_0) \leq K^{-1} \quad (6.20b121)$$

such that, outside $\mathcal{A} \cup \mathcal{B}_0$, both ordered pairs (μ, ν) and (ν, μ) have

$$(\eta_0, Kr_0^{-\eta_0})\text{-thin } (d-1)\text{-plates down from scale } r_0. \quad (6.20b122)$$

Proof. We record the scale argument because it is later used in the bootstrap. At a fixed $r \leq r_0$, call an r -hyperplate heavy when its ν -mass is at least $K(r/r_0)^\eta$. Lemma III.6.24.52 puts all heavy plates into $O((K(r/r_0)^\eta)^{-N})$ clusters whose enlarged widths are a positive power of r/r_0 , provided $\eta \ll \kappa/N$.

For a base point x , two cluster plates through x whose central hyperplanes make angle at least r/r_0 have intersection inside an $O(r/r_0)$ -neighbourhood of an affine $(d-2)$ -plane. The μ -bound (6.20b119), followed by a union bound over pairs of clusters, shows that the set of such base points has μ -mass

$$\lesssim K^{-2}(r/r_0)^\eta$$

after fixing, for example, $\eta = \kappa/(4N)$ and increasing K_0 . Hence outside a set E_r of that mass, every heavy r -hyperplate through x lies in one $(r_0, d-1)$ -plate $P_r(x)$.

If x is not already in the concentrated-pair set $\mathcal{A}$, the mass outside $P_r(x)$ is therefore bounded by $K(r/r_0)^\eta$. Apply this at the superexponential scales $r_n = r_0 K^{-2^n}$ and telescope the nested exceptional hyperplates. The first $O(\log \eta^{-1})$ terms cost $O(K^{-1})$ by the definition of $\mathcal{A}$; the tail is bounded by $\sum_n K^{1-2^{n-2}\eta} \lesssim K^{-1}$. Summing the base exceptional sets gives (6.20b121), and the resulting fibre estimate is (6.20b122). Interchanging μ, ν and taking the union of the two exceptional sets gives the symmetric statement. This is the full content of Ren's Proposition 7.3 specialized to $k = d-1$. $\square$

Lemma III.6.24.54 (Radial L^p base case inside a hyperplate). *Let $m = d-1$ and $m-1 < s_* < m$. Suppose that two compactly supported probability measures λ, τ are separated and have finite s_* -energy after orthogonal projection to an m -plane. Then, for some $p = p(s_*, m) > 1$,*

$$\int \|\pi_x \tau\|_{L^p(S^{m-1})}^p d\lambda(x) < \infty. \quad (6.20b123)$$

Consequently, if both measures lie in an r_0 -neighbourhood of that m -plane and have uniform s -Frostman constants for some $s > s_ > m-1$, then for every $K \geq K_0$ there is a relation $\mathcal{B}_{\text{base}}$ of product mass at most K^{-1} on whose complement every r -tube through a base point, $0 < r \leq r_0$, has opposite-fibre mass*

$$\lesssim K^N r_0^{m-1}. \quad (6.20b124)$$

Proof. We include the analytic estimate to avoid any forward dependency on Chapter III.10. Put $n = m-1$. Choose

$$2n - s_* < t < n, \quad 1 < p < \min \left\{ 2 - \frac{t}{n}, \frac{t}{2n - s_*} \right\}.$$

For a direction $e \in S^n$, let P_e denote orthogonal projection to $e^\perp$. The Fourier-energy identity and Cauchy-Schwarz give the fractional trace inequality

$$\left| \int g d\eta \right| \lesssim I_{n-2\sigma}(\eta)^{1/2} \left(\int |\hat{g}(\xi)|^2 |\xi|^{2\sigma} d\xi \right)^{1/2}.$$

Moreover, if a spherical measure ζ has ball growth $\zeta(B(e, r)) \lesssim r^\alpha$ with $\alpha > t$, then dyadic cap summation gives

$$\int I_t(P_e \lambda) d\zeta(e) \lesssim I_t(\lambda).$$

Apply the trace inequality to $g = P_e \tau$ with $\sigma = (s_* - n)/2$, dualize against $L^{p'}(P_e \lambda)$, and use $p(2n - s_*) \leq t$. A second duality on S^n , with $d\zeta = |F|^p d\mathcal{H}^n$, is admissible because $n(2-p) > t$. Coarea then yields (6.20b123). The usual weighted-ray measure $d\tau_x(y) = c|x-y|^{1-m} d\tau(y)$ converts line projections to radial projections; separation makes τ_x uniformly comparable with τ . Approximation by smooth measures and weak L^p compactness remove regularity assumptions.

For the finite-scale consequence, first project to the central m -plane. At scales $\rho \geq r_0$ the projected measures retain the s -Frostman estimate up to a fixed constant. Replace each occupied r_0 -cell by normalized Lebesgue measure on that cell. The resulting probability measures are uniformly s -Frostman at every scale and hence have uniformly finite s_* -energy for any fixed $m-1 < s_* < s$. Since the original supports are separated by a fixed amount and r_0 is below a fixed geometric threshold, the regularized projections are still separated. Apply (6.20b123) to these regularizations; radial arcs at angular resolution r_0 differ from those of the original projected measures only by a fixed enlargement.

Now cover all r_0 -tubes through x carrying mass at least $K^N r_0^{m-1}$ by finitely overlapping $2r_0$ -tubes. Their number is $O(K^{-N} r_0^{1-m})$, so their radial images have total spherical measure $O(K^{-N})$. Hölder and (6.20b123) make the product mass of pairs lying in such tubes $O(K^{-N(1-1/p)})$. Choose $N > 1 + (1 - 1/p)^{-1}$. Outside this exceptional relation no r_0 -tube is heavy, and smaller tubes are contained in one of the covering r_0 -tubes, proving (6.20b124). This is Ren's Proposition 8.2 with its Orponen input reconstructed locally. $\square$

Lemma III.6.24.55 (One-step radial bootstrap from the confined kernel). *Fix $m = d - 1$ and parameters*

$$0 \leq u \leq m, \quad \max\{u, m-1\} < s \leq m, \quad \kappa > 0.$$

There are $\eta_1 = \eta_1(u, s, \kappa, d) > 0$ and K_0 with the following property. Suppose that separated s -Frostman measures μ, ν , supported in one (r_0, m) -plate, have symmetrically on a Borel relation G

$$(u, Kr_0^{-(u-(m-1))})\text{-thin tubes}, \quad (\kappa, Kr_0^{-\kappa})\text{-thin } m\text{-plates} \quad (6.20b125)$$

down from scale r_0 . Assuming Proposition III.6.24.51, there is $\mathcal{B}_1$ with product mass at most K^{-1} such that both ordered pairs have

$$(u + \eta_1, K^{d+1} r_0^{-(u+\eta_1-(m-1))})\text{-thin tubes} \quad (6.20b126)$$

on $G \setminus \mathcal{B}_1$. The gain is uniform on compact subsets of $\{(u, s, \kappa) : \max\{u, m-1\} < s \leq m, \kappa > 0\}$.

Proof. This is the measure-theoretic bootstrap of Ren's Proposition 8.3; we give the two contradiction mechanisms because they are the only places where the finite incidence kernel enters.

For dyadic $r \leq r_0$, call an r -tube through x *new-heavy* when its G -fibre mass is at least

$$K^{d+1} \frac{r^{u+\eta}}{r_0^{u+\eta-(m-1)}}.$$

Let H_r be the relation swept out by essentially distinct new-heavy tubes. The old u -thin estimate and bounded overlap away from the base point give

$$|\mathcal{T}'_{x,r}| \lesssim r^{-(u+\eta)} r_0^{u+\eta-(m-1)}. \quad (6.20b127)$$

It suffices to prove

$$(\mu \times \nu)(H_r) \leq 2(r/r_0)^\eta \quad \text{whenever } r/r_0 < K^{-1/\eta}, \quad (6.20b128)$$

because summing the dyadic H_r below $r_0 K^{-d/\eta}$ gives product mass $O(K^{-d})$, while above that scale new-heaviness contradicts the old tube estimate.

Assume (6.20b128) fails. Fubini and dyadic pigeonholing leave a base set of mass $\gtrsim (r/r_0)^{2\eta}$ and, for each surviving base point, a family $\mathcal{T}_x$ of new-heavy tubes whose masses lie between the lower threshold and the old u -thin upper bound. The tube family is therefore a $(r, u, (r/r_0)^{-5\eta})$ -set; the m -plate hypothesis gives at the same time a $(r, \kappa, (r/r_0)^{-5\eta}, m-1)$ -set after the natural powers of r_0 are restored. Choose

$$\gamma = \frac{15\eta}{s - \max\{u, m-1\}}.$$

Split the tubes according as less than or at least one third of their fibre mass is contained in some ball of radius $(r/r_0)^\gamma$.

In the non-concentrated case, discretize a Frostman-positive set of base points at scale r . The preceding two nonconcentration estimates are precisely the hypotheses of Proposition III.6.24.51 with plate index $m-1 = d-2$ and point exponent s . If $\eta \leq \epsilon_*^2/4$, the kernel gives a gain $(r/r_0)^{-\sqrt{\eta}}$ in the total number of tubes. But every retained tube contains two fibre points separated by at least $(r/r_0)^\gamma$; double counting such pairs shows that some separated pair lies in $\gtrsim (r/r_0)^{-\sqrt{\eta}+6\eta}$ essentially distinct $2r$ -tubes. Geometry permits only $O((r/r_0)^{-(d-1)\gamma})$ such tubes. Taking η so small that $\sqrt{\eta} - 6\eta > (d-1)\gamma$ is a contradiction.

In the concentrated case, retain the heavy balls and use the s -Frostman bound to remove the central r -ball. A summable annular pigeonholing produces one scale $r \leq \xi \leq (r/r_0)^\gamma$ and a relation of product mass

$$\gtrsim (r/r_0)^{8\eta} \left(\frac{\xi}{(r/r_0)^\gamma} \right)^\eta \quad (6.20b129)$$

such that every selected tube contains at annular distance $\asymp \xi$ a fibre mass

$$\gtrsim (r/r_0)^{u+4\eta} \left(\frac{\xi}{(r/r_0)^\gamma} \right)^\eta r_0^m. \quad (6.20b130)$$

Fubini fixes one annular centre y . Maximal separation of tube directions at angular scale r/ξ , together with the old u -thin estimate (when $\xi > r/r_0$) or the trivial covering by r_0 -tubes (when $\xi < r/r_0$), gives the same lower bounds as in Ren's equations (8.38)–(8.39). The selected tubes have bounded overlap outside $B(y, \xi)$, so summing (6.20b130) is bounded above by the s -Frostman estimate $\nu(B(y, 2\xi)) \lesssim \xi^s$. In the first regime the power comparison reduces to the strict gap $u < s$ after the $O(\eta + \eta\gamma)$ losses are absorbed. In the

second regime the ratio of the lower and upper bounds is monotone increasing in ξ because $m + 2\eta - s > 0$; it is therefore enough to test the endpoint $\xi = r/r_0$, where the same strict gap $u < s$ absorbs the $O(\eta + \eta\gamma)$ losses. Taking η sufficiently small yields a contradiction in both cases. Hence (6.20b128) holds, and the dyadic union gives (6.20b126). All inequalities are strict on compact parameter subsets, giving the asserted uniformity. $\square$

Proposition III.6.24.56 (Local co-rank-one Ren thin-tube theorem relative to the confined kernel). *Let $d \geq 3$, $d - 2 < \sigma < s \leq d - 1$, and $\varepsilon > 0$. Assuming Proposition III.6.24.51, there exist N, K_0 depending on σ, s, d and $\eta = \eta(\varepsilon) > 0$ such that, for $r_0 \leq 1$, $K \geq K_0$, and separated s -Frostman measures μ, ν of mass at most one in a fixed bounded set, the concentrated-pair set $\mathcal{A}$ at scale r_0 admits a Borel exceptional relation $\mathcal{B}$ with*

$$(\mu \times \nu)(\mathcal{B}) \leq K^{-\eta} \quad (6.20b131)$$

such that every r -tube T through x obeys

$$\nu(T \setminus (\mathcal{A}_x \cup \mathcal{B}_x)) \leq r^\sigma r_0^{-[\sigma - (d-2) + N\varepsilon]} K^N. \quad (6.20b132)$$

Proof. We reconstruct Ren's Sections 7–8 in the co-rank-one case.

Maximal hyperplate concentration. Fix

$$0 < \kappa_0 < s - (d - 2).$$

An (s, C) Frostman measure assigns mass $O(\rho^{s-(d-2)})$, hence in particular $O(\rho^{\kappa_0})$, to every $(\rho, d - 2)$ -plate: cover the plate inside the bounded ambient ball by $O(\rho^{-(d-2)})$ balls of radius $O(\rho)$. Thus Lemma III.6.24.53, after deleting product mass $O(K_1^{-1})$ both ordered pairs have a positive-power thin-hyperplate estimate outside the K_1^{-1} -concentrated relation. Lemma III.6.24.54 gives the $u = 0$ thin-tube estimate. Starting from $u = 0$, iterate Lemma III.6.24.55 with $\kappa = \kappa_0$. The bootstrap gain is bounded below while u ranges in a compact subset of $[0, \sigma]$ because $s > \sigma > d - 2$; hence finitely many iterations reach $u \geq \sigma$. This proves the analogue of Ren's Theorem 8.1 inside any fixed r_0 -hyperplate, with product exceptional mass $O(K_1^{-1})$ and the correct factor $r_0^{-(\sigma-(d-2))}$.

General configuration. It remains to remove the hypothesis that both measures are already confined to one r_0 -hyperplate. It is enough to treat $\varepsilon = 1/M$; a smaller exponent only weakens the final power. Let N_0 be the polynomial heavy-hyperplate counting exponent in Lemma III.6.24.52, put $\eta_n = (N_0 + 2)^{n-M}$, and let $\mathcal{A}_n$ be the pairs lying in a $K^{-\eta_n}$ -concentrated $(r_0^{n\varepsilon}, d - 1)$ -plate. Thus $\mathcal{A}_M = \mathcal{A}$.

At stage n , cover the relevant concentrated plates by essentially distinct plates. The heavy-plate count gives $O(K^{N_0\eta_n})$ such plates. Apply the already proved maximal-concentration theorem inside each one, with next threshold $K^{-\eta_{n+1}}$ and the rescaled parameter $r_0^{n\varepsilon}$. Since $(N_0 + 1)\eta_n < \eta_{n+1}$, the union of the new exceptional relations has product mass $O(K^{-\eta_n})$. Moreover every tube meeting $\mathcal{A}_n$ satisfies the required σ -thin estimate after removal of $\mathcal{A}_{n+1}$ and that stage's exceptional relation. This is the scale-exhaustion lemma corresponding to Ren's Lemma 8.7.

Take the union over $n = 1, \dots, M$. Its product mass is $O(K^{-\eta_0})$. For a fixed tube, choose the last stage at which it meets $\mathcal{A}_n$; the next concentrated set then disappears from that tube, while the accumulated rescalings contribute only $r_0^{-N\varepsilon}$. This yields (6.20b132) with

$\eta = \eta_0$ and proves the proposition. No measure-level theorem of Ren is used. The confined co-rank-one reconstruction proves the confined kernel Proposition III.6.24.51 locally, and the unconfined co-rank-one reconstruction proves the unconfined M -sensitive finite gain Proposition III.6.24.46 locally relative only to the endpoint multilinear Keakey estimate (6.20b116a). $\square$

The measure-level thin-tube proof parallels Ren [53, Theorem 1.13], while the confined co-rank-one argument gives the $k = d - 2$ specialization of Ren's Theorem 1.11 from Section 6. The unconfined argument supplies the remaining Section 3–5 chain for every effective plate index needed here: Furstenberg gain, robust slicing, quasi-product incidence, angular-to-ball transfer, regular-set gain, and general-set branching assembly. Hence both Propositions III.6.24.46 and III.6.24.51 follow from the proofs above and the endpoint multilinear Keakey estimate (6.20b116a).

Lemma III.6.24.57 (Concentrated-pair cleanup in the zero-hyperplane centre branch). *Let $a > d - 2$. Let μ be a compactly supported a -Frostman probability measure and let ν be a compactly supported probability measure satisfying $\nu(P) = 0$ for every affine hyperplane P . There is a normalized positive-mass restriction $\tilde{\mu}$ of μ , chosen independently of K and ϵ_0 , such that for every fixed $K > 1$ and $\epsilon_0 > 0$ there is $r_0 = r_0(K, \epsilon_0, \tilde{\mu}, \nu) > 0$ with the following property. For readability rename $\tilde{\mu}$ as μ . Let $\mathcal{A}_K^\vee(r_0)$ be the set of pairs (x, y) which lie in a common closed $2r_0$ -neighbourhood of an affine hyperplane Q for which*

$$\max\{\mu(Q^{(2r_0)}), \nu(Q^{(2r_0)})\} \geq (2K)^{-1}. \quad (6.20b120)$$

Then $\mathcal{A}_K^\vee(r_0)$ is Borel and

$$(\mu \times \nu)(\mathcal{A}_K^\vee(r_0)) < \epsilon_0. \quad (6.20b121)$$

Moreover Ren's source concentrated-pair set at scale r_0 and threshold K^{-1} is contained in $\mathcal{A}_K^\vee(r_0)$.

Proof. First note that two distinct affine hyperplanes meet in an affine set of dimension at most $d - 2$. Lemma III.6.24.61, with exponent $a > d - 2$, shows that this intersection is μ -null. Hence the affine hyperplanes of positive μ -mass are pairwise disjoint modulo μ and are therefore countable. If the complement of their union has positive μ -mass, restrict μ to that complement; then the new μ gives zero mass to every affine hyperplane. Otherwise restrict μ to one positive-mass affine hyperplane P . The Frostman exponent a is preserved, up to the harmless normalization constant.

In the first case both μ and ν are affine-hyperplane-null. The compactness argument of Lemma III.6.24.7 gives, for all sufficiently small r_0 ,

$$\sup_Q \mu(Q^{(2r_0)}) < (2K)^{-1}, \quad \sup_Q \nu(Q^{(2r_0)}) < (2K)^{-1}.$$

Thus $\mathcal{A}_K^\vee(r_0) = \emptyset$.

Suppose now that μ is supported on an affine hyperplane P . The same compactness argument gives $\sup_Q \nu(Q^{(2r_0)}) < (2K)^{-1}$ for small r_0 , so only μ -heavy plates need be considered. Let Q be their central hyperplane and let ϑ be the acute angle between the direction spaces of P and Q . If $\vartheta \lesssim r_0$ there is nothing to prove. Otherwise $P \cap Q^{(2r_0)}$, inside one

fixed ball containing the supports, is contained in a slab about an affine $(d - 2)$ -plane of width $w \lesssim r_0/\vartheta$. Covering that slab by $O(w^{-(d-2)})$ balls of radius $O(w)$ gives

$$\mu(Q^{(2r_0)}) \lesssim w^{a-(d-2)}. \quad (6.20b122)$$

The lower bound $(2K)^{-1}$ therefore forces $w \gtrsim_K 1$, hence $\vartheta \lesssim_K r_0$. Since the plate has positive μ -mass, it meets P within $O(r_0)$; on the fixed bounded support the angle estimate then implies

$$Q^{(2r_0)} \cap \text{spt } \nu \subset P^{(C_K r_0)}. \quad (6.20b123)$$

Consequently

$$\mathcal{A}_K^\vee(r_0) \subset \text{spt } \mu \times (\text{spt } \nu \cap P^{(C_K r_0)}).$$

Because $\nu(P) = 0$, the right-hand product mass tends to zero as $r_0 \downarrow 0$, which proves (6.20b121).

For the measurability assertion, restrict the affine Grassmannian to hyperplanes meeting one fixed bounded ball. It is compact. After using closed plates, the set of heavy hyperplanes in (6.20b120) is closed by upper semicontinuity of the mass of a closed plate, and the incidence relation “ x, y lie in the same plate” is closed. Projection over the compact hyperplane parameter therefore gives a closed, hence Borel, set. Finally every Ren K^{-1} -concentrated r_0 -plate is contained in the doubled plate appearing above, and its mass for the relevant measure is at least $(2K)^{-1}$; hence the source cleanup is contained in $\mathcal{A}_K^\vee(r_0)$. $\square$

Lemma III.6.24.58 (Ren radial ball-Frostman bridge). *Let $d \geq 3$ and*

$$d - 2 < \sigma < s \leq d - 1.$$

Let μ, ν be compactly supported probability measures. Assume that μ is a-Frostman for some $a > s$, that ν is b-Frostman for some $b > s$, and that $\nu(P) = 0$ for every affine hyperplane P . Then, after a positive-mass restriction of μ , there are a Borel set E with $\mu(E) > 0$ and a Borel relation $\Lambda \subset E \times \text{spt } \nu$ such that

$$\nu(\Lambda_x) \geq \frac{1}{2} \quad (6.20b124)$$

and

$$(e_x)_\# \nu_{\Lambda_x}(B_{S^{d-1}}(\omega, r)) \leq C_x r^\sigma, \quad \omega \in S^{d-1}, \quad 0 < r \leq 1. \quad (6.20b125)$$

Here $e_x(y) = (y - x)/|y - x|$ and $C_x < \infty$.

Proof. First make the K -independent positive-mass restriction of μ furnished by the first paragraph of the proof of Lemma III.6.24.57. Then restrict μ and ν to small positive-mass balls with separated closures and apply one fixed similarity so that the resulting supports lie in $B(0, 1)$ and are ~ 1 separated. Such balls exist because positive Frostman exponents exclude atoms, so one may choose distinct support points and then shrink around them. These choices are now frozen. Since $a, b > s$, both normalized restrictions are s -dimensional in the sense of Ren’s Definition 2.1, with fixed Frostman constants.

Apply Proposition III.6.24.56 and, for concreteness, $\varepsilon = 1$. Choose $K \geq K_0$ so large, in terms of these fixed Frostman constants, that the product mass bound for the resulting exceptional relation $\mathcal{B}$ is less than $1/8$. The hyperplane-null/supported-on-one-hyperplane

dichotomy fixed above is preserved by the subsequent positive-mass localization. With this K fixed, apply the small-scale conclusion of Lemma III.6.24.57 within that already fixed branch to choose r_0 so that $(\mu \times \nu)(\mathcal{A}_K^\vee(r_0)) < 1/8$. Now apply Proposition III.6.24.56 with these fixed K, r_0 . Since the source set $\mathcal{A}$ is contained in $\mathcal{A}_K^\vee(r_0)$, (6.20b132) yields, with a fixed constant C_* ,

$$\nu(T \setminus (\mathcal{A}_{K,x}^\vee(r_0) \cup \mathcal{B}_x)) \leq C_* r^\sigma \quad (6.20b126)$$

for every tube T through x .

By regularity of the finite product measure, enlarge the source exceptional set $\mathcal{B}$, if necessary, to a Borel set while keeping its product mass below $1/8$. Let $\mathcal{R} = \mathcal{A}_K^\vee(r_0) \cup \mathcal{B}$. Fubini and Markov give a positive- μ Borel set E on which $\nu(\mathcal{R}_x) \leq 1/2$ (after replacing the measurable level set by a Borel representative modulo a μ -null set). The relation $\Lambda = \{(x, y) : x \in E, y \notin \mathcal{R}_x\}$ is then Borel, and (6.20b124) holds.

Because the supports are separated and bounded, there is $r_* > 0$, independent of x , such that for $0 < r \leq r_*$ the inverse image under e_x of a spherical cap of angular radius r is contained in a Euclidean Cr -tube through x . Apply (6.20b126) and normalize by $\nu(\Lambda_x) \geq 1/2$. For $r_* < r \leq 1$, the probability bound $1 \leq r_*^{-\sigma} r^\sigma$ supplies the same estimate after enlarging the constant. This proves (6.20b125); similarity normalization changes only the constant. $\square$

Lemma III.6.24.59 (Ball-Frostman transfer through the norm Gauss map). *Let N be a C^2 norm whose unit sphere has everywhere positive Gaussian curvature, and put*

$$\psi_N(e) = \frac{DN(e)}{|DN(e)|}, \quad e \in S^{d-1}.$$

If a probability measure ρ on S^{d-1} satisfies

$$\rho(B_{S^{d-1}}(\omega, r)) \leq Cr^\sigma \quad (0 < r \leq 1) \quad (6.20b127)$$

for some $\sigma > d - 2$, then

$$(\psi_N)_\# \rho(H^{(r)}) \leq C'_N r^{\sigma - (d-2)}, \quad H \in G(d, d-1), \quad 0 < r \leq 1. \quad (6.20b128)$$

Proof. Positive Gaussian curvature makes the spherical Gauss map ψ_N a C^1 diffeomorphism. Its differential and inverse differential are bounded on the compact sphere, so ψ_N is bi-Lipschitz. Therefore $(\psi_N)_\# \rho$ satisfies the same spherical ball-Frostman exponent σ , with a norm-dependent constant. The intersection of a linear hyperplane with S^{d-1} is a $(d-2)$ -sphere; its spherical r -neighbourhood is covered by $O_d(r^{-(d-2)})$ spherical balls of radius $O(r)$. Summing the ball estimates proves (6.20b128). $\square$

Lemma III.6.24.60 (Bare- C^2 curved-norm closure via Ren radial preconditioning). *In the residual zero-hyperplane branch of Theorem III.6.25, let $d \geq 3$ and $d - 2 < \alpha, \kappa < d$, with $\kappa \leq d - 1$. Under the finite incidence input Proposition III.6.24.51, every C^2 positively curved norm satisfies the pinned conclusion with a positive gain depending only on d, α, κ and not on the norm.*

Proof. Set

$$\Delta = \min\{\alpha - (d-2), \kappa - (d-2), 1\} > 0, \quad s = d-2 + \frac{\Delta}{2}, \quad \sigma = d-2 + \frac{\Delta}{4}, \quad (6.20b129)$$

and put

$$\zeta = \sigma - (d-2) = \frac{\Delta}{4} > 0. \quad (6.20b130)$$

The continuous nonlinear gain $\eta_d(\cdot, \zeta)$ from Theorem III.6.24 is positive. Choose $a < \alpha$ so close to α that $a > s$ and

$$\frac{a}{d} + \eta_d(a, \zeta) \geq \frac{\alpha}{d} + \frac{1}{4}\eta_d(\alpha, \zeta). \quad (6.20b131)$$

Argue by contradiction with a compact bad-centre set Y of Hausdorff dimension $> \kappa$. Choose an a -Frostman probability μ on a compact subset of A and a b -Frostman probability ν on Y with $\kappa < b < \dim_{\text{H}} Y$. In the present residual branch $\nu(P) = 0$ for every affine hyperplane. Since $a, b > s$, apply Lemma III.6.24.58. It gives a Borel base set E of positive μ -mass, hence $\dim_{\text{H}} E \geq a$, and positive- ν fibres Λ_x satisfying the spherical ball estimate (6.20b125).

The preceding localization keeps E a positive distance from the support of the restricted parameter measure, so $F_y(x) = N(x-y)$ has no singular point on the relevant product. Its direction map is

$$\theta_x(y) = \pm \psi_N(e_x(y)).$$

Lemma III.6.24.59 therefore gives

$$(\theta_x)_\# \nu_{\Lambda_x}(H^{(r)}) \leq C_{x,N} r^\zeta, \quad (6.20b132)$$

which is exactly condition (6.14). Theorem III.6.24, applied on E , now produces a centre $y \in \text{spt } \nu \subset Y$ with

$$\dim_{\text{H}} N(E-y) \geq \frac{a}{d} + \eta_d(a, \zeta) \geq \frac{\alpha}{d} + \frac{1}{4}\eta_d(\alpha, \zeta). \quad (6.20b133)$$

Since $E \subset A$, this contradicts the definition of the bad-centre set whenever the theorem gain is chosen below the last positive quantity.

The exponent $\zeta = \Delta/4$ depends continuously only on (α, κ) . Ren's constants, the cleanup scale, and the bi-Lipschitz constant of ψ_N enter only through $C_{x,N}$ and the starting scale, which Theorem III.6.24 allows. For the present residual branch one may take, for example,

$$c_{\text{Ren}}(d, \alpha, \kappa) := \frac{1}{8} \eta_d(\alpha, \zeta) > 0.$$

This is continuous in (α, κ) and independent of the norm; the auxiliary choice of a is used only to obtain the contradiction in (6.20b131). Taking the minimum with the already proved hyperplane-centre, planar/ellipsoidal, and supercritical gains therefore gives the continuous norm-independent theorem gain. $\square$

Ren reduction and the bare- C^2 closure. The finite-incidence input in Proposition III.6.24.51, together with Proposition III.6.24.56, resolves the final residual branch. The higher-dimensional thin-tube argument supplies, after Lemma III.6.24.57, a positive

parameter fibre whose *ordinary radial direction measure* is Frostman on spherical balls with an exponent $\sigma > d - 2$. This is the stronger estimate missing from the direct one-scale Gauss-map analysis. Ball-Frostman regularity is genuinely invariant under the bi-Lipschitz norm Gauss map, and only afterwards does Lemma III.6.24.59 convert it into the hyperplane decay required by Theorem III.6.24. The Gauss-map counterexample therefore still rules out the direct hyperplane-decay transfer, while the ball-Frostman route avoids that obstruction.

Lemma III.6.24.60 closes the arbitrary positively curved C^2 zero-hyperplane branch, including the superflat moduli left open by the critical modulus balance. The confined co-rank-one argument proves the corresponding Theorem 1.11 specialization, and the unconfined co-rank-one argument proves the specialization of Ren [53, Theorem 5.1] required by the co-rank bootstrap, including the effective- k reduction, from the harmonic-analysis estimate (6.20b116a). This supplies the radial-preconditioning bridge needed for Theorem III.6.25 without using the obstructed direct Gauss-map transfer. **Persistent common-**

track obstruction. Lemma III.6.24.9 makes each same-vertex bad-cone family exactly affine, while Lemma III.6.24.12 shows that the Appendix-B cross-scale telescope needs no alignment of exceptional hyperplanes. The exact-belt exclusion also remains valid in eliminating exact atoms: by Lemma III.6.24.14, positive exact Legendre-belt mass can occur only at a μ -null set of vertices in the zero-hyperplane centre branch.

Lemma III.6.24.15 is likewise still a valid pointwise dichotomy: failure of one-exceptional-belt pruning forces two separated heavy same-base Legendre belts. What the persistent-belt counterexample changes is the status of the proposed global estimate. Lemma III.6.24.17 constructs an admissible smooth positively curved norm and Frostman measures with zero mass on every affine hyperplane for which

$$\mu(T_\delta) = 1$$

for every sufficiently small δ . In the same example, no single exceptional $O(\delta^{\tau/2})$ belt can satisfy the one-scale pruning hypothesis (6.20b17). Thus (6.20b21m) is not merely unproved: it is false in the generality in which the exact-belt exclusion proposed to use it.

The isolated-contact warning in Lemma III.6.24.16 is therefore upgraded to a persistent open-sector obstruction. Two distinct Legendre directions can carry logarithmically heavy mass along the same physical hyperplane track at all small scales. Lemma III.6.24.19 remains consistent with this phenomenon. Thus the one-exceptional-belt architecture cannot handle the low-centre general-norm zero-hyperplane case. The radial-Frostman bridge above supplies the different mechanism that is needed.

Multibelt telescope and multiplicity obstruction. Lemma III.6.24.20 shows that the same-base telescope lemma residue-class argument survives with several exceptional belts. The only new transition cost is $M_n \delta_{n-N}^\tau$, but for unbounded multiplicity one must also retain a small seed union in each residue class; transition control alone cannot prevent the exceptional families from covering all of Y . In particular every uniformly bounded multiplicity is harmless at the telescopic level.

Lemma III.6.24.21 shows that bounded multiplicity is not universal. A single smooth positively curved norm and a single hyperplane-null 2-Frostman centre measure carry, for every M , $M + 1$ separated heavy Legendre belts on one fixed positive-mass base set

once the scale is sufficiently small. No union of at most M shrinking exceptional belts can prune them all. Thus the first telescope architecture not refuted by the present examples must allow scale-dependent multiplicity and simultaneously prove both the seed condition (6.20b25) and the transition summability (6.20b24).

Finally, a direct check of the published Section 6.2 argument in Shmerkin [16] confirms that the normed extension is justified there precisely by asserting that local bi-Lipschitzness of the Gauss map preserves the required decay with the same exponent. The Gauss-map counterexample disproves that implication in $d \geq 3$. Hence the remaining low-centre general-norm clause is best not available from that transfer step alone. The radial-Frostman argument below supplies an independent route to the final normed conclusion.

Why the higher-dimensional transfer is not automatic. For $d \geq 3$, a nonlinear sphere diffeomorphism may send a curved $(d - 2)$ -dimensional submanifold into a great hyperplane section. Hyperplane decay of a measure supported on the curved submanifold can then be positive, while the push-forward gives full mass to one hyperplane. This phenomenon can occur for the Gauss map of a C^2 norm with everywhere positive Gaussian curvature. The counterexample below gives the explicit construction and its $O(r^{1/2})$ strip estimate. Thus the general $d \geq 3$ normed clause cannot be made local from the Section 6.2 bi-Lipschitz sentence alone. The preceding lemma bypasses this obstruction when $\kappa > d - 1$ by using the Frostman dimension of the parameter measure directly; it does not assert invariance of decay under ψ .

Lemma III.6.24.61 (Frostman decay near affine planes). *Let σ be a compactly supported t -Frostman probability measure in $\mathbb{R}^d$. If V is an affine m -plane and $t > m$, then*

$$\sigma(V^{(r)}) \lesssim r^{t-m}, \quad 0 < r \leq 1. \quad (6.20c)$$

Proof. Inside a fixed bounded ball, $V^{(r)}$ is covered by $O(r^{-m})$ balls of radius $O(r)$. Summing the Frostman estimate over those balls gives the claim. $\square$

Theorem III.6.25 (Pinned Euclidean and curved-norm distance sets). *Let $d \geq 2$ and let $d - 2 < \alpha, \kappa < d$. There is*

$$c = c_d(\alpha, \kappa) > 0,$$

continuous in (α, κ) , such that if $A \subset \mathbb{R}^d$ is Borel with $\dim_H A \geq \alpha$, then

$$\dim_H \{y \in \mathbb{R}^d : \dim_H \{|x - y| : x \in A\} < \alpha/d + c\} \leq \kappa. \quad (6.20a)$$

The same conclusion holds with $|x - y|$ replaced by $N(x - y)$ for every C^2 norm N whose unit ball has everywhere positive Gaussian curvature, with a gain independent of the norm.

Proof. We prove the dual statement and keep track of a gain which is continuous in the two dimension parameters. Namely, if Y is compact with $\dim_H Y > \kappa$, we show that some $y \in Y$ satisfies

$$\dim_H \{|x - y| : x \in A\} \geq \alpha/d + c_d(\alpha, \kappa). \quad (6.20d)$$

If the exceptional set in the theorem had dimension $> \kappa$, compact capacitability would supply such a set Y inside it, contradicting (6.20d).

We first prove the dual statement under the strict hypothesis $\dim_{\mathbb{H}} A > \alpha$, by induction on the ambient dimension in the one place where the parameter measure is supported on a hyperplane. Compact selection and separation allow us to assume that A, Y are disjoint compact subsets of a bounded ball. Choose an α -Frostman probability measure μ on A . Choose also κ_1 with $\kappa < \kappa_1 < \dim_{\mathbb{H}} Y$ and a κ_1 -Frostman probability measure ν on Y . We shall use only the weaker κ -Frostman estimates of ν , so all quantitative gains below depend on κ , not on the auxiliary margin $\kappa_1 - \kappa$. For $y \in Y$ set $F_y(x) = |x - y|$. On the separated product this is a C^2 nonsingular family and

$$\theta_x(y) = \text{dir } D_x F_y = \pm e_x(y), \quad e_x(y) = \frac{y - x}{|y - x|}. \quad (6.20e)$$

Case 1: ν charges an affine hyperplane. Suppose $\nu(P) > 0$ for an affine hyperplane P , and replace ν by its normalized restriction to P . It remains κ_1 -Frostman, hence also κ -Frostman, and its support has Hausdorff dimension at least $\kappa_1 > \kappa$. Since this support lies in the $(d - 1)$ -plane P , necessarily $\kappa_1 \leq d - 1$, hence $\kappa < d - 1$; together with the standing bound $\kappa > d - 2$, this places the parameter exponent in the admissible range for the $(d - 1)$ -dimensional induction below. There are two subcases.

If $\dim_{\mathbb{H}}(A \cap P) \geq \alpha$, then necessarily $\alpha \leq d - 1$. For $d = 2$, identify P with $\mathbb{R}$. For every $y \in P$, one of $A \cap P \cap (-\infty, y]$ and $A \cap P \cap [y, \infty)$ has the full Hausdorff dimension of $A \cap P$, and on either half-line the map $x \mapsto |x - y|$ is an isometry. Hence

$$\dim_{\mathbb{H}}\{|x - y| : x \in A \cap P\} \geq \alpha,$$

which is stronger than the required ambient two-dimensional baseline $\alpha/2$.

For $d \geq 3$, apply the already established $(d - 1)$ -dimensional theorem inside $P \simeq \mathbb{R}^{d-1}$ if $\alpha < d - 1$. It gives, for some $y \in Y$,

$$\dim_{\mathbb{H}}\{|x - y| : x \in A \cap P\} \geq \frac{\alpha}{d - 1} + c_{d-1}(\alpha, \kappa) > \frac{\alpha}{d} + \frac{\alpha}{d(d - 1)}. \quad (6.20f)$$

If $\alpha = d - 1$, apply the lower-dimensional theorem with any $a < d - 1$ sufficiently close to $d - 1$; already the baseline $a/(d - 1)$ exceeds $(d - 1)/d + 1/(2d)$. Thus in every hyperplane-intersection subcase we may reserve the continuous ambient gain

$$c_{\text{hyp}}(\alpha) = \frac{\alpha}{2d(d - 1)}. \quad (6.20g)$$

(The branch cannot occur when $\alpha > d - 1$.)

Assume instead that $\dim_{\mathbb{H}}(A \cap P) < \dim_{\mathbb{H}} A$. Then $A \setminus P$ has full Hausdorff dimension, and from

$$A \setminus P = \bigcup_{n \geq 1} \{x \in A : \text{dist}(x, P) \geq 1/n\}$$

we may replace A by a compact subset of dimension $> \alpha$ at positive distance from P . For fixed x in this subset, the map $e_x : P \rightarrow S^{d-1}$ is bi-Lipschitz on bounded pieces, uniformly after a finite decomposition. Therefore $e_x \nu$ is κ -Frostman on the sphere. A spherical strip $S^{d-1} \cap H^{(r)}$ is covered by $O(r^{-(d-2)})$ spherical r -balls, so

$$e_x \nu(H^{(r)}) \lesssim r^{\kappa - (d-2)}. \quad (6.20h)$$

Since $\kappa - (d - 2) > 0$, Theorem III.6.24 gives a parameter $y \in Y$ with

$$\dim_{\mathbb{H}}\{|x - y| : x \in A\} \geq \frac{\alpha}{d} + \eta_d(\alpha, \kappa - (d - 2)). \quad (6.20i)$$

Case 2: ν gives zero mass to every affine hyperplane. Lemma III.6.24.61, applied with $m = d - 2$, yields uniform power decay near affine $(d - 2)$ -planes with exponents

$$a_0 = \alpha - (d - 2) > 0, \quad b_0 = \kappa - (d - 2) > 0. \quad (6.20j)$$

The local spherical-direction theorem above, with $k = d - 1$, therefore provides a number

$$q_0 = q_d(a_0, b_0) > 0, \quad (6.20k)$$

continuous in (a_0, b_0) , a set E of positive μ -measure and, for each $x \in E$, a set $K(x)$ of positive ν -measure such that

$$e_x(\nu_{K(x)})(H^{(r)}) \leq C_x r^{q_0}, \quad H \in G(d, d - 1), \quad 0 < r \leq 1. \quad (6.20l)$$

A positive- μ set still has Hausdorff dimension at least α , so (6.20e) and (6.20l) meet the hypotheses of Theorem III.6.24. Hence some $y \in Y$ satisfies

$$\dim_{\mathbb{H}}\{|x - y| : x \in A\} \geq \frac{\alpha}{d} + \eta_d(\alpha, q_0). \quad (6.20m)$$

The strict-case proof is therefore valid with any positive continuous function below

$$g_d(\alpha, \kappa) = \frac{1}{4} \min\{c_{\text{hyp}}(\alpha), \eta_d(\alpha, \kappa - (d - 2)), \eta_d(\alpha, q_d(\alpha - (d - 2), \kappa - (d - 2)))\}. \quad (6.20n)$$

All three entries are positive on $(d - 2, d)^2$, and the continuity statements in Theorem III.6.24 and in the local spherical-direction theorem make g_d continuous.

We now remove the strict inequality $\dim_{\mathbb{H}} A > \alpha$. Choose $a < \alpha$, still with $a > d - 2$, so close to α that

$$\frac{a}{d} + g_d(a, \kappa) \geq \frac{\alpha}{d} + \frac{1}{2} g_d(\alpha, \kappa). \quad (6.20o)$$

This is possible by continuity. Since $\dim_{\mathbb{H}} A \geq \alpha > a$, the strict case applied with a gives (6.20d) with

$$c_d(\alpha, \kappa) = \frac{1}{2} g_d(\alpha, \kappa).$$

Thus the equality endpoint is obtained from one nearby exponent; no convergence of centres y is needed.

For a C^2 norm N with everywhere positive Gaussian curvature, put

$$\psi(e) = \text{dir}(DN(e)), \quad e \in S^{d-1}.$$

Then

$$\text{dir } D_x N(x - y) = \psi(\pm e_x(y)). \quad (6.20p)$$

The Gauss map is odd because N is even. If $d = 2$, positive curvature makes $\psi : S^1 \rightarrow S^1$ a diffeomorphism, hence a bi-Lipschitz homeomorphism on the compact circle. Lemma III.6.24.1 therefore transports (6.20l) to the normed direction measure

with the same power q_0 . The proof above then repeats verbatim with $F_y(x) = N(x - y)$. This proves the full planar normed clause, with the same gain $c_2(\alpha, \kappa)$ independent of the norm; the norm changes only the harmless constants in the direction-decay estimate.

For ellipsoidal norms, the same conclusion is local in every dimension; this follows from Lemma III.6.24.2.

For an arbitrary positively curved norm, the entire branch in which the chosen centre measure ν charges an affine hyperplane is also local. Repeat Case 1 above with $F_y(x) = N(x - y)$ and normalize ν on the charged hyperplane P . As noted there, this branch necessarily has $\kappa < d - 1$. If $\dim_{\mathbb{H}}(A \cap P) \geq \alpha$, restrict the norm to $P - P$. Lemma III.6.24.4 keeps it inside the same positively curved class. In dimension $d - 1$ the centre exponent satisfies $\kappa > d - 2 = (d - 1) - 1$, so the direct the supercritical covering argument supercritical argument applies there: Lemma III.6.24.3, in dimension $d - 1$, followed by Theorem III.6.24 gives the required pinned image. Its baseline $\alpha/(d - 1)$ is strictly larger than α/d . The endpoint $\alpha = d - 1$ is handled, as before, by using an exponent $a < d - 1$ close to $d - 1$. For $d = 2$ the restriction is one-dimensional and every norm is a constant multiple of absolute value.

If instead $A \setminus P$ has full relevant dimension, pass to a compact subset at positive distance from P . Lemma III.6.24.5 gives the normed direction decay

$$(\psi \circ e_x)_{\#} \nu(H^{(r)}) \lesssim r^{\kappa - (d - 2)},$$

with a positive exponent independent of N . Theorem III.6.24 then gives the same norm-independent gain as in (6.20i). The remaining branch, where ν gives zero mass to every affine hyperplane, is closed below by the radial-Frostman input.

There is also a direct local argument for every positively curved norm when $\kappa > d - 1$. Repeat the initial dual setup with a compact hypothetical exceptional set Y of dimension $> \kappa$ and choose a κ -Frostman probability measure ν on Y . Lemma III.6.24.3 therefore gives (6.14), uniformly in x , with the fixed exponent

$$q = \kappa - (d - 1) > 0.$$

Apply Theorem III.6.24 directly to $F_y(x) = N(x - y)$. Some $y \in Y$ then satisfies

$$\dim_{\mathbb{H}} N(A - y) \geq \frac{\alpha}{d} + \eta_d(\alpha, \kappa - (d - 1)). \quad (6.20q)$$

The gain depends only on d, α, κ and not on N . This proves the general curved-norm clause locally throughout $d - 1 < \kappa < d$.

For a general positively curved norm in the residual band $d - 2 < \kappa \leq d - 1$, the argument for centre measures charging a hyperplane has removed that branch. It remains only to treat the case in which ν gives zero mass to every affine hyperplane. The published proof invokes a transfer through ψ based on local bi-Lipschitzness, and the Gauss-map counterexample shows that this implication is not valid in that generality, even for admissible norm Gauss maps. The supercritical covering argument controls fixed-angle intersections of the translated bad cones $\{y : n \cdot DN(y - x) = 0\}$ by the expected codimension-two Frostman estimate, while Lemma III.6.24.6 identifies exact tangency by a common tangent hyperplane through the vertices.

the two-contact train-track lemma makes the near-tangent geometry quantitative without increasing the smoothness assumption. Lemma III.6.24.8 shows that two separated normal directions among near-tangency contacts force the moving vertex into a thin neighbourhood of an affine $(d - 2)$ -plane. This is precisely the codimension for which the available Frostman exponent $\alpha - (d - 2) > 0$ supplies a power saving. If the contact normals do not separate, then all the contacts and the moving vertex lie in a single thin affine hyperplane: a genuine curved train track. In the centre variable, Lemma III.6.24.7 records the uniform qualitative smallness of thin hyperplane neighbourhoods under the zero-hyperplane hypothesis.

the homogeneous Legendre linearisation changes the form of the remaining problem. Put $\mathcal{G} = D(N^2/2) = N DN$. Lemma III.6.24.9 proves that $\mathcal{G}$ is a global bi-Lipschitz homogeneous map and sends every cone C_n exactly to $n^\perp$. Hence all same-vertex transversality is literally affine in the $\mathcal{G}_x$ coordinates, with the polynomial estimate (6.20b13); moreover (6.20b14) freezes this affine picture on a small base cell at a linear cost in the cell diameter. Thus no power modulus for D^2N is needed for the same-vertex geometry.

The telescope at a fixed base point corrects the remaining multiscale diagnosis. Lemma III.6.24.12 shows that the Appendix-B superexponential telescope is pointwise in the base point and needs no alignment of exceptional hyperplanes at different scales. Once a one-scale pruning statement of the form (6.20b17) is available for the curved direction belts, the identity $\delta_n^{\tau/2} = \delta_{n-N}$ gives the same residue-class telescope as in the Euclidean argument, now in the exact same-base coordinates $\mathcal{G}_x$.

The atomic-remainder analysis further reduced exact atoms to a pure atomic remainder and ruled out the naive proof mechanism in which one freezes each base cell and copies the Euclidean Step 1/Step 2 estimates independently. The exact-belt exclusion closes the atom branch completely: Lemma III.6.24.14 shows that, in the zero-hyperplane centre branch, vertices carrying any positive-mass exact Legendre belt form a μ -null set. Hence after restricting the base to a positive-mass compact subset, all exact belts are null and the same-base telescope has the exact atom-freeness it needs.

Lemma III.6.24.15 remains useful as a diagnostic: outside T_δ , one exceptional $O(\delta^{\tau/2})$ Legendre belt captures every heavy $O(\delta)$ belt. The persistent-belt counterexample proves that this diagnostic cannot be completed by a power bound on T_δ in the full admissible class. Indeed, Lemma III.6.24.17 gives a smooth positively curved norm and hyperplane-null Frostman measures for which $\mu(T_\delta) = 1$ at all small scales and, more strongly, no single exceptional belt satisfies (6.20b17).

the multibelt telescope gives the sharp telescopic refinement of this diagnosis. Multiple exceptional belts are compatible with the same-base telescope if their transition cost (6.20b24) is summable and one synchronized seed block satisfies (6.20b25); bounded multiplicity automatically has both properties after starting sufficiently deep. However Lemma III.6.24.21 shows that no fixed finite multiplicity is a universal consequence of the residual hypotheses. The tracked-birth telescope then replaces total active multiplicity by genuinely new birth mass and weakens the final telescope endpoint to a positive-mass restriction. The superflat pointwise obstruction proves that even this seed/birth input cannot be forced pointwise in the open band $d - 2 < \kappa < d - 1$, so the base variable must be used jointly. The conditioned-witness reduction closes the conditioning and selection part of the joint determinant route: Lemmas III.6.24.25 and III.6.24.26 give polynomially

conditioned witness tuples of positive product mass and reduce the two-heavy branch to the determinant incidence set (6.20b57). The transverse determinant estimate now closes the raw transverse sublevel analysis: Lemma III.6.24.27 gives the exact modulus-aware codimension-two Frostman bound, and Lemma III.6.24.28 removes the transverse-dominated base branch summably at bare C^2 and with a genuine power loss under $C^{2,\beta}$. Lemma III.6.24.29 further identifies the normalized gradients with barycentric averages of physical cone normals. The robust-anchor decomposition sharpens this structural edge via Lemmas III.6.24.30–III.6.24.32: robust anchors plus one test point turn the normalized gradient into an affine-normal defect with error $O(\delta^{1-3\tau/q})$, and homogeneity splits normalized degeneracy into a physical affine-hyperplane track or a one-radius-per-direction radial graph. The latter has the exact modulus-aware mass bound (6.20b84); under $C^{2,\beta}$ and $\beta(2\kappa - (d - 2)) > \kappa$ it is $o(m)$ relative to a heavy belt, so only track-track geometry can carry a positive witness fraction. the Euclidean-radial preconditioning now removes that physical track in a stronger but still explicit Hölder range: apply the already-local Euclidean spherical-direction theorem first, obtaining the x -dependent radial restriction (6.20b88); then Lemma III.6.24.33 makes track, radial and small-defect witness classes simultaneously $o(m)$ whenever $\beta > \beta_*(\alpha, \kappa)$. The transverse branch is power-small by the transverse determinant estimate, so the same-base telescope gives normed direction decay and Lemma III.6.24.34 closes this residual subcase for each fixed admissible β , with a gain allowed to depend on β . At $\beta = 1$ the theorem-form gain depends only on d, α, κ , so every $C^{2,1}$ positively curved norm is local throughout $d - 2 < \kappa \leq d - 1$.

the critical modulus balance reaches the critical endpoint of the Euclidean-radial preconditioning exponent balance. Lemma III.6.24.35 inserts a slowly varying loss into the critical choices $s_0 = q/q_0$ and $c_0 = (\kappa + (d - 2)s_0)/q$; under the little-Hölder condition (6.20b103), physical tracks, radial non-track fibres, and small defects are still all $o(m)$, while the logarithmic loss is absorbed by the the transverse determinant estimate transverse power bound. Hence Lemma III.6.24.37 closes the critical little-Hölder class with a gain depending only on d, α, κ .

Lemma III.6.24.36 remains a sharp obstruction to the robust-anchor one-scale architecture: logarithmically slow continuous Hessian moduli can violate its scalar compatibility, so retuning σ , R , or the synchronized scales cannot close arbitrary C^2 by that mechanism alone. The radial-Frostman bypass and measure-level thin-tube argument reduce the problem to the confined finite kernel; the co-rank-one argument in turn derives that kernel from the endpoint multilinear Takeya estimate, using the $(k_u + 2)$ -linear member at ordinary fibre exponent u . Proposition III.6.24.56 and Lemmas III.6.24.57–III.6.24.60 produce ordinary radial direction fibres which are Frostman on spherical balls with an exponent strictly larger than $d - 2$. This ball-Frostman property passes legitimately through the bi-Lipschitz norm Gauss map; only then is it converted to great-hyperplane decay by a spherical cap cover. Condition (6.14) therefore holds on a Borel base set of the required dimension for every positively curved C^2 norm. Applying Theorem III.6.24 proves the remaining normed branch. $\square$

III.6.5 Sharp incidences and Furstenberg sets

The projection theorems above become considerably stronger when the direction set is allowed to vary with the base point. The correct finite-scale language is that of point-tube incidence configurations. We work in a fixed bounded slope/intercept chart. If $\delta \in 2^{-\mathbb{N}}$ and $p = I \times J$ is a dyadic δ -square in parameter space, write

$$\mathbf{D}(p) = \{(x, y) : y = ax + b \text{ for some } (a, b) \in p\}$$

for the corresponding dyadic δ -tube. The family of all such tubes is denoted $\mathcal{T}^\delta$.

A bounded nonempty set $P \subset \mathbb{R}^d$ is a (δ, s, C) -**set** if

$$|P \cap Q|_\delta \leq C |P|_\delta r^s, \quad Q \in \mathcal{D}_r^d, \quad \delta \leq r \leq 1. \quad (6.41)$$

A tube family is a (δ, s, C) -set when its parameter squares form such a set. Notice that (6.41), applied to an occupied δ -cell, already implies $|P|_\delta \geq C^{-1} \delta^{-s}$. Thus the definition contains both a global lower-cardinality condition and a relative nonconcentration condition.

A pair $(P, \mathcal{T}) \subset \mathcal{D}_\delta^2 \times \mathcal{T}^\delta$ is a (δ, s, C, M) -**nice configuration** if for each $p \in P$ there is a (δ, s, C) -subfamily $\mathcal{T}(p) \subset \mathcal{T}$, every member of which meets p , with $|\mathcal{T}(p)| = M$. We shall repeatedly replace a configuration by a subset losing only a fixed power of $\log(1/\delta)$; such losses are harmless compared with the positive powers of δ that drive the arguments below.

When $\delta^{1/2}$ is dyadic, a nonempty $P \subset \mathcal{D}_\delta^2$ is called (δ, b, C, K) -**regular** if it is a δ, b, C -set and, in addition,

$$|P|_{\delta^{1/2}} \leq K \delta^{-b/2}. \quad (6.41a)$$

Thus “regular” here is the square-root-scale condition of Orponen–Shmerkin, not an assertion of Ahlfors regularity at every intermediate scale. This distinction is important in the bootstrap below [22].

Imported theorem (Two-scale refinement of nice configurations). *Let $0 < \delta < \Delta \leq 1$ be dyadic. Optionally fix a dyadic scale $\bar{\Delta}$ with $\Delta \leq \bar{\Delta} \leq 1$. Let $(P_0, \mathcal{T}_0)$ be a (δ, s, C, M) -nice configuration. For $Q \in \mathcal{D}_\Delta^2$, let S_Q be the homothety taking Q onto $[0, 1]^2$, and set*

$$\mathcal{D}_\Delta(P_0) = \{Q : P_0 \cap Q \neq \emptyset\}, \quad \mathcal{T}^\Delta(\mathcal{T}_0) = \{\mathbf{T} \in \mathcal{T}^\Delta : \mathcal{T}_0 \cap \mathbf{T} \neq \emptyset\}.$$

There are $P \subset P_0$ and subfamilies $\mathcal{T}(p) \subset \mathcal{T}_0(p)$, with $\mathcal{T} = \bigcup_{p \in P} \mathcal{T}(p)$, such that, up to a fixed power of $\log(1/\delta)$,

- (i) $|\mathcal{D}_\Delta(P)| \asymp |\mathcal{D}_\Delta(P_0)|$, and $|P \cap Q| \asymp |P_0 \cap Q|$ for every occupied Δ -square Q ;
- (ii) $|\mathcal{T}(p)| \geq M$ for every $p \in P$;
- (iii) there are $C_\Delta \leq (1 + \log(1/\delta))^{O(1)} C$, $M_\Delta \geq 1$, and a family

$$\mathcal{T}_\Delta \subset \mathcal{T}^\Delta(\mathcal{T}_0)$$

such that $(\mathcal{D}_\Delta(P), \mathcal{T}_\Delta)$ is a $(\Delta, s, C_\Delta, M_\Delta)$ -nice configuration. Writing $\mathcal{T}_\Delta(Q)$ for its associated fibre at Q , one has

$$\bigcup \mathcal{T}(p) \subset \bigcup \mathcal{T}_\Delta(Q) \quad (p \in P \cap Q);$$

if $\bar{\Delta}$ has been fixed, the covering fibres may in addition be chosen so that one integer $H_{\bar{\Delta}} \geq 1$, independent of Q and of the ancestor, satisfies

$$|\mathcal{T}_{\Delta}(Q) \cap \mathbf{S}| = H_{\bar{\Delta}} \quad (Q \in \mathcal{D}_{\Delta}(P), \mathbf{S} \in \mathcal{T}^{\bar{\Delta}}(\mathcal{T}_{\Delta}(Q))); \quad (6.42b)$$

(iv) for every occupied Q , there are $C_Q \leq (1 + \log(1/\delta))^{O(1)}C$, $M_Q \geq 1$, and $\mathcal{T}_Q \subset \mathcal{T}^{\delta/\Delta}$ such that $(S_Q(P \cap Q), \mathcal{T}_Q)$ is $(\delta/\Delta, s, C_Q, M_Q)$ -nice. Moreover, writing $\bar{\delta} = \delta/\Delta$, for every retained fine square $p \in P \cap Q$ the slope ancestors satisfy

$$\mathcal{D}_{\bar{\delta}}[\sigma(\mathcal{T}_Q(S_Q p))] = \mathcal{D}_{\bar{\delta}}[\sigma(\mathcal{T}(p))]. \quad (6.42a)$$

Moreover these objects may be chosen so that

$$\frac{|\mathcal{T}_0|}{M} \gtrsim_{\log, \delta} \frac{|\mathcal{T}_{\Delta}|}{M_{\Delta}} \max_{Q \in \mathcal{D}_{\Delta}(P)} \frac{|\mathcal{T}_Q|}{M_Q}. \quad (6.42)$$

This is Shmerkin–Wang, Proposition 2.7, equivalently the two-scale refinement of Orponen–Shmerkin, Proposition 5.2 [17, 22]. A proof is given immediately below. By itself, the proposition supplies no sharp incidence gain.

Local proof of the two-scale refinement. Write $\bar{\delta} = \delta/\Delta$, and allow the symbol L below to denote a fixed power of $\log(1/\delta)$. Every loss by a factor L is absorbed by the notation in the statement. The proof has two independent tasks. First we regularise the families of Δ -ancestors of the fine tubes. Then, inside every occupied Δ -square, we regularise the packets of fine tubes having a common $\bar{\delta}$ -ancestor. The product estimate follows by multiplying the two resulting multiplicities.

Step 1: coarse regularisation inside each Δ -square. Fix $Q \in \mathcal{D}_{\Delta}(P_0)$. Apply the already proved coarse-regularisation Lemma III.6.29.5 to the index set $P_0 \cap Q$, the fine families $\mathcal{T}_0(p)$, and the scale $R = \Delta$. It gives a subset $P_Q^0 \subset P_0 \cap Q$, with

$$|P_Q^0| \geq L^{-1}|P_0 \cap Q|,$$

and a family $\mathcal{U}_{\Delta}(Q)$ of dyadic Δ -tubes meeting Q such that $\mathcal{U}_{\Delta}(Q)$ is a (Δ, s, LC) -set. Moreover every $\mathbf{T} \in \mathcal{U}_{\Delta}(Q)$ carries at least

$$H_Q \geq L^{-1} \frac{M|P_Q^0|}{|\mathcal{U}_{\Delta}(Q)|}$$

fine incidences (p, T) with $p \in P_Q^0$, $T \in \mathcal{T}_0(p)$, and $T \subset \mathbf{T}$.

All Δ -tubes under consideration meet a fixed bounded region, so there are at most $O(\Delta^{-2})$ of them. Pigeonhole the cardinality $|\mathcal{U}_{\Delta}(Q)|$ over the occupied squares and retain a family $\mathcal{Q} \subset \mathcal{D}_{\Delta}(P_0)$ with

$$|\mathcal{Q}| \geq L^{-1}|\mathcal{D}_{\Delta}(P_0)|, \quad M_{\Delta}^0 \leq |\mathcal{U}_{\Delta}(Q)| < 2M_{\Delta}^0 \quad (Q \in \mathcal{Q}).$$

Let $\mathcal{U}_{\Delta} = \bigcup_{Q \in \mathcal{Q}} \mathcal{U}_{\Delta}(Q)$. For a coarse tube $\mathbf{T} \in \mathcal{U}_{\Delta}$, put

$$n(\mathbf{T}) = |\mathcal{T}_0 \cap \mathbf{T}|.$$

Since a dyadic Δ -tube contains at most $O((\Delta/\delta)^2) \leq O(\delta^{-2})$ dyadic δ -tubes, dyadic pigeonholing in $n(\mathbf{T})$ costs only L . Thus there are a dyadic number $N_\Delta \geq 1$ and a subfamily $\mathcal{U}'_\Delta \subset \mathcal{U}_\Delta$ such that

$$N_\Delta/2 < n(\mathbf{T}) \leq N_\Delta \quad (\mathbf{T} \in \mathcal{U}'_\Delta),$$

and the incidence set $\{(Q, \mathbf{T}) : Q \in \mathcal{Q}, \mathbf{T} \in \mathcal{U}_\Delta(Q) \cap \mathcal{U}'_\Delta\}$ retains at least an L^{-1} -fraction of its original size. A second popularity pruning in Q therefore leaves $\mathcal{Q}' \subset \mathcal{Q}$, $|\mathcal{Q}'| \geq L^{-1}|\mathcal{Q}|$, for which

$$|\mathcal{U}_\Delta(Q) \cap \mathcal{U}'_\Delta| \geq L^{-1}M_\Delta^0 \quad (Q \in \mathcal{Q}').$$

If a scale $\bar{\Delta}$ was fixed, now partition each family $\mathcal{U}_\Delta(Q) \cap \mathcal{U}'_\Delta$ into its two-dimensional dyadic $\bar{\Delta}$ -tube ancestors. The number of Δ -tubes in one such ancestor is at most $O((\bar{\Delta}/\Delta)^2)$, while the number of occupied ancestors is at most $O(\bar{\Delta}^{-2})$. Both numbers therefore have $O(1 + \log(1/\delta))$ dyadic size classes. For each Q , choose a class of ancestor occupancies carrying at least an L^{-1} -fraction of the Δ -tubes, trim every ancestor in that class to the same number $H(Q)$, and discard the other classes. The retained family has at least an L^{-1} -fraction of the former size. Pigeonhole the dyadic size of $H(Q)$ and the number of retained ancestors over $Q \in \mathcal{Q}'$. This keeps an L^{-1} -fraction of the squares; trim the surviving families by deleting whole ancestor blocks, and then an equal number of tubes inside every surviving block. We obtain common positive integers $H_{\bar{\Delta}}$ and $K_{\bar{\Delta}}$ for which every retained square has exactly $K_{\bar{\Delta}}$ occupied $\bar{\Delta}$ -ancestors, each containing exactly $H_{\bar{\Delta}}$ retained Δ -tubes. All losses, including the loss in the number of squares, are fixed powers of $1 + \log(1/\delta)$. Relabel the retained family of squares $\mathcal{Q}'$, and put $M_\Delta = H_{\bar{\Delta}}K_{\bar{\Delta}}$. If no $\bar{\Delta}$ was fixed, instead trim the families below to a common size in the usual way.

Trim every one of these families to a common cardinality $M_\Delta \geq L^{-O(1)}M_\Delta^0$, call the result $\mathcal{T}_\Delta(Q)$, and set $\mathcal{T}_\Delta = \bigcup_{Q \in \mathcal{Q}'} \mathcal{T}_\Delta(Q)$. In the optional case, the preceding block trimming has already performed this last step, so no further individual tubes are deleted. In particular (6.42b) holds. Because each $\mathcal{T}_\Delta(Q)$ is a large subset of the (Δ, s, LC) -set $\mathcal{U}_\Delta(Q)$, it is a (Δ, s, L^2C) -set. Hence $(\mathcal{Q}', \mathcal{T}_\Delta)$ is a $(\Delta, s, C_\Delta, M_\Delta)$ -nice configuration with $C_\Delta \leq L^2C$.

For $Q \in \mathcal{Q}'$ and $p \in P_Q^0$, define the preliminary fine family

$$\tilde{\mathcal{T}}(p) = \bigcup_{\mathbf{T} \in \mathcal{T}_\Delta(Q)} (\mathcal{T}_0(p) \cap \mathbf{T}).$$

Every retained coarse tube still has the incidence lower bound H_Q , and

$$|\mathcal{T}_\Delta(Q)| \geq L^{-1}|\mathcal{U}_\Delta(Q)|.$$

Consequently

$$\sum_{p \in P_Q^0} |\tilde{\mathcal{T}}(p)| \geq L^{-2}M|P_Q^0|.$$

Since every summand is at most M , another popularity pruning leaves $P_Q^1 \subset P_Q^0$, $|P_Q^1| \geq L^{-2}|P_Q^0|$, such that

$$|\tilde{\mathcal{T}}(p)| \geq L^{-2}M \quad (p \in P_Q^1).$$

At this stage the occupied coarse squares have changed by only logarithmic factors and the point counts in each retained square remain logarithmically comparable to the original

ones. Also, by construction,

$$\bigcup \tilde{\mathcal{T}}(p) \subset \bigcup \mathcal{T}_\Delta(Q), \quad p \in P_Q^1.$$

This already gives the geometric inclusion required in part (iii).

Step 2: slope-packet regularisation at the relative scale $\bar{\delta}$. Fix $Q \in \mathcal{Q}'$. For $p \in P_Q^1$, partition $\tilde{\mathcal{T}}(p)$ according to the dyadic $\bar{\delta}$ -slope interval. Call the non-empty pieces *slope packets*. Dyadic pigeonholing in their sizes gives a packet size $m(p)$, a number $K(p)$ of retained packets, and a union $\hat{\mathcal{T}}(p) \subset \tilde{\mathcal{T}}(p)$ such that every retained packet has between $m(p)/2$ and $m(p)$ elements and

$$m(p)K(p) \geq L^{-1}|\tilde{\mathcal{T}}(p)| \geq L^{-3}M.$$

Pigeonhole both $K(p)$ and $m(p)$ over $p \in P_Q^1$. After passing to $P_Q \subset P_Q^1$, with $|P_Q| \geq L^{-2}|P_Q^1|$, trim packet collections and their contents to common cardinalities M_Q and m_Q , respectively. Then every retained $p \in P_Q$ has exactly M_Q slope packets of size m_Q , with

$$m_Q M_Q \geq L^{-O(1)}M.$$

Define $\mathcal{T}(p)$ to be the union of these packets. Then $|\mathcal{T}(p)| \geq L^{-O(1)}M$, so after a final harmless trimming the families satisfy part (ii). Being subsets of $\mathcal{T}_0(p)$, they retain the fine $(\delta, s, L^{O(1)}C)$ -nonconcentration estimate.

Represent every retained slope packet by a dyadic δ -tube through p whose slope is the left endpoint of that packet's $\bar{\delta}$ -interval; choose its intercept on the δ -grid so that the tube meets p . Such a tube always exists. We do *not* require its two-dimensional tube-parameter ancestor to agree with the original packet: that requirement is false in general and is unnecessary here. The representatives form a family $\mathcal{R}(p)$ of exactly M_Q tubes with $\bar{\delta}$ -separated slopes. If I is a dyadic interval of length $|I| \geq \bar{\delta}$, each representative slope in I accounts for exactly m_Q original slopes from $\mathcal{T}(p)$ lying in an $O(\bar{\delta})$ -enlargement of I . Therefore

$$|I \cap \sigma(\mathcal{R}(p))| \lesssim L m_Q^{-1} C |\mathcal{T}(p)| |I|^s \lesssim L^6 C M_Q |I|^s.$$

Thus the representative slopes form a $(\bar{\delta}, s, C_Q)$ -set with $C_Q \leq L^6 C$.

Let $q = S_Q(p)$. If $R \in \mathcal{R}(p)$, the image $S_Q(R)$ has slope equal to $\sigma(R) \in \bar{\delta}\mathbb{Z}$ and width comparable to $\bar{\delta}$. Hence it is covered by a bounded number of dyadic $\bar{\delta}$ -tubes, at least one of which meets q . Select one such tube for every representative and call the resulting family $\mathcal{T}_Q(q)$. Put $\mathcal{T}_Q = \bigcup_{q \in S_Q(P_Q)} \mathcal{T}_Q(q)$. The slope separation shows that $|\mathcal{T}_Q(q)| \asymp M_Q$, and the preceding interval estimate shows that $(S_Q(P_Q), \mathcal{T}_Q)$ is a $(\bar{\delta}, s, C'_Q, M'_Q)$ -nice configuration after bounded trimming, where $C'_Q \leq L^{O(1)}C$ and $M'_Q \asymp M_Q$. Relabel C'_Q, M'_Q as C_Q, M_Q .

The construction also proves the slope-ancestor identity (6.42a). Indeed, each representative slope is the left endpoint of the $\bar{\delta}$ -interval containing its packet's slopes, while every retained packet is represented. Therefore, for $p \in P_Q$,

$$\mathcal{D}_{\bar{\delta}}[\sigma(\mathcal{T}_Q(S_Q p))] = \mathcal{D}_{\bar{\delta}}[\sigma(\mathcal{T}(p))].$$

Step 3: the product inequality. By the global multiplicity pruning in Step 1,

$$|\mathcal{T}_0| \geq \frac{1}{2} |\mathcal{T}_\Delta| N_\Delta.$$

For fixed Q , every fine tube in $\bigcup_{p \in P_Q} \mathcal{T}(p)$ lies in one of the M_Δ coarse tubes of $\mathcal{T}_\Delta(Q)$, each of which contains at most N_Δ fine tubes from $\mathcal{T}_0$. Hence

$$\left| \bigcup_{p \in P_Q} \mathcal{T}(p) \right| \leq M_\Delta N_\Delta.$$

We claim conversely that

$$\left| \bigcup_{p \in P_Q} \mathcal{T}(p) \right| \geq L^{-O(1)} \frac{|\mathcal{T}_Q|}{M_Q} M. \quad (*)$$

To prove this, choose for every representative tube $R \in \bigcup_{p \in P_Q} \mathcal{R}(p)$ one packet that it represents. If an original fine tube T belongs to such a packet and both T and its representative R meet the same fine square $p \subset Q$, then their slope difference is at most $\tilde{\delta} = \delta/\Delta$. Over a square of diameter $O(\Delta)$, the corresponding central lines separate by only $O(\delta)$. Thus, inside Q , a fixed $O(\delta)$ -enlargement of R contains the $O(\delta)$ -enlargement of every tube in the represented packet.

Call two representative tubes separated if either their slopes are different, or their fixed $O(\delta)$ -enlargements are disjoint inside $[0, 1]^2$. Packets represented by separated tubes are disjoint: distinct representative slope intervals contain disjoint fine slope sets, while in the equal-slope case a common fine tube would meet the two defining squares and force the two enlarged representatives to intersect. Finally, at any fixed representative slope the intercepts lie on the δ -grid, and a tube can fail the separation condition with only $O(1)$ other tubes of that slope. A greedy selection therefore produces a separated subfamily of representatives of cardinality comparable to the whole representative union. Since every selected packet contains $\gtrsim_L m_Q \gtrsim_L M/M_Q$ original fine tubes, and since the selected packets are disjoint,

$$\left| \bigcup_{p \in P_Q} \mathcal{T}(p) \right| \gtrsim_L \left| \bigcup_{p \in P_Q} \mathcal{R}(p) \right| \frac{M}{M_Q} \gtrsim_L \frac{|\mathcal{T}_Q|}{M_Q} M,$$

which is (*).

Combining the last three displays gives, for every retained Q ,

$$\frac{|\mathcal{T}_0|}{M} \gtrsim_L \frac{|\mathcal{T}_\Delta|}{M_\Delta} \frac{|\mathcal{T}_Q|}{M_Q}.$$

Taking the maximum in Q proves (6.42). Finally set $P = \bigcup_{Q \in \mathcal{Q}'} P_Q$. All refinements of $\mathcal{Q}'$ and P_Q cost only powers of $\log(1/\delta)$, so part (i) holds; parts (ii)–(iv) were established above. This completes the local proof. $\square$

Lemma III.6.26 (one rich tube forces many tubes). *Let $(P, \mathcal{T})$ be a (δ, s, C, M) -nice configuration with $s > 0$. For every δ -tube T_0 , not necessarily belonging to $\mathcal{T}$,*

$$|\mathcal{T}| \gtrsim C^{-1/s} |P \cap T_0|_\delta M. \quad (6.43)$$

Proof. Fix $p \in P \cap T_0$. Since the slopes of $\mathcal{T}(p)$ form a $(\delta, s, O(C))$ -set, an interval of slope length $cC^{-1/s}$ contains at most half of $\mathcal{T}(p)$, for a suitable absolute $c > 0$. Hence at least $M/2$ selected tubes through p make angle $\gtrsim C^{-1/s}$ with T_0 . Any one such tube can meet T_0 in only $O(C^{1/s})$ δ -squares. Double counting the pairs (p, T) over these transverse subfamilies gives (6.43). $\square$

Lemma III.6.27 (bounded richness forces many tubes). *If, in addition, every $T \in \mathcal{T}$ contains at most K squares of P , then*

$$|\mathcal{T}| \geq K^{-1}|P|M. \quad (6.44)$$

Proof. The incidence set $I = \{(p, T) : p \in P, T \in \mathcal{T}(p)\}$ has cardinality $|P|M$. On the other hand,

$$|I| \leq \sum_{T \in \mathcal{T}} |P \cap T| \leq K|\mathcal{T}|.$$

$\square$

Corollary III.6.28 (elementary square-root incidence bound).

Under the hypotheses of Lemma III.6.26,

$$|\mathcal{T}| \gtrsim C^{-1/s}|P|^{1/2}M. \quad (6.45)$$

Proof. If one tube contains at least $|P|^{1/2}$ points, use Lemma III.6.26; otherwise use Lemma III.6.27 with $K < |P|^{1/2}$. $\square$

The square-root estimate is the baseline. The next proposition obtains a uniform power gain when the point set has dimension $t > s$. Its proof has two logically distinct parts. The first is the Shmerkin–Wang rigid reduction: failure of the desired gain forces a dual Furstenberg configuration. The second, longer part proves locally that this rigid configuration cannot occur. The latter is the Orponen–Shmerkin sharp incidence mechanism, reconstructed here rather than imported as a theorem.

Proposition III.6.29 (Shmerkin–Wang discretized incidence gain). *Let $s \in (0, 1)$ and $t \in (s, 2)$. There are $\epsilon, \eta > 0$ such that, for all sufficiently small dyadic δ , the following holds. Let $P \subset B(0, 1)$ be a $(\delta, t, \delta^{-\epsilon})$ -set. For each $p \in P$, let $\mathcal{T}(p)$ be a $(\delta, s, \delta^{-\epsilon})$ -set of at least M dyadic tubes through p . Then*

$$\left| \bigcup_{p \in P} \mathcal{T}(p) \right| \geq M\delta^{-(t/2+\eta)}. \quad (6.46)$$

Proof. We first isolate and prove the sharp dual incidence statement needed at the terminal stage of the argument. Throughout the inner proof, $0 < a < 1$ and $a < b \leq 2$.

Lemma III.6.29.1 (point–line duality on a bounded dyadic chart). *For the line $D(\alpha, \beta) = \{y = \alpha x + \beta\}$ and a dyadic tube $T = D(I \times J)$, put*

$$D^*(T) = \{(-u, v) : (u, v) \in I \times J\}.$$

If p is a dyadic δ -square and T a dyadic δ -tube, then

$$T \cap p \neq \emptyset \implies D(p) \cap D^*(T)^{(C\delta)} \neq \emptyset, \quad (6.47)$$

and conversely after changing the absolute constant. On a fixed bounded chart, duality preserves (δ, u, C) -nonconcentration up to an absolute multiplicative change of C .

Proof. Write $p = I_\alpha \times I_\beta$ and $T = D(I_\alpha \times I_\beta)$. If $(x, y) \in p \cap T$, then $y = \alpha x + \beta$ for some $(\alpha, \beta) \in I_\alpha \times I_\beta$, equivalently $\beta = y - \alpha x$. Thus $(-\alpha, \beta)$ lies within $O(\delta)$ of the dual line with parameter (x, y) . Reversing the algebra gives the converse. The coordinate changes involved are affine maps with uniformly bounded condition number on the chosen chart; replacing their images by standard dyadic cells costs only bounded multiplicity. $\square$

Proposition III.6.29.2 (product-like terminal configuration). *Fix $0 < a < 1$ and $\tau > 0$. There exists $\eta = \eta(a, \tau) > 0$ such that the following holds for all sufficiently small dyadic δ . Let $Y \subset \delta\mathbb{Z} \cap [0, 1]$ be a $(\delta, \tau, \delta^{-\eta})$ -set and, for each $y \in Y$, let $X_y \subset \delta\mathbb{Z} \cap [0, 1]$ be a $(\delta, a, \delta^{-\eta})$ -set. Put*

$$Z = \bigcup_{y \in Y} X_y \times \{y\}.$$

If every $z \in Z$ carries a $(\delta, a, \delta^{-\eta})$ -set $\mathcal{T}(z)$ of dyadic tubes incident to z , then

$$\left| \bigcup_{z \in Z} \mathcal{T}(z) \right| \geq \delta^{-2a-\eta}. \quad (6.48)$$

Proof. Assume the contrary and denote the tube union by $\mathcal{T}$. Trimming by dyadic cardinality classes, with a $\delta^{O(\eta)}$ loss, we may suppose $|X_y| \leq \delta^{-a}$ and $|\mathcal{T}(z)| \leq \delta^{-a}$. For fixed y , identify a tube $D([u, u + \delta) \times [v, v + \delta))$ with its parameter square (u, v) . Incidence with (x, y) gives

$$\pi_y(u, v) := uy + v = x + O(\delta). \quad (6.49)$$

Consequently $\mathcal{T}_y = \bigcup_{x \in X_y} \mathcal{T}(x, y)$ has cardinality $\delta^{-2a+O(\eta)}$. Indeed, a fixed δ -tube meets the horizontal line $\{(x, y) : x \in \mathbb{R}\}$ in an interval of length $O(\delta)$; since $X_y \subset \delta\mathbb{Z}$, the same tube belongs to only $O(1)$ of the fibres $\mathcal{T}(x, y)$. Thus bounded overlap, together with $|X_y|, |\mathcal{T}(x, y)| = \delta^{-a+O(\eta)}$, gives both upper and lower cardinality bounds for $\mathcal{T}_y$.

Moreover $\mathcal{T}_y$ is a $(\delta, 2a, \delta^{-O(\eta)})$ -set. If R is a parameter square of side r , then (6.49) confines the possible x 's to an $O(r)$ -interval. The (δ, a) -bound for X_y contributes one factor r^a , and for each such x the (δ, a) -bound for $\mathcal{T}(x, y)$ contributes a second factor r^a .

Now double count the incidence relation

$$\mathcal{I} = \{(T, y) : y \in Y, T \in \mathcal{T}_y\}.$$

Since every $|\mathcal{T}_y| = \delta^{-2a+O(\eta)}$ while, under the contradictory assumption, $|\mathcal{T}| \leq \delta^{-2a-\eta}$, dyadic pigeonholing in the multiplicity $m(T) = |\{y : T \in \mathcal{T}_y\}|$, followed by a second pigeonhole in y , gives $\mathcal{T}_0 \subset \mathcal{T}$ and $Y_0 \subset Y$, with

$$|\mathcal{T}_0| = \delta^{-2a+O(\eta)}, \quad |Y_0| = \delta^{-\tau+O(\eta)},$$

such that $m(T)$ is constant up to a factor two on $\mathcal{T}_0$ and

$$|\mathcal{T}_0 \cap \mathcal{T}_y| \geq \delta^{O(\eta)} |\mathcal{T}_0|, \quad N_\delta(\pi_y(\mathcal{T}_0 \cap \mathcal{T}_y)) \leq \delta^{-a+O(\eta)} \quad (6.50)$$

for every $y \in Y_0$. Finally, for a parameter square R of side r , the uniform multiplicity and the preceding $(\delta, 2a)$ -estimate give

$$m|\mathcal{T}_0 \cap R| \lesssim \sum_{y \in Y_0} |\mathcal{T}_y \cap R| \lesssim |Y_0| \delta^{-2a+O(\eta)} r^{2a}.$$

The same double count with R equal to the whole parameter chart gives $m|\mathcal{T}_0| \gtrsim |Y_0| \delta^{-2a+O(\eta)}$. Dividing the two inequalities shows that $\mathcal{T}_0$ is a $(\delta, 2a, \delta^{-O(\eta)})$ -set.

Give the directions $e_y = (y, 1)/|(y, 1)|$, $y \in Y_0$, equal mass on their δ -intervals. Since Y_0 is a $(\delta, \tau, \delta^{-O(\eta)})$ -set, this direction measure satisfies the hyperplane nonconcentration required by Theorem III.6.6 with $\kappa = \min\{\tau, a, 1/2\} > 0$. Apply that theorem in dimension two to the parameter set $\mathcal{T}_0$, whose effective dimension is $2a$. The robust-subset conclusion applies to $\mathcal{T}_0 \cap \mathcal{T}_y$ by the first inequality in (6.50). For all but a power-small set of directions one obtains

$$N_\delta(\pi_y(\mathcal{T}_0 \cap \mathcal{T}_y)) \geq \delta^{-a-c(a, \tau)},$$

contradicting the second inequality in (6.50) once $\eta \ll c(a, \tau)$. This proves (6.48). $\square$

The preceding proposition is the only place in the sharp incidence reconstruction where Bourgain's projection theorem enters. We next develop the elementary and potential-theoretic tools for the regular-set bootstrap.

Lemma III.6.29.3 (slope parametrisation and coarse-tube occupancy). *Let p be a dyadic ρ -square and let $\mathcal{S}$ be dyadic ρ -tubes all meeting p . The map from $\mathcal{S}$ to the left endpoints of its slope intervals has bounded multiplicity. Hence if $\mathcal{S}(p)$ is a (ρ, a, C) -family through p , $\rho \leq R \leq 1$, and T_R is a dyadic R -tube, then*

$$|\mathcal{S}(p) \cap T_R| \lesssim C |\mathcal{S}(p)| R^a. \quad (6.51)$$

Proof. Once the slope cell is fixed, the requirement that the tube meet p leaves only $O(1)$ possible intercept cells. A fine tube contained in T_R has slope in an interval of length $O(R)$, so the slope nonconcentration estimate gives (6.51). $\square$

Lemma III.6.29.4 (asymmetric elementary incidence estimate). *Let $0 \leq a \leq u \leq 1$. Let P be a (ρ, u, C_P) -family of dyadic ρ -squares. For each $p \in P$, let $\mathcal{S}(p)$ be a (ρ, a, C_T) -family of exactly M incident ρ -tubes and put $\mathcal{S} = \bigcup_p \mathcal{S}(p)$. Up to a fixed power of $\log(1/\rho)$,*

$$|\mathcal{S}| \gtrsim (C_P C_T)^{-1} M \rho^{-a} (M \rho^a)^{(u-a)/(1-a)}. \quad (6.52)$$

For $u = a$, the final factor is one.

Proof. Let $I = \{(p, T) : T \in \mathcal{S}(p)\}$, so $|I| = M|P|$. Cauchy–Schwarz gives

$$|I|^2 \leq |\mathcal{S}| \left(|I| + \sum_{p \neq q} |\mathcal{S}(p) \cap \mathcal{S}(q)| \right). \quad (6.53)$$

If $r = d(p, q)$, a tube meeting both squares has slope in an interval of length $O(\rho/r)$. Lemma III.6.29.3 and the geometric bound that only $O(r^{-1})$ slope cells can meet both squares yield

$$|\mathcal{S}(p) \cap \mathcal{S}(q)| \lesssim \min\{C_T M(\rho/r)^a, r^{-1}\}. \quad (6.54)$$

Put $\theta = (1 - u)/(1 - a)$, so $u = a\theta + (1 - \theta)$. Interpolating the two terms in (6.54),

$$|\mathcal{S}(p) \cap \mathcal{S}(q)| \lesssim (C_T M \rho^a)^\theta r^{-u}.$$

The (ρ, u, C_P) -condition and a dyadic annulus sum imply

$$\sum_{q \neq p} d(p, q)^{-u} \lesssim_{\log} C_P |P|.$$

Insert this in (6.53). If the diagonal term dominates, then $|\mathcal{S}| \gtrsim M|P|$, which is at least the right side of (6.52): indeed (6.41) at scale ρ gives $|P| \geq C_P^{-1} \rho^{-u}$, while $M \rho^a \geq C_T^{-1}$. Otherwise solving the quadratic inequality gives

$$|\mathcal{S}| \gtrsim_{\log} C_P^{-1} C_T^{-\theta} M^{2-\theta} \rho^{-a\theta},$$

which implies (6.52) because $C_T \geq 1$ and $1 - \theta = (u - a)/(1 - a)$. $\square$

Lemma III.6.29.5 (coarse regularisation of tube fibres). *Let $\rho \leq R \leq 1$ be dyadic. Let E be finite and, for every $p \in E$, let $\mathcal{S}(p)$ be a (ρ, a, C) -family of fine tubes with $M/2 < |\mathcal{S}(p)| \leq M$. If $\mathcal{S}_R$ is a minimal dyadic R -tube cover of $\bigcup_p \mathcal{S}(p)$, then there are $E' \subset E$ with $|E'| \gtrsim_{\log R} |E|$ and $\mathcal{S}'_R \subset \mathcal{S}_R$ such that*

- (i) $\mathcal{S}'_R$ is an (R, a, C') -set with $C' \lesssim_{\log R} C$;
- (ii) every $T_R \in \mathcal{S}'_R$ carries at least

$$H \gtrsim_{\log R} \frac{M|E'|}{|\mathcal{S}'_R|} \quad (6.55)$$

fine incidences (p, T) with $p \in E'$, $T \in \mathcal{S}(p)$, $T \subset T_R$.

Proof. For each p , dyadically classify coarse tubes by the number m_1 of fine tubes of $\mathcal{S}(p)$ they contain. Cells with $m_1 \ll MR^2$ cannot carry a fixed proportion of $\mathcal{S}(p)$, because only $O(R^{-2})$ coarse tubes meet the reference bounded region. Pigeonholing gives, for a large class of points, $m_1 m_2 \asymp_{\log R} M$, where m_2 is the number of retained coarse tubes through the point. Pigeonhole once more so m_1, m_2 are independent of p .

Classify the retained coarse tubes by the number m_3 of selected points whose coarse family contains them. A further pigeonholing gives $\mathcal{S}'_R$ and

$$m_3 |\mathcal{S}'_R| \asymp_{\log R} m_2 |E'|.$$

Each retained coarse tube then contains $\gtrsim m_1 m_3$ fine incidences, which is (6.55).

For nonconcentration, fix a dyadic r -tube T_r , $r \geq R$. Averaging in p gives some $p_0 \in E'$ for which

$$|\mathcal{S}'_R \cap T_r| \lesssim_{\log R} |\mathcal{S}'_R| M^{-1} |\{T \in \mathcal{S}(p_0) : T \subset T_r\}|.$$

The fine (ρ, a, C) -bound makes the last factor at most $C M r^a$, proving (i). $\square$

Lemma III.6.29.6 (floor-slope branching-profile decomposition). *Fix $0 < a < b \leq 2$ and $\varepsilon > 0$. There is $\tau = \tau(a, b, \varepsilon) > 0$ with the following property. If m is sufficiently large and $f : [0, m] \rightarrow \mathbb{R}$ is continuous, piecewise affine, 2-Lipschitz, $f(0) = 0$, and*

$$f(x) \geq bx - \varepsilon m \quad (0 \leq x \leq m), \quad (6.56)$$

then there are pairwise disjoint intervals $[c_j, d_j]$, each of length at least τm , covering all but $O_{a,b}(\varepsilon)m$ of $[0, m]$, such that on each interval either

- (a) *f is $O(\varepsilon)$ -affine with endpoint slope at least a ; or*
- (b) *f is $O(\varepsilon)$ -superlinear and its endpoint slope is exactly a .*

Proof. Apply Lemma III.6.15 to $f/2$, with tolerance $\varepsilon/4$, and rescale. This produces long almost-affine intervals leaving total gap length $O(\varepsilon)m$. Keep those whose endpoint slope is at least a . Consider a maximal remaining interval $[c, d]$ of slope below a . Except for an initial $O_{a,b}(\varepsilon)m$ segment, the lower bound (6.56) forces $f(d)/d > a$. By continuity there is a largest $c' < c$ such that

$$\frac{f(d) - f(c')}{d - c'} = a.$$

Maximality of c' , together with almost-affinity on $[c, d]$, implies

$$f(x) \geq f(c') + a(x - c') - O(\varepsilon)m, \quad c' \leq x \leq d.$$

Thus $[c', d]$ is of type (b). Choosing the leftmost possible predecessor intervals successively keeps the selected pieces disjoint. The original gaps, the initial exceptional segment and integer endpoint rounding account for only $O_{a,b}(\varepsilon)m$ total length. $\square$

Lemma III.6.29.7 (many robust good slopes). *Fix $0 < a < 1$, $a < b \leq 2$, $A, B \geq 1$, and $\varepsilon > 0$. For all sufficiently small dyadic Δ , let E be a $(\Delta, b, \Delta^{-A\varepsilon})$ -set of Δ -squares in the unit square with $|E| \asymp \Delta^{-b}$, and let Σ be a $(\Delta, a, \Delta^{-A\varepsilon})$ -set of slope cells. There is $\Sigma_{\text{good}} \subset \Sigma$, $|\Sigma_{\text{good}}| \geq |\Sigma|/2$, such that for every $\sigma \in \Sigma_{\text{good}}$ and every $E' \subset E$ with $|E'| \geq \Delta^{B\varepsilon}|E|$, the rescaled projection of $\bigcup E'$ in direction σ contains a $(\Delta, a, \Delta^{-C(A+B)\varepsilon})$ -set, where $C = C(a, b)$ and the small-scale threshold may depend on all fixed parameters.*

Proof. Put normalized Lebesgue probability measures μ and ν on the unions of the squares of E and the slope intervals of Σ . Discrete nonconcentration gives

$$\mu(B(x, r)) \lesssim \Delta^{-CA\varepsilon} r^b, \quad \nu(I) \lesssim \Delta^{-CA\varepsilon} |I|^a. \quad (6.57)$$

For $x \neq y$, the function $\sigma \mapsto \pi_\sigma(x) - \pi_\sigma(y)$ is affine with derivative comparable to $|x - y|$ on the bounded chart. Dyadic layer-cake in σ gives

$$\int |\pi_\sigma(x) - \pi_\sigma(y)|^{-a} d\nu(\sigma) \lesssim \Delta^{-CA\varepsilon} |x - y|^{-a}.$$

Integrating in x, y , and using $b > a$ in a second annular sum, yields

$$\int I_a((\pi_\sigma)_\# \mu) d\nu(\sigma) \lesssim \Delta^{-C_1 A\varepsilon}. \quad (6.58)$$

Chebyshev leaves at least half the slope cells containing a representative with energy $\leq \Delta^{-2C_1 A \varepsilon}$. Restricting μ to E' and renormalizing multiplies energy by at most $\Delta^{-2B\varepsilon}$. The earlier-volume energy-to-content and Frostman extraction theorems then give an a -Frostman probability measure on a compact projected subset with quantitative loss $\Delta^{-C(A+B)\varepsilon}$. Discretize it into Δ -cells and dyadically pigeonhole the cell masses; the Frostman growth inequality produces the required $(\Delta, a, \Delta^{-C'(A+B)\varepsilon})$ -subset. Moving from the representative slope to the left endpoint of its Δ -cell changes the projection by only $O(\Delta)$. $\square$

Theorem III.6.29.8 (square-root-scale alternative). *There exists $\varepsilon = \varepsilon(a, b) > 0$ such that the following holds for all sufficiently small dyadic δ , with dyadic $\Delta = \delta^{1/2}$. Let P be a $(\delta, b, \delta^{-\varepsilon})$ -set of dyadic squares satisfying*

$$|P|_{\Delta} \leq \delta^{-b/2-\varepsilon}. \quad (6.59)$$

If every $p \in P$ carries a $(\delta, a, \delta^{-\varepsilon})$ -family of incident tubes and $\mathcal{T}$ is their union, then either

$$|\mathcal{T}| \geq \delta^{-2a-\varepsilon} \quad \text{or} \quad |\mathcal{T}|_{\Delta} \geq \delta^{-a-\varepsilon}. \quad (6.60)$$

Proof. Assume both alternatives fail. Pigeonhole so that every retained point carries M tubes up to a factor two, and trim P to $|P| \leq \delta^{-b}$. Applying Lemma III.6.29.4 with $u = \min\{b, 1\}$ gives

$$\delta^{-2a+O(\varepsilon)} \lesssim |\mathcal{T}| \leq \delta^{-2a-\varepsilon}, \quad M \leq \delta^{-a-O(\varepsilon)}. \quad (6.61)$$

All $O(\varepsilon)$ losses below are fixed multiples depending only on a, b .

Step 1: balanced parents and coarse tubes. Let $\mathcal{Q}$ be the occupied dyadic Δ -squares. The hypothesis (6.59), together with the fine (δ, b) -condition and dyadic pigeonholing, permits a restriction for which

$$|\mathcal{Q}| = \Delta^{-b+O(\varepsilon)}, \quad |P \cap Q| = \Delta^{-b+O(\varepsilon)} \quad (Q \in \mathcal{Q}), \quad (6.62)$$

and $\mathcal{Q}$ itself is a $(\Delta, b, \delta^{-O(\varepsilon)})$ -set. Let $\mathcal{T}_{\Delta}$ be a minimal Δ -tube cover of $\mathcal{T}$. By the second failed alternative, $|\mathcal{T}_{\Delta}| \leq \Delta^{-2a-O(\varepsilon)}$. Apply Lemma III.6.29.5 in every Q , then pigeonhole the coarse fibre size $\bar{M}$. Lemma III.6.29.4 at scale Δ gives

$$\bar{M} \Delta^{-a+O(\varepsilon)} \lesssim |\mathcal{T}_{\Delta}| \leq \Delta^{-2a-O(\varepsilon)}.$$

Since each coarse fibre is a (Δ, a) -set and hence has size at least $\Delta^{-a+O(\varepsilon)}$,

$$\bar{M} = \Delta^{-a+O(\varepsilon)}, \quad |\mathcal{T}_{\Delta}| = \Delta^{-2a+O(\varepsilon)}. \quad (6.63)$$

Pigeonholing also the number of fine tubes inside a coarse tube produces a refined configuration with the same estimates and with uniform fine incidence mass in every retained parent/coarse-tube pair.

Step 2: robust projection directions. For a fixed Q , rescale $P \cap Q$ to the unit square and apply Lemma III.6.29.7 to it and the slopes of the coarse tubes through Q . Retain a half-sized good family $\mathcal{T}_{\Delta}^{\pi}(Q)$. For every subset $P'_Q \subset P \cap Q$ of relative size at least $\delta^{B\varepsilon}$, every represented good slope projects the rescaled P'_Q onto a set containing a $(\Delta, a, \delta^{-C\varepsilon})$ -subset.

Step 3: a $(b - a)$ -dimensional family of parents along one coarse tube. Write $d(Q, Q')$ for the distance of the parent centres. A coarse tube in both $\mathcal{T}_\Delta^\pi(Q)$ and $\mathcal{T}_\Delta^\pi(Q')$ has slope constrained to an interval of length $O(\Delta/d(Q, Q'))$. By Lemma III.6.29.3,

$$\sum_{S \in \mathcal{T}_\Delta} \sum_{Q \neq Q'} \frac{\mathbf{1}_{S \in \mathcal{T}_\Delta^\pi(Q) \cap \mathcal{T}_\Delta^\pi(Q')}}{d(Q, Q')^{b-a}} \lesssim_\varepsilon \sum_{Q \neq Q'} d(Q, Q')^{-b} \lesssim_\varepsilon \Delta^{-2b}. \quad (6.64)$$

Chebyshev and the elementary incidence estimate allow us to discard only a fixed fraction and arrange that the corresponding pair-energy is at most $\Delta^{a-b-O(\varepsilon)}$. Averaging the incidence relation using (6.63) gives one coarse tube S_0 meeting a family $\mathcal{Q}(S_0)$ of $\Delta^{-(b-a)+O(\varepsilon)}$ parents. After one further Chebyshev restriction,

$$\sum_{Q' \in \mathcal{Q}(S_0) \setminus \{Q\}} d(Q, Q')^{a-b} \leq \Delta^{a-b-O(\varepsilon)} \quad (Q \in \mathcal{Q}(S_0)). \quad (6.65)$$

Dyadic annuli show that $\mathcal{Q}(S_0)$ is a $(\Delta, b - a, \delta^{-O(\varepsilon)})$ -set. The uniform coarse-incidence mass and (6.51) then yield, in every such parent, a subset $P_Q \subset P \cap Q$ satisfying

$$|P_Q| = \Delta^{-b+O(\varepsilon)}, \quad |\mathcal{T}(p) \cap S_0| = \Delta^{-a+O(\varepsilon)} \quad (p \in P_Q). \quad (6.66)$$

Step 4: normalize the distinguished tube and extract a product set. In coordinates $x = \sigma y + h$, the affine shear

$$F(x, y) = (x - \sigma_0 y - h_0, y)$$

sends $S_0 = D([\sigma_0, \sigma_0 + \Delta] \times [h_0, h_0 + \Delta])$ to the vertical coarse tube $D([0, \Delta]^2)$. Its condition number is bounded, and replacing images of dyadic squares by $O(1)$ standard cells preserves all exponents. Thus assume S_0 is vertical. Write the retained parents as $Q = [x_Q, x_Q + \Delta] \times [y_Q, y_Q + \Delta]$. Thinning by an absolute factor makes $Q \mapsto y_Q$ injective, and (6.65) shows that

$$Y = \{y_Q : Q \in \mathcal{Q}(S_0)\}$$

is a $(\Delta, b - a, \delta^{-O(\varepsilon)})$ -set. The good-slope conclusion applied to P_Q provides, after rescaling, a $(\Delta, a, \delta^{-O(\varepsilon)})$ -set X_{y_Q} of horizontal coordinates. Hence

$$Z = \bigcup_{y \in Y} X_y \times \{y\} \quad (6.67)$$

has exactly the product-like form of Proposition III.6.29.2 at scale Δ .

For $z \in Z$, represented by $p \in P_Q$, equation (6.66) gives $\Delta^{-a+O(\varepsilon)}$ fine tubes through p inside S_0 . Under the anisotropic map $(x, y) \mapsto (\Delta^{-1}x, y)$, each becomes a rectangle in line-parameter space of dimensions Δ by $O(\Delta)$, hence is covered by $O(1)$ dyadic Δ -tube cells. The slope multiplicity bound gives a $(\Delta, a, \delta^{-O(\varepsilon)})$ -family through each z , while all these tubes arise from fine tubes inside the single S_0 . Therefore

$$\left| \bigcup_{z \in Z} \mathcal{T}_Z(z) \right| \leq \Delta^{-2a+O(\varepsilon)}. \quad (6.68)$$

Proposition III.6.29.2, now at scale Δ and with vertical exponent $b - a > 0$, gives the opposite estimate

$$\left| \bigcup_{z \in Z} \mathcal{T}_Z(z) \right| \geq \Delta^{-2a - \eta(a, b - a)}.$$

Taking ε smaller than a fixed multiple of $\eta(a, b - a)$ contradicts (6.68). Hence the two alternatives in (6.60) cannot both fail. $\square$

Theorem III.6.29.9 (regular-set incidence gain).

For $0 < a < 1$ and $a < b \leq 2$, there exists $\varepsilon_0 = \varepsilon_0(a, b) > 0$ such that if P is a $(\delta, b, \delta^{-\varepsilon_0}, \delta^{-\varepsilon_0})$ -regular set and every $p \in P$ carries a $(\delta, a, \delta^{-\varepsilon_0})$ -family of incident tubes, then their union satisfies

$$|\mathcal{T}| \geq \delta^{-2a - \varepsilon_0}. \quad (6.69)$$

Regularity is exactly Definition (6.41a): besides the $(\delta, b, \delta^{-\varepsilon_0})$ -condition it requires only the square-root cover bound $|P|_{\delta^{1/2}} \leq \delta^{-b/2 - \varepsilon_0}$.

Proof. Trim to a common fibre size $M = \delta^{-a + O(\varepsilon_0)}$ and $|P| = \delta^{-b + O(\varepsilon_0)}$. Put $\Delta = \delta^{1/2}$ and discard light Δ -parents, leaving $\Delta^{-b + O(\varepsilon_0)}$ points in each retained parent. Apply the two-scale interface III.6.5. It provides a refined global configuration, a coarse configuration with fibre size M_Δ , and rescaled fine configurations with fibre size M_Q , satisfying

$$|\mathcal{T}_0| \gtrsim_{\log, \delta} M \frac{|\mathcal{T}_\Delta|}{M_\Delta} \frac{|\mathcal{T}_Q|}{M_Q}. \quad (6.70)$$

The parent family is (Δ, b) . Lemma III.6.29.4, with $u = \min\{b, 1\}$, gives

$$|\mathcal{T}_\Delta| \gtrsim M_\Delta \Delta^{-a} (M_\Delta \Delta^a)^{(u-a)/(1-a)}, \quad (6.71)$$

while its $u = a$ case inside each parent gives

$$|\mathcal{T}_Q| \gtrsim M_Q \Delta^{-a}. \quad (6.72)$$

Apply Theorem III.6.29.8 to the refined global configuration. If its first branch holds, we already have (6.69) after shrinking the bookkeeping exponent. In the second branch, $|\mathcal{T}_\Delta| \geq \Delta^{-2a - O(\varepsilon_0)}$. Combining this with (6.71) yields, in either size regime for M_Δ ,

$$|\mathcal{T}_\Delta| \geq M_\Delta \Delta^{-a - C_1 \varepsilon_0}.$$

Substitute this and (6.72) into (6.70):

$$|\mathcal{T}_0| \gtrsim M \Delta^{-2a - C_1 \varepsilon_0} = M \delta^{-a - C_1 \varepsilon_0} \geq \delta^{-2a - C_2 \varepsilon_0}.$$

Choose the input loss smaller by a factor larger than C_2 , then rename the resulting positive exponent ε_0 . $\square$

Definition (regularity and uniformity along a scale chain). Let $0 < r < R \leq 1$ be dyadic, and let $P \subset \mathcal{D}_\delta^2$, where $\delta \leq r$. We say that P is an (a, C) -set between r and R if, for every occupied $Q \in \mathcal{D}_R^2$, the rescaled family $S_Q(P \cap Q)$, observed at resolution r/R , is an $(r/R, a, C)$ -set in the sense of (6.41). It is (a, C, K) -regular between those scales if in addition

$$|S_Q(P \cap Q)|_{(r/R)^{1/2}} \leq K(r/R)^{-a/2} \quad (Q \in \mathcal{D}_R^2, P \cap Q \neq \emptyset).$$

If the square-root resolution is not dyadic, use an adjacent dyadic resolution; this changes the covering bound only by a dimensional factor. For a chain $1 = \Delta_0 > \Delta_1 > \dots > \Delta_n = \delta$, the family P is uniform along the chain if, for each j , every occupied Δ_{j-1} -square has the same number N_j of occupied Δ_j -descendants. Thus $|P| = \prod_{j=1}^n N_j$.

Lemma III.6.29.10 (uniformizing a dense restriction of a chain). Let $P \subset \mathcal{D}_\delta^2$ be uniform along a chain of length n , and let $P' \subset P$ with $|P'| \geq h|P|$, $0 < h \leq 1$. Writing $L = 1 + \log(1/\delta)$, there is a subset $P'' \subset P'$ uniform along the same chain and satisfying

$$|P''| \geq h(4L)^{-n}|P|. \quad (6.72a)$$

If P was an (a, C) -set, or an (a, C, K) -regular set, between any one pair of successive scales, then P'' has the corresponding property with C replaced by $h^{-1}(4L)^n C$ and with the same K . The assertion also holds after rescaling an occupied ancestor square to the unit square.

Proof. Work from the finest generation upward. At generation j , group occupied Δ_{j-1} -parents according to the dyadic class of their number of retained Δ_j -children. There are at most $2L$ classes. Keep a class carrying at least a $(2L)^{-1}$ fraction of the current terminal squares, and within each of its parents keep the same number of whole child subtrees; this costs at most another factor two. Repeat for $j = n, n-1, \dots, 1$. Deleting whole subtrees preserves the branching numbers already made equal at finer generations. The final family is uniform and obeys (6.72a).

Let $N_j'' \leq N_j$ be its branching numbers. Since $\prod_j N_j''/N_j = |P''|/|P| \geq h(4L)^{-n}$ and each ratio is at most one, every ratio is at least $h(4L)^{-n}$. In each occupied parent the new normalized counting measure is therefore dominated by $h^{-1}(4L)^n$ times the old one at the next scale. Applying the relative estimate (6.41) proves the asserted Frostman constant. Square-root covering numbers only decrease under restriction, so K is unchanged. Rescaling commutes with this construction. $\square$

Proposition III.6.29.11 (uniform-set multiscale decomposition). Fix $0 < a < 1$, $a < b \leq 2$, a dyadic base Δ , and $\varepsilon > 0$. For large m , put $\delta = \Delta^m$. If $P \subset [0, 1)^2$ is uniform on the scales $\Delta, \dots, \Delta^m$ and is a $(\delta, b, \delta^{-\varepsilon})$ -set, then there are scales

$$\delta = \Delta_n < \Delta_{n-1} < \dots < \Delta_1 < \Delta_0 = 1,$$

exponents $b_j \in [a, 2]$, and a partition $\{1, \dots, n\} = S \sqcup B$ such that, for some $\tau \in (0, \varepsilon]$,

- (i) $\Delta_{j-1}/\Delta_j \geq \delta^{-\tau}$ for $j \in S$, while $\prod_{j \in B} (\Delta_{j-1}/\Delta_j) \leq \delta^{-\varepsilon}$;
- (ii) on a structured interval $j \in S$, every rescaled descendant set is a $(b_j, (\Delta_{j-1}/\Delta_j)^\varepsilon)$ -set, and it is regular when $b_j > a$;
- (iii) $\prod_{j \in S} (\Delta_{j-1}/\Delta_j)^{b_j} \geq \delta^{-b+\varepsilon}$;

(iv) no two bad indices are adjacent.

Proof. Let N_i be the common branching number between levels Δ^{i-1} and Δ^i , and define the piecewise affine profile

$$f(0) = 0, \quad f(k) = \frac{\log(N_1 \cdots N_k)}{\log(1/\Delta)}.$$

Then f is 2-Lipschitz, and the $(\delta, b, \delta^{-\varepsilon})$ -condition implies $f(x) \geq bx - O(\varepsilon)m$. Apply Lemma III.6.29.6. Structured intervals have either endpoint slope $b_j > a$ and almost-affine branching, which gives regularity, or endpoint slope exactly a and superlinear branching, which gives the a -Frostman estimate. Round endpoints to integers and declare the gaps bad. Their total length is $O(\varepsilon)m$, which gives (i), and maximality prevents adjacent gaps. Finally,

$$\sum_{j \in S} b_j \log(\Delta_{j-1}/\Delta_j) \geq (b - O(\varepsilon)) \log(1/\delta),$$

which gives (iii) after starting with a smaller tolerance. $\square$

Proposition III.6.29.12 (combining incidence estimates across scales). *Fix $0 < a < 1$, $a < b \leq 2$, $0 < \tau < 1$, and an integer $n \geq 1$. There are $\varepsilon_G = \varepsilon_G(a, b) > 0$ and $\eta = \eta(a, b) > 0$ with the following property. Given $0 < \varepsilon_N \leq \varepsilon_G$ and $C_P \geq 1$, there is $\lambda_0 = \lambda_0(a, b, \tau, n, \varepsilon_N, C_P) > 0$ such that, whenever $0 < \lambda < \lambda_0$ and δ is sufficiently small, the following assertion holds. Let $(P, \mathcal{T})$ be a $(\delta, a, \delta^{-\lambda}, M)$ -nice configuration, and let $1 = \Delta_0 > \Delta_1 > \cdots > \Delta_n = \delta$ be a dyadic scale chain along which P is uniform. Partition $\{1, \dots, n\}$ into normal, good, and bad indices N, G, B , respectively, and put $R_j = \Delta_{j-1}/\Delta_j$. Assume:*

- (i) $R_j \geq \delta^{-\tau}$ for $j \in N \cup G$;
- (ii) if $j \in N$, then P is an $(a, L^{C_P} R_j^{\varepsilon_N})$ -set between Δ_j and Δ_{j-1} ;
- (iii) if $j \in G$, then for some $b_j \in [b, 2]$, P is $(b_j, L^{C_P} R_j^{\varepsilon_G}, L^{C_P} R_j^{\varepsilon_G})$ -regular between those scales.

Here $L = 1 + \log(1/\delta)$, and no condition is imposed at bad indices. There are finite constants $A = A(n, a, b, \tau, C_P)$ and $D = D(n, \tau)$ such that

$$|\mathcal{T}| \geq L^{-A} M \delta^{D\lambda} \delta^{-a+\varepsilon_N} \prod_{j \in G} R_j^\eta \prod_{j \in B} R_j^{-1}. \quad (6.73)$$

The constants and small-scale threshold are uniform over the configuration and the chosen chain. This is a quantitative multiscale form of Orponen–Shmerkin’s Proposition 7.5 [22], with their (s, t) written as (a, b) .

Proof. We first record one-scale bounds with a margin that can be spent in the induction. Write ρ for a one-scale resolution and $u_0 = \min\{b, 1\} > a$. We make the regular-set gain uniform in the exponent. For each $v \in [b, 2]$, let $e_v > 0$ be an admissible input tolerance and output gain in Theorem III.6.29.9. If $|v' - v| < e_v/8$, then a $(\rho, v', \rho^{-e_v/8}, \rho^{-e_v/8})$ -regular set is (ρ, v, ρ^{-e_v}) -regular: its relative Frostman constant increases by at most $\rho^{-|v-v'|}$, and its square-root covering constant by at most $\rho^{-|v-v'|/2}$. The open intervals so obtained cover $[b, 2]$. Choose a finite subcover, and let $e_* > 0$ be smaller than all the corresponding e_v and all their input tolerances divided by eight. Thus, whenever both the point

and fibre constants are at most ρ^{-e_*} , the regular-set theorem gives $|\mathcal{T}| \geq \rho^{-2a-e_*}$, uniformly for every exponent in $[b, 2]$. We decrease e_* once more below if necessary, without changing this conclusion. Put $\vartheta = (u_0 - a)/(1 - a) > 0$, choose $\gamma = e_*/4$, and then fix $0 < \eta < \min\{\vartheta e_*/16, e_*/16\}$ and $0 < \varepsilon_G < \min\{e_*/8, \vartheta e_*/100, b - a\}$. Shrink the permitted λ , depending on τ , so that $2\lambda/\tau < \min\{\varepsilon_G, \vartheta e_*/100\}$.

For a first scale $\rho = \Delta_1$ satisfying (i), the normal-case application of Lemma III.6.29.4 with $u = a$ gives, after the fixed logarithmic losses,

$$\frac{|\mathcal{T}_\rho|}{M_\rho} \geq L^{-A_0} \delta^\lambda \rho^{-a+\varepsilon_N} \quad (1 \in N). \quad (6.73a)$$

Indeed the coarse point-set constant is at most $L^{C_P+O(1)} \rho^{-\varepsilon_N}$, and its tube-fibre constant is at most $L^{O(1)} \delta^{-\lambda}$. For a good first scale, put $u = \min\{b_1, 1\}$, so $(u - a)/(1 - a) \geq \vartheta$. If $M_\rho \geq \rho^{-a-\gamma}$, the final factor in (6.52) is at least $\rho^{-\vartheta\gamma}$. The point and tube constants cost at most $L^{A_0} \rho^{-\varepsilon_G}$ and $L^{A_0} \delta^{-\lambda}$, respectively. Since $\rho \leq \delta^\tau$, we have $\delta^\lambda \geq \rho^{\lambda/\tau}$, the preceding lower bound is at least $L^{-A_0} \rho^{-a-\vartheta\gamma+\varepsilon_G+\lambda/\tau}$. Our choices give $\vartheta\gamma - \varepsilon_G - \lambda/\tau > 2\eta$; also $L^{-A_0} \geq \rho^\eta$ for sufficiently small δ . Hence $|\mathcal{T}_\rho|/M_\rho \geq \rho^{-a-\eta}$. If $M_\rho < \rho^{-a-\gamma}$, the uniform regular-set theorem applies: the point constant is at most $L^{A_0} \rho^{-\varepsilon_G}$, and the fibre constant is at most $L^{A_0} \rho^{-\lambda/\tau}$, both bounded by ρ^{-e_*} after our small-parameter choices and a decrease of δ . It gives $|\mathcal{T}_\rho| \geq \rho^{-2a-e_*}$, hence $|\mathcal{T}_\rho|/M_\rho \geq \rho^{-a-e_*+\gamma} \geq \rho^{-a-\eta}$. Thus in both cases

$$\frac{|\mathcal{T}_\rho|}{M_\rho} \geq \rho^{-a-\eta} \quad (1 \in G). \quad (6.73b)$$

If $1 \in B$, the only bound needed is $|\mathcal{T}_\rho|/M_\rho \geq 1$. All three statements also apply directly when $n = 1$. In the good case the surplus $\rho^{-\varepsilon_N}$ in (6.73) is harmless; in the bad case its required right side is $\delta^{1-a+\varepsilon_N} \leq 1$.

Now induct on n . Suppose $n \geq 2$, put $\Delta = \Delta_1$ and $q = \delta/\Delta$, and apply the proved two-scale interface III.6.5. Its logarithmic loss has the form L^{-A_1} , for a fixed finite A_1 , and for every retained parent Q it gives

$$|\mathcal{T}| \geq L^{-A_1} M \frac{|\mathcal{T}_\Delta|}{M_\Delta} \frac{|\mathcal{T}_Q|}{M_Q}. \quad (6.74)$$

The coarse family $\mathcal{D}_\Delta(P')$ has, up to a $L^{O(1)}$ factor, the same number of cells and the same first-scale relative and square-root bounds as P , by clause (i) of that interface. Its tube constant is at most $L^{O(1)} \delta^{-\lambda}$. Consequently (6.73a)–(6.73b), or the trivial bound in the bad case, control the first factor $|\mathcal{T}_\Delta|/M_\Delta$.

If all indices $2, \dots, n$ are bad, use $|\mathcal{T}_Q| \geq M_Q$ in (6.74). There is no hidden tail estimate in this case: the bad product telescopes to q . If $1 \in N$, the desired right side, apart from $L^{-A} M \delta^{D\lambda}$, is $\Delta^{-a+\varepsilon_N} q^{1-a+\varepsilon_N}$, which is no greater than the normal bound in (6.73a). If $1 \in G$, it is $\Delta^{-a-\eta+\varepsilon_N} q^{1-a+\varepsilon_N}$, which is no greater than (6.73b). If $1 \in B$, it is $\delta^{1-a+\varepsilon_N} \leq 1$. This proves the claim for this tail, after increasing A, D .

Otherwise at least one tail index is normal or good, and hence $q \leq \delta^\tau$. For every retained Q , the rescaled original descendant set $S_Q(P \cap Q)$ is uniform on the remaining chain. Clause (i) of the two-scale interface retains at least a L^{-A_2} fraction of its terminal

squares. Apply Lemma III.6.29.10 to the retained descendants. It gives a uniform family of terminal squares inside $S_Q(P' \cap Q)$ of relative size at least $L^{-A_2}(4L)^{-(n-1)}$. Its normal-scale and good-scale conditions persist with their constants enlarged only by $L^{A_2+O(n)}$; the square-root covering bound is unchanged. Restricting the point family in the fine nice configuration to this subset preserves its tube fibres and M_Q .

Let $L_q = 1 + \log(1/q)$. Since $q \leq \delta^\tau$, $L \leq \tau^{-1}L_q$. Thus the enlarged point constants have the required form $L_q^{C'_P} R_j^{\varepsilon_N}$ or $L_q^{C'_P} R_j^{\varepsilon_G}$, for a fixed $C'_P = C'_P(C_P, n, \tau)$. Clause (iv) of the interface bounds the fine tube constant by $L^{A_3} \delta^{-\lambda}$. After decreasing δ , this is at most $q^{-\lambda'}$, with $\lambda' = 2\lambda/\tau$: $L^{A_3} \leq \delta^{-\lambda}$ and $\delta^{-2\lambda} \leq q^{-2\lambda/\tau}$. Choose λ_0 recursively so that λ' lies below the permitted $(n-1)$ -scale threshold. All hypotheses of the induction assertion now hold at scale q .

Write $G_* = G \setminus \{1\}$ and $B_* = B \setminus \{1\}$. The induction hypothesis, with logarithmic exponent enlarged to cover the restriction, yields

$$\frac{|\mathcal{T}_Q|}{M_Q} \geq L_q^{-A_{n-1}} q^{D_{n-1}\lambda'} q^{-a+\varepsilon_N} \prod_{j \in G_*} R_j^\eta \prod_{j \in B_*} R_j^{-1}. \quad (6.74a)$$

The first-scale estimates and this tail have an exact exponent ledger. If $1 \in N$, their scale factors multiply to $\Delta^{-a+\varepsilon_N} q^{-a+\varepsilon_N} = \delta^{-a+\varepsilon_N}$. If $1 \in G$, they give $\Delta^{-a-\eta} q^{-a+\varepsilon_N} = \delta^{-a+\varepsilon_N} \Delta^{-\eta-\varepsilon_N} \geq \delta^{-a+\varepsilon_N} \Delta^{-\eta}$. If $1 \in B$, they give $q^{-a+\varepsilon_N} = \delta^{-a+\varepsilon_N} \Delta^{a-\varepsilon_N} \geq \delta^{-a+\varepsilon_N} \Delta$, because $a - \varepsilon_N < 1$. These are precisely the three first factors in (6.73). Moreover $q \geq \delta$, so $q^{D_{n-1}\lambda'} \geq \delta^{(2D_{n-1}/\tau)\lambda}$. The two-scale, coarse, uniformization, and induction logarithmic losses multiply to at worst L^{-A_n} for a finite recursively chosen A_n . Taking, for example, $D_1 = 2$ and $D_n = 2 + 2D_{n-1}/\tau$, we may absorb every displayed δ^λ -loss in $\delta^{D_n\lambda}$. Substitution into (6.74) proves (6.73) for n , completing the induction. $\square$

Proposition III.6.29.13 (primal sharp incidence assembly). *For $0 < a < 1$ and $a < b \leq 2$, there is $\zeta > 0$ such that if P is a $(\delta, b, \delta^{-\zeta})$ -set and every $p \in P$ carries a $(\delta, a, \delta^{-\zeta})$ -set of incident tubes, then the union $\mathcal{T}$ satisfies*

$$|\mathcal{T}| \geq \delta^{-2a-\zeta}. \quad (6.75)$$

Proof. Fix the constants ε_G, η in Proposition III.6.29.12, decreasing ε_G so that $b - \varepsilon_G/2 > a$, and write $\Gamma = \varepsilon_G \eta/8$. Choose a profile tolerance $0 < \varepsilon < \min\{\varepsilon_G/4, \Gamma/16\}$. Choose a fixed dyadic base Δ sufficiently small that

$$\kappa_\Delta := \frac{\log(4(1 + 2\log(1/\Delta)))}{\log(1/\Delta)} < \varepsilon/4.$$

For the moment take $\delta = \Delta^m$. Pigeonhole the sizes of the incident tube families, retaining at least a $(1 + \log(1/\delta))^{-1}$ fraction of P , and trim the retained fibres to a common size M . Since each original fibre is a $(\delta, a, \delta^{-\zeta})$ -set, $M \geq \frac{1}{2}\delta^{-a+\zeta}$, and the relative tube constant grows by at most an absolute factor. Next uniformize the retained point family at the m powers of Δ . At each level bottom-up dyadic pigeonholing has at most $1 + 2\log(1/\Delta)$ branching classes and loses at most twice that many in terminal mass. Hence its total loss is at most δ^{κ_Δ} . The resulting point family is a $(\delta, b, L\delta^{-\zeta-\kappa_\Delta})$ -set, which is a $(\delta, b, \delta^{-\varepsilon})$ -set once

$\zeta < \varepsilon/4$ and δ is small. Apply Proposition III.6.29.11 to this uniform family with tolerance ε . Its structured indices number at most τ^{-1} , and its bad indices are nonadjacent, so the total number of blocks is bounded by $n_0 = 2\lceil \tau^{-1} \rceil + 1$, independently of δ . Declare a structured index good if its exponent $b_j \geq b - \varepsilon_G/2$, and normal otherwise. On a normal block the b_j -set condition implies the a -set condition. On a good block put $\hat{b}_j = \max\{b, b_j\}$. If $b_j < b$, upgrading the relative set exponent from b_j to b costs at most $R_j^{b-b_j} \leq R_j^{\varepsilon_G/2}$, whereas its square-root covering bound becomes only easier. If $b_j \geq b$, no upgrade is needed. Consequently each good block is $(\hat{b}_j, R_j^{\varepsilon_G}, R_j^{\varepsilon_G})$ -regular, after taking $\varepsilon \leq \varepsilon_G/2$. These are precisely the normal and good hypotheses of Proposition III.6.29.12.

Write $W = \log(1/\delta)$ and $W_G = \sum_{j \in G} \log R_j$. Since $\sum_{j \in S} b_j \log R_j \geq (b - \varepsilon)W$, normal exponents are below $b - \varepsilon_G/2$, and every exponent is at most two,

$$(b - \varepsilon)W \leq (b - \varepsilon_G/2)(W - W_G) + 2W_G.$$

Our choice $\varepsilon \leq \varepsilon_G/4$, together with $b > \varepsilon_G/2$, implies $W_G \geq \varepsilon_G W/8$. Equivalently,

$$\prod_{j \in G} (\Delta_{j-1}/\Delta_j) \geq \delta^{-\varepsilon_G/8} \quad (6.76)$$

The bad-block product is at least δ^ε . Choose $\lambda > 0$ below the finitely many admissible thresholds of Proposition III.6.29.12 for $1 \leq n \leq n_0$, with $D(n, \tau)\lambda < \Gamma/8$ throughout, and then choose the input loss $\zeta > 0$ so small that $2\zeta < \min\{\lambda, \varepsilon/2\}$ and $3\zeta < \Gamma/8$. The preceding uniformization leaves a nice configuration with common fibre size $M \geq \delta^{-a+2\zeta}$; its point-set loss is at most $\delta^{-\varepsilon}$ and its tube-fibre loss at most $\delta^{-\lambda}$ for sufficiently small δ . Proposition III.6.29.12, with $\varepsilon_N = \varepsilon$, now gives

$$|\mathcal{T}| \geq L^{-A} \delta^{-2a+2\zeta+D\lambda+2\varepsilon-\Gamma}, \quad D = \max_{n \leq n_0} D(n, \tau).$$

As $2\varepsilon < \Gamma/8$, and the logarithmic factor is at least $\delta^{\Gamma/8}$ at sufficiently small scales, the last display is at least $\delta^{-2a-\zeta}$. Replacing an arbitrary dyadic δ by an adjacent power of the fixed base changes only bounded comparison constants, which are absorbed by the same strict exponent margin. $\square$

Theorem III.6.29.14 (sharp dual Furstenberg incidence theorem). *Let $0 < a < 1$ and $a < b \leq 2$. There exists $\zeta = \zeta(a, b) > 0$ such that, for all sufficiently small dyadic δ , if $\mathcal{L}$ is a $(\delta, b, \delta^{-\zeta})$ -set of dyadic tubes and every $T \in \mathcal{L}$ carries a $(\delta, a, \delta^{-\zeta})$ -set $P(T)$ of incident dyadic squares, then*

$$\left| \bigcup_{T \in \mathcal{L}} P(T) \right| \geq \delta^{-2a-\zeta}. \quad (6.77)$$

Proof. Apply Lemma III.6.29.1. The tube family $\mathcal{L}$ becomes a $(\delta, b, O(\delta^{-\zeta}))$ -point family, while every point fibre $P(T)$ becomes a $(\delta, a, O(\delta^{-\zeta}))$ -family of incident dual tubes. Proposition III.6.29.13 therefore gives (6.77), after decreasing ζ to absorb the bounded chart multiplicity. $\square$

We now finish the proof of Proposition III.6.29. Choose a small $\eta > 0$, set $N = \lceil \eta^{-1} \rceil$, and $\rho = \delta^{1/N}$. Uniformize P on the scales $\rho, \dots, \rho^N$. Iterating the two-scale interface III.6.5, and pigeonholing the fibre sizes after each refinement, gives nested configurations $(P_j, \mathcal{T}_j)$, $0 \leq j \leq N$, with $|P_{j+1}| \geq \delta^{O(\epsilon)}|P_j|$, fibre sizes $M_j \geq \delta^{O(\epsilon)}M$, and the following decisive estimate: for every $Q \in \mathcal{D}_{\rho^j}(P_j)$ and every δ -tube T ,

$$|\mathcal{T}| \geq \delta^{O(\epsilon)} M |P_j|_{\rho^j}^{1/2} |T \cap P_j \cap Q|_{\delta}. \quad (6.78)$$

This follows by combining the coarse square-root estimate (6.45) with the fine rich-tube estimate (6.43) in the product inequality (6.42).

If for some j, Q, T ,

$$|T \cap P_N \cap Q|_{\delta} \geq \delta^{-2\eta} |P_N \cap Q|_{\delta}^{1/2},$$

then (6.78), uniformity, and the $(\delta, t, \delta^{-\epsilon})$ -bound for P_j imply (6.46), once $\epsilon \ll \eta$. We may therefore assume the reverse inequality at every scale. If at least half of the surviving point squares have at least half of their tubes satisfying

$$|T \cap P_N|_{\delta} \leq \delta^{2\eta} |P_N|_{\delta}^{1/2},$$

then Lemma III.6.27, after restriction, already yields (6.46). Hence, after another fixed-fraction refinement,

$$|T \cap P_N|_{\delta} \geq \delta^{2\eta} |P_N|_{\delta}^{1/2} \quad (6.79)$$

for every surviving tube.

Combining the all-scale reverse inequality with (6.79) and the (δ, t) -nonconcentration of P_N gives, whenever $\rho^{j+1} \leq r < \rho^j$,

$$|T \cap P_N \cap B_r|_{\delta} \leq \delta^{-6\eta} r^{t/2} |T \cap P_N|_{\delta}. \quad (6.80)$$

Thus each surviving section $T \cap P_N$ is a $(\delta, t/2, \delta^{-7\eta})$ -set. Corollary III.6.28, followed by the usual discrete Frostman extraction in line space, yields a subfamily $\mathcal{T}_2$ which is a $(\delta, s + t/2, \delta^{-\eta})$ -set of tubes. The sharp dual theorem III.6.29.14, applied with

$$a = t/2, \quad b = s + t/2,$$

is legal because $s < t < 2$, hence $0 < a < 1$, while $a < b < 2$. It gives

$$|P_N|_{\delta} \geq \delta^{-t-3\eta}$$

after choosing η below the local sharp gain and $\epsilon \ll \eta$. But the original $(\delta, t, \delta^{-\epsilon})$ -condition and the refinements give $|P_N|_{\delta} \leq \delta^{-t-O(\epsilon)}$, a contradiction. Therefore one of the previous non-rigid alternatives must hold, and each of them gives (6.46). $\square$

Theorem III.6.30 (single-scale extension of Bourgain's incidence estimate). *For $s, u \in (0, 1)$ there are $\epsilon, \eta > 0$ such that the following holds for all sufficiently small dyadic δ . Suppose $P \subset B(0, 1)$ satisfies the natural-scale anti-clustering condition*

$$|P \cap B(x, \delta) P|_{\delta}^{1/2} |_{\delta} \leq \delta^u |P|_{\delta} \quad (x \in B(0, 1)). \quad (6.81)$$

If each $p \in P$ carries a $(\delta, s, \delta^{-\epsilon})$ -family of at least M incident dyadic tubes, then

$$|\mathcal{T}| \geq \delta^{-\eta} M |P|_{\delta}^{1/2}. \quad (6.82)$$

Proof. Choose $\xi = \xi(u) > 0$ from Theorem III.6.21, and put $\vartheta = \frac{1}{4} \min\{s, \xi\}$. If an intermediate scale has local point exponent $\alpha \in [\xi, 2 - \xi]$, replace the tube exponent s , when necessary, by

$$s_\alpha := \min\{s, \alpha\} - \vartheta.$$

Then $0 < s_\alpha < \alpha < 2$, the gap $\alpha - s_\alpha$ is at least ϑ , and every (λ, s, C) -tube family is automatically a (λ, s_α, C) -family. Subdivide the compact interval $[\xi, 2 - \xi]$ into finitely many sufficiently short pieces and, on each piece, freeze slightly smaller/larger exponents so that Proposition III.6.29 applies with fixed margins. Taking the minimum of the finitely many resulting gains yields a uniform $\eta_0 > 0$ for every intermediate scale. Choose $\epsilon \ll \xi\eta_0, u$.

Uniformize P on powers of a sufficiently small base ρ , losing only a δ^ϵ -factor. The hypothesis (6.81) survives with $u/2$. Apply Theorem III.6.21. It provides scales

$$1 = \Delta_0 > \Delta_1 > \dots > \Delta_N = \delta, \quad \lambda_j = \Delta_{j+1}/\Delta_j,$$

and local exponents $\alpha_j \in [0, 2]$. The weighted sum of the α_j 's recovers $\log |P|_\delta$ up to $O(\epsilon \log(1/\delta))$, and a total logarithmic length at least $\xi \log(1/\delta)$ lies on long intervals with $\alpha_j \in [\xi, 2 - \xi]$.

Iterate the two-scale interface III.6.5. For any chain of occupied parent squares, its product inequality gives

$$\frac{|\mathcal{T}|}{M} \gtrsim_{\log, \delta} \prod_{j=0}^{N-1} \frac{|\mathcal{T}_{Q_j}|}{M_{Q_j}}. \quad (6.83)$$

On an intermediate long interval, Proposition III.6.29 gives

$$\frac{|\mathcal{T}_{Q_j}|}{M_{Q_j}} \geq \lambda_j^{-\alpha_j/2 - \eta_0};$$

on the remaining long intervals, Corollary III.6.28 gives the baseline $\lambda_j^{-\alpha_j/2}$, up to the already budgeted losses; short gaps contribute the trivial bound one. Multiplication in (6.83) gives

$$\frac{|\mathcal{T}|}{M} \geq \delta^{O(\epsilon)} |P|_\delta^{1/2} \exp\left(\eta_0 \sum_{j \in \mathcal{E}} \log(1/\lambda_j)\right) \geq \delta^{O(\epsilon) - \xi\eta_0} |P|_\delta^{1/2}.$$

Taking ϵ sufficiently small and absorbing the initial uniformization loss gives (6.82). $\square$

We now pass from finite incidence configurations to Hausdorff dimension. Recall that an (s, t) -**Furstenberg set** is a set $K \subset \mathbb{R}^2$ for which there exists a family of lines $\mathcal{L}$ with $\dim_{\text{H}} \mathcal{L} \geq t$ and $\dim_{\text{H}}(K \cap \ell) \geq s$ for every $\ell \in \mathcal{L}$. The space of lines is covered by countably many bounded slope/intercept charts; Hausdorff dimension is invariant under the resulting bilipschitz changes of coordinates.

From discrete overlap to arbitrary covers. The low-parameter bound will follow from three elementary lemmas. The first makes finite Frostman selection available even for nonmeasurable sets; the second selects one scale from an arbitrary Hausdorff cover. This is essential: a lower bound for the covering numbers of K along a sequence of scales alone would not give the desired Hausdorff bound. For a bounded set F , write

$$\mathcal{H}_\infty^u(F) = \inf \left\{ \sum_i (\text{diam } U_i)^u : F \subset \bigcup_i U_i \right\}.$$

Only countable subadditivity of this outer content is used below. In a bounded line chart $\ell_{a,b} = \{(x, ax + b) : x \in \mathbb{R}\}$ we use the metric $d(\ell_{a,b}, \ell_{a',b'}) = \max\{|a - a'|, |b - b'|\}$. All chart and bounded-region constants are fixed before the scale is chosen.

Lemma III.6.30.1 (Finite selection from Hausdorff content). *Let $0 < u \leq d$, let $F \subset [0, 1]^d$ be arbitrary, and let $\delta = 2^{-k}$. There is a finite δ -separated set $P \subset F$ such that*

$$c_{d,u} \mathcal{H}_\infty^u(F) \delta^{-u} \leq |P| \leq C_{d,u} \delta^{-u}, \quad |P \cap B(x, r)| \leq C_{d,u} (r/\delta)^u \quad (\delta \leq r \leq 1).$$

The same assertion holds in a fixed bounded cube, with constants depending on that cube.

Proof. Consider the finite dyadic tree down to side δ , retaining the leaves meeting F . Give a cube Q capacity $c(Q) = \lfloor (\ell(Q)/\delta)^u \rfloor$. At an occupied leaf set $m(Q) = 1$, at an empty leaf set $m(Q) = 0$, and upwards set

$$m(Q) = \min \left\{ c(Q), \sum_{Q' \text{ child of } Q} m(Q') \right\}.$$

Induction shows that $m(Q)$ points can be selected from distinct occupied descendant leaves without exceeding any descendant capacity. Indeed, combine the children's selections and, if necessary, discard points until the parent capacity is met.

The same recursion says that $m([0, 1]^d)$ is the minimum sum of capacities over a cut covering all occupied leaves: at each vertex one either takes that vertex or cuts all its children. Any such cut covers F . Since $\lfloor z \rfloor \geq z/2$ for $z \geq 1$ and $\text{diam } Q = \sqrt{d} \ell(Q)$, every cut has total capacity at least

$$\frac{1}{2} \delta^{-u} \sum_Q \ell(Q)^u \geq \frac{1}{2} d^{-u/2} \delta^{-u} \mathcal{H}_\infty^u(F).$$

Choose a point of F in each selected leaf. A ball of radius $r \geq \delta$ meets only $O_d(1)$ dyadic cubes of a scale between r and $2r$, so the capacities give the stated ball bound. Finally colour the leaf indices by their residues modulo 3 in each coordinate and retain a largest colour class. Its representatives are δ -separated, at a loss of at most 3^d , and the upper bounds survive. A fixed dilation gives the last assertion. $\square$

Lemma III.6.30.2 (One-scale configurations selected from an arbitrary cover). *Suppose K is an (s, t) -Furstenberg set. Fix $0 < \alpha < s$ and $0 < \beta < t$. There are a bounded piece $K_0 \subset K$ and a bounded line chart with the following property. For every sufficiently fine countable*

ball cover $\{B_i = B(x_i, r_i)\}$ of K_0 with finite positive-power cost, one can choose an arbitrarily small scale $\delta = 2^{-k}$ as the maximum allowed covering radius tends to zero, a δ -separated family Ω of lines, and finite families $\mathcal{P}(\ell)$ of dyadic δ -squares, such that

$$\begin{aligned} |\Omega| &\geq ck^{-2}\delta^{-\beta}, & |\Omega \cap B_d(\ell, r)| &\leq C(r/\delta)^\beta, \\ |\mathcal{P}(\ell)| = M &\geq ck^{-2}\delta^{-\alpha}, & \#\{p \in \mathcal{P}(\ell) : p \cap B(x, r) \neq \emptyset\} &\leq C(r/\delta)^\alpha \end{aligned}$$

for $\delta \leq r \leq 1$. All the squares meet ℓ and are drawn from at most CN_k squares, where $N_k = \#\{i : \delta/2 < r_i \leq \delta\}$. Moreover $|\Omega| \leq C\delta^{-\beta}$ and $M \leq C\delta^{-\alpha}$. The constants may depend on K_0, α, β , but not on the cover or k . No measurability assumption on K or its original line family is needed.

Proof. For each line in the original family, some bounded piece of its intersection with K has positive α -content, since its dimension is at least $s > \alpha$. Decompose the line family according to the bounded spatial region, a bounded chart, and an integer lower bound for that positive content. This is a countable cover of the line family. Because its dimension is greater than β , countable subadditivity of content gives a piece Λ and a bounded $K_0 = K \cap [-R, R]^2$ for which

$$h := \mathcal{H}_\infty^\beta(\Lambda) > 0, \quad \mathcal{H}_\infty^\alpha(K_0 \cap \ell) > c_0 > 0 \quad (\ell \in \Lambda).$$

We may rotate to use the displayed slope/intercept chart. All constants below can depend on this fixed normalization.

Choose k_0 so large that $\sum_{k \geq k_0} k^{-2} < \min\{c_0, h\}$, and consider a cover with $r_i \leq 2^{-k_0}$. Put $U_k = \bigcup \{B_i : 2^{-k-1} < r_i \leq 2^{-k}\}$ and

$$\Lambda_k = \{\ell \in \Lambda : \mathcal{H}_\infty^\alpha(K_0 \cap \ell \cap U_k) > k^{-2}\}.$$

Subadditivity on each fibre shows $\Lambda \subset \bigcup_{k \geq k_0} \Lambda_k$. Subadditivity in line space then gives a $k \geq k_0$ such that $\mathcal{H}_\infty^\beta(\Lambda_k) > k^{-2}$. Apply Lemma III.6.30.1 in the parameter chart to obtain Ω . Apply it again in the plane to each $K_0 \cap \ell \cap U_k$, and retain the dyadic δ -squares containing the selected points. The tree construction uses distinct leaves, so square counts equal point counts before discarding. A ball meeting such a square lies within $O(\delta)$ of its selected point, which gives the asserted ball bound. Each ball in U_k meets $O(1)$ squares; hence the union contains at most CN_k squares. Finite positive-power cost ensures $N_k < \infty$.

The fibre cardinalities lie between $ck^{-2}\delta^{-\alpha}$ and $C\delta^{-\alpha}$. Retain exactly $M = \lfloor ck^{-2}\delta^{-\alpha}/2 \rfloor$ squares in each fibre, increasing k_0 if necessary. Discarding preserves the absolute upper bounds and gives all the conclusions. Taking the maximum covering radius smaller forces $k \geq k_0$ larger. $\square$

Lemma III.6.30.3 (Overlap bound in a bounded line chart). *Let $0 < \alpha \leq 1$, $0 < \beta \leq 2$, $A \geq 1$, and let Ω be a δ -separated family in a fixed bounded line chart. Suppose*

$$|\Omega| \geq A^{-1}\delta^{-\beta}, \quad |\Omega \cap B_d(\ell, r)| \leq A(r/\delta)^\beta.$$

For each $\ell \in \Omega$, let $\mathcal{P}(\ell)$ consist of dyadic δ -squares meeting ℓ in a fixed bounded region, with

$$A^{-1}\delta^{-\alpha} \leq |\mathcal{P}(\ell)| \leq A\delta^{-\alpha}, \quad \#\{p \in \mathcal{P}(\ell) : p \cap B(x, r) \neq \emptyset\} \leq A(r/\delta)^\alpha.$$

The ball estimates are required for $\delta \leq r \leq 1$; fixed larger radii are absorbed in the constants. Then

$$\left| \bigcup_{\ell \in \Omega} \mathcal{P}(\ell) \right| \geq \frac{cA^{-5}}{1 + \log(1/\delta)} \delta^{-\alpha - \min\{\alpha, \beta\}}.$$

Proof. First consider two distinct lines $y = ax + b$ and $y = a'x + b'$ at distance $d \geq \delta$. If a square meets both, its centre (x, y) satisfies $|(a - a')x + (b - b')| \leq C\delta$. Since x is bounded, nonempty overlap implies $|b - b'| \leq C|a - a'| + C\delta$. If $d \leq C'\delta$, the entire bounded region is contained in a ball of radius $O(\delta/d)$. Otherwise $|a - a'| \geq cd$, and the allowed x -coordinates lie in an interval of length $O(\delta/d)$. The centres are also within $O(\delta)$ of the first line, whose slope is bounded. Consequently the common squares lie in a ball of radius $O(\delta/d)$, and

$$|\mathcal{P}(\ell) \cap \mathcal{P}(\ell')| \leq CA d^{-\alpha}.$$

This includes nearly parallel lines; parallel lines with well-separated intercepts have no common squares. When the displayed radius exceeds 1, use the total-cardinality bound.

For fixed ℓ , split the other lines into dyadic distance annuli $d \asymp r$. The diagonal term is at most $A\delta^{-\alpha}$, and the annuli give

$$\begin{aligned} \sum_{\ell' \in \Omega} |\mathcal{P}(\ell) \cap \mathcal{P}(\ell')| &\leq A\delta^{-\alpha} + CA^2\delta^{-\beta} \sum_{r=\delta, 2\delta, \dots, O(1)} r^{\beta-\alpha} \\ &\leq CA^2(1 + \log(1/\delta))\delta^{-\max\{\alpha, \beta\}}. \end{aligned}$$

The logarithm covers the critical case $\alpha = \beta$ as well as the two geometric-sum cases. Let $n(p)$ count the fibres containing p . Cauchy–Schwarz yields

$$\left| \bigcup_t \mathcal{P}(\ell) \right| \geq \frac{(\sum_p n(p))^2}{\sum_p n(p)^2} \geq \frac{|\Omega| A^{-4} \delta^{-2\alpha + \max\{\alpha, \beta\}}}{C(1 + \log(1/\delta))}.$$

Use the lower bound for $|\Omega|$ and $2\alpha + \beta - \max\{\alpha, \beta\} = \alpha + \min\{\alpha, \beta\}$. □

Proposition III.6.30.4 (Low-parameter Furstenberg bound). *For $0 < s \leq 1$ and $0 < t \leq 2$, every (s, t) -Furstenberg set satisfies*

$$\dim_{\text{H}} K \geq s + \min\{s, t\}. \quad (6.84)$$

Proof. Fix $0 < \alpha < s$, $0 < \beta < t$, and put $q = \alpha + \min\{\alpha, \beta\}$. Apply Lemma III.6.30.2 to an arbitrary sufficiently fine ball cover of its bounded piece K_0 . In Lemma III.6.30.3 take $A = Ck^2$. Since $\delta = 2^{-k}$ and at most CN_k squares occur, we obtain

$$N_k \geq ck^{-11} \delta^{-q}.$$

For any $0 < v < q$, that single scale contributes

$$\sum_i (2r_i)^v \geq N_k \delta^v \geq ck^{-11} 2^{k(q-v)}.$$

For all sufficiently large k_0 , this is at least 1 for every $k \geq k_0$. Covers of infinite v -cost already satisfy the same lower bound. Thus every sufficiently fine cover has v -cost at least 1, so $\dim_{\text{H}} K \geq \dim_{\text{H}} K_0 \geq v$. Let $v \uparrow q$, then $\alpha \uparrow s$ and $\beta \uparrow t$. □

The preceding proof gives the planar content of Héra–Shmerkin–Yavicoli, Appendix Theorem A.1 [23]. It includes the $s = 1$ endpoint and uses neither a measurable selection of fibre measures nor a later radial-projection theorem. The higher-dimensional version is not needed here.

Theorem III.6.31 ((s, t)-Furstenberg improvement). *For every $s \in (0, 1]$ and $t \in (0, 2)$ there exists $\epsilon(s, t) > 0$ such that every (s, t) -Furstenberg set $K \subset \mathbb{R}^2$ satisfies*

$$\dim_{\text{H}} K \geq s + \frac{t}{2} + \epsilon(s, t). \quad (6.85)$$

Proof. If $t \leq s$, the local bound (6.84) gives $\dim_{\text{H}} K \geq s + t$, which is stronger than (6.85); for example one may take any $\epsilon < t/2$.

Assume $t > s$, first with $s < 1$. Fix the positive constants ϵ_0, η in Proposition III.6.29 for the original parameters (s, t) . Choose $\alpha < s$ and $\beta < t$ so close to them that

$$s - \alpha < \min\{\epsilon_0/4, \eta/4\}, \quad t - \beta < \epsilon_0/4.$$

Apply Lemma III.6.30.2 to every sufficiently fine cover of the resulting bounded piece K_0 . Its absolute counting bounds and cardinality lower bounds imply relative nonconcentration constants at most Ck^2 . For $\delta \leq r \leq 1$,

$$r^\alpha \leq \delta^{-(s-\alpha)} r^s, \quad r^\beta \leq \delta^{-(t-\beta)} r^t.$$

For large k these losses and Ck^2 fit strictly inside $\delta^{-\epsilon_0}$, including the bounded chart and duality constants. Thus the configuration satisfies Corollary III.6.32 with the *original* exponents s, t , giving

$$CN_k \geq M\delta^{-(t/2+\eta)} \geq ck^{-2}\delta^{-\alpha-t/2-\eta}.$$

This use of Corollary III.6.32 is noncircular: its proof below uses only point–line duality and Proposition III.6.29, not the present theorem. For every $v < \alpha + t/2 + \eta$, the selected scale of the original cover therefore has v -cost at least $ck^{-2}2^{k(\alpha+t/2+\eta-v)}$. As in the proof of Proposition III.6.30.4, arbitrary sufficiently fine covers have cost at least 1. Hence

$$\dim_{\text{H}} K \geq \alpha + t/2 + \eta > s + t/2 + 3\eta/4.$$

One may take $\epsilon(s, t) = \eta/2$ in this branch. This argument does not assume continuity of the incidence gain as the parameters vary.

If $s = 1$, no limiting argument is needed. The local bound (6.84) gives

$$\dim_{\text{H}} K \geq 1 + \min\{1, t\}.$$

For $0 < t < 2$ this exceeds $1 + t/2$ by $t/2$ when $t \leq 1$, and by $1 - t/2$ when $1 < t < 2$. Hence a positive gain exists throughout the stated endpoint $s = 1$ range. $\square$

Corollary III.6.32 (dual union form). *Let $0 < s < 1$ and $s < t < 2$. For the $\epsilon, \eta > 0$ supplied by Proposition III.6.29, let $\mathcal{L}$ be a $(\delta, t, \delta^{-\epsilon})$ -family of lines in a bounded chart. If every $\ell \in \mathcal{L}$ carries at least M point squares forming a $(\delta, s, \delta^{-\epsilon})$ -set inside $\ell^{(O(\delta))}$, then*

$$\left| \bigcup_{\ell \in \mathcal{L}} P(\ell) \right|_{\delta} \geq M\delta^{-(t/2+\eta)}. \quad (6.86)$$

Proof. Apply Lemma III.6.29.1. The line family becomes a $(\delta, t, O(\delta^{-\epsilon}))$ -point set in dual space and each point fibre becomes a $(\delta, s, O(\delta^{-\epsilon}))$ -family of incident dual tubes. Proposition III.6.29 is exactly (6.86) after decreasing ϵ to absorb bounded chart constants. $\square$

The excluded endpoint $t = 2$. A positive improvement over $s + t/2 = s + 1$ is false at $t = 2$. Indeed, choose an s -Ahlfors-regular Cantor set $C \subset \mathbb{R}$ (an interval when $s = 1$) and put $K = C \times \mathbb{R}$. Then $\dim_{\mathbb{H}} K = s + 1$, while every nonvertical line meets K in a set bi-Lipschitz to C . The nonvertical lines form a two-dimensional family, so K is an $(s, 2)$ -Furstenberg set of dimension exactly $s + 1$. Thus the closed endpoint $t = 2$ appearing in the cited Shmerkin–Wang formulation cannot support a strictly positive gain and is therefore stated here as the open range $t < 2$ [17]; at $t = 2$ the sharp bound is $s + 1$. The same example explains the open endpoint in Proposition III.6.29 and Corollary III.6.32.

III.6.6 Consequences and scope

The chapter has established three linked groups of results, with the conditions recorded at their statement loci. First, Theorem III.6.6 and its continuous consequence provide robust linear projection growth. Second, Theorems III.6.23– III.6.25 transport that growth through nonsingular smooth families and yield pinned Euclidean and curved-norm distance estimates using Theorem III.6.6; the moving-fibre measurability is proved in Lemma III.6.12. Third, Proposition III.6.29, Theorem III.6.30, and Theorem III.6.31 turn the projection mechanism into a sharp point–tube incidence gain and its Furstenberg consequences.

The sharp Orponen–Shmerkin incidence theorem, the two-scale assembly proposition, the low-parameter Furstenberg bound, and the arbitrary-cover passage are all proved above. The potential-theoretic steps use only the stated earlier-volume energy and Frostman results. Thus later chapters may use the incidence and Furstenberg conclusions without a circular dependency on radial projection or visibility theorems.

Part III

Incidences and Furstenberg Sets

Chapter III.7

Discretized Incidence Geometry

Incidence estimates convert geometric nonconcentration into quantitative restrictions on how many point–tube pairs can meet. In this chapter we build the finite-scale language needed for that conversion and then develop two complementary sharp packages. The first is the Fu–Ren incidence theorem for fractally distributed balls and tubes, together with the normalized and fibrewise forms used in projection and Furstenberg arguments. The fibrewise conversion is proved locally by removing heavy point pockets with an explicit product replacement model; its sharp high-parameter Furstenberg conclusion follows from the weighted Fu–Ren induction proved below. The second package is the Fu–Gan–Ren rich-incidence theory, which controls the number of tubes containing many well-spaced balls and, by duality, the number of balls meeting many well-spaced tubes.

The proofs are organised so that bookkeeping is separated from the genuinely analytic steps. Double counting, Katz–Tao decomposition, duality, shadings, two-ends reduction, and finite multiscale regularisation are proved first. Fu–Ren’s high-dimensional argument then uses a weighted Fourier high–low thickening inequality, reconstructed locally below from the same Plancherel mechanism that drives the rich-incidence branch. The transition from that weighted recurrence to the refined theorem is proved below with multiplicity retained at every coarse scale. The Fu–Gan–Ren bootstrap and rich-tube induction are proved below using a selected-family attachment argument and a translation-uniform spacing hypothesis. The latter is essential: the weaker fixed-grid hypothesis in the printed source statement has a grid-corner counterexample, explained below. No later chapter is used as an incoming premise.

Prerequisites and source boundary. We use the covering and Frostman conventions from Chapters III.1–III.2, the bounded-scale projection/incidence conventions already fixed in Chapter III.6, and standard Fourier facts from the earlier harmonic-analysis foundation. The weighted thickening estimate associated with Bradshaw [25, Proposition 2.1] is proved locally in Theorem III.7.23. Fu–Ren’s refined theorem [24], its fibrewise Furstenberg conversion, and the Fu–Gan–Ren rich-incidence results [26] are studied below. The weighted Fu–Ren branch and the corrected, translation-uniform Fu–Gan–Ren branch are proved locally. Thus the displayed $\dim_{\text{H}} F \geq s + 1$ conclusion for $s + t \geq 2$ follows from the locally proved Fu–Ren branch. The formally similar result of Demeter–Guth–Wang is mentioned for comparison and is not used.

III.7.1 Geometric currencies at scale δ

Throughout the chapter, $\delta = 2^{-N} \in (0, 1]$ is dyadic unless explicitly stated otherwise. All point and line families are contained in a fixed bounded region, and all implicit constants may depend on the fixed chart conventions but not on δ .

Definition (dyadic cells, covering numbers, and nets). *A δ -cell is a half-open dyadic square of side length δ . For a bounded set E , let $N_\delta(E)$ denote the minimum number of radius- δ balls required to cover E . A maximal δ -separated subset of E is called a δ -net. In the plane, net cardinality and covering number are comparable up to an absolute constant.*

Definition (bounded line charts, tubes, and thickening). *In a bounded nonvertical affine chart we write*

$$\ell_{a,b} = \{(x, y) : y = ax + b\}, \quad d_{\mathcal{A}}(\ell_{a,b}, \ell_{a',b'}) = \max\{|a - a'|, |b - b'|\}.$$

A δ -tube is a fixed-constant δ -neighbourhood of a unit-scale line segment. Two tubes are essentially identical if their axes are $O(\delta)$ -close in line space; fixed enlargement then makes their geometric supports comparable. A family described as distinct or δ -separated is always understood in the standard continuous-incidence sense: after a fixed colouring, two members whose directions differ by $O(\delta)$ have disjoint fixed shrinkings (equivalently, in each bounded line chart their parameters form a $c\delta$ -separated set for a fixed $c > 0$). Replacing one convention by the other costs only fixed colour/enlargement constants. If $S \geq 1$, P^S and $\mathcal{T}^S$ denote thickening from scale δ to $S\delta$. Whenever a weighted incidence estimate is used, multiplicity created by thickening is retained. For subsegments of length ℓ , the natural local line-parameter metric in a bounded-slope chart is $d_\ell((\theta, b), (\theta', b')) := \ell|\theta - \theta'| + |b - b'|$, with the offsets measured at the centre of the longitudinal cell. Short segments are separated at scale δ in this metric within each cell; the cell is part of the label, and different cells may carry subsegments of the same long tube. This convention is used in the short-rectangle reduction below.

Definition (Katz–Tao and relative regularity). *A δ -separated set P is a Katz–Tao (δ, s, C) -set if*

$$|P \cap B(x, r)| \leq C(r/\delta)^s, \quad \delta \leq r \leq 1. \quad (7.1)$$

It is a relative (δ, s, C) -set if

$$|P \cap B(x, r)| \leq Cr^s |P|, \quad \delta \leq r \leq 1. \quad (7.2)$$

The same definitions are used in bounded line space with the chart metric $d_{\mathcal{A}}$. For families of δ -balls or δ -tubes we use the equivalent containment formulation of Fu–Ren: a (δ, s, K) -family has at most $K(w/\delta)^s$ members inside every scale- w ball or scale- w tube, respectively.

Definition (incidences and richness). *For a finite point/cell family P and tube family $\mathcal{T}$,*

$$I(P, \mathcal{T}) = \#\{(p, T) \in P \times \mathcal{T} : p \cap T \neq \emptyset\}.$$

If the two families carry nonnegative weights w_p, v_T , the same symbol denotes the multiplicity-counted incidence

$$I(P, \mathcal{T}) = \sum_{p \in P} \sum_{T \in \mathcal{T}} w_p v_T \mathbf{1}_{\{p \cap T \neq \emptyset\}}, \quad |P| = \sum_p w_p, \quad |\mathcal{T}| = \sum_T v_T.$$

The weighted (δ, s, K) condition counts the sum of weights, not just the number of distinct supports, in every test ball or line-space cell. It is preserved under merging coincident objects. The r -rich point and tube levels are

$$P_r(\mathcal{T}) = \{p \in P : \deg_{\mathcal{T}}(p) \geq r\}, \quad \mathcal{T}_r(P) = \{T \in \mathcal{T} : \deg_P(T) \geq r\}.$$

When richness is dyadically pigeonholed, the resulting logarithmic loss will always be displayed or absorbed into an explicitly available $\delta^{-\epsilon}$ factor.

III.7.2 Incidence bookkeeping and geometric stability

Lemma III.7.1 (double counting). For finite P and $\mathcal{T}$,

$$I(P, \mathcal{T}) = \sum_{p \in P} \deg_{\mathcal{T}}(p) = \sum_{T \in \mathcal{T}} \deg_P(T).$$

Proof. Each incidence pair is counted exactly once in each sum. $\square$

Lemma III.7.2 (rich-level bound). For every integer $r \geq 1$,

$$r|P_r(\mathcal{T})| \leq I(P, \mathcal{T}), \quad r|\mathcal{T}_r(P)| \leq I(P, \mathcal{T}).$$

Proof. Restrict either identity in [Theorem III.7.1](#) to objects of degree at least r . $\square$

Lemma III.7.3 (dyadic richness decomposition). Let $D_P = \max_{p \in P} \deg_{\mathcal{T}}(p)$. If $I(P, \mathcal{T}) > 0$, then for some dyadic $r \in [1, 2D_P]$,

$$r \#\{p : r \leq \deg_{\mathcal{T}}(p) < 2r\} \geq \frac{I(P, \mathcal{T})}{1 + \lceil \log_2(2D_P) \rceil}. \quad (7.5)$$

The dual statement holds for tube degrees.

Proof. Partition the positive degrees into dyadic intervals and pigeonhole their sum. $\square$

Lemma III.7.4 (cell-centre conversion). Replacing each δ -cell by a chosen centre changes incidence statements only after enlarging all δ -tubes by a fixed constant factor.

Proof. Every point of a δ -cell is within $O(\delta)$ of its centre. A cell meeting a tube therefore has its centre in a fixed enlargement, and the converse follows after another fixed enlargement. $\square$

Lemma III.7.5 (finite chart reduction). Every unit-scale line segment meeting $[0, 1]^2$ belongs, after one of finitely many fixed rotations, to a bounded slope/intercept chart. On such a chart, $d_{\mathcal{A}}$ is bi-Lipschitz equivalent to the usual angular/offset line metric.

Proof. Cover the projective direction circle by finitely many arcs avoiding the vertical direction after the corresponding rotations. On a compact slope interval the coordinate map and its inverse have uniformly bounded derivatives. $\square$

III.7.3 Duality, regularity, and decomposition

In a bounded affine chart set

$$\mathbf{D}(a, b) = \{(x, y) : y = ax + b\}, \quad \mathbf{D}^*(\ell_{c,d}) = (-c, d). \quad (7.4)$$

Thus points are sent to lines and lines to points. On compact parameter ranges both maps are bi-Lipschitz up to absolute constants.

Lemma III.7.6 (exact chart duality and thickened incidence stability). *For the duality maps defined above,*

$$(a, b) \in \ell_{c,d} \iff \mathbf{D}^*(\ell_{c,d}) \in \mathbf{D}(a, b). \quad (7.6)$$

Moreover, on a fixed compact chart, a primal δ -incidence becomes a dual $C\delta$ -incidence and conversely, for an absolute chart-dependent C .

Proof. The algebraic identity is $b = ca + d$ if and only if $d = -ac + b$. Bi-Lipschitz control on the fixed chart and the mean-value theorem transfer $O(\delta)$ perturbations of points and axes into $O(\delta)$ perturbations after duality. Apply [Theorem III.7.4](#) to pass between cells and their centres. $\square$

Lemma III.7.7 (Katz–Tao regularity survives duality). *If E is a Katz–Tao (δ, s, K) -set in one bounded point or line chart, then its image under $\mathbf{D}$ or $\mathbf{D}^*$, after changing δ by a fixed multiplicative constant, is a Katz–Tao $(C\delta, s, C_s K)$ -set.*

Proof. A bi-Lipschitz map sends an r -ball into an Lr -ball and preserves separation at scale δ/L . Substitution in the defining counting estimate gives the result. $\square$

Lemma III.7.8 (conversion between relative and Katz–Tao currencies). *Suppose a nonempty δ -separated finite set P satisfies*

$$|P \cap B(x, r)| \leq Cr^s |P|, \quad \delta \leq r \leq 1.$$

Then P is a Katz–Tao (δ, s, K) -set with

$$K = C|P|\delta^s. \quad (7.7)$$

Conversely, a Katz–Tao (δ, s, K) -set is a relative $(\delta, s, K/(|P|\delta^s))$ -set.

Proof. Rewrite $Cr^s |P| = (C|P|\delta^s)(r/\delta)^s$; the converse is the same identity reversed. $\square$

Lemma III.7.9 (Katz–Tao decomposition in a doubling space). *Let (X, d) have doubling constant D_0 . For every $\eta > 0$ and $t \geq 0$, there is $\delta_0 = \delta_0(\eta, D_0, t) > 0$ such that the following holds. If $0 < \delta \leq \delta_0$ and $P \subset B(x_0, 1)$ is nonempty, δ -separated, and*

$$|P \cap B(x, r)| \leq Cr^t |P| \quad (\delta \leq r \leq 1), \quad (7.8)$$

then there is a disjoint partition

$$P = P_1 \sqcup \cdots \sqcup P_M, \quad M \leq C|P|\delta^{t-\eta}, \quad (7.9)$$

and every P_j is a Katz–Tao $(\delta, t, 1)$ -set.

Proof. If $t = 0$, partition P into singletons; since $\delta^{-\eta} \geq 1$, (7.9) follows. Assume $t > 0$. At every dyadic $r \in [\delta, 2]$, choose a maximal $(r/2)$ -separated family of centres in $B(x_0, 1)$, and let $\mathcal{B}_r$ be the associated radius- r balls. Doubling gives bounded overlap, and every ball of radius at most $r/2$ meeting $B(x_0, 1)$ is contained in a member of $\mathcal{B}_r$.

Put

$$H_0 = 4^{t+1}C|P|\delta^t, \quad h = \lceil H_0 \rceil. \quad (7.10)$$

The inequality with $r = \delta$, applied at a point of P , shows $H_0 \geq 1$ after harmlessly enlarging the numerical factor. For every $B \in \mathcal{B}_r$, divide $P \cap B$ into groups of size at most h , all but possibly one having size exactly h . Form a graph on P by joining two points when they lie in the same group for at least one ball and one dyadic scale. For a fixed point and scale, bounded overlap places that point in only $O_{D_0}(1)$ balls, and each group contributes fewer than h neighbours. Hence

$$\Delta(G) \lesssim_{D_0} h \log(1/\delta).$$

Greedy colouring partitions P into at most that many independent colour classes. Since $h \leq 2H_0$ and logarithms are absorbed by $\delta^{-\eta}$ for sufficiently small δ , this gives (7.9).

Fix one colour class P_j , a ball $B(x, \rho)$ with $\delta \leq \rho \leq 1$, and a dyadic r with $r/4 \leq \rho \leq r/2$. Choose $B \in \mathcal{B}_r$ containing $B(x, \rho)$. By (7.8),

$$|P \cap B| \leq Cr^t|P| \leq 4^t C \rho^t |P|.$$

Thus the number of groups in $P \cap B$ is at most

$$\left\lceil \frac{|P \cap B|}{h} \right\rceil \leq \left\lceil \frac{1}{4}(\rho/\delta)^t \right\rceil \leq (\rho/\delta)^t.$$

An independent colour class contains at most one point from each group, proving the unit-constant Katz–Tao estimate. $\square$

III.7.4 Shadings, two ends, and finite multiscale regularisation

A *shading* $Y(T)$ is a union of δ -cells meeting T . Put

$$M(Y) = \sum_{T \in \mathcal{T}} |Y(T)|, \quad m_Y(p) = \#\{T : p \in Y(T)\}.$$

A dyadic density class satisfies

$$\lambda \leq \frac{|Y(T)|}{N_\delta(T)} < 2\lambda.$$

For $\sigma > 0$ and $A \geq 1$, the shading has the (σ, A) *two-ends property* if

$$N_\delta(Y(T) \cap B(x, r)) \leq Ar^\sigma N_\delta(Y(T)) \quad (7.3)$$

for every T , every Euclidean ball, and every $\delta \leq r \leq 1$.

Let $1 = \rho_0 > \rho_1 > \dots > \rho_m = \delta$ be nested dyadic scales. A family has *branching profile* $(B_0, \dots, B_{m-1})$ if, after refinement, every occupied ρ_j -parent contains between B_j and $2B_j$ occupied ρ_{j+1} -children.

Lemma III.7.10 (weighted two-ends pruning). *Let $Y(T)$ be finite nonempty δ -shadings, fix $\sigma > 0$ and $A \geq 1$, and put*

$$L_\delta = 1 + \lceil \log_2(\delta^{-1}) \rceil.$$

After discarding at most half of the shaded incidence mass, exactly one of the following alternatives can be retained.

- (i) *There is a subfamily $\mathcal{G}$ carrying at least $M(Y)/2$ shaded incidences, and every $T \in \mathcal{G}$ obeys the (σ, A) two-ends condition at all dyadic radii.*
- (ii) *There are a common dyadic $r \in [\delta, 1/2]$, a subfamily $\mathcal{B}_r$, and balls $B_T = B(x_T, r)$ such that*

$$\sum_{T \in \mathcal{B}_r} |Y(T)| \geq \frac{M(Y)}{2L_\delta}, \quad |Y(T) \cap B_T| > Ar^\sigma |Y(T)|. \quad (7.11)$$

Proof. Call a tube good if the two-ends estimate holds at every dyadic radius. If good tubes carry at least half of $M(Y)$, retain them. Otherwise the bad tubes carry more than half. For each bad tube choose the largest dyadic violating radius r_T ; it is at most $1/2$, since the radius-one inequality is automatic for $A \geq 1$. Pigeonhole the bad tubes by r_T , using the weights $|Y(T)|$. There are at most L_δ choices. $\square$

Lemma III.7.11 (local rescaling of a two-ends failure). *Suppose the concentrated alternative in Theorem III.7.10 holds at radius r . On each selected ball $B = B(x_B, r)$, let $\Phi_B(x) = (x - x_B)/r$. Then δ -cells become $O(\delta/r)$ -cells, clipped portions of δ -tubes become $O(\delta/r)$ -tubes after unit-scale extension, incidences are preserved up to fixed enlargement, and Katz–Tao estimates transform by replacing δ with δ/r .*

Proof. All Euclidean distances are divided by r . Hence widths, cell diameters, and covering radii are divided by r , while affine lines remain affine. The regularity statement follows by writing $\rho = r\rho'$ in the defining inequality. $\square$

Lemma III.7.12 (uniform shading-density refinement). *For a nonempty family of shaded unit-scale δ -tubes, put*

$$\eta_T = \frac{|Y(T)|}{N_\delta(T)}, \quad M = \sum_T |Y(T)|.$$

There are a dyadic $\lambda \in (0, 1]$ and a subfamily $\mathcal{T}_\lambda$ such that

$$\lambda \leq \eta_T < 2\lambda \quad (T \in \mathcal{T}_\lambda), \quad \sum_{T \in \mathcal{T}_\lambda} |Y(T)| \geq \frac{M}{CL_\delta}. \quad (7.12)$$

Proof. A unit-scale δ -tube contains $\asymp \delta^{-1}$ cells, so a nonempty shading has $c\delta \leq \eta_T \leq 1$. Pigeonhole the dyadic density intervals with weights $|Y(T)|$. $\square$

Corollary III.7.13 (common density and common two-ends scale). *Fix $\sigma > 0$ and $A \geq 1$. From any nonempty shaded family one may retain a subfamily with common density λ up to a factor two and either*

- (i) *the (σ, A) two-ends property, while retaining at least $M/(CL_\delta)$ shaded incidences; or*
- (ii) *a common violating dyadic radius, while retaining at least $M/(CL_\delta^2)$ shaded incidences.*

Proof. Apply Theorem III.7.12 and then Theorem III.7.10. $\square$

Lemma III.7.14 (shaded multiplicity layer). *Let $P = \bigcup_T Y(T)$ and $M = \sum_T |Y(T)|$. Put $L_{\mathcal{T}} = 1 + \lceil \log_2(2|\mathcal{T}|) \rceil$. There are a dyadic $\mu \geq 1$ and*

$$P_\mu = \{p : \mu \leq m_Y(p) < 2\mu\}$$

such that

$$\sum_{p \in P_\mu} m_Y(p) \geq \frac{M}{L_{\mathcal{T}}}, \quad \mu |P_\mu| \geq \frac{M}{2L_{\mathcal{T}}}. \quad (7.13)$$

Proof. Use $M = \sum_p m_Y(p)$ and pigeonhole positive multiplicities into dyadic classes. On the selected class $m_Y(p) < 2\mu$, yielding the second inequality. $\square$

Lemma III.7.15 (local two-ends density increment). *Fix $0 < \sigma < 1$. There is $A_0(\sigma)$ such that, whenever the (σ, A) two-ends condition fails for $A \geq A_0(\sigma)$ at a dyadic radius $r \leq 1/2$, restricting to the violating ball and rescaling as in [Theorem III.7.11](#) produces a new shading density λ' satisfying*

$$\lambda' \geq cAr^{\sigma-1}\lambda \geq 2\lambda. \quad (7.14)$$

Hence an iterated local two-ends reduction terminates after $O(\log(1/\lambda))$ violating rescalings.

Proof. The violating ball contains more than $Ar^\sigma |Y(T)|$ shaded cells. A unit δ -tube has at least $c_0\delta^{-1}$ cells, whereas the rescaled clipped portion is contained in a unit tube with at most C_0r/δ cells at the new scale. Thus

$$\lambda' \geq cAr^\sigma |Y(T)| \frac{\delta}{r} \geq c'Ar^{\sigma-1}\lambda.$$

Choose A_0 so that the last factor is at least two. Density is bounded above by an absolute constant, so it can double only logarithmically many times. $\square$

Lemma III.7.16 (biregular incidence pruning). *Let $G = (P, \mathcal{T}, E)$ be a finite bipartite incidence graph with $I = |E| > 0$, and put*

$$L = 1 + \lceil \log_2(2 \max\{|P|, |\mathcal{T}|\}) \rceil.$$

There are $P' \subset P$, $\mathcal{T}' \subset \mathcal{T}$, and dyadic $d_P, d_T \geq 1$ such that

$$I(P', \mathcal{T}') \geq \frac{I}{2L^2}, \quad (7.15)$$

all retained point degrees are at most $2d_P$, all retained tube degrees are at most $2d_T$, and

$$\deg_{\mathcal{T}'}(p) \geq \frac{I(P', \mathcal{T}')}{4|P'|}, \quad \deg_{P'}(T) \geq \frac{I(P', \mathcal{T}')}{4|\mathcal{T}'|}. \quad (7.16)$$

Proof. Bin the original degrees on both sides into dyadic intervals. Some pair of bins carries at least I/L^2 edges. Inside that induced graph, repeatedly delete a point whose current degree is below $E_0/(4|P_0|)$ or a tube whose current degree is below $E_0/(4|\mathcal{T}_0|)$. Point deletions remove fewer than $E_0/4$ edges in total, and tube deletions remove fewer than $E_0/4$. The terminal graph therefore retains at least $E_0/2$ edges and has the claimed lower-degree bounds. The dyadic upper bounds survive restriction. $\square$

Lemma III.7.17 (shading incidence lower bound). *Let $Y(T)$ be δ -cell shadings and $P = \bigcup_T Y(T)$. Then*

$$I(P, \mathcal{T}) \geq \sum_{T \in \mathcal{T}} |Y(T)|. \quad (7.17)$$

If every shading is λ -dense, then

$$I(P, \mathcal{T}) \geq \lambda \sum_{T \in \mathcal{T}} N_\delta(T). \quad (7.18)$$

Proof. Every membership $p \in Y(T)$ is a distinct incidence pair; sum them. $\square$

Lemma III.7.18 (branching-profile uniformisation). *Let $\delta = 2^{-N}$, fix $m \geq 1$, and choose nested dyadic scales*

$$1 = \rho_0 > \rho_1 > \dots > \rho_m = \delta, \quad \rho_j = 2^{-n_j},$$

where $0 = n_0 < \dots < n_m = N$ and the increments $n_{j+1} - n_j$ differ by at most one. For every nonempty family P of δ -cells in $[0, 1]^2$, there are $P' \subset P$ and dyadic integers $B_0, \dots, B_{m-1} \geq 1$ such that

$$|P'| \geq |P|/L^m, \quad L = 1 + 2\lceil N/m \rceil, \quad (7.19)$$

and every occupied ρ_j -square contains between B_j and $2B_j$ occupied ρ_{j+1} -children. Consequently every occupied ρ_j -square has between

$$\prod_{i=j}^{k-1} B_i \quad \text{and} \quad 2^{k-j} \prod_{i=j}^{k-1} B_i \quad (7.20)$$

occupied ρ_k -descendants.

Proof. Work from the finest block upward. At level j , bin the number of occupied ρ_{j+1} -children of each occupied ρ_j -parent dyadically. There are at most L bins. Keep the bin carrying the largest δ -cell mass and discard the other parents. Proceed for $j = m-1, m-2, \dots, 0$. Later coarser restrictions discard whole already-regularised parents and therefore do not alter child counts inside surviving parents. The total loss is at most L^m , and multiplication of one-step bounds yields (7.20). $\square$

Lemma III.7.19 (covering/net and continuum/discrete dictionary). *Let $E \subset B(0, 2)$ be bounded and let $P \subset E$ be a maximal δ -separated set. Then*

$$N_{C\delta}(E) \leq |P| \leq CN_\delta(E). \quad (7.21)$$

The same assertion holds in a bounded affine line chart. If E meets a δ -tube T , then the selected net contains a point in a fixed enlargement CT ; conversely a selected-point incidence with T is a genuine incidence of E with the same fixed enlargement.

Proof. Maximality gives a δ -cover of E by balls centred at P . Conversely a δ -ball contains only $O(1)$ points of a δ -separated planar set. The line-chart statement follows by bi-Lipschitz equivalence. For incidences, choose a net point within δ of an intersection point and apply Theorem III.7.4. $\square$

III.7.5 Low-dimensional Fu–Ren estimates

Proposition III.7.20 (low-dimensional incidence bound). *Let P be a Fu–Ren (δ, β, K_β) -family of balls and $\mathcal{T}$ a $(\delta, \alpha, K_\alpha)$ -family of tubes. Put $D = \delta^{-1}$ and $b = \min\{\beta, 1\}$. If $0 < \alpha \leq b$, then for every $\epsilon > 0$,*

$$I(P, \mathcal{T})^{\alpha+b} \lesssim_\epsilon D^{ab+\epsilon} K_\beta^\alpha K_\alpha^b |P|^b |\mathcal{T}|^\alpha. \quad (7.22)$$

Proof. Write $\mathcal{T}(p) = \{T : p \cap T \neq \emptyset\}$ and

$$J = \sum_{p \in P} |\mathcal{T}(p)|^{1+b/\alpha}.$$

Hölder gives

$$J^\alpha \geq |P|^{-b} I(P, \mathcal{T})^{\alpha+b}. \quad (7.23)$$

For a fixed T , define

$$j(T) = \sum_{p \sim T} |\mathcal{T}(p)|^{b/\alpha}, \quad J = \sum_{T \in \mathcal{T}} j(T).$$

Decompose tubes meeting T into dyadic angular annuli $\mathcal{T}_w(T)$, $w \in \{\delta, 2\delta, \dots, O(1)\}$. Tube regularity gives

$$|\mathcal{T}_w(T)| \leq K_\alpha (w/\delta)^\alpha. \quad (7.24)$$

For $S \in \mathcal{T}_w(T)$, the intersection of the two thick tubes is contained in a rectangle of dimensions $O(\delta) \times O(\delta/w)$. If $\beta \geq 1$, cover it by $O(w^{-1})$ radius- $O(\delta)$ balls; if $\beta < 1$, contain it in one ball of radius $O(\delta/w)$. In either case,

$$\sum_{p \sim T} |\mathcal{T}(p) \cap \mathcal{T}_w(T)| \leq |\mathcal{T}_w(T)| K_\beta w^{-b}. \quad (7.25)$$

Set $q = b/\alpha \geq 1$ and let $L = O(\log D)$ be the number of angular scales. Convexity and $a^q \leq aM^{q-1}$ for $0 \leq a \leq M$ give

$$\begin{aligned} j(T) &\leq L^{q-1} \sum_w \sum_{p \sim T} |\mathcal{T}(p) \cap \mathcal{T}_w(T)|^q \\ &\leq L^{q-1} \sum_w |\mathcal{T}_w(T)|^{q-1} \sum_{p \sim T} |\mathcal{T}(p) \cap \mathcal{T}_w(T)| \\ &\lesssim L^q K_\alpha^q K_\beta \delta^{-b} \lesssim_\epsilon K_\alpha^{b/\alpha} K_\beta D^{b+\epsilon/\alpha}. \end{aligned}$$

Summing over T , combining with (7.23), and raising to the power α proves (7.22). $\square$

Proposition III.7.21 (dual low-dimensional branch). *Put $a = \min\{\alpha, 1\}$ and assume $0 < \beta \leq a$. Then*

$$I(P, \mathcal{T})^{a+\beta} \lesssim_\epsilon D^{a\beta+\epsilon} K_\beta^a K_\alpha^\beta |P|^\beta |\mathcal{T}|^a. \quad (7.26)$$

Proof. Use [Theorems III.7.6](#) and [III.7.7](#) to interchange points and tubes, apply [Theorem III.7.20](#) with the parameters exchanged, and transfer the estimate back. $\square$

Proposition III.7.22 (extreme separated-object branches). *If $\beta \geq \alpha + 1$, then*

$$I(P, \mathcal{T}) \lesssim K_\alpha K_\beta D^{\alpha+1}. \quad (7.27)$$

If $\alpha \geq \beta + 1$, then

$$I(P, \mathcal{T}) \lesssim K_\alpha K_\beta D^{\beta+1}. \quad (7.28)$$

Proof. Assume $\beta \geq \alpha + 1$. The intersection graph of the δ -balls has degree $O(K_\beta)$, so colour it with $O(K_\beta)$ colours. Balls of one colour are pairwise disjoint, and a unit δ -tube meets $O(D)$ pairwise disjoint δ -balls in a bounded region. Hence one colour contributes $O(|\mathcal{T}|D)$, while $|\mathcal{T}| \lesssim K_\alpha D^\alpha$. Sum the colours. The second estimate follows by bounded-chart duality. $\square$

III.7.6 The refined Fu–Ren theorem

Proposition III.7.23 (two-sided weighted Fourier thickening estimate; Bradshaw provenance). *Fix $0 < \theta < 1$. Let A be a family of distinct weighted δ -atoms in $[0, 1]^2$ with weights w , and let $\mathcal{T}$ be a family of distinct weighted δ -tubes with weights v . If $\delta^{-\theta} \lesssim S \lesssim \delta^{-1}$, then*

$$I(A, \mathcal{T}) \lesssim_\theta \left(S \delta^{-1} \sum_{T \in \mathcal{T}} v(T)^2 \sum_{a \in A} w(a)^2 \right)^{1/2} + \delta^{-\theta} S^{-1} I(A^S, \mathcal{T}^S). \quad (7.29)$$

When thickening merges atoms or tubes, the resulting objects carry summed multiplicities. The unit-weight tube case is the two-dimensional specialization of Bradshaw [25, Proposition 2.1]; the two-sided weighted proof below is local.

Proof. Write $R = \delta^{-1}$ and dilate space by R . Thus the atoms have unit diameter, the tubes have unit width and length $O(R)$, and the incidence count is unchanged up to the fixed bump conventions. Choose nonnegative smooth functions ϕ_a, ϕ_T , equal to one on sufficiently large fixed enlargements of the corresponding atom or tube and supported in still larger fixed enlargements, and set

$$f = \sum_{a \in A} w(a) \phi_a, \quad g = \sum_{T \in \mathcal{T}} v(T) \phi_T.$$

Distinctness of the atoms gives bounded overlap after a fixed colouring, hence

$$\|f\|_2^2 \lesssim \sum_{a \in A} w(a)^2, \quad I(A, \mathcal{T}) \lesssim \int f g.$$

Put

$$L = \delta^{-\theta/2}, \quad \rho = L/S.$$

Because $S \gtrsim \delta^{-\theta} = L^2$ and $S \lesssim R$, we have $R^{-1} \lesssim \rho \lesssim 1$, while the spatial low-frequency scale $r := \rho^{-1} = S/L$ satisfies $1 \lesssim r \lesssim S$. Choose a smooth radial cutoff η , equal to one on $B(0, \rho)$ and supported in $B(0, 2\rho)$, and split the Plancherel pairing into low and high frequencies.

For the high-frequency term, divide directions into arcs of length comparable with R^{-1} and write $g = \sum_\omega g_\omega$. The Fourier transform of a unit-width, length- R tube is rapidly

concentrated in its dual $R^{-1} \times 1$ slab. After enlarging these slabs by the factor L , the omitted tails are summable to arbitrary polynomial order. If $|\xi| \geq \rho$, the angular condition $|\xi \cdot v_T| \lesssim LR^{-1}$ places the possible tube directions in an arc of length $O(LR^{-1}|\xi|^{-1})$. Consequently ξ belongs to only

$$O(L|\xi|^{-1}) \leq O(L\rho^{-1}) = O(S)$$

of the enlarged dual slabs. Cauchy–Schwarz over the direction classes and Plancherel therefore give

$$\int |\hat{g}(\xi)|^2 |1 - \eta(\xi)|^2 d\xi \lesssim_\theta S \sum_\omega \|g_\omega\|_2^2 \lesssim_\theta SR \sum_{T \in \mathcal{T}} v(T)^2.$$

For the last inequality, subdivide the direction arcs by a fixed factor if necessary: distinct tubes in one class are then disjoint before a fixed enlargement, so their bumps have bounded overlap. Thus Cauchy–Schwarz yields

$$\left| \int \widehat{f \tilde{g}}(1 - \eta) \right| \lesssim_\theta \left(SR \sum_{T \in \mathcal{T}} v(T)^2 \sum_{a \in A} w(a)^2 \right)^{1/2}.$$

For the low-frequency term, factor the cutoff as $\eta = |\hat{h}_\rho|^2$, where $h_\rho(x) = \rho^2 h(\rho x)$ and h is a fixed Schwartz function. Plancherel gives

$$\left| \int \widehat{f \tilde{g}} \eta \right| \leq \int |f * h_\rho| |g * h_\rho|.$$

The scaled bump bounds imply, for every fixed large N ,

$$|\phi_a * h_\rho(x)| \lesssim_N \rho^2 (1 + \rho \operatorname{dist}(x, a))^{-N}, \quad |\phi_T * h_\rho(x)| \lesssim_N \rho (1 + \rho \operatorname{dist}(x, T))^{-N}.$$

Integrating these bounds, a pair at separation h contributes at most $C_N \rho (1 + \rho h)^{-N} w(a) v(T)$, after decreasing N by a fixed amount. Split at $h = S/2$. Every near pair is counted by $I(A^S, \mathcal{T}^S)$ and contributes at most $C \rho w(a) v(T)$. For a far pair, $\rho h \geq L/2$, so its contribution is $O_N(\rho L^{-N} w(a) v(T))$. There are $O(R^2)$ distinct atom supports and $O(R^2)$ distinct tube supports in the bounded spatial and line-space charts. Thus

$$\sum_a w(a) \sum_T v(T) \lesssim R^2 \left(\sum_a w(a)^2 \sum_T v(T)^2 \right)^{1/2}.$$

Since $L = R^{\theta/2}$, choosing $N = N(\theta)$ large makes the total far-pair contribution no larger than the high-frequency bound. Therefore

$$\left| \int \widehat{f \tilde{g}} \eta \right| \lesssim_\theta \rho I(A^S, \mathcal{T}^S) + \left(SR \sum_{T \in \mathcal{T}} v(T)^2 \sum_{a \in A} w(a)^2 \right)^{1/2}.$$

Finally

$$\rho = \delta^{-\theta/2} S^{-1} \leq \delta^{-\theta} S^{-1}, \quad R = \delta^{-1},$$

which proves (7.29). □

Proposition III.7.24 (weighted Fu–Ren coarse-scale recurrence). *Fix $0 < \epsilon < 1/2$, put $D = \delta^{-1}$ and $S = D^{\epsilon/20}$. Let $P, \mathcal{T}$ be finite families of δ -balls and δ -tubes carrying nonnegative weights. Suppose the weighted mass in a fixed enlargement of any one δ -ball is at most K_β , and that in any one essential-identity class of tubes is at most K_α . Then, with $|P|, |\mathcal{T}|$ denoting total weights,*

$$I(P, \mathcal{T}) \lesssim_\epsilon S^{1/2} D^{1/2} (K_\alpha K_\beta)^{1/2} |P|^{1/2} |\mathcal{T}|^{1/2} + S^{-1+\epsilon/2} I(P^S, \mathcal{T}^S). \quad (7.30)$$

Proof. Snap ball centres and tube parameters to fixed δ -grid charts, merging objects in a common grid cell and summing their weights. Fixed enlargement dominates the original incidences. The snapped supports have bounded overlap in the spatial and line-space charts. Their maximum atom and tube weights are $O(K_\beta)$ and $O(K_\alpha)$, respectively. Apply the proof of Theorem III.7.23 to these two weighted families with $\theta = \epsilon^2/40$. Since

$$\delta^{-\theta} = D^{\epsilon^2/40} = S^{\epsilon/2},$$

the low-frequency spatial scale in that proof is $r = SD^{-\theta/2} = S/L$, with $L = D^{\theta/2}$. Split its weighted atom–tube pairs at distance $S/2$ in the scaled geometry. Moving a near pair back from its snapped supports by $O(1)$ puts the original pair inside the S -thickenings for large D ; these pairs contribute at most $C\rho I(P^S, \mathcal{T}^S)$, where $\rho = L/S \leq D^\theta/S$. For a far pair the Schwartz-kernel estimate in the preceding proof is at most $C_N \rho L^{-N} w_p v_T$. There are $O(D^2)$ snapped atom supports and $O(D^2)$ snapped tube supports, so Cauchy–Schwarz gives

$$\sum_p w_p \sum_T v_T \lesssim D^2 \left(\sum_p w_p^2 \sum_T v_T^2 \right)^{1/2}.$$

Choose $N = N(\epsilon)$ so large that $C_N \rho L^{-N} D^2$ is bounded by $(SD)^{1/2}$ for all sufficiently large D . The far-pair contribution is then absorbed by the high-frequency term. This margin-and-tail split, rather than the false transfer of every snapped S -incidence, gives the low-frequency term displayed in (7.30). For the high-frequency term use

$$\sum_{p \in P} w_p^2 \leq (\max_p w_p) |P| \lesssim K_\beta |P|, \quad \sum_{T \in \mathcal{T}} v_T^2 \lesssim K_\alpha |\mathcal{T}|.$$

This proves the weighted recurrence, including multiplicities created by all later thickenings. $\square$

Theorem III.7.25 (Fu–Ren refined incidence theorem). *Assume $0 < \alpha, \beta \leq 2$ and define*

$$c^{-1} = \max\{\alpha + \beta - 1, 2\}. \quad (7.31)$$

If P is a Fu–Ren (δ, β, K_β) -family of balls and $\mathcal{T}$ a $(\delta, \alpha, K_\alpha)$ -family of tubes, then for every $\epsilon > 0$,

$$I(P, \mathcal{T}) \leq C_\epsilon \delta^{-c-\epsilon} (K_\alpha K_\beta)^c |P|^{1-c} |\mathcal{T}|^{1-c}. \quad (7.32)$$

The same inequality holds for weighted multisets satisfying the weighted (δ, β, K_β) and $(\delta, \alpha, K_\alpha)$ conditions of Definition III.7.1, with $|P|, |\mathcal{T}|$ interpreted as total weights. In particular, coincident objects created by thickening do not require an unweighted reduction.

Proof. This is the refined high-dimensional theorem of Fu–Ren [24, Theorem 1.5]. We prove the stronger weighted statement from [Theorem III.7.24](#); the unweighted assertion is its unit-weight case. Write

$$Q = |P||\mathcal{T}|, \quad K = K_\alpha K_\beta,$$

and take $0 < \epsilon < 1/2$. Let $C_R(\epsilon)$ dominate the implicit constant in [Theorem III.7.24](#), including the bounded loss from replacing $S\delta$ by an adjacent dyadic scale. We induct on dyadic scale. Choose the base threshold $N = N(\epsilon)$ so large that

$$2C_R(\epsilon) 2^{-N\epsilon^2/40} \leq 1. \quad (7.32a)$$

This is the quantitative contraction choice hidden in the usual phrase “take the base scale sufficiently small.”

At the finitely many base scales $\delta \geq 2^{-N}$, weighted regularity gives $|P| \lesssim K_\beta D^\beta$ and $|\mathcal{T}| \lesssim K_\alpha D^\alpha$. Therefore

$$Q \lesssim D^{c(\alpha+\beta)} K^c Q^{1-c} \leq C_N D^c K^c Q^{1-c}, \quad (7.33)$$

because $c(\alpha + \beta - 1) \leq 1$ and $D \leq 2^N$. Enlarging C_ϵ settles the base.

For the inductive step, if $Q \leq DK$, then

$$I(P, \mathcal{T}) \leq Q \leq D^c K^c Q^{1-c}. \quad (7.34)$$

Assume therefore $Q \geq DK$. The spacing conditions imply the overlap hypotheses in [Theorem III.7.24](#), up to absolute constants. Its first term is controlled by

$$D^{1/2} K^{1/2} Q^{1/2} \leq D^c K^c Q^{1-c}, \quad (7.35)$$

since $c \leq 1/2$ and $Q \geq DK$. The extra factor $S^{1/2} = D^{\epsilon/40}$ is absorbed by the target D^ϵ slack.

For the thickened term, P^S is a weighted $(S\delta, \beta, CS^\beta K_\beta)$ -family and $\mathcal{T}^S$ a weighted $(S\delta, \alpha, CS^\alpha K_\alpha)$ -family, with multiplicity retained. Indeed a coarse test cell pulls back to one fixed enlargement of a fine test cell, and the weights of merged objects are summed. The harmless fixed C is absorbed into the recurrence constant. Replace $S\delta$ by the adjacent larger dyadic scale if necessary. Since $S > 1$, the inductive hypothesis applies and gives

$$\begin{aligned} I(P^S, \mathcal{T}^S) &\leq C_\epsilon \delta^{-c-\epsilon} K^c Q^{1-c} S^{c(\alpha+\beta-1)-\epsilon} \\ &\leq C_\epsilon \delta^{-c-\epsilon} K^c Q^{1-c} S^{1-\epsilon}, \end{aligned} \quad (7.36)$$

using $c(\alpha + \beta - 1) \leq 1$. Multiplication by the recurrence factor $S^{-1+\epsilon/2}$ leaves

$$S^{-\epsilon/2} = D^{-\epsilon^2/40}. \quad (7.36a)$$

On every non-base scale we have $D \geq 2^N$, so [Equation \(7.32a\)](#) makes the coefficient of the inductive contribution at most $1/2$. The first term of the recurrence does not carry the induction constant; after [Equation \(7.35\)](#), its remaining factor is $S^{1/2} D^{-\epsilon} = D^{-39\epsilon/40}$, and it is absorbed by increasing C_ϵ once. The induction is over the full weighted class, so the coarser family is admissible and the contraction closes. $\square$

Theorem III.7.26 (normalized all-range Fu–Ren interface). *Let $0 \leq s, t \leq 2$ and $\epsilon > 0$. For all sufficiently small δ , suppose P is a δ -separated relative $(\delta, s, \delta^{-\epsilon})$ -set and $\mathcal{T}$ is a δ -separated relative $(\delta, t, \delta^{-\epsilon})$ -family of tubes. If $s + t > 1$, then*

$$I(P, \mathcal{T}) \leq |P||\mathcal{T}| \delta^{\kappa(s,t)(s+t-1)-5\epsilon}, \quad \kappa(s, t) = \min \left\{ \frac{1}{2}, \frac{1}{s+t-1} \right\}. \quad (7.37)$$

For $s + t \leq 1$, the trivial estimate $I(P, \mathcal{T}) \leq |P||\mathcal{T}|$ is used instead.

Proof. Apply [Theorem III.7.9](#) to both families with decomposition loss ϵ . We obtain partitions

$$P = \bigsqcup_{i=1}^M P_i, \quad \mathcal{T} = \bigsqcup_{j=1}^N \mathcal{T}_j,$$

where

$$M \leq |P| \delta^{s-2\epsilon}, \quad N \leq |\mathcal{T}| \delta^{t-2\epsilon}, \quad (7.38)$$

and all pieces are unit-constant Katz–Tao sets. For a pair $(P_i, \mathcal{T}_j)$, use [Theorems III.7.20](#) to [III.7.22](#) and [III.7.25](#) according to the parameter region. These alternatives cover $[0, 2]^2$ and imply the common estimate

$$I(P_i, \mathcal{T}_j) \leq \delta^{-s-t+\kappa(s,t)(s+t-1)-\epsilon}. \quad (7.39)$$

Indeed, when $s, t \leq 1$, the low-dimensional gain is $st/(s+t)$, and

$$\frac{st}{s+t} \geq \frac{s+t-1}{2}$$

because $s+t-s^2-t^2 \geq 0$. If, for example, $t < 1 < s < t+1$, [Theorem III.7.20](#) gives gain $st/(1+t)$, which is at least $(s+t-1)/2$ because

$$2st - (1+t)(s+t-1) = (1-t)(1+t-s) \geq 0.$$

The dual case is identical. In the extreme region $s \geq t+1$, [Theorem III.7.22](#) gives exponent $t+1$, no larger than $(s+t+1)/2$, which is exactly the desired exponent when $\kappa = 1/2$; again the dual case is identical. Finally, when $s, t \geq 1$, [Theorem III.7.25](#) gives the exponent encoded by κ .

Summing (7.39) over i, j and using (7.38) gives the loss accounting

$$\underbrace{\delta^{-\epsilon}}_{\text{one local cell estimate}} \underbrace{\delta^{-2\epsilon} \delta^{-2\epsilon}}_{\text{two Katz–Tao decompositions}} = \delta^{-5\epsilon}. \quad (7.39a)$$

The powers δ^{s+t} from MN cancel the δ^{-s-t} in the cell estimate, leaving exactly (7.37). This is the normalized form appearing as Orponen–Shmerkin–Wang [7, Theorem 3.7]; this deduction uses the refined incidence theorem proved locally above and no additional research black box. $\square$

III.7.6.1 A fibrewise incidence conversion and the high Furstenberg branch

The refined incidence theorem controls a globally regular point family. A Furstenberg configuration is naturally *fibrewise*: each tube carries a regular set, but their union can have heavy pockets. The next local argument removes those pockets. It is the finite replacement mechanism behind Fu–Ren’s fibrewise estimate, but we include it because it lets later chapters use the high-parameter Furstenberg branch without importing another research theorem.

Lemma III.7.27 (product replacement inside a dyadic square). *Fix $0 < \beta \leq 1$. Let Q be a dyadic square of side $w \geq \delta$, where w/δ is dyadic. There is a family $\mathcal{M}_{\delta,w}(Q)$ of δ -balls, supported in the concentric square $5Q$, with the following properties.*

(i) *For every ball $B(x, r)$ with $\delta \leq r \leq 5w$,*

$$\#(\mathcal{M}_{\delta,w}(Q) \cap B(x, r)) \lesssim (r/\delta)^{\beta+1}. \quad (7.37a)$$

In particular $|\mathcal{M}_{\delta,w}(Q)| \lesssim (w/\delta)^{\beta+1}$.

(ii) *Let T be a δ -tube meeting a δ -ball centred in Q . Extend its unit-scale axis by a fixed amount within the bounded ambient chart. Then*

$$\#\{p \in \mathcal{M}_{\delta,w}(Q) : p \cap CT \neq \emptyset\} \gtrsim (w/\delta)^\beta. \quad (7.37b)$$

Here CT is a fixed transverse enlargement of the extended tube. All axes are extended to the same fixed length, sufficient to cross every $5Q$ in the bounded ambient chart. A single fixed similarity then turns them into unit-scale tubes; their line-space regularity changes only by an absolute constant.

The family is a union of $O(1)$ $c\delta$ -separated colour classes, which is enough for every incidence application below.

Proof. Translate and dilate only for the construction, so $5Q = [0, 5w]^2$ and $Q = [2w, 3w]^2$. Put $H = 5w$ and $N = \lfloor (H/\delta)^\beta \rfloor$ and

$$C_0 = \{H(j/N)^{1/\beta} : 0 \leq j \leq N\}, \quad C = C_0 \cup (H - C_0),$$

with the evident interpretation $C_0 = \{0, H\}$ if $N = 1$. Convexity of $x \mapsto x^{1/\beta}$ gives $c\delta$ separation. If I is an interval of length $r \geq \delta$, concavity of $x \mapsto x^\beta$ gives

$$\#(C \cap I) \lesssim (r/\delta)^\beta, \quad (7.37c)$$

whereas for $\delta \leq r \leq H$,

$$\#(C \cap [0, r]) + \#(C \cap [H - r, H]) \gtrsim (r/\delta)^\beta. \quad (7.37d)$$

Let G be a $c\delta$ -grid of $[0, H]$ and take the δ -balls centred at

$$(C \times G) \cup (G \times C). \quad (7.37e)$$

Add one ball centred at the centre of Q . This does not change any upper estimate. Using the two product pieces in (7.37c) proves (7.37a). Their overlap has bounded multiplicity, so a fixed colouring gives the separation assertion.

For (7.37b), if $w \leq C_* \delta$ for a sufficiently large absolute C_* , the added central ball supplies the required bounded number of incidences after a fixed enlargement. Otherwise the axis is within $O(\delta)$ of Q and its extended line crosses $3Q$. Consequently its chord in $5Q$ has length $\geq w$. Every chord of a square has one coordinate projection of length comparable to the chord that reaches a side of that square. If it is the horizontal projection, $C \times G$ and (7.37d) give at least $c(w/\delta)^\beta$ centres within $O(\delta)$ of the extended line; for a vertical projection use $G \times C$. The fixed transverse enlargement of T absorbs grid rounding and the $O(\delta)$ displacement of its axis from Q . $\square$

Proof status. The fibrewise deduction below uses the normalized refined theorem, whose weighted induction is proved above. Its product-replacement lemma is also local.

Proposition III.7.28 (fibrewise Fu–Ren incidence estimate). *Let $0 < \alpha \leq 2$, $0 < \beta \leq 1$, $K_\alpha, K_\beta \geq 1$, and put*

$$c^{-1} = \max\{\alpha + \beta, 2\}. \quad (7.37f)$$

Let $\mathcal{T}$ be a Fu–Ren $(\delta, \alpha, K_\alpha)$ -family of tubes in a fixed bounded region. For each $T \in \mathcal{T}$, let P_T be a Fu–Ren (δ, β, K_β) -family of δ -balls, every member of which meets T , and set $P = \bigcup_T P_T$. Then, for every $\epsilon > 0$ and all sufficiently small δ ,

$$\sum_{T \in \mathcal{T}} |P_T| \lesssim_\epsilon \delta^{-c-\epsilon} (K_\alpha K_\beta)^c |P|^{1-c} |\mathcal{T}|^{1-c}. \quad (7.37g)$$

In particular, when $\alpha + \beta \leq 2$ we have $c = 1/2$ and obtain the square-root Katz–Tao incidence currency with the product $K_\alpha K_\beta$ under the square root.

Proof. Split the tube directions into finitely many fixed angular charts. It is enough to prove the estimate in one chart and sum, so all geometric constants below are absolute.

Call a dyadic square Q of side $w \in [\delta, 1]$ *heavy* if

$$|P \cap Q| > AK_\beta(w/\delta)^{\beta+1}, \quad (7.37h)$$

where A is a sufficiently large absolute constant. Let $\mathcal{Q}$ be the maximal heavy squares. Here and below $P \cap Q$ means the balls whose chosen centres lie in the half-open square Q ; the same convention applies to each $P_T \cap Q$. The squares in $\mathcal{Q}$ are pairwise disjoint. In each $Q \in \mathcal{Q}$ replace $P \cap Q$ by the model $\mathcal{M}_{\delta,w}(Q)$ of Theorem III.7.27, assigning weight K_β to every model ball. Use finitely many rotated copies if necessary to cover the fixed angular charts, and leave P unchanged with unit weight outside $\bigcup \mathcal{Q}$. Call the resulting weighted family P' .

The choice of A and (7.37a) give $K_\beta |\mathcal{M}_{\delta,w}(Q)| \leq |P \cap Q|$, hence $|P'| \leq |P|$ in total weight. Moreover P' is a weighted Fu–Ren $(\delta, \beta + 1, CK_\beta \log(2/\delta))$ -family. Here is the boundary accounting necessitated by the support $5Q$. Fix a dyadic square R of side $r \geq \delta$. If R is heavy, it lies in a maximal heavy square, so it contains no unreplaced original point; if it is not heavy, the unreplaced mass in R is at most $AK_\beta(r/\delta)^{\beta+1}$. For maximal heavy squares Q with side $w < r$ whose models meet R , Q lies in one of $O(1)$ dyadic r -squares adjoining R . Each such r -square containing one of these Q is nonheavy, since otherwise that Q would not be maximal. The disjointness of the maximal Q and the mass inequality

above therefore bound all small-model mass in R by $O(K_\beta(r/\delta)^{\beta+1})$. For $w \geq r$, at most $O(1)$ dyadic w -squares have $5Q$ meeting R . Each such model contributes at most $O(K_\beta(r/\delta)^{\beta+1})$ by (7.37a); there are at most $O(\log(2/\delta))$ possible dyadic values of w . Summing proves the dyadic-square bound. A ball meets $O(1)$ dyadic squares at an adjacent scale, yielding the weighted ball condition with the same logarithmic loss.

Fix $T \in \mathcal{T}$ and $Q \in \mathcal{Q}$. Put $m = |P_T \cap Q|$. All centres counted by m lie in a ball of radius $O(w)$, so the (δ, β, K_β) condition gives

$$m \lesssim K_\beta(w/\delta)^\beta. \quad (7.37i)$$

By (7.37b), the weighted replacement model contributes at least $cK_\beta(w/\delta)^\beta \gtrsim m$ incidences with a fixed enlargement of T . Outside the heavy squares every original fibre point remains. Consequently

$$I(P', C\mathcal{T}) \gtrsim \sum_T |P_T|. \quad (7.37j)$$

All extended tubes have the same fixed length. Apply one fixed similarity to the entire configuration to make that length one; the new scale is comparable to δ , and all point and line-space regularity constants change by fixed factors. Fixed transverse enlargement and finite colouring have the same property. Apply Theorem III.7.25 to P' with point exponent $\beta + 1$ and tube exponent α . Since

$$\max\{\alpha + (\beta + 1) - 1, 2\} = \max\{\alpha + \beta, 2\} = c^{-1},$$

we obtain

$$I(P', C\mathcal{T}) \lesssim_\epsilon \delta^{-c-\epsilon/2} (K_\alpha K_\beta \log(2/\delta))^c |P'|^{1-c} |\mathcal{T}|^{1-c}.$$

Start with the refined theorem at loss $\epsilon/2$ and absorb $\log(2/\delta)^c \lesssim_\epsilon \delta^{-\epsilon/2}$. Combine this with (7.37j) and $|P'| \lesssim |P|$. $\square$

Corollary III.7.29 (Katz–Tao fibre incidence in the smoothing range). *Let $0 < s \leq 1$, $0 < t \leq 2$, and $s + t \leq 2$. Suppose $\mathcal{T}$ is a Katz–Tao (δ, t, C_1) family of δ -tubes in one bounded chart and every $T \in \mathcal{T}$ carries a Katz–Tao (δ, s, C_2) family $F(T)$ of incident δ -squares. With $F = \bigcup_T F(T)$, for every $\epsilon > 0$,*

$$\sum_{T \in \mathcal{T}} |F(T)| \lesssim_{s,t,\epsilon} \delta^{-\epsilon} (\delta^{-1} C_1 C_2 |F||\mathcal{T}|)^{1/2}. \quad (7.37ka)$$

The bound is invariant under a fixed change of chart and fixed enlargement of the incident squares or tubes.

Proof. Replace each square by a comparable ball, using finitely many colour classes to restore essential distinctness. These changes multiply the Katz–Tao constants by fixed factors. Apply Proposition III.7.28 with $(\alpha, \beta) = (t, s)$. Since $s + t \leq 2$, its exponent is $c = 1/2$, and the resulting inequality is exactly (7.37ka), after absorbing the finite chart and colouring factors into the constant. $\square$

Corollary III.7.30 (Fu–Ren high-parameter Furstenberg branch). *Let $0 < s \leq 1$ and $1 \leq t \leq 2$. Every (s, t) -Furstenberg set $F \subset \mathbb{R}^2$ satisfies*

$$\dim_{\mathrm{H}} F \geq \min\{2s + t - 1, s + 1\}. \quad (7.37k)$$

In particular, if $s + t \geq 2$, then

$$\dim_{\mathrm{H}} F \geq s + 1. \quad (7.37l)$$

Proof. If $t = 1$, the already-local low-parameter bound (6.84) gives $\dim_{\mathrm{H}} F \geq 2s$, which is the claimed minimum. Hence assume $t > 1$, and fix $0 < a < s$ and $1 < b < t$.

Apply the arbitrary-cover selection Lemma III.6.30.2 to a sufficiently fine ball cover of a bounded piece of F . At a dyadic scale $\delta = 2^{-k}$ it produces a tube family $\mathcal{T}$ with

$$|\mathcal{T}| = \delta^{-b+o(1)}$$

and, for every $T \in \mathcal{T}$, a fibre P_T of incident δ -squares with

$$|P_T| = \delta^{-a+o(1)}.$$

All tube and fibre regularity constants are $\delta^{-o(1)}$, and every square in $P = \bigcup_T P_T$ meets a fixed enlargement of a ball from the scale- δ layer of the original cover. Replace squares by comparable balls and apply Theorem III.7.28, choosing its ϵ and then the cover sufficiently fine so that all $o(1)$ losses are smaller than a prescribed $\eta > 0$.

If $|P| = \delta^{-d+o(1)}$, comparison of the lower incidence mass $\sum_T |P_T| = \delta^{-a-b+o(1)}$ with (7.37g) gives

$$d \geq \begin{cases} 2a + b - 1 - O(\eta), & a + b \leq 2, \\ a + 1 - O(\eta), & a + b \geq 2. \end{cases} \quad (7.37m)$$

Indeed, when $a + b \leq 2$ one has $c = 1/2$, while when $a + b \geq 2$ one has $c = (a + b)^{-1}$; solving the resulting linear inequality for d gives exactly the two displayed expressions.

Let $q < \min\{2s + t - 1, s + 1\}$. Choose $a \uparrow s$ and $b \uparrow t$ so that the right-hand side of (7.37m), before the $O(\eta)$ loss, exceeds q , and then choose η smaller than the remaining gap. Hence the scale- δ layer of every sufficiently fine cover contains at least $\delta^{-q-\eta'}$ balls for some $\eta' > 0$. Its q -cost is bounded below by $\delta^{-\eta'}$, contradicting arbitrarily small total q -cost. The common-scale selection also says that the union P is drawn from at most CN_k dyadic squares, where N_k is the number of balls in that cover layer. Hence the same lower bound holds for N_k . Thus (7.37k) follows by the arbitrary-cover criterion. Finally, if $s + t \geq 2$, then $2s + t - 1 \geq s + 1$, yielding (7.37l). $\square$

III.7.7 A local high–low lemma for rich incidences

For the rich-incidence branch we freeze the loss budget once. Because only finitely many pigeonholing and high–low estimates occur below, choose an absolute constant $C_* \geq 1$ large enough that every aggregate starred estimate used in the two Fu–Gan–Ren inductions may be read as

$$A \lesssim_* B \implies A \leq C(\epsilon) \delta^{-C_* \epsilon^7} B. \quad (7.39b)$$

Polylogarithmic factors are absorbed into the same budget after decreasing the scale threshold. Put

$$\epsilon_* := \min \left\{ \frac{1}{4}, \frac{1}{40C_*} \right\}. \quad (7.39c)$$

It is enough to run the inductions for $0 < \epsilon \leq \epsilon_*$. For larger requested ϵ , the result with ϵ_* is stronger: its $\delta^{-\epsilon_*}$ loss is smaller, and in the rich-tube theorem the hypothesis with the larger ϵ imposes a larger richness threshold. This reduction prevents the phrase “ $O(\epsilon^7)$ is harmless” from concealing the sign of a closing exponent.

Lemma III.7.31 (planar Fourier high-low dichotomy). *Let $\mathcal{B}$ be essentially distinct unit balls in $[0, D]^2$ and $\mathcal{T}$ essentially distinct $1 \times D$ tubes. Suppose every ball meets between E and $2E$ tubes. Set*

$$S = D^{\epsilon/20}, \quad \rho = D^{\epsilon^8} S^{-1}.$$

For D sufficiently large, either

$$|\mathcal{B}| \lesssim_* S^2 E^{-2} |\mathcal{T}| D, \quad (7.40)$$

or there are pairwise disjoint S -balls whose fixed enlargements contain a $\gtrsim_$ -fraction of $\mathcal{B}$, and every selected S -ball meets $\gtrsim_* SE$ tubes.*

Proof. Choose nonnegative smooth bumps ψ_B supported in fixed enlargements of the unit balls and equal to one on them; choose analogous ψ_T for the tubes. Set

$$f = \sum_{B \in \mathcal{B}} \psi_B, \quad g = \sum_{T \in \mathcal{T}} \psi_T.$$

Bounded overlap gives $\|f\|_2^2 \lesssim |\mathcal{B}|$, and the incidence hypothesis gives

$$c_I E |\mathcal{B}| \leq \int f g \quad (7.41)$$

for a fixed $c_I > 0$; fixed enlargements of intersecting ball-tube pairs ensure this positive overlap. Let η be a smooth Fourier cutoff equal to one on $B(0, \rho)$ and supported in $B(0, 2\rho)$. Split (7.41) into high and low frequencies using Plancherel.

For the high-frequency piece, decompose tube directions into D^{-1} -arcs and write $g = \sum_{\theta} g_{\theta}$. The Fourier transform of a $1 \times D$ tube is rapidly concentrated in the dual $D^{-1} \times 1$ slab perpendicular to its direction. At frequency $|\xi| \geq \rho$, one frequency belongs to at most $O(\rho^{-2})$ of these slabs. Hence

$$\int (1 - \eta) |\hat{g}|^2 \lesssim \rho^{-2} \sum_{\theta} \|g_{\theta}\|_2^2 + O(D^{-100}). \quad (7.42)$$

Essential distinctness gives bounded overlap within each direction class, so

$$\sum_{\theta} \|g_{\theta}\|_2^2 \lesssim D |\mathcal{T}|. \quad (7.43)$$

The high-frequency estimate is available for every subfamily $\mathcal{A} \subset \mathcal{B}$: with $f_{\mathcal{A}} = \sum_{B \in \mathcal{A}} \psi_B$,

$$\left| \int f_{\mathcal{A}} (g - g * \check{\eta}) \right| \lesssim |\mathcal{A}|^{1/2} (\rho^{-2} D |\mathcal{T}|)^{1/2}. \quad (7.43a)$$

For one tube, $\psi_T * \check{\eta}$ decays rapidly outside an $O(\rho^{-1}) \times D$ enlargement and has pointwise size $O(\rho)$. Since $\rho^{-1} < S$, for a unit ball B ,

$$\left| \int \psi_B(g * \check{\eta}) \right| \lesssim \rho \# \{T : T \cap N_S(B) \neq \emptyset\} + O(D^{-100}). \quad (7.44)$$

The negligible remainder is uniform because essential distinctness gives $|\mathcal{T}| = O(D^2)$, and the bump functions are fixed. Let $\mathcal{L}$ be the *light* balls for which the tube count in (7.44) is less than cE/ρ , with $c > 0$ chosen so that the absolute low-frequency integral for each light ball is at most $c_I E/2$. Each ball in $\mathcal{L}$ meets at least E original tubes, so applying (7.43a) to $\mathcal{L}$ gives

$$|\mathcal{L}| \lesssim E^{-2} \rho^{-2} D |\mathcal{T}|.$$

As $\rho^{-1} \leq S$, either (7.40) holds (after enlarging its fixed constant) or $|\mathcal{L}| \leq |\mathcal{B}|/2$. In the latter case a fixed fraction of the balls are heavy and each meets at least $cE/\rho = cSED^{-\epsilon^8} \gtrsim SE$ tubes within its S -neighbourhood. Choose a maximal disjoint family of S -balls centred at heavy balls. Their fixed enlargements cover every heavy ball, hence cover a fixed fraction of $\mathcal{B}$; this is the claimed thick alternative. $\square$

Corollary III.7.32 (dual heavy-tube form). *Suppose $\mathcal{B}$ is a set of essentially distinct unit balls in $[0, D]^2$ and $\mathcal{T}$ a set of essentially distinct $1 \times D$ tubes, each containing between E and $2E$ balls. With $S = D^{\epsilon/20}$, either*

$$|\mathcal{T}| \lesssim_* S^2 E^{-2} |\mathcal{B}| D, \quad (7.45)$$

or a $\gtrsim_$ -fraction of $\mathcal{T}$ lies in fixed enlargements of a line-parameter-separated family of $O(S) \times D$ tubes, each containing $\gtrsim_* SE$ balls. Each enlarged thick tube contains $O(S^2)$ members of $\mathcal{T}$. In particular, if $\tilde{\mathcal{T}}$ denotes a maximal parameter-separated family of all $\gtrsim_* SE$ -rich $CS \times D$ tubes, for one fixed absolute C , then*

$$|\mathcal{T}| \lesssim_* S^2 (E^{-2} |\mathcal{B}| D + |\tilde{\mathcal{T}}|). \quad (7.46)$$

Here separation in a bounded-slope chart means separation by $\gtrsim S$ in the line-parameter metric $d_D((\theta, b), (\theta', b')) := D|\theta - \theta'| + |b - b'|$.

Proof. Work in finitely many bounded-slope line charts and apply Theorem III.7.6 after normalizing the long scale. The normalization is explicit: for primal lines $y = mx + b$ in $[0, D]^2$, use dual coordinates $(a', b') = (Dm, b)$. Uncertainty $O(D^{-1})$ in direction and $O(1)$ in offset for a primal $1 \times D$ tube becomes an $O(1)$ -ball in these dual coordinates. A primal unit ball centred at (x, y) becomes an $O(1) \times D$ tube about $b' = y - (x/D)a'$. Thus the normalized dual objects have exactly the scales required by Theorem III.7.31. The dual unit-ball family represents $\mathcal{T}$, and the dual long-tube family represents $\mathcal{B}$; both are essentially distinct by hypothesis. Apply Theorem III.7.31 to the dual families. In its thick alternative the selected dual S -balls are disjoint, and their fixed enlargements cover a $\gtrsim_*$ -fraction of the dual unit balls. Dualizing back gives the stated separated family of thick primal tubes, each meeting $\gtrsim_* SE$ original balls. Map each selected dual S -ball to a nearest member of the maximal rich family; enlarging the fixed width constant C preserves richness, and parameter separation makes this map $O(1)$ -to-one. Separation is in line-parameter space,

not spatial bounded overlap. A fixed $O(S) \times D$ tube contains only $O(S^2)$ essentially distinct $1 \times D$ tubes by packing their angle-offset parameters; this gives (7.46). Fixed chart and enlargement constants are absorbed in the starred notation. $\square$

Lemma III.7.33 (selected short-rectangle high-low reduction). *Let $D \geq D_0(\epsilon)$, $D\delta \leq 1$, and $S = D^{\epsilon/20} \leq D/100$. Let $\mathcal{U}$ be a family of $\delta \times 1$ rectangles in $[0, D]^2$ that is essentially distinct with the longitudinal-cell convention above. Let $\mathcal{T}$ be an essentially distinct family of $\delta \times D$ rectangles, each containing at least r members of $\mathcal{U}$ after one fixed enlargement. There exists a family $\mathcal{H}$ of $CS\delta \times D$ rectangles such that*

$$|\mathcal{T}| \lesssim_* S^2(r^{-2}|\mathcal{U}|D + |\mathcal{H}|), \quad (7.47)$$

every $H \in \mathcal{H}$ contains at least $\tilde{r} = c_\epsilon D^{-2\epsilon^8} Sr$ members of $\mathcal{U}$, and every H contains an original member $T_H \in \mathcal{T}$ after fixed enlargement. The family $\mathcal{H}$ is essentially distinct at scale $S\delta$. More precisely, its members have attachment-cell labels j , and a fixed thick tube can arise from only $O(1)$ labels among the selected members of $\mathcal{H}$.

Proof. Work in finitely many bounded-slope charts. Partition directions into arcs of length $\asymp \delta$, and, for each direction arc, partition transverse offsets into intervals of length $\asymp D\delta$. Let R_j be the corresponding $O(D\delta) \times O(D)$ attachment strip. A long tube $T \in \mathcal{T}$ belongs to $O(1)$ labels j , and all short rectangles it contains belong to the associated enlarged R_j . Assign each T to one of its $O(1)$ admissible labels. Define $\mathcal{U}_j$ to contain a short rectangle only when it is incident to an assigned long tube, its direction is within $O(\delta)$ of the label, and its support lies in a fixed enlargement of R_j . A fixed-colour partition makes every assigned tube contain at least cr of the rectangles in its cell. Each short rectangle belongs to $O(1)$ enlarged cells, because its direction has only $O(1)$ nearby direction labels and its transverse extent over length one is $O(\delta)$. Consequently

$$\sum_j |\mathcal{U}_j| \lesssim |\mathcal{U}|, \quad \sum_j |\mathcal{T}_j| = |\mathcal{T}|.$$

The finitely many degree layers are absorbed in $\lesssim_*$.

Stretch the transverse coordinate in R_j by δ^{-1} and leave the longitudinal coordinate unchanged. Then R_j becomes an $O(D)$ -square, its short rectangles become bounded-eccentricity unit parallelograms, and its long tubes become $O(1) \times O(D)$ tubes. A bounded colouring replaces the parallelograms by essentially distinct unit balls and the long rectangles by essentially distinct long tubes, preserving all incidences after fixed enlargement. To see bounded overlap of the unit objects, consider those meeting one unit ball in the normalized cell. Only $O(1)$ longitudinal unit cells contribute; the angle lies in an interval of length $O(\delta)$ before normalization; and for each of its $O(1)$ δ -separated angle classes, only $O(1)$ transverse offset classes can pass through that ball. This is exactly where essential distinctness is used.

Dyadically partition the assigned long tubes by their *actual* unit-ball incidence degree after this replacement. Every degree layer has a parameter $E_j \geq r$, and there are only $O(\log(1/\delta))$ layers because the families are finite scale-separated nets. Apply [Theorem III.7.32](#) to each layer. Its thin alternative contributes at most $\lesssim_* S^2 r^{-2} |\mathcal{U}_j| D$. In its thick alternative retain only the separated thick tubes H_j selected by that corollary, discarding any other rich thick tubes. Every retained H_j contains $\gtrsim_* SE_j \gtrsim_* Sr$ short rectangles in

$\mathcal{U}_j$ and contains, after fixed enlargement, at least one original D -long tube $T_j \in \mathcal{T}_j$. Each H_j covers $O(S^2)$ original long tubes in its cell. The constant in the starred richness is fixed, and the only power loss in [Theorem III.7.31](#) is $D^{-\epsilon^8}$; after the finite cell partition the lower threshold may be chosen as $c_\epsilon D^{-2\epsilon^8} Sr$.

It remains to count selected tubes across cells, not all globally rich tubes. Fix a thick tube H and suppose a selected H_j is within $O(S\delta)$ of it in the long-tube parameter metric. Its witness T_j lies in a fixed enlargement of H . Since T_j has length D , its direction differs from that of H by $O(S\delta/D)$. Its direction label therefore has only $O(1)$ choices, as the labels are δ -separated and $S/D \leq 1/100$. For each such direction, the offset label of the $D\delta$ -wide attachment strip has only $O(1)$ choices: the witness lies inside a $CS\delta$ -wide strip, while distinct offset labels differ by $\asymp D\delta$. Within one label and one dyadic degree layer the selected thick tubes are separated. There are only $O(\log(1/\delta))$ layers. Take a maximal $S\delta$ -separated subfamily of the union of all selected thick tubes, retaining each member's original cell label and witness. A fixed member of this net has at most $O(\log(1/\delta))$ nearby selected tubes: only $O(1)$ cell labels are possible by the witness argument, and each layer contributes $O(1)$ tubes per label. The logarithm is absorbed in $\lesssim_*$. Every retained net member keeps its own $\gtrsim_* Sr$ short rectangles in its own cell. Summing the cell estimates proves (7.47).

The witness restriction is indispensable. An arbitrary globally rich thick tube can collect short rectangles from $\Theta(S)$ different direction labels, so the same cross-cell assertion is false for the unrestricted family. $\square$

III.7.8 The Fu–Gan–Ren induction

Let $1 \leq W \leq X \leq \delta^{-1}$. In the rectangular-spacing currency, “a ball lies in a cell” means that its *centre* lies in that cell. The *translation-uniform* (W, X) -spacing condition means that every translate of an axis-parallel $W^{-1} \times X^{-1}$ rectangle contains at most one centre. It is stronger than one point per cell of a single fixed grid. Dually, a tube family satisfies (W, X) -spacing if every W^{-1} -tube contains at most X/W retained δ -tubes and their directions are X^{-1} -separated. We write $|\mathcal{B}_{\max}| = |\mathcal{T}_{\max}| = WX$ for the maximal model cardinality.

The quantitative bootstrap below includes the bounded-richness range and displays the thick-branch contraction explicitly.

Proposition III.7.34 (three-parameter rich-tube bootstrap). *Let $1 \leq W \leq \delta^{-1}$. Tile $[0, 1]^2$ by $W^{-1} \times \delta$ rectangles and assume every such rectangle contains the centres of at most $N \geq 1$ balls of an essentially distinct δ -ball family $\mathcal{B}$. Fix C_0 strictly larger than an absolute packing constant for the number of essentially distinct δ -balls meeting a $\delta \times 1$ tube. For every $1 \leq r \leq 2C_0\delta^{-1}$ and $\epsilon > 0$, there are $1 \leq s \leq \delta^{-1}$ and an integer M_s such that*

$$|\mathcal{T}_r(\mathcal{B})| \leq C_\epsilon \delta^{-\epsilon} \frac{|\mathcal{B}| M_s \delta^{-1}}{sr^2}, \quad (7.48)$$

$$r \leq C_\epsilon \delta^{-\epsilon} \frac{M_s \delta^{-1}}{s^2}, \quad (7.49)$$

and

$$M_s \leq C_\epsilon \delta^{-\epsilon} N s \max\{1, sW\delta\}. \quad (7.50)$$

Proof. We use simultaneous induction upward in richness and upward in scale: $(2r, \delta)$ and $(r, 2\delta)$ are earlier than (r, δ) . Although the ambient chapter convention starts with dyadic δ , this induction is run simultaneously for every scale in a dyadic band, ordered by $\lceil \log_2(1/\delta) \rceil$. The coarser scale $S\delta$ below exceeds 2δ outside the base range, so it lies in an earlier band; no assumption that S is an integer is needed.

The packing constant exists because essential distinctness separates ball centres by a fixed multiple of δ , while the $O(\delta)$ -neighbourhood of a unit tube has area $O(\delta)$. If δ is bounded below in terms of ϵ , enlarge C_ϵ and take $s = M_s = 1$. If $r \geq C_0\delta^{-1}$, the rich family is empty; take $s = M_s = 1$. The remaining two inequalities then follow from $r \leq 2C_0\delta^{-1}$ and $N \geq 1$. The cap on r also makes the richness-doubling call $2r$ legitimate whenever that branch is used. If

$$NW \geq \delta^{-1+\epsilon/2}, \quad (7.51)$$

take $s = \delta^{-1}$, $M_s = \lceil s^2 \rceil$. Counting ordered pairs of incident balls in a rich tube gives

$$r^2 |\mathcal{T}_r(\mathcal{B})| \lesssim |\mathcal{B}| \delta^{-2}, \quad (7.52)$$

which gives (7.48); (7.49) is trivial and (7.50) follows from (7.51).

Assume no base case applies. Let $\mathcal{T}$ be tubes containing between r and $2r$ balls. If $|\mathcal{T}_r(\mathcal{B})| \geq 10|\mathcal{T}|$, the induction hypothesis at richness $2r$ closes the estimate with extra r^{-2} slack. Hence we may assume

$$|\mathcal{T}_r(\mathcal{B})| \leq 10|\mathcal{T}|. \quad (7.53)$$

Set $D = \delta^{-1}$ and $S = \delta^{-\epsilon/20}$, and apply [Theorem III.7.32](#).

In the thin branch, take $s = M_s = 1$. Then [Equations \(7.45\) and \(7.53\)](#) give (7.48); the remaining two estimates are immediate from $r \leq \delta^{-1}$.

In the thick branch there is a family of $CS\delta$ -tubes, each $\geq_* S$ -rich, such that

$$|\mathcal{T}| \lesssim_* S^2 |\tilde{\mathcal{T}}|. \quad (7.54)$$

Choose a maximal $S\delta$ -separated family of centres from $\mathcal{B}$. Assign every original centre to a nearest selected centre, breaking ties once; denote the resulting clusters by C . The radius- $S\delta$ balls at selected centres are essentially distinct, and the clusters form a partition. A heavy tube meeting A original balls meets, after a fixed enlargement and a bounded cover by standard $S\delta$ -tubes, one standard coarse tube carrying at least cA assigned incidences. This bounded cover affects only absolute constants and retains a bounded-to-one map from heavy tubes to coarse tubes. Let $m(C)$ be the number of original balls assigned to cluster C . Dyadically pigeonhole $M \leq m(C) < 2M$. There are $O(\log(1/\delta))$ levels, since essential distinctness gives $|\mathcal{B}| = O(\delta^{-2})$. For each heavy tube one level contributes at least a reciprocal-logarithmic fraction of its assigned incidences; a second pigeonhole retains a reciprocal-logarithmic fraction of the heavy tubes at one common level M . Since $\log(1/\delta) \leq C(\epsilon)\delta^{-\epsilon^7}$, the starred budget (7.39b) now gives, with $a := (C_* + 1)\epsilon^7$,

$$M|\tilde{\mathcal{B}}_M| \leq |\mathcal{B}|, \quad |\mathcal{T}| \leq C(\epsilon)\delta^{-a}S^2|\tilde{\mathcal{T}}_M|.$$

Every retained heavy tube contains at least

$$\tilde{r} = c_\epsilon \delta^a S r / M \quad (7.55)$$

members of $\tilde{\mathcal{B}}_M$. Every $W^{-1} \times S\delta$ rectangle, enlarged by a fixed factor, meets $O(S)$ original $W^{-1} \times \delta$ cells: indeed $WS\delta \ll 1$ outside (7.51). Consequently it contains at most

$$\tilde{N} := \max\{1, \lceil CNS/M \rceil\} \asymp NS/M \quad (7.56)$$

coarse balls, for a fixed absolute C . The equivalence uses $M \lesssim NS$, obtained by applying the same cell count to one nonempty cluster. Since the base case (7.51) is absent, $W \leq (S\delta)^{-1}$ for small δ . If $\tilde{r} < 1$, replace it by $\hat{r} = 1$; otherwise put $\hat{r} = \tilde{r}$. Every retained heavy tube is $\hat{r}$ -rich, and $\hat{r} \geq \tilde{r}$. Because its coarse-ball family is essentially distinct and has a nonempty $\hat{r}$ -rich tube, $\hat{r} \leq C_0(S\delta)^{-1}$; thus this call lies in the quantified richness range. Apply the induction hypothesis at

$$\tilde{\delta} = S\delta, \quad \tilde{W} = W, \quad \tilde{N} \asymp M^{-1}NS, \quad \hat{r},$$

producing $\tilde{s}, \tilde{M}_s$. Define

$$s = S\tilde{s}, \quad M_s = M\tilde{M}_s. \quad (7.57)$$

The scale bound is valid: $1 \leq s \leq S(S\delta)^{-1} = \delta^{-1}$, and M_s is an integer. Using $\hat{r} \geq \tilde{r}$, the retained-cardinality estimate above, and the inductive (7.48), we obtain

$$\begin{aligned} |\mathcal{T}_r(\mathcal{B})| &\leq 10|\mathcal{T}| \leq C(\epsilon)\delta^{-a}S^2|\tilde{\mathcal{T}}_M| \\ &\leq C(\epsilon)\delta^{-a}S^2C_\epsilon(S\delta)^{-\epsilon} \frac{|\tilde{\mathcal{B}}_M|\tilde{M}_s(S\delta)^{-1}}{\tilde{s}\tilde{r}^2} \\ &\leq C(\epsilon)\delta^{-3a}S^{-\epsilon}C_\epsilon\delta^{-\epsilon} \frac{M|\tilde{\mathcal{B}}_M|M_s\delta^{-1}}{sr^2}. \end{aligned}$$

Here one power δ^{-a} comes from retained cardinality and two from $\tilde{r}^{-2}$; fixed covering constants are in $C(\epsilon)$. Since $M|\tilde{\mathcal{B}}_M| \leq |\mathcal{B}|$, the coefficient in front of the desired right-hand side in (7.48) is

$$C(\epsilon)\delta^{-3(C_*+1)\epsilon^7}S^{-\epsilon}. \quad (7.58)$$

Since $S^{-\epsilon} = \delta^{\epsilon^2/20}$, the coefficient in (7.58) is

$$C(\epsilon)\delta^{\epsilon^2/20-3(C_*+1)\epsilon^7}.$$

For $0 < \epsilon \leq \epsilon_*$, the choice (7.39c) gives

$$\frac{\epsilon^2}{20} - 3(C_* + 1)\epsilon^7 \geq \frac{\epsilon^2}{40} > 0, \quad (7.58a)$$

so the coefficient is at most $1/2$ once δ is sufficiently small. This displays the strict contraction rather than absorbing it into starred notation. For the inductive richness estimate, use $r = c_\epsilon^{-1}\delta^{-a}MS^{-1}\tilde{r} \leq c_\epsilon^{-1}\delta^{-a}MS^{-1}\hat{r}$ and (7.49) at the coarser scale. It gives

$$r \leq C(\epsilon)\delta^{-a}S^{-\epsilon}C_\epsilon\delta^{-\epsilon} \frac{M_s\delta^{-1}}{s^2} \leq C_\epsilon\delta^{-\epsilon} \frac{M_s\delta^{-1}}{s^2},$$

since $\delta^{-a}S^{-\epsilon} \rightarrow 0$. Fixed constants in the cardinality estimate below are also absorbed after the same scale threshold is chosen. Finally the inductive cardinality estimate gives

$$\begin{aligned} M_s &= M\tilde{M}_s \\ &\leq C_\epsilon (S\delta)^{-\epsilon} N S \tilde{s} \max\{1, \tilde{s} W S \delta\} \\ &\leq C_\epsilon \delta^{-\epsilon} N s \max\{1, s W \delta\} \end{aligned}$$

and proves (7.50), completing the simultaneous induction. $\square$

Lemma III.7.35 (the $X = \delta^{-1}$ endpoint). *Assume at most one δ -ball centre lies in each $W^{-1} \times \delta$ rectangle and put $|\mathcal{B}_{\max}| = W\delta^{-1}$. If*

$$r > \max\{\delta^{1-3\epsilon} |\mathcal{B}_{\max}|, 1\},$$

then

$$|\mathcal{T}_r(\mathcal{B})| \lesssim_\epsilon \delta^{-\epsilon} \frac{|\mathcal{B}| \delta^{-1}}{r^2}. \quad (7.59)$$

Proof. If $r \geq C_0 \delta^{-1}$, the rich family is empty by the packing bound above. Otherwise apply Theorem III.7.34 with $N = 1$ and $\epsilon/2$ in place of ϵ . If $sW\delta > 1$, then Equations (7.49) and (7.50) imply $r \lesssim_\epsilon \delta^{-\epsilon} W$, contradicting $r > \delta^{-3\epsilon} W$ for sufficiently small δ . Thus $sW\delta \leq 1$, and $M_s \lesssim_\epsilon \delta^{-\epsilon/2} s$. Inserting this in (7.48), with its own $\delta^{-\epsilon/2}$ factor, proves (7.59). $\square$

The distinction between fixed-grid and translation-uniform spacing is substantive. Take $\delta = 2^{-4n}$, $W = X = \delta^{-1/4}$, and four disjoint radius- δ balls centred at $z + (\pm 1.1\delta, \pm 1.1\delta)$, where z is an interior vertex of the fixed $W^{-1} \times X^{-1}$ grid. Each ball lies wholly in its own grid cell. Nevertheless a fixed angular sector of unit δ -tubes through z meets all four balls; selecting their slopes $K\delta$ apart yields $\gtrsim \delta^{-1}$ essentially distinct 3-rich tubes. For $\epsilon = 1/10$, the right-hand side of (7.61) under the fixed-grid hypothesis would be $O(\delta^{-3/5})$. Thus the literal fixed-grid version of [26, Theorem 3] cannot hold, and the next theorem must be read with its stronger hypothesis.

Theorem III.7.36 (translation-uniform Fu–Gan–Ren rich-tube theorem). *Let $1 \leq W \leq X \leq \delta^{-1}$, and let $\mathcal{B}$ be a family of essentially distinct δ -balls satisfying the translation-uniform (W, X) -spacing condition. If*

$$r > \max\{\delta^{1-2\epsilon} W X, 1\}, \quad (7.60)$$

then

$$|\mathcal{T}_r(\mathcal{B})| \lesssim_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} (r^{-1} + W^{-1}). \quad (7.61)$$

Proof. We again use a two-parameter induction in richness and scale.

There are three base regimes. If $r \geq C_0 \delta^{-1}$, the rich family is empty. If $X \geq \delta^{-1+\epsilon/2}$, apply Theorem III.7.35 with $\epsilon/2$ in place of ϵ : since $WX \geq W\delta^{-1+\epsilon/2}$,

$$|\mathcal{T}_r(\mathcal{B})| \lesssim_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} W^{-1}. \quad (7.62)$$

Finally suppose $r \lesssim \delta^{-\epsilon/4}$ or $W \lesssim \delta^{-\epsilon/4}$. Count triples (B_1, B_2, T) with both balls incident to the same rich tube. For fixed B_1 and distinct B_2 , translation-uniform spacing gives

$\text{dist}(B_1, B_2) \gtrsim X^{-1}$: otherwise their centres would lie in a common translated $W^{-1} \times X^{-1}$ rectangle. Thus no close-pair contribution is omitted. For fixed B_1 and dyadic distance $w \in [X^{-1}, 1]$, rectangular spacing gives

$$\#\{B_2 : \text{dist}(B_1, B_2) \asymp w\} \lesssim wX(1 + wW). \quad (7.63)$$

Two such balls lie in only $O(w^{-1})$ essentially distinct δ -tubes. Therefore the number of triples is

$$\lesssim (\log X)|\mathcal{B}|WX. \quad (7.64)$$

It is also $\gtrsim r^2|\mathcal{T}_r(\mathcal{B})|$. Since in this base regime $r^{-1} + W^{-1} \gtrsim \delta^{\epsilon/4}$, the logarithm is absorbed by $\delta^{-\epsilon}$, proving (7.61).

Assume now that no base case applies. Put

$$D = \delta^{-\epsilon^4}, \quad S = D^{\epsilon/20}. \quad (7.65)$$

Then $1 \leq D \leq W$. Cover $[0, 1]^2$ by finitely overlapping D^{-1} -squares. Let $\mathcal{T}$ be the tubes meeting between r and $2r$ balls. If the higher richness layer dominates, use the induction hypothesis at $2r$; hence we may assume

$$|\mathcal{T}_r(\mathcal{B})| \leq 10|\mathcal{T}|. \quad (7.66)$$

A tube may cross a square only in a tiny corner, so its literal intersection with Q is not necessarily a length- D^{-1} segment. For every pair (T, Q) with a ball of $\mathcal{B}$ incident to T and assigned to Q , centre a length- $2D^{-1}$ segment of the axis of T at the orthogonal projection of the centre of Q onto that axis. This segment lies inside a fixed enlargement of Q . Split it into two length- D^{-1} halves. Assign each ball incident to T to one of the $O(1)$ squares meeting it, and then to one of the two halves in that square which meets the ball after fixed thickening. Retain the more populated half and let $U(T, Q)$ be its δ -neighbourhood; let $m(T, Q)$ be its assigned ball count. Thus every $U(T, Q)$ has the required full length D^{-1} , and

$$\frac{1}{2}\#\{B \in \mathcal{B} : B \text{ meets } T\} \leq \sum_Q m(T, Q) \leq \#\{B \in \mathcal{B} : B \text{ meets } T\}.$$

In a fixed square Q and dyadic M -class $M \leq m(T, Q) < 2M$, choose a maximal short-segment net in the metric $d_{D^{-1}} = D^{-1}|\Delta\theta| + |\Delta b_Q|$ at separation $c\delta$, where b_Q is the offset at the centre of Q . Thus its angular resolution is $\asymp D\delta$, not δ . Let $\mathcal{U}_M$ be its members over all squares, with the longitudinal square retained as part of the label. Assign each (T, Q) in this class to a nearby net representative $U \in \mathcal{U}_M$. The two short segments are at Hausdorff distance $O(\delta)$ within Q , so one fixed enlargement of T contains the representative. Call these *designated* edges G_M . A representative itself comes from an original segment in the same M -class, so it contains $\asymp M$ original balls after a fixed enlargement. Each original pair produces one designated edge, and no tube produces more than one edge to the same longitudinal square. Consequently the exact graph-counting relation is

$$\sum_M M \deg_{G_M}(T) \asymp \#\{B \in \mathcal{B} : B \text{ meets } T\} \quad (T \in \mathcal{T}).$$

This is a statement about *designated* edges, not all geometric short-segment incidences; counting the latter here would be false. We may choose M , and then dyadically pigeonhole designated tube degrees, so that a subfamily $\mathcal{T}_E$ satisfies

$$|\mathcal{T}_E| \gtrsim_* |\mathcal{T}|, \quad r \asymp_* ME, \quad (7.67)$$

each retained tube has $\asymp E$ designated members of $\mathcal{U}_M$, hence at least this many geometric short-segment incidences after fixed enlargement.

Apply [Theorem III.7.33](#) at richness E , after the homothety by D . Its short width is $D\delta$, and $D(D\delta) = D^2\delta \leq 1$ for sufficiently small δ . Let $\mathcal{H}$ be the resulting *selected, attachment-labelled* family, returned to the original unit square. For $\tilde{r} = c_\epsilon D^{-2\epsilon^8} SE$,

$$|\mathcal{T}_E| \lesssim_* S^2 \left(E^{-2} |\mathcal{U}_M| D + |\mathcal{H}| \right) =: I + II. \quad (7.68)$$

Within one attachment cell j , a single original ball meets only $O(1)$ distinct short segments of $\mathcal{U}_M(j)$. Indeed, after the transverse normalization in the proof of [Theorem III.7.33](#), those segments form an essentially distinct family of bounded-eccentricity unit parallelograms, with bounded overlap at a point. Every $H \in \mathcal{H}$ is selected in one such cell and contains at least $\tilde{r}$ of its short segments. All the $\asymp M$ balls attached to each such segment lie in a fixed enlargement of H . Counting pairs (ball, segment) inside this *one* cell therefore gives at least $cM\tilde{r}$ distinct original balls in that enlargement. The loss in $ME \asymp_* r$ here is a fixed power of $\log(1/\delta)$, which is at most $D^{C_*\epsilon^7 - 2\epsilon^8}$ once δ is sufficiently small. Thus, after choosing C_* large enough at the outset, we obtain

$$\mathcal{H} \subset \mathcal{T}_{r_1}(\mathcal{B}), \quad r_1 = c_\epsilon D^{-C_*\epsilon^7} Sr, \quad (7.69)$$

where the tubes on the right have width $CS\delta$, and the inclusion is understood after fixed enlargement. The selected family is essentially distinct at that width by the last assertion of [Theorem III.7.33](#). Crucially, (7.69) does *not* hold with $\mathcal{H}$ replaced by the unrestricted family of all globally rich thick tubes. To see why, take $W = X \asymp \delta^{-1/2}$, $r \asymp \delta^{-1/4}$, and the D, S in (7.65). Put r balls, separated by $\asymp (SDr)^{-1} \gg W^{-1}$, in an interval of length $c/(SD)$ inside one D^{-1} -square. There are $\asymp S$ essentially distinct original δ -tubes through all these balls, with directions separated by a sufficiently large multiple of $D\delta$. Their $\delta \times D^{-1}$ segments are essentially distinct, each with $M \asymp r$ balls, and one $2S\delta$ -tube contains all the short segments. Yet that thick tube meets only the same r original balls, not $\asymp Sr$. This configuration is outside all three base regimes. It does not contradict (7.69): the thick tube in this example has no witness original long tube in its narrow direction class as required of a selected member of $\mathcal{H}$.

For the first term, fix one D^{-1} -square Q , let $\mathcal{B}_Q$ consist of the original balls meeting a fixed $O(\delta)$ -enlargement of Q , and rescale Q to the unit square. Write $\mathcal{U}_M(Q)$ for the short segments assigned to Q . Bounded overlap gives $\sum_Q |\mathcal{B}_Q| \leq |\mathcal{B}|$. Each member of $\mathcal{U}_M$ in Q becomes a tube meeting $\asymp M$ rescaled balls. Translation-uniform spacing is preserved by this homothety: the inverse image of every translated $(W/D)^{-1} \times (X/D)^{-1}$ rectangle is a translated $W^{-1} \times X^{-1}$ rectangle. The fixed enlargements used for incidence do not move ball centres. The new parameters are

$$\delta' = D\delta, \quad r' = M, \quad W' = W/D, \quad X' = X/D. \quad (7.70)$$

Here $1 \leq W' \leq X' \leq (\delta')^{-1}$: the first inequality follows from $D \leq W$, and the last from $X \leq \delta^{-1}$. A tube meets at most $O(D)$ of the D^{-1} -squares, so $E \lesssim D$. The absence of the small- r base case and Equations (7.60) and (7.67) give

$$M \gtrsim_* r/E \gtrsim_* r/D \gg 1, \quad \frac{M}{(\delta')^{1-2\epsilon} W' X'} \gtrsim_* D^{2\epsilon} \frac{r}{\delta^{1-2\epsilon} W X}.$$

Indeed $(\delta')^{1-2\epsilon} W' X' = \delta^{1-2\epsilon} W X D^{-1-2\epsilon}$. Since $D^{2\epsilon} \asymp \delta^{-2\epsilon^5}$, the strict power margin absorbs the starred loss in (7.39b); thus $M > \max\{(\delta')^{1-2\epsilon} W' X', 1\}$ for sufficiently small δ . The induction hypothesis applies in every Q , and gives

$$|\mathcal{U}_M(Q)| \lesssim_* C_\epsilon (D\delta)^{-\epsilon} |\mathcal{B}_Q| \frac{WX}{D^2} M^{-2} (M^{-1} + D/W).$$

Summing over Q and using $ME \asymp_* r$ and $E \lesssim D$, we obtain the explicit normalized estimate

$$\begin{aligned} I &= S^2 E^{-2} D |\mathcal{U}_M| \\ &\lesssim_* C_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X (ME)^{-2} S^2 D^{-\epsilon} ((MD)^{-1} + W^{-1}) \\ &\lesssim_* C_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} (r^{-1} + W^{-1}) S^2 D^{-\epsilon}. \end{aligned}$$

Since $S^2 D^{-\epsilon} = D^{-9\epsilon/10}$, the aggregate starred loss is at most $D^{2\epsilon/5}$ under (7.39b) and $0 < \epsilon \leq \epsilon_*$. Hence

$$I \lesssim_* D^{-\epsilon/2} C_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} (r^{-1} + W^{-1}). \quad (7.71)$$

For the second term, thicken every original ball to an $S\delta$ -ball. The absence of the X -endpoint base case gives $X < \delta^{-1+\epsilon/2} < (S\delta)^{-1}$ for small δ , since $S \asymp \delta^{-\epsilon^5/20}$. The original centres are $\gtrsim X^{-1}$ -separated by translation-uniform spacing, and $S\delta = o(X^{-1})$. Hence the thick balls are already essentially distinct, and their centres retain exactly the same translation-uniform spacing. Cover each selected $CS\delta$ -tube by $O_C(1)$ standard $S\delta$ -tubes, and retain one carrying at least a fixed fraction of its r_1 incidences. Because $\mathcal{H}$ is $S\delta$ -separated in line parameter space, this selection has $O_C(1)$ fibres. Thus the selected family is bounded, up to a fixed factor, by an essentially distinct family of standard coarse tubes with richness $\gtrsim r_1$. Apply the induction hypothesis at

$$\delta' = S\delta, \quad r' = r_1 \gtrsim_* S r, \quad W' = W, \quad X' = X.$$

Admissibility follows from (7.67)–(7.69):

$$\frac{r_1}{(S\delta)^{1-2\epsilon} W X} \gtrsim_* S^{2\epsilon} \frac{r}{\delta^{1-2\epsilon} W X} > 1,$$

because $S^{2\epsilon} \asymp \delta^{-\epsilon^6/10}$ absorbs the starred loss in (7.39b); also $r_1 > 1$ outside the small- r base case. The inductive estimate gives

$$\begin{aligned} II &\lesssim_* S^2 C_\epsilon (S\delta)^{-\epsilon} |\mathcal{B}| W X r_1^{-2} (r_1^{-1} + W^{-1}) \\ &\lesssim_* S^{-\epsilon} C_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} (r^{-1} + W^{-1}). \end{aligned}$$

Thus

$$II \lesssim_* S^{-\epsilon} C_\epsilon \delta^{-\epsilon} |\mathcal{B}| W X r^{-2} (r^{-1} + W^{-1}). \quad (7.72)$$

Finally,

$$D^{-\epsilon/2} = \delta^{\epsilon^5/2}, \quad S^{-\epsilon} = \delta^{\epsilon^6/20}.$$

The chosen loss budget (7.39b) gives the following closing calculation. Since $0 < \epsilon \leq \epsilon_*$,

$$\frac{\epsilon^5}{2} - C_* \epsilon^7 = \epsilon^5 \left(\frac{1}{2} - C_* \epsilon^2 \right) \geq \frac{\epsilon^5}{4}, \quad (7.72a)$$

and, more critically,

$$\frac{\epsilon^6}{20} - C_* \epsilon^7 = \epsilon^6 \left(\frac{1}{20} - C_* \epsilon \right) \geq \frac{\epsilon^6}{40}. \quad (7.72b)$$

The two normalized inductive coefficients would then be bounded by $C(\epsilon)\delta^{\epsilon^5/4}$ and $C(\epsilon)\delta^{\epsilon^6/40}$, respectively. Choose the induction constant first, then choose the scale threshold so that each coefficient is at most $1/4$. Their sum is strictly less than one. This closes the induction. The selected-family formulation is necessary: the unrestricted intermediate inclusion in the published argument [26] is invalid, as the example following (7.69) shows. $\square$

Proposition III.7.37 (((W, X) -spacing transfer under duality). *In a bounded line chart, a (W, X) -spaced family of δ -tubes dualises, after fixed thickening and a fixed affine coordinate change, to a δ -ball family at most one centre in every translate of a $cW^{-1} \times cX^{-1}$ rectangle, for a fixed $c > 0$. Equivalently it is translation-uniformly $(c^{-1}W, c^{-1}X)$ -spaced. Conversely, translation-uniform rectangular ball spacing dualises to the corresponding tube-spacing condition, up to fixed constants.*

Proof. Inside a primal W^{-1} -tube there are at most X/W retained axes, and their directions are X^{-1} -separated. In dual coordinates these become points lying in a strip of width $O(W^{-1})$, separated by $\gtrsim X^{-1}$ in the transverse parameter. This conclusion holds for any translated strip, because the primal spacing hypothesis quantifies over *every* W^{-1} -tube, rather than one fixed tube grid. Consequently any translated $cW^{-1} \times cX^{-1}$ rectangle contains at most one point once c is smaller than the separation constants: two such points would correspond to two primal tubes in a common W^{-1} -tube with directions closer than the allowed X^{-1} separation. The converse is the same argument reversed. When $c^{-1}X > \delta^{-1}$, one may instead use $\delta' = c\delta$, for which $c^{-1}X \leq (\delta')^{-1}$. The incidence-preserving finite-cover argument for this replacement is given in the proof of the rich-ball theorem below; merely shrinking the original dual objects would not preserve incidences. The finite chart cover and fixed-thickening constants are absorbed by Theorems III.7.5 and III.7.6. $\square$

Theorem III.7.38 (Fu–Gan–Ren rich-ball theorem). *Let $1 \leq W \leq X \leq \delta^{-1}$, and let $\mathcal{T}$ be a family of essentially distinct δ -tubes satisfying the (W, X) -spacing condition. If*

$$r > \max\{\delta^{1-2\epsilon}WX, 1\},$$

then the family $\mathcal{B}_r(\mathcal{T})$ of r -rich δ -balls satisfies

$$|\mathcal{B}_r(\mathcal{T})| \lesssim_{\epsilon} \delta^{-\epsilon} |\mathcal{T}| W X r^{-2} (r^{-1} + W^{-1}). \quad (7.73)$$

Proof. Choose a fixed finite cover of the projective direction circle by overlapping bounded-slope charts such that any pair of directions with angular distance below a fixed $c_0 > 0$ lies in one common chart. Each tube is assigned to only a bounded number of charts. By the spacing hypothesis, $|\mathcal{T}| \leq WX$. Fix constants $c_s > 0$ and $C_d \geq 1$ from the dual-spacing and incidence-stability lemmas. In a chart, the dual point centres are one-per-translate for rectangles of sides $c_s W^{-1} \times c_s X^{-1}$, while an original incidence becomes incidence with a dual $C_d \delta$ -ball. These are fixed constants, but the radius conversion must retain richness.

First suppose r exceeds a sufficiently large absolute constant R_0 . A ball which meets r tubes meets at least r/C tubes in one chart. In that chart, apply [Theorems III.7.6](#) and [III.7.37](#). Cover every dual $C_d \delta$ -ball by $K = K(c_s, C_d)$ balls of radius $c_s \delta$, with centres $p + v_1, \dots, p + v_K$, where the offsets v_k do not depend on p . For each k , simultaneous translation by v_k leaves all spacing differences unchanged. Every dual line meeting r/C original balls meets at least $r/(CK)$ radius- $c_s \delta$ balls in one of these K translated families. Apply [Theorem III.7.36](#) to that family with $(\delta', W', X') = (c_s \delta, c_s^{-1} W, c_s^{-1} X)$ and exponent $\epsilon/2$. Here $X' \leq (\delta')^{-1}$ follows from $X \leq \delta^{-1}$. Choose $R_0 > 2CK$. For sufficiently small δ , the hypothesis $r > \delta^{1-2\epsilon} WX$ implies $r/(CK) > (c_s \delta)^{1-\epsilon} (c_s^{-1} W)(c_s^{-1} X)$; the fixed factors in richness and spacing are absorbed by the final $\delta^{-\epsilon}$ loss. Sum the fixed chart family and K translated covers. The map from original output balls to the selected coarse dual tubes has bounded fibres by the same line-parameter packing argument given explicitly in the small- r case below.

If $1 < r \leq R_0$, every rich ball meets two distinct tubes. Pairs whose directions differ by at least c_0 meet in only $O(1)$ essentially distinct δ -balls; they contribute at most $O(|\mathcal{T}|^2) \leq |\mathcal{T}|WX$ balls. Every remaining pair lies in a common chart, so its intersection ball is 2-rich for that chart's tube subfamily. Here the threshold itself gives $X \leq WX < R_0 \delta^{-1+2\epsilon}$; hence $c_s^{-1} X \leq (C_d \delta)^{-1}$ for sufficiently small δ . Keep the dual balls at their original $C_d \delta$ radius, so both incidences survive together, and apply [Theorem III.7.36](#) at richness 2, exponent $\epsilon/2$, and parameters $(C_d \delta, c_s^{-1} W, c_s^{-1} X)$. Its threshold is below 2, because $(C_d \delta)^{1-\epsilon} c_s^{-2} WX \leq C(c_s, C_d) R_0 \delta^\epsilon \rightarrow 0$. The dual balls are essentially distinct at radius $C_d \delta$: translated occupancy separates their centres by $\geq c_s X^{-1}$, while $X \delta \leq R_0 \delta^{2\epsilon} \rightarrow 0$. Assign each original output ball to one coarse dual tube whose fixed enlargement contains its dual line. Conversely, a fixed coarse dual tube has only $O_{c_s, C_d}(1)$ original essentially distinct δ -ball outputs within that enlargement, by packing in the two-dimensional line-parameter metric. Thus this assignment has bounded fibres and the dual count controls the desired original count. Consequently each chart contributes $\leq_\epsilon \delta^{-\epsilon/2} |\mathcal{T}|WX$, which is bounded by the right-hand side of (7.73) because $r \leq R_0$. This proves the small-richness range and avoids losing the threshold when the finite chart cover splits a two-tube incidence. $\square$

III.7.9 Consequences and scope

The incidence theorems have different roles. The refined Fu–Ren theorem [Theorem III.7.25](#) is strongest when both fractal exponents are at least one, whereas [Theorem III.7.26](#) packages all parameter regimes into the relative regularity currency. The local fibrewise conversion (7.37g) turns that global incidence theorem into the continuum Furstenberg estimate (7.37k). In particular, the high branch $\dim_{\mathbb{H}} F \geq s+1$ for $s+t \geq 2$ is available as an internally proved result. The rich-tube and rich-ball theorems [Theorems III.7.36](#) and [III.7.38](#) keep more geometric information: they control high multiplicity under the translation-

uniform (W, X) -spacing model. The two packages therefore complement rather than duplicate one another. The printed fixed-grid rich-tube hypothesis is refuted by the grid-corner example above; it is not silently treated as the theorem proved here.

The shading and multiscale lemmas in [Section III.7.4](#) are deliberately independent of the headline estimates. They can be combined with any later incidence theorem that is stable under fixed thickening. This separation will be used in the Furstenberg-set chapters that follow.

Chapter III.8

Classical Furstenberg Sets

A Furstenberg set is small in the plane but large along a prescribed family of lines. There are several inequivalent ways to measure the size of that family, and the distinction is essential. The classical direction-fractal problem assigns a set of directions of dimension $\beta \leq 1$, whereas the modern affine problem measures a set of lines in the two-dimensional affine line space and therefore has a parameter $t \leq 2$.

This chapter develops the classical theory in both currencies. We prove the two elementary lower-bound mechanisms

$$\dim_{\mathrm{H}} E \geq 2s + \beta - 1, \quad \dim_{\mathrm{H}} E \geq s + \frac{\beta}{2},$$

combine them in the Molter–Rela direction-fractal form, and record the affine pair-counting baseline $\dim_{\mathrm{H}} E \geq s + t/2$. We then reconstruct the Wolff–Molter–Rela upper example. The construction is presented from the ground up: an irrational slope net, rational coalescence, a superlacunary rational-approximation Cantor fibre, and a nested G -set refinement. The modern sharp Ren–Wang theorem is compared with these bounds, with its three-branch proof ownership and remaining middle-branch dependency identified at the end of the chapter.

Prerequisites and source boundary. All ingredients needed for the proofs are developed in this chapter. In particular, the finite-scale Frostman selection needed for the directional maximal argument is proved below by a finite dyadic max-flow/min-cut construction; no prior-volume Frostman or capacitability theorem is required. The line-space metric conventions agree with Chapter III.7, but the chart and separation facts used here are proved again locally. The classical lower bounds agree with Molter–Rela [30], while the upper construction follows the Wolff mechanism [29] in the quantitative Molter–Rela form [31]. These references identify the classical antecedents; no theorem from another chapter is used here.

III.8.1 Line space and the two parameter currencies

Definition (affine line space and bounded charts). *Let $\mathcal{A}(2, 1)$ be the space of unoriented affine lines in $\mathbb{R}^2$. In a bounded nonvertical chart we write*

$$\ell_{a,b} = \{(x, y) : y = ax + b\}, \quad |a|, |b| \leq M,$$

and use

$$d_{\mathcal{A}}(\ell_{a,b}, \ell_{a',b'}) = |a - a'| + |b - b'|. \quad (8.1)$$

A finite collection of such charts, after rotations, covers every compact family of lines.

Lemma III.8.1 (bounded-chart invariance). *On the overlap of two bounded affine-line charts, the coordinate change and its inverse are bi-Lipschitz. Consequently Hausdorff dimension, Hausdorff content up to fixed constants, and δ -separation in line space are independent of the chosen bounded chart.*

Proof. Write a line in normal coordinates as

$$n \cdot x = c, \quad (n, c) \sim (-n, -c), \quad n \in \mathbb{S}^1.$$

On a chart where $|n_2| \geq c_0 > 0$, the slope-intercept coordinates are

$$a = -\frac{n_1}{n_2}, \quad b = \frac{c}{n_2},$$

and similarly after rotating the coordinate axes. On a compact chart all denominators remain uniformly separated from zero, so the coordinate maps and their inverses have uniformly bounded derivatives. The mean-value theorem gives the bi-Lipschitz conclusion. The metric consequences follow immediately. $\square$

Definition (classical s -Furstenberg set). *For $0 < s \leq 1$, a set $E \subset \mathbb{R}^2$ is a classical s -Furstenberg set if for every unoriented direction $e \in \mathbb{RP}^1$ there is a unit line segment ℓ_e in direction e such that*

$$\dim_{\mathrm{H}}(E \cap \ell_e) \geq s.$$

Definition (direction-fractal class). *For $0 < s \leq 1$ and $0 < \beta \leq 1$, write $E \in \mathcal{F}_{s,\beta}^{\mathrm{dir}}$ if there is a set of directions $L \subset \mathbb{RP}^1$ with $\dim_{\mathrm{H}} L \geq \beta$ and, for every $e \in L$, a unit segment ℓ_e in direction e satisfying*

$$\dim_{\mathrm{H}}(E \cap \ell_e) \geq s.$$

Here β measures directions only; it never denotes dimension in the full affine line space.

III.8.2 Uniform fibre reduction

For a bounded set A , $N_{\delta}(A)$ denotes the least number of radius- δ balls required to cover A . For $q > 0$,

$$\mathcal{H}_{\infty}^q(A) = \inf \left\{ \sum_i (\mathrm{diam} U_i)^q : A \subset \bigcup_i U_i \right\}$$

is the q -dimensional Hausdorff content. Fixed multiplicative changes in radii are harmless.

A δ -tube is a fixed-constant δ -neighbourhood of a unit line segment. A compact set G is a G -set in the working chart if for every slope $m \in [0, 1]$ there is an intercept b such that

$$\{(x, mx + b) : 0 \leq x \leq 1\} \subset G.$$

A finite union of rotated copies of a working-chart G -set supplies all unoriented directions.

Lemma III.8.2 (finite-scale Frostman selection). *Let $A \subset [-M, M]^d$ be bounded, let $q > 0$, and let $0 < \delta < 1$. There is a finite atomic Borel measure ν_δ , concentrated on finitely many points of A , such that*

$$\|\nu_\delta\| \geq c_{d,q,M} \mathcal{H}_\infty^q(A) \quad \text{and} \quad \nu_\delta(B(x, r)) \leq C_{d,q,M} r^q \quad (r \geq \delta). \quad (8.2)$$

Thus a positive q -content set supplies, at every working scale, exactly the Frostman growth needed by the arguments below.

Proof. Choose a dyadic cube Q_0 of fixed side length, depending only on M and d , which contains $[-M, M]^d$. Stop its dyadic tree at the first generation J whose cubes have side length at most $\delta/2$, and retain only cubes having a descendant at level J which meets A . For every retained cube Q , write $\ell(Q)$ for its side length and assign the vertex capacity

$$c(Q) = \ell(Q)^q.$$

Define a number $F(Q)$ from the leaves upward by

$$F(Q) = c(Q) \quad (Q \text{ a leaf}), \quad F(Q) = \min \left\{ c(Q), \sum_{Q' \in \text{ch}(Q)} F(Q') \right\}$$

for an internal cube. This recursion is exactly the finite min-cut dynamic program. Indeed, for the subtree rooted at an internal cube Q , a cut either contains Q , at cost $c(Q)$, or omits Q , in which case it must contain a cut in every retained child subtree, at minimum total cost $\sum_{Q'} F(Q')$. The two alternatives are attainable, so induction from the leaves proves that $F(Q_0)$ is the minimum of $\sum_{Q \in \mathcal{C}} \ell(Q)^q$ over all dyadic antichains $\mathcal{C}$ which meet every root-to-leaf path. Every such antichain covers A , so

$$F(Q_0) \geq d^{-q/2} \mathcal{H}_\infty^q(A). \quad (8.3)$$

Now send mass $F(Q_0)$ downward. If an internal cube Q receives mass $m(Q) \leq F(Q)$, then $F(Q) \leq \sum_{Q'} F(Q')$, so its mass can be distributed among the children with $0 \leq m(Q') \leq F(Q')$ and $\sum m(Q') = m(Q)$. At a leaf Q , place its received mass at one point of $A \cap Q$. The resulting atomic measure satisfies

$$\nu_\delta(Q) \leq F(Q) \leq \ell(Q)^q \quad (8.4)$$

for every retained cube, and its total mass is $F(Q_0)$, giving the first part of (8.2).

For $r \geq \delta$, choose a dyadic generation with side length ℓ satisfying $r \leq \ell < 2r$, unless r already exceeds the side length of Q_0 . In that case $\|\nu_\delta\| = F(Q_0) \leq c(Q_0) = \ell(Q_0)^q \leq r^q$, so the claim is immediate. A Euclidean ball of radius r meets only $O_d(1)$ dyadic cubes of side ℓ . Summing (8.4) over those cubes proves $\nu_\delta(B(x, r)) \lesssim_{d,q,M} r^q$. $\square$

We shall also use the elementary consequence that

$$\dim_{\mathcal{H}} A > q \implies \mathcal{H}_{\infty}^q(A) > 0. \quad (8.5)$$

Indeed, if $\mathcal{H}_{\infty}^q(A) = 0$, then for every $\eta > 0$ one may choose a cover of total q -cost below η^q ; every member of such a cover has diameter below η , forcing $\mathcal{H}^q(A) = 0$ and hence $\dim_{\mathcal{H}} A \leq q$.

Lemma III.8.3 (uniform fibre/content and scale reduction). *Let $0 < s_0 < s$, and suppose $\dim_{\mathcal{H}}(E \cap \ell_{\lambda}) \geq s$ for every parameter λ in a set Λ . After replacing Λ by a subset of any prescribed strictly smaller Hausdorff dimension, there is $c_0 > 0$ such that*

$$\mathcal{H}_{\infty}^{s_0}(E \cap \ell_{\lambda}) \geq c_0 \quad (\lambda \in \Lambda). \quad (8.6)$$

Let $\{B_j\}$ be any cover of E by balls of radii below 2^{-k_0} , put

$$E_k = \bigcup \{5B_j : 2^{-k-1} < r(B_j) \leq 2^{-k}\}, \quad \delta_k = 2^{-k},$$

and choose

$$a_k = \frac{c}{(k+1)^2} > 0, \quad \sum_{k \geq k_0} a_k < c_0/2.$$

Then

$$\Lambda_k = \{\lambda : \mathcal{H}_{\infty}^{s_0}(E_k \cap \ell_{\lambda}) \geq a_k\} \quad (8.7)$$

cover Λ .

Proof. For $m \geq 1$, let

$$\Lambda^{(m)} = \{\lambda : \mathcal{H}_{\infty}^{s_0}(E \cap \ell_{\lambda}) \geq m^{-1}\}.$$

Because $\dim_{\mathcal{H}}(E \cap \ell_{\lambda}) > s_0$, these sets cover Λ . Hausdorff dimension is countably stable, so after lowering the target parameter dimension strictly, one of the $\Lambda^{(m)}$ has the required dimension. Restrict to that set and take $c_0 = m^{-1}$.

For a fixed λ , the sets $E_k \cap \ell_{\lambda}$ cover $E \cap \ell_{\lambda}$. If $\lambda \notin \Lambda_k$ for every k , subadditivity of Hausdorff content would give

$$\mathcal{H}_{\infty}^{s_0}(E \cap \ell_{\lambda}) \leq \sum_{k \geq k_0} \mathcal{H}_{\infty}^{s_0}(E_k \cap \ell_{\lambda}) < \sum_{k \geq k_0} a_k < c_0,$$

contradicting (8.6). □

III.8.3 The directional maximal-function mechanism

Fix $0 < \beta_0 < 1$ and $0 < \delta < 1/10$. Let ν be any finite Borel measure on $\mathbb{RP}^1$ satisfying the scale-truncated growth estimate

$$\nu(B(e, r)) \leq C_{\nu} r^{\beta_0} \quad (r \geq \delta). \quad (8.8)$$

For each e , choose an arbitrary unit δ -tube T_e in a fixed bounded region and set

$$A_{\delta} f(e) = \frac{1}{|T_e|} \int_{T_e} f(x) dx.$$

Lemma III.8.4 (directional L^2 tube maximal estimate). *Under the scale-truncated growth hypothesis (8.8),*

$$\|A_\delta f\|_{L^2(\nu)}^2 \lesssim_{C_v, \beta_0} \delta^{\beta_0-1} \|f\|_2^2. \quad (8.9)$$

The same estimate, with only a fixed change of constant, holds for any fixed multiple of δ as the tube width.

Proof. The kernel of $A_\delta A_\delta^*$ is

$$K_\delta(e, e') = \frac{|T_e \cap T_{e'}|}{|T_e| |T_{e'}|}.$$

If $\theta = d(e, e')$, two unit δ -tubes overlap in area at most $C\delta^2/(\delta + \theta)$, while $|T_e| \asymp \delta$. Hence

$$K_\delta(e, e') \lesssim \frac{1}{\delta + d(e, e')}. \quad (8.10)$$

Using (8.8) and dyadic annuli,

$$\begin{aligned} \int K_\delta(e, e') d\nu(e') &\lesssim \delta^{-1} \nu(B(e, \delta)) + \sum_{j=0}^{O(\lceil \log \delta \rceil)} (2^j \delta)^{-1} \nu(B(e, 2^{j+1} \delta)) \\ &\lesssim_{C_v} \delta^{\beta_0-1} \left(1 + \sum_{j \geq 0} 2^{j(\beta_0-1)} \right) \lesssim_{C_v, \beta_0} \delta^{\beta_0-1}. \end{aligned}$$

The same bound holds after interchanging e, e' . Schur's test proves (8.9). Replacing δ by $C\delta$ only changes the implicit constant. $\square$

The endpoint $\beta_0 = 1$ carries a logarithmic loss. Dimension arguments avoid it by choosing $\beta_0 < \beta$ and then letting $\beta_0 \uparrow \beta$.

III.8.4 The $2s + \beta - 1$ lower bound

Theorem III.8.5 (first direction-fractal lower bound). *If $E \in \mathcal{F}_{s, \beta}^{\text{dir}}$, then*

$$\dim_H E \geq 2s + \beta - 1. \quad (8.11)$$

Proof. There is nothing to prove when the right-hand side is nonpositive. Fix

$$s_0 < s, \quad \beta_0 < \beta_1 < \beta, \quad d < 2s_0 + \beta_0 - 1. \quad (8.12)$$

Use Lemma III.8.3 to pass to a direction witness L of dimension strictly larger than β_1 , with the uniform fibre content (8.6). By (8.5), $\mathcal{H}_\infty^{\beta_0}(L) > 0$.

Let $\{B_j\}$ be an arbitrary sufficiently fine cover of E , and form E_k, a_k, L_k as in Lemma III.8.3. Put

$$H_k = \mathcal{H}_\infty^{\beta_0}(L_k).$$

Since the L_k cover L , subadditivity gives

$$\sum_k H_k \geq \mathcal{H}_\infty^{\beta_0}(L) > 0. \quad (8.13)$$

For every k with $H_k > 0$, identify $\mathbb{RP}^1$ with a Euclidean circle by doubling the angle coordinate; this is bi-Lipschitz up to an absolute constant for the projective angular metric. Apply Lemma III.8.2 at scale δ_k . We obtain a finite atomic measure ν_k , supported on L_k , such that

$$\|\nu_k\| \gtrsim H_k, \quad \nu_k(B(e, r)) \lesssim r^{\beta_0} \quad (r \geq \delta_k), \quad (8.14)$$

with constants independent of k . Fix $e \in L_k$. A maximal $C_1 \delta_k$ -separated subset of $E_k \cap \ell_e$, with C_1 a sufficiently large absolute constant, has cardinality

$$M_e \gtrsim a_k \delta_k^{-s_0}. \quad (8.15)$$

Indeed, otherwise the intervals of diameter $O(\delta_k)$ around the maximal set would cover the fibre piece with s_0 -content smaller than a_k .

For every chosen point $x_i \in E_k \cap \ell_e$, choose an original scale- k cover ball $B_{j(i)}$ with $x_i \in 5B_{j(i)}$. If C_1 is large, the centres of these original balls are $c\delta_k$ -separated; hence the balls $B_{j(i)}$ are pairwise disjoint after changing an absolute constant. Since each centre lies $O(\delta_k)$ from ℓ_e , all these balls are contained in a fixed-multiple δ_k -tube about ℓ_e . They are also contained in E_k . Therefore

$$|E_k \cap T_e^C| \gtrsim M_e \delta_k^2 \gtrsim a_k \delta_k^{2-s_0},$$

and hence

$$A_{C\delta_k} \mathbf{1}_{E_k}(e) \gtrsim a_k \delta_k^{1-s_0}. \quad (8.16)$$

Apply Lemma III.8.4 with $\nu = \nu_k$ and use $|E_k| \lesssim \delta_k^2 \#J_k$, where J_k is the family of original cover balls at scale δ_k . By (8.14),

$$H_k a_k^2 \delta_k^{2-2s_0} \lesssim \|\nu_k\| a_k^2 \delta_k^{2-2s_0} \lesssim \delta_k^{\beta_0-1} |E_k| \lesssim \delta_k^{\beta_0+1} \#J_k,$$

so

$$\#J_k \gtrsim H_k a_k^2 \delta_k^{-(2s_0+\beta_0-1)}. \quad (8.17)$$

Multiplying by δ_k^d gives

$$\#J_k \delta_k^d \gtrsim H_k a_k^2 2^{k(2s_0+\beta_0-1-d)}.$$

The exponential factor dominates the polynomial decay of a_k^2 . Thus for all large k , $\#J_k \delta_k^d \gtrsim H_k$. Summing and using (8.13), every sufficiently fine cover of E has positive d -dimensional cost. Let $d \uparrow 2s_0 + \beta_0 - 1$, then $s_0 \uparrow s$ and $\beta_0 \uparrow \beta$. $\square$

III.8.5 Separated fibres and the $s + \beta/2$ bound

Lemma III.8.6 (balanced splitting of a fibre). *Let $0 < s_0 \leq 1$. If A is contained in a unit line segment and $\mathcal{H}_\infty^{s_0}(A) \geq a$, then there are subsets A^- and A^+ of A such that*

$$\mathcal{H}_\infty^{s_0}(A^\pm) \geq c_{s_0} a, \quad \text{dist}(A^-, A^+) \geq c_{s_0} a^{1/s_0}. \quad (8.18)$$

Proof. Identify the segment with $[0, 1]$, let $r = (a/100)^{1/s_0}$, and partition it into consecutive intervals I_j of length at most r . Put $P_m = I_1 \cup \dots \cup I_m$, and choose the least m with

$$\mathcal{H}_\infty^{s_0}(A \cap P_m) \geq a/3.$$

By minimality and subadditivity,

$$a/3 \leq \mathcal{H}_\infty^{s_0}(A \cap P_m) \leq a/3 + r^{s_0} < 0.35a.$$

Set $A^- = A \cap P_m$, omit the next interval I_{m+1} , and let A^+ be the part of A strictly to its right. Since I_{m+1} has s_0 -content at most $a/100$, subadditivity and $\mathcal{H}_\infty^{s_0}(A) \geq a$ give $\mathcal{H}_\infty^{s_0}(A^+) > 0.64a$. The omitted interval separates A^- and A^+ by at least r . Adjusting constants proves the claim. $\square$

Lemma III.8.7 (separated-pair multiplicity). *Let B, B' be planar balls of radius $C\delta$ satisfying $\text{dist}(B, B') \geq \sigma > 10C\delta$. Then at most C'/σ members of a δ -separated family of directions can have a unit line segment meeting both balls. More generally, in a bounded affine-line chart, at most C'/σ members of a δ -separated line family can meet both balls.*

Proof. If a line meets both balls, its direction differs by at most $C_1\delta/\sigma$ from the direction of the vector joining their centres. Thus all admissible directions lie in an arc of length $O(\delta/\sigma)$, which contains $O(\sigma^{-1})$ elements of a δ -separated direction family.

In a bounded affine-line chart, the direction coordinate is restricted to the same interval, and meeting one of the balls restricts the intercept to an interval of length $O(\delta)$. Hence the admissible line parameters lie in a rectangle of side lengths $O(\delta/\sigma)$ and $O(\delta)$. A δ -separated family contains $O(\sigma^{-1})$ points in such a rectangle. $\square$

Theorem III.8.8 (second direction-fractal lower bound). *If $E \in \mathcal{F}_{s,\beta}^{\text{dir}}$, then*

$$\dim_{\text{H}} E \geq s + \frac{\beta}{2}. \quad (8.19)$$

Proof. Fix

$$s_0 < s, \quad \beta_0 < \beta, \quad d < s_0 + \beta_0/2.$$

Apply Lemma III.8.3 to an arbitrary sufficiently fine cover, retaining a direction witness L with $\dim_{\text{H}} L > \beta_0$, and let L_k be the resulting scale sets. Set

$$H_k = \mathcal{H}_\infty^{\beta_0}(L_k).$$

By (8.5) and subadditivity of content,

$$\sum_k H_k \geq \mathcal{H}_\infty^{\beta_0}(L) > 0. \quad (8.20)$$

Choose a maximal δ_k -separated subset of L_k . Since the corresponding $O(\delta_k)$ -balls cover L_k , its cardinality satisfies

$$N_k \gtrsim H_k \delta_k^{-\beta_0}. \quad (8.21)$$

For each selected direction, apply Lemma III.8.6 to $E_k \cap \ell_e$. This gives two packets of s_0 -content $\gtrsim a_k$, separated by

$$\sigma_k \gtrsim a_k^{1/s_0}. \quad (8.22)$$

Each packet requires at least $ca_k\delta_k^{-s_0}$ of the enlarged balls $5B_j$, $j \in J_k$, because their intersections with the witness line have diameter $O(\delta_k)$. Hence the number of triples (e, B, B') , where the witness line meets an enlarged ball from each packet, is at least

$$cN_k a_k^2 \delta_k^{-2s_0}. \quad (8.23)$$

For all sufficiently large k , $\delta_k \ll \sigma_k$, because a_k decays only polynomially. Therefore Lemma III.8.7, applied to the fixed enlargements $5B, 5B'$, shows that each ordered pair contributes to at most C/σ_k selected directions. Thus

$$(\#J_k)^2 \gtrsim \sigma_k N_k a_k^2 \delta_k^{-2s_0},$$

and by (8.21) and (8.22),

$$\#J_k \gtrsim H_k^{1/2} a_k^{1+1/(2s_0)} \delta_k^{-(s_0+\beta_0/2)}. \quad (8.24)$$

Multiplication by δ_k^d introduces the exponentially growing factor $2^{k(s_0+\beta_0/2-d)}$, which dominates the polynomial loss in a_k . After multiplying all Hausdorff contents by a harmless fixed normalising constant we may assume $H_k \leq 1$, and hence $H_k^{1/2} \geq H_k$. For all sufficiently large k ,

$$\#J_k \delta_k^d \gtrsim H_k.$$

Summing and using (8.20) gives a positive lower bound for the d -dimensional cost of every sufficiently fine cover. Finally let the auxiliary parameters tend to their endpoints. $\square$

Corollary III.8.9 (combined classical direction-fractal bound). *If $E \in \mathcal{F}_{s,\beta}^{\text{dir}}$, then*

$$\dim_H E \geq \max \left\{ s + \frac{\beta}{2}, 2s + \beta - 1 \right\}. \quad (8.25)$$

In particular, every classical s -Furstenberg set satisfies

$$\dim_H E \geq \max \left\{ s + \frac{1}{2}, 2s \right\}. \quad (8.26)$$

Proof. Take the maximum of Theorems III.8.5 and III.8.8; for the all-directions class put $\beta = 1$. $\square$

III.8.6 The affine pair-counting baseline

For $0 < s \leq 1$ and $0 < t \leq 2$, a set $E \subset \mathbb{R}^2$ is an *affine (s, t) -Furstenberg set* if there is a family $\mathcal{L} \subset \mathcal{A}(2, 1)$ with $\dim_H \mathcal{L} \geq t$ such that

$$\dim_H(E \cap \ell) \geq s \quad (\ell \in \mathcal{L}).$$

A direction-fractal witness induces an affine line family, but the converse is false: affine line-space dimension can come from intercept variation while the direction set remains small.

The maximal-function proof of [Theorem III.8.5](#) uses the local finite-scale Frostman measure on the one-dimensional direction space. It must not be reinterpreted with the two-dimensional affine parameter t . The separated-pair mechanism, by contrast, depends only on separation in the parameter metric and therefore has a clean affine version.

Theorem III.8.10 (affine pair-counting baseline). *If E is an affine (s, t) -Furstenberg set, $0 < s \leq 1$ and $0 < t \leq 2$, then*

$$\dim_{\text{H}} E \geq s + \frac{t}{2}. \quad (8.27)$$

Proof. Fix $s_0 < s$, $t_0 < t$, and $d < s_0 + t_0/2$. By [Lemma III.8.1](#) and a finite/countable bounded chart reduction, restrict the witnessing line family to one bounded chart while retaining Hausdorff dimension larger than t_0 . Apply [Lemma III.8.3](#) in line space while retaining a witnessing family $\mathcal{L}$ with $\dim_{\text{H}} \mathcal{L} > t_0$, and set

$$H_k = \mathcal{H}_{\infty}^{t_0}(\mathcal{L}_k).$$

By (8.5) and subadditivity, $\sum_k H_k \geq \mathcal{H}_{\infty}^{t_0}(\mathcal{L}) > 0$. A maximal δ_k -separated subfamily of $\mathcal{L}_k$ has cardinality $N_k \gtrsim H_k \delta_k^{-t_0}$. Split each fibre with [Lemma III.8.6](#). The line-space form of [Lemma III.8.7](#) then gives exactly the estimate

$$\#J_k \gtrsim H_k^{1/2} a_k^{1+1/(2s_0)} \delta_k^{-(s_0+t_0/2)}.$$

As in the proof of [Theorem III.8.8](#), the exponential gain after multiplication by δ_k^d dominates the polynomial loss in a_k , and summing in k gives positive d -dimensional covering cost. Let $d \uparrow s_0 + t_0/2$, then $s_0 \uparrow s$ and $t_0 \uparrow t$. $\square$

III.8.7 The one-step Wolff construction

We now turn from lower bounds to a construction. Fix the irrational number $\sqrt{2}$. For $n \geq 2$, define

$$\ell_{j,k}(x) = (1-x)\frac{j}{n} + x\sqrt{2}\frac{k}{n}, \quad -3n \leq j \leq 3n, \quad 0 \leq k < n, \quad 0 \leq x \leq 1, \quad (8.28)$$

and put

$$\delta_n = C \frac{\log n}{n^2}. \quad (8.29)$$

For this recursive construction, let G_n be the union of the closed *vertical graph strips*

$$\{(x, y) : 0 \leq x \leq 1, |y - \ell_{j,k}(x)| \leq \delta_n\} \quad (8.29a)$$

over the indices in (8.28). Each strip is contained in a fixed-constant Euclidean δ_n -tube, but its exact horizontal domain is $[0, 1]$; this is needed for nested refinement.

Lemma III.8.11 (continued-fraction slope net). *For a sufficiently large absolute C in (8.29), G_n is a compact G -set in the working chart for all sufficiently large n .*

Proof. The slopes of the graph segments are

$$m_{j,k} = \frac{\sqrt{2}k - j}{n}.$$

Let p_m/q_m be the continued-fraction convergent of $\sqrt{2}$ such that $q_m \leq n < q_{m+1}$. Since $\sqrt{2} = [1; 2]$, the partial quotients are bounded, so $q_m \geq cn$. For $0 \leq k < q_m$,

$$\left| k\sqrt{2} - \frac{kp_m}{q_m} \right| < \frac{k}{q_m q_{m+1}} \leq \frac{1}{q_{m+1}}. \quad (8.30)$$

Multiplication by p_m permutes the residue classes modulo q_m . Thus the fractional parts of kp_m/q_m , $0 \leq k < q_m$, are exactly the q_m^{-1} -grid, and (8.30) shows that $\{k\sqrt{2} : 0 \leq k < q_m\}$ is a C_1/n -net modulo one.

Given $m_0 \in [0, 1]$, choose $k < q_m$ so that the fractional part of $\sqrt{2}k$ differs by at most C_1/n from that of nm_0 , then choose $j \in \mathbb{Z}$ with

$$|\sqrt{2}k - j - nm_0| \leq C_1/n.$$

For large n , necessarily $|j| \leq 3n$, and therefore

$$|m_0 - m_{j,k}| \lesssim n^{-2}. \quad (8.31)$$

For every indexed line, assign the closed slope interval

$$J_{j,k}^{(n)} = [m_{j,k} - \delta_n/4, m_{j,k} + \delta_n/4] \cap [0, 1]. \quad (8.31a)$$

Since $n^{-2} = o(\delta_n)$, these intervals cover $[0, 1]$ for large n . For every $m \in J_{j,k}^{(n)}$, the graph $y = j/n + mx$ stays within $\delta_n/4$ of $\ell_{j,k}$ on $[0, 1]$, and hence lies in its closed δ_n -tube. This proves the G -set assertion with an interior margin that will be preserved in the recursion. $\square$

Fix $0 < s \leq 1$, and set

$$\Gamma_n = \left\{ \frac{p}{q} \in [1/4, 3/4] : (p, q) = 1, 1 \leq q \leq n^s \right\}. \quad (8.32)$$

The elementary estimate $\#\Gamma_n \leq n^{2s}$ will be used repeatedly. Define

$$\phi(t) = \frac{1-t}{t\sqrt{2}}. \quad (8.33)$$

Choose a compact interval $D \subset (0, 1)$ on which ϕ is bi-Lipschitz and $\phi(D) = [1/4, 3/4]$, and put

$$Q_n = \phi^{-1}(\Gamma_n).$$

If $t \in Q_n$ and $\phi(t) = p/q$, then

$$\ell_{j,k}(t) = \frac{t\sqrt{2}}{nq}(pj + qk). \quad (8.34)$$

Since $|j| \leq 3n$, $0 \leq k < n$, and $p \asymp q$, the integer $pj + qk$ assumes only $O(nq) \leq O(n^{1+s})$ values. Thus, at each $t \in Q_n$, all n^2 -scale line values coalesce into at most $O(n^{1+s})$ vertical positions.

Lemma III.8.12 (one-step Wolff coalescence and covering). *Let*

$$\Lambda_n = G_n \cap \{(x, y) : \text{dist}(x, Q_n) \leq n^{-2}\}.$$

Then Λ_n can be covered by at most

$$C_s n^{1+3s} \tag{8.35}$$

balls of radius $C_s \delta_n$.

Proof. Fix $t \in Q_n$. Equation (8.34) gives $O(n^{1+s})$ distinct line values at $x = t$. Centre a rectangle of horizontal width $O(n^{-2})$ and vertical width $O(\delta_n)$ at each such value. Every graph in (8.28) has uniformly bounded slope, so moving x by at most n^{-2} changes its height by $O(n^{-2})$, which is absorbed by δ_n . Thus these rectangles cover the portion of G_n above the horizontal n^{-2} -neighbourhood of t , and each is covered by $O(1)$ balls of radius $C\delta_n$.

Finally $\#Q_n = \#\Gamma_n \lesssim n^{2s}$. Multiplying the two counts gives $O(n^{1+s})O(n^{2s}) = O(n^{1+3s})$. $\square$

III.8.8 A rational-approximation Cantor fibre

Choose integers $n_1 < n_2 < \dots$ so rapidly increasing that, at every stage,

$$\log n_{j-1} = o(\log n_j), \quad \log M_{j-1} = o(\log n_j), \tag{8.36}$$

and, in addition, every parent interval I from stage $j-1$ satisfies

$$|I| n_j^s / \log n_j \longrightarrow \infty. \tag{8.37}$$

Here M_{j-1} denotes any finite combinatorial or Lipschitz constant generated before stage j . Such a choice is possible recursively.

Lemma III.8.13 (primitive rationals and the superlacunary Cantor fibre). *There is a compact set $T_s \subset [1/4, 3/4]$ with*

$$\dim_{\text{H}} T_s = s \tag{8.38}$$

such that, for every $x \in T_s$ and every j ,

$$\text{dist}(x, \Gamma_{n_j}) \leq n_j^{-2}. \tag{8.39}$$

Proof. Write $Q_j = n_j^s$. At stage j , inside every retained parent interval I , retain closed intervals of radius cn_j^{-2} centred at reduced fractions $p/q \in I$ whose distance from both endpoints of I is at least cn_j^{-2} , with

$$Q_j/2 \leq q \leq Q_j. \tag{8.40}$$

Take $c > 0$ small and fixed.

Primitive rational count. For an interval $I \subset [1/4, 3/4]$, Möbius inversion gives

$$\#\{(p, q) : (p, q) = 1, Q/2 \leq q \leq Q, p/q \in I\} = c_0 |I| Q^2 + O(Q \log Q), \quad (8.41)$$

with $c_0 > 0$. Indeed, insert $\mathbf{1}_{(p,q)=1} = \sum_{d|p,q} \mu(d)$, write $p = dp'$, $q = dq'$, and use

$$\#\{p' : p'/q' \in I\} = |I|q' + O(1).$$

The endpoint trim removes length $O(n_j^{-2}) = o(|I|)$ from each parent; the same count on the trimmed interval has the same main term. The main coefficient is a positive constant multiple of $\sum_{d \geq 1} \mu(d) d^{-2} = 1/\zeta(2)$; the accumulated endpoint error is $O(Q \log Q)$. By (8.37), for every stage- $(j-1)$ parent I ,

$$c|I|n_j^{2s} \leq B_j(I) \leq C|I|n_j^{2s}, \quad (8.42)$$

where $B_j(I)$ is the number of retained children.

Distinct reduced fractions with denominators at most Q_j are separated by at least $Q_j^{-2} = n_j^{-2s}$. Since $s \leq 1$, the child intervals of radius cn_j^{-2} can be chosen pairwise disjoint, including when $s = 1$ by taking c sufficiently small. The nested intersection is therefore a nonempty compact set T_s , and (8.39) is automatic.

Mass and endpoint scales. Give every child of a fixed parent equal conditional mass, starting from total mass one. Let $I_j(x)$ be the stage- j interval containing x , and let $\ell_j \asymp n_j^{-2}$ be the common child length. From (8.42) and superlacunarity,

$$-\log \mu(I_j(x)) = 2s \log n_j + o(\log n_j), \quad -\log \ell_j = 2 \log n_j + O(1), \quad (8.43)$$

uniformly in x . Hence for every $\varepsilon > 0$, after discarding finitely many levels,

$$\mu(I_j(x)) \leq \ell_j^{s-\varepsilon/2}. \quad (8.44)$$

Intermediate radii. We verify the estimate at all scales rather than only at cylinder lengths. The reduced-fraction separation implies that any interval J of length r contains at most

$$1 + CrQ_j^2 \quad (8.45)$$

centres from (8.40). Fix a small r , and choose j so that $\ell_j \leq r < \ell_{j-1}$. Let I be a stage- $(j-1)$ parent meeting $B(x, r)$. There are only $O(1)$ such parents after a fixed enlargement.

If

$$\ell_j \leq r \leq Q_j^{-2} = n_j^{-2s},$$

then (8.45) shows that $B(x, r)$ meets only $O(1)$ stage- j children. By (8.44),

$$\mu(B(x, r)) \leq \ell_j^{s-\varepsilon/2} \leq r^{s-\varepsilon}. \quad (8.46)$$

If instead $Q_j^{-2} < r < \ell_{j-1}$, then (8.45) and (8.42) give

$$\mu(B(x, r) \cap I) \leq \mu(I) \frac{rQ_j^2}{|I|Q_j^2} = \mu(I) \frac{r}{|I|}. \quad (8.47)$$

For large j , (8.43) at level $j - 1$ gives $\mu(I) \leq |I|^{s-\varepsilon/2}$. Since $r \leq |I|$ and $1 - s + \varepsilon > 0$,

$$\mu(I) \frac{r}{|I|} \leq r^{s-\varepsilon} \left(\frac{r}{|I|} \right)^{1-s+\varepsilon} |I|^{\varepsilon/2} \leq r^{s-\varepsilon}. \quad (8.48)$$

Absorbing the finitely many initial levels into the constant yields

$$\mu(B(x, r)) \leq C_\varepsilon r^{s-\varepsilon}$$

for every sufficiently small r . The mass-distribution principle gives $\dim_{\text{H}} T_s \geq s$.

For the reverse inequality, the stage- j set has

$$n_j^{2s+o(1)}$$

intervals of length $n_j^{-2+o(1)}$, again by (8.42) and (8.36). For every $q > s$, their total q -cost tends to zero. Hence $\dim_{\text{H}} T_s \leq s$, proving (8.38). $\square$

Set

$$T'_s = \phi^{-1}(T_s) \subset D. \quad (8.49)$$

Since ϕ is bi-Lipschitz on D , $\dim_{\text{H}} T'_s = s$, and (8.39) implies

$$\text{dist}(x, Q_{n_j}) \lesssim n_j^{-2} \quad (x \in T'_s). \quad (8.50)$$

III.8.9 Recursive G-set refinement and the upper example

The direction condition must survive refinement inside each individual strip. We record the extra invariant explicitly. A strip with data (b, m, w) is

$$S(b, m, w) = \{(x, y) : 0 \leq x \leq 1, |y - (b + mx)| \leq w\}.$$

Its assigned slope interval J is a closed subset of $[0, 1]$ contained in $[m - w/4, m + w/4]$. For every $u \in J$, the full graph $y = b + ux$ lies in the strip with vertical margin at least $3w/4$.

Lemma III.8.14 (slope-interval refinement invariant). *Suppose F_j is a finite union of strips $S_i^j = S(b_i^j, m_i^j, w_i^j)$ with assigned intervals J_i^j as above, and $\bigcup_i J_i^j \supset [0, 1]$. For all sufficiently large n there is a finite union $F_{j+1} \subset F_j$ of affine copies of the individual tubes of G_n , each with an assigned slope interval satisfying the same invariant. The new intervals cover $[0, 1]$, and each parent uses at most a fixed number of affine maps applied to subfamilies of G_n , independent of j, n , and the parent width.*

Proof. All G_n lie in $[0, 1] \times [-B, B]$ for one absolute $B \geq 4$: the endpoint heights of $\ell_{j,k}$ lie in $[-3, 3]$ and $[0, \sqrt{2}]$, while $\delta_n < 1$ for large n . Fix a parent S_i^j , abbreviate its data by (b, m, w, J) , and put $\rho = w/(4B)$. Since $J \subset [m - w/4, m + w/4]$, its length is at most $w/2 = 2B\rho$. Choose at most $2B + 2$ base slopes $u_k \in J$ such that the intervals $u_k + \rho[0, 1]$

cover J . For example, start at the left endpoint and advance by ρ , using the right endpoint as the final base when necessary. Define

$$A_{i,k}^j(x, y) = (x, b + u_k x + \rho y). \quad (8.51)$$

For $(x, y) \in G_n$, the distance from its image to the parent centre line is bounded by

$$|u_k - m|x + \rho|y| \leq w/4 + w/4 = w/2. \quad (8.51a)$$

Thus $A_{i,k}^j(G_n)$ lies inside S_i^j , with margin $w/2$.

The image of the source vertical strip about $\ell_{a,c}$ is a strip of width $w' = \rho\delta_n$, centre slope $m' = u_k + \rho m_{a,c}$, and centre intercept $b' = b + \rho a/n$. Assign it the interval

$$J' = (u_k + \rho J_{a,c}^{(n)}) \cap [0, 1]. \quad (8.51b)$$

Discard a child tube when $J' = \emptyset$; denote by $G_n^{i,k}$ the union of the source tubes retained for this (i, k) . Thus $G_n^{i,k} \subset G_n$. Equation (8.31a) implies $J' \subset [m' - w'/4, m' + w'/4]$. These child intervals cover $(u_k + \rho[0, 1]) \cap [0, 1]$ as (a, c) ranges over the source tubes. The base windows cover J , and the parent intervals cover $[0, 1]$; hence all child intervals cover $[0, 1]$. The bound $2B + 2$ proves the asserted uniform map count. $\square$

Start with the single strip $F_0 = S(0, 1/2, 4)$ assigned $J_0 = [0, 1]$. Its data satisfy the lemma's invariant. Choose n_{j+1} sufficiently large for both the lemma and the superlacunarity conditions (8.36)–(8.37), then apply the lemma to every parent. With its affine maps,

$$F_{j+1} = \bigcup_{i,k} A_{i,k}^j(G_{n_{j+1}}^{i,k}) \subset F_j. \quad (8.52)$$

Each F_j is compact, and each assigned interval supports an entire graph segment inside its strip. Since those intervals cover $[0, 1]$, every F_j is a G -set.

Put

$$F = \bigcap_{j \geq 0} F_j, \quad E_s^{(0)} = F \cap \{(x, y) : x \in T_s'\}. \quad (8.53)$$

For a fixed slope $m \in [0, 1]$, let $I_j(m)$ be the set of intercepts b for which the full graph segment $\{(x, mx + b) : 0 \leq x \leq 1\}$ is contained in F_j . Each $I_j(m)$ is nonempty and compact, and the nesting $F_{j+1} \subset F_j$ gives $I_{j+1}(m) \subset I_j(m)$. Thus $\bigcap_j I_j(m) \neq \emptyset$, proving that F itself contains a full graph segment of every slope in $[0, 1]$.

Theorem III.8.15 (Wolff–Molter–Rela construction). *For every $0 < s \leq 1$, there exists a classical s -Furstenberg set $E_s \subset \mathbb{R}^2$ such that*

$$\dim_{\text{H}} E_s \leq \frac{1 + 3s}{2}. \quad (8.54)$$

Proof. First consider the working-chart set $E_s^{(0)}$ from (8.53). For every slope $m \in [0, 1]$, the preceding nested-intercept argument gives a graph segment $\ell_m \subset F$. Its intersection with $E_s^{(0)}$ contains

$$\{(x, \ell_m(x)) : x \in T_s'\},$$

which is a bi-Lipschitz image of T'_s . Hence

$$\dim_{\mathbf{H}}(E_s^{(0)} \cap \ell_m) \geq s. \quad (8.55)$$

We next estimate $\dim_{\mathbf{H}} E_s^{(0)}$. At stage j , (8.50) says that every point of $E_s^{(0)}$ has horizontal coordinate within $O(n_j^{-2})$ of Q_{n_j} . Inside each of the finitely many stage- $(j-1)$ parent strips, apply the affine image of Lemma III.8.12 to each of the at most $2B+2$ subfamilies supplied by Lemma III.8.14. The maps (8.51) have finite Lipschitz constants determined before n_j is chosen; these constants and the fixed copy factor are included in M_{j-1} in the superlacunarity condition (8.36). Consequently the relevant part of $E_s^{(0)}$ has a stage- j cover by

$$M_{j-1} C_s n_j^{1+3s} = n_j^{1+3s+o(1)} \quad (8.56)$$

balls of radius

$$r_j = n_j^{-2+o(1)}, \quad (8.57)$$

where the logarithm in δ_{n_j} and all previously generated finite constants are absorbed into the $o(1)$. If $q > (1+3s)/2$, then

$$n_j^{1+3s+o(1)} r_j^q = n_j^{1+3s-2q+o(1)} \longrightarrow 0.$$

Thus

$$\dim_{\mathbf{H}} E_s^{(0)} \leq \frac{1+3s}{2}. \quad (8.58)$$

Finally choose finitely many rotations $R_1, \dots, R_N$ such that the rotated slope chart $[0, 1]$ covers all of $\mathbb{RP}^1$, and define

$$E_s = \bigcup_{v=1}^N R_v E_s^{(0)}. \quad (8.59)$$

For every unoriented direction, one rotated component supplies a full graph segment whose intersection has dimension at least s by (8.55). That graph has length at most $\sqrt{2}$, so two unit segments cover it; the dimension of a finite union is the maximum of the component dimensions. One of those unit segments therefore has intersection dimension at least s . Hence E_s is a classical s -Furstenberg set. A finite union has Hausdorff dimension equal to the maximum of the component dimensions, so (8.58) yields (8.54). $\square$

Molter–Rela obtain a logarithmically stronger gauge statement at the critical exponent [31]. The dimension-only conclusion (8.54) is the exact upper-bound statement needed in this volume.

III.8.10 Examples and sharpness bookkeeping

Example III.8.16 (radial fan). Let $C_s \subset (1, 2)$ and $L_\beta \subset \mathbb{RP}^1$ be compact sets of dimensions s and β . On finitely many angular charts the map $(r, e) \mapsto re$ is bi-Lipschitz, so

$$E = \{re : r \in C_s, e \in L_\beta\}$$

has Hausdorff dimension $s + \beta$ and belongs to $\mathcal{F}_{s,\beta}^{\text{dir}}$.

Example III.8.17 (parallel stack). *Let $C_t, C_s \subset [0, 1]$ be standard product-compatible Cantor sets with dimensions $t \leq 1$ and s . Then*

$$E = C_t \times C_s$$

is affine (s, t) , witnessed by the vertical lines $\{x\} \times \mathbb{R}$, $x \in C_t$, and $\dim_{\text{H}} E = s + t$.

The sharp affine theorem and its present proof boundary

The classical all-directions lower estimate (8.26) and the Wolff upper construction (8.54) historically left a gap for $0 < s < 1$. This is no longer an open dimension problem. Ren and Wang proved that every affine (s, t) -Furstenberg set satisfies

$$\dim_{\text{H}} E \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}. \quad (8.60)$$

See [32]. At $t = 1$ and $0 < s \leq 1$, the minimum is $(3s + 1)/2$, exactly the exponent in [Theorem III.8.15](#). The three parameter ranges have distinct proof owners in this volume: for $t \leq s$, the locally proved low-parameter theorem (6.84) gives $s + t$; for $t \geq 2 - s$, the locally proved Fu–Ren theorem (7.371) gives $s + 1$; and for $s < t < 2 - s$, [Theorem III.11.7.14](#) gives $(3s + t)/2$ from the regular-set estimate (9.46). That estimate uses the locally proved robust-projection input X-III.9.ABC49. Thus the middle-range argument and the resulting sharp minimum have local proof chains; the parameter regimes remain distinct.

III.8.11 Distinguishing the parameter regimes

The notation $\mathcal{F}_{s,\beta}^{\text{dir}}$ concerns a β -dimensional set of *directions*, so $\beta \leq 1$. The parameter t concerns a set in the full affine line space, so $t \leq 2$. In particular, the formula $2s + \beta - 1$ from [Theorem III.8.5](#) must never be silently reinterpreted with an affine line-space parameter $t > 1$.

The word “Furstenberg” here refers to incidence sets of lines. Furstenberg stationary measures for random matrix or group actions belong to a different part of the series.

The results needed later are the combined classical direction-fractal lower bound [Corollary III.8.9](#), the affine elementary baseline [Theorem III.8.10](#), and the upper construction [Theorem III.8.15](#). The sharper modern estimate (8.60) is included with its dependency audit. Its middle-range proof uses the locally proved Chapter 9 robust-projection theorem, including its high-multiplicity bootstrap.

Chapter III.9

Orponen–Shmerkin Advances

This chapter has two logically different halves. The first returns to the sharp incidence theorem proved in Chapter III.6 and derives the continuum Furstenberg and exceptional projection theorems of Orponen–Shmerkin. Here the research-scale discrete core is already owned by this volume, so the remaining work is a careful Hausdorff-content and common-scale passage.

The second half studies the later *ABC* sum-product, equal Hausdorff/packing projection, and regular-line Furstenberg theorems of Orponen–Shmerkin, published in *Journal of the American Mathematical Society* 39 (2026), 857–913 [18]. The discretised *ABC* theorem is proved below from the locally established thin-tube and finite combinatorial machinery; the robust projection theorem is then proved by the high-multiplicity bootstrap. The regular-set Furstenberg theorem follows from that projection theorem and the two-scale refinement of Chapter 6. The passages from these finite-scale statements to arbitrary sets of equal Hausdorff and packing dimension are proved below, using the disjoint regularisation of Chapter 6.

Prerequisites and source boundary. The sharp dual incidence theorem used below is the Chapter 6 estimate (6.77); the point–line duality machinery is proved in the same incidence section of Chapter III.6. We also use the cover-sensitive uniform fibre reduction from Lemma III.8.3. The Duke continuum deductions are proved below from these ingredients. In the later branch, the published Orponen–Shmerkin discretised *ABC* theorem is proved locally, using the Chapter 6 sharp incidence estimate and the Volume II asymmetric BSG lemma. The robust projection theorem is also proved locally by the high-multiplicity bootstrap below; the published result [18] supplies attribution, not an additional proof premise. The regular-set Furstenberg estimate is proved below from the robust projection corollary and the locally proved two-scale refinement; its continuum passage is also local. Equal Hausdorff and packing dimension is never replaced by Ahlfors regularity. The low-parameter Furstenberg estimate (6.84) is reused only in the branch $t \leq s$. The Chapter 6 multiscale combination Proposition III.6.29.12 is proved locally with its normal, good, and bad scale losses quantified; the deductions here use that proved supplier.

III.9.1 Exceptional sets and endpoint conventions

Definition (exceptional orthogonal-projection set). For $K \subset \mathbb{R}^2$ and $u \geq 0$, put

$$\text{Exc}_u(K) = \{\theta \in \mathbb{S}^1 : \dim_{\text{H}} \pi_{\theta}(K) < u\}. \quad (9.1)$$

The strict threshold in (9.1) is the convention used in the sharp 2026 theorem. When a theorem is stated with the closed threshold $\dim_{\mathbb{H}} \pi_{\theta}(K) \leq s$, we write that set out explicitly and pass to a slightly larger strict threshold when needed. The two endpoint conventions will not be conflated.

Definition (equal Hausdorff and packing dimension). A set K has equal Hausdorff and packing dimension d if

$$\dim_{\mathbb{H}} K = \dim_{\mathbb{P}} K = d. \quad (9.2)$$

This hypothesis does not assert Ahlfors regularity, uniform covering estimates, or the existence of an Ahlfors-regular measure.

Definition (all-scales finite regular set). Let $\delta = 2^{-m}$, $0 < t \leq 2$, and $C \geq 1$. A nonempty family $P \subset \mathcal{D}_{\delta}^2([0, 1]^2)$ is a (δ, t, C) -regular set if it is a (δ, t, C) -set in the sense of (6.41) and, for all dyadic $\delta \leq r \leq R \leq 1$ and $Q \in \mathcal{D}_R^2$,

$$|P \cap Q|_r \leq C(R/r)^t. \quad (9.2a)$$

Here $|P \cap Q|_r$ is the number of dyadic r -cubes meeting at least one δ -cube of $P \cap Q$. This all-scales notion is the one in the Orponen–Shmerkin ABC interfaces below. It is not the square-root-scale (δ, t, C, K) -regularity of (6.41a); that earlier definition does not imply (9.2a). The unit-square normalization is part of the finite-scale definition; a fixed bounded region may be covered by finitely many unit squares before this definition is applied.

III.9.2 A finite-scale extraction lemma

The next elementary device is useful twice: first in passing a cover-sensitive Furstenberg statement to a discrete configuration, and then in selecting a common set of bad directions in the projection argument.

Lemma III.9.1 (finite-scale Frostman extraction from content). Let X be a bounded subset of a Euclidean space or a bounded affine-line chart, let $0 < q \leq \dim X$, and suppose $\mathcal{H}_{\infty}^q(X) \geq h > 0$. For every $\eta > 0$, all sufficiently small dyadic δ admit a δ -separated subset $P \subset X$ such that

$$\delta^{-q+\eta} \leq |P| \leq \delta^{-q-\eta} \quad (9.7)$$

and

$$|P \cap B(x, r)| \leq \delta^{-\eta} r^q |P|, \quad \delta \leq r \leq 1. \quad (9.8)$$

The same conclusion holds inside a Borel set carrying a probability measure μ with $\mu(B(x, r)) \leq Cr^q$, after restricting to any subset of μ -mass at least h ; the threshold for δ may then depend on C, h, η .

Proof. We give the content version, since it also explains why no measurability of X is needed. Intersect a sufficiently large fixed dyadic cube with the dyadic tree down to scale δ . Put capacity $c(Q) = A \ell(Q)^q$ on every occupied dyadic cube Q , where A is a fixed dimensional constant. Connect each occupied cube to its occupied children and regard

the scale- δ cubes as sinks. By the max-flow/min-cut theorem, the maximal flow from the root to the leaves is the minimum capacity of a dyadic cutset. Every cutset, after a bounded enlargement of its cubes, covers X ; hence its capacity is at least $c_q \mathcal{H}_\infty^q(X) \geq c_q h$. We therefore obtain nonnegative leaf weights w_Q with

$$\sum_Q w_Q \geq c_q h, \quad \sum_{Q \subset R} w_Q \leq Al(R)^q \quad (9.9)$$

for every dyadic ancestor R .

Here is a simultaneous thinning argument, including the small-expectation cubes. Write D for the ambient dimension and $m = \sum_Q w_Q$, so $m \geq c_q h$. Colour the scale- δ grid by the parity of its D integer coordinates. One colour carries weight at least $m/2^D$. Retain that colour, renormalise its weights to a probability π , and choose a point $x_Q \in X \cap Q$ from every retained leaf. Distinct chosen points are δ -separated. The capacity bound gives, for each retained leaf and each dyadic ancestor R of side $r \geq \delta$,

$$\pi(Q) \leq C_{D,q} h^{-1} \delta^q, \quad \sum_{Q \subset R} \pi(Q) \leq C_{D,q} h^{-1} r^q. \quad (9.9a)$$

It is enough to treat $0 < \eta < q$, since a single point works when $\eta \geq q$. Put $N = \lceil \delta^{-q+\eta/2} \rceil$. Independently retain each representative x_Q with probability $p_Q = N\pi(Q)$. By (9.9a), all $p_Q \leq 1$ once δ is small enough; the expected cardinality is exactly N . The standard exponential-moment bound for a sum of independent Bernoulli variables gives

$$\Pr\{\#P \notin [N/2, 2N]\} \leq 2e^{-cN}. \quad (9.9b)$$

For a dyadic cube R of side $r \in [\delta, 1]$, the number Z_R of retained representatives in R has mean at most $C_{D,q} h^{-1} N r^q$. Set $T_R = N \delta^{-3\eta/4} r^q$. For sufficiently small δ , $T_R \geq \delta^{-\eta/4}$ and $eEZ_R/T_R \leq eC_{D,q} h^{-1} \delta^{3\eta/4} \leq e^{-1}$. The same exponential-moment calculation yields

$$\Pr\{Z_R > T_R\} \leq \left(\frac{eEZ_R}{T_R} \right)^{T_R} \leq e^{-T_R}. \quad (9.9c)$$

There are at most $C_D \delta^{-D} \log(1/\delta)$ cubes at these scales. Their union-bound failure probability tends to zero, by the lower bound on T_R . We may therefore choose one outcome for which $N/2 \leq |P| \leq 2N$ and $Z_R \leq T_R$ for every such R .

Any ball of radius $r \in [\delta, 1]$ meets at most C_D cubes of dyadic side between r and $2r$. The preceding estimate and $|P| \geq N/2$ give

$$|P \cap B(x, r)| \leq C_{D,q} \delta^{-3\eta/4} r^q |P| \leq \delta^{-\eta} r^q |P|$$

for all sufficiently small δ . The bounds on $|P|$ likewise imply (9.7). The displayed conditions require only that the Frostman constant $C_{D,q}/h$ grow more slowly than a fixed small power of δ^{-1} ; in particular they hold for the polynomial-in- k content losses used below when $\delta = 2^{-k}$.

For the measure version, restrict and renormalise μ on the stated measurable subset. Its dyadic leaf masses obey (9.9a) with $C_{D,q}/h$ replaced by a constant depending on C/h . Choose representatives from its positive-mass leaves, then use the same parity selection and independent Bernoulli argument. This proves both versions. $\square$

Lemma III.9.2 (common-scale Furstenberg discretisation). *Let $0 < a < s$ and $0 < b < t$, with $a < b$. Suppose $K \subset \mathbb{R}^2$ and a line family $\mathcal{L}$ satisfy*

$$\dim_{\text{H}} \mathcal{L} > b, \quad \dim_{\text{H}}(K \cap \ell) > a \quad (\ell \in \mathcal{L}). \quad (9.10)$$

Fix $\eta > 0$. Given an arbitrarily fine ball cover of K , after restricting to one bounded line chart and one bounded spatial ball, there is a dyadic scale δ tending to zero with the fineness of the cover, a $(\delta, b, \delta^{-\eta})$ -set of dyadic tubes $\mathcal{T}$, and for every $T \in \mathcal{T}$ a $(\delta, a, \delta^{-\eta})$ -set $P(T)$ of incident dyadic squares. Moreover, every square in $\bigcup_T P(T)$ meets a fixed enlargement of one ball from the scale- δ layer of the original cover.

Proof. First decompose line space into bounded affine charts and K into bounded balls. Countable stability of Hausdorff dimension and (9.10) allow a restriction for which $\dim_{\text{H}} \mathcal{L} > b$ and

$$\mathcal{H}_{\infty}^a(K \cap \ell) \geq c_0 > 0 \quad (\ell \in \mathcal{L}). \quad (9.11)$$

This is the same uniformisation as in Lemma III.8.3.

Let E_k, δ_k, a_k be the cover layers from that lemma, with $a_k = c(k+1)^{-2}$, and put

$$\mathcal{L}_k = \{\ell \in \mathcal{L} : \mathcal{H}_{\infty}^a(E_k \cap \ell) \geq a_k\}.$$

These sets cover $\mathcal{L}$. Choose a bounded subset of $\mathcal{L}$ with positive b -content. By subadditivity of Hausdorff content, $\sum_k \mathcal{H}_{\infty}^b(\mathcal{L}_k) > 0$. Consequently, for at least one k beyond the initial cover scale,

$$\mathcal{H}_{\infty}^b(\mathcal{L}_k) \gtrsim (k+1)^{-2}. \quad (9.12)$$

Set $\delta = \delta_k$. Apply Lemma III.9.1 in line space to $\mathcal{L}_k$ with internal loss $\eta/2$, and then replace each selected line by a boundedly enlarged dyadic δ -tube. Because $(k+1)^C = \delta^{-o(1)}$, for every prescribed $\eta > 0$ and all sufficiently fine covers the resulting tube family is a $(\delta, b, \delta^{-\eta})$ -set.

For a selected line ℓ , the Borel set $E_k \cap \ell$ has a -content at least a_k . Apply the one-dimensional instance of Lemma III.9.1 with internal loss $\eta/2$ at the same scale δ , and replace selected points by dyadic squares meeting the corresponding enlarged cover balls. This produces a $(\delta, a, \delta^{-\eta})$ -set $P(T)$. Fixed geometric enlargement changes only absolute constants. Finally $a_k^{-1} = (k+1)^{O(1)} = \delta^{-o(1)}$, so the unused half of the η -slack absorbs the fixed and subpower losses. The final assertion follows from the definition of E_k . $\square$

III.9.3 The Duke incidence theorem and its continuum consequences

Proposition III.9.3 (sharp dual incidence theorem, reused). *Let $0 < a < 1$ and $a < b \leq 2$. There exists $\zeta = \zeta(a, b) > 0$ such that, for all sufficiently small dyadic δ , if $\mathcal{T}$ is a $(\delta, b, \delta^{-\zeta})$ -set of dyadic tubes and every $T \in \mathcal{T}$ carries a $(\delta, a, \delta^{-\zeta})$ -set $P(T)$ of incident dyadic squares, then*

$$\left| \bigcup_{T \in \mathcal{T}} P(T) \right| \geq \delta^{-2a-\zeta}. \quad (9.13)$$

Moreover, ζ may be chosen uniformly when (a, b) ranges over a compact subset of $\{(a, b) : 0 < a < 1, a < b < 2\}$.

Proof. The estimate is exactly the locally proved sharp dual Furstenberg incidence estimate (6.77), which is the dual form of Orponen–Shmerkin’s Duke Theorem 1.3/3.2 [22]. Uniformity on compact parameter sets follows by decreasing the gain, as in the source theorem’s parameter-stability remark. No proof is duplicated here. $\square$

Theorem III.9.4 (Orponen–Shmerkin Furstenberg gain). *For every $s \in (0, 1)$ and $t \in (s, 2]$, there exists $\epsilon = \epsilon(s, t) > 0$ such that every (s, t) -Furstenberg set $K \subset \mathbb{R}^2$ satisfies*

$$\dim_{\mathrm{H}} K \geq 2s + \epsilon. \quad (9.14)$$

Proof. If $t < 2$, put $t' = t$; if $t = 2$, first choose $s < t' < 2$. Choose a compact neighbourhood $\mathcal{K}$ of the interior pair (s, t') contained in

$$\{(a, b) : 0 < a < 1, a < b < 2\}.$$

By the uniform clause of Theorem III.9.3, one number $\zeta_0 > 0$ is admissible throughout $\mathcal{K}$. Now choose $(a, b) \in \mathcal{K}$ with $a < s$, $b < t$, and $2(s - a) < \zeta_0/2$, and put $\zeta = \zeta_0$. Then

$$2a + \frac{\zeta}{2} > 2s. \quad (9.15)$$

We prove $\dim_{\mathrm{H}} K \geq 2a + \zeta/2$.

Suppose instead that $\dim_{\mathrm{H}} K < q < 2a + \zeta/2$. For every $\tau > 0$ there is an arbitrarily fine ball cover $\{B_j\}$ of K with

$$\sum_j r(B_j)^q < \tau. \quad (9.16)$$

Apply Lemma III.9.2 with a loss parameter $\eta < \zeta/10$. For a sufficiently fine cover it produces a scale δ , a tube family $\mathcal{T}$, and fibres $P(T)$ whose nonconcentration constants are at most $\delta^{-\zeta}$. Hence Theorem III.9.3 gives

$$\left| \bigcup_T P(T) \right| \geq \delta^{-2a-\zeta}. \quad (9.17)$$

Each square counted in (9.17) meets a fixed enlargement of a cover ball with radius comparable to δ , and one such enlarged ball meets only $O(1)$ dyadic δ -squares. Thus the scale- δ layer of (9.16) contains at least $c\delta^{-2a-\zeta}$ balls. Its q -cost is therefore

$$\geq \delta^{q-2a-\zeta}. \quad (9.18)$$

Since $q < 2a + \zeta/2$, the right-hand side tends to infinity as the initial cover scale tends to zero, contradicting (9.16). Therefore $\dim_{\mathrm{H}} K \geq 2a + \zeta/2$, and (9.15) yields (9.14) with a positive final gain. $\square$

Corollary III.9.5 (classical gain above the critical fibre exponent). *If $s \in (1/2, 1)$, every classical s -Furstenberg set satisfies*

$$\dim_{\mathrm{H}} K \geq 2s + \epsilon(s)$$

for some $\epsilon(s) > 0$.

Proof. A classical Furstenberg set has a one-dimensional family of witnessing lines. Apply Theorem III.9.4 with $t = 1 > s$. $\square$

III.9.4 Exceptional projections from the primal incidence theorem

Lemma III.9.6 (bad projections create a forbidden incidence configuration). *Fix $0 < a < 1$ and $a < b < 2$, and let $\zeta = \zeta(a, b) > 0$ be a uniform gain for the primal form of [Theorem III.9.3](#). Let $c > a$ satisfy*

$$c - a < \frac{\zeta}{4}. \quad (9.19)$$

There do not exist an analytic set $K \subset [0, 1]^2$ with $\dim_{\mathbb{H}} K > b$ and a direction set $E \subset \mathbb{S}^1$ with $\dim_{\mathbb{H}} E > a$ such that

$$\dim_{\mathbb{H}} \pi_e(K) < c \quad (e \in E). \quad (9.20)$$

Proof. Choose a compact subset of K carrying a b -Frostman probability measure μ ; this is legitimate because K is analytic and $b < \dim_{\mathbb{H}} K$. For every $e \in E$, condition (9.20) gives an interval cover $\mathcal{I}_e$ of $\pi_e(K)$, with lengths below an arbitrarily small common upper bound, such that

$$\sum_{I \in \mathcal{I}_e} |I|^c \leq 1. \quad (9.21)$$

Group these intervals into dyadic length classes. If $U_{e,k}$ denotes the inverse projection of the class with $|I| \asymp 2^{-k}$, then $\bigcup_k U_{e,k}$ contains K . Choose positive numbers $b_k \asymp (k+1)^{-2}$ with $\sum b_k < 1/2$, and set

$$E_k = \{e \in E : \mu(U_{e,k}) \geq b_k\}. \quad (9.22)$$

The sets E_k cover E . Since $\dim_{\mathbb{H}} E > a$, after restricting to a bounded angular chart we may assume $\mathcal{H}_{\infty}^a(E) > 0$. Subadditivity of content then yields, for some arbitrarily large k ,

$$\mathcal{H}_{\infty}^a(E_k) \gtrsim (k+1)^{-2}. \quad (9.23)$$

Put $\delta = 2^{-k}$. By [Lemma III.9.1](#), after choosing the initial covers sufficiently fine, E_k contains a $(\delta, a, \delta^{-\eta})$ -set D of directions with

$$\delta^{-a+\eta} \leq |D| \leq \delta^{-a-\eta}, \quad (9.24)$$

where $\eta > 0$ will be much smaller than ζ .

For $e \in D$, the scale- δ part of (9.21) contains at most $C\delta^{-c}$ intervals. Replace their inverse images in $[0, 1]^2$ by a bounded number of dyadic δ -tubes; denote this family by $\mathcal{T}_e$. Then

$$|\mathcal{T}_e| \lesssim \delta^{-c}, \quad \mu\left(\bigcup_{T \in \mathcal{T}_e} T\right) \gtrsim b_k. \quad (9.25)$$

Define $n(x) = \#\{e \in D : x \in \bigcup_{T \in \mathcal{T}_e} T\}$. Integrating (9.25) gives $\int n d\mu \gtrsim b_k |D|$. Since $n \leq |D|$, the set

$$X = \{x : n(x) \geq c_1 b_k |D|\} \quad (9.26)$$

has $\mu(X) \gtrsim b_k$, after fixing a sufficiently small absolute $c_1 > 0$. The normalised restriction μ_X is still b -Frostman with constant $O(b_k^{-1})$. Apply the measure version of [Lemma III.9.1](#) to obtain a $(\delta, b, \delta^{-\eta})$ -set P of dyadic point squares meeting X .

For every $p \in P$, choose $x_p \in X \cap p$. By (9.26), at least $c_1 b_k |D|$ directions $e \in D$ have a selected strip containing x_p . For one such tube per direction, enlarge by a fixed factor so that it meets the whole square p . Any subset of D inherits its upper nonconcentration estimate, while its cardinality is only a polynomial factor in k smaller than $|D|$. Hence, after another harmless thinning, the tubes through p contain a $(\delta, a, \delta^{-3\eta})$ -set $\mathcal{T}(p)$.

Let $\mathcal{T} = \bigcup_{e \in D} \mathcal{T}_e$. On one hand,

$$|\mathcal{T}| \leq |D| \delta^{-c} \leq \delta^{-(a+c+2\eta)}. \quad (9.27)$$

On the other hand, the locally proved primal sharp-incidence estimate (6.75) gives, for η sufficiently small,

$$|\mathcal{T}| \geq \delta^{-2a-\zeta/2}. \quad (9.28)$$

The exponents are incompatible because (9.19) and $\eta \ll \zeta$ imply $2a + \zeta/2 > a + c + 2\eta$. This contradiction proves the lemma. $\square$

Theorem III.9.7 (Orponen–Shmerkin exceptional projection gain). *For every $s \in (0, 1)$ and $t \in (s, 2]$, there exists $\epsilon = \epsilon(s, t) > 0$ such that every analytic $K \subset \mathbb{R}^2$ with $\dim_{\text{H}} K = t$ satisfies*

$$\dim_{\text{H}} \{\theta \in \mathbb{S}^1 : \dim_{\text{H}} \pi_{\theta}(K) \leq s\} \leq s - \epsilon. \quad (9.29)$$

Proof. Choose $\rho > 0$ so small that

$$a = s - 2\rho > 0, \quad b = t - \rho > a, \quad c = s + \rho < \min\{1, t\} + \rho,$$

and, using the compact-parameter uniformity of the Duke discrete theorem,

$$3\rho = c - a < \frac{1}{4}\zeta(a, b). \quad (9.30)$$

If $t = 2$, the choice $b = 2 - \rho$ keeps the discrete pair in the interior. Let

$$E = \{\theta : \dim_{\text{H}} \pi_{\theta}(K) \leq s\}.$$

If $\dim_{\text{H}} E > a$, then every $e \in E$ satisfies $\dim_{\text{H}} \pi_e(K) < c$, while $\dim_{\text{H}} K = t > b$. This contradicts Lemma III.9.6. Hence $\dim_{\text{H}} E \leq a = s - 2\rho$, proving (9.29) with $\epsilon = 2\rho$. $\square$

III.9.5 The published asymmetric ABC theorem

The continuous theorem uses

$$0 < \beta \leq \alpha < 1, \quad \kappa > 0, \quad (9.3)$$

and returns a gain $\eta = \eta(\alpha, \beta, \kappa) > 0$. Its discretised form uses a coefficient exponent

$$\gamma \in (\alpha - \beta, 1] \quad (9.4)$$

and returns a gain $\chi = \chi(\alpha, \beta, \gamma) > 0$. Thus α and β are dimensions of the summand sets, whereas γ controls the coefficient set.

For $G \subset A \times B$ and $c \in \mathbb{R}$, set

$$\Phi_c(G) = \{a + cb : (a, b) \in G\}. \quad (9.5)$$

The discretised theorem is robust: its expansion conclusion holds for every graph G of relative size at least δ^χ , not merely for the full Cartesian product.

Theorem III.9.8 (Orponen–Shmerkin asymmetric *ABC* theorem). *Let $0 < \beta \leq \alpha < 1$ and $\kappa > 0$. There exists $\eta = \eta(\alpha, \beta, \kappa) > 0$ such that, whenever Borel sets $A, B \subset \mathbb{R}$ satisfy $\dim_{\text{H}} A = \alpha$ and $\dim_{\text{H}} B = \beta$,*

$$\dim_{\text{H}}\{c \in \mathbb{R} : \dim_{\text{H}}(A + cB) \leq \alpha + \eta\} \leq (\alpha - \beta) + \kappa. \quad (9.31)$$

The proof of this continuous statement follows the discretised theorem below; the common-scale reduction is adapted from [19, Section 5.6]. The key point is that a small *Hausdorff* image need not have small covering number at every scale: one must first select a common scale and a positive-mass graph. We give that passage after stating its finite-scale input.

The proof of the finite-scale input itself uses an endpoint-logarithmic projection estimate. We record and prove this ingredient now; it is valid independently of the robust *ABC* theorem proved later in this chapter.

The second elementary ingredient is the passage from discrete Cartesian products to measures with both ball and tube control. It explains the initial exponent in the thin-tube bootstrap used in [18, Section 3.1]; the bootstrap itself is a separate research-level step.

Lemma III.9.9 (Cartesian products have thin tubes at the coordinate exponent). *Let $\delta \in 2^{-\mathbb{N}} \cap (0, 1/4)$, let $0 < s, t \leq 1$, and let $A, B \subset \delta\mathbb{Z} \cap [-2, 2]$ be nonempty (δ, s, C_A) - and (δ, t, C_B) -sets, respectively, in the relative sense of (6.41). Write λ_A for the probability measure which is uniform on the union of the intervals $a + [-\delta/2, \delta/2]$, $a \in A$, with equal mass $|A|^{-1}$ on each interval, and define λ_B similarly. The product $\rho = \lambda_A \otimes \lambda_B$ obeys, for every $z \in \mathbb{R}^2$ and $r > 0$,*

$$\rho(B(z, r)) \leq C_{s,t} C_A C_B r^{s+t}, \quad \rho(T_r) \leq C_{s,t} \max\{C_A, C_B\} r^{\min\{s,t\}}, \quad (9.30a)$$

where T_r is the radius- r neighbourhood of any affine line. The constants are uniform in δ, A, B .

Proof. The relative set condition implies, for $\delta \leq r \leq 1$, $\lambda_A(I) \leq CC_A r^s$ and $\lambda_B(I) \leq CC_B r^t$ for every interval I of length at most $2r$. At $r < \delta$, the densities on the occupied δ -intervals give the same estimates:

$$\lambda_A(I) \leq CC_A \delta^{s-1} |I| \leq CC_A |I|^s, \quad \lambda_B(I) \leq CC_B |I|^t.$$

The first inequality in (9.30a) follows by placing a ball inside the product of its coordinate intervals.

If the unit direction of the line defining T_r has first coordinate of absolute value at most $2^{-1/2}$, every horizontal fibre of $T_r \cap [-3, 3]^2$ is an interval of length $O(r)$. Fubini and the interval bound for λ_A give $\rho(T_r) \leq CC_A r^s$. In the complementary case vertical fibres have length $O(r)$, and $\rho(T_r) \leq CC_B r^t$. For $r \leq 1$, either estimate is at most $C \max\{C_A, C_B\} r^{\min\{s,t\}}$, since $r^s, r^t \leq r^{\min\{s,t\}}$. For $r > 1$, enlarge the constant and use $\rho(T_r) \leq 1$. $\square$

The thin-tube argument needs the two product measures to have separated supports. The following localisation does that while retaining a quantified part of any putative bad set of quadruples.

Lemma III.9.10 (separation of two Frostman coordinates). *Let $0 < s \leq 1$, $C \geq 1$, and let λ_1, λ_2 be probability measures supported in $[-2, 2]$ such that $\lambda_i(I) \leq C|I|^s$ for every interval I of length at most 1. Let η_1, η_2 be arbitrary probability measures on $[-2, 2]$, and let $E \subset ([-2, 2]^2)^2$ be Borel with $(\lambda_1 \otimes \eta_1 \otimes \lambda_2 \otimes \eta_2)(E) \geq m > 0$. There are subprobability restrictions λ'_1, λ'_2 of λ_1, λ_2 , each of mass at least $c_s m h$, where $h = c_s(m/C)^{1/s}$, such that their supports are separated by at least h , and their normalisations $\bar{\lambda}_i$ satisfy*

$$(\bar{\lambda}_1 \otimes \eta_1 \otimes \bar{\lambda}_2 \otimes \eta_2)(E) \geq c_s m h, \quad \bar{\lambda}_i(I) \leq \frac{C}{c_s m h} |I|^s. \quad (9.30b)$$

Proof. Partition $[-2, 2]$ into intervals I_j of length between $2h$ and $4h$, with at most C_0/h intervals, where C_0 is absolute. For each I_j , let $3I_j$ be the concentric interval of triple length. The portion of the product measure with $a_1 \in I_j$, $a_2 \in 3I_j$, summed over j , is at most

$$\sum_j \lambda_1(I_j) \lambda_2(3I_j) \leq C(12h)^s.$$

Choose the dimensional constant in $h = c_s(m/C)^{1/s}$ so this is at most $m/2$. Hence at least $m/2$ of the mass of E lies in the union of the disjoint events $a_1 \in I_j$, $a_2 \notin 3I_j$. One of them carries mass at least cmh . Restrict λ_1 to that I_j , and λ_2 to the complement of $3I_j$. The two masses are each at least cmh , because their product is at least the mass just found. Their supports have distance at least $2h$, hence at least h . Normalisation increases each Frostman constant by at most $(cmh)^{-1}$. Finally the restricted product mass of E is at least cmh , while the product of the two restricting masses is at most one. Thus the normalised product mass of E is also at least cmh . $\square$

For two compactly supported probability measures μ, ν on $\mathbb{R}^2$, we say that (μ, ν) has (u, K, c) -thin tubes if there is a Borel set $G \subset \text{spt } \mu \times \text{spt } \nu$ with $(\mu \otimes \nu)(G) \geq c$ such that

$$\nu(G_x \cap T_r) \leq K r^u \quad \text{for every } x \in \text{spt } \mu \text{ and every radius-}r \text{ tube } T_r \ni x. \quad (9.30c)$$

Here $G_x = \{y : (x, y) \in G\}$, and a tube is the closed radius- r neighbourhood of an affine line. The reversed condition is defined by interchanging μ, ν . This quantitative form is needed because the later ABC argument must retain almost all coefficient quadruples, not merely one positive-mass subfamily.

The next proof reconstructs the incidence-driven thin-tube improvement of [7, Lemma 2.8] using the primal sharp incidence result already proved in Chapter 6. The two alternatives in the proof distinguish mass spread along a tube from mass concentrated in a shorter piece; the displayed inequalities, rather than this description, carry the quantitative argument.

Lemma III.9.11 (incidence-driven improvement of thin tubes). *Let $0 < \sigma < s \leq 1$, $0 < \varepsilon < 1/20$, and let μ, ν be probability measures supported in $B(0, 2)$. Assume, for $0 < r \leq 1$,*

$$\mu(B(z, r)), \nu(B(z, r)) \leq C r^s, \quad \text{dist}(\text{spt } \mu, \text{spt } \nu) \geq \Delta > 0, \quad (9.30d)$$

where $C, K \geq 1, 0 < \Delta \leq 1$. If both ordered pairs $(\mu, \nu), (\nu, \mu)$ have $(\sigma, K, 1-\varepsilon)$ -thin tubes, then there are $\eta = \eta(s, \sigma) > 0$ and $M = M(s, \sigma) < \infty$ such that both pairs have $(\sigma + \eta, K', 1 - 5\varepsilon)$ -thin tubes with

$$K' \leq (CK\Delta^{-1}\varepsilon^{-1})^M. \quad (9.30e)$$

Proof. We prove the assertion for (μ, ν) ; interchanging the measures gives the other assertion with the same constants. Intersect the two Borel graphs supplied by the hypothesis, transposing the second. The resulting graph G has $(\mu \otimes \nu)(G) \geq 1 - 2\varepsilon$ and obeys (9.30c) in both directions. All constants below depend only on s, σ unless an explicit parameter is displayed.

Fix $u = (s + \sigma)/2$. Let $\zeta = \zeta(\sigma, u) > 0$ be the gain in the locally proved primal sharp incidence theorem III.6.29.13. Choose $\eta > 0$ so small that

$$\sigma + 4\eta < s, \quad \kappa := \frac{16\eta}{s - \sigma} < 1, \quad 10\eta + \kappa < \zeta. \quad (9.30f)$$

We will show that a sufficiently large auxiliary threshold K' , of the form (9.30e), makes failure of the improved thin-tube assertion with constant $4K'$ impossible. This fixed factor is absorbed into the final choice of M .

For each dyadic r , choose a finite r -net of lines meeting $B(0, 3)$, and let $\mathcal{N}_r$ be their $10r$ -neighbourhoods. It has $O(r^{-2})$ members; every radius- r tube meeting $B(0, 2)$ is covered there by one of them. For $x \in \text{spt } \mu$, mark $V \in \mathcal{N}_r$ if its axis is within $O(r)$ of x and $\nu(V \cap G_x) \geq K'r^{\sigma+\eta}$. Let $\mathcal{N}_r(x)$ be the marked family and let $H_r = \{(x, y) \in G : y \in \bigcup \mathcal{N}_r(x)\}$. The finite net makes H_r Borel. A point $y \in \text{spt } \nu$, at distance at least Δ from x , belongs to at most $O(\Delta^{-1})$ members of $\mathcal{N}_r$ whose axes pass within $O(r)$ of x . Consequently

$$|\mathcal{N}_r(x)| \leq \frac{C_0\Delta^{-1}}{K'r^{\sigma+\eta}} \leq r^{-\sigma-\eta} \quad (9.30g)$$

once $K' \geq C_0\Delta^{-1}$.

If $(\mu \otimes \nu)(\bigcup_r H_r) < 3\varepsilon$, the graph $G \setminus \bigcup_r H_r$ has mass greater than $1 - 5\varepsilon$. Every tube through x whose fibre mass in that graph exceeds $4K'r^{\sigma+\eta}$ is contained, after enlarging to the next dyadic width, in a marked net tube, which is impossible. Thus failure of the improved assertion with constant $4K'$ forces $(\mu \otimes \nu)(\bigcup_r H_r) \geq 3\varepsilon$. For $r > r_0$, no tube is marked if $K'r_0^{\sigma+\eta} > 1$. Choose r_0 small enough that

$$\sum_{\substack{r \leq r_0 \\ r \text{ dyadic}}} 4r^\eta < \varepsilon, \quad Kr_0^\eta \leq c, \quad C\Delta^{-1}r_0^\eta \leq c, \quad r_0^K \leq c\Delta, \quad (9.30h)$$

and below the fixed threshold of Proposition III.6.29.13. Enlarging K' so that $K'r_0^{\sigma+\eta} > 1$, we obtain a dyadic $r \leq r_0$ with $(\mu \otimes \nu)(H_r) \geq 4r^\eta$. Put $X = \{x : \nu((H_r)_x) \geq 2r^\eta\}$; then $\mu(X) \geq 2r^\eta$. Discard from $\mathcal{N}_r(x)$ the tubes meeting $(H_r)_x$ in ν -mass less than $r^{\sigma+3\eta}$, and call the survivors $\mathcal{T}_x$. By (9.30g), the discarded part has mass at most $r^{2\eta}$. Let Y_x be the part of $(H_r)_x$ covered by $\mathcal{T}_x$. Since $Y_x \subset G_x$, the old thin-tube bound and (9.30h) give

$$\nu(Y_x) \geq r^{2\eta}, \quad r^{\sigma+3\eta} \leq \nu(V \cap Y_x) \leq r^{\sigma-\eta}, \quad r^{-\sigma+3\eta} \leq |\mathcal{T}_x| \leq r^{-\sigma-\eta}. \quad (9.30i)$$

The net tubes have bounded overlap on $\text{spt } \nu$. If $r \leq \rho \leq 1$, the members of $\mathcal{T}_x$ with axes in a line-space ρ -ball lie in an $O(\rho)$ -tube through x . Applying the old thin-tube bound and then the lower bound in (9.30i) shows that their number is at most $C_0 \Delta^{-1} K r^{-\sigma-3\eta} \rho^\sigma$. The lower bound on $|\mathcal{T}_x|$, together with (9.30h), therefore makes every $\mathcal{T}_x$ an $(r, \sigma, r^{-8\eta})$ -set of tubes in the relative sense of Chapter 6.

Fix a finite $r^\kappa/10$ -net of centres in $B(0, 3)$. Call $V \in \mathcal{T}_x$ *concentrated* if a ball of radius $2r^\kappa$ centred on this net contains at least one third of $\nu(V \cap Y_x)$ inside V . This is a Borel condition. If V is not concentrated, then every ball of radius r^κ has less than one third of that mass, since it lies in a radius- $2r^\kappa$ net ball. First suppose that for a set $X_1 \subset X$ of μ -mass at least $\mu(X)/2$, at least half of $\mathcal{T}_x$ is nonconcentrated for every $x \in X_1$. The proof of the finite Frostman extraction Lemma III.9.1, applied with exponent u , content mass $\mu(X_1) \gtrsim r^\eta/C$, and slack 9η , gives an $(r, u, r^{-9\eta})$ -set P of r -squares meeting X_1 . (In its Bernoulli step, the largest retention probability is $O(Cr^{9\eta/2-\eta}) < 1$; its final relative concentration constant is $O(Cr^{-\eta-27\eta/4}) \leq r^{-9\eta}$ for sufficiently small r . This verifies the scale-dependent use of that lemma.) For each $p \in P$, choose $x_p \in X_1 \cap p$ and the nonconcentrated half of $\mathcal{T}_{x_p}$. Converting their line parameters to the bounded dyadic charts of Chapter 6 changes the concentration constants only by fixed factors. Because $10\eta < \zeta$, the sharp incidence proposition III.6.29.13 gives for their union $\mathcal{T}$

$$|\mathcal{T}| \geq r^{-2\sigma-\zeta}. \quad (9.30j)$$

For each $V \in \mathcal{T}$, choose one x_p for which it was selected and put $m_V = \nu(V \cap Y_{x_p})$. Nonconcentration implies directly that

$$(\nu \otimes \nu)\{(y_1, y_2) \in (V \cap Y_{x_p})^2 : |y_1 - y_2| \geq r^\kappa\} \geq \frac{2}{3} m_V^2 \geq cr^{2\sigma+6\eta}.$$

For two points separated by $L \geq r^\kappa$, at most C_0/L members of the common line net contain both. Integrating over the two-point measure therefore gives

$$cr^{2\sigma+6\eta}|\mathcal{T}| \leq C_0 r^{-\kappa}, \quad |\mathcal{T}| \leq C_1 r^{-2\sigma-6\eta-\kappa}, \quad (9.30k)$$

contrary to (9.30f) and (9.30j) once r_0 is sufficiently small.

It remains to treat the alternative: on a set $X_1 \subset X$ of μ -mass at least $\mu(X)/2$, at least half of $\mathcal{T}_x$ is concentrated. For each such tube choose the first witnessing net ball, which is a Borel choice. The union $H_1 \subset G$ of the corresponding restricted fibres has, by (9.30i) and the bounded overlap $O(\Delta^{-1})$, mass at least

$$(\mu \otimes \nu)(H_1) \geq c\Delta r^\eta r^{-\sigma+3\eta} r^{\sigma+3\eta} \geq r^{8\eta}. \quad (9.30l)$$

For $(x, y) \in H_1$, the chosen tube contains at least $cr^{\sigma+3\eta}$ of ν in $B(y, 4r^\kappa)$. The ball estimate in (9.30d) gives $\nu(B(y, Cr)) \leq C_1 r^s = o(r^{\sigma+3\eta})$. Pigeonholing the dyadic annuli between Cr and $4r^\kappa$ assigns to each $(x, y) \in H_1$ one radius $\xi(x, y) \in [Cr, 4r^\kappa]$ for which

$$\nu(V_{x,y} \cap \{z : \xi(x, y) \leq |z - y| < 2\xi(x, y)\}) \geq r^{\sigma+4\eta}. \quad (9.30m)$$

There are $O(\log(1/r))$ choices, so one common ξ has a subgraph $H_2 \subset H_1$ of mass at least $r^{9\eta}$. Fix y with $\mu((H_2)^y) \geq r^{9\eta}$.

Since $\xi \leq 4r^\kappa \leq \Delta$, a tube of width $O(r/\xi)$ through y contains every x whose chosen net-tube direction lies in one $O(r/\xi)$ -arc. The reverse old thin-tube condition says that any tube through y of radius r/ξ contains at most $K(r/\xi)^\sigma \leq r^{-\eta}(r/\xi)^\sigma$ of μ -mass from G^ν . A maximal family of directions of the tubes $V_{x,y}$, $x \in (H_2)^\nu$, separated by $C_0 r/\xi$ for a sufficiently large absolute C_0 , therefore contains at least $cr^{10\eta}(\xi/r)^\sigma$ members. Within the annulus $\xi \leq |z - y| < 2\xi$, the selected tubes have bounded overlap. Sum (9.30m) over this family and use the s -Frostman bound on ν :

$$cr^{14\eta}\xi^\sigma \leq \nu(B(y, 3\xi)) \leq C_1\xi^s.$$

Absorbing $C_1 \leq r^{-\eta}$ yields $cr^{15\eta} \leq \xi^{s-\sigma} \leq C_0 r^{\kappa(s-\sigma)} = C_0 r^{16\eta}$, again impossible for small r_0 . This excludes both alternatives and proves the gain.

Finally, the contradiction establishes the improved property with constant $4K'$; rename this constant in the statement. All smallness requirements on r_0 above are of the form $r_0 \leq c(CK\Delta^{-1}\varepsilon^{-1})^{-M_0}$, with M_0 depending only on s, σ : the sole non-elementary threshold is the fixed one in the local incidence theorem. Taking $K' \geq \max\{C_0\Delta^{-1}, 2r_0^{-\sigma-\eta}\}$ gives (9.30e) after increasing M . $\square$

The bootstrap is particularly efficient for product measures: the coordinate argument supplies the first thin-tube exponent on the whole product, so there is no separate positive-exponent seed to import.

Corollary III.9.12 (finite-scale strong thin tubes from a tube seed). *Let $0 < b \leq 1$, $b < a \leq 2$, and $b < \sigma < \min\{a, 1\}$. There are $L = L(a, b, \sigma) < \infty$ and $\varepsilon_0 = \varepsilon_0(a, b, \sigma) > 0$ such that for each $0 < \varepsilon < \varepsilon_0$ there is $\delta_0 = \delta_0(a, b, \sigma, \varepsilon) > 0$ with the following property: if $0 < \delta < \delta_0$ and ρ_1, ρ_2 are probability measures on $B(0, 2)$ satisfying, for all $0 < r \leq 1$,*

$$\rho_i(B(z, r)) \leq \delta^{-\varepsilon} r^a, \quad \rho_i(T_r) \leq \delta^{-\varepsilon} r^b, \quad \text{dist}(\text{spt } \rho_1, \text{spt } \rho_2) \geq \delta^\varepsilon, \quad (9.30n)$$

then both ordered pairs have $(\sigma, \delta^{-L\varepsilon}, 1 - L\delta^\varepsilon)$ -thin tubes.

Proof. Choose s_0 strictly between σ and $\min\{a, 1\}$. The ball estimate in (9.30n) implies the s_0 -Frostman estimate with the same constant. The tube estimate gives both ordered pairs $(b, \delta^{-\varepsilon}, 1 - \delta^\varepsilon)$ -thin tubes, using the full product as graph. Starting at exponent b , apply Lemma III.9.11 successively. At an exponent $\nu < \sigma$, its gain $\eta(s_0, \nu) > 0$ is fixed by a, b, σ . If the next exponent crosses σ , weaken the exponent to σ ; otherwise repeat. There are finitely many steps. Here is why the gains have a positive common lower bound on $[b, \sigma]$, even though Proposition III.6.29.13 is stated pointwise in its exponents. At a fixed parameter pair $(\nu_0, (s_0 + \nu_0)/2)$, let the proposition's gain be $\zeta_0 > 0$. For ν near ν_0 , a relative (r, ν, r^{-e}) -condition implies a $(r, \nu_0, r^{-e-|\nu-\nu_0|})$ -condition, since $\rho^{\nu-\nu_0} \leq r^{-|\nu-\nu_0|}$ for $r \leq \rho \leq 1$. The same conversion for the point exponent $(s_0 + \nu)/2$ loses at most $r^{-|\nu-\nu_0|/2}$. Choose the neighbourhood and e so these losses fit below $r^{-\zeta_0}$, and $2|\nu - \nu_0| < \zeta_0/2$. The exact- ν_0 theorem then yields at least $r^{-2\nu-\zeta_0/2}$, a fixed gain throughout this neighbourhood. A finite cover of $[b, \sigma]$ gives a common gain and a common scale threshold. The strict margin $\sigma < s_0$ also bounds $\kappa = 16\eta/(s_0 - \nu)$ uniformly. At each step the exceptional mass is multiplied by at most five. If K_j is the current tube constant and the bad mass is $5^j \delta^\varepsilon$, then (9.30e) gives $K_{j+1} \leq (\delta^{-3\varepsilon} K_j)^{M_j}$ up to a fixed multiplicative constant, with

$M_j = M_j(a, b, \sigma)$. Thus after the fixed number $N = N(a, b, \sigma)$ of steps the bad mass is at most $5^N \delta^\varepsilon$ and the constant at most $\delta^{-L\varepsilon}$ for a sufficiently large L . Choose $L \geq 20 \cdot 5^N$, and then choose δ_0 , depending on the fixed ε , so $\delta^\varepsilon < 1/L$ and every intermediate bad mass is below $1/20$. Enlarge L once more to absorb 5^N and the fixed multiplicative constants, shrinking δ_0 if necessary. This proves the nonvacuous assertion. $\square$

Lemma III.9.13 (robust finite-scale Kaufman energy estimate). *Let $0 < s \leq 1$, $\delta \in 2^{-\mathbb{N}}$ with $0 < \delta < 1/4$, and let $P \subset \mathcal{D}_\delta^2([0, 1]^2)$ be a nonempty (δ, s, C_1) -set in the sense of (6.41). Let ν be a probability measure on $\mathbb{S}^1$ satisfying $\nu(B(\theta, r)) \leq C_2 r^s$ for $0 < r \leq 1$, where $C_1, C_2 \geq 1$. There is $\theta \in \text{spt } \nu$ such that, simultaneously for every $0 < a \leq 1$ and every $P' \subset P$ with $|P'| \geq a|P|$,*

$$|\pi_\theta(P')|_\delta \geq \frac{c_s a^2}{C_1 C_2 \log(2/\delta)} \delta^{-s}. \quad (9.31a)$$

Here $\pi_\theta(P')$ means the union of the projections of the squares in P' . In particular this proves the slightly weaker $\log^2(1/\delta)$ version of [18, Proposition 3.9].

Proof. Let $N = |P|$, and let μ be the probability measure that puts mass N^{-1} uniformly on each square of P . For $Q \in P$, the projection of normalized area on Q has a density $g_{Q,\theta}$ bounded by $C\delta^{-1}$, supported in an interval of length $C\delta$ centred at $\pi_\theta(c_Q)$, where c_Q is the centre of Q . Thus the density $f_\theta = d(\pi_\theta \mu)/d\mathcal{L}^1$ satisfies

$$\|f_\theta\|_2^2 \leq \frac{C}{N^2 \delta} \sum_{Q, Q' \in P} \mathbf{1}_{\{|\pi_\theta(c_Q - c_{Q'})| \leq C\delta\}}.$$

For $\nu \neq 0$, the set of directions with $|\pi_\theta \nu| \leq C\delta$ lies in two arcs of length $O(\min\{1, \delta/|\nu|\})$. The Frostman condition therefore gives, also for $\nu = 0$,

$$\nu\{\theta : |\pi_\theta \nu| \leq C\delta\} \leq CC_2 \left(\frac{\delta}{\max\{|\nu|, \delta\}} \right)^s.$$

Consequently

$$\int \|f_\theta\|_2^2 d\nu(\theta) \leq \frac{CC_2 \delta^{s-1}}{N^2} \sum_{Q, Q' \in P} \max\{|c_Q - c_{Q'}|, \delta\}^{-s}. \quad (9.31b)$$

For fixed Q , split the other centres into dyadic distance shells from δ to 2 . The (δ, s, C_1) -set condition, extended from dyadic cells to Euclidean balls by a fixed covering, gives $\#\{Q' : |c_Q - c_{Q'}| \leq r\} \leq CC_1 N r^s$ for $\delta \leq r \leq 2$. It also gives $1 \leq CC_1 N \delta^s$ by taking an occupied δ -cell. Each shell therefore contributes at most $CC_1 N$ to the inner sum in (9.31b). There are $O(\log(2/\delta))$ shells, so

$$\int \|f_\theta\|_2^2 d\nu(\theta) \leq CC_1 C_2 \delta^{s-1} \log(2/\delta).$$

Choose $\theta \in \text{spt } \nu$ for which the integrand does not exceed twice this average. The choice is made from P, ν alone and therefore works for every subfamily P' and every a at once. Indeed the projection of $\mu|_{\cup P'}$ has total mass at least a , has density bounded pointwise

by f_θ , and is supported in a union of at most $C|\pi_\theta(P')|_\delta$ intervals of length $C\delta$. Cauchy–Schwarz yields

$$a^2 \leq C\delta|\pi_\theta(P')|_\delta \|f_\theta\|_2^2,$$

which is exactly (9.31a). $\square$

We can now prove the expansion step that converts thin tubes into a robust assertion about four coefficients. This is the finite-scale mechanism behind [18, Proposition 3.7]; its proof uses the uniformity in the subfamily P' in the preceding Kaufman lemma.

Proposition III.9.14 (four-coefficient robust expansion). *Let $0 < s, t < 1$ and $0 \leq \sigma < \min\{s + t, 1\}$. There are $\varepsilon = \varepsilon(s, t, \sigma) > 0$ and $\delta_0 = \delta_0(s, t, \sigma) > 0$ with the following property for dyadic $0 < \delta < \delta_0$. Let $A_1, A_2, B_1, B_2 \subset \delta\mathbb{Z} \cap [-2, 2]$ be nonempty relative $(\delta, s, \delta^{-\varepsilon})$ - and $(\delta, t, \delta^{-\varepsilon})$ -sets, respectively, and let $P \subset \delta\mathbb{Z}^2 \cap [0, 1]^2$ be a nonempty relative $(\delta, s + t, \delta^{-\varepsilon})$ -set. For each fixed choice of signs in*

$$L_{\mathbf{a}, \mathbf{b}}(u, v) = (b_1 \pm b_2)u + (a_1 \pm a_2)v, \quad (9.31c)$$

there is a set $\mathcal{G} \subset A_1 \times A_2 \times B_1 \times B_2$ with $|\mathcal{G}| \geq (1 - \delta^\varepsilon)|A_1||A_2||B_1||B_2|$ such that, for every $(a_1, a_2, b_1, b_2) \in \mathcal{G}$ and every $P' \subset P$ with $|P'| \geq \delta^\varepsilon|P|$,

$$|L_{\mathbf{a}, \mathbf{b}}(P')|_\delta \geq \delta^{-\sigma}. \quad (9.31d)$$

Proof. We treat the two minus signs. Replacing A_2 and/or B_2 by their reflections proves the other cases, with the same relative concentration constants. Put $q = \min\{s + t, 1\}$, $v = (q + \sigma)/2$, and $b = \min\{s, t\}$. Then $\sigma < v < q$. All constants denoted D_j below depend only on s, t, σ , never on δ or ε .

Suppose the set of bad coefficient quadruples has relative cardinality at least δ^ε . Smooth the uniform measures on A_i, B_i by convolution with the uniform probability measure on $[-\delta/2, \delta/2]$, and call the results λ_i, τ_i , respectively. For this proof, order a quadruple as (a_1, b_1, a_2, b_2) , the permutation of the statement's (a_1, a_2, b_1, b_2) order. For each bad discrete quadruple, select one witnessing P' ; there are only finitely many choices, so the selection is Borel. If each coefficient is changed by at most $\delta/2$, the image of $P' \subset [0, 1]^2$ changes pointwise by at most 2δ . Let E be the union of the four-dimensional smoothing boxes around the bad quadruples, in this proof order. Then

$$(\lambda_1 \otimes \tau_1 \otimes \lambda_2 \otimes \tau_2)(E) \geq \delta^\varepsilon, \quad (9.31e)$$

and every $\mathbf{q} \in E$ has a subfamily $P'(\mathbf{q})$ of the required size for which $|L_{\mathbf{q}}(P'(\mathbf{q}))|_{5\delta} \leq \delta^{-\sigma}$.

Apply Lemma III.9.10 to the two A -coordinate measures, with $m = \delta^\varepsilon$ and $C \leq C_0\delta^{-\varepsilon}$. It yields normalised restrictions $\bar{\lambda}_1, \bar{\lambda}_2$ whose supports have distance $h \geq c\delta^{2\varepsilon/s}$, whose one-dimensional Frostman constants are at most $C_0\delta^{-(2+2/s)\varepsilon}$, and for which

$$(\bar{\lambda}_1 \otimes \tau_1 \otimes \bar{\lambda}_2 \otimes \tau_2)(E) \geq c\delta^{(1+2/s)\varepsilon}. \quad (9.31f)$$

Set $\rho_i = \bar{\lambda}_i \otimes \tau_i$. The coordinate-interval Frostman bounds and the fibrewise argument in the proof of Lemma III.9.9 give, for all $r > 0$,

$$\rho_i(B(z, r)) \leq \delta^{-D_0\varepsilon} r^{s+t}, \quad \rho_i(T_r) \leq \delta^{-D_0\varepsilon} r^b, \quad \text{dist}(\text{spt } \rho_1, \text{spt } \rho_2) \geq \delta^{D_0\varepsilon}, \quad (9.31g)$$

for a fixed sufficiently large $D_0 = D_0(s)$, after decreasing δ_0 to absorb fixed factors. If $v \leq b$, the tube estimate itself gives both ordered pairs $(v, \delta^{-D_0\epsilon}, 1)$ -thin tubes. If $v > b$, apply Corollary III.9.12 with input loss $D_0\epsilon$ to the pushforwards of ρ_1, ρ_2 under $z \mapsto z/3$. Their supports lie in $B(0, 2)$. Pull its thin-tube graph back under the same dilation: ball and tube radii and support separation change by fixed factors only, which are absorbed into D_0, D_1 . Take ϵ small enough to meet its threshold. In both cases there is a Borel graph H of $(\rho_2 \otimes \rho_1)$ -mass at least $1 - D_1\delta^{D_0\epsilon}$ such that for every $z \in \text{spt } \rho_2$ the restricted radial-direction measure from z to the ρ_1 -fibre of H is $(v, \delta^{-D_1\epsilon})$ -Frostman. Choose $D_0 > 2 + 2/s$. Then (9.31f) and the graph-mass bound imply $(\rho_1 \otimes \rho_2)(E \cap H^\top) > 0$, with mass at least $c\delta^{(1+2/s)\epsilon}$. Fubini supplies $z = (a_2, b_2) \in \text{spt } \rho_2$ such that the set

$$F = \{(a_1, b_1) : (a_1, b_1, a_2, b_2) \in E \cap H^\top\}$$

has $\rho_1(F) \geq c\delta^{(1+2/s)\epsilon}$. Normalise $\rho_1|_F$ and push it forward by $(a_1, b_1) \mapsto \theta(a_1, b_1) = (b_1 - b_2, a_1 - a_2)/|(b_1 - b_2, a_1 - a_2)|$. The separation in the first coordinate makes this map defined everywhere on F . The resulting probability measure ω on $\mathbb{S}^1$ satisfies

$$\omega(B(\theta, r)) \leq \delta^{-D_2\epsilon} r^v \quad (0 < r \leq 1), \quad (9.31h)$$

for a fixed D_2 : the inverse image of an angular r -arc is contained in an $O(r)$ -tube through z , and normalising costs $\rho_1(F)^{-1}$.

Associate to P its dyadic δ -squares. The relative $(\delta, s + t, \delta^{-\epsilon})$ condition implies the same condition with exponent v . Apply Lemma III.9.13 with this square family, ω , and $a = \delta^\epsilon$. It selects one direction $\theta \in \text{spt } \omega$ such that, simultaneously for every eligible P' ,

$$|\pi_\theta(P')|_\delta \geq c\delta^{-v+(D_2+3)\epsilon} / \log(2/\delta). \quad (9.31i)$$

Choose $(a_1, b_1) \in F$ whose direction is within δ of θ ; such a point exists by the support condition. Its bad witness P' has $|L_{\mathbf{q}}(P')|_{5\delta} \leq \delta^{-\sigma}$. Because $P \subset [0, 1]^2$, perturbing the direction by δ changes each unit projection by at most 2δ . The Kaufman projection here is of the associated δ -squares, not just their lattice corners; passing from a corner to its square enlarges its projected interval by at most $\sqrt{2}\delta$. Also $|(b_1 - b_2, a_1 - a_2)| \geq h$, so an interval of length $O(\delta/h)$ can be covered by at most C/h intervals of length δ . It follows that

$$|\pi_\theta(P')|_\delta \leq Ch^{-1}|L_{\mathbf{q}}(P')|_{5\delta} \leq C\delta^{-\sigma-(2/s)\epsilon}. \quad (9.31j)$$

Since $v > \sigma$, choose ϵ small in terms of s, t, σ, D_2 ; then (9.31i) exceeds (9.31j) for small δ . This contradiction proves the proposition. $\square$

The preceding expansion becomes useful for a small-set contradiction only after two additional operations: choose one good coefficient pair by averaging, then use a Plünnecke–Ruzsa estimate to append the two small sums $A + c_1B$. The next lemma performs both operations.

Lemma III.9.15 (large perturbation by almost every difference). *Let $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\chi_0, \delta_0 > 0$, depending only on these parameters, with the following property. Suppose $A, B \subset \delta\mathbb{Z} \cap [0, 1]$, $C \subset \delta\mathbb{Z} \cap [1, 2]$, $B, C \neq \emptyset$, $\delta < \delta_0$ is dyadic, $|A| \leq \delta^{-\alpha}$, and B, C obey the relative concentration bounds (9.33)–(9.34) with χ_0 . If*

$$|A + cB|_\delta \leq \delta^{-\chi_0}|A| \quad \text{for every } c \in C, \quad (9.31k)$$

then there are $c_1, c_2 \in C$ and $E \subset B \times B$ with $|E| \leq \delta^{\chi_0} |B|^2$ such that

$$|A' + (b_1 - b_2)C'|_\delta \geq \delta^{-\chi_0} |A| \quad (9.31l)$$

for every $(b_1, b_2) \in (B \times B) \setminus E$ and every $A' \subset A, C' \subset C$ satisfying $|A'| |C'| \geq \delta^{\chi_0} |A| |C|$.

Proof. Choose $\gamma_0 < \min\{\gamma, 1\}$ with $\beta + \gamma_0 > \alpha$, and choose $\alpha < \sigma_0 < \min\{\beta + \gamma_0, 1\}$. Let $\varepsilon_0 > 0$ be the gain of Proposition III.9.14 with $(s, t, \sigma) = (\beta, \gamma_0, \sigma_0)$. Take $\chi_0 > 0$ much smaller than ε_0 and $\sigma_0 - \alpha$, for example $20\chi_0 < \min\{\varepsilon_0, \sigma_0 - \alpha, \beta, \gamma_0\}$. The relative γ -condition on C implies the γ_0 -condition. The relative product $B \times (C - 1)$ has exponent $\beta + \gamma_0$ and concentration constant at most $C_0 \delta^{-2\chi_0}$.

For the application of the preceding proposition, remove the endpoints $1 \in B$ and $2 \in C$, if present, and call the resulting sets B_0, C_0 . The relative conditions at scale δ imply $|B| \geq \delta^{-\beta + \chi_0}$ and $|C| \geq \delta^{-\gamma + \chi_0}$; hence $|B_0| \geq |B|/2$, $|C_0| \geq |C|/2$ for small δ . Their concentration constants increase by at most a fixed factor. A set $C' \subset C$ with $|C'| \geq \delta^{\chi_0} |C|$ satisfies $|C' \cap C_0| \geq \frac{1}{2} \delta^{\chi_0} |C|$, which is at least $\delta^{\varepsilon_0} |C_0|$ for small δ . Pairs from B^2 involving the removed endpoint number at most $2|B|$, which is $o(\delta^{\chi_0} |B|^2)$ by $20\chi_0 < \beta$. We apply the expansion proposition to B_0, C_0 , then add these pairs to the exceptional set. Since the desired sumset for B, C' contains the one for $B_0, C' \cap C_0$, this proves the assertion for the original sets. To simplify notation, write B, C for B_0, C_0 in the rest of the proof.

Apply Proposition III.9.14 with coefficient sets $A_1 = A_2 = B$, $B_1 = B_2 = C$, point set $P = B \times (C - 1)$, and the signs giving $(c_1 + c_2)b + (b_1 - b_2)(c - 1)$. The point set and both coefficient families meet its hypotheses because $2\chi_0 < \varepsilon_0$. Average its exceptional quadruples over (c_1, c_2) . For some pair $(c_1, c_2) \in C^2$, at most $\delta^{\varepsilon_0} |B|^2$ pairs (b_1, b_2) are bad. Enlarge this bad-pair set to E of the permitted size. For each $(b_1, b_2) \notin E$ and $C' \subset C$ with $|C'| \geq \delta^{\chi_0} |C|$, the uniform-in- P' conclusion applied to $P' = B \times (C' - 1)$ yields

$$|(c_1 + c_2)B + (b_1 - b_2)C'|_\delta \geq c\delta^{-\sigma_0}. \quad (9.31m)$$

The translation by $b_1 - b_2$ and the fixed covering constant are harmless; after decreasing δ_0 the right-hand side may be replaced by $\delta^{-\sigma_0 + \chi_0}$.

Fix such a pair (b_1, b_2) and dense $A' \times C'$. Put $K = |A' + (b_1 - b_2)C'|_\delta / |A|$. The density condition gives $|A'| \geq \delta^{\chi_0} |A|$ and $|C'| \geq \delta^{\chi_0} |C|$. Apply the Plünnecke–Ruzsa inequalities in the allowed additive-combinatorics background to A' and the three summands $(b_1 - b_2)C', c_1B, c_2B$. The first sum has covering number $K|A|$, while (9.31k) bounds each of the other two sums by $\delta^{-\chi_0} |A|$. There is a nonempty $A'' \subset A'$ such that

$$|A'' + (b_1 - b_2)C' + c_1B + c_2B|_\delta \leq CK\delta^{-5\chi_0} |A|. \quad (9.31n)$$

Fix $a'' \in A''$. The set on the left contains $a'' + (b_1 - b_2)C' + (c_1 + c_2)B$, so (9.31m) and (9.31n) imply

$$K \geq c\delta^{-(\sigma_0 - \alpha - 6\chi_0)}.$$

Since $20\chi_0 < \sigma_0 - \alpha$, the right-hand side exceeds $\delta^{-\chi_0}$ for small δ , proving (9.31l). $\square$

The weak ABC theorem asks only for expansion of the full Cartesian sumset $A + cB$. Its proof is shorter than the robust theorem, but the distinction matters: no argument below upgrades the conclusion to every dense graph $G \subset A \times B$. We use the following result

proved in Volume II, Chapter II.12 (the exponential-scale asymmetric BSG lemma). For every $\tau > 0$ there is $\zeta > 0$ such that, for all sufficiently small dyadic δ , if $X, Y \subset [-2, 2] \cap \delta\mathbb{Z}$ and $G \subset X \times Y$ obey

$$|X| \leq \delta^{-1}|Y|, \quad |G| \geq \delta^\zeta |X||Y|, \quad |\{x + y : (x, y) \in G\}| \leq \delta^{-\zeta} |X|,$$

then there are $X' \subset X, Y' \subset Y$ with $|X'| \geq \delta^\tau |X|, |Y'| \geq \delta^\tau |Y|$, and $|X' + Y'| \leq \delta^{-\tau} |X'|$.

Theorem III.9.16 (weak discretised ABC theorem). *Let $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\chi, \delta_0 > 0$, depending on (α, β, γ) , such that for every dyadic $0 < \delta < \delta_0$, if $A, B \subset \delta\mathbb{Z} \cap [0, 1], C \subset \delta\mathbb{Z} \cap [1, 2], B, C \neq \emptyset, |A| \leq \delta^{-\alpha}$, and*

$$|B \cap B(x, r)| \leq \delta^{-\chi} r^\beta |B|, \quad |C \cap B(x, r)| \leq \delta^{-\chi} r^\gamma |C| \quad (\delta \leq r \leq 1),$$

then some $c \in C$ satisfies

$$|A + cB|_\delta \geq \delta^{-\chi} |A|. \quad (9.31o)$$

Proof. If $A = \emptyset$, the conclusion is immediate. Henceforth assume $|A| > 0$. Let χ_0 be the gain in Lemma III.9.15. Put $\tau = \chi_0/4$, and let $\zeta = \zeta(\tau) > 0$ be the exponent in the Volume II exponential-scale asymmetric Balog–Szemerédi–Gowers lemma recalled above. Choose $\chi > 0$ so small that

$$\chi < \chi_0/10, \quad \rho := 4\chi/\beta < \chi_0/4, \quad 3\chi + \rho < \zeta. \quad (9.31p)$$

The hypotheses with χ imply those of the perturbation lemma with χ_0 . Suppose, contrary to the conclusion, that $|A + cB|_\delta < \delta^{-\chi} |A|$ for every $c \in C$. Lemma III.9.15 provides $E \subset B^2$, $|E| \leq \delta^{\chi_0} |B|^2$, with its lower bound (9.31l).

For each $c \in C$, the energy of the two finite sets A, cB at tolerance 2δ obeys, by Cauchy–Schwarz and a δ -interval cover of $A + cB$,

$$\#\{(a, a', b_1, b_2) : |a + cb_1 - a' - cb_2| \leq 2\delta\} \geq c \frac{|A|^2 |B|^2}{|A + cB|_\delta} \geq c\delta^\chi |A| |B|^2. \quad (9.31q)$$

Indeed an interval of length δ contributes at least the square of the number of pairs it contains, and any two sums in that interval differ by at most δ . Summing (9.31q) over c , and grouping by (b_1, b_2) , yields a set $\mathcal{P} \subset B^2$ of at least $c\delta^\chi |B|^2$ pairs for which

$$\#\{(a, a', c) \in A^2 \times C : |a - (a' + c(b_2 - b_1))| \leq 2\delta\} \geq c\delta^\chi |A| |C|. \quad (9.31r)$$

The required upper bound for each grouped count is $C|A||C|$: for a fixed (a', c) , at most a fixed number of δ -separated $a \in A$ lie in the displayed interval.

The proportion of pairs with $|b_1 - b_2| < \delta^\rho$ is at most $\delta^{-\chi+\rho\beta} = \delta^{3\chi}$ by the relative Frostman condition on B . Also $|E^\top| = |E| \ll \delta^\chi |B|^2$. Thus we may fix $(b_1, b_2) \in \mathcal{P} \setminus (E^\top \cup \{|b_1 - b_2| < \delta^\rho\})$. Put $t = b_2 - b_1$, so $\delta^\rho \leq |t| \leq 1$ and $(b_2, b_1) \notin E$. From (9.31r), let

$$G = \{(a', c) \in A \times C : \text{dist}(a' + tc, A) \leq 2\delta\}.$$

Each (a', c) accounts for at most a fixed number of triples in (9.31r); hence $|G| \geq c\delta^\chi |A| |C|$, while the restricted sumset $\{a' + tc : (a', c) \in G\}$ has δ -covering number at most $C|A|$.

To apply the lattice BSG lemma without assuming $tC \subset \delta\mathbb{Z}$, round each tc to a nearest point $R(tc) \in \delta\mathbb{Z}$, and put $Y = R(tC) \subset [-2, 2] \cap \delta\mathbb{Z}$. Every rounding fibre contains at most $C|t|^{-1}$ points of the δ -separated set C ; therefore

$$|Y| \geq c|t||C|, \quad |\tilde{G}| \geq c|t||G| \geq c\delta^{\chi+\rho}|A||Y|, \quad (9.31s)$$

where $\tilde{G} = \{(a', R(tc)) : (a', c) \in G\}$. Moreover its restricted sumset has cardinality at most $C|A|$, since it lies within 3δ of A . The size condition $|A| \leq \delta^{-1}|Y|$ in the Volume II BSG input follows from $|A| \leq \delta^{-\alpha}$ and $|Y| \geq 1$. By (9.31p), all BSG hypotheses hold for small enough δ . It gives $A' \subset A, Y' \subset Y$ with

$$|A'| \geq \delta^\tau|A|, \quad |Y'| \geq \delta^\tau|Y|, \quad |A' + Y'| \leq \delta^{-\tau}|A'|.$$

Lift Y' to $C' = \{c \in C : R(tc) \in Y'\}$. Then $|C'| \geq |Y'| \geq c\delta^{\tau+\rho}|C|$ and

$$|A' + tC'|_\delta \leq C|A' + Y'| \leq C\delta^{-\tau}|A'|.$$

Since $2\tau + \rho < \chi_0$, the product $|A'||C'|$ is at least $\delta^{\chi_0}|A||C|$, while the last covering number is at most $\delta^{-\chi_0}|A|$, for small δ . This contradicts (9.31l) for $(b_2, b_1) \notin E$. Hence (9.31o) holds. $\square$

The gain above also bounds the number of coefficients that are bad for any prescribed dense subset of B . It does not yet select one coefficient that works for all such subsets at once; that quantifier change is a separate part of the robust theorem.

Corollary III.9.17 (few bad coefficients for a fixed fibre). *Let $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\eta, \delta_0 > 0$ such that the following holds for all sufficiently small dyadic $\delta < \delta_0$. Suppose A, B, C have the ambient locations and size bound in Theorem III.9.16, and the two relative concentration inequalities there hold with η in place of χ . If $B' \subset B$ and $|B'| \geq \delta^\eta|B|$, then*

$$|\{c \in C : |A + cB'|_\delta < \delta^{-\eta}|A|\}| < \delta^\eta|C|. \quad (9.31t)$$

Proof. Let χ_w be the exponent supplied by Theorem III.9.16, and take $0 < \eta < \chi_w/4$. Write C' for the coefficient set on the left of (9.31t). If $|C'| \geq \delta^\eta|C|$, normalising the concentration estimates by the sizes of B' and C' gives

$$|B' \cap B(x, r)| \leq \delta^{-2\eta}r^\beta|B'|, \quad |C' \cap B(x, r)| \leq \delta^{-2\eta}r^\gamma|C'| \quad (\delta \leq r \leq 1).$$

For sufficiently small δ , these imply the hypotheses of Theorem III.9.16 with exponent χ_w , so there is a $c \in C'$ with $|A + cB'|_\delta \geq \delta^{-\chi_w}|A|$. This contradicts the definition of C' , because $\eta < \chi_w$. $\square$

Definition III.9.18 (tight finite sumset). *For a finite $D \subset \delta\mathbb{Z}$, write $kD = D + \dots + D$ with k summands. Given $\tau > 0, T \geq 1$, and an integer $N \geq 2$, we say that D is (τ, T, N) -tight if*

$$|kD| \leq T|kD'| \quad \text{for every } 1 \leq k \leq N \text{ and every } D' \subset D \text{ with } |D'| \geq \delta^\tau|D|. \quad (9.31u)$$

Tightness is a finite-scale way to say that no dense subfamily has a dramatically smaller short iterated sumset.

Lemma III.9.19 (extraction of a tight subset). *Let $N \geq 2$, $\tau > 0$, and $D \subset \delta\mathbb{Z} \cap [0, 1]$ with $|D| \geq \delta^{-N^2\tau}$. For all sufficiently small δ , there is $D_* \subset D$ such that*

$$|D_*| \geq \delta^{N^2\tau}|D|, \quad D_* \text{ is } (\tau, 2\delta^{-1/N}, N)\text{-tight.} \quad (9.31v)$$

Proof. Set $D_0 = D$. If D_j is not tight, choose $D_{j+1} \subset D_j$ with $|D_{j+1}| \geq \delta^\tau|D_j|$ and some $k_j \in \{1, \dots, N\}$ for which

$$|k_j D_{j+1}| < (2\delta^{-1/N})^{-1} |k_j D_j|. \quad (9.31w)$$

Suppose this selection runs for N^2 steps. Every D_j is nonempty, since $|D_{N^2}| \geq \delta^{N^2\tau}|D| \geq 1$. By pigeonholing, one value k occurs in (9.31w) at least N times. The sumset cardinalities never increase as j increases, so

$$1 \leq |kD_{N^2}| < (2\delta^{-1/N})^{-N} |kD| \leq 2^{-N} \delta(N\delta^{-1} + 1) < 1$$

for small δ , a contradiction. Thus some D_j , $j < N^2$, is tight. Its cardinality is at least $\delta^{N^2\tau}|D|$, proving the claim. $\square$

Lemma III.9.20 (restricted-to-full sumset transfer). *Let $H, D \subset \delta\mathbb{Z}$ be finite nonempty sets, $c \in [0, 2]$, K_1, K_2, K_3 be finite numbers at least 1, and $G \subset H \times D$. If*

$$|H + H| \leq K_1|H|, \quad |D + D| \leq K_2|D|, \quad |G| \geq \frac{|H||D|}{K_3},$$

then

$$|H + cD|_\delta \leq CK_1K_2K_3|\{h + cd : (h, d) \in G\}|_\delta, \quad (9.31x)$$

where C is absolute.

Proof. Let $S_G = \{h + cd : (h, d) \in G\}$, and count grid points within distance 2δ of $H + cD$. Fix such a grid point z , and choose $(h_0, d_0) \in H \times D$ with $|z - (h_0 + cd_0)| \leq 2\delta$. For every $(h, d) \in G$, the distinct point $(x, y) = (h_0 + h, d_0 + d)$ belongs to $(H + H) \times (D + D)$, and

$$|z - (x + cy) + (h + cd)| \leq 2\delta.$$

Therefore the indicator of the 2δ -neighbourhood of $H + cD$, evaluated at a grid point z , is at most

$$\frac{1}{|G|} \sum_{(x,y) \in (H+H) \times (D+D)} \mathbf{1}_{(x+cy) - S_G + [-2\delta, 2\delta]}(z).$$

Sum over grid points. A translate of a set covered by M intervals of length δ has at most CM grid points in its 2δ -neighbourhood. Hence the number of grid points counted on the left is at most $C|H + H||D + D||S_G|_\delta/|G|$. The δ -covering number of $H + cD$ is at most the same grid-point count, up to an absolute factor, yielding (9.31x). $\square$

Proposition III.9.21 (a favourable fibre with few bad coefficients). *Fix $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\eta, \delta_0 > 0$ such that, if $0 < \delta < \delta_0$ is dyadic and A, B, C satisfy the location and size hypotheses of Theorem III.9.16 and its relative concentration bounds with exponent η , then there is a nonempty $D \subset B$ for which*

$$\left| \left\{ c \in C : \begin{array}{l} \text{some } D' \subset D, \quad |D'| \geq \delta^\eta |D|, \\ |A + cD'|_\delta < \delta^{-\eta} |A| \end{array} \right\} \right| \leq \delta^\eta |C|. \quad (9.31y)$$

Proof. The conclusion is immediate when $A = \emptyset$, so assume $A \neq \emptyset$. Choose $\alpha < \alpha_1 < \min\{\beta + \gamma, 1\}$, and let $\vartheta > 0$ be the gain of Theorem III.9.16 for $(\alpha_1, \beta, \gamma)$. Choose an integer N so large, with $m = \lfloor \log_2 N \rfloor$, that

$$\frac{3}{N} + \frac{1}{m} < \frac{\vartheta}{4}. \quad (9.31z)$$

Put $L = N + 1$, $Q = N^2 + 2$, and $\tau_j = Q^{L-j}\eta$ for $0 \leq j \leq L$. After these choices, take $\eta > 0$ so small that

$$L\tau_0 + \tau_0 + 2\eta < \beta, \quad L\tau_0 + \eta < \vartheta/4, \quad N^L\eta < (\alpha_1 - \alpha)/4. \quad (9.31aa)$$

All fixed constants below may depend on $(\alpha, \beta, \gamma, N)$, but not on δ .

Suppose that (9.31y) fails for every nonempty $D \subset B$. Thus each such D has a bad-coefficient set $\mathcal{E}(D) \subset C$ of cardinality $> \delta^\eta|C|$, and each $c \in \mathcal{E}(D)$ has a witness $\bar{D}_c \subset D$ with

$$|\bar{D}_c| \geq \delta^\eta|D|, \quad |A + c\bar{D}_c|_\delta < \delta^{-\eta}|A|. \quad (9.31ab)$$

All choices here are finite. We construct $B = B_0 \supset B_1 \supset \dots \supset B_L$, finite lattice sets $H_0, \dots, H_L$, and $c_j \in C$, so that

$$\begin{aligned} H_0 &= \{0\}, & |B_{j+1}| &\geq \delta^{\tau_j}|B_j| \quad (0 \leq j < L), \\ B_j &\text{ is } (\tau_j, 2\delta^{-1/N}, N)\text{-tight}, \\ |A + c_j B_j|_\delta &< \delta^{-\eta}|A| \quad (1 \leq j \leq L). \end{aligned} \quad (9.31ac)$$

The last two assertions are required only for $j \geq 1$. In particular,

$$|B_j| \geq \delta^{L\tau_0}|B|. \quad (9.31ad)$$

The relative concentration bound for B , applied at radius δ around any $b \in B$, gives $|B| \geq \delta^{-\beta+\eta}$.

Here is the induction step. Given B_j, H_j , put $C_{j+1} = \mathcal{E}(B_j)$; then $|C_{j+1}| > \delta^\eta|C|$. For each $c \in C_{j+1}$, choose a witness $\bar{D}_c$ as in (9.31ab). By (9.31ad) and (9.31aa),

$$|\bar{D}_c| \geq \delta^\eta|B_j| \geq \delta^{-\beta+2\eta+L\tau_0} \geq \delta^{-N^2\tau_{j+1}}.$$

Lemma III.9.19 supplies $D_c \subset \bar{D}_c$ that is $(\tau_{j+1}, 2\delta^{-1/N}, N)$ -tight and obeys

$$|D_c| \geq \delta^{N^2\tau_{j+1}}|\bar{D}_c| \geq \delta^{N^2\tau_{j+1}+\eta}|B_j| \geq \delta^{\tau_j}|B_j|. \quad (9.31ae)$$

The last inequality uses $\tau_j = (N^2 + 2)\tau_{j+1}$ and $\tau_{j+1} \geq \eta$.

Write $R_\delta(S)$ for rounding each member of a bounded real set S to a nearest point of $\delta\mathbb{Z}$, and put $S_c = R_\delta(cD_c)$. Among the finitely many pairs (c, D_c) above, choose one maximising $|H_j + NS_c|$. If this maximum is at least $|H_j + H_j|$, let H_{j+1} be the corresponding sumset, set $B_{j+1} = D_c$, and put $c_{j+1} = c$. Otherwise set $H_{j+1} = H_j + H_j$, $B_{j+1} = B_j$, and $c_{j+1} = c_j$. The latter alternative cannot occur for $j = 0$: by (9.31ae), D_c has more than one point for small δ , so $|NS_c| > 1 = |H_0 + H_0|$. Consequently B_1 is tight; in every later doubling alternative it

remains tight because $\tau_{j+1} < \tau_j$. All properties in (9.31ac) follow. The maximal choice also ensures, for every $c \in C_{j+1}$ and its selected candidate D_c ,

$$|H_j + NS_c| \leq |H_{j+1}|, \quad |H_j + H_j| \leq |H_{j+1}|. \quad (9.31af)$$

Each H_j is contained in a fixed interval $[0, M_N]$, where $M_N = 2N^{L+1}$ will do. Since $H_1 \neq \emptyset$, $|H_L|/|H_1| \leq C_N\delta^{-1}$. Pigeonholing the N ratios $|H_{j+1}|/|H_j|$, $1 \leq j < L$, gives an index $1 \leq j < L$ with

$$|H_{j+1}| \leq C_N\delta^{-1/N}|H_j|, \quad |H_j + H_j| \leq C_N\delta^{-1/N}|H_j|. \quad (9.31ag)$$

Fix this j , and write $D = B_j$. The sets $2^k D$, $0 \leq k \leq m$, are nonempty subsets of $\delta\mathbb{Z} \cap [0, N]$. Another pigeonhole gives $0 \leq k < m$ such that

$$\frac{|2^{k+1}D|}{|2^k D|} \leq (C_N\delta^{-1})^{1/m}. \quad (9.31ah)$$

For an arbitrary $c \in C_{j+1}$, choose its tight candidate D_c as above. The set D is $(\tau_j, 2\delta^{-1/N}, N)$ -tight and $|D_c| \geq \delta^{\tau_j}|D|$, hence

$$\frac{|2^k D|}{|2^k D_c|} \leq 2\delta^{-1/N}. \quad (9.31ai)$$

Apply Lemma III.9.20 with $H = H_j$, its second set $2^k D$, and the full Cartesian graph $H_j \times 2^k D_c$. Using (9.31ag)–(9.31ai), and the inclusions of D into a translate of $2^k D$ and of $2^k D_c$ into a translate of ND_c , we obtain

$$\begin{aligned} |H_j + cD|_\delta &\leq C_N\delta^{-2/N-1/m}|H_j + c2^k D_c|_\delta \\ &\leq C_N\delta^{-2/N-1/m}|H_j + NS_c| \\ &\leq C_N\delta^{-3/N-1/m}|H_j|. \end{aligned} \quad (9.31aj)$$

Rounding changes the middle covering number only by a fixed factor (depending on N), absorbed into C_N .

We check the inputs needed to contradict the weak theorem. Every H_j is a sum of at most N^L sets of the form $R_\delta(c_i B_i)$; this follows inductively from the two definitions of H_{i+1} . For each such summand,

$$|A + R_\delta(c_i B_i)| \leq C|A + c_i B_i|_\delta \leq C\delta^{-\eta}|A|.$$

The Plünnecke–Ruzsa inequality for distinct summands, available in the additive-combinatorics background, therefore gives

$$|H_j| \leq C_N\delta^{-N^L\eta}|A| \leq \delta^{-\alpha_1} \quad (9.31ak)$$

for small δ , by (9.31aa). Select a most populous unit interval in $[0, M_N]$, translate its H_j -points by an integer, and call the result $H \subset \delta\mathbb{Z} \cap [0, 1]$. Then $|H| \geq C_N^{-1}|H_j|$, while (9.31aj) implies

$$|H + cD|_\delta < \delta^{-\vartheta}|H| \quad \text{for every } c \in C_{j+1} \quad (9.31al)$$

once δ is small, by (9.31z). Finally, (9.31ad) and the original concentration assumptions imply

$$|D \cap B(x, r)| \leq \delta^{-(\eta + L\tau_0)} r^\beta |D|, \quad |C_{j+1} \cap B(x, r)| \leq \delta^{-2\eta} r^\gamma |C_{j+1}|.$$

Both losses are smaller than ϑ by (9.31aa). Thus (H, D, C_{j+1}) satisfies Theorem III.9.16 with parameters $(\alpha_1, \beta, \gamma)$, contradicting (9.31al). $\square$

The preceding proposition is deliberately asymmetric: it produces one favourable subfamily of B , not yet a coefficient good for all large subfamilies of the original B . The finite exhaustion below is the step that reverses these quantifiers. It follows the strategy of [19, Sections 5.3–5.4], with the present relative Frostman losses absorbed directly into the weak theorem.

Theorem III.9.22 (uniform expansion over dense fibres). *Fix $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\varepsilon, \delta_0 > 0$ such that, for every small dyadic $\delta < \delta_0$, if A, B, C obey the location, size, and relative concentration hypotheses of Theorem III.9.16 with ε in place of χ , then some $c \in C$ satisfies*

$$|A + cB'|_\delta \geq \delta^{-\varepsilon} |A| \quad \text{for every } B' \subset B \text{ with } |B'| \geq \delta^\varepsilon |B|. \quad (9.31am)$$

Proof. If $A = \emptyset$, there is nothing to prove. Let $\eta > 0$ be supplied by Proposition III.9.21, and choose $0 < \varepsilon < \eta/4$. Construct disjoint nonempty blocks $D_1, D_2, \dots \subset B$ inductively. Suppose the blocks already chosen leave a remainder $R \subset B$. If $|R| < \delta^{2\varepsilon} |B|$, stop. Otherwise the relative concentration estimate for B gives

$$|R \cap B(x, r)| \leq \delta^{-3\varepsilon} r^\beta |R| \leq \delta^{-\eta} r^\beta |R|,$$

and C meets the corresponding hypothesis with exponent η . Apply Proposition III.9.21 to (A, R, C) , and choose as the next block a nonempty $D_j \subset R$ whose exceptional coefficient set

$$E_j = \{c \in C : \text{some } D' \subset D_j, \quad |D'| \geq \delta^\eta |D_j|, \quad |A + cD'|_\delta < \delta^{-\eta} |A|\}$$

has $|E_j| \leq \delta^\eta |C|$. The process terminates, since a nonempty block is removed at every step. Write $B_0 = \bigcup_j D_j$; its complement has size less than $\delta^{2\varepsilon} |B|$.

Suppose $c \in C$ violates (9.31am), witnessed by a set $B_c \subset B$ of size at least $\delta^\varepsilon |B|$ and with $|A + cB_c|_\delta < \delta^{-\varepsilon} |A|$. Let $J(c) = \{j : |B_c \cap D_j| \geq \delta^\eta |D_j|\}$. For every $j \in J(c)$, the set $B_c \cap D_j$ witnesses $c \in E_j$, since $\varepsilon < \eta$ and $A + c(B_c \cap D_j) \subset A + cB_c$. On the other hand, for sufficiently small δ ,

$$\begin{aligned} \delta^\varepsilon |B| &\leq |B_c| \leq |B \setminus B_0| + \sum_{j \notin J(c)} |B_c \cap D_j| + \sum_{j \in J(c)} |D_j| \\ &\leq (\delta^{2\varepsilon} + \delta^\eta) |B| + \sum_{j \in J(c)} |D_j| \leq \frac{1}{2} \delta^\varepsilon |B| + \sum_{j \in J(c)} |D_j|. \end{aligned}$$

Thus every bad c satisfies $\sum_j (|D_j|/|B|) \mathbf{1}_{E_j}(c) \geq \frac{1}{2} \delta^\varepsilon$. Averaging this inequality over $c \in C$ and using $|E_j| \leq \delta^\eta |C|$ gives

$$\frac{|\{c \in C : c \text{ is bad}\}|}{|C|} \leq 2\delta^{-\varepsilon} \sum_j \frac{|D_j|}{|B|} \frac{|E_j|}{|C|} \leq 2\delta^{\eta-\varepsilon} < 1.$$

Some $c \in C$ is therefore good for every eligible B_c , as claimed. $\square$

Theorem III.9.23 (robust discretised ABC theorem). *Let $0 < \beta \leq \alpha < 1$ and $\gamma \in (\alpha - \beta, 1]$. There exist $\chi, \delta_0 \in (0, 1/2]$ such that, for $0 < \delta \leq \delta_0$, if*

$$A, B \subset \delta\mathbb{Z} \cap [0, 1], \quad C \subset \delta\mathbb{Z} \cap [1, 2],$$

with $B, C \neq \emptyset$,

$$|A| \leq \delta^{-\alpha}, \quad (9.32)$$

$$|B \cap B(x, r)| \leq \delta^{-\chi} r^\beta |B|, \quad \delta \leq r \leq 1, \quad (9.33)$$

and

$$|C \cap B(x, r)| \leq \delta^{-\chi} r^\gamma |C|, \quad \delta \leq r \leq 1, \quad (9.34)$$

then there exists $c \in C$ such that

$$|\Phi_c(G)|_\delta \geq \delta^{-\chi} |A| \quad (9.35)$$

for every $G \subset A \times B$ with $|G| \geq \delta^\chi |A||B|$. This is the finite-coefficient form of Theorem 1.7 of [18].

Proof. The case $A = \emptyset$ is immediate, so assume $A \neq \emptyset$. We first prove the assertion for dyadic δ , with a provisional gain denoted by χ ; the final all-scale gain will be selected after this argument. Choose $\alpha < \alpha_1 < \min\{\beta + \gamma, 1\}$, and let $\vartheta > 0$ be the gain of Theorem III.9.22 for $(\alpha_1, \beta, \gamma)$. Fix an integer $N \geq 3$ such that

$$\frac{1}{N-1} < \frac{\vartheta}{4}. \quad (9.35d)$$

We use the Volume II asymmetric BSG lemma stated above, with an output exponent $\zeta > 0$ chosen sufficiently small in terms of $N, \vartheta, \alpha_1 - \alpha$. Let $\xi > 0$ be its input exponent. Finally choose the present theorem's $\chi > 0$ so small that $\chi < \xi/4$, and so that every inequality below with a fixed positive exponent margin is valid for sufficiently small δ . For definiteness one may require, with $\lambda = 4\zeta$,

$$N\lambda + \chi < \vartheta/4, \quad N(N+1)\lambda < (\alpha_1 - \alpha)/4, \quad \lambda < \vartheta/(8N). \quad (9.35e)$$

The two bounds involving λ are secured by choosing ζ first; then choose χ below both $\xi/4$ and $\vartheta/4 - N\lambda$.

Suppose the desired coefficient does not exist. For each $c \in C$, select a graph $G_c \subset A \times B$ with

$$|G_c| \geq \delta^\chi |A||B|, \quad |\Phi_c(G_c)|_\delta < \delta^{-\chi} |A|. \quad (9.35f)$$

Round cB to $\delta\mathbb{Z}$. Since $c \in [1, 2]$ and $B \subset \delta\mathbb{Z}$, the rounding map is injective after a fixed choice of ties. Its image Y_c has size $|B|$, and the induced graph in $A \times Y_c$ has the same size as G_c , while its restricted sumset has cardinality at most $C|\Phi_c(G_c)|_\delta$. Also $|A| \leq \delta^{-1}|Y_c|$. The Volume II BSG lemma applies, by $\chi < \xi/4$ and smallness of δ , and yields $A_c \subset A$, $B_c \subset B$ such that

$$|A_c| \geq \delta^\zeta |A|, \quad |B_c| \geq \delta^\zeta |B|, \quad |A_c + cB_c|_\delta \leq C\delta^{-\zeta} |A_c|. \quad (9.35g)$$

Decreasing δ_0 absorbs the fixed constants into the exponent $\lambda = 4\zeta$; in particular, for every $c \in C$,

$$|A_c||B_c| \geq \delta^\lambda |A||B|, \quad |A_c + cB_c|_\delta \leq \delta^{-\lambda} |A|. \quad (9.35h)$$

Let $P_c = A_c \times B_c \subset A \times B$, and take N independent uniformly distributed coefficients from C . For $(a, b) \in A \times B$, write $p(a, b) = |\{c : (a, b) \in P_c\}|/|C|$. Jensen's inequality gives

$$\frac{1}{|C|^N} \sum_{(c_1, \dots, c_N) \in C^N} \left| \bigcap_{i=1}^N P_{c_i} \right| = \sum_{(a, b) \in A \times B} p(a, b)^N \geq \delta^{N\lambda} |A||B|. \quad (9.35i)$$

Consequently the set

$$\Omega = \left\{ (c_1, \dots, c_N) \in C^N : \left| \bigcap_{i=1}^N P_{c_i} \right| \geq \frac{1}{2} \delta^{N\lambda} |A||B| \right\}$$

has relative size at least $q := \frac{1}{2} \delta^{N\lambda}$. Indeed the integrand in (9.35i) is bounded by $|A||B|$.

For a prefix $(c_1, \dots, c_j)$, let $f_j(c_1, \dots, c_j)$ be the proportion of its completions in C^{N-j} that belong to Ω ; thus $f_0 \geq q$ and f_j is the average of f_{j+1} over the next coefficient. Starting from the empty prefix, choose $c_1, \dots, c_N$ successively so that

$$f_j(c_1, \dots, c_j) \geq 2^{-j} q. \quad (9.35j)$$

At stage $j < N$, the finite set

$$C_{j+1} = \{c \in C : f_{j+1}(c_1, \dots, c_j, c) \geq 2^{-j-1} q\}$$

has cardinality at least $2^{-j-1} q|C|$: otherwise the average of $f_{j+1} \in [0, 1]$ would be less than $2^{-j} q$. Every $c \in C_{j+1}$ extends the current prefix to some member of Ω . Hence, with

$$A_j = \bigcap_{i=1}^j A_{c_i}, \quad B_j = \bigcap_{i=1}^j B_{c_i}, \quad B_{j,c} = B_j \cap B_c,$$

we have, for every such c ,

$$|A_j| \geq q|A|, \quad |B_{j,c}| \geq q|B| \geq q|B_j|. \quad (9.35k)$$

At each stage choose $c_{j+1} \in C_{j+1}$ to maximise $|H_j + R_\delta(cB_{j,c})|$, with $H_0 = \{0\}$ and $H_{j+1} = H_j + R_\delta(c_{j+1}B_{j+1})$. Here R_δ is the rounding map used in the preceding proposition. By construction,

$$|H_j + R_\delta(cB_{j,c})| \leq |H_{j+1}| \quad (c \in C_{j+1}). \quad (9.35l)$$

All these sets are finite and nonempty.

We first bound their size. The selected prefix extends to Ω , so $|A_j| \geq q|A|$. Since $B_i \subset B_{c_i}$ and $A_j \subset A_{c_i}$ for $i \leq j$, (9.35h) gives

$$|A_j + R_\delta(c_i B_i)| \leq C \delta^{-\lambda} |A| \leq C q^{-1} \delta^{-\lambda} |A_j|.$$

The different-summand Plünnecke–Ruzsa inequality, applied with base A_j to the j summands of H_j , therefore yields

$$|H_j| \leq C_N \delta^{-j(N+1)\lambda} |A_j| \leq C_N \delta^{-N(N+1)\lambda} |A| \leq \delta^{-\alpha_1} \quad (9.35m)$$

for small δ , by (9.35e). Also $H_j \subset [0, 3N]$, so $|H_N| \leq C_N \delta^{-1}$. Pigeonholing the $N - 1$ ratios from H_1 to H_N , choose $1 \leq j < N$ such that

$$|H_{j+1}| \leq C_N \delta^{-1/(N-1)} |H_j|. \quad (9.35n)$$

For each $c \in C_{j+1}$, rounding and (9.35l) give

$$|H_j + cB_{j,c}|_\delta \leq C |H_j + R_\delta(cB_{j,c})| \leq C_N \delta^{-1/(N-1)} |H_j|. \quad (9.35o)$$

Select a most populous unit interval in $[0, 3N]$, translate its H_j -points into $[0, 1]$, and call the result $\tilde{H}_j$. It is a subset of $\delta\mathbb{Z}$, satisfies $|\tilde{H}_j| \geq C_N^{-1} |H_j|$, and obeys $|\tilde{H}_j| \leq \delta^{-\alpha_1}$. By (9.35d), (9.35o) implies

$$|\tilde{H}_j + cB_{j,c}|_\delta < \delta^{-\vartheta} |\tilde{H}_j| \quad (c \in C_{j+1}) \quad (9.35p)$$

for small δ . Moreover (9.35k) and $\lambda < \vartheta/(8N)$ ensure $|B_{j,c}| \geq \delta^\vartheta |B_j|$. The concentration assumptions for B_j and C_{j+1} , after normalisation by their sizes, have constants at most $C_N \delta^{-(\lambda+N\lambda)}$, hence at most $\delta^{-\vartheta}$ by (9.35e) and smallness of δ . Thus $(\tilde{H}_j, B_j, C_{j+1})$ meets every hypothesis of Theorem III.9.22 with parameters $(\alpha_1, \beta, \gamma)$. That theorem gives a $c \in C_{j+1}$ for which $|\tilde{H}_j + cB_{j,c}|_\delta \geq \delta^{-\vartheta} |\tilde{H}_j|$, contradicting (9.35p).

To pass from dyadic to arbitrary scales, note that the dyadic argument was valid for every admissible triple of exponents. Choose $\alpha < \alpha_+ < \min\{\beta + \gamma, 1\}$ and let $\chi_d > 0$ be its dyadic gain for $(\alpha_+, \beta, \gamma)$. Choose the final gain $0 < \chi < \chi_d/4$ and reduce δ_0 as needed. Given any $0 < \delta < \delta_0$, choose a dyadic Δ with $\delta \leq \Delta < 2\delta$, and round each member of A, B, C to a nearest point of $\Delta\mathbb{Z}$, obtaining $A_\Delta, B_\Delta, C_\Delta$. The endpoints 0, 1, 2 belong to $\Delta\mathbb{Z}$, so these sets retain their respective ambient intervals. Every rounding fibre has at most three members, whence

$$|A_\Delta| \leq |A| \leq \Delta^{-\alpha_+}, \quad |A_\Delta| \geq |A|/3, \quad |B_\Delta| \geq |B|/3, \quad |C_\Delta| \geq |C|/3 \quad (9.35q)$$

for small δ . For $\Delta \leq r \leq 1/2$, the preimages of $B_\Delta \cap B(x, r)$ lie in $B \cap B(x, 2r)$; the same is true for C . At larger radii their cardinalities are trivially at most $|B_\Delta|$ and $|C_\Delta|$. Consequently the two relative concentration inequalities transfer to B_Δ, C_Δ with constants $C\Delta^{-\chi}$, hence with the required $\Delta^{-\chi_d}$ for small δ .

For $G \subset A \times B$, let $G_\Delta \subset A_\Delta \times B_\Delta$ be its rounded image. Its fibres have at most nine pairs, so

$$|G_\Delta| \geq |G|/9 \geq \frac{1}{9} \delta^\chi |A||B| \geq \Delta^{\chi_d} |A_\Delta||B_\Delta| \quad (9.35r)$$

for small δ . The dyadic assertion therefore supplies $c_\Delta \in C_\Delta$ that works for every such G_Δ . Fix any $c \in C$ rounding to c_Δ . For $(a, b) \in G$, its rounded image (a_Δ, b_Δ) satisfies

$$|(a + cb) - (a_\Delta + c_\Delta b_\Delta)| \leq 3\Delta.$$

Thus each δ -interval covering $\Phi_c(G)$ accounts for only $O(1)$ Δ -intervals covering $\Phi_{c_\Delta}(G_\Delta)$. The dyadic lower bound and (9.35q) yield

$$|\Phi_c(G)|_\delta \geq C^{-1} |\Phi_{c_\Delta}(G_\Delta)|_\Delta \geq C^{-1} \Delta^{-\chi_d} |A_\Delta| \geq \delta^{-\chi} |A|$$

after one last decrease of δ_0 . The chosen lift c is independent of G , which completes the all-scale proof. $\square$

Corollary III.9.24 (Frostman coefficient version). *Let $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$. There are $\chi, \delta_0 > 0$ such that for $0 < \delta < \delta_0$, if $A, B \subset \delta\mathbb{Z} \cap [0, 1]$, $|A| \leq \delta^{-\alpha}$, $B \neq \emptyset$,*

$$|B \cap B(x, r)| \leq \delta^{-\chi} r^\beta |B| \quad (\delta \leq r \leq 1),$$

and ν is a Borel probability measure on $[1, 2]$ satisfying

$$\nu(B(x, r)) \leq \delta^{-\chi} r^\gamma \quad (\delta \leq r \leq 1),$$

then some $c \in \text{spt } \nu$ satisfies

$$|\Phi_c(G)|_\delta \geq \delta^{-\chi} |A| \quad \text{for every } G \subset A \times B \text{ with } |G| \geq \delta^\chi |A| |B|. \quad (9.35s)$$

Proof. Let χ_d be the gain in Theorem III.9.23, and choose $0 < \chi < \min\{\chi_d/8, \gamma/4\}$. Partition $[1, 2]$ into Voronoi cells of the grid $\delta\mathbb{Z} \cap [1, 2]$. There are at most $C\delta^{-1}$ cells, each of diameter at most 2δ . Write w_j for its ν -mass and discard those with $w_j < \delta^3$; their total mass is at most $C\delta^2$. The remaining weights lie in at most $C \log(2/\delta)$ dyadic size classes. For one class of grid points C_δ , the corresponding cells have total mass

$$m := \sum_{j \in C_\delta} w_j \geq c / \log(2/\delta),$$

and all their weights lie in an interval $[w/2, w]$ for some $w > 0$. For $\delta \leq r \leq 1/3$, each cell whose grid point belongs to $B(x, r)$ lies in $B(x, 3r)$. Hence

$$\frac{|C_\delta \cap B(x, r)|}{|C_\delta|} \leq \frac{2\nu(B(x, 3r))}{m} \leq C \log(2/\delta) \delta^{-\chi} r^\gamma. \quad (9.35t)$$

For $r > 1/3$, the same estimate follows from the trivial upper bound 1. Thus C_δ satisfies the relative coefficient condition of Theorem III.9.23 with exponent χ_d for sufficiently small δ ; the same is true of B . That theorem supplies $c_\delta \in C_\delta$ working uniformly for all graphs of density at least δ^{χ_d} .

The cell corresponding to c_δ has positive mass, so choose $c \in \text{spt } \nu$ in that cell. Then $|c - c_\delta| \leq 2\delta$. For every $G \subset A \times B$, the two projected sets $\Phi_c(G)$ and $\Phi_{c_\delta}(G)$ lie within 2δ pointwise; consequently

$$|\Phi_c(G)|_\delta \geq C^{-1} |\Phi_{c_\delta}(G)|_\delta.$$

If $|G| \geq \delta^\chi |A| |B|$, then $|G| \geq \delta^{\chi_d} |A| |B|$. The finite theorem and $\chi < \chi_d$ now give (9.35s) after decreasing δ_0 . $\square$

Proof of Theorem III.9.8. If $(\alpha - \beta) + \kappa \geq 1$, the conclusion follows from the ambient dimension bound. Otherwise put $\gamma' = \alpha - \beta + \kappa/2 < 1$. Choose $d > 0$ sufficiently small that

$$\bar{\alpha} := \alpha + d < 1, \quad \beta' := \beta - d > 0, \quad \gamma' > \bar{\alpha} - \beta'. \quad (9.35a)$$

Apply the proved discretised ABC theorem with $(\bar{\alpha}, \beta', \gamma')$, and call its gain $\chi > 0$. Choose $\alpha_0 < \alpha$ so close to α that $\alpha - \alpha_0 < \chi/4$, and then fix

$$0 < \eta < \min\{d/4, \chi/8\}, \quad s := \alpha + 2\eta. \quad (9.35b)$$

Thus $s < \bar{\alpha}$ and $s < \alpha_0 + \chi$.

Suppose for contradiction that the exceptional set in (9.31) has Hausdorff dimension greater than $(\alpha - \beta) + \kappa$. Restricting to a compact subset and making fixed affine changes of variables, we may work with compact $A, B \subset [0, 1]$ of dimensions greater than α_0 and β_0 , where $\beta' < \beta_0 < \beta$, and a compact exceptional coefficient set $E \subset [1, 2]$ carrying a probability measure ν with

$$\nu(B(c, r)) \leq Cr^{\gamma'}, \quad \gamma' = \alpha - \beta + \kappa/2, \quad \gamma' < \gamma < (\alpha - \beta) + \kappa. \quad (9.35c)$$

The compact sets A, B carry probability measures μ_A, μ_B with $\mu_A(B(x, r)) \leq Cr^{\alpha_0}$ and $\mu_B(B(x, r)) \leq Cr^{\beta_0}$. To justify the reduction, first take compact Frostman subsets of the original Borel sets. Replacing them by subsets preserves the exceptional inequality. Divide the coefficient line away from zero into countably many intervals of bounded multiplicative width, and rescale B and, if necessary, reflect it. Since a countable union has dimension equal to the supremum of its pieces, one such interval retains dimension greater than $(\alpha - \beta) + \kappa$. Finally use Frostman's lemma there. The exceptional coefficient set is Borel for compact A, B : for each $u > 0$, the condition $\mathcal{H}^u(A + cB) = 0$ has a countable description by finite dyadic covers of the compact set $A + cB$.

Fix a large integer j_0 . Since $\dim_H(A + cB) \leq \alpha + \eta < s$ for $c \in E$, choose for each such c a finite cover of $A + cB$ by enlarged dyadic intervals of lengths 2^{-j} , $j \geq j_0$, whose total s -cost is at most 1. The cover may be selected measurably: enumerate the countable family of finite dyadic covers and take the first admissible one. Write $\mathcal{J}_{c,j}$ for its intervals of length 2^{-j} , and

$$G_{c,j} := \{(a, b) \in A \times B : a + cb \in \bigcup \mathcal{J}_{c,j}\}.$$

For every c , $\sum_{j \geq j_0} (\mu_A \times \mu_B)(G_{c,j}) \geq 1$. Integrating in c and using $\sum_{j \geq j_0} j^{-2} < \infty$ gives an arbitrarily large $j \geq j_0$ for which, with $\delta = 2^{-j}$,

$$\int_E (\mu_A \times \mu_B)(G_{c,j}) d\nu(c) \geq c_1 j^{-2}, \quad |\mathcal{J}_{c,j}| \leq \delta^{-s}. \quad (9.35d)$$

Here $c_1 > 0$ is absolute, and the second bound follows from the s -cost of the selected cover. Markov's inequality, applied to the complement of a superlevel set, gives a Borel $E_1 \subset E$ with $\nu(E_1) \geq c_2 j^{-2}$ and $(\mu_A \times \mu_B)(G_{c,j}) \geq c_2 j^{-2}$ for $c \in E_1$.

Partition the dyadic δ -cells meeting $[0, 1]$ into mass classes $\rho \leq \mu(Q) < 2\rho$, with dyadic $\rho \geq \delta^3$. The omitted cells have total mass $O(\delta^2)$ for either measure. There are $O(j)$ classes for each measure. Pigeonholing the integral in (9.35d) first over these classes and then over c yields fixed classes $\mathcal{A}, \mathcal{B}$, their masses ρ_A, ρ_B , a Borel $E_2 \subset E_1$, and a constant L independent of j such that

$$\nu(E_2) \geq j^{-L}, \quad (\mu_A \times \mu_B)(G_{c,j} \cap (\cup \mathcal{A} \times \cup \mathcal{B})) \geq j^{-L} \quad (c \in E_2). \quad (9.35e)$$

Write $q = j^{-L}$, decreasing it by a fixed factor if needed. Choose a compact $E_3 \subset E_2$ of ν -mass at least $q/2$, and normalize $\nu|_{E_3}$ to ν_3 . By (9.35c), for all sufficiently large j ,

$$\nu_3(B(c, r)) \leq \delta^{-\chi} r^{\gamma'}, \quad \delta \leq r \leq 1. \quad (9.35f)$$

Let $A_\delta, B_\delta \subset \delta\mathbb{Z}$ be the left endpoints of the cells in $\mathcal{A}, \mathcal{B}$, and let $N_A = |A_\delta|, N_B = |B_\delta|$. Because both selected classes have mass at least q , the Frostman bound on μ_A gives

$$N_A \geq \frac{q}{2\rho_A} \geq cq\delta^{-\alpha_0}. \quad (9.35g)$$

On the other hand, for any fixed $c \in E_3$, Fubini applied to (9.35e) gives a b for which at least q of μ_A -mass in the selected class is carried by $\{a : a + cb \in \bigcup \mathcal{J}_{c,j}\}$. This set meets at most $C\delta^{-s}$ dyadic δ -cells. Consequently

$$q \leq C\rho_A\delta^{-s}, \quad N_A \leq \rho_A^{-1} \leq Cq^{-1}\delta^{-s} \leq \delta^{-\tilde{\alpha}} \quad (9.35h)$$

for all sufficiently large j . Similarly, if $\delta \leq r \leq 1$, the lower mass ρ_B of every selected cell and the Frostman bound on μ_B imply

$$|B_\delta \cap B(x, r)| \leq C\rho_B^{-1}r^{\beta_0} \leq Cq^{-1}r^{\beta_0}N_B \leq \delta^{-\chi}r^{\beta'}N_B. \quad (9.35i)$$

The finite-scale theorem therefore applies to $(A_\delta, B_\delta, v_3)$.

For each $c \in E_3$, form a graph $H_c \subset A_\delta \times B_\delta$ by retaining the pair of cell endpoints whenever the corresponding product cell meets $G_{c,j} \cap (\bigcup \mathcal{A} \times \bigcup \mathcal{B})$. Every product cell has product mass at most $4\rho_A\rho_B$; as $N_A\rho_A, N_B\rho_B \leq 1$, (9.35e) gives

$$|H_c| \geq \frac{q}{4\rho_A\rho_B} \geq \frac{q}{4}N_A N_B \geq \delta^\chi N_A N_B. \quad (9.35j)$$

The selected dyadic cover also gives $|\Phi_c(H_c)|_\delta \leq C\delta^{-s}$: moving from a point of a product cell to its endpoints changes $a + cb$ by at most 3δ , uniformly for $c \in [1, 2]$. The measure form of the discretised ABC theorem supplies some $c \in E_3$ with

$$C\delta^{-s} \geq |\Phi_c(H_c)|_\delta \geq \delta^{-\chi}N_A \geq cq\delta^{-\chi-\alpha_0}.$$

This is impossible as $j \rightarrow \infty$, since $q = j^{-L} = \delta^{o(1)}$ and $s < \alpha_0 + \chi$ by (9.35b). The contradiction proves (9.31). $\square$

Corollary III.9.25 (qualitative ABC consequence). *Let $A, B, C \subset \mathbb{R}$ be Borel sets satisfying*

$$0 < \dim_H B \leq \dim_H A < 1, \quad \dim_H C > \dim_H A - \dim_H B. \quad (9.36)$$

Then some $c \in C$ satisfies

$$\dim_H(A + cB) > \dim_H A. \quad (9.37)$$

Proof. Set $\alpha = \dim_H A$ and $\beta = \dim_H B$. Choose $\kappa > 0$ such that

$$\alpha - \beta + \kappa < \dim_H C.$$

By Theorem III.9.8, for some $\eta > 0$ the exceptional coefficient set

$$Z = \{c : \dim_H(A + cB) \leq \alpha + \eta\}$$

has Hausdorff dimension at most $\alpha - \beta + \kappa$. Hence $C \not\subset Z$. Any $c \in C \setminus Z$ satisfies $\dim_H(A + cB) > \alpha + \eta > \alpha$, proving (9.37). $\square$

III.9.6 Sharp exceptional projections for equal dimensions

The robust projection statement is best understood through the number of small squares met by a line in the support. We give the precise intermediate assertion before deriving the stated projection form. Its base case already follows from Bourgain's theorem proved in Chapter III.6. The remaining improvement is the research-scale bootstrap proved below.

Definition (regular measure and high-multiplicity set). *Let $0 < \delta \leq 1$, $0 < t < 2$, and $C \geq 1$. A finite measure μ with support K is (δ, t, C) -regular if $\mu(B(x, r)) \leq Cr^t$ for $\delta \leq r \leq 8$ and*

$$|K \cap B(x, R)|_r \leq C(R/r)^t \quad (\delta \leq r \leq R \leq 8). \quad (9.37a)$$

Here $|S|_r$ denotes the number of axis-parallel half-open squares $r(k + [0, 1)^2)$, $k \in \mathbb{Z}^2$, meeting S ; for dyadic r this is the usual dyadic grid. The support need not be confined to the unit ball. For a direction θ , $x \in K$, and $\delta \leq r \leq R \leq 8$, put

$$\begin{aligned} m_{K, \theta}(x \mid [r, R]) &= |B(x, R) \cap K^{(r)} \cap \pi_\theta^{-1}\{\pi_\theta(x)\}|_r, \\ H_\theta(K, M; [r, R]) &= \{x \in K : m_{K, \theta}(x \mid [r, R]) \geq M\}. \end{aligned} \quad (9.37b)$$

The r -neighbourhood $K^{(r)}$ makes this count stable under moving a centre by $O(r)$; $|\cdot|_r$ counts squares meeting the displayed line, not squares whose centres happen to lie on it.

Lemma III.9.25.1 (the locally proved Bourgain base case). *Fix $0 < t < 2$ and $s > 0$. There is $\eta = \eta(s, t) > 0$ such that, for sufficiently small δ , every $(\delta, t, \delta^{-\eta})$ -regular measure μ and $(\delta, s, \delta^{-\eta})$ direction set E admit $\theta \in E$ with*

$$\mu(B(0, 1) \cap H_\theta(\text{spt } \mu, \delta^{-t/2+\eta}; [\delta, 1])) \leq \delta^\eta. \quad (9.37c)$$

Proof. Choose $0 < \kappa < \min\{s, t, 1\}$. Apply the planar case of the locally proved robust Bourgain theorem III.6.6 with the fixed parameters (t, κ) ; write $\tau, \gamma > 0$ for its allowed hypothesis loss and its projection gain. Choose λ so small that $10\lambda < \min\{\tau, \gamma, t\}$, and set $\eta = \lambda$.

Suppose, contrariwise, that for every $\theta \in E$ the high-multiplicity set B_θ in (9.37c) has $\mu(B_\theta) > \delta^\lambda$. Let X be one representative of every δ -square meeting $K \cap B(0, 2)$. The regularity assumptions give

$$\delta^{-t+2\lambda} \lesssim |X| \lesssim \delta^{-t-\lambda}, \quad |X \cap B(z, r)| \lesssim \delta^{-3\lambda} r^t |X| \quad (\delta \leq r \leq 1). \quad (9.37d)$$

The first lower bound follows from $\mu(B_\theta) > \delta^\lambda$ and the mass bound $\mu(Q) \lesssim \delta^{t-\lambda}$ for a terminal square Q ; the upper and local bounds follow from support regularity. Fixed changes of δ and of the containing ball do not affect the chosen exponent margins. The uniform probability on a maximal δ -separated subset of E gives a direction law ρ satisfying $\rho(\ell^{(r)} \cap S^1) \lesssim \delta^{-2\lambda} r^\kappa$ for every line ℓ and $\delta \leq r \leq 1$. After a fixed similarity placing X in $B(0, 1)$, we may apply Theorem III.6.6 directly: (9.37d) supplies its source cardinality and concentration bounds because $\kappa < t$, and the preceding bound supplies its direction condition. For each θ , let $X_\theta \subset X$ consist of the previously chosen representatives of terminal squares meeting B_θ . Then $|X_\theta| \gtrsim \delta^{-t+2\lambda} \geq \delta^\tau |X|$ after reducing λ .

On the other hand, if J is the size of a 10δ -separated set in $\pi_\theta(B_\theta)$, each selected line meets at least $\delta^{-t/2+\lambda}$ distinct terminal squares of $K^{(\delta)} \cap B(0, 2)$. These families of squares have bounded overlap as the projected coordinates vary by at least 10δ . Since their total number is $O(\delta^{-t-\lambda})$, we obtain

$$|\pi_\theta(X_\theta)|_\delta \lesssim \delta^{-t/2-2\lambda}. \quad (9.37e)$$

Theorem III.6.6, with source parameter t , instead gives for some $\theta \in E$ outside a ρ -set of mass < 1 the lower bound $|\pi_\theta(X_\theta)|_\delta \geq \delta^{-t/2-\gamma}$. This contradicts (9.37e) when $2\lambda < \gamma$ and δ is small. $\square$

The next lemma separates two phenomena that are easy to conflate. A line may encounter many squares of the *support* even when those squares carry very little measure. Consequently upper Frostman growth cannot convert geometric multiplicity into a lower mass bound. Orponen's later tube argument instead sorts projection tubes by the mass they actually carry. We give its finite-scale form, including the small additive error that is essential under upper regularity; compare [21, Lemma 2.4].

Lemma III.9.25.2 (mass-aware multiplicity splitting). *Let $0 < t < 2$, $C \geq 1$, $0 < \delta \leq \Delta/8 \leq 1/8$, $1 \leq M \leq N$, and $0 < \kappa \leq 1$. Suppose that μ is a (δ, t, C) -regular measure with support K . Put $\mu_1 = \mu|_{B(0,1)}$. There is an absolute $c_0 > 0$ such that, for every direction θ ,*

$$\begin{aligned} \mu_1(H_\theta(K, N; [\delta, 1])) &\leq (1 + \kappa)\mu_1(H_\theta(K, M; [3\delta, 3\Delta])) + \kappa \\ &\quad + \mu_1(H_\theta(K, c_0\kappa^2C^{-3}N/M; [\Delta, 3])). \end{aligned} \quad (9.37d')$$

The high-multiplicity sets retain the support K throughout; no restriction or reweighting of μ is implicit.

Proof. Fix θ and abbreviate the three high-multiplicity sets in (9.37d') by G, L, D , in their displayed order. Partition the projection axis into half-open intervals of length δ , and let $\mathcal{T}$ consist of the corresponding disjoint tubes $T = \pi_\theta^{-1}(I)$ which meet $G \cap B(0, 1)$. For each such tube choose $x_T \in G \cap T \cap B(0, 1)$. The line through x_T meets at least N distinct δ -squares of $K^{(\delta)} \cap B(x_T, 1)$. Each of these squares meets T , and a fixed δ -square meets only $O(1)$ members of $\mathcal{T}$. The support-covering clause of regularity therefore gives

$$N|\mathcal{T}| \leq C_1|K^{(\delta)} \cap B(0, 3)|_\delta \leq C_2C\delta^{-t}, \quad (9.37d'')$$

where C_1, C_2 are absolute. The last inequality follows by enlarging by a fixed number of squares every δ -square in a cover of $K \cap B(0, 4)$.

Choose $\eta = c_1\kappa/C$, where the absolute $c_1 > 0$ is so small that $(1 - \eta)^{-1} \leq 1 + \kappa$ and $C_2C\eta \leq \kappa$. Call $T \in \mathcal{T}$ good when

$$\mu_1(T \cap L) \geq (1 - \eta)\mu_1(T \cap G),$$

and bad otherwise. Summing over good tubes gives at most $(1 + \kappa)\mu_1(L)$. A bad tube is light if $\mu_1(T \cap G) \leq \eta N\delta^t$; by (9.37d''), all light bad tubes together contribute at most κ . It remains to show that, for every heavy bad tube,

$$B(0, 1) \cap T \cap K \subset D. \quad (9.37d''')$$

Fix a heavy bad T . Badness and heaviness imply

$$\mu_1(T \setminus L) > \eta \mu_1(T \cap G) > \eta^2 N \delta^t.$$

Take a maximal δ -separated family $x_1, \dots, x_m$ in $K \cap T \cap B(0, 1) \setminus L$. Its δ -balls cover this set. The Frostman clause at scale δ hence gives

$$m \geq \eta^2 N / C. \quad (9.37e')$$

Fix any $x \in K \cap T \cap B(0, 1)$, and let $n = m_{K, \theta}(x \mid [\Delta, 3])$. Since the projections of $x, x_1, \dots, x_m$ lie in the same interval of length δ , for each j there is a point y_j on the line through x with $|y_j - x_j| < \delta$. Thus $y_j \in K^{(\Delta)} \cap B(x, 3)$; assign to it one of the n grid squares of side Δ meeting this intersection.

We claim $n \geq c_2 m / M$ for an absolute $c_2 > 0$. Otherwise one of those Δ -squares is assigned more than $C_3 M$ indices, where C_3 is a sufficiently large absolute constant. Fix one of them, say i . For every other assigned index j , the points y_i, y_j lie in the same square, so $|x_j - x_i| \leq \sqrt{2}\Delta + 2\delta$. Also, because x_i, x_j lie in T , there is a point z_j on the line through x_i with $|z_j - x_j| < \delta$. Since $\Delta \geq 8\delta$,

$$z_j \in K^{(3\delta)} \cap B(x_i, 3\Delta).$$

If a grid square of side 3δ contains several z_j , their associated x_j lie in its δ -neighbourhood; that neighbourhood contains only $O(1)$ points of a δ -separated family. Hence the line through x_i meets at least $c_3 C_3 M$ distinct 3δ -squares of the displayed set. Taking C_3 large yields $m_{K, \theta}(x_i \mid [3\delta, 3\Delta]) \geq M$, contrary to $x_i \notin L$. This proves the claim. Together with (9.37e') and $\eta = c_1 \kappa / C$, it gives

$$n \geq c_2 \eta^2 N / (CM) \geq c_0 \kappa^2 C^{-3} N / M.$$

Thus (9.37d'') holds. The heavy bad tubes contribute at most $\mu_1(G \cap D) \leq \mu_1(D)$, which proves the lemma. $\square$

Here is the exact point at which the additive error in Lemma III.9.25.2 can be absorbed in the Orponen–Shmerkin bootstrap. The hypotheses below isolate the local-scale estimate supplied there by the inductive projection assertion; they do not assume the desired improvement.

Lemma III.9.25.3 (upper-regular coarse-scale transfer). *Fix $0 < t < 2$ and $0 < \sigma < 1$. There is $\zeta_0(\sigma) > 0$ with the following property. Suppose $0 < \zeta < \zeta_0$, $0 < \epsilon \leq \zeta$, $0 < \delta < \bar{\delta} < 1$ are dyadic scales satisfying*

$$\delta / \bar{\delta} \leq \delta^{1/3}, \quad \delta \geq \bar{\delta}^6 / \sqrt{\zeta},$$

μ is $(\delta, t, \delta^{-\epsilon})$ -regular, and E is a finite set of directions. Write $K = \text{spt } \mu$, $\mu_1 = \mu|_{B(0,1)}$, and $\lambda = \delta / \bar{\delta}$. Assume, for every $\theta \in E$,

$$\mu_1(H_\theta(K, \delta^{-\sigma+\zeta}; [\delta, 1])) \geq \delta^\epsilon, \quad (9.37f')$$

and assume the local-scale estimate

$$\sum_{\theta \in E} \mu_1(H_\theta(K, \lambda^{-\sigma}; [4\delta, 4\bar{\delta}])) \leq \delta^{7\epsilon} |E|. \quad (9.37f'')$$

Then, for sufficiently small δ depending on the displayed parameters, a subset $E' \subset E$ of cardinality $|E'| \geq c\delta^{2\epsilon} |E|$ satisfies

$$\mu_1(H_\theta(K, \bar{\delta}^{-\sigma+50\sqrt{\zeta}}; [\bar{\delta}, 3])) \geq \delta^{2\epsilon}, \quad \theta \in E'. \quad (9.37f''')$$

Here $c > 0$ is absolute. Replacing the outer radius 3 by 5 only weakens the conclusion.

Proof. Covering numbers at radii 3δ and 4δ differ by an absolute factor. Hence a sufficiently large absolute L gives

$$H_\theta(K, L\lambda^{-\sigma}; [3\delta, 3\bar{\delta}]) \subset H_\theta(K, \lambda^{-\sigma}; [4\delta, 4\bar{\delta}]). \quad (9.37g')$$

Set $N = \delta^{-\sigma+\zeta}$, $M = L\lambda^{-\sigma}$, $C = \delta^{-\epsilon}$, and $\kappa = \delta^{2\epsilon}$. The scale assumptions imply $M \leq N$ once $\zeta_0(\sigma)$ and then δ are small: indeed, writing $\bar{\delta} = \delta^a$, one has $a \geq \sqrt{\zeta}/6$, so

$$\frac{N}{M} = L^{-1} \delta^\zeta \bar{\delta}^{-\sigma} = L^{-1} \bar{\delta}^{-\sigma+\zeta/a} \geq L^{-1} \bar{\delta}^{-\sigma+6\sqrt{\zeta}} \longrightarrow \infty.$$

The same hypotheses give $\bar{\delta}/\delta \geq \delta^{-1/3}$, so $\bar{\delta} \geq 8\delta$ for small δ . Apply Lemma III.9.25.2 with $\Delta = \bar{\delta}$. Its coarse threshold is

$$c_0 \kappa^2 C^{-3} \frac{N}{M} = \frac{c_0}{L} \delta^{\zeta+7\epsilon} \bar{\delta}^{-\sigma} \geq \bar{\delta}^{-\sigma+50\sqrt{\zeta}}. \quad (9.37h')$$

For the last inequality, $a \geq \sqrt{\zeta}/6$ and $\epsilon \leq \zeta$ imply $(\zeta + 7\epsilon)/a \leq 48\sqrt{\zeta}$; the remaining $2\sqrt{\zeta}$ absorbs c_0/L when δ is small. Summing the splitting inequality over E , and using (9.37f'), (9.37f''), and (9.37g'), we obtain

$$\sum_{\theta \in E} \mu_1(H_\theta(K, \bar{\delta}^{-\sigma+50\sqrt{\zeta}}; [\bar{\delta}, 3])) \geq (\delta^\epsilon - (1 + \delta^{2\epsilon})\delta^{7\epsilon} - \delta^{2\epsilon})|E| \geq \frac{1}{2}\delta^\epsilon |E|.$$

Finally $\mu_1(K) \leq C = \delta^{-\epsilon}$. Discarding directions for which the summand is smaller than $\delta^{2\epsilon}$ can account for at most $\delta^{2\epsilon}|E|$; the remaining directions number at least $c\delta^{2\epsilon}|E|$. $\square$

Theorem III.9.25.4 (high-multiplicity improvement). *Let $0 < t < 2$ and $0 < s < \min\{t, 2-t\}$. For every $\sigma > (t-s)/2$ there exist $\epsilon, \delta_0 > 0$ such that if $0 < \delta < \delta_0$, μ is a $(\delta, t, \delta^{-\epsilon})$ -regular measure, and $E \subset \mathbb{S}^1$ is a $(\delta, s, \delta^{-\epsilon})$ -set, then some $\theta \in E$ satisfies*

$$\mu(B(0, 1) \cap H_\theta(\text{spt } \mu, \delta^{-\sigma}; [\delta, 1])) \leq \delta^\epsilon. \quad (9.37f)$$

This is Theorem 4.7 of [18].

III.9.6.1 The sharp high-multiplicity bootstrap

The gain in the preceding Bourgain base case cannot simply be iterated at the original scale. A direction set may put nearly all its branching above the square-root scale. We first select a scale at which every surviving direction fibre has useful branching, then transfer the counterassumed high multiplicity to that scale. The mass-aware splitting lemma above makes this transfer valid for upper-regular measures; it is important that no lower mass bound for an occupied ball is used. Orponen's earlier arithmetic-sum argument [20] makes a related scale selection but ends with an inverse theorem for convolution. Here the terminal contradiction is the robust ABC theorem already proved in this chapter. The two routes share the geometry of high-multiplicity fibres, not the final additive-combinatorial input.

Definition III.9.26 (uniform direction packet and branching function). *Fix an integer $T \geq 1$, put $q = 2^{-T}$, and let $m \geq 1$. A finite q^m -separated set $F \subset [0, 1]$ is (m, T) -uniform if there are integers $N_1, \dots, N_m \in [1, q^{-1}]$ such that each occupied dyadic interval J of length q^{j-1} has exactly N_j occupied children of length q^j , for $1 \leq j \leq m$. Its branching function $b_F : [0, m] \rightarrow [0, m]$ is the piecewise-affine interpolation of*

$$b_F(0) = 0, \quad b_F(j) = \frac{1}{T} \log_2 |F|_{q^j} = \frac{1}{T} \sum_{i=1}^j \log_2 N_i.$$

Thus b_F is nondecreasing and 1-Lipschitz. For an occupied dyadic interval J , F_J denotes the affine renormalisation of $F \cap J$ onto $[0, 1]$.

Lemma III.9.27 (uniformisation and restriction). *Let $0 < s \leq 1$, $0 < h < 1/20$, and $F \subset [0, 1]$ be a (q^m, s, q^{-mh}) -set. If T is chosen sufficiently large in terms of h , there is an (m, T) -uniform $F_0 \subset F$ with*

$$|F_0| \geq q^{mh} |F|, \quad F_0 \text{ a } (q^m, s, q^{-2mh})\text{-set.} \quad (9.37i)$$

If F_0 is (m, T) -uniform and a (q^m, s, C) -set, $q^m \leq r = q^j \leq 1$, and $F_1 \subset F_0$ has $|F_1| \geq p|F_0|$, then

$$F_1 \text{ is an } (r, s, C_1 C/p)\text{-set,} \quad (9.37j)$$

where C_1 is absolute. The same conclusion holds at an intermediate dyadic scale with an additional factor $O(2^T)$.

Proof. Work upwards from the terminal generation. At a given generation, the numbers of retained children of occupied parents lie in $[1, 2^T]$. Divide these numbers into the dyadic bands $[2^k, 2^{k+1})$, $0 \leq k \leq T$, and retain a band carrying at least $1/(T+1)$ of the terminal points. In every parent in that band retain exactly 2^k children. This loses at most an additional factor 2 of terminal points. Repeat at all m generations. The resulting set is uniform and retains at least $[2(T+1)]^{-m} |F|$ points. Taking $\log_2(2(T+1))/T < h$, and then decreasing the initial h slightly if necessary, proves the first cardinality bound. Restricting a relative Frostman set to a subset of density p changes its constant by at most p^{-1} , proving the other part of (9.37i).

For (9.37j), each occupied interval of length r in F_0 contains the same number D of terminal points. Hence $|F_0| = D|F_0|_r$ and, for a larger interval I , $|F_0 \cap I| = D|F_0 \cap I|_r$. The fine-scale Frostman inequality gives $|F_0 \cap I|_r \leq C|I|^s|F_0|_r$. Moreover $|F_1|_r \geq |F_1|/D \geq p|F_0|_r$. Therefore

$$|F_1 \cap I|_r \leq |F_0 \cap I|_r \leq (C/p)|I|^s|F_1|_r.$$

Replacing intervals by balls and moving an intermediate scale to the next block scale gives the asserted fixed factors. $\square$

Lemma III.9.28 (normalising a coefficient packet near zero). *Fix $0 < \beta \leq \alpha < 1$ and $\alpha - \beta < \gamma \leq 1$, and let $\chi > 0$ be the gain in Theorem III.9.23 for these parameters. Suppose $0 < v, d$ are sufficiently small in terms of β, γ, χ . Let $A \subset [0, 1]$ and $B \subset [0, 1]$ be Δ -separated, $|A| \leq \Delta^{-\alpha}$; suppose B is a $(\Delta, \beta, \Delta^{-v})$ -set. Let $S \subset [-1, 1]$ be a Δ -separated $(\Delta, \gamma, \Delta^{-v})$ -set. For each $c \in S$ let $G_c \subset A \times B$ satisfy $|G_c| \geq \Delta^d|A||B|$. Then for some $c \in S$,*

$$|\{a + cb : (a, b) \in G_c\}|_{\Delta} \geq \Delta^{-\chi/2}|A|. \quad (9.37q')$$

The same holds if the displayed set hypotheses have exponents $\beta' > \beta, \gamma' > \gamma$ and constants at most Δ^{-v} .

Proof. Put $r_0 = \Delta^{2v/\gamma}$ and take $v < \gamma/4$. Relative nonconcentration gives

$$|S \cap [-r_0, r_0]| \leq C\Delta^{-v}r_0^\gamma|S| \leq C\Delta^v|S|.$$

One of the $O(\log(1/\Delta))$ signed dyadic annuli $\{a \leq |c| < 2a\}$, $a \geq r_0/2$, therefore contains a subset S_0 of at least $c|S|/\log(1/\Delta)$ points. If this is a negative annulus, replace B by $1 - B$ and each graph edge (x, b) by $(x, 1 - b)$. Then $x + cb$ is a translate, by the fixed quantity c , of $x + (-c)(1 - b)$. Hence we may assume the annulus is positive and put $c' = c/a \in [1, 2]$ for $c \in S_0$. Round these coefficients to the nearest point of $\Delta\mathbb{Z} \cap [1, 2]$ and keep one original coefficient per rounded value. Since $a \leq 1$ and S_0 is Δ -separated, the rounding has bounded fibres. We retain the notation c' and S_0 ; this costs only a fixed factor in cardinality and relative regularity. This coefficient set is $(\Delta, \gamma, C\Delta^{-v}\log(1/\Delta))$: the map $c \mapsto c/a$ expands distances, and the restriction to S_0 loses at most a logarithmic factor in relative nonconcentration.

Take one point b' on the Δ -grid in each Δ -cell meeting aB , and call the resulting set B' . A cell of aB comes from at most C/a points of B , because B is Δ -separated. Therefore

$$|B'| \geq ca|B|, \quad B' \text{ is a } (\Delta, \beta, C\Delta^{-v}a^{-\beta-1})\text{-set.} \quad (9.37r)$$

For the second assertion, pull an interval of radius $r \geq \Delta$ back under multiplication by a , use the relative Frostman bound for B , and divide by $|B'|$. If $r/a > 1$, the same inequality follows from the trivial bound $|B' \cap I| \leq |B'|$.

Map each edge $(x, b) \in G_c$ to $(x, b') \in A \times B'$, where b' is the cell representative of ab ; denote the image graph by G'_c . The fibre multiplicity of this map is at most C/a , hence

$$|G'_c| \geq ca|G_c| \geq ca\Delta^d|A||B'|. \quad (9.37s)$$

The projections $x + cb$ and $x + c'b'$ of corresponding edges differ by at most $O(\Delta)$, including the coefficient-rounding error since $b' \in [0, 1]$. Grid-rounding A , if required, changes covering numbers and cardinalities by only absolute factors.

Since $a \geq \frac{1}{2}\Delta^{2v/\gamma}$, choose v, d so that

$$v + \frac{2v(\beta + 1)}{\gamma} < \frac{\chi}{4}, \quad d + \frac{2v}{\gamma} < \frac{\chi}{4}.$$

For small Δ , (9.37r) and the coefficient bound meet Theorem III.9.23 with its gain χ , and (9.37s) is denser than its Δ^χ threshold. The theorem supplies one c' which expands every eligible graph, including G'_c . The $O(\Delta)$ comparison transfers its conclusion to G_c , with fixed constants absorbed by replacing χ with $\chi/2$. $\square$

Lemma III.9.29 (a square-root branching scale). *Let F be (m, T) -uniform and a (q^m, s, q^{-mh}) -set. Fix $0 < \eta < s/2$ and assume $h < \eta/20$. For all sufficiently large m there is an integer*

$$\frac{\eta m}{12} \leq k \leq \frac{m}{3}$$

such that, for every occupied interval J of length q^k , F_J , viewed at scale q^k , is a $(q^k, s - \eta, C_T)$ -set. One may take $C_T = 2^{6T}$. With $\delta = q^m$, $\bar{\delta} = q^{2k}$, and $\bar{\Delta} = q^k = \sqrt{\bar{\delta}}$, this gives

$$\delta/\bar{\delta} \leq \delta^{1/3}, \quad \delta \geq \bar{\delta}^{6/\eta}. \quad (9.37k)$$

Proof. Write $b = b_F$. Uniformity and the relative Frostman bound imply

$$b(j) \geq sj - hm - 2, \quad 0 \leq j \leq m : \quad (9.37l)$$

indeed, an occupied q^j -interval contains $|F|/|F|_{q^j}$ terminal points, while Frostman bounds this number by $q^{-mh}q^{js}|F|$. Interpolation and the 1-Lipschitz property give the same bound, with an additional absolute constant, between integers. Set $f(x) = m^{-1}b(mx)$. For large m , $f(x) \geq sx - 3h$, $f(0) = 0$, and $f(x) \leq x$.

Put $c = \eta/10$ and $g(x) = f(x) - (s - \eta)x$. For $x \geq 1/3$,

$$g(x) \geq \eta/3 - 3h > \eta/10 \geq g(c).$$

Choose a minimiser a of g on $[c, 1]$. Then $c \leq a < 1/3$ and $g(x) \geq g(a)$ for $x \geq a$. Let $k = \lfloor am \rfloor$. The 2-Lipschitz property of g shows, for $k \leq j \leq 2k$,

$$b(j) - b(k) \geq (s - \eta)(j - k) - 4. \quad (9.37m)$$

For large m , $k \geq \eta m/12$. The equal-descendant identity for uniform sets says

$$\frac{|F_J \cap I|_{q^{2k}}}{|F_J|_{q^{2k}}} = \frac{|F|_{q^{2k}}/|F|_{q^j}}{|F|_{q^{2k}}/|F|_{q^k}} = 2^{-T(b(j)-b(k))}$$

for any occupied q^j -subinterval $I \subset J$, $k \leq j \leq 2k$. Inserting (9.37m), and comparing adjacent block scales, proves the $(q^k, s - \eta, C_T)$ -set property with $C_T = 2^{6T}$. Finally $k \leq m/3$ gives the first inequality in (9.37k); $k \geq \eta m/12$ gives the second. $\square$

Definition III.9.30 (local high multiplicity). For $0 < \delta < 1$, $0 < \rho < 1$, and $0 < \sigma < 1$, set

$$H_\theta^{\text{loc}}(K, \sigma; \delta, \rho) = \bigcup_{\substack{\delta \leq r \leq R \leq 8 \\ r, R \text{ dyadic}, r/R \leq \rho}} H_\theta(K, 4(R/r)^\sigma; [r, R]). \quad (9.37n)$$

As elsewhere in this chapter, replacing a dyadic radius by a fixed multiple changes a covering number by at most a fixed factor; the numerical 4 leaves room for that comparison.

Definition III.9.31 (the projection predicate). For $0 < s \leq 1$, $0 < \sigma < 1$, and $0 < t < 2$, the statement $\text{Projection}(s, \sigma, t)$ means that there exist $\epsilon_0, \Delta_0 > 0$ such that for every $0 < \Delta \leq \Delta_0$, every $(\Delta, t, \Delta^{-\epsilon_0})$ -regular measure μ , and every $(\Delta, s, \Delta^{-\epsilon_0})$ -set E of directions, some $\theta \in E$ satisfies

$$\mu(B(0, 1) \cap H_\theta(\text{spt } \mu, \Delta^{-\sigma}; [\Delta, 1])) \leq \Delta^{\epsilon_0}.$$

The predicate is monotone in σ : it remains true with the same constants when σ increases.

Lemma III.9.32 (small slices are sparse at every separated pair of scales). Assume the high-multiplicity conclusion $\text{Projection}(s, \sigma, t)$: there are $\epsilon_0, \Delta_0 > 0$ such that (9.37f) holds with exponent σ for every $\Delta \leq \Delta_0$, with ϵ_0 in place of ϵ . For every $0 < \eta < 1$ there is $h_0 = h_0(\eta, \epsilon_0) > 0$ such that, for $0 < h < h_0$ and sufficiently small $\delta = q^m$, every (δ, t, δ^{-h}) -regular measure μ and every (m, T) -uniform (δ, s, δ^{-h}) -set E admit $\theta \in E$ with

$$\mu(B(0, 1) \cap H_\theta^{\text{loc}}(\text{spt } \mu, \sigma; \delta, \delta^\eta)) \leq \delta^h. \quad (9.37o)$$

For fixed T , one can take $h_0 = c\eta\epsilon_0$ with an absolute $c > 0$; the sufficiently-small-scale threshold may depend on T .

Proof. Suppose the left side exceeds δ^h for every $\theta \in E$. There are $O(\log^2(1/\delta))$ possible pairs (r, R) . First pigeonhole such a pair for each direction, then a common pair for a subset $E_0 \subset E$ with $|E_0| \geq \delta^{2h}|E|$, so that the corresponding high set has mass at least δ^{2h} for every direction in E_0 . The support-covering bound permits a cover of $\text{spt } \mu \cap B(0, 1)$ by at most $C\delta^{-h} \max\{1, R^{-t}\}$ balls of radius R . Double counting their masses and then discarding directions with below-average mass produces a ball $B = B(z, R)$ and $E_1 \subset E_0$ satisfying

$$|E_1| \geq \delta^{6h}|E|, \quad \mu(B \cap H_\theta(K, 4(R/r)^\sigma; [r, R])) \geq c\delta^{6h}R^t, \quad \theta \in E_1, \quad (9.37p)$$

where $K = \text{spt } \mu$. If $R > 1$, the fixed factor $R^t \leq 8^t$ is absorbed by decreasing δ_0 . Indeed the average over directions for some ball is at least $c\delta^{3h}R^t$, while its mass is at most $C\delta^{-h}R^t$. Keeping directions whose contribution is at least $c\delta^{4h}R^t$ retains $\geq \delta^{4h}|E_0|$ of them; the weaker powers $6h$ in (9.37p) absorb the fixed constants and the earlier pigeonhole loss.

Restrict the measure to $B(z, 3R)$, dilate $B(z, 4R)$ to the unit ball, and set

$$\mu_B = (4R)^{-t} T_{B(z, 4R)}(\mu|_{B(z, 3R)}), \quad \Delta = r/(4R).$$

This localisation is needed when R is not small: regularity of the original measure is assumed only up to radius 8. The support of μ_B lies in $B(0, 3/4)$, and μ_B is $(\Delta, t, C'\delta^{-h})$ -regular with an absolute C' . To check this, rescale the two regularity inequalities whenever the

original upper radius is at most 8. If the rescaled lower radius a is at least $1/4$, bounded support gives the inequalities directly. Otherwise its original value $4Ra$ is less than $R \leq 8$. Cover the restricted support by a fixed number of balls of radius R , apply the original covering inequality inside those balls at the lower radius $4Ra$, and use that an original upper radius above 8 corresponds to a rescaled upper radius $b \geq 1/4$. The resulting count is at most $C'a^{-t} \leq C'(b/a)^t$. The Frostman bound in this case follows from $\mu(B(z, 3R)) \leq CR^t$, obtained by a fixed cover with R -balls. These cases give the same constant C' , independent of r, R, δ .

Points of the high set in B remain high for the rescaled restricted support: their r -multiplicity in $B(x, R)$ comes from $K \cap B(z, 2R)$, which is inside the restriction, and

$$4(R/r)^\sigma \geq (4R/r)^\sigma = \Delta^{-\sigma}$$

because $\sigma \leq 1$. Thus (9.37p) gives

$$\mu_B(B(0, 1) \cap H_\theta(\text{spt } \mu_B, \Delta^{-\sigma}; [\Delta, 1])) \geq c'\delta^{6h}, \quad \theta \in E_1. \quad (9.37q)$$

Here $\Delta \leq \delta^\eta/4$. By Lemma III.9.27, E_1 is a $(\Delta, s, C_T\delta^{-7h})$ -set, after moving Δ to an adjacent block scale. Since $\delta^{-7h} \leq \Delta^{-7h/\eta}$, choose $h_0 < \eta\epsilon_0/20$. The fixed C_T is absorbed by the exponent margin once δ is sufficiently small. Then the inductive projection assertion applies to μ_B, E_1 . It gives some $\theta \in E_1$ for which the left side of (9.37q) is at most $\Delta^{\epsilon_0} \leq \delta^{\eta\epsilon_0}$, contradicting (9.37q) when $6h < \eta\epsilon_0$ and δ is small. $\square$

Proposition III.9.32.1 (uniform high-multiplicity improvement). *Fix $0 < t < 2$, $0 < s < \min\{t, 2 - t\}$, and $(t - s)/2 < \sigma < t/2$. Suppose $\text{Projection}(s, \sigma, t)$ holds in the sense of Lemma III.9.32. There exists $\zeta = \zeta(s, \sigma, t) > 0$ such that $\text{Projection}(s, \sigma - \zeta, t)$ holds. The choice of ζ can be bounded below as (s, σ, t) ranges over a compact subset of the displayed open parameter range.*

Proof. We give the exponent bookkeeping because it is essential to the upper-regular version of this argument. Choose $\rho > 0$ so small that

$$0 < \beta := \sigma - \rho \leq \alpha := t - \sigma + \rho < 1, \quad \gamma := s - \rho > \alpha - \beta. \quad (9.37t)$$

This is possible precisely because $(t - s)/2 < \sigma < t/2$ and $s < 2 - t$. Write $\chi > 0$ for the gain of the locally proved robust ABC theorem III.9.23 at (α, β, γ) . Choose $\zeta > 0$ with

$$z := 50\sqrt{\zeta} < 10^{-8} \min\{\rho, \chi, s - \rho, \sigma, t - \sigma\}^3. \quad (9.37u)$$

We prove the asserted improvement with gain $\zeta/2$; the spare half of ζ absorbs fixed changes of the initial scale. All losses denoted by $O_{s,t,\sigma,z}(h)$ below are bounded by a fixed multiple of h , independent of the scale and the objects. After z has been fixed, choose $h > 0$ so small that every such loss is below z^4 , $h < \epsilon_0/1000$, and Lemma III.9.32 applies at the values used below. Here ϵ_0, Δ_0 are the constants of the assumed $\text{Projection}(s, \sigma, t)$ assertion. Fix a block length T large enough for Lemma III.9.27 with loss $h/2$, and put $q = 2^{-T}$.

Suppose, contrary to the improvement with gain $\zeta/2$, that there are arbitrarily small scales δ_0 with counterexamples of regularity constant $\delta_0^{-h/4}$ and bad-set mass greater than

$\delta_0^{h/4}$ in every direction, at threshold $\delta_0^{-\sigma+\zeta/2}$. Pass to a maximal separated direction set, then round δ_0 up to a block scale $\delta = q^m \in [\delta_0, 2^T \delta_0)$. At most a fixed number of old terminal cells form one new cell; hence the high multiplicity drops by at most a fixed factor. The spare exponent $\zeta/2$ absorbs this factor, and the gaps between $h/4$ and $h/2$ absorb the fixed regularity and mass constants. We therefore obtain a $(\delta, t, \delta^{-h/2})$ -regular measure μ with support K and a $(\delta, s, \delta^{-h/2})$ -set E such that

$$\mu(B(0, 1) \cap H_\theta(K, \delta^{-\sigma+\zeta}; [\delta, 1])) \geq \delta^{h/2}, \quad \theta \in E. \quad (9.37v)$$

Cover the direction circle by a fixed number of short arcs on which angle is a bi-Lipschitz coordinate. One arc contains a fixed fraction of E ; keep that arc, identify it with a subinterval of $[0, 1)$, and absorb the fixed loss in h . Apply Lemma III.9.27 to E , retaining the notation E . It is now (m, T) -uniform and a (δ, s, δ^{-h}) -set. Formula (9.37v) remains valid for every retained direction, and its lower bound is at least δ^h .

Branching-scale selection and transfer. Apply Lemma III.9.29 with $\eta = \sqrt{\zeta}$, obtaining $\bar{\delta} = q^{2k}$, $\bar{\Lambda} = q^k$ and the scale relations (9.37k). Every E_J , J an occupied $\bar{\Lambda}$ -interval, is a $(\bar{\Lambda}, s - \eta, C_T)$ -set. Put $\lambda = \delta/\bar{\delta}$. We next verify rather than assume the local estimate

$$\sum_{\theta \in E} \mu(B(0, 1) \cap H_\theta(K, \lambda^{-\sigma}; [4\delta, 4\bar{\delta}])) \leq \delta^{7h}|E|. \quad (9.37w)$$

Cover $K \cap B(0, 2)$ by at most $C\delta^{-h}\bar{\delta}^{-t}$ balls $B = B(w, 4\bar{\delta})$ with bounded overlap. Set $\mu_B = (4\bar{\delta})^{-t}T_B\mu$. The measure μ_B is $(\lambda, t, C\delta^{-h})$ -regular and $\mu_B(B(0, 1)) \leq C\delta^{-h}$. If for one such B

$$\sum_{\theta \in E} \mu_B(B(0, 1) \cap H_\theta(T_B K, \lambda^{-\sigma}; [\lambda, 1])) > \delta^{10h}|E|,$$

then a subset $E_B \subset E$ of size at least $\delta^{20h}|E|$ has the summand at least δ^{20h} for each of its directions. Lemma III.9.27 makes E_B a $(\lambda, s, C_T\delta^{-21h})$ -set. Because $\lambda \leq \delta^{1/3}$, both its constant and that of μ_B are at most $\lambda^{-\epsilon_0}$ when $h < \epsilon_0/100$ and δ is small. The assumed projection assertion at scale $\lambda \leq \Delta_0$ then gives a direction in E_B whose summand is at most $\lambda^{\epsilon_0} \leq \delta^{\epsilon_0/3}$, contrary to δ^{20h} . Thus every ball contributes at most $\delta^{10h}|E|$ after renormalisation. Rescaling the local high set exactly as in (9.37f") and summing the covering balls yields

$$C\delta^{-h}\bar{\delta}^{-t}(4\bar{\delta})^t\delta^{10h}|E| \leq \delta^{7h}|E|$$

for small δ . This proves (9.37w).

Apply Lemma III.9.25.3 with $\epsilon = h$, E as now uniform, and (9.37v), (9.37w). The two scale requirements of that lemma are (9.37k). We obtain $E' \subset E$, $|E'| \geq c\delta^{2h}|E|$, such that

$$\mu(B(0, 1) \cap H_\theta(K, \bar{\delta}^{-\sigma+z}; [\bar{\delta}, 3])) \geq \delta^{2h}, \quad \theta \in E'. \quad (9.37x)$$

Choose one point of E' in each occupied $\bar{\delta}$ -cell. Retain one colour class of a fixed colouring of adjacent cells, so these representatives are literally $\bar{\delta}$ -separated; this loses only a fixed factor. Discard the $\bar{\Lambda}$ -cells whose relative density of these cells is below half the global density, then apply Lemma III.9.27 at scale $\bar{\delta}$. This gives a surviving set, again called E ,

with the following properties. For some loss $h \leq L = O_z(h) > 0$, as small as desired after choosing h ,

$$\begin{aligned} \mu &\text{ is } (\bar{\delta}, t, \bar{\delta}^{-L})\text{-regular, } E \text{ is an } (\bar{\delta}, s, \bar{\delta}^{-L})\text{-set,} \\ E_J &\text{ is a } (\bar{\Delta}, s - \eta, \bar{\Delta}^{-L})\text{-set for every occupied } \bar{\Delta}\text{-interval } J, \\ \mu(B(0, 1) \cap H_\theta(K, \bar{\delta}^{-\sigma+z}; [\bar{\delta}, 3])) &\geq \bar{\delta}^L \quad (\theta \in E). \end{aligned} \quad (9.37y)$$

For completeness, the density of E' in the fine uniform set is at least $c\delta^{2h}$. At each coarser block scale its covering number is therefore at least this fraction of that of E . All occupied $\bar{\delta}$ -cells of the original uniform set contain the same number of terminal points. Likewise all occupied $\bar{\Delta}$ -cells contain the same number of $\bar{\delta}$ -cells. The discarded $\bar{\Delta}$ -cells contain at most half the retained $\bar{\delta}$ -cells. Within every remaining $\bar{\Delta}$ -cell the relative density is at least $c\delta^{2h}/2$. The uniformisation loses at most a factor $\bar{\delta}^h$. Since its output and the original set both have equal descendant counts in each occupied $\bar{\Delta}$ -cell, the global retention lower bound also holds relatively in each surviving cell. Thus all relative constants worsen by at most $C\delta^{-3h}\bar{\delta}^{-h}$. Finally $\delta \geq \bar{\delta}^{6/\eta}$, whence all these losses are $\bar{\delta}^{-O_\eta(h)}$. This proves every clause of (9.37y), including the mass inequality; no lower density of μ is used. From this point we erase bars and write $\delta = \bar{\delta}$, $\Delta = \sqrt{\delta}$.

Removal of anomalously dense local slices. Take L in (9.37y) so small that $3L < h_0(\sqrt{L}, \epsilon_0)$ in Lemma III.9.32, and apply that lemma with its exponent $3L$ and $\eta = \sqrt{L}$. It gives a direction with local-high-set mass at most $\delta^{3L} < \delta^{2L}$. The weaker bound δ^{2L} holds for at least half the directions of E . Otherwise the bad half is a subset of relative density at least $1/2$. Apply Lemma III.9.27 to that subset, choosing the fixed block size T large enough that the resulting uniformisation loses at most δ^L . The new direction packet and the unchanged measure both have regularity constant at most δ^{-3L} for small δ . Applying Lemma III.9.32 to this packet with exponent $3L$ produces a direction whose local-high-set mass is at most δ^{3L} , contrary to the definition of the bad half. Restrict to these directions, discard the Δ -cells whose relative density is less than half the global density, and uniformise once more. Equal descendant counts, as above, show that at least a fixed fraction of the retained directions remain. All constants in (9.37y) worsen by $\delta^{-O_z(L)}$, but its pointwise mass lower bound is unchanged. Define

$$K_\theta = B(0, 1) \cap H_\theta(K, \delta^{-\sigma+z}; [\delta, 3]) \setminus H_\theta^{\text{loc}}(K, \sigma; \delta, \delta^{\sqrt{L}}). \quad (9.37z)$$

Then

$$\mu(K_\theta) \geq \frac{1}{2}\delta^L \geq \delta^{2L}, \quad \theta \in E. \quad (9.37aa)$$

Let $v_0 = O_z(L)$ bound the actual regularity and direction-density losses after these restrictions. Choose once and for all a larger bookkeeping exponent $v = C_z(L + v_0)$, where the fixed constant C_z dominates the finitely many losses below. At the initial choice of h , require it to be small enough that this v satisfies

$$v + \sqrt{v} < 10^{-4}z^2. \quad (9.37ab)$$

The measure and direction packet satisfy every bound stated with v , since $v \geq v_0$. Fixed geometric constants are absorbed by reducing the final scale threshold.

Two projection estimates. The definition of the local-high set, applied at $(r, R) = (8\delta, 8\Delta)$, implies that for a disc B of radius Δ and every δ -projection strip U meeting $B \cap K_\theta$,

$$|K \cap B \cap U|_\delta \lesssim (\Delta/\delta)^\sigma.$$

Indeed, take $x \in K_\theta \cap B \cap U$, move one point of $K \cap B \cap U$ from every occupied δ -cell, and project these points to the fibre through x . Their displacement is at most $C\delta$, they lie within 8Δ of x , and each 8δ -cell on that fibre receives at most a fixed number of original cells. If the displayed bound failed by a large absolute factor, x would belong to the $(8\delta, 8\Delta)$ term of H_θ^{loc} , since $\delta/\Delta = \sqrt{\delta} \leq \delta^{\sqrt{L}}$. The Frostman bound $\mu(Q) \lesssim \delta^{t-v}$ for a terminal cell gives

$$|\pi_\theta(B \cap K_\theta)|_\delta \gtrsim \delta^{v-t} (\delta/\Delta)^\sigma \mu(B \cap K_\theta). \quad (9.37ac)$$

Conversely, every δ -strip meeting K_θ contains a point of multiplicity at least $\delta^{-\sigma+z}$ at radius 3. The corresponding families of terminal cells of $K^{(\delta)} \cap B(0, 4)$ have bounded overlap as the strips vary. The support-covering bound therefore gives

$$|\pi_\theta(K_\theta)|_\delta \lesssim \delta^{\sigma-t-z-v}. \quad (9.37ad)$$

If U_Δ is a strip of width Δ parallel to a fibre of π_θ , take $x_0 \in K_\theta \cap U_\Delta$. The pair $(8\Delta, 8)$ is admissible in H_θ^{loc} , since $\Delta \leq \delta^{\sqrt{L}}$. Hence the part of $K \cap B(0, 4)$ in U_Δ can be covered by $O(\Delta^{-\sigma})$ discs of radius Δ . Each contains at most $O(\delta^{-v}(\Delta/\delta)^t)$ terminal cells. Repeating the preceding disjoint-strip count inside U_Δ yields

$$|\pi_\theta(K_\theta \cap U_\Delta)|_\delta \lesssim \delta^{-z-v} (\Delta/\delta)^{t-\sigma} = \Delta^{\sigma-t-2z-2v}. \quad (9.37ae)$$

For a strip parallel to a fibre of a direction within $O(\Delta)$ of θ , its intersection with $B(0, 1)$ lies in a fixed enlargement of a π_θ -fibre strip; the same bound holds after changing the absolute constant.

A well-populated coarse fibre. Fix an occupied Δ -interval I of the direction packet. By (9.37aa), Cauchy-Schwarz and the sharper $\mu(B(0, 1)) \leq \delta^{-v_0}$ give

$$\sum_{\theta, \phi \in E \cap I} \mu(K_\theta \cap K_\phi) \geq \frac{(\sum_{\theta \in E \cap I} \mu(K_\theta))^2}{\mu(B(0, 1))} \geq \delta^{4L+v_0} |E \cap I|^2. \quad (9.37af)$$

Choose $\theta_0 \in E \cap I$ with the corresponding lower average. The summands are at most δ^{-v_0} . Removing those below $\frac{1}{2}\delta^{4L+v_0}$ leaves at least $\frac{1}{2}\delta^{4L+2v_0} |E \cap I|$ directions. The choice of C_z in $v = C_z(L + v_0)$ absorbs these fixed losses, so we may write

$$\mu(K_0 \cap K_\theta) \geq \delta^v \quad (\theta \in E \cap I), \quad K_0 := K_{\theta_0}, \quad (9.37ag)$$

while retaining a δ^v -fraction of the directions and an $(\Delta, s - \eta, \Delta^{-v})$ -set. The initial choice of C_z has room for the relative Frostman loss and fixed grid factors.

Rotate the plane to make θ_0 horizontal, and divide each orthogonal projection by its nonzero horizontal component. Directions in I then have the form

$$\pi_\theta(x, y) = x + c_\theta y, \quad |c_\theta| \lesssim \Delta, \quad (9.37ah)$$

and $c_{\theta_0} = 0$. The coordinate map $\theta \mapsto c_\theta$ is bi-Lipschitz on I ; hence $\{c_\theta/\Delta : \theta \in E \cap I\}$ has the same relative Frostman bounds, up to fixed constants.

Cover $K \cap B(0, 1)$ by a boundedly overlapping family $\mathcal{B}_\Delta$ of Δ -discs. There are at most Δ^{-t-2v} such discs. Call B heavy when

$$\frac{1}{|E \cap I|} \sum_{\theta \in E \cap I} \mu(B \cap K_0 \cap K_\theta) \geq \Delta^{t+10v}.$$

The total contribution of the light discs is at most $\Delta^{8v}|E \cap I|$, smaller than half the lower bound in (9.37ag). Every heavy B has $\mu(B \cap K_0) \geq \Delta^{t+10v}$; applying (9.37ac) at θ_0 gives

$$|\pi_{\theta_0}(B \cap K_0)|_\delta \geq \Delta^{\sigma-t+12v}. \quad (9.37ai)$$

Cover the heavy discs by disjoint vertical Δ -strips. Their projected intervals have bounded overlap, so (9.37ad) and (9.37ai) show that the number of strips is at most

$$C\Delta^{\sigma-t-2z-14v} \leq \Delta^{\sigma-t-3z}. \quad (9.37aj)$$

Count a disc in each of the at most three strips it meets. By (9.37ag), the sum over all these disc-direction incidences is at least $\delta^{O(v)}|E \cap I|$. A strip for which this sum is less than $\Delta^{t-\sigma+4z}|E \cap I|$ contributes, by (9.37aj), at most $\Delta^z|E \cap I|$ in total. This is smaller than the preceding lower bound because $v \ll z$. Fix one remaining strip T_0 , and let $\mathcal{B}(T_0)$ be the heavy discs meeting it. Then

$$\sum_{\theta \in E \cap I} \sum_{B \in \mathcal{B}(T_0)} \mu(B \cap K_0 \cap K_\theta) \geq \Delta^{t-\sigma+4z}|E \cap I|. \quad (9.37ak)$$

For $\Delta\delta^{-\sqrt{L}} \leq R \leq 1$, choose $B \in \mathcal{B}(T_0)$ meeting $B(x, R)$, and $x_0 \in B \cap K_0$. The pair $(8\Delta, 8R)$ is admissible in the local-high set for x_0 . Moving the vertical fibre through x_0 across the width- Δ tube T_0 , and grouping nearby discs, gives

$$|\{B \in \mathcal{B}(T_0) : B \cap B(x, R) \neq \emptyset\}| \lesssim (R/\Delta)^\sigma. \quad (9.37al)$$

The argument is a packing count: the intersecting discs yield 8Δ -cells on the vertical fibre through x_0 with bounded multiplicity. Indeed, choose $y_B \in B \cap K$ for each such disc, and project y_B horizontally to that fibre. Because the discs meet the same width- Δ strip and one meets $B(x, R)$, the projected points lie in $K^{(8\Delta)} \cap B(x_0, 8R)$. If more than a fixed number were in one 8Δ -cell, their discs would all meet one ball of radius $C\Delta$, contrary to the bounded-overlap choice of $\mathcal{B}_\Delta$. Since $x_0 \notin H_{\theta_0}^{\text{loc}}$, the number of such cells is at most $4(R/\Delta)^\sigma$, proving (9.37al). In particular, $|\mathcal{B}(T_0)| \lesssim \Delta^{-\sigma}$. Together with $\mu(B) \lesssim \Delta^{t-2v}$, (9.37ak) implies that a subset of $E \cap I$ of size at least $\Delta^{5z}|E \cap I|$ satisfies

$$\sum_{B \in \mathcal{B}(T_0)} \mu(B \cap K_0 \cap K_\theta) \geq \Delta^{t-\sigma+5z}. \quad (9.37am)$$

Restrict to this subset. Its rescaled direction set remains a $(\Delta, s-\eta, \Delta^{-v-5z})$ -set. The upper Frostman bound on each disc and (9.37am) also give

$$\Delta^{-\sigma+6z} \leq |\mathcal{B}(T_0)| \lesssim \Delta^{-\sigma}. \quad (9.37an)$$

The product configuration and the ABC contradiction. Let T_0^* be the concentric vertical strip of width 5Δ . Every disc of $\mathcal{B}(T_0)$ lies in T_0^* . Let $\mathcal{A}$ be the intervals of length δ meeting $\pi_{\theta_0}(K_0 \cap T_0^*)$, and use their left endpoints as representatives. By (9.37ae),

$$|A| := |\mathcal{A}| \leq \Delta^{\sigma-t-3z}. \quad (9.37ao)$$

For $\theta \in E \cap I$, call $J \in \mathcal{A}$ popular when its vertical strip meets $B \cap K_0 \cap K_\theta$ for at least $\Delta^{-\sigma+11z}$ discs $B \in \mathcal{B}(T_0)$. Write A_θ for the popular representatives. The estimate behind (9.37ac), applied at θ_0 , gives for every $B \in \mathcal{B}(T_0)$ and $J \in \mathcal{A}$

$$\mu(B \cap K_0 \cap \pi_{\theta_0}^{-1}(J)) \lesssim \delta^{t-v}(\Delta/\delta)^\sigma = \Delta^{2t-\sigma-2v}. \quad (9.37ap)$$

Consequently the contribution of unpopular strips to the sum in (9.37am) is at most

$$C|A|\Delta^{-\sigma+11z}\Delta^{2t-\sigma-2v} \leq C\Delta^{t-\sigma+8z-2v} < \frac{1}{2}\Delta^{t-\sigma+5z}. \quad (9.37aq)$$

The contribution of one popular strip is at most $C|\mathcal{B}(T_0)|\Delta^{2t-\sigma-2v}$. Thus (9.37am), (9.37an), and (9.37aq) imply

$$|A_\theta| \gtrsim \Delta^{\sigma-t+5z+2v} \geq \Delta^{\sigma-t+6z}. \quad (9.37ar)$$

Let B_0 be the vertical coordinates of the centres of the discs in $\mathcal{B}(T_0)$, rounded to a Δ -grid. The discs have bounded overlap and their centres lie in a fixed-width multiple of T_0 ; therefore this rounding has bounded multiplicity. In particular,

$$\Delta^{-\sigma+6z} \lesssim |B_0| \lesssim \Delta^{-\sigma}. \quad (9.37as)$$

For $\theta \in E \cap I$, let $G_\theta \subset A \times B_0$ consist of the pairs (a, b) for which the strip represented by a meets $B \cap K_0 \cap K_\theta$ for a disc B whose rounded vertical coordinate is b . Bounded multiplicity and (9.37ar) yield

$$|G_\theta| \gtrsim \Delta^{-t+17z} \geq \Delta^{21z}|A||B_0|. \quad (9.37at)$$

For every $(a, b) \in G_\theta$ choose $(x, y) \in B \cap K_0 \cap K_\theta$ in the indicated strip. Then $|a - x| \leq C\delta$, $|b - y| \leq C\Delta$, and $|c_\theta| \leq C\Delta$ by (9.37ah). Therefore

$$|\{a + c_\theta b : (a, b) \in G_\theta\}|_\delta \leq |\pi_\theta(K_\theta \cap T_0^*)|_\delta \leq \Delta^{\sigma-t-3z} \leq \Delta^{-9z}|A|. \quad (9.37au)$$

The fixed enlargement of T_0 changes only absolute constants in (9.37ae).

The packing estimate (9.37al) implies that for $\Delta\delta^{-\sqrt{L}} \leq r \leq 1$,

$$\frac{|B_0 \cap B(y, r)|}{|B_0|} \lesssim \Delta^{-6z}r^\sigma.$$

For $\Delta \leq r < \Delta\delta^{-\sqrt{L}}$, Δ -separation gives the bound Cr/Δ , so (9.37as) gives

$$\frac{|B_0 \cap B(y, r)|}{|B_0|} \lesssim \Delta^{-6z-2\sqrt{L}}r^\sigma \quad (\Delta \leq r \leq 1). \quad (9.37av)$$

Choose a fixed integer $C_* \geq 100$ divisible by 4, large enough for the geometric constants in (9.37ah), and put $d = \Delta/C_*$. Translate the horizontal coordinates by the left endpoint a_0 of T_0^* and write

$$a' = (a - a_0)/(C_*\Delta), \quad b' = (b + 2)/4, \quad c'_\theta = 4c_\theta/(C_*\Delta). \quad (9.37aw)$$

The transformed sets A' and B' lie in $[0, 1]$, are d -separated, and the coefficients c'_θ lie in $[-1, 1]$. Also,

$$a' + c'_\theta b' = \frac{a + c_\theta b - a_0 + 2c_\theta}{C_* \Delta}. \quad (9.37ax)$$

The term depending only on θ is a translation, and $\delta/(C_* \Delta) = d$. Thus (9.37au) is exactly the corresponding projection estimate for the transformed graph at scale d , up to a fixed covering constant. The map $\theta \mapsto c'_\theta$ is a bi-Lipschitz renormalisation of $E \cap I$ at scale d . Grid-rounding the coefficients and, if needed, the horizontal representatives moves all sums by $O(d)$; a fixed separated subpacket avoids any collision. Consequently the coefficients form a $(d, s - \eta, d^{-v-6z})$ -set. By (9.37t), (9.37u), (9.37ao), (9.37as), (9.37at), and (9.37av), all the hypotheses of Lemma III.9.28 hold at the scale d , with its parameters $(\alpha, \beta, \gamma) = (t - \sigma + \rho, \sigma - \rho, s - \rho)$. Indeed, the losses $6z + 2\sqrt{L}$, $v + 5z$, and $21z$ are smaller than the fixed margins ρ and χ . The lemma supplies a coefficient for which the left side of (9.37au), after dilation, is at least $d^{-\chi/2}|A|$. This contradicts $C\Delta^{-9z}|A|$ because $9z < \chi/2$. The contradiction proves the asserted uniform gain. To justify the stated compact uniformity, let $J = [\sigma_-, \sigma_+]$ be a compact subinterval of $((t - s)/2, t/2)$. Choose ρ from the smallest parameter margin on J , and apply the robust ABC theorem once with

$$\alpha_* = t - \sigma_- + \rho, \quad \beta_* = \sigma_- - \rho, \quad \gamma_* = s - \rho.$$

These are admissible. The horizontal cardinality bound for any $\sigma \in J$ is no weaker than the one with α_* , and the vertical nonconcentration bound is no weaker than the one with β_* . Thus a single ABC gain $\chi_* > 0$ works throughout J ; all subsequent exponent choices are uniform. A finite covering by such parameter boxes gives the same conclusion when (s, σ, t) varies over a compact set. $\square$

Proof of Theorem III.9.25.4. Fix s, t in the displayed range and write $p = (t - s)/2$. It suffices to consider $p < \sigma < 1$: for larger σ , choose a smaller exponent in $(p, 1)$ and use monotonicity of the high sets. The predicate $\text{Projection}(s, \sigma, t)$ was defined in Definition III.9.31; it is monotone in σ . Lemma III.9.25.1 shows that the set

$$\mathcal{S} = \{p < \sigma < 1 : \text{Projection}(s, \sigma, t)\}$$

is nonempty and contains a point strictly below $t/2$. Put $\Sigma = \inf \mathcal{S}$. If $\Sigma > p$, choose a compact interval $J \subset (p, t/2)$ whose interior contains Σ . Proposition III.9.32.1 gives a gain bounded below by a positive number ζ_J for $\sigma \in J$. We may decrease this gain so that $\zeta_J < (\Sigma - p)/2$. Choose $\sigma \in \mathcal{S} \cap J$ with $\Sigma < \sigma < \Sigma + \zeta_J/2$. The proposition gives $\text{Projection}(s, \sigma - \zeta_J, t)$, where $p < \sigma - \zeta_J < \Sigma$, contradicting the definition of Σ . Thus $\Sigma = p$. Every $\sigma > p$ belongs to $\mathcal{S}$ by monotonicity, which is exactly the assertion. $\square$

Theorem III.9.32.2 (robust finite-scale projection). *Let $s \in (0, 1]$, $t \in [s, 2]$, and $0 \leq u < \min\{(s + t)/2, 1\}$. There are $\epsilon, \delta_0 > 0$ such that, for $0 < \delta \leq \delta_0$, whenever $\mathcal{P}$ is a $(\delta, t, \delta^{-\epsilon})$ -regular set and $E \subset \mathbb{S}^1$ is a $(\delta, s, \delta^{-\epsilon})$ -set, some $\theta \in E$ satisfies*

$$|\pi_\theta(\mathcal{P}')|_\delta \geq \delta^{-u} \quad (9.38)$$

for every $\mathcal{P}' \subset \mathcal{P}$ with $|\mathcal{P}'| \geq \delta^\epsilon |\mathcal{P}|$. This is Corollary 4.9 of [18].

Deduction from the high-multiplicity improvement. If $t = 2$, the nonconcentration condition gives $|\mathcal{P}| \geq \delta^{-2+\epsilon}$. Each radius- $O(\delta)$ projection strip inside the fixed unit square meets $O(\delta^{-1})$ terminal squares. Hence every $\mathcal{P}'$ with $|\mathcal{P}'| \geq \delta^\epsilon |\mathcal{P}|$ has $|\pi_\theta(\mathcal{P}')|_\delta \gtrsim \delta^{-1+2\epsilon} \geq \delta^{-u}$ for sufficiently small ϵ and every direction θ . We may therefore suppose $t < 2$.

Choose s_0 with

$$0 < s_0 < \min\{s, t, 2 - t\}, \quad u < (s_0 + t)/2. \quad (9.37g)$$

Such a choice exists even when $s = t$ or $s \geq 2 - t$, because $u < \min\{(s + t)/2, 1\}$. A (δ, s, C) -set is also a (δ, s_0, C) -set. Choose $(t - s_0)/2 < \sigma < t - u$, and let $\epsilon_0 > 0$ be the exponent given by (9.37f). Take the conclusion's $\epsilon > 0$ with $2\epsilon < \epsilon_0$, much smaller than $t - \sigma - u$.

Translate the dyadic unit square to $[-1/2, 1/2]^2$; this preserves its δ -grid, regularity, projection covering numbers, and the conclusion, and places it inside $B(0, 1)$. Let $Y_{\mathcal{P}}$ be the union of the translated terminal squares and let μ be normalized planar Lebesgue measure on it. The two clauses in the all-scales regularity definition (9.2a) imply, by counting squares meeting balls, that μ is $(\delta, t, C_0 \delta^{-2\epsilon})$ -regular. Absorb the fixed C_0 by the gap between ϵ and ϵ_0 . The high-multiplicity theorem gives a direction $\theta \in E$ for which the bad set H in (9.37f) has $\mu(H \cap B(0, 1)) \leq \delta^{\epsilon_0}$.

Fix $\mathcal{P}' \subset \mathcal{P}$ with $|\mathcal{P}'| \geq \delta^\epsilon |\mathcal{P}|$. Then $Y_{\mathcal{P}'} \setminus H$ has μ -mass at least $\frac{1}{2}\delta^\epsilon$. Partition the unit square into a fixed number of squares of diameter below $1/4$; one contains a subset G of this good set with $\mu(G) \gtrsim \delta^\epsilon$. Consider the δ -intervals meeting $\pi_\theta(G)$. For an interval I choose $x \in G$ projecting into I . Every terminal square of $\mathcal{P}'$ meeting both the chosen small spatial square and $\pi_\theta^{-1}(I)$ lies within $O(\delta)$ of the line through x . Because $Y_{\mathcal{P}}$ contains whole terminal squares, the line through x meets the δ -neighbourhood of $Y_{\mathcal{P}}$ in at least a fixed multiple of the number of such squares, up to bounded overlap. All of them lie within $B(x, 1)$; since $x \notin H$, their number is $O(\delta^{-\sigma})$. Consequently

$$|\pi_\theta(G)|_\delta \gtrsim \delta^\sigma |\mathcal{P}| \mu(G) \gtrsim \delta^{-t+\sigma+2\epsilon} \geq \delta^{-u}, \quad (9.37h)$$

where $|\mathcal{P}| \geq \delta^{-t+\epsilon}$ follows from the $(\delta, t, \delta^{-\epsilon})$ -set condition. The parameter choice makes the last inequality valid for small δ . Since $G \subset Y_{\mathcal{P}'}$, this proves (9.38) for every admissible $\mathcal{P}'$ with the same direction θ . $\square$

Lemma III.9.33 (finite regular pieces from content and packing control). *Fix $0 < t \leq 2$ and $0 < \epsilon < 1/10$. Let $0 < t_- < t < t_+$ with $t - t_- < \epsilon/10^4$ and $t_+ - t < \epsilon/10^4$. Suppose $K \subset [0, 1]^2$ has $\mathcal{H}_\infty^{t_-}(K) \geq h > 0$ and $|K|_r \leq Cr^{-t_+}$ for all sufficiently small dyadic r . There is a block length $T = T(\epsilon)$ such that, for every sufficiently fine $\delta = 2^{-mT}$, the family P_0 of occupied δ -squares admits a subfamily P and disjoint subfamilies $P_1, \dots, P_N \subset P$ with:*

$$\mathcal{H}_\infty^{t_-}\left(K \setminus \bigcup_{p \in P} p\right) \leq \delta^{\epsilon/200}, \quad c_h \delta^{-t_-} \leq |P| \leq C' \delta^{-t_+}; \quad (9.38a)$$

$$P_i \text{ is } (\delta, t, \delta^{-\epsilon})\text{-regular,} \quad |P_i| \geq \delta^{\epsilon/50} |P|, \quad (9.38b)$$

and

$$\left|P \setminus \bigcup_{i=1}^N P_i\right| \leq \delta^{\epsilon/100} |P|, \quad N \leq \delta^{-\epsilon/50}. \quad (9.38c)$$

The set K need not be measurable. The same conclusions, with a slightly smaller fixed power in the first content-loss bound if needed, hold when $h \geq (\log(1/\delta))^{-A}$ for fixed A ; all factors depending polynomially on h^{-1} are absorbed by the displayed power margins.

Proof. Let $L = \delta^{-\varepsilon/100}$. Call a dyadic square Q of side $R \in [\delta, 1]$ crowded if it contains more than $L(R/\delta)^t$ members of P_0 . Remove every δ -square lying in a crowded square, and call the remaining family P . The maximal crowded squares are pairwise disjoint. Counting their distinct terminal squares gives

$$\sum_{Q \text{ maximal crowded}} R(Q)^t \leq L^{-1} \delta^t |P_0| \leq CL^{-1} \delta^{t-t_+}.$$

Since $R(Q) \geq \delta$, the same sum with exponent t_- is at most $CL^{-1} \delta^{t_- - t_+}$, which is at most $\delta^{\varepsilon/200}$ for sufficiently small δ . This proves the content-loss assertion. Subadditivity of Hausdorff content leaves $\mathcal{H}_\infty^{t_-}(K \cap \bigcup P) \geq h/2$. Covering that set by its occupied terminal squares gives $|P| \geq c_h \delta^{-t_-}$; the upper bound is inherited from K . No occupied dyadic R -square now contains more than $L(R/\delta)^t$ members of P . Consequently P is a $(\delta, t, \delta^{-\varepsilon/8})$ -set after fixed constants are absorbed, because

$$\frac{L(R/\delta)^t}{|P|} \leq C_h \delta^{-\varepsilon/100 - (t-t_-)R^t}. \quad (9.38d)$$

Put the uniform probability on the terminal squares of P and apply the disjoint measure regularisation Theorem III.6.6 with its loss parameter $\varepsilon/100$. It gives disjoint terminal subfamilies P_i with the cardinality and residual bounds in (9.38b)–(9.38c). We verify the all-scales regularity, which is stronger than the block regularity returned by that theorem.

For a dyadic scale $r \in [\delta, 1]$, write $N_r(P_i)$ for the number of occupied r -squares. The global covering bound on K and the crowded-square bound give, uniformly in i ,

$$c_h \delta^{3\varepsilon/100 + (t-t_-)r^{-t}} \leq N_r(P_i) \leq C' \delta^{-(t_+ - t_-)r^{-t}}. \quad (9.38e)$$

For the lower bound, divide $|P_i| \geq \delta^{\varepsilon/50} |P|$ by the maximal number $L(r/\delta)^t$ of its terminal squares in one r -square; the displayed exponent has a harmless extra $\varepsilon/100$ margin. At block levels $r = 2^{-jT}$, the $(\sigma; T)$ -regularity says that every occupied parent has between $2^{T\sigma_j}$ and $2^{T\sigma_j+1}$ occupied children. Iterating and comparing with the global average yields, for any $\delta \leq r \leq R \leq 1$ and any dyadic R -square Q ,

$$N_r(P_i \cap Q) \leq C_T \delta^{-1/T} \frac{N_r(P_i)}{N_R(P_i)} \leq C_{h,T} \delta^{-1/T - 4\varepsilon/100 - (t_+ - t_-) - (t-t_-)} (R/r)^t. \quad (9.38f)$$

Here passage from arbitrary dyadic levels to neighbouring block levels costs only C_T . Choose T so large that $1/T < \varepsilon/10$. All fixed constants and the small exponents in (9.38f) fit within $\delta^{-\varepsilon}$ for sufficiently small δ . Likewise (9.38d) and $|P_i| \geq \delta^{\varepsilon/50} |P|$ show that P_i is a $(\delta, t, \delta^{-\varepsilon})$ -set. Both clauses of the [all-scales regular-set definition](#) follow. $\square$

Proposition III.9.34 (continuum regularisation for projections). *Assume the finite-scale statement (9.38) holds with fixed parameters (s, t, u) and some $\varepsilon, \delta_0 > 0$. If $K \subset \mathbb{R}^2$ satisfies $\dim_{\mathbb{H}} K = \dim_{\mathbb{P}} K = t$ and $E \subset \mathbb{S}^1$ has $\dim_{\mathbb{H}} E \geq s$, then some $\theta \in E$ satisfies*

$$\dim_{\mathbb{H}} \pi_\theta(K) \geq u. \quad (9.39)$$

Neither K nor E is assumed Borel or Ahlfors regular.

Proof. The case $u = 0$ is immediate. Suppose, to the contrary, that $\dim_{\text{H}} \pi_{\theta}(K) < u$ for every $\theta \in E$. Reduce the finite-scale loss ε if needed; a statement valid with loss ε is also valid for a smaller loss, since its hypotheses are stronger and it asks only about larger subsets than the original conclusion. Choose $t_- < t < t_+$ and $s_- < s$ so close to t, s that all three gaps are below $\varepsilon/10^4$.

By the countable-cover characterisation of packing dimension, one bounded piece of K has positive t_- -content and upper box dimension below t_+ . Replace K by this piece, translate and dilate it into $[0, 1)^2$, and write $h = \mathcal{H}_{\infty}^{t_-}(K) > 0$. We then have $|K|_r \leq Cr^{-t_+}$ at every sufficiently small r . Since $\mathcal{H}_{\infty}^{s_-}(E) > 0$ and

$$E = \bigcup_{j \geq 1} \{\theta \in E : \dim_{\text{H}} \pi_{\theta}(K) \leq u - 1/j\},$$

one member E' of this union has positive s_- -content. Choose v strictly between $u - 1/j$ and u for that member. For every $\theta \in E'$, cover $\pi_{\theta}(K)$ by arbitrarily short dyadic intervals I with $\sum_I |I|^v < 1$. Partition these intervals by dyadic length 2^{-k} and let $K_{\theta,k}$ be the preimage in K of their union. The sets $K_{\theta,k}$ cover K , so subadditivity of t_- -content and the summable weights $a_k = c(k+1)^{-2}$ imply that, for some k ,

$$\mathcal{H}_{\infty}^{t_-}(K_{\theta,k}) \geq ha_k, \quad |\pi_{\theta}(K_{\theta,k})|_{2^{-k}} \leq C2^{kv}. \quad (9.39a)$$

Let E_k be the directions for which this holds. Another application of content subadditivity yields an arbitrarily large k with $\mathcal{H}_{\infty}^{s_-}(E_k) \geq c'a_k$. Increase k by fewer than T units, refining the scale to $\delta = 2^{-mT}$; this changes the covering bounds by only a constant depending on T . Write E_{δ} for the selected direction set and K_{θ} for its selected target subset. All polynomial factors in k are $\delta^{-o(1)}$.

Apply Lemma III.9.33 to K at this scale. Its discarded set has t_- -content at most $\delta^{\varepsilon/200}$, which is smaller than $ha_k/2$ for sufficiently large k . Thus, for each $\theta \in E_{\delta}$, the family P_{θ} of δ -squares meeting K_{θ} and retained in P has

$$|P_{\theta}| \geq ca_k \delta^{-t_-} \geq \delta^{\varepsilon/1000} |P|, \quad |\pi_{\theta}(P_{\theta})|_{\delta} \leq C_T \delta^{-v}. \quad (9.39b)$$

The second bound uses the $O(\delta)$ diameter of one terminal square. By (9.38c), P_{θ} cannot lie mostly in the residual family. Since $N \leq \delta^{-\varepsilon/50}$, for each direction some index $i(\theta)$ satisfies

$$|P_{\theta} \cap P_{i(\theta)}| \geq \delta^{\varepsilon/10} |P_{i(\theta)}|. \quad (9.39c)$$

Partition E_{δ} by this index. Subadditivity of s_- -content yields one index i whose direction part has content at least $c'a_k/N \geq \delta^{\varepsilon/40}$. The max-flow and thinning argument in Lemma III.9.1, with internal loss $\varepsilon/4$, extracts from that part a $(\delta, s, \delta^{-\varepsilon})$ direction set D : the content loss $\delta^{\varepsilon/40}$ and the exponent gap $s - s_- < \varepsilon/10^4$ fit in its explicit sampling slack. Every $\theta \in D$ has $P'_{\theta} = P_{\theta} \cap P_i$ of size at least $\delta^{\varepsilon} |P_i|$, but its projection meets at most $C_T \delta^{-v} < \delta^{-u}$ terminal intervals for small δ . This contradicts (9.38), since P_i is a $(\delta, t, \delta^{-\varepsilon})$ -regular set. $\square$

Lemma III.9.35 (parameter conversion to the sharp exceptional-set exponent). *Let $K \subset \mathbb{R}^2$ satisfy $\dim_{\text{H}} K = \dim_{\text{P}} K = d > 0$. Assume that for every $0 < \sigma \leq \min\{d, 1\}$, every $E \subset \mathbb{S}^1$ with $\dim_{\text{H}} E \geq \sigma$, and every*

$$v < \min\{(\sigma + d)/2, 1\}, \quad (9.40)$$

there exists $\theta \in E$ with $\dim_{\text{H}} \pi_{\theta}(K) \geq v$. Then, for every $0 \leq u \leq \min\{d, 1\}$,

$$\dim_{\text{H}} \text{Exc}_u(K) \leq \max\{2u - d, 0\}. \quad (9.41)$$

Proof. Write $q = \max\{2u - d, 0\}$. The case $u = 0$ is immediate. First assume $0 < u < \min\{d, 1\}$. Then $q < \min\{d, 1\}$. If $\dim_{\text{H}} \text{Exc}_u(K) > q$, choose

$$q < \sigma < \min\{\dim_{\text{H}} \text{Exc}_u(K), d, 1\}. \quad (9.42)$$

The inequality $\sigma > 2u - d$ gives $u < (\sigma + d)/2$, and $u < 1$, so one may choose

$$u < v < \min\{(\sigma + d)/2, 1\}. \quad (9.43)$$

Take $E \subset \text{Exc}_u(K)$ with $\dim_{\text{H}} E \geq \sigma$. The assumed property produces $\theta \in E$ with $\dim_{\text{H}} \pi_{\theta}(K) \geq v > u$, contradicting the definition of $\text{Exc}_u(K)$.

It remains to treat the endpoint $u = \min\{d, 1\}$. If $d = 1$, the desired bound is simply $\dim_{\text{H}} \text{Exc}_1(K) \leq 1$, which follows from the dimension of the direction circle. If $d < 1$, then $u = d$ and the desired bound is d . Were $\dim_{\text{H}} \text{Exc}_d(K) > d$, the decomposition

$$\text{Exc}_d(K) = \bigcup_{m \geq 1} \{\theta : \dim_{\text{H}} \pi_{\theta}(K) \leq d - 1/m\}$$

would give an m for which the set on the right has dimension $> d$. Apply the assumed property with $\sigma = d$ and a number $v \in (d - 1/m, d)$, a contradiction. Finally, if $d > 1$, then $u = 1$ and $q = \max\{2 - d, 0\} < 1$. If the exceptional set had dimension $> q$, one of the sets

$$E_m = \{\theta : \dim_{\text{H}} \pi_{\theta}(K) \leq 1 - 1/m\}$$

would also have dimension $> q$. Choose $q < \sigma < \min\{\dim_{\text{H}} E_m, 1\}$. Since $\sigma > 2 - d$, we have $\min\{(\sigma + d)/2, 1\} = 1$. Taking $v \in (1 - 1/m, 1)$ contradicts the assumed property. This proves (9.41) in every endpoint case. $\square$

Theorem III.9.36 (sharp exceptional-set bound for equal Hausdorff and packing dimension). *Let $K \subset \mathbb{R}^2$ be an arbitrary set with $\dim_{\text{H}} K = \dim_{\text{P}} K = d$. Then, for $0 \leq u \leq \min\{d, 1\}$,*

$$\dim_{\text{H}} \text{Exc}_u(K) \leq \max\{2u - d, 0\}. \quad (9.44)$$

This is Theorem 1.13 of [18]. Its remaining research-scale input is the robust finite-scale projection estimate above. The continuum regularisation is proved as Proposition III.9.34, and the final parameter conversion is Lemma III.9.35.

Deduction from the finite-scale input. If $d = 0$, only $u = 0$ occurs and the assertion is immediate. Assume $d > 0$. Fix $0 < \sigma \leq \min\{d, 1\}$, a set $E \subset \mathbb{S}^1$ with $\dim_{\text{H}} E \geq \sigma$, and $v < \min\{(\sigma + d)/2, 1\}$. Apply Proposition III.9.34, whose proof above uses the robust finite-scale projection interface Section III.9.6.1. It gives some $\theta \in E$ with $\dim_{\text{H}} \pi_{\theta}(K) \geq v$. Thus the hypothesis of Lemma III.9.35 holds, and (9.44) follows. $\square$

III.9.7 Regular-line Furstenberg sets

An (s, t) -Furstenberg set has a *regular-dimensional associated line family* when its witnessing line family $\mathcal{L}$, measured in the affine-line metric fixed in Chapters III.7–III.8, satisfies

$$\dim_{\mathcal{H}} \mathcal{L} = \dim_{\mathcal{P}} \mathcal{L} = t. \quad (9.6)$$

This is weaker than Ahlfors regularity.

The next elementary pruning step is needed whenever a two-scale refinement is iterated: the number of fine tubes in a fixed coarse *slope* interval must be uniform across point fibres. It is a separate assertion from uniformity of the total number of tubes in a two-dimensional tube-parameter ancestor.

Lemma III.9.37 (uniform fixed-scale slope multiplicity). *Let $0 < \delta \leq \tau \leq 1$ be dyadic, let $0 < s \leq 1$, and let $(P, \mathcal{T})$ be a (δ, s, C, M) -nice configuration in one bounded slope chart. Put $L = 1 + \log(1/\delta)$. There is a subconfiguration $(P', \mathcal{T}')$ and integers $H, K \geq 1$ such that*

$$|P'| \geq L^{-C_0} |P|, \quad |\mathcal{T}'(p)| = M' = HK \geq L^{-C_0} M \quad (p \in P'), \quad (9.44a)$$

and every occupied dyadic τ -slope interval contains exactly H members of $\mathcal{T}'(p)$, for every $p \in P'$. Each $\mathcal{T}'(p)$ is a $(\delta, s, L^{C_0}C)$ -set, and its family of occupied τ -slope intervals is a $(\tau, s, L^{C_0}C)$ -set. Here C_0 is an absolute exponent. If P is (δ, t, C) -regular, then P' is $(\delta, t, L^{C_0}C)$ -regular.

Proof. For each p , group the M tubes of $\mathcal{T}(p)$ by their dyadic τ -slope interval. There are at most $O(L)$ dyadic ranges of nonzero occupancies. Choose a range $[H(p), 2H(p))$ carrying at least $M/O(L)$ tubes; trim every interval in that range to exactly $H(p)$ tubes and discard all other intervals. The retained number of intervals $K(p)$ obeys $H(p)K(p) \geq M/O(L)$. There are at most $O(L^2)$ pairs of dyadic ranges for $H(p)$ and $K(p)$. Pigeonhole that pair over $p \in P$, retain at least $|P|/O(L^2)$ points, and trim the number of intervals and their occupancies to common integers K, H . This costs only another absolute factor, giving (9.44a).

The original relative (δ, s, C) nonconcentration passes to each retained fibre because it lost only an $L^{-O(1)}$ fraction of its tubes. For an arc I of length $r \geq \tau$, all occupied τ -intervals meeting I lie in an arc of length at most $3r$. Each contributes exactly H retained tubes, so the original fine-scale bound gives

$$\#\{\tau\text{-intervals of } \mathcal{T}'(p) \text{ meeting } I\} \leq H^{-1}C(3r)^s M \leq L^{C_0}Cr^s K.$$

The bounded-chart equivalence of slope and tube-parameter nonconcentration for tubes through one square, proved in Chapter 6, handles the fixed geometric constants. Finally a large subset P' inherits the all-scales upper regularity of P and its relative (δ, t, C) condition with the constant multiplied by $|P|/|P'| \leq L^{C_0}$. $\square$

Theorem III.9.38 (regular-set Furstenberg estimate). *For*

$$s \in (0, 1], \quad t \in [s, 2], \quad 0 \leq u < \min\{(s+t)/2, 1\}, \quad (9.45)$$

there exist $\epsilon, \delta_0 > 0$ such that every $(\delta, s, \delta^{-\epsilon}, M)$ -nice configuration whose point family is a $(\delta, t, \delta^{-\epsilon})$ -regular set satisfies

$$\left| \bigcup_{p \in P} \mathcal{T}(p) \right| \geq M\delta^{-u}, \quad 0 < \delta \leq \delta_0. \quad (9.46)$$

This is Theorem 5.2 of [18].

The proof illustrates why the regularity hypothesis is stronger than an ordinary finite Frostman bound. It allows a family of points to be rescaled at each stage without losing the hypotheses of the robust projection theorem. The fixed-scale ancestor multiplicity in (6.42b) is equally important: it prevents a refinement from concentrating all its surviving tubes in a few directions.

Proof. Fix $u < v < \min\{(s+t)/2, 1\}$. Apply the robust projection interface (9.38) at exponent v , and write $\kappa > 0$ for one permissible input exponent. We shall also use its following immediate strengthened form: if the regularity and direction-set constants are at most $h^{-\kappa}/2$, at least half the directions of a finite $(h, s, h^{-\kappa}/2)$ -set have the projection conclusion for *every* subset of relative size h^κ . Indeed, if fewer than half did, the remaining bad directions would still form an $(h, s, h^{-\kappa})$ -set, contrary to (9.38). A bounded change between orthogonal projection and projection along a slope changes only fixed constants.

Choose an integer R sufficiently large, put $\epsilon = R^{-1/3}$, and require

$$8\epsilon < v - u, \quad 10\epsilon < \kappa. \quad (9.46a)$$

First suppose $\delta = 2^{-Rn}$, and set $h = 2^{-n} = \delta^{1/R}$, $J = R - 1$, and $\delta_j = \delta h^{-j}$ for $0 \leq j \leq J$. Thus $\delta_J = h$ and $\delta_j/\delta_{j+1} = h$. All logarithmic losses below are bounded by $L = (1 + \log(1/\delta))^B$, for a fixed exponent $B = B(s, t, u)$; its value may increase a fixed number of times. Since R is fixed, choose a special-scale input exponent $\alpha > 0$ so small that $R5^R\alpha < \min\{\epsilon, \kappa/10\}/2$. For sufficiently small δ , every fixed polylogarithm $L^{O(3^R)}$ is absorbed in the margin between $\delta^{-3^R\alpha}$ and $\delta^{-5^R\alpha}$. The latter is at most both $h^{-\epsilon}$ and $h^{-\kappa/10}$. We first prove the theorem at scales 2^{-Rn} for input constant $\delta^{-\alpha}$; the unrestricted input exponent will be smaller.

Initial equalization. For each $p \in P$, partition $\mathcal{T}(p)$ by its two-dimensional dyadic h -tube ancestor. A dyadic occupancy class retains at least $M/O(\log(1/\delta))$ tubes. In each retained ancestor trim to the lower endpoint of that class. Pigeonhole the occupancy and the number of retained ancestors over p , and trim whole ancestor blocks to common values. This costs only a fixed power of $\log(1/\delta)$, and gives a subconfiguration $(P_0, \mathcal{T}_0)$ with a common fibre size $M_0 \geq L^{-1}M$, $|P_0| \geq L^{-1}|P|$, and an integer $H_0 \geq 1$ such that

$$|\mathcal{T}_0(p) \cap \mathbf{S}| = H_0 \quad (p \in P_0, \mathbf{S} \in \mathcal{T}^h(\mathcal{T}_0(p))). \quad (9.46b)$$

The original all-scales estimates pass to P_0 and its fibres with at most a factor L . It is enough to prove $|\mathcal{T}_0|/M_0 \geq \delta^{-v+6\epsilon}$, because $|\mathcal{T}|/M \geq L^{-1}|\mathcal{T}_0|/M_0$ and $v - 6\epsilon > u$.

The chain of configurations. Inductively construct $(P_j, \mathcal{T}_j)$ at scale δ_j , with equal fibre degree M_j , for $0 \leq j \leq J$. For $j < J$, first retain a dyadic occupancy class of $|P_j \cap Q|$, where

Q ranges over the δ_{j+1} -parents. Call the resulting point family $\bar{P}_j$; it contains at least $|P_j|/L$ points and every occupied parent contains the same number A_j of them. Apply the proved two-scale refinement (6.42) to $(\bar{P}_j, \mathcal{T}_j)$ at scales $\delta_j < \delta_{j+1}$, using its optional fixed ancestor scale $\bar{\Delta} = h$. Denote its fine refinement by $(P'_j, \mathcal{T}'_j)$, its covering configuration by $(P_{j+1}, \mathcal{T}_{j+1})$, and its rescaled configuration inside a parent $q \in P_{j+1}$ by $(S_q(P'_j \cap q), \mathcal{U}_q)$. Write $N_q = |\mathcal{U}_q(S_q p)|$, independent of $p \in P'_j \cap q$. The two-scale statement gives, with polylogarithmic losses uniform in j ,

$$\begin{aligned} P_{j+1} &= \mathcal{D}_{\delta_{j+1}}(P'_j), \quad |P'_j \cap q| \asymp_L A_j, \quad |\mathcal{T}'_j(p)| \geq L^{-1} M_j, \\ \mathcal{T}^h(\mathcal{T}'_j(p)) &\subset \mathcal{T}^h(\mathcal{T}_{j+1}(q)), \quad \mathcal{D}_h[\sigma(\mathcal{U}_q(S_q p))] = \mathcal{D}_h[\sigma(\mathcal{T}'_j(p))], \\ \frac{|\mathcal{T}_j|}{M_j} &\geq L^{-1} \frac{|\mathcal{T}_{j+1}|}{M_{j+1}} \max_{q \in P_{j+1}} \frac{|\mathcal{U}_q|}{N_q}. \end{aligned} \quad (9.46c)$$

The covering fibres $\mathcal{T}_{j+1}(q)$ have a common number H_{j+1} of δ_{j+1} -tubes in each occupied two-dimensional h -ancestor, by (6.42b). This is precisely the uniformity that will be used at the next stage.

For clarity, here is the regularity bookkeeping behind these applications. If P_j is (δ_j, t, C_j) -regular, then $|P_j| \geq C_j^{-1} \delta_j^{-t}$, and every δ_{j+1} -parent contains at most $C_j h^{-t}$ points. Since $|\bar{P}_j| \geq |P_j|/L$, the number of its occupied parents is at least $L^{-1} C_j^{-2} \delta_{j+1}^{-t}$. The two-scale refinement retains an L^{-1} -fraction of these parents and an L^{-1} -fraction of their points. Consequently

$$|P_{j+1}| \geq L^{-2} C_j^{-2} \delta_{j+1}^{-t}, \quad |P'_j \cap q| \geq L^{-2} C_j^{-2} h^{-t}. \quad (9.46d)$$

All-scale upper counts for P_{j+1} and the rescaled $S_q(P'_j \cap q)$ follow by counting the corresponding subfamilies of P_j . Dividing those upper counts by the lower bounds in (9.46d) gives regularity constants at most $L^3 C_j^3$. The tube-fibre constants gain only further powers of L . Starting at $C_0 \leq L \delta^{-\alpha}$, induction gives constants $L^{O(3^R)} \delta^{-3^R \alpha} \leq \delta^{-5^R \alpha}$; by the choice of α , every rescaled point family used below is $(h, t, h^{-\kappa/10})$ -regular, and all relevant h -slope families have constants at most $h^{-\kappa/10}$.

Choose a nested chain $q_0, \dots, q_J$ with $q_j \in P'_j$ for $j < J$ and $q_j \subset q_{j+1}$. Such a chain exists by the covering property; in the final position choose any $q_J \in P_J$. Put $N_j = N_{q_j}$ for $1 \leq j \leq J$, where N_{q_j} is the degree of the rescaled configuration attached to the parent q_j at the preceding step. Iterating the last line of (9.46c), and using $|\mathcal{T}_J| \geq M_J$, gives

$$\frac{|\mathcal{T}_0|}{M_0} \geq L^{-R} \prod_{j=1}^J \frac{|\mathcal{U}_{q_j}|}{N_j}. \quad (9.46e)$$

Since tubes through a common square have comparable numbers of two-dimensional h -ancestors and h -slope ancestors, the equality of slope ancestors in (9.46c) and the h -ancestor uniformity imply

$$N_{j+1} \asymp_L |\mathcal{T}_j(q_j)|_h \quad (0 \leq j < J), \quad N_j \lesssim_L N_{j+1} \quad (1 \leq j < J). \quad (9.46f)$$

For the second inequality at index j , the retained family $\mathcal{T}'_{j-1}(q_{j-1})$ has at least M_{j-1}/L tubes, hence at least an L^{-1} -fraction of the h -ancestors of $\mathcal{T}_{j-1}(q_{j-1})$; its ancestors are

included among those of $\mathcal{T}_j(q_j)$. Also $N_j \lesssim h^{-1}$, because a family of h -tubes through one h -square has only $O(h^{-1})$ possible slopes. Therefore the set

$$\mathcal{E} = \{1 \leq j < J : N_{j+1} > h^{-\varepsilon} N_j\}$$

has cardinality at most $2/\varepsilon$ when δ is small. Indeed, multiply the ratios N_{j+1}/N_j , use the lower bound L^{-1} at every nonexceptional index, and absorb the fixed power L^R into $h^{-\varepsilon/2}$. *A good level and the projection gain.* Fix $j \in \{1, \dots, J-1\} \setminus \mathcal{E}$, set $q = q_j$, and put $Q = P'_{j-1} \cap q$. Define

$$\Theta = \mathcal{D}_h[\sigma(\mathcal{T}_j(q))], \quad \Theta_p = \mathcal{D}_h[\sigma(\mathcal{U}_q(S_q p))] = \mathcal{D}_h[\sigma(\mathcal{T}'_{j-1}(p))] \quad (p \in Q). \quad (9.46g)$$

The ancestor inclusion gives $\Theta_p \subset \Theta$. Uniform h -ancestor multiplicity, fibre retention, and (9.46f) give

$$|\Theta| \asymp_L N_{j+1}, \quad |\Theta_p| \asymp_L N_j \quad (p \in Q). \quad (9.46h)$$

Since $j \notin \mathcal{E}$, every Θ_p occupies at least an $L^{-1}h^\varepsilon$ -fraction of Θ . Counting pairs (p, θ) gives a subset $\Theta_1 \subset \Theta$ with

$$|\Theta_1| \geq L^{-1}h^\varepsilon|\Theta|, \quad |Q_\theta| \geq h^{2\varepsilon}|Q| \quad (\theta \in \Theta_1), \quad Q_\theta = \{p \in Q : \theta \in \Theta_p\}. \quad (9.46i)$$

To see that Θ is an $(h, s, L^{O(R)}\delta^{-5R\alpha})$ -set, consider a slope interval I of length at least h . Each occupied h -slope interval represents between one and a fixed number of two-dimensional h -tube ancestors of $\mathcal{T}_j(q)$, since all tubes meet q . Each such ancestor contains exactly H_j tubes. The (δ_j, s, C_j) estimate for $\mathcal{T}_j(q)$ therefore gives

$$|\Theta \cap I| \lesssim H_j^{-1}|\mathcal{T}_j(q) \cap I^{+h}| \lesssim C_j|I|^s|\Theta|. \quad (9.46j)$$

Passing to Θ_1 costs at most $Lh^{-\varepsilon}$. By our parameter choices, Θ_1 is an $(h, s, h^{-\kappa}/2)$ -set. The rescaled $\bar{Q} = S_q(Q)$ is $(h, t, h^{-\kappa}/2)$ -regular by (9.46d). Since $2\varepsilon < \kappa$, each $S_q(Q_\theta)$ has at least $h^\kappa|\bar{Q}|$ points. The many-direction form of (9.38) gives $\Theta_2 \subset \Theta_1$, $|\Theta_2| \geq |\Theta_1|/2$, such that

$$|\pi_\theta(S_q(Q_\theta))|_h \geq h^{-v} \quad (\theta \in \Theta_2), \quad (9.46k)$$

where $\pi_\theta(x, y) = y - \theta x$. For every $p \in Q_\theta$, the fibre $\mathcal{U}_q(S_q p)$ contains an h -tube of slope θ . One such tube meets the h -neighbourhood of only a bounded number of distinct h -projection cells. Thus (9.46k) forces at least ch^{-v} distinct slope- θ tubes in $\mathcal{U}_q$. Different slopes give disjoint tube collections. Using (9.46i), (9.46h), and $N_{j+1} \gtrsim_L N_j$, we obtain

$$\frac{|\mathcal{U}_q|}{N_j} \gtrsim L^{-O(1)}h^{\varepsilon-v}\frac{N_{j+1}}{N_j} \geq h^{-v+2\varepsilon} \quad (9.46l)$$

for sufficiently small δ .

There are at least $J-1-2/\varepsilon$ good indices. The remaining factors in (9.46e) are at least one, since every $\mathcal{U}_q$ contains a fibre of size N_q . Consequently

$$\frac{|\mathcal{T}_0|}{M_0} \geq L^{-R}h^{(-v+2\varepsilon)(J-1-2/\varepsilon)} \geq \delta^{-v+6\varepsilon}$$

after decreasing δ_0 ; the last inequality uses $h = \delta^{1/R}$, $R = \varepsilon^{-3}$, and $v \leq 1$. This proves the claimed strict gain.

Finally, set the exponent in the theorem's unrestricted hypothesis to $\alpha/4$. An arbitrary dyadic δ lies within a factor 2^R of a scale 2^{-Rn} . Coarsening the point and tube grids to the latter scale changes all cardinalities, relative nonconcentration constants, and fibre degrees by factors depending only on R . For sufficiently small δ , these fixed factors turn an input constant $\delta^{-\alpha/4}$ into at most $(2^{-Rn})^{-\alpha}$, so the special-scale estimate applies. Apply the proved estimate at exponent v and absorb these fixed factors in the strict gap $v-u$. $\square$

Proposition III.9.39 (packing-to-continuum Furstenberg passage). *Assume the finite-scale regular-set Furstenberg estimate (9.46). If $t \geq s$, F is an (s, t) -Furstenberg set, and its associated line family has packing dimension t , then*

$$\dim_{\mathrm{H}} F \geq s + u \quad \text{for every } u < \min\{(s+t)/2, 1\}. \quad (9.47)$$

No Ahlfors-regularity hypothesis is imposed on the line family.

Proof. Fix $u < u_1 < \min\{(s+t)/2, 1\}$ and put $\chi = s + u$. Choose the finite-scale loss $\varepsilon > 0$ in (9.46) for (s, t, u_1) , decreasing it so that $\varepsilon < (u_1 - u)/100$. Also require $\varepsilon < s/10$, which ensures the common fibre degree below tends to infinity. Suppose $\dim_{\mathrm{H}} F < \chi$. We shall contradict a dyadic cover of F by squares $\mathcal{Q}$ with arbitrarily small side lengths and

$$\sum_{Q \in \mathcal{Q}} \ell(Q)^\chi < 1. \quad (9.47a)$$

The witnessing line family $\mathcal{L}$ has $\dim_{\mathrm{H}} \mathcal{L} \geq t$ and $\dim_{\mathrm{p}} \mathcal{L} = t$; thus both dimensions equal t . By the packing-dimension countable-cover characterisation, restrict to one bounded line chart and a subfamily $\mathcal{L}_0$ with positive t_- -content and upper box dimension below t_+ , where $t_- < t < t_+$ differ from t by less than $\varepsilon/10^4$. Choose $s_- < s$ equally close to s . Countable stability of Hausdorff dimension of the fibres, followed by a bounded spatial restriction, yields a further line subfamily of positive t_- -content and a constant $c_0 > 0$ such that

$$\mathcal{H}_\infty^{s_-}(F \cap \ell) \geq c_0 \quad (\ell \in \mathcal{L}_0). \quad (9.47b)$$

The upper box bound is inherited by this subfamily.

Group $\mathcal{Q}$ into dyadic side-length layers $\mathcal{Q}_k$ of side 2^{-k} . Set $F_k = \bigcup \mathcal{Q}_k$ and choose summable weights $a_k = c(k+1)^{-2}$. By (9.47b) and content subadditivity, every $\ell \in \mathcal{L}_0$ belongs to at least one line set

$$\mathcal{L}_k = \{\ell \in \mathcal{L}_0 : \mathcal{H}_\infty^{s_-}(F_k \cap \ell) \geq c_0 a_k\}.$$

Applying t_- -content subadditivity in line space, and making the cover arbitrarily fine, selects an arbitrarily large k for which

$$\mathcal{H}_\infty^{t_-}(\mathcal{L}_k) \geq c_1 a_k. \quad (9.47c)$$

Let $\delta_0 = 2^{-k}$. Refine this scale by at most a fixed number T of dyadic generations to $\delta = 2^{-mT}$; replacing each square of $\mathcal{Q}_k$ by its $O_T(1)$ terminal subsquares changes all cardinalities

by fixed factors only. In particular, the corresponding family P_F of δ -squares meeting F_k obeys, by (9.47a),

$$|P_F| \leq C_T |\mathcal{C}_k| < C_T \delta^{-\chi}. \quad (9.47d)$$

Identify the bounded line chart bi-Lipschitzly with a bounded parameter square, and apply Lemma III.9.33 to $\mathcal{L}_k$ at scale δ . Its content lower bound $c_1 a_k \gtrsim k^{-2}$ is allowed by the polynomial-loss clause of that lemma. Retain any one nonempty $(\delta, t, \delta^{-\varepsilon})$ -regular line piece P_L . Choose one representative $\ell_p \in \mathcal{L}_k$ from each of its parameter squares. The finite Frostman extraction Lemma III.9.1, applied on each ℓ_p to (9.47c), gives a separated family of points in $F_k \cap \ell_p$ of cardinality between $\delta^{-s_- + \varepsilon/10}$ and $\delta^{-s_- - \varepsilon/10}$, with $(\delta, s_-, \delta^{-\varepsilon/10})$ nonconcentration. The polynomially small content a_k is absorbed in the sampling slack. Keep exactly

$$M = \left\lfloor c \delta^{-s_- + \varepsilon/10} \right\rfloor \quad (9.47e)$$

points from each family. Arbitrary thinning to this cardinality worsens its relative nonconcentration by at most $O(\delta^{-\varepsilon/5})$. Since $s - s_- < \varepsilon/10^4$, the retained fibres are $(\delta, s, \delta^{-\varepsilon})$ -sets with exactly M members.

Apply the bounded-chart point–line duality of Chapter III.6 to this configuration. Each parameter square of P_L becomes a point of a $(\delta, t, \delta^{-\varepsilon})$ -regular set, and the M selected points along its representative line become an $(\delta, s, \delta^{-\varepsilon})$ -family of M essentially distinct tubes through that dual point. An original δ -square of P_F produces only $O(1)$ dual tubes in the fixed chart, so

$$\left| \bigcup_{p \in P_L} \mathcal{T}(p) \right| \leq C |P_F|. \quad (9.47f)$$

After a fixed enlargement of tubes to absorb grid rounding, this is a nice configuration in the exact sense of (6.41). The finite-scale theorem (9.46) at u_1 gives

$$|P_F| \geq c M \delta^{-u_1} \geq c' \delta^{-(s+u_1)+(s-s_-)+\varepsilon/10}.$$

The exponent on the right is strictly larger than $\chi = s + u$ once ε and $s - s_-$ are chosen below the gap $u_1 - u$. This contradicts (9.47d) for small δ . Hence $\dim_H F \geq s + u$. $\square$

Lemma III.9.40 (piecewise form of the sharp minimum). *For $0 < s \leq 1$ and $0 \leq t \leq 2$,*

$$\min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\} = \begin{cases} s + t, & 0 \leq t \leq s, \\ \frac{3s + t}{2}, & s \leq t \leq 2 - s, \\ s + 1, & 2 - s \leq t \leq 2. \end{cases} \quad (9.48)$$

Proof. The two relevant differences are

$$(s + t) - \frac{3s + t}{2} = \frac{t - s}{2}, \quad \frac{3s + t}{2} - (s + 1) = \frac{s + t - 2}{2}.$$

Their signs change precisely at $t = s$ and $t = 2 - s$, giving the three regions in (9.48). $\square$

Imported theorem (low-parameter Furstenberg branch). *For $0 < s \leq 1$ and $0 \leq t \leq s$, the earlier Furstenberg estimate (6.84) gives $\dim_{\text{H}} F \geq s + t$. This is the exact historical low-parameter input used in the first branch below; it contributes no additional hidden hypothesis.*

Theorem III.9.41 (sharp Furstenberg numerology for regular-dimensional line families). *Let $F \subset \mathbb{R}^2$ be an (s, t) -Furstenberg set with $0 < s \leq 1$ and $0 \leq t \leq 2$. Suppose the associated t -dimensional line family satisfies $\dim_{\text{H}} \mathcal{L} = \dim_{\text{P}} \mathcal{L} = t$. Then*

$$\dim_{\text{H}} F \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}. \quad (9.49)$$

This is Theorem 1.18 of [18]. For $t \geq s$, its remaining research-scale core is the finite-scale Theorem 5.2; Proposition 6.4's continuum passage is proved above. For $t \leq s$ we reuse the earlier low-parameter estimate (6.84).

Deduction from the displayed interfaces. If $t \leq s$, the bound (6.84) gives $\dim_{\text{H}} F \geq s + \min\{s, t\} = s + t$, which is the first branch of Lemma III.9.40.

Assume now $t \geq s$. By (9.47), for every $u < \min\{(s + t)/2, 1\}$,

$$\dim_{\text{H}} F \geq s + u.$$

Letting $u \nearrow \min\{(s + t)/2, 1\}$ gives

$$\dim_{\text{H}} F \geq s + \min\{(s + t)/2, 1\} = \min \left\{ \frac{3s + t}{2}, s + 1 \right\}.$$

Together with the first branch and Lemma III.9.40, this is exactly (9.49). $\square$

Corollary III.9.42 (all-directions exponent). *If $t = 1$, then under the hypotheses of Theorem III.9.41,*

$$\dim_{\text{H}} F \geq \frac{3s + 1}{2}. \quad (9.50)$$

Proof. For $0 < s \leq 1$, $s \leq 1 \leq 2 - s$, so the middle branch of Lemma III.9.40 applies. $\square$

At $t = 1$, the exponent $(3s + 1)/2$ is exactly the upper-construction exponent from Theorem III.8.15. Thus the regular-line theorem closes the classical all-directions dimension numerology represented by that construction.

III.9.8 Comparison of hypotheses and consequences

There are three different advances in play. Orponen–Shmerkin's Duke theorem [22] gives a non-explicit positive gain over $2s$ when the line-family parameter satisfies $t > s$. The Shmerkin–Wang theorem proved in Theorem III.6.31 gives the stronger $s + t/2 + \epsilon$ lower bound in its stated range. The later Orponen–Shmerkin JAMS theorem [18] reaches the conjectured piecewise numerology when the relevant planar set or line family has equal Hausdorff and packing dimension.

These hypotheses must remain distinct. Equal Hausdorff and packing dimension does not imply Ahlfors regularity. Conversely, an arbitrary t -dimensional line family need not

have packing dimension t . The sharp formulas (9.44) and (9.49) therefore do not supersede the general results merely by notation.

For later chapters, the relevant conclusions are the continuum Duke deductions [Theorems III.9.4](#) and [III.9.7](#), the qualitative *ABC* consequence [Theorem III.9.25](#), the parameter conversion Lemma [III.9.35](#), and the algebraic minimum identity Lemma [III.9.40](#). The three JAMS headline theorems retain the exact source boundaries displayed above.

Part IV

Distances, Slices, and Fourier Bridges

Chapter III.10

Distance and Pinned-Distance Sets

This chapter studies the Euclidean distance set

$$\Delta(X) = \{|x - y| : x, y \in X\}$$

and its pinned variants. It has two complementary layers. The first is the classical Fourier layer: distance measures, spherical Fourier averages, the Mattila integral, and the distinction between full Hausdorff dimension and positive Lebesgue measure. The second is the endpoint multiscale layer of Shmerkin–Wang [33]. There equal Hausdorff and packing dimension supply scale-uniform upper control, radial non-concentration is encoded by thin tubes, and robust linear projections are transported through the nonlinear distance map.

The endpoint theorem developed here is deliberately narrower than the strongest theorem in the source. We prove that a Borel set in $\mathbb{R}^d$ with $\dim_{\text{H}} X = \dim_{\text{P}} X = d/2$ has $\dim_{\text{H}} \Delta(X) = 1$. The stronger source statement about the Hausdorff dimension of the exceptional set of centres is discussed at the end of the chapter.

Prerequisites and source boundary. Frostman, energy and capacitability results from Volume I may be used as declared interfaces. The Fourier-energy and Plancherel identities are part of the harmonic-analysis foundation. The finite-scale projection, robustness and smooth-image entropy tools are those proved in Chapters III.2 and III.6. The planar endpoint argument also uses the sharp Orponen–Shmerkin incidence block III.6.29.14, including the locally proved multiscale combination Proposition III.6.29.12. The radial L^p slicing estimate calibrated against Orponen [34] is reconstructed locally below from the Fourier–energy identity, a fractional trace estimate, and a Kaufman-type directional energy estimate; The co-rank-one robust projection engine of Shmerkin–Wang [33] is also proved below: in the hyperplane case He’s higher-rank reduction [15] terminates at the already-local rank-one theorem of Chapter III.6, while the remaining intersection and multiscale steps are proved here. The planar 5/4 pinned positive-measure escape of Guth–Iosevich–Ou–Wang [39] is reconstructed locally as well; the endpoint decoupling estimate is proved in a separate subsection below. The higher-dimensional Peres–Schlag positive-measure escape is proved from the generalized-projection theorem of Chapter III.4. The planar adaptedness gain and the general and planar nonlinear entropy engines are proved locally below. Thus the endpoint distance deduction uses the stated local results and Category 1–2 background, subject to that incidence gate; the adaptedness extraction itself is not an omitted step.

III.10.1 Distance sets, measures, and Fourier normalisation

Definition (distance and pinned-distance sets). For $X \subset \mathbb{R}^d$ and $y \in \mathbb{R}^d$, define

$$\Delta(X) = \{|x - z| : x, z \in X\}, \quad \Delta^y(X) = \{|x - y| : x \in X\}. \quad (10.1)$$

If $y \in X$, then $\Delta^y(X) \subset \Delta(X)$.

Definition (distance measures). For finite Borel measures μ, ν on $\mathbb{R}^d$, the two-measure distance measure is

$$\Delta(\mu, \nu) = ((x, z) \mapsto |x - z|)_\#(\mu \times \nu). \quad (10.2)$$

For fixed y , put $\Delta^y \mu = (x \mapsto |x - y|)_\# \mu$. When $\mu = \nu$ we write $\Delta(\mu)$.

Definition (Fourier normalisation). For a finite Borel measure μ ,

$$\hat{\mu}(\xi) = \int e^{-2\pi i x \cdot \xi} d\mu(x), \quad (10.3)$$

and the same convention is used for integrable functions. All Plancherel constants below correspond to this normalisation.

III.10.2 Distance measures and the Mattila integral

Let σ be normalised surface measure on $\mathbb{S}^{d-1}$ and define the spherical average

$$A_\mu(r) = \int_{\mathbb{S}^{d-1}} |\hat{\mu}(r\omega)|^2 d\sigma(\omega). \quad (10.4)$$

When unnormalised surface measure is used, the factor $|\mathbb{S}^{d-1}|$ is displayed.

The associated Mattila integral is

$$\mathcal{M}(\mu) = \int_1^\infty A_\mu(r)^2 r^{d-1} dr. \quad (10.5)$$

For a finite measure the omitted interval $(0, 1)$ contributes only a bounded amount depending on the total mass.

For $0 < s < d$, the Riesz energy is

$$I_s(\mu) = \iint |x - z|^{-s} d\mu(x) d\mu(z). \quad (10.6)$$

Its Fourier representation is

$$I_s(\mu) = c_{d,s} \int_{\mathbb{R}^d} |\hat{\mu}(\xi)|^2 |\xi|^{s-d} d\xi. \quad (10.7)$$

Falconer's dimension conjecture predicts $\dim_H \Delta(X) = 1$ whenever $\dim_H X \geq d/2$; the positive-measure conjecture predicts $|\Delta(X)| > 0$ when $\dim_H X > d/2$. These assertions remain distinct, especially at the critical threshold.

Lemma III.10.1 (support of a distance measure). *For finite Borel measures μ, ν ,*

$$\text{supp } \Delta(\mu, \nu) \subset \overline{\{|x - z| : x \in \text{supp } \mu, z \in \text{supp } \nu\}}. \quad (10.13)$$

If the supports are compact and both measures have full support on them, equality holds.

Proof. The map $(x, z) \mapsto |x - z|$ is continuous, so the support of its push-forward is contained in the closure of the image of the original support. Conversely, if $t = |x - z|$ with $x \in \text{supp } \mu$ and $z \in \text{supp } \nu$, every product neighbourhood of (x, z) has positive $\mu \times \nu$ mass and maps into an arbitrarily small neighbourhood of t . Hence t lies in the push-forward support. Compactness makes the distance image closed. $\square$

Let dR be Haar probability on $SO(d)$ and define

$$(\mathcal{R}f)(x) = \int_{SO(d)} f(Rx) dR. \quad (10.14)$$

Then $(\mathcal{R}f)(x) = \int_{\mathbb{S}^{d-1}} f(|x|\omega) d\sigma(\omega)$.

Lemma III.10.2 (radialisation–Plancherel and the Mattila identity). *For $f \in L^2(\mathbb{R}^d)$,*

$$\widehat{\mathcal{R}f} = \mathcal{R}\hat{f} \quad (10.15)$$

and

$$\int_0^\infty \left| \int_{\mathbb{S}^{d-1}} f(r\omega) d\sigma(\omega) \right|^2 r^{d-1} dr = \int_0^\infty \left| \int_{\mathbb{S}^{d-1}} \hat{f}(r\omega) d\sigma(\omega) \right|^2 r^{d-1} dr. \quad (10.16)$$

*If $\mu_\varepsilon = \mu * \varphi_\varepsilon$ is a smooth compactly supported approximation of a compactly supported probability measure and g_ε is the density of $\Delta(\mu_\varepsilon)$, then*

$$\int_0^\infty |g_\varepsilon(t)|^2 t^{-(d-1)} dt = C_d \int_0^\infty A_{\mu_\varepsilon}(r)^2 r^{d-1} dr. \quad (10.17)$$

Proof. Rotations commute with the Fourier transform, which gives (10.15) after averaging over $SO(d)$. Plancherel applied to $\mathcal{R}f$, followed by polar coordinates, gives (10.16); the same factor $|\mathbb{S}^{d-1}|$ occurs on both sides.

Put $h_\varepsilon = \mu_\varepsilon * \tilde{\mu}_\varepsilon$. Then $\hat{h}_\varepsilon = |\hat{\mu}_\varepsilon|^2$, while polar coordinates in the physical variable give

$$g_\varepsilon(t) = |\mathbb{S}^{d-1}| t^{d-1} \int_{\mathbb{S}^{d-1}} h_\varepsilon(t\omega) d\sigma(\omega).$$

Insert $f = h_\varepsilon$ in (10.16). This is exactly (10.17). $\square$

Theorem III.10.3 (Mattila integral criterion). *Let μ be a compactly supported probability measure. If $\mathcal{M}(\mu) < \infty$, then $\Delta(\mu)$ is absolutely continuous on $(0, \infty)$ with a nonzero locally L^2 density. Consequently*

$$|\Delta(\text{supp } \mu)| > 0. \quad (10.18)$$

Proof. Choose a nonnegative smooth approximate identity with $|\hat{\varphi}(\xi)| \leq 1$. Then $|\hat{\mu}_\varepsilon(\xi)| \leq |\hat{\mu}(\xi)|$, hence

$$A_{\mu_\varepsilon}(r) \leq A_\mu(r)$$

for every r . On $0 < r < 1$ we also have $A_{\mu_\varepsilon}(r) \leq 1$. Therefore (10.17) yields a uniform bound

$$\sup_{\varepsilon > 0} \int_0^\infty |g_\varepsilon(t)|^2 t^{-(d-1)} dt < \infty. \quad (10.19)$$

Fix $0 < a < R < \infty$. Since $t^{-(d-1)}$ is bounded below on $[a, R]$, (10.19) gives a uniform $L^2([a, R])$ bound. Passing to a weakly convergent subsequence and using weak convergence $\Delta(\mu_\varepsilon) \rightarrow \Delta(\mu)$ shows that the restriction of $\Delta(\mu)$ to $[a, R]$ has an L^2 density. Exhausting $(0, \infty)$ proves local absolute continuity there.

It remains to see that the density is not identically zero. If $\Delta(\mu)((0, \infty)) = 0$, then $|x - z| = 0$ for $(\mu \times \mu)$ -almost every pair (x, z) , which forces μ to be a Dirac mass. But for a Dirac mass $A_\mu(r) \equiv 1$, so (10.5) diverges. Hence some compact interval $[a, R] \subset (0, \infty)$ carries positive distance mass. On that interval write $d\Delta(\mu) = g(t)dt$ with $g \in L^2$ and $\int g > 0$. Then

$$0 < \left(\int_{\Delta(\text{supp } \mu) \cap [a, R]} g(t) dt \right)^2 \leq |\Delta(\text{supp } \mu)| \|g\|_2^2,$$

which proves (10.18). $\square$

The energy identity (10.7) and polar coordinates also give

$$I_s(\mu) = c'_{d,s} \int_0^\infty A_\mu(r) r^{s-1} dr. \quad (10.20)$$

Thus finite energy controls a first spherical average, whereas the Mattila integral contains its square with the much heavier weight r^{d-1} . At the critical dimension $d/2$ this mismatch is exactly why the Fourier criterion does not by itself prove Theorem III.10.30.

III.10.3 Regular pieces and two elementary reductions

The endpoint hypothesis is

$$\dim_H X = \dim_p X = d/2. \quad (10.8)$$

It is strictly weaker than Ahlfors regularity. Its role is to permit positive-mass restrictions whose upper box dimension is arbitrarily close to $d/2$.

Lemma III.10.4 (positive-mass regular-piece extraction). *Let $X \subset \mathbb{R}^d$ be Borel with $\dim_p X = s$. If λ is a nonzero finite Borel measure supported on X , then for every $\varepsilon > 0$ there is a compact set $Y \subset X$ with $\lambda(Y) > 0$ and*

$$\overline{\dim}_B Y \leq s + \varepsilon. \quad (10.21)$$

If in addition $\dim_H X = s > 0$, then for every $\varepsilon > 0$ one may choose two compact sets $Y_1, Y_2 \subset X$ with $\text{dist}(Y_1, Y_2) > 0$ and probability measures μ_i on Y_i such that

$$\mu_i(B(x, r)) \lesssim r^{s-\varepsilon} \quad (0 < r \leq 1), \quad \overline{\dim}_B Y_i \leq s + \varepsilon. \quad (10.22)$$

Proof. Use the standard packing-dimension characterisation

$$\dim_{\text{p}} X = \inf \left\{ \sup_i \overline{\dim}_B X_i : X \subset \bigcup_{i=1}^{\infty} X_i \right\}. \quad (10.23)$$

Choose such a countable cover with every upper box dimension at most $s + \epsilon$. Some X_i has positive outer λ -measure. Replacing X_i by its closure does not increase upper box dimension: a cover of X_i by balls of radius $r/2$ induces a cover of $\overline{X_i}$ by the corresponding closed balls of radius r . Since λ is supported on the Borel set X , $X \cap \overline{X_i}$ is a Borel subset of X with positive λ -mass. Intersect it with a sufficiently large closed ball and use inner regularity to obtain a compact positive-mass subset $Y \subset X \cap \overline{X_i}$. Monotonicity of covering numbers gives $\overline{\dim}_B Y \leq s + \epsilon$, proving (10.21).

For the second statement, if $\epsilon \geq s$, first apply the assertion with $\epsilon' = s/2 < \epsilon$; the resulting Frostman and box bounds are stronger than the bounds requested for ϵ . We may therefore assume $0 < \epsilon < s$. Choose, by the Volume I Frostman interface, a non-atomic $(s - \epsilon)$ -Frostman probability λ on a compact subset of X . Since $\lambda \times \lambda$ gives zero mass to the diagonal, there are disjoint balls of positive λ -mass. Restrict to compact subsets of those balls and normalise. Apply the first part to each restriction. Frostman upper bounds survive normalised restriction, with only a change of constant. $\square$

Lemma III.10.5 (pinned images suffice for the unpinned theorem). *If for every $\eta > 0$ there is $y_\eta \in X$ with*

$$\dim_{\text{H}} \Delta^{y_\eta}(X) \geq 1 - \eta, \quad (10.24)$$

then $\dim_{\text{H}} \Delta(X) = 1$.

Proof. For $y_\eta \in X$, inclusion (10.1) gives $\Delta^{y_\eta}(X) \subset \Delta(X)$. Hence $\dim_{\text{H}} \Delta(X) \geq 1 - \eta$ for every $\eta > 0$; the reverse inequality follows from $\Delta(X) \subset \mathbb{R}$. $\square$

Lemma III.10.6 (critical lattice scaling). *Let $q \geq 1$, let $P_N = \{0, 1, \dots, N-1\}^d$, and rescale it into a bounded cube. Then $|P_N| = N^d$, while the number of Euclidean distances determined by P_N is at most $O_d(N^2)$. At spatial scale $\delta \asymp N^{-2}$ this has the critical relation*

$$|P_N| \asymp \delta^{-d/2}, \quad |\Delta(P_N)| \lesssim \delta^{-1}. \quad (10.25)$$

Consequently the exponent $d/2$ is the natural endpoint at which pure counting alone cannot force more than one-dimensional distance growth.

Proof. A squared lattice distance has the form $m_1^2 + \dots + m_d^2$ with $|m_j| \leq N-1$, so it is an integer in $[0, dN^2]$. Hence there are $O_d(N^2)$ possible distances. Substituting $N \asymp \delta^{-1/2}$ gives (10.25). $\square$

The lemma is a scaling obstruction, not a counterexample to Theorem III.10.30. The classical Cantorisations of such lattices explain why the threshold $d/2$ in Falconer's conjecture cannot be lowered; see the discussion in Shmerkin–Wang [33].

III.10.4 Planar thin-tube bootstrap

Let μ, ν be probability measures. For $x \in \mathbb{R}^d$, say that (x, ν) has (s, K, c, δ) -thin tubes if there is a Borel function $0 \leq \phi_x \leq 1$ such that

$$\int \phi_x d\nu \geq c, \quad \int_T \phi_x d\nu \leq Kr^s \quad (10.9)$$

for every open width- r tube T through x and every $r > \delta$. The pair (μ, ν) has such thin tubes if the set of admissible centres has μ -mass at least c . The pair has *strong* thin tubes if the witnesses are indicators of compact fibres Y_x , the incidence graph is compact, both the centre set and each fibre have mass at least c , and

$$\nu(Y_x \cap T) \leq Kr^s. \quad (10.10)$$

The distance map satisfies

$$F_y(x) = |x - y|, \quad \text{dir } \nabla_x F_y(x) = \frac{x - y}{|x - y|}. \quad (10.26)$$

Thus non-concentration of radial directions is exactly the direction information needed to linearise pinned distance maps.

Lemma III.10.7 (thin tubes can be made strong). *Suppose (μ, ν) has $(s, K, 1 - c, \delta)$ -thin tubes. If $c < 1/4$, then after discarding at most $O(c)$ mass one obtains strong $(s, 2K, c', \delta)$ -thin tubes with $c' > 1 - 3c$. For arbitrary $c \in (0, 1)$ one obtains, after positive-mass restriction, a strong thin-tube configuration with the same exponent and a finite enlarged constant.*

Proof. For each good centre choose a witness $\nu_x = \phi_x \nu \leq \nu$ and write $g_x = d\nu_x/d\nu$. For $a \in (0, 1)$ put $Y_x = \{g_x \geq a\}$. Since $\int g_x d\nu \geq 1 - c$,

$$\nu(Y_x) \geq 1 - \frac{c}{1 - a}, \quad \nu(Y_x \cap T) \leq a^{-1}Kr^s. \quad (10.27)$$

Take $a = 1/2$ when $c < 1/4$. We now justify the measurable witness selection and compact incidence graph. The admissible densities $g_x \in L^\infty(\nu)$ lie in the weak-* compact metrizable set $\{0 \leq g \leq 1\}$; their mass and tube inequalities are weak-* closed. It suffices to impose the latter for a countable dense set of tube directions and rational widths, since any open tube is contained in slightly wider members of that family and the width can then decrease to its original value. The constraints have a Borel graph in (x, g) , so standard compact-fibre measurable selection supplies a Borel choice $x \mapsto g_x$ on the Borel set of admissible centres. As in Lemma III.6.12, finite-partition conditional expectations give a jointly Borel representative $g(x, z)$, hence a Borel graph $G = \{(x, z) : g(x, z) \geq 1/2\}$.

The good centres have μ -mass at least $1 - c$, and each section G_x has ν -mass at least $1 - 2c$. Choose a compact $K \subset G$ whose $(\mu \times \nu)$ -mass differs from that of G by less than ε^2 . Markov's inequality applied to $\nu(G_x \setminus K_x)$ shows that, outside a set of centre mass at most ε , one has $\nu(K_x) \geq 1 - 2c - \varepsilon$. Retain a compact subset of these centres after an additional mass loss ε and restrict K to it. The resulting compact incidence graph has centre mass at least $1 - c - 2\varepsilon$, fibre mass at least $1 - 2c - \varepsilon$, and tube constant $2K$. Choosing $\varepsilon < c/2$ gives a common $c' > 1 - 3c$. For general c , choose any $a < 1 - c$ and repeat the same compact selection with a sufficiently small positive mass loss. $\square$

Lemma III.10.8 (a pointwise regular core). *Let μ be a $(t - \epsilon/10)$ -Frostman probability measure on a compact set with upper box dimension at most $t + \epsilon/10$. For every $q > 0$ there are a compact set Y with $\mu(Y) > 1 - q$, a constant $C_q \geq 1$, and m_0 such that for every $x \in Y$ and $m \geq m_0$, the dyadic m -cube $Q_m(x)$ satisfies*

$$C_q^{-1} 2^{-m(t+\epsilon/4)} \leq \mu(Q_m(x)) \leq C_q 2^{-m(t-\epsilon/4)}. \quad (10.28)$$

The estimate concerns the original measure μ , not its restriction μ_Y . It is the form needed when conditionals are taken before a positive-mass incidence restriction.

Proof. The Frostman bound, by enclosing a dyadic cube in a ball of comparable radius, gives the upper estimate for every cube. For all large m the upper-box assumption bounds the number of cubes meeting $\text{supp } \mu$ by $2^{m(t+\epsilon/8)}$. Call such a cube light when its original mass is less than $2^{-m(t+\epsilon/4)}$. The union of light cubes has μ -mass at most $2^{-m\epsilon/8}$. Choose m_0 so that the sum of these bounds for $m \geq m_0$ is less than $q/2$. Outside the union of the light cubes the lower estimate holds simultaneously at every level $m \geq m_0$; inner regularity supplies a compact subset of this Borel set with mass greater than $1 - q$. No estimate for the mass of a *further restriction* inside each cube is asserted. $\square$

Proposition III.10.9 (initial planar radial thin tubes). *Let μ, ν be separated compactly supported Frostman probabilities in $\mathbb{R}^2$, each with some positive Frostman exponent, and suppose $\nu(\ell) = 0$ for every affine line ℓ . Then (μ, ν) has s_0 -thin tubes for some $s_0 > 0$. If both Frostman exponents are at least $q > 0$, the exponent $s_0 = s_0(q) > 0$ can be chosen independently of the measures; only the constants and starting scale depend on them.*

Proof. Apply the spherical-direction decay interface from Chapter 6 with $d = 2$ and $k = 1$. It gives a positive- μ set of centres and positive- ν fibres on which the radial direction measure has a power bound on neighbourhoods of hyperplanes. On S^1 such a hyperplane meets the sphere in two antipodal points, so the conclusion is an angular Frostman estimate. Dyadic pigeonholing makes the exponent, constant and fibre mass uniform on a positive-mass centre set. Because the supports are separated, angular balls pull back to tubes through the centre with comparable widths. This is exactly an s_0 -thin-tube estimate. For the uniform assertion, the local spherical-direction theorem in Chapter 6 has $k = 1$ and plane-neighbourhood exponents $p = q$ when both measures are q -Frostman. Its output exponent $q_{2,1}(q, q) > 0$ depends only on q . $\square$

Definition (rough dyadic regularity). *Let $0 < t \leq d$, $\epsilon > 0$, $C \geq 1$, and $m_0 \leq m_1$ be integers. A probability μ on $[0, 1]^d$ is $(t, \epsilon, C; [m_0, m_1])$ -roughly regular if, for every integer $k \in [m_0, m_1]$ and every dyadic k -cube Q with $\mu(Q) > 0$,*

$$C^{-1} 2^{-k(t+\epsilon)} \leq \mu(Q) \leq C 2^{-k(t-\epsilon)}. \quad (10.28a)$$

The all-scales version means that the same t, ϵ, C, m_0 work for every $k \geq m_0$. Normalised restrictions are allowed only after their changed constant C has been recorded. In the statements below, “roughly t -regular” refers to this definition with the loss ϵ and range specified in that statement.

Definition (upper regularity with abundant lower cores). Fix $0 < t \leq 2$, $\sigma > 0$ and $C \geq 1$. A compactly supported probability μ is (t, σ, C) -core-regular if every occupied dyadic k -cube satisfies

$$\mu(Q) \leq C2^{-k(t-\sigma)}$$

for all sufficiently large k , and for every $q \in (0, 1)$ there are $C_q \geq 1$, k_q , and a compact $E_q \subset \text{supp } \mu$ with $\mu(E_q) > 1 - q$ such that

$$\mu(Q_k(x)) \geq C_q^{-1} 2^{-k(t+\sigma)} \quad (x \in E_q, k \geq k_q). \quad (10.28b)$$

All masses here belong to the original measure μ . The constants C_q, k_q may depend on q ; neither the exponent nor the ambient measure changes when a core is chosen.

Definition (finite-scale measure and direction measure). For $\delta \in (0, 1)$, $a > 0$ and $K \geq 1$, a probability ω on $\mathbb{R}^d$ is a (δ, a, K) measure if $\omega(B(x, r)) \leq Kr^a$ for every x and $\delta \leq r \leq 1$. A probability on $\mathbb{S}^{d-1}$ is a (δ, a, K) direction measure if the same bound holds for spherical balls (with the spherical metric). Fixed changes of the ball-radius convention alter K by a dimensional factor, not a .

Definition (quantitatively adapted direction law). Fix $0 < t \leq d$, $0 < u \leq 1$, $0 < \epsilon < u$, $\gamma > 0$, $K \geq 1$, and a dyadic scale δ . A probability ρ on $\mathbb{S}^{d-1}$ is $(\delta, t \rightarrow u; \epsilon, \gamma, K)$ -adapted if, for every probability ω on $[0, 1]^d$ satisfying

$$\omega(B(x, r)) \leq K\delta^{-\epsilon}r^t \quad (\delta \leq r \leq 1),$$

there is a Borel exceptional set $E_\omega \subset \mathbb{S}^{d-1}$ with $\rho(E_\omega) \leq \delta^\gamma$ such that $P_\theta \omega$ is $(\delta, u - \epsilon, \delta^\gamma)$ -robust for every $\theta \notin E_\omega$. Here (δ, a, p) -robust means that every Borel set A of image mass greater than p meets more than δ^{-a} half-open dyadic intervals of length δ . Passing to arbitrary length- δ intervals changes the count by at most a fixed factor, which is absorbed only after an explicit exponent slack. For $\delta = 2^{-m}$ and $p = \delta^\gamma$, this is the (a, γ, m) convention of Chapter III.6. A family of laws is uniformly adapted on a range of scales if the same ϵ, γ, K work at every scale in that range.

The last step in the planar projection argument is a finite-sampling issue. A positive proportion of bad directions need not itself be a well-spaced finite family, and the points incident to many selected directions must be sampled from the *same* positive-mass core. The following lemma makes both selections without changing their Frostman exponents.

Lemma III.10.10 (sampling a finite-scale Frostman law). Fix $0 < u \leq d \leq 2$, $0 < \eta < u/20$, and a fixed bounded dyadic chart in $\mathbb{R}^d$. For all sufficiently small dyadic δ , suppose a probability ν on the chart satisfies

$$\nu(B(z, r)) \leq C_0 \delta^{-2\eta} r^u \quad (\delta \leq r \leq 1). \quad (10.28c)$$

There is a finite sample $F \subset \text{spt } \nu$, with no two members in the same dyadic δ -cell, such that

$$\frac{1}{2}N \leq |F| \leq N, \quad N = \lceil \delta^{-u+4\eta} \rceil, \quad (10.28d)$$

and, for every dyadic cell Q of side $r \in [\delta, 1]$,

$$|F \cap Q| \leq C_1 \delta^{-7\eta} |F| r^u. \quad (10.28e)$$

Here C_1 depends only on the chart, C_0 , and d . The same conclusion holds in a spherical direction chart after a fixed bi-Lipschitz parametrisation.

Proof. Draw N independent points with law ν . For each dyadic r -cell Q , its sampled occupancy X_Q has mean at most $CN\delta^{-2\eta}r^u$, and the factorial-moment bound gives

$$\Pr\{X_Q \geq N\delta^{-6\eta}r^u\} \leq (eC\delta^{4\eta})^{\lfloor N\delta^{-6\eta}r^u \rfloor}.$$

At $r = \delta$, the exponent on the right is at least $c\delta^{-2\eta}$; it is larger at all coarser scales. There are $O(\delta^{-2} \log(1/\delta))$ relevant cells, so with probability tending to one all occupancies obey the claimed bound with $\delta^{-6\eta}N$ in place of the right side of (10.28e).

For each pair of sampled points, the probability of occupying the same δ -cell is at most $C\delta^{u-2\eta}$. Thus the expected number of indices which must be deleted to leave one point per cell is at most $CN^2\delta^{u-2\eta} \leq C'N\delta^{2\eta}$. Markov's inequality shows that, with probability tending to one, at most $N\delta^{3\eta/2}$ indices need deletion. Choose a sample with both properties and delete the repeated-cell indices. For small δ , its size is at least $N/2$, and the occupancy bound becomes (10.28e), with one spare power $\delta^{-\eta}$ to absorb fixed chart and dyadic/ball comparisons. A direction chart is handled by its bi-Lipschitz coordinates. $\square$

Proposition III.10.11 (planar adaptedness gain). *Fix $0 < s < \min\{t, 1\}$ with $t < 2$. There is $0 < \eta = \eta(s, t) < 1 - s$ such that, for all sufficiently small dyadic δ , if μ is a $(\delta, t, \delta^{-\eta})$ measure on $[0, 1]^2$ and ρ is a $(\delta, s, \delta^{-\eta})$ direction measure on $\mathbb{S}^1$, then outside a set of ρ -mass at most δ^η , the projected measure $P_\theta\mu$ is*

$$(\delta, s + \eta, \delta^\eta)\text{-robust}. \quad (10.29)$$

The gain is uniform on compact subsets of $\{(s, t) : 0 < s < \min\{t, 1\}, t < 2\}$.

Proof. Let $\zeta = \zeta(s, t) > 0$ be the gain in the sharp dual incidence theorem III.6.29.14. Choose $\eta > 0$ so small that

$$20\eta < \min\{s, t, 1 - s, \zeta\}. \quad (10.29a)$$

All fixed chart and thickening constants below are absorbed by decreasing δ . Suppose the conclusion fails. The set B of bad directions has $\rho(B) > \delta^\eta$. Partition the circle into a fixed number of slope charts and retain a chart on which $\rho(B) \geq c\delta^\eta$. Replace B by its intersection with that chart, retaining the same notation. In both applications of the sampling lemma below, the sample is chosen among the actual draws from the restricted measure, so it lies in the measured Borel set even when the support of that restriction meets its boundary. For each direction θ in this chart, order the finitely many dyadic δ -intervals meeting $P_\theta([0, 1]^2)$ by decreasing $P_\theta\mu$ -mass, breaking ties by their grid index. Let $\mathcal{J}_\theta$ be the first $M = \lceil \delta^{-s-\eta} \rceil$ intervals. This is a Borel choice in θ . Failure of robustness means that

$$\mu\{y : P_\theta y \in \cup \mathcal{J}_\theta\} > \delta^\eta \quad (\theta \in B). \quad (10.29b)$$

Indeed, if any collection of at most M intervals has mass greater than δ^η , the M heaviest do too.

Normalise $\rho|_B$ to a probability ρ_B . Its direction Frostman constant is at most $C\delta^{-2\eta}$. Apply Lemma III.10.10 with $u = s, d = 1$ to obtain a δ -separated family $E \subset B$ of size $N \asymp \delta^{-s+4\eta}$ and

$$|E \cap J| \leq C\delta^{-7\eta}|E|r^s \quad \text{for every direction arc } J \text{ of length } r \geq \delta. \quad (10.29c)$$

In the sampling proof at most $N\delta^{3\eta/2}$ repeated-cell draws were deleted. Since every original sampled direction obeys (10.29b), this deletion changes the incidence average by at most $\delta^{3\eta/2}$. Hence

$$\int f_E(y) d\mu(y) \geq \frac{1}{2}\delta^\eta, \quad f_E(y) := |E|^{-1} \sum_{\theta \in E} \mathbf{1}_{\{P_\theta y \in \cup \mathcal{J}_\theta\}}. \quad (10.29d)$$

The common high-incidence core $Y_0 = \{y : f_E(y) \geq \delta^\eta/4\}$ therefore has $\mu(Y_0) \geq \delta^\eta/4$. The normalized restriction μ_{Y_0} has Frostman constant at most $C\delta^{-2\eta}$ at scales $r \geq \delta$. Apply the same sampling lemma with $u = t, d = 2$ to obtain $Y \subset Y_0$, one point per δ -cell, $|Y| \asymp \delta^{-t+4\eta}$, and

$$|Y \cap Q| \leq C\delta^{-7\eta}|Y|r^t \quad (Q \text{ a dyadic } r\text{-cell, } r \geq \delta). \quad (10.29e)$$

Every $y \in Y$ is incident to at least $\delta^\eta|E|/4$ of the selected direction strips.

In the chosen bounded slope chart, the exact line through y with normal θ has offset $P_\theta y$. When $P_\theta y \in I$, its slope and offset lie in one of $O(1)$ dyadic line-parameter squares determined by (θ, I) . Let $\mathcal{Q}$ be the union of these candidate squares over all selected (θ, I) . Thus

$$|\mathcal{Q}| \leq C|E|M \lesssim \delta^{-2s+3\eta}. \quad (10.30)$$

For $y \in Y$, retain one parameter square for each $\theta \in E$ whose selected strip contains y , and call this set P_y . Distinct selected directions give distinct squares up to fixed multiplicity, so $|P_y| \gtrsim \delta^\eta|E|$. If a parameter ball has radius r , its direction coordinates lie in an arc of length Cr . By (10.29c),

$$|P_y \cap Q| \leq C\delta^{-8\eta}|P_y|r^s \quad (\delta \leq r \leq 1). \quad (10.30a)$$

The point–line duality lemma III.6.29.1 turns Y into a family $\mathcal{L}$ of dyadic δ -tubes and each P_y into incident dyadic points: choosing the square containing the exact line through y makes incidence literal, up to the fixed dyadic-cell enlargement. Equations (10.29e) and (10.30a), with bounded chart changes, make these respectively a $(\delta, t, \delta^{-\zeta})$ -tube family and $(\delta, s, \delta^{-\zeta})$ -point fibres; the budget (10.29a) absorbs all fixed constants. The sharp dual incidence theorem therefore gives

$$|\mathcal{Q}| \geq \delta^{-2s-\zeta}. \quad (10.31)$$

This contradicts (10.30) for small δ . Here is the compact-uniformity argument without assuming continuity of the Chapter 6 gain. At a fixed pair (a_0, b_0) , let $\zeta_0 > 0$ be its incidence gain. For (s, t) sufficiently close to (a_0, b_0) , any $(\delta, s, \delta^{-\lambda})$ -set is also a $(\delta, a_0, \delta^{-\lambda-(a_0-s)_+})$ -set: for $r \geq \delta$, $r^s \leq \delta^{-(a_0-s)_+} r^{a_0}$. The same transfer applies to the exponent t of the other family.

If $|s - a_0|, |t - b_0| < \zeta_0/10$ and $10\eta < \zeta_0/2$, the fixed (a_0, b_0) theorem therefore applies to the sampled families and gives a lower bound with exponent $2a_0 + \zeta_0 > 2s + \zeta_0/2$. Every parameter pair has such an open neighbourhood. A compact set is covered by finitely many of them; choose η smaller than all their admissible values and the uniform positive parameter margins. This yields one gain and one small-scale threshold throughout that compact set. $\square$

Proposition III.10.12 (roughly regular nonlinear image lemma). *Fix $0 < s \leq 1$, $0 < t \leq 2$, and $0 < \epsilon < \min\{s, t, 1/16\}$. Let μ have compact support in $(0, 1)^2$ and be $(t, \epsilon/4, C_{\text{reg}}; [k_0, \infty))$ -roughly regular. Let $F_y : U \rightarrow \mathbb{R}$ be a compact nonsingular C^2 family with U an open neighbourhood of $\text{supp } \mu$ and compact parameter space carrying a Borel probability ν . Suppose $G \subset \text{supp } \mu \times \text{supp } \nu$ is compact with $(\mu \times \nu)(G) > 0$. For μ -almost every x with $\nu(G_x) > 0$, suppose the derivative-direction law associated to ν_{G_x} is $(2^{-k}, t - \epsilon \rightarrow s; \epsilon, \gamma, K)$ -adapted for every $k \geq k_0$, with fixed $\gamma > 0$ and $K \geq C_2 C_{\text{reg}}^2$, where C_2 is the planar ball-cube comparison constant. Then there are a Borel positive- ν -measure set Y , a number $b > 0$, compact sets $Z_y \subset G^y := \{x : (x, y) \in G\}$ with $\mu(Z_y) \geq b$ for every $y \in Y$, and $\eta > 0$ such that, provided $s > C_* \sqrt{\epsilon}$, for every $y \in Y$ and every sufficiently large m (uniformly in y),*

$$F_y(\mu_{Z_y}) \text{ is } (2^{-m}, s - C_* \sqrt{\epsilon}, 2^{-\eta m})\text{-robust.} \quad (10.32)$$

Here $C_* > 0$ is dimensional. If $s \leq C_* \sqrt{\epsilon}$, only the trivial nonnegative dimension bound is asserted.

Proof. Let

$$A = \{x \in \text{supp } \mu : \mu(Q_k(x)) > 0 \text{ for every } k \geq k_0\}.$$

The union of the empty dyadic cubes at all these levels has μ -measure zero, so $\mu(A) = 1$. The finite Borel measure $\sigma(B) = (\mu \times \nu)(G \cap (B \times \text{supp } \nu))$ is concentrated on A . By inner regularity of σ , choose a compact $E \subset A$ with $\sigma(E) > 0$. Restrict the graph only in its centre coordinate; its complete original fibres G_x are unchanged for $x \in E$, so the adaptedness premise survives. Apply Proposition III.10.27 below with $d = 2$, $u = s$, this E , and the restricted graph. The all-scales rough-regularity assumption gives its global upper cube inequality and the pointwise lower inequality at every $x \in E$. Its source constant is exactly the one stated here. The proposition's simultaneous-scale proof retains one positive-measure parameter set and compact fibres uniformly at every output scale; it does not select new parameters as the scale changes. Its uniform positive-parameter conclusion is precisely (10.32). $\square$

Lemma III.10.13 (one-step thin-tube improvement). *Fix $0 < s < t \leq 1$. There are $\eta = \eta(s, t) > 0$ and $\epsilon_0 = \epsilon_0(s, t) > 0$ with the following property. Let $0 < \epsilon < \epsilon_0$, and let μ, ν be separated $(t, \epsilon/4, C_{\text{reg}})$ -core-regular probabilities in $\mathbb{R}^2$. If one common compact incidence graph witnesses strong (s, K, c) -thin tubes at every sufficiently small scale, then, after interchanging their roles and retaining a compact positive-product-mass incidence subgraph, the original pair (ν, μ) has strong $(s + \eta/3, K', c')$ -thin tubes at all sufficiently small scales on one common compact incidence graph. Here K', c' and the starting scale may depend on $C_{\text{reg}}, K, c, \mu, \nu$; the gain η does not.*

Proof. Choose $\eta > 0$ below one quarter of the uniform gain in Proposition III.10.11 on the compact parameter range $t - \epsilon \in [(s + 3t)/4, t]$, and small enough that $s + \eta < 1$. Choose ϵ_0 so that, for $\epsilon < \epsilon_0$,

$$\epsilon < (t - s)/4, \quad \epsilon < \eta/10, \quad C_*\sqrt{\epsilon} < \eta/6, \quad \epsilon < \min\{s, t, 1/16\}. \quad (10.32a)$$

Apply a common dyadic similarity (a power-of-two dilation followed by a dyadic translation) to place both compact supports inside $(0, 1)^2$; fine-scale core regularity and strong thin tubes survive with changed fixed constants. Let G be the common compact strong incidence graph and put $p = (\mu \times \nu)(G) > 0$. Choose lower-regular compact cores for the two ambient measures with individual complementary mass below $p/8$. The intersection of G with their product still has positive product mass; take a compact positive-mass subgraph and select a centre subset with uniformly positive parameter-fibre mass. Keep its full fibres during this centre selection. The strong s -tube bound persists on the subgraph with a changed fixed constant.

The map needed for tubes is the *angular* map, not the distance map. Cover the separated compact product supports by finitely many products $Q \times V$, with Q a sufficiently small dyadic cube and V a compact parameter patch, on each of which a smooth real branch $A_y(x) = \arg(x - y)$ exists on a neighbourhood of $\bar{Q} \times V$. Retain one product meeting the strong incidence graph in positive product mass. Use the normalised original source $\mu|_Q/\mu(Q)$ and the normalised parameter measure $\nu|_V/\nu(V)$; the global upper bound for the source persists from the level of Q , and the pointwise lower bound persists on the selected centre core, with only changed fixed constants. Fibre-mass pigeonholing and compact selection give a positive-mass compact subgraph H with uniform positive fibre masses. Its centre projection E is compact and lies in the selected lower-regular core. Write C_{loc} for a common upper/lower cube constant of the normalised source on E . Its conditional direction laws have a uniform s -Frostman bound; after the fixed normalising factors are absorbed at sufficiently fine scales, Proposition III.10.11 makes these laws uniformly $(2^{-k}, t - \epsilon \rightarrow s + 2\eta/3; \epsilon, \eta, K_*)$ -adapted, where $K_* \geq C_2 C_{\text{loc}}^2$ is fixed. Relabel these normalised measures μ, ν for the remainder of this one-step argument. On this patch

$$\text{dir } \nabla_x A_y(x) = R_{\pi/2} \frac{x - y}{|x - y|} \quad (10.32b)$$

up to a fixed sign. Rotation preserves the directional Frostman bound, so the planar adaptedness gain applies to these derivative-direction laws as well. Apply the pointwise-core Proposition III.10.27 with $d = 2$, $F_y = A_y$, the selected compact centre core E , and $u = s + 2\eta/3$. It gives a positive- ν set Y , a uniform $b > 0$, and compact fibres Z_y with $\mu(Z_y) \geq b$ on which the angular images of μ_{Z_y} are $(2^{-m}, a, 2^{-\eta_1 m})$ -robust at every sufficiently large m , uniformly in y , where $a = s + 2\eta/3 - C_*\sqrt{\epsilon} > s + \eta/2$ and $\eta_1 > 0$.

Put $a' = s + \eta/3$. For each $y \in Y$ and large m , let $\mathcal{H}_m(y)$ consist of the dyadic intervals of length 2^{-m} whose angular-image mass under μ_{Z_y} exceeds $2^{-a'm}$. There are fewer than $2^{a'm} < 2^{am}$ of them, so robustness gives

$$\mu_{Z_y}(A_y^{-1}(\cup \mathcal{H}_m(y))) \leq 2^{-\eta_1 m}. \quad (10.32c)$$

Choose the starting level so that the sum of these losses is less than $1/2$. The Borel graph obtained by deleting all the heavy preimages has positive product mass. Inner regularity gives a compact positive-mass subgraph; Fubini and another fibre-mass selection give

uniform positive centre and fibre masses. On each retained fibre, every angular dyadic interval at level m has μ -mass at most $2^{-a'm}$. An arbitrary angular interval of comparable size meets at most two such dyadic intervals; the finitely many coarser scales are absorbed in K' . Since angular intervals pull back to tubes through γ with comparable width on separated supports, this is exactly the strong (a', K', c') -thin-tube estimate for the swapped pair. Translating the graph and constants back through the two normalisations gives a strong incidence subgraph for the original pair (ν, μ) ; the ambient measures have not been replaced for the next bootstrap step. $\square$

Proposition III.10.14 (iteration to the source dimension). *Let $0 < t \leq 1$ and $\zeta > 0$. There is $\epsilon_0(t, \zeta) > 0$ such that the following holds. Suppose $0 < \epsilon < \epsilon_0$, and μ_1, μ_2 are separated (t, ϵ, C) -core-regular probabilities and neither charges a line. After positive-mass compact restriction they have strong*

$$(t - \zeta, K, 1 - c)\text{-thin tubes} \quad (10.33)$$

for any prescribed small $c > 0$.

Proof. Proposition III.10.9 and Lemma III.10.7 provide a common compact all-scale strong incidence graph with a starting exponent $s_0 > 0$. Indeed, $\epsilon < t/4$ makes each measure $(t/2)$ -Frostman with a fixed constant, so the exponent $s_0 = s_0(t/2)$ from the initial proposition depends only on t . Decrease it if necessary to have $s_0 < t - \zeta$. The one-step gain and its admissible rough-regularity slack have positive minima on the compact exponent interval $[s_0, t - \zeta]$. Choose $\epsilon_0(t, \zeta)$ so that $4\epsilon_0$ is below this minimum slack. Apply Lemma III.10.13 finitely many times, using 4ϵ as its loss parameter and alternating the *original* core-regular measures. Each step carries one compact all-scale incidence graph; no ambient measure is restricted during the iteration. This reaches exponent $t - \zeta$.

To obtain the advertised $1 - c$ masses, use product dyadic differentiation on the final positive-product-mass compact graph H . There are dyadic squares Q, R whose product conditional measure gives H density greater than $1 - c^2/100$. Normalise $\mu_1|_Q$ and $\mu_2|_R$. Fubini–Markov discards at most $c/10$ of the centres whose relative fibre mass is below $1 - c/10$. Inner regularity then gives a compact subgraph losing less than another $c^2/100$ in product measure, and a second Fubini–Markov selection ensures both centre and fibre masses exceed $1 - c$. The tube estimate on H survives these final restrictions, with its constant multiplied by the reciprocal of the appropriate square mass. Thus the normalised dyadic restrictions give (10.33). $\square$

Lemma III.10.15 (subcritical Kaufman adaptedness). *Fix $0 < u < \min\{q, t_*\}$, where $0 < q \leq 1$ and $0 < t_* \leq 2$. Let ρ be a q -Frostman probability on $\mathbb{S}^1$. For every $K \geq 1$ and $0 < \beta < u$ there are $\gamma > 0$ and $\delta_0 > 0$ such that ρ is uniformly $(\delta, t_* \rightarrow u; \beta, \gamma, K)$ -adapted for every dyadic $0 < \delta \leq \delta_0$.*

Proof. Choose $u < w < \min\{q, t_*\}$ and then $0 < \gamma < (w - u)/4$. For a probability ω in the source class of the adaptedness definition, put

$$I_w^{(\delta)}(\omega) = \iint \max\{|x - x'|, \delta\}^{-w} d\omega(x) d\omega(x').$$

The layer-cake formula and its $(\delta, t_*, K\delta^{-\beta})$ ball bound give

$$I_w^{(\delta)}(\omega) \leq C_{w,t_*} K\delta^{-\beta}, \quad (10.34a)$$

since $t_* > w$. For each $v \neq 0$, the q -Frostman bound on ρ and a dyadic angular-annulus sum show

$$\int_{S^1} \max\{|P_\theta v|, \delta\}^{-w} d\rho(\theta) \leq C_{q,w,\rho} \max\{|v|, \delta\}^{-w}.$$

Indeed, for $|v| \geq \delta$ the directions with $|P_\theta v| \leq 2^{-j}|v|$ lie in two arcs of total ρ -mass at most $C2^{-jq}$; their weighted sum is $\sum_{j \geq 0} O(2^{j(w-q)}) < \infty$. The case $|v| < \delta$ is immediate. Fubini and Markov therefore give, outside a ρ -exceptional set of mass at most δ^γ ,

$$I_w^{(\delta)}(P_\theta \omega) \leq CK\delta^{-\beta-\gamma}. \quad (10.34b)$$

Let a Borel image set A have projected mass $p > \delta^\gamma$, and suppose it meets N dyadic intervals of length δ . The normalised restriction of $P_\theta \omega$ to A has truncated energy at most $CK\delta^{-\beta-\gamma}p^{-2} \leq CK\delta^{-\beta-3\gamma}$. Pairs in the same one of the N intervals contribute at least δ^{-w} , and the sum of squared interval masses is at least $1/N$. Thus

$$N \geq cK^{-1}\delta^{-(w-\beta-3\gamma)} > \delta^{-(u-\beta)}$$

for all sufficiently small δ , because $w - u > 4\gamma$. All constants are independent of the test source ω , so the same exceptional-mass and robustness parameters work for every source in the class. This is exactly the claimed uniform adaptedness. $\square$

Theorem III.10.16 (planar equal-Hausdorff/packing endpoint). *Let $0 < t \leq 1$ and let $X \subset \mathbb{R}^2$ be Borel with $\dim_H X = \dim_p X = t$. For every $\zeta > 0$ there is $y \in X$ such that*

$$\dim_H \Delta^\gamma(X) \geq t - \zeta. \quad (10.34)$$

Proof. If $\zeta \geq t$, the assertion is the trivial nonnegative dimension bound. Assume $0 < \zeta < t$. Choose $\epsilon > 0$ sufficiently small relative to t, ζ and put $\lambda = 10\epsilon$. Lemma III.10.4 gives separated compact pieces supporting $(t - \epsilon)$ -Frostman measures and having upper box dimension at most $t + \epsilon$. If one measure gives positive mass to a line, restrict to that line: on either half-line the distance map from a centre on the line is bi-Lipschitz, giving $\dim_H \Delta^\gamma(X) \geq t - \epsilon$.

Otherwise apply Lemma III.10.8 with every $q \in (0, 1)$. Its fixed Frostman and box hypotheses give global upper cube bounds at exponent $t - \lambda/4$ and compact lower cores of mass $> 1 - q$ at exponent $t + \lambda/4$. Thus both original measures are $(t, \lambda/4, C)$ -core-regular, with the same C after enlargement. Proposition III.10.14, with its regularity parameter $\lambda/4$, gives strong $(t - \zeta/3)$ -thin tubes on a common compact graph after the final dyadic restrictions. Those restricted ambient measures remain core-regular, with changed fixed constants: an original lower core with complementary mass below $q\mu(Q)$ yields, by inner regularity inside the half-open dyadic square Q , a compact $> 1 - q$ core after normalisation. Apply a common dyadic similarity so that the two restricted compact supports lie inside $(0, 1)^2$.

Put $q_* = t - \zeta/3$ and $u = t - \zeta/2$; then $u < q_*$. Choose λ still smaller if necessary so $u < t - \lambda$ and $u - C_*\sqrt{\lambda} \geq t - \zeta$. On the final strong graph the derivative-direction laws of $F_y(x) = |x - y|$ are uniformly q_* -Frostman. Choose a compact centre core whose complement has mass smaller than one quarter of the graph's product mass, restrict the graph only in its centre coordinate, and retain its original parameter fibres. Let C_{loc} be a common global-upper and core-lower cube constant for this restricted ambient source, and take $K_* \geq C_2 C_{\text{loc}}^2$. The derivative laws retain their q_* -Frostman bound. Lemma III.10.15, with $t_* = t - \lambda$ and $\beta = \lambda$, gives the $(2^{-k}, t - \lambda \rightarrow u; \lambda, \gamma, K_*)$ adaptedness required by the general scalar nonlinear Proposition III.10.27. The selected core-lower bound and the global upper bound meet that proposition's regularity hypotheses with loss $\lambda/4$. It yields some $y \in X$ whose pinned distance measure is robust of exponent at least $t - \zeta$ at all sufficiently fine scales. The robustness-to-dimension lemma III.6.3 gives (10.34). $\square$

For $t = 1$, letting $\zeta \downarrow 0$ already proves Theorem III.10.30 in the plane.

III.10.5 Higher-dimensional slicing

The higher-dimensional reduction begins with Orponen's radial L^p statement, proved below from the Fourier-energy identity and a Kaufman-type directional energy estimate. The co-rank-one robust projection engine follows from the rank-one theorem of Chapter III.6, an elementary co-rank-one intersection reduction, and the vector-valued entropy linearisation. The planar 5/4 pinned positive-measure escape follows from the endpoint decoupling statement recorded below.

Quoted source theorem (Orponen radial L^p slicing). *Let $k \geq 2$ and $k - 1 < s < k$. If λ, τ are separated compactly supported probability measures on $\mathbb{R}^k$ with finite s -energy, then for some $p = p(k, s) > 1$,*

$$\int \|\pi_x \tau\|_{L^p(\mathbb{S}^{k-1})}^p d\lambda(x) < \infty. \quad (10.35)$$

The same remains true after positive-mass compact restriction and normalisation. This is the exact consequence of Theorem 1.13 and the estimate around equation (3.6) in Orponen [34]; the proof needed here is reconstructed below.

Lemma (fractional energy trace). *Let $m \geq 1$, $0 < \sigma < m/2$, and let η be a compactly supported finite measure on $\mathbb{R}^m$ with $I_{m-2\sigma}(\eta) < \infty$. For every Schwartz function g ,*

$$\left| \int g d\eta \right| \lesssim_{m,\sigma} I_{m-2\sigma}(\eta)^{1/2} \left(\int_{\mathbb{R}^m} |\hat{g}(\xi)|^2 |\xi|^{2\sigma} d\xi \right)^{1/2}. \quad (10.35a)$$

Proof. Mollify η and use Plancherel. Cauchy-Schwarz with weights $|\xi|^\sigma$ and $|\xi|^{-\sigma}$ gives

$$\left| \int \hat{g}(\xi) \overline{\hat{\eta}(\xi)} d\xi \right| \leq \left(\int |\hat{g}(\xi)|^2 |\xi|^{2\sigma} d\xi \right)^{1/2} \left(\int |\hat{\eta}(\xi)|^2 |\xi|^{-2\sigma} d\xi \right)^{1/2}.$$

The second factor is comparable to $I_{m-2\sigma}(\eta)^{1/2}$ by the Fourier-energy identity. Pass to the original measure by weak convergence and lower semicontinuity. $\square$

Lemma (direction-weighted projected energy). *Let $m \geq 1$, $0 < t < \alpha \leq m$, and let ς be a finite measure on $\mathbb{S}^m$ satisfying $\varsigma(B(e, r)) \leq Cr^\alpha$. If $P_e : \mathbb{R}^{m+1} \rightarrow e^\perp$ is orthogonal projection, then for every compactly supported finite measure λ ,*

$$\int_{\mathbb{S}^m} I_t(P_e \lambda) d\varsigma(e) \lesssim_{m,t,\alpha,C} I_t(\lambda). \quad (10.35b)$$

Proof. For $v \neq 0$, $|P_e v| = |v| \sqrt{1 - |\langle e, v/|v| \rangle|^2}$, so the set on which $|P_e v| \leq 2^{-j}|v|$ lies in two spherical balls of radius $O(2^{-j})$ centred at $\pm v/|v|$. Dyadic summation therefore gives

$$\int_{\mathbb{S}^m} |P_e v|^{-t} d\varsigma(e) \lesssim |v|^{-t} \sum_{j \geq 0} 2^{jt} 2^{-j\alpha} \lesssim |v|^{-t}. \quad (10.35c)$$

Integrate this estimate with $v = x - y$ against $d\lambda(x)d\lambda(y)$. $\square$

Proposition (local radial L^p slicing). *The conclusion (10.35) holds locally from the Fourier-energy foundation; no independent Orponen theorem is required.*

Proof. Put $m = k - 1$. Choose

$$2m - s < t < m, \quad 1 < p < \min \left\{ 2 - \frac{t}{m}, \frac{t}{2m - s} \right\}, \quad q = p'. \quad (10.35d)$$

Both upper bounds for p exceed 1, since $2m - s < t < m$. Because $t < s$ and the supports are compact, $I_t(\lambda) < \infty$ follows from $I_s(\lambda) < \infty$.

First suppose that τ has a smooth compactly supported density and, for $e \in \mathbb{S}^m$, write $P_e : \mathbb{R}^k \rightarrow e^\perp$. For a centre x define the weighted measure

$$d\tau_x(y) = c_k |x - y|^{1-k} d\tau(y).$$

The density of $\pi_x \tau_x$ in direction e is a constant multiple of the integral of the smooth density of τ over the ray $x + \mathbb{R}_+ e$. Hence it is bounded by the full line integral, which is $(P_e \tau)(P_e x)$. Fubini therefore yields

$$\int \|\pi_x \tau_x\|_{L^p(\mathbb{S}^m)}^p d\lambda(x) \lesssim \int_{\mathbb{S}^m} \|P_e \tau\|_{L^p(P_e \lambda)}^p d\mathcal{H}^m(e). \quad (10.35e)$$

Fix e and abbreviate $v_e = P_e \lambda$. For $h \in L^q(v_e)$ with $\|h\|_{L^q(v_e)} = 1$, Hölder on $v_e \times v_e$ and $p(2m - s) \leq t$ give

$$I_{2m-s}(|h|v_e) \lesssim I_t(v_e)^{1/p}. \quad (10.35f)$$

Apply the fractional energy trace lemma above in $e^\perp \cong \mathbb{R}^m$ with $\sigma = (s - m)/2$ to $g = P_e \tau$. Since $\widehat{P_e \tau}(\xi) = \hat{\tau}(\xi)$ for $\xi \in e^\perp$, duality gives

$$\|P_e \tau\|_{L^p(v_e)} \lesssim I_t(v_e)^{1/(2p)} A(e), \quad A(e)^2 := \int_{e^\perp} |\hat{\tau}(\xi)|^2 |\xi|^{s-m} d\xi. \quad (10.35g)$$

We now take the $L^p(\mathbb{S}^m)$ norm of the left side of (10.35g). Let $F \in L^q(\mathbb{S}^m)$ with $\|F\|_q = 1$. Cauchy-Schwarz gives a product of two factors. By the standard coarea identity for $\{(e, \xi) : e \in \mathbb{S}^m, \xi \in e^\perp\}$ and the Fourier-energy formula,

$$\int_{\mathbb{S}^m} A(e)^2 d\mathcal{H}^m(e) \asymp \int_{\mathbb{R}^k} |\hat{\tau}(\xi)|^2 |\xi|^{s-k} d\xi \asymp I_s(\tau).$$

For the other factor, Hölder gives

$$\int I_t(P_e \lambda)^{1/p} |F(e)|^2 d\mathcal{H}^m(e) \leq \left(\int I_t(P_e \lambda) |F(e)|^p d\mathcal{H}^m(e) \right)^{1/p}.$$

The measure $d\zeta_F = |F|^p d\mathcal{H}^m$ satisfies

$$\zeta_F(B(e, r)) \lesssim r^{m(2-p)}$$

because $q = p/(p-1)$ and $\|F\|_q = 1$. The first inequality in (10.35d) gives $m(2-p) > t$, so the direction-weighted projected-energy lemma above implies

$$\int I_t(P_e \lambda) |F(e)|^p d\mathcal{H}^m(e) \lesssim I_t(\lambda).$$

Consequently, duality on the sphere yields

$$\|e \mapsto \|P_e \tau\|_{L^p(P_e \lambda)}\|_{L^p(\mathbb{S}^m)} \lesssim I_t(\lambda)^{1/(2p)} I_s(\tau)^{1/2} < \infty. \quad (10.35h)$$

Together with (10.35e), this proves a uniform L^p estimate for the weighted radial projections $\pi_x \tau_x$.

For a general τ , convolve with probability approximate identities whose Fourier multipliers have modulus at most one. The s -energies do not increase, and for all sufficiently fine mollifications the two supports remain separated. The resulting joint radial densities on $\text{supp } \lambda \times \mathbb{S}^m$ are uniformly bounded in $L^p(\lambda \times \mathcal{H}^m)$, hence a subsequence converges weakly in L^p . Testing against continuous functions of (x, e) and using separation identifies the weak limit with the joint radial pushforward of the original measure; thus its conditional densities have the same L^p bound for λ -almost every centre. Finally, on the separated compact supports there are constants $0 < c < C < \infty$ with $c\tau \leq \tau_x \leq C\tau$ for every $x \in \text{supp } \lambda$; pushing forward by π_x therefore shows that the ordinary radial projection $\pi_x \tau$ has an L^p density dominated by a fixed multiple of that of $\pi_x \tau_x$. This proves (10.35). $\square$

Quoted source theorem (co-rank-one robust projection engine). *Fix $d \geq 2$, $\kappa > 0$ and $0 < \alpha < d$. There is $\eta = \eta_d(\kappa, \alpha) > 0$ such that a regular family of co-rank-one projections whose source measure has α -Frostman control and whose Grassmannian direction law obeys*

$$\rho_x\{H : |\langle n_H, \theta \rangle| < r\} \leq Cr^\kappa \quad (10.36)$$

on an almost-full compact incidence graph has, for almost every parameter, robust image exponent

$$\frac{d-1}{d}\alpha + \eta \quad (10.37)$$

at all sufficiently fine scales, up to errors tending to zero with the discarded mass. This is Theorem 6.10 of Shmerkin–Wang [33]. The proof needed here is reconstructed below. Its finite-scale higher-rank step is the co-rank-one specialization of He [15]; in this specialization the induction remainder has rank one, so the only arithmetic projection theorem at the bottom is exactly Theorem III.6.6, already proved locally in Chapter III.6.

Lemma (refined Loomis–Whitney fibre inequality). *Let $Z \subset \mathbb{Z}^d$ be finite, let $p_0(z) = z_d$, and for $1 \leq j \leq d-1$ let p_j delete the j th coordinate. Put*

$$E_0(Z) = \sum_{t \in p_0(Z)} |Z \cap p_0^{-1}(t)|^2.$$

Then

$$|Z|^d \leq E_0(Z) \prod_{j=1}^{d-1} |p_j(Z)|. \quad (10.36a)$$

Consequently, if $K \geq 1$ and $|p_j(Z)| < K|Z|^{(d-1)/d}$ for every j , then either some p_0 -fibre has at least $K|Z|^{(d-1)/d}$ points, or there is $Z' \subset Z$ such that

$$|Z'| \geq \frac{|Z|}{2K^d}, \quad |p_0(Z')| \leq 2K^{d-1}|Z|^{1/d}. \quad (10.36b)$$

Proof. Write $Z_t = Z \cap p_0^{-1}(t)$ and $n_t = |Z_t|$. On the copy of $\mathbb{Z}^{d-1}$ given by $p_0 = t$, the finite Loomis–Whitney inequality gives

$$n_t^{d-2} \leq \prod_{j=1}^{d-1} |p_j(Z_t)|.$$

For completeness, this form follows by induction on d : apply Cauchy–Schwarz in the last coordinate to the incidence function of Z_t , and then the induction hypothesis to each of the two resulting $(d-2)$ -coordinate projections. Raising to the power $1/(d-1)$, summing in t , and using Hölder gives

$$\sum_t n_t^{(d-2)/(d-1)} \leq \prod_{j=1}^{d-1} \left(\sum_t |p_j(Z_t)| \right)^{1/(d-1)} = \prod_{j=1}^{d-1} |p_j(Z)|^{1/(d-1)}.$$

Since

$$\sum_t n_t \leq \left(\sum_t n_t^2 \right)^{1/d} \left(\sum_t n_t^{(d-2)/(d-1)} \right)^{(d-1)/d},$$

we obtain (10.36a).

Assume the first alternative in (10.36b) fails. From (10.36a) and the bounds for $p_j(Z)$,

$$E_0(Z) \geq K^{-(d-1)}|Z|^{1+(d-1)/d}.$$

Put $\tau = E_0(Z)/(2|Z|)$ and let $T = \{t : n_t \geq \tau\}$. The fibres outside T contribute at most $E_0(Z)/2$ to the energy, while every fibre has size at most $K|Z|^{(d-1)/d}$. Hence, for $Z' = Z \cap p_0^{-1}(T)$,

$$|Z'| \geq \frac{E_0(Z)}{2K|Z|^{(d-1)/d}} \geq \frac{|Z|}{2K^d}, \quad |T| \leq \frac{|Z|}{\tau} \leq 2K^{d-1}|Z|^{1/d}.$$

This is (10.36b). □

Lemma (intersection law for co-rank-one directions). *Let $q = d - 1$, and let ρ be a probability measure on $G(d, d - 1)$ such that, for some $K \geq 1$ and $\kappa > 0$,*

$$\rho\{H : |\langle n_H, \theta \rangle| < r\} \leq Kr^\kappa \quad (\theta \in \mathbb{S}^{d-1}, \delta \leq r \leq 1). \quad (10.36c)$$

Choose $H_1, \dots, H_q$ independently with law ρ . For $\tau > 0$ let $\mathcal{T}_\tau$ be the event that

$$\text{dist}(n_{H_j}, \text{span}(n_{H_1}, \dots, n_{H_{j-1}})) \geq \delta^\tau \quad (2 \leq j \leq q).$$

Then

$$\rho^{\otimes q}(\mathcal{T}_\tau^c) \leq qK\delta^{\tau\kappa}, \quad |n_{H_1} \wedge \dots \wedge n_{H_q}| \geq \delta^{q\tau} \quad \text{on } \mathcal{T}_\tau.$$

If $\mathcal{A} \subset \mathcal{T}_\tau$ is Borel with $\rho^{\otimes q}(\mathcal{A}) \geq \delta^u$ and $\rho_{L, \mathcal{A}}$ is the pushforward of the normalised restriction of $\rho^{\otimes q}$ to $\mathcal{A}$ under $L = H_1 \cap \dots \cap H_q = \mathbb{R}\ell$, then

$$\rho_{L, \mathcal{A}}\{\ell : |\langle \ell, \theta \rangle| < r\} \leq qK\delta^{-u}r^{\kappa/q} \quad (10.36d)$$

for $\delta \leq r \leq 1$ and every unit θ .

Proof. Put $S_{j-1} = \text{span}(n_{H_1}, \dots, n_{H_{j-1}})$. If $\text{dist}(n_{H_j}, S_{j-1}) < \delta^\tau$, choose any unit $\vartheta \in S_{j-1}^\perp$; then $|\langle n_{H_j}, \vartheta \rangle| < \delta^\tau$. Conditioning on the previous normals and using (10.36c), followed by a union bound in j , gives the first estimate. The wedge lower bound is the Gram–Schmidt product of the successive distances.

For the direction law fix a unit vector w . Ordering w first in Gram–Schmidt gives the exact factorisation

$$|\det(n_{H_1}, \dots, n_{H_q}, w)| = \prod_{j=1}^q \text{dist}(n_{H_j}, \text{span}(w, n_{H_1}, \dots, n_{H_{j-1}})).$$

Hence a determinant smaller than r forces one factor below $r^{1/q}$. Conditional on the preceding normals, (10.36c) bounds each such event by $Kr^{\kappa/q}$, and therefore

$$\rho^{\otimes q}\{|\det(n_{H_1}, \dots, n_{H_q}, w)| < r\} \leq qKr^{\kappa/q}.$$

On $\mathcal{T}_\tau$, if ℓ is the unit normal to $\text{span}(n_{H_1}, \dots, n_{H_q})$, then

$$|\langle \ell, w \rangle| = \frac{|\det(n_{H_1}, \dots, n_{H_q}, w)|}{|n_{H_1} \wedge \dots \wedge n_{H_q}|}.$$

In particular $|\langle \ell, w \rangle| < r$ implies the determinant is $< r$. Intersecting with $\mathcal{A}$ and dividing by $\rho^{\otimes q}(\mathcal{A}) \geq \delta^u$ proves (10.36d). This explicit denominator is the only cost of conditioning. $\square$

Proposition (local co-rank-one discretized projection theorem). *Fix $d \geq 2$, $0 < a < d$, and $\kappa > 0$. There is $\gamma = \gamma(d, a, \kappa) > 0$ such that, for all sufficiently small dyadic δ , the following holds. Let $A \subset B(0, 1)$ be a union of δ -cubes and let $\rho \in \mathcal{P}(G(d, d - 1))$ satisfy, for $\delta \leq r \leq 1$,*

$$\begin{aligned} N_\delta(A) &\geq \delta^{-a+\gamma}, \\ N_\delta(A \cap B(x, r)) &\leq \delta^{-\gamma}r^\kappa N_\delta(A), \\ \rho\{H : |\langle n_H, \theta \rangle| < r\} &\leq \delta^{-\gamma}r^\kappa. \end{aligned} \quad (10.36e)$$

Then there is $\mathcal{G} \subset G(d, d-1)$ with $\rho(\mathcal{G}) \geq 1 - \delta^\gamma$ such that, for every $H \in \mathcal{G}$ and every $A' \subset A$ with $N_\delta(A') \geq \delta^\gamma N_\delta(A)$,

$$N_\delta(P_H A') \geq \delta^{-((d-1)a/d+\gamma)}. \quad (10.36f)$$

Proof. We give the co-rank-one reduction completely, because this is the only part of He's higher-rank projection theorem required in Volume III. Put $q = d-1$ and $c = q/d$.

Controlled-size thinning. The lower bound in (10.36e) may be replaced, at a loss $O(\gamma)$ in all exponents, by

$$\delta^{-a+O(\gamma)} \leq N_\delta(A) \leq \delta^{-a-O(\gamma)}. \quad (10.36g)$$

Indeed choose uniformly a fixed number $M \asymp \delta^{-a+2\gamma}$ of occupied terminal cubes. For dyadic balls at the $O(\log(1/\delta))$ relevant radii, hypergeometric concentration and a union bound give simultaneously

$$N_\delta(A_0 \cap B(x, r)) \leq \delta^{-O(\gamma)} r^{\kappa_0} M, \quad \kappa_0 = \min\{\kappa/2, a/4\} > 0.$$

The threshold on the right is at least a fixed positive power of δ^{-1} even at $r = \delta$, so the union bound is harmless. Thus some realization A_0 has the asserted properties. In the weak argument below failure is inherited by every subset, so we may replace A by this A_0 .

Weak exceptional-set statement. Use Theorem III.6.6 on a finite parameter net in a small compact interval around a , with spatial exponent κ_0 and direction exponent κ/q . Let $\eta_1 > 0$ be the minimum rank-one gain and exceptional exponent on this net. Choose parameters in the order

$$0 < \gamma \ll_{d,a,\kappa} \varepsilon \ll_{d,a,\kappa} \eta_1.$$

For every nonempty $B \subset A$ let $\mathcal{E}(B)$ consist of those H for which some $B' \subset B$ obeys

$$N_\delta(B') \geq \delta^\varepsilon N_\delta(B), \quad N_\delta(P_H B') < \delta^{-ca-\varepsilon}, \quad (10.36h)$$

We first show that some nonempty $B \subset A$ has $\rho(\mathcal{E}(B)) \leq \delta^\varepsilon$.

Otherwise $\rho(\mathcal{E}(B)) > \delta^\varepsilon$ for every nonempty $B \subset A$. There are only finitely many subsets of the occupied terminal cubes, so for each $H \in \mathcal{E}(B)$ we may choose the first witness in a fixed ordering; all choices below can therefore be made Borel. Starting with $A_0 = A$, choose successively $H_j \in \mathcal{E}(A_{j-1})$ and its witness $A_j \subset A_{j-1}$, while excluding those H_j whose normal lies within δ^b of the span of the preceding normals. Taking $b = 3\varepsilon/\kappa$ and then $\gamma \ll \varepsilon$, the excluded ρ -mass is at most $\delta^{-\gamma} \delta^{b\kappa} \leq \delta^{2\varepsilon}$, so at every stage the admissible set has conditional ρ -mass at least $\frac{1}{2}\delta^\varepsilon$. Iterated Fubini therefore gives a Borel set $\mathcal{C} \subset G(d, d-1)^q$ of product measure at least $2^{-q} \delta^{q\varepsilon}$ on which

$$|n_{H_1} \wedge \cdots \wedge n_{H_q}| \geq \delta^{O_{d,\kappa}(\varepsilon)}, \quad N_\delta(A_q) \geq \delta^{q\varepsilon} N_\delta(A), \quad (10.36i)$$

and $N_\delta(P_{H_j} A_q) < \delta^{-ca-\varepsilon}$ for all j . Here $L = H_1 \cap \cdots \cap H_q = \mathbb{R}\ell$, and the determinant with the unit vector ℓ equals the wedge norm displayed above.

Apply an $O(\delta^{-O_{d,\kappa}(\varepsilon)})$ -conditioned linear change of variables sending L to the last coordinate axis and the normal lines $H_j^\perp$ to the first q coordinate axes. Covering numbers

change only by powers $\delta^{-O_{d,\kappa}(\varepsilon)}$. The preceding refined Loomis–Whitney lemma, with $K = \delta^{-C_{d,\kappa}\varepsilon}$ and $C_{d,\kappa}$ fixed sufficiently large, gives a dichotomy.

If a P_L -fibre contains at least $KN_\delta(A_q)^c$ terminal cubes, let B be this thick hyperplane slice. Its supporting hyperplane is $W = \ell^\perp$. Whenever $|\langle n_H, \ell \rangle| \geq \delta^b$, the restriction $P_H|_W$ is invertible with condition number $\delta^{-O(b)}$. Hence every $B' \subset B$ with $N_\delta(B') \geq \delta^\varepsilon N_\delta(B)$ has $N_\delta(P_H B') \geq \delta^{-ca-\varepsilon}$ once $C_{d,\kappa}$ is fixed large and then ε small. Thus

$$\mathcal{E}(B) \subset \{H : |\langle n_H, \ell \rangle| < \delta^b\},$$

whose ρ -mass is $< \delta^\varepsilon$ by (10.36e), contradicting our assumption.

We are therefore in the second alternative. It yields $A' \subset A_q$ with

$$N_\delta(A') \geq \delta^{O_d(\varepsilon)} N_\delta(A), \quad N_\delta(P_L A') \leq \delta^{-a/d-O_d(\varepsilon)}. \quad (10.36j)$$

Let $\mathcal{T}$ be the product transversality event $|n_{H_1} \wedge \cdots \wedge n_{H_q}| \geq \delta^{O_{d,\kappa}(\varepsilon)}$. Its product probability is bounded below independently of δ once δ is small. Apply the preceding intersection-law lemma to the normalised restriction of $\rho^{\otimes q}$ to $\mathcal{T}$. The resulting line law has the rank-one hyperplane-neighbourhood bound with exponent κ/q and only $\delta^{-O_{d,\kappa}(\varepsilon+\gamma)}$ loss. The source set still has spatial nonconcentration exponent κ_0 , and (10.36g) fixes its size exponent at $a + O(\varepsilon)$. Hence Theorem III.6.6, by the finite net used when choosing η_1 , says that the line-law mass of directions for which *some* $\delta^{O_d(\varepsilon)}$ -large subset has a projection as small as (10.36j) is at most $\delta^{\eta_1/2}$, after decreasing ε . Every tuple in $\mathcal{C}$ which reaches the second alternative produces exactly such a bad line, whereas the first alternative has already contradicted the standing assumption. Thus the preimage of the rank-one bad-line set has product measure at least $2^{-q}\delta^{q\varepsilon}$, contradicting the upper bound $O(\delta^{\eta_1/2})$ because $q\varepsilon < \eta_1/2$. This proves the weak statement.

Uniformity over all large subsets. We now perform the exhaustion behind He’s formal upgrade explicitly. Apply the weak statement to the current remainder, remove a 2δ -neighbourhood of the resulting nonempty piece, and iterate until the remainder R has $N_\delta(R) \leq \delta^{\varepsilon/2} N_\delta(A)$. Before stopping, the remainder still has the required size and spatial nonconcentration with only fixed $O(\varepsilon)$ losses. The weak proof above is stable under such fixed losses—one merely increases the constants in b and in the condition number and keeps $\varepsilon \ll \eta_1$ —so it remains applicable at every stage. Since only finitely many terminal cubes are present, the process terminates. The selected pieces B_i have disjoint δ -neighbourhoods. Put

$$a_i = \frac{N_\delta(B_i)}{\sum_j N_\delta(B_j)}, \quad E_i = \mathcal{E}(B_i), \quad \rho(E_i) \leq \delta^\varepsilon.$$

Suppose H admits $A' \subset A$ with $N_\delta(A') \geq \delta^{\varepsilon/3} N_\delta(A)$ and $N_\delta(P_H A') < \delta^{-ca-\varepsilon/3}$. Let I be the indices for which the 2δ -thickening of A' occupies at least a δ^ε fraction of B_i . Bounded multiplicity of terminal cubes under 2δ -thickening, the bound on R , and the disjointness of the $B_i^{(\delta)}$ imply, for small δ ,

$$\sum_{i \in I} a_i \gtrsim \delta^{\varepsilon/2}.$$

For every $i \in I$, the relevant part of B_i is a valid witness in (10.36h); the harmless thickening constant is swallowed by the gap between $\varepsilon/3$ and ε . Hence $H \in E_i$ for every $i \in I$ and

$$\delta^{\varepsilon/2} \rho\{H : H \text{ bad for } A\} \lesssim \int \sum_i a_i \mathbf{1}_{E_i}(H) d\rho(H) = \sum_i a_i \rho(E_i) \leq \delta^\varepsilon.$$

Thus the bad set has mass $O(\delta^{\varepsilon/2})$. Renaming a smaller exponent as γ proves (10.36f). All ingredients used in the reduction are now local: the rank-one projection theorem is Theorem III.6.6, and the co-rank-one reduction is the two elementary lemmas above. $\square$

Lemma (co-rank-one adaptedness for Frostman pieces). *Let $c = (d - 1)/d$. Fix a compact interval $J \subset (0, d)$ and $\kappa > 0$. There is $\eta_0 = \eta_0(d, J, \kappa) > 0$ such that the following is uniform for $a \in J$. If ω is a $(\delta, a, \delta^{-\varepsilon})$ -measure on $[0, 1]^d$ and $\rho \in \mathcal{P}(G(d, d - 1))$ satisfies (10.36c) with $K \leq \delta^{-\varepsilon}$, then, outside a set of ρ -mass $\leq \delta^{\eta_0}$,*

$$P_H \omega \quad \text{is} \quad (\delta, ca + \eta_0, \delta^{\eta_0})\text{-robust}, \quad (10.36k)$$

after decreasing η_0 to absorb $O(\varepsilon)$ losses.

Proof. Suppose robustness fails for H . Pull back a witnessing projected set and split its occupied terminal cubes according to dyadic mass. Since the witness has mass at least a fixed power of δ , one mass level carries a power-sized fraction. If A is the set of representatives of that level, then

$$N_\delta(A) \geq \delta^{-a+O(\varepsilon)}, \quad N_\delta(A \cap B(x, r)) \leq \delta^{-O(\varepsilon)} r^a N_\delta(A),$$

and the projected covering number of A is bounded by that of the witness. Apply the local co-rank-one discretized theorem with spatial exponent $\min\{a, \kappa, 1\}/2$; the logarithmic number of mass levels is swallowed by the power slack. Compactness of J gives one positive minimum of the resulting gain and exceptional exponent. This is exactly (10.36k). $\square$

Proposition (local co-rank-one all-scales nonlinear engine). *Fix $d \geq 2$, $0 < \alpha < d$, $\kappa > 0$ and $c_0 > 0$. There is $\eta = \eta(d, \alpha, \kappa) > 0$ with the following property. Let $\mathcal{F} = (Y, \nu, U, F)$ be a compact regular C^2 family of rank- $(d - 1)$ maps, and let μ be an α -Frostman probability measure on U . Suppose that for μ -almost every x there is a Borel $Y_x \subset Y$ with $\nu(Y_x) \geq c_0$ such that, uniformly after restriction to a compact incidence graph,*

$$(V_x)_\# \nu_{Y_x} \{H : | \langle n_H, \theta \rangle | < r\} \leq Cr^\kappa \quad (0 < r \leq 1, \theta \in \mathbb{S}^{d-1}). \quad (10.36l)$$

Then there is $Y_0 \subset Y$ of positive ν -measure such that, for every $y \in Y_0$, there is a positive- μ set X_y for which

$$F_y(\mu_{X_y}) \quad \text{is} \quad (2^{-m}, c\alpha + \eta, 2^{-\eta m})\text{-robust} \quad (10.36m)$$

for every sufficiently large m , where $c = (d - 1)/d$.

Proof. This is the vector-valued analogue of the all-scales argument already proved in Proposition III.6.23.1; we spell out the only changes. The entropy linearisation needed for

rank $d - 1$ is already available verbatim as Proposition III.6.22, whose statement allows every $1 \leq k < d$.

Fix a small profile error $\zeta > 0$. Choose $\xi > 0$ so small that $ca - \zeta \leq 0$ for $0 \leq a \leq 2\xi$ and $a - 1 \geq ca - \zeta$ for $d - 2\xi \leq a \leq d$. On the compact middle interval $[\xi, d - \xi]$, the preceding adaptedness lemma gives a uniform $\eta_0 > 0$ and exponentially small exceptional sets of hyperplanes with robust exponent $ca + \eta_0$. At the low endpoint use the trivial exponent 0. At the high endpoint use the elementary fibre estimate: a union of M terminal $(d - 1)$ -cubes in the image has inverse image coverable by $O_d(M2^m)$ terminal d -cubes, so an $(2^{-m}, a, 2^{O(\zeta m)})$ source gives robust exponent $a - 1 - O(\zeta) \geq ca - O(\zeta)$. Thus on every multiscale block we have

$$\Gamma(a) \geq ca - O(\zeta) \quad (0 \leq a \leq d), \quad \Gamma(a) \geq ca + \eta_0 \quad (\xi \leq a \leq d - \xi). \quad (10.36n)$$

Now discretise μ at $m = T\ell$ and apply the proved measure-regularisation Theorem III.6.6 from Chapter III.6; then use the Shmerkin Frostman-piece decomposition Theorem III.6.21. On each selected block, apply (10.36n) to the conditional measure and the law $(V_x)_{\#} \nu_{Y_x}$. The exceptional parameter mass decays exponentially in the block length; there are only $O(1)$ selected blocks at each output scale. Exactly as in (6.40c)–(6.40d), the union over all sufficiently large output scales is summable. Fubini therefore selects one positive- ν set of parameters and, for each selected parameter, one positive- μ base set on which all large scales are good simultaneously.

It remains only to read the entropy telescope. Write $\bar{\sigma} = \ell^{-1} \sum_{j=1}^{\ell} \sigma_j$ for the mean branching exponent of the selected regular piece and put $\bar{\alpha} = (d + \alpha)/2$.

In the moderate-branching case $\bar{\sigma} \leq \bar{\alpha}$. The natural volume radius of the support satisfies

$$r_0 = |\text{supp } \omega|^{1/d} = 2^{-(1-\bar{\sigma}/d)m+O(\ell)}.$$

The selected piece has original μ -mass $2^{-O(\zeta m)}$, while μ is α -Frostman. Hence, after decreasing the regularisation error,

$$\omega(B(x, r_0)) \leq 2^{O(\zeta m)} r_0^\alpha \leq 2^{-u_0 m}, \quad u_0 = \frac{\alpha(d - \alpha)}{4d} > 0.$$

Thus the hypothesis of Theorem III.6.21 is satisfied with a scale-independent u_0 . Its fourth conclusion puts at least ξm total block length in the middle interval $[\xi, d - \xi]$, while its third conclusion gives $\sum_j \alpha_j m_j \geq \log_2 N(\text{supp } \omega, m) - O(\zeta m)$. Combining these two statements with Proposition III.6.22 and (10.36n) yields

$$m^{-1} H_m(F_Y \omega_Z) \geq c m^{-1} \log_2 N(\text{supp } \omega, m) + \eta_1 - O(\zeta) \geq c\alpha + \eta_1 - O(\zeta)$$

for some $\eta_1 = \eta_1(d, \alpha, \kappa) > 0$; the last inequality follows from the Frostman bound and the power-sized mass of the selected piece exactly as in (6.40h).

If $\bar{\sigma} > \bar{\alpha}$, use instead Proposition III.6.21.1. Its two-sided block regularity supplies the endpoint hypotheses in (10.36n), and (6.27c) together with Proposition III.6.22 gives

$$m^{-1} H_m(F_Y \omega_Z) \geq c\bar{\sigma} - O(\zeta) > c \frac{d + \alpha}{2} - O(\zeta) = c\alpha + \frac{c(d - \alpha)}{2} - O(\zeta).$$

This branch therefore has a fixed surplus as well. Choose ζ below both surpluses and then choose the robustness parameter smaller than all regularisation errors. Entropy is bounded above by the logarithm of the terminal covering number of any projected set carrying the prescribed mass. Hence (10.36m) follows. This is precisely the local multiscale assembly using Theorem III.6.6; no higher-rank arithmetic theorem remains as an additional premise. $\square$

The co-rank-one single-scale gain reduces to the rank-one theorem III.6.6 through the intersection-line law and the refined Loomis–Whitney fibre inequality. The all-scales passage then uses the proved multiscale regularisation Theorem III.6.6, the profile, and rank- $(d-1)$ entropy-linearisation machinery of Chapter III.6.

Theorem III.10.16.1 (planar endpoint decoupling). *Let $S \subset \mathbb{R}^2$ be a compact C^2 curve with curvature bounded above and below by positive constants. If $R \geq 2$, the R^{-1} -neighbourhood of S is decomposed into finitely overlapping $R^{-1/2} \times R^{-1}$ caps θ , $\widehat{f}_\theta \subset 2\theta$, and $f = \sum_\theta f_\theta$, then for every $\varepsilon > 0$ and every radius- R ball B_R , with $w_{B_R}(x) = (1 + |x - c(B_R)|/R)^{-100}$,*

$$\|f\|_{L^6(B_R)} \lesssim_\varepsilon R^\varepsilon \left(\sum_\theta \|f_\theta\|_{L^6(w_{B_R})}^2 \right)^{1/2}. \quad (10.36o)$$

The implicit constant may depend on the curve and its C^2 atlas. The case needed below is a circle. We prove the estimate, including its weighted localisation, in the next subsection.

A proof of the planar endpoint decoupling estimate

We give the bilinear proof of the parabola estimate, in a form uniform for a family of C^2 curves with curvature bounded away from zero. The proof is a two-dimensional specialisation of the bilinear method of Guo–Li–Yung–Zorin–Kranich [38]. Its important feature here is that the only lower-degree decoupling step is one-dimensional Plancherel, which is among our allowed harmonic-analysis foundations. We include the kernel and localisation details because both matter at the endpoint.

After finitely many graph charts and fixed affine changes of coordinates, it is enough to consider

$$\Gamma(t) = (t, \varphi(t)), \quad 0 \leq t \leq 1, \quad \varphi(0) = \varphi'(0) = 0, \quad 0 < c \leq \varphi''(t) \leq C. \quad (10.36o.1)$$

The constants in this subsection are uniform over this class with fixed c, C . For a dyadic interval $J \subset [0, 1]$, let U_J be the frequency parallelogram centred at $\Gamma(c_J)$ whose sides are $K|J|\Gamma'(c_J)$ and $K|J|^2\Gamma''(c_J)$; $K \geq K_0(c, C)$ is fixed. In particular, $\Gamma(J) \subset U_J$. Write

$$P_J = \{x \in \mathbb{R}^2 : |x \cdot \Gamma'(c_J)| \leq |J|^{-1}, |x \cdot \Gamma''(c_J)| \leq |J|^{-2}\}, \quad \phi_J(x) = \frac{c_A}{|P_J|} (1 + \|x\|_{P_J})^{-A}, \quad A > 10, \quad (10.36o.2)$$

where $\|x\|_{P_J} = \inf\{r > 0 : x \in rP_J\}$ and c_A normalises $\int \phi_J = 1$. These are spatial kernels, not the graph function φ . The fixed factor K is suppressed from the notation.

Lemma III.10.16.2 (the two kernel facts). *If $J \subset I$ are dyadic intervals and $\text{supp } \hat{g} \subset C_0 U_J$, then for $p \geq 1$*

$$|g(x)|^p \leq C_{p,C_0}(|g|^p * \phi_J)(x), \quad \phi_I * \phi_J \leq C\phi_J. \quad (10.360.3)$$

If I' has length h and v is a unit normal to $\Gamma'(c_{I'})$, then

$$\phi_{I'}(x + uv) \leq C(1 + |u|h^2)^A \phi_{I'}(x). \quad (10.360.4)$$

Proof. Choose a Schwartz function ψ_J whose Fourier transform is one on $C_0 U_J$, is supported in a fixed enlargement of U_J , and whose absolute value is bounded by $C\phi_J$. Since $g = g * \psi_J$, Hölder's inequality with the measure $|\psi_J(z)| dz$ proves the first inequality.

For the second, parabolically rescale I to unit length using the map

$$(u, v) \mapsto (a + |I|u, \varphi(a) + |I|\varphi'(a)u + |I|^2 v), \quad I = [a, a + |I|]. \quad (10.360.5)$$

The rescaled graph still satisfies $c \leq \varphi'' \leq C$. After one further bounded linear change sending the two sides of P_J to coordinate axes, P_I has bounded diameter, while P_J has sides comparable to $\lambda^{-1}, \lambda^{-2}$, $\lambda = |J|/|I|$. Consequently the required convolution inequality reduces, up to uniform constants, to

$$\int_{\mathbb{R}^2} (1 + |x - y|)^{-A} \lambda^3 (1 + \max\{\lambda|y_1|, \lambda^2|y_2|\})^{-A} dy \lesssim \lambda^3 (1 + \max\{\lambda|x_1|, \lambda^2|x_2|\})^{-A}. \quad (10.360.6)$$

On the region where the second kernel at y is at most 2^A times its value at x , integrate the first kernel. On the complement, a coordinate $i \in \{1, 2\}$ attaining the maximum at x obeys $|x_i - y_i| \geq |x_i|/2$; there the first kernel is at most $C\lambda^{iA-3}$ times the second kernel at x . Integrating the second kernel proves (10.360.6), since $A > 3$. Finally, displacement by uv changes the $P_{I'}$ gauge by at most $C|u|h^2$; the elementary inequality $1 + \|x\|_{P_{I'}} \leq (1 + C|u|h^2)(1 + \|x + uv\|_{P_{I'}})$ gives (10.360.4). $\square$

For $0 < \delta < 1/16$, let $\mathcal{P}(\delta)$ denote the dyadic partition into intervals of length comparable to δ . If $\text{supp } \hat{f}_J \subset U_J$, put $f_I = \sum_{J \subset I} f_J$. Define $D(\delta)$ as the least constant, uniform over the graph class, in

$$\|f_{[0,1]}\|_6 \leq D(\delta) \left(\sum_{J \in \mathcal{P}(\delta)} \|f_J\|_6^2 \right)^{1/2}. \quad (10.360.7)$$

The trivial bound is $D(\delta) \lesssim \delta^{-1/2}$. Changing the fixed enlargement factor K changes constants in the following arguments but not their exponents. The affine map (10.360.5) maps the boxes for subintervals of I exactly to boxes of the same form for the rescaled graph, so it gives

$$\|f_I\|_6 \leq D(\delta/|I|) \left(\sum_{J \subset I} \|f_J\|_6^2 \right)^{1/2}. \quad (10.360.8)$$

All norms in this subsection without a domain are over $\mathbb{R}^2$.

We also need a bilinear constant. For two separated quarter-intervals $I, I' \subset [0, 1]$, set $F = f_I, G = f_{I'}$ and let $B(\delta)$ be the least uniform constant in

$$\|FG\|_3 \leq B(\delta)^2 \left(\sum_{J \subset I} \|f_J\|_6^2 \right)^{1/2} \left(\sum_{J \subset I'} \|f_J\|_6^2 \right)^{1/2}. \quad (10.360.9)$$

Here and below “separated” means that the distance between the intervals is at least $1/4$; finite changes in the quarter-interval convention only change absolute constants.

Lemma III.10.16.3 (linear-to-bilinear reduction). *For dyadic $\delta = 2^{-N}$,*

$$D(\delta)^2 \lesssim 1 + \sum_{n=2}^N B(2^{n-2}\delta)^2. \quad (10.360.10)$$

Proof. Decompose $[0, 1]^2$ into its δ -squares touching the diagonal and the maximal dyadic squares $I \times I'$ off the diagonal for which $|I| = |I'| = h = 2^{-n}$ and $\text{dist}(I, I')/h \in \{1, 2\}$. Start with the largest eligible squares and, at each finer scale, retain only squares not already covered. Every remaining δ -square touches the diagonal, so this gives a disjoint cover up to boundaries. At each n , every I participates in only a bounded number of pairs. Since $f_{[0,1]}^2$ is the sum of the products $f_I f_{I'}$ over this decomposition, Minkowski in L^3 bounds its norm by the sum of the L^3 norms of these products. The diagonal contribution is at most $C \sum_J \|f_J\|_6^2$. Each off-diagonal pair lies in an h -grid interval of length $4h$ contained in $[0, 1]$: start at the leftmost member unless this crosses the right endpoint of $[0, 1]$, in which case shift left by h . Rescale this interval by (10.360.5). The two members become quarter-intervals at distance $1/4$ or $1/2$, so (10.360.9) bounds its contribution by $CB(2^{n-2}\delta)^2$ times the product of the two corresponding square sums. The inequality $2ab \leq a^2 + b^2$, followed by the bounded participation count, gives (10.360.10). $\square$

To exploit transversality we deliberately use different exponents on the two sides. For $0 < a, b < 1$, let $B_{a,b}(\delta)$ be the least constant for which, whenever I, I' are separated dyadic intervals of respective lengths comparable to δ^a, δ^b ,

$$\int_{\mathbb{R}^2} (|f_I|^2 * \phi_I)(|f_{I'}|^4 * \phi_{I'}) \leq B_{a,b}(\delta)^6 \left(\sum_{J \subset I} \|f_J\|_6^2 \right) \left(\sum_{J \subset I'} \|f_J\|_6^2 \right)^2. \quad (10.360.11)$$

The final sums run over the δ -intervals. The smoothed factors are nonnegative; this is essential for the next orthogonality argument.

Lemma III.10.16.4 (one-dimensional gain). *If $0 < a \leq 2b < 1$, then*

$$B_{a,b}(\delta) \leq CB_{2b,b}(\delta). \quad (10.360.12)$$

The constant is uniform in a, b, δ while $2b < 1$.

Proof. Fix I, I' as in (10.360.11), write $F = f_I, G = f_{I'}$, and set $H = |G|^4 * \phi_{I'}$ and $W = H * \phi_I$. Partition I into intervals J of length comparable to δ^{2b} , and put $F_J = f_J$. Let v be the unit normal to $\Gamma'(c_{I'})$ and $L = \delta^{-2b}$. By (10.360.4), convolution with $|G|^4$ and then with ϕ_I gives

$$W(x + uv) \leq C(1 + |u|/L)^A W(x). \quad (10.360.13)$$

On the other hand, $v \cdot \Gamma'(t)$ is bounded away from zero on I : the mean-value theorem, $\phi'' \geq c$, and separation of I, I' give $|v \cdot \Gamma'(t)| \geq c_1 > 0$. Thus the Fourier supports of the restrictions of the F_J to almost every line $x + \mathbb{R}v$ lie in intervals of length $O(L^{-1})$ with bounded overlap as J varies.

Here is the weighted Plancherel step explicitly. Cover each such line by intervals Q of length L and choose translated Schwartz cutoffs ψ_Q with $|\psi_Q| \geq 1$ on Q , Fourier support in an interval of length c_2/L , and arbitrarily rapid spatial decay away from Q . The projected frequency intervals of $F_J\psi_Q$ still have bounded overlap. Plancherel therefore gives

$$\int_Q \left| \sum_J F_J \right|^2 W \leq (\sup_Q W) \int \left| \sum_J F_J \psi_Q \right|^2 \lesssim (\sup_Q W) \sum_J \int |F_J \psi_Q|^2. \quad (10.36o.14)$$

The slow-variation bound (10.36o.13) and decay of ψ_Q imply $\sum_Q (\sup_Q W) |\psi_Q(t)|^2 \lesssim W(t)$; choose the decay order greater than $A+2$. Integrating over the parallel lines and summing Q yields $\int |F|^2 W \lesssim \sum_J \int |F_J|^2 W$. Finally the first inequality in (10.36o.3), applied to F_J , and the second give

$$\begin{aligned} \int H(|F|^2 * \phi_I) &= \int |F|^2 W \\ &\lesssim \sum_J \int H(|F_J|^2 * \phi_I) \\ &\lesssim \sum_J \int H(|F_J|^2 * \phi_J * \phi_I) \lesssim \sum_J \int H(|F_J|^2 * \phi_J). \end{aligned} \quad (10.36o.15)$$

Apply (10.36o.11) with exponents $(2b, b)$ to every summand. Their first square sums add exactly to the first square sum for I , proving (10.36o.12). $\square$

Lemma III.10.16.5 (the exponent bootstrap). *For every $\varepsilon > 0$, $D(\delta) \lesssim_\varepsilon \delta^{-\varepsilon}$.*

Proof. We record all three inequalities that drive the bootstrap. First, Hölder and Jensen applied to (10.36o.11), followed by (10.36o.8), give

$$B_{a,b}(\delta) \lesssim D(\delta^{1-a})^{1/3} D(\delta^{1-b})^{2/3}. \quad (10.36o.16)$$

Second, the uncertainty inequality (10.36o.3), Hölder, and subdivision of the two separated quarter-intervals into δ^a - and δ^b -intervals give

$$B(\delta) \lesssim \delta^{-(a+2b)/3} B_{a,b}(\delta). \quad (10.36o.17)$$

Indeed $|\sum_{i=1}^m z_i|^p \leq m^{p-1} \sum_i |z_i|^p$ for $p = 2, 4$; using the slightly larger factors m^p and $m_a \lesssim \delta^{-a}$, $m_b \lesssim \delta^{-b}$ makes the displayed exponent immediate. Apply the asymmetric estimate once with powers $(2, 4)$. For the $(4, 2)$ estimate, partition the original first source interval at scale δ^b and the second at scale δ^a , then interchange their roles; this invokes the same $B_{a,b}$, not $B_{b,a}$. Cauchy-Schwarz recovers the powers $(3, 3)$ in (10.36o.9).

Third, Hölder inside each convolution and then in the integral gives, with $F = f_I, G = f_{I'}$,

$$\int (|F|^2 * \phi_I)(|G|^4 * \phi_{I'}) \leq \left[\int (|F|^4 * \phi_I)(|G|^2 * \phi_{I'}) \right]^{1/2} \left[\int (|G|^6 * \phi_{I'}) \right]^{1/2},$$

so that

$$B_{a,b}(\delta) \lesssim B_{b,a}(\delta)^{1/2} D(\delta^{1-b})^{1/2}. \quad (10.36o.18)$$

Combining this for $a = 2b$ with the one-dimensional gain (10.36o.12) at $(a, b) = (b, 2b)$ gives, for $0 < b < 1/4$,

$$B_{2b,b}(\delta) \lesssim B_{4b,2b}(\delta)^{1/2} D(\delta^{1-b})^{1/2}. \quad (10.36o.19)$$

Let η be the infimum of exponents $u \geq 0$ for which $D(\delta) \lesssim_u \delta^{-u}$, and let $A(b)$ be the analogous infimum for $B_{2b,b}(\delta)$. Both are finite by the triangle inequality. Equations (10.36o.16), (10.36o.17), (10.36o.10), and (10.36o.19) imply, respectively,

$$A(b) \leq \eta(1 - 4b/3), \quad \eta \leq 4b/3 + A(b), \quad A(b) \leq \frac{1}{2}A(2b) + \frac{1}{2}\eta(1 - b). \quad (10.36o.20)$$

The logarithm from summing Whitney scales is absorbed by an arbitrarily small power before taking infima. Set $q(b) = (\eta - A(b))/b$. The first two inequalities show $4\eta/3 \leq q(b) \leq 4/3$; in particular the $\liminf q_* := \liminf_{b \downarrow 0} q(b)$ is finite. The last inequality gives $q(b) \geq q(2b) + \eta/2$. Taking the $\liminf$ as $b \downarrow 0$ yields $q_* \geq q_* + \eta/2$. Hence $\eta = 0$. $\square$

Proof of Theorem III.10.16.1. The preceding lemma is the global L^6 decoupling theorem for each graph chart. A fixed finite chart partition handles S ; its pieces are combined by the triangle and Cauchy–Schwarz inequalities. Set $\delta = R^{-1/2}$. Choose a Schwartz function χ_B whose Fourier transform is supported in a ball of radius cR^{-1} , such that $|\chi_B| \geq 1$ on B_R and $|\chi_B(x)|^6 \lesssim (1 + |x - c_B|/R)^{-A_0}$, where A_0 is the exponent in the weight w_{B_R} . Multiplication by χ_B enlarges each $R^{-1/2} \times R^{-1}$ cap by only a fixed factor; the global estimate just proved permits any such fixed factor. Therefore

$$\|f\|_{L^6(B_R)} \leq \|f\chi_B\|_6 \lesssim_\varepsilon R^\varepsilon \left(\sum_\theta \|f_\theta \chi_B\|_6^2 \right)^{1/2} \lesssim_\varepsilon R^\varepsilon \left(\sum_\theta \|f_\theta\|_{L^6(w_{B_R})}^2 \right)^{1/2}.$$

We applied the global estimate with exponent 2ε to $\delta = R^{-1/2}$. A finitely overlapping cap family may be split into a bounded number of disjoint families. This proves (10.36o). $\square$

Quoted source theorem (planar $5/4$ pinned positive-measure escape). *If $A \subset \mathbb{R}^2$ is Borel with $\dim_{\mathbb{H}} A > 5/4$, then some $y \in A$ satisfies $|\Delta^y(A)| > 0$ [39, Theorem 1.2]. The proof is reconstructed below from the locally proved radial L^p estimate, the Fourier–energy foundation and the locally proved endpoint decoupling estimate (10.36o).*

Lemma (train-track removal with enlarged goodness strips). *Let $E_1, E_2 \subset B(0, 1) \subset \mathbb{R}^2$ be compact with $\text{dist}(E_1, E_2) \geq c_0 > 0$, assume after a rotation that all directions $(x - y)/|x - y|$ with $y \in E_1, x \in E_2$ lie in one fixed sufficiently short arc of $\mathbb{S}^1$, and let μ_i be probability measures supported on E_i satisfying*

$$\mu_i(B(z, r)) \leq Cr^\alpha, \quad \alpha > 1. \quad (10.36p)$$

Fix $0 < \delta \ll 1$. For $R_j = 2^j R_0$ one can decompose μ_1 , in the sense of distributions, into a low-frequency part and wave packets $M_T \mu_1$ indexed by tubes T of length $O(1)$ and width $\rho_j \asymp R_j^{-1/2+\delta}$. Write ℓ_T for the axis of T , and fix a constant $A_{\text{good}} > 3$. Put

$$T^{\text{act}} := 3T, \quad T^{\text{good}} := \{z : \text{dist}(z, \ell_T) \leq A_{\text{good}} \rho_j\},$$

where R_0 is chosen so large that $(A_{\text{good}} - 3)R_j^\delta > 300$ for every $j \geq 1$. Call T bad when

$$\mu_2(T^{\text{good}}) \geq R_j^{-1/2+100\delta}, \quad (10.36q)$$

and let $\mu_{1,g}$ be the sum of the low-frequency part and all good packets. If R_0 is large enough, there is $E'_2 \subset E_2$ with $\mu_2(E'_2) > 999/1000$ such that

$$\|d_*^x(\mu_1) - d_*^x(\mu_{1,g})\|_{L^1(\mathbb{R})} < 10^{-3} \quad (x \in E'_2), \quad (10.36r)$$

where $d^x(z) = |z - x|$.

Proof. Choose a smooth annular partition of unity $1 = \psi_0 + \sum_{j \geq 1, \tau} \psi_{j,\tau}$, where τ has radial length $\asymp R_j$ and tangential length $\asymp R_j^{1/2}$. For each (j, τ) take a bounded-overlap smooth partition $\{\eta_T\}_T$ by tubes of width ρ_j parallel to the radial direction of τ , and put

$$M_T \mu_1 = \eta_T(\psi_{j,\tau} \hat{\mu}_1)^\vee, \quad M_0 \mu_1 = (\psi_0 \hat{\mu}_1)^\vee.$$

The partitions may be chosen on all of $\mathbb{R}^2$, so the resulting packet expansion is exact as a tempered distribution. The inverse Fourier kernel of $\psi_{j,\tau}$ has anisotropic rapid decay on the dual $R_j^{-1/2} \times R_j^{-1}$ scale.

We need two elementary consequences. First, if $x \in E_2$ and $x \notin 2T$, then for every N ,

$$\|d_*^x(M_T \mu_1)\|_1 \lesssim_N R_j^{-N}. \quad (10.36s)$$

Indeed, on any circle about x meeting the physical packet, the separation from the packet axis keeps the circle tangent a definite multiple of $R_j^{-1/2+\delta}$ away from the stationary direction. Repeated integration by parts in the tangential variable gains an arbitrary power of $R_j^{-\delta}$; separation of E_1, E_2 handles the complementary radial range.

Second, total variation cannot increase under a push-forward, and the same anisotropic kernel estimate, with the packet cutoff chosen with a fixed collar, gives

$$\|d_*^x(M_T \mu_1)\|_1 \leq \|M_T \mu_1\|_1 \lesssim \mu_1(2T) + O_N(R_j^{-N}). \quad (10.36t)$$

Define

$$\text{Bad}_j(x) := \{y \in E_1 : \text{there is a bad } T \text{ with } x, y \in 2T\}.$$

For fixed separated x, y , the admissible tube directions occupy an $O(R_j^{-1/2+\delta})$ angular window in an $R_j^{-1/2}$ -net, hence at most $O(R_j^\delta)$ labels occur. Summing (10.36t) over the bad tubes for which $x \in 2T$, and using (10.36s) for all remaining bad packets, gives

$$\sum_{T \text{ bad}} \|d_*^x(M_T \mu_1)\|_1 \lesssim R_j^\delta \mu_1(\text{Bad}_j(x)) + O_N(R_j^{-N}).$$

It remains to estimate the bad-pair set. Choose s with $1 < s < \alpha$. Then $I_s(\mu_i) < \infty$, so the local radial L^p proposition above gives some $p > 1$ with

$$\int \|\pi_y \mu_2\|_{L^p(\mathbb{S}^1)}^p d\mu_1(y) < \infty. \quad (10.36u)$$

Fix $y \in E_1$. If a bad tube T has $y \in 2T$, then $P_y(T^{\text{good}} \cap E_2)$ is contained in an arc of length $O(R_j^{-1/2+\delta})$, while (10.36q) assigns at least $R_j^{-1/2+100\delta}$ radial mass to that arc. A Vitali subfamily therefore shows that the union of bad direction arcs has arclength $O(R_j^{-99\delta})$. Hölder in direction and then in y using (10.36u) gives, for some $c = c(\alpha, s, p) > 0$,

$$(\mu_1 \times \mu_2)(\text{Bad}_j) \lesssim R_j^{-c\delta}, \quad (10.36v)$$

where Bad_j consists of pairs (y, x) lying in $2T \times 2T$ for some bad T . Markov's inequality supplies sets $B_j \subset E_2$ with $\mu_2(B_j) \lesssim R_j^{-c\delta/2}$ and $\mu_1(\text{Bad}_j(x)) \lesssim R_j^{-c\delta/2}$ for $x \notin B_j$. After decreasing c to absorb the bounded-overlap loss, the geometric series is summable. Taking $E'_2 = E_2 \setminus \bigcup_j B_j$ and then R_0 sufficiently large proves the estimate first for finite packet truncations. For $x \in E'_2$ the bad-packet push-forwards are absolutely summable in L^1 by (10.36s)–(10.36v). Their L^1 sum therefore defines $d_*^x \mu_1 - d_*^x \mu_{1,g}$ and yields (10.36r). $\square$

Lemma (enlargement-stable refined planar $p = 6$ packet estimate). *For every $\varepsilon > 0$ there is $\delta_0(\varepsilon) > 0$ with the following property. Let $0 < \delta \leq \delta_0$, r be sufficiently large, and let $\mathbb{W}$ be a family of W unit length tubes $T \subset B(0, 3)$ of common half-width $\rho \asymp r^{-1/2+\delta}$, with bounded overlap for each fixed direction. Associate to each T a tangential arc $\theta(T)$ of the radius- r circle of length $O(r^{1/2})$, normal to T , and a function f_T satisfying*

$$\text{supp } \widehat{f_T} \subset \mathcal{N}_{r^{3\delta/4}}(\theta(T)), \quad \|f_T\|_{L^6(\mathbb{R}^2 \setminus 2.1T)} \leq r^{-200} \|f_T\|_6.$$

Assume the L^6 norms are within a factor 2 of one another. If Y is a union of $r^{-1/2}$ -squares q and

$$\#\{T \in \mathbb{W} : 3T \cap 60q \neq \emptyset\} \leq M \quad (q \subset Y),$$

then

$$\left\| \sum_{T \in \mathbb{W}} f_T \right\|_{L^6(Y)} \lesssim_{\varepsilon, \delta} r^\varepsilon \left(\frac{M}{W} \right)^{1/3} \left(\sum_{T \in \mathbb{W}} \|f_T\|_6^2 \right)^{1/2}. \quad (10.36w)$$

Proof. The proof keeps separate the tube on which a packet is concentrated and the strictly larger tube used to count multiplicity. This is the point at which the literal arbitrary-packet formulation of the original refined-decoupling argument fails; the repaired induction below is calibrated to [40, Theorems 1.1–1.2 and Appendix A] but is proved here from the Bourgain–Demeter estimate (10.36o).

Cover the circle by finitely many graph charts and make a fixed affine normalization on one chart. It suffices to prove the following radius- R statement, uniformly for normalized phases g with positive curvature and bounded derivatives. Let intervals I have length $R^{-1/2}$, let

$$\Theta_I^\Lambda = \{\xi : \xi_1 \in 32I, |\xi_2 - g(\xi_1)| \leq 4\Lambda R^{-1}\},$$

and let

$$T_{I,v} = \{x \in [-25R, 25R] \times [-R, R] : |x_1 + g'(c_I)x_2 - v| \leq \Lambda R^{1/2}\}.$$

Here $R^{\vartheta/2} \leq \Lambda \leq R^{2\vartheta}$, $1 \leq a \leq 10$, and $a + \frac{1}{2} \leq b \leq 20$. Suppose

$$F = \sum_{T \in \mathbb{W}} F_T, \quad \text{supp } \widehat{F_T} \subset \Theta_I^\Lambda, \quad \|F_T\|_{L^6(\mathbb{R}^2 \setminus aT)} \leq R^{-100} \|F_T\|_6,$$

with bounded overlap for each label I , comparable packet norms, and at most M of the counted bT meeting any $R^{1/2}$ -square in Y . We prove

$$\|F\|_{L^6(Y)} \leq \mathfrak{A}(R, \Lambda, a, b) \left(\frac{M}{|\mathcal{W}|} \right)^{1/3} \left(\sum_T \|F_T\|_6^2 \right)^{1/2}, \quad (10.36x)$$

with $\mathfrak{A} \lesssim_{\varepsilon, \vartheta} R^\varepsilon$.

Set $R' = R^{1/2}$. Cover the frequency parameter interval by parents J of length $2R^{-1/4}$ and assign every I to one $J(I)$. For each J cover the radius- R box by cylinders

$$\square_{J,u} = \{x : |x_1 + g'(c_J)x_2 - u| \leq 25R^{3/4}, |x_2| \leq R\},$$

assign each packet to one such $\square$, and put $F_\square = \sum_{T: \square(T)=\square} F_T$. Because the support dilation satisfies $a \leq 10$, the assignment may be made so that $(a+1)T \subset \square(T)$ once $\Lambda \leq R^{1/4}/100$. The parabolic map

$$\Phi(x) = (R^{-1/4}(x_1 + g'(c_J)x_2 - u), R^{-1/2}x_2)$$

sends $\square$ to a radius- R' box and sends every sT exactly to the corresponding sT' at scale R' with the same Λ ; the rescaled phase remains in the same normalized curvature class. Thus both the physical support dilation and the frequency support hypothesis survive the rescaling without changing their constants.

The only delicate point is multiplicity. Put $b' = b - 12\Lambda^{-3/4}$. If a rescaling cell $C \subset \square$ meets $b'T$ and C meets an $R^{1/2}$ -square Q'' whose distance from an original square Q is at most $\Lambda^{1/4}R^{1/2}$, then the centre of Q belongs to bT . Indeed, in the adapted coordinates $y_1 = x_1 + g'(c_J)x_2$, the displacement from a point of $C \cap b'T$ to the centre of Q is at most $9\Lambda^{1/4}R^{1/2}$ in y_1 plus at most $1.1R^{3/4}$ in the long coordinate; the slope error is $O(R^{-1/4})$. Dividing the resulting transverse error by $\Lambda R^{1/2}$ gives at most $12\Lambda^{-3/4}$. Hence lower-scale activity counted with $b'T$ transfers to the original bT count. This is the dilation margin missing from the literal arbitrary-packet formulation.

We now prove the one-step recurrence rather than suppressing the packet algebra. Let $\mathfrak{R}$ denote the right-hand side of (10.36x) without the factor $\mathfrak{A}$. All rapid-decay errors below are chosen $O(R^{-40}\mathfrak{R})$; since only polynomially many packets and cells occur, their total contribution is absorbed by the final additive constant. First pigeonhole the $R^{1/2}$ -squares according to $\|F\|_{L^6(Q)}$. For one retained union Y_0 ,

$$\|F\|_{L^6(Y)}^6 \lesssim (\log R) \|F\|_{L^6(Y_0)}^6.$$

For a parent frequency interval J write $F_{T_J} = \sum_{\square=\square_{J,u}} F_\square$. The parent Fourier support has thickness $R^{-1/2}$. Apply the weighted Bourgain–Demeter estimate on one $Q \subset Y_0$ and put

$$\mathcal{N}^*(Q) = \{x : \text{dist}(x, Q) \leq \Lambda^{1/4}R^{1/2}\}.$$

Choose the weight-decay order $N_1 \geq 2000/\vartheta$. Since $\Lambda \geq R^{\vartheta/2}$, the decoupling weight is $O(R^{-250})$ outside $\mathcal{N}^*(Q)$. A parent cylinder not meeting this neighbourhood is also negligible: by $(a+1)T \subset \square(T)$ and the packet-tail hypothesis,

$$\|F_\square\|_{L^6(\mathbb{R}^2 \setminus \square)} \leq \sum_{T \in \mathcal{W}_\square} \|F_T\|_{L^6(\mathbb{R}^2 \setminus aT)} \leq R^{-100} \sum_{T \in \mathcal{W}_\square} \|F_T\|_6.$$

For fixed J , the adapted transverse diameter of $\mathcal{N}^*(Q)$ is $O(R^{3/4})$, while the parent cylinders have transverse width $\asymp R^{3/4}$ and uniformly separated centres. Hence only $O(1)$ such cylinders meet $\mathcal{N}^*(Q)$, and Cauchy–Schwarz gives

$$\|F_{T_J}\|_{L^6(w_Q)} \lesssim \left(\sum_{\square=\square_{J,u}} \|F_{\square}\|_{L^6(w_Q)}^2 \right)^{1/2} + O(R^{-40}\mathfrak{R}).$$

Cover $\mathcal{N}^*(Q)$ by the $R^{1/2}$ -squares

$$\mathcal{N}(Q) = \{Q'' : \text{dist}(Q, Q'') \leq \Lambda^{1/4} R^{1/2}\}, \quad |\mathcal{N}(Q)| \lesssim \Lambda^{1/2},$$

and subdivide each parent cylinder into the rescaling cells C . Every Q'' meets only $O(1)$ such cells, so the weighted decoupling estimate yields

$$\|F\|_{L^6(Q)} \lesssim R^{\varepsilon'/2} \left(\sum_{(\square, Q'', C)} \|F_{\square}\|_{L^6(Q'' \cap C)}^2 \right)^{1/2} + O(R^{-40}\mathfrak{R}), \quad (10.36y)$$

where $Q'' \in \mathcal{N}(Q)$ and $C \cap Q'' \neq \emptyset$.

Put $b' = b - 12\Lambda^{-3/4}$ and $m_{\square, b'}(C) = \#\{T \in \mathcal{W}_{\square} : b'T \cap C \neq \emptyset\}$. Discarding negligible zero-multiplicity terms, pigeonhole the triples in (10.36y) according to

$$m_{\square, b'}(C) \sim M', \quad |\mathcal{W}_{\square}| \sim W',$$

and then pigeonhole the original squares by the number $n(Q) \sim n$ of retained triples. Together with the first square-mass pigeonhole this costs at most $O((\log R)^4)$. Since $(\sum_i a_i^2)^3 \leq n(Q)^2 \sum_i a_i^6$, summing over the retained original squares gives

$$\|F\|_{L^6(Y)}^6 \lesssim (\log R)^4 R^{3\varepsilon'} n^2 \Lambda^{1/2} \sum_{\square: |\mathcal{W}_{\square}| \sim W'} \|F_{\square}\|_{L^6(Y_{\square, M'})}^6 + O(R^{-30}\mathfrak{R}^6), \quad (10.36z)$$

where $Y_{\square, M'}$ is the union of cells C with $m_{\square, b'}(C) \sim M'$. Here the factor $\Lambda^{1/2}$ appears because a fixed retained triple can be attached to only $O(\Lambda^{1/2})$ original squares.

Parabolic rescaling sends each retained parent to a scale- $R' = R^{1/2}$ configuration with the same Λ , support dilation a , counting dilation b' , packet count $|\mathcal{W}_{\square}|$ and multiplicity $O(M')$. Therefore

$$\|F_{\square}\|_{L^6(Y_{\square, M'})} \lesssim \mathfrak{A}(R', \Lambda, a, b') \left(\frac{M'}{|\mathcal{W}_{\square}|} \right)^{1/3} \left(\sum_{T \in \mathcal{W}_{\square}} \|F_T\|_6^2 \right)^{1/2}. \quad (10.36aa)$$

The families $\mathcal{W}_{\square}$ are disjoint, their cardinalities lie in one dyadic W' -class, and the original packet norms are comparable. If their common norm scale is λ , then

$$\sum_{T \in \mathcal{W}_{\square}} \|F_T\|_6^2 \asymp W' \lambda^2, \quad \#\{\square : |\mathcal{W}_{\square}| \sim W'\} \lesssim W/W'.$$

Consequently, after taking sixth powers in (10.36aa),

$$\begin{aligned} \sum_{\square} \|F_{\square}\|_{L^6(Y_{\square, M'})}^6 &\lesssim \mathfrak{A}(R', \Lambda, a, b')^6 \left(\frac{M'}{W'} \right)^2 \frac{W}{W'} (W' \lambda^2)^3 \\ &\lesssim \mathfrak{A}(R', \Lambda, a, b')^6 \left(\frac{M'}{W} \right)^2 \left(\sum_{T \in \mathcal{W}} \|F_T\|_6^2 \right)^3. \end{aligned} \quad (10.36ab)$$

It remains to transfer the local multiplicity to the original count. For one original square Q , a fixed cylinder contributes to at most $O(\Lambda^{1/2})$ of the retained triples, so at least $c n \Lambda^{-1/2}$ distinct parent cylinders occur. Select one triple from each such cylinder. The transfer estimate proved above sends every one of its $\sim M'$ lower-scale packets to an original packet whose bT meets Q . Distinct cylinders have disjoint packet families; hence the original multiplicity hypothesis gives

$$nM' \lesssim \Lambda^{1/2} M. \quad (10.36ac)$$

Substitute (10.36ab) and (10.36ac) into (10.36z) and take sixth roots. The exact remaining power is no worse than $\Lambda^{1/4}$, which we harmlessly enlarge to $\Lambda^{1/2}$. This proves

$$\mathfrak{A}(R, \Lambda, a, b) \lesssim (\log R)^4 R^{\varepsilon'/2} \Lambda^{1/2} \mathfrak{A}(R^{1/2}, \Lambda, a, b - 12\Lambda^{-3/4}) + 1. \quad (10.36ad)$$

The terminal estimate is elementary: $\mathfrak{A}(R, \Lambda, a, b) \lesssim R^{1/4}$, and when $\Lambda \geq R^{1/4}/100$ this is $O(\Lambda)$, by Hölder in the packet sum and the $O(R^{1/2})$ available direction labels.

Iterate (10.36ad) through $R_k = R^{2^{-k}}$ until $\Lambda \geq R_k^{1/4}/100$. Since $\Lambda \geq R^{\vartheta/2}$, only $O(\log(1/\vartheta))$ steps occur. The total loss of counting dilation is $O(\log(1/\vartheta)\Lambda^{-3/4}) < 1/4$ for large R , while initially $b - a \geq 1/2$; thus the strict support/counting margin persists throughout the induction. We obtain

$$\mathfrak{A}(R, \Lambda, a, b) \lesssim_{\varepsilon', \vartheta} \Lambda^{1+O(\log(1/\vartheta))} (\log R)^{O(\log(1/\vartheta))} R^{O(\varepsilon')}.$$

Choose first ϑ sufficiently small in terms of the requested ε and then ε' sufficiently small. Since $\Lambda \leq R^{2\vartheta}$, the right side is $O(R^\varepsilon)$.

Finally return to the radius- r circle. On each fixed chart the affine frequency normalization has $R \asymp r$ and sends the physical tubes to the normalized strips with $\Lambda \asymp r^\delta$. The exact frequency neighbourhood above maps into Θ_I^Λ ; moreover $2.1T$ maps inside a 2.2 -dilate, while the normalized counted 2.8 -dilate pulls back inside $3T$. An $R^{1/2}$ -square pulls back into a fixed enlargement $60q$ of an $r^{-1/2}$ -square. Apply the normalized estimate with $a = 2.2$, $b = 2.8$ and absorb the finite chart/color decomposition. This proves (10.36w). $\square$

Lemma (exact preparation of the Falconer parent packets). *Fix $r > 10R_0$, a scale $R_j \asymp r$, and a parent packet T . Let $\eta_1 \in C_c^\infty(B(0, 3/2))$ equal one on $B(0, 1)$ and put*

$$f_T = \eta_1(M_T \mu_1 * \widehat{\sigma}_r).$$

There are smooth frequency cutoffs χ_T and $\widetilde{f}_T = (\chi_T \widehat{f}_T)^\vee$ such that for every N ,

$$\text{supp } \widehat{\widetilde{f}}_T \subset \mathcal{N}_{r^{3\delta/4}}(\theta(T)), \quad \|f_T - \widetilde{f}_T\|_6 \lesssim_N r^{-N}, \quad (10.36ae)$$

and

$$\|\widetilde{f}_T\|_{L^6(\mathbb{R}^2 \setminus 2.1T)} + \|\widetilde{f}_T\|_{L^6(\mathbb{R}^2 \setminus B(0,3))} \lesssim_N r^{-N}, \quad \|f_T\|_2 + \|\widetilde{f}_T\|_2 + \|\check{\chi}_T\|_1 \lesssim 1. \quad (10.36af)$$

At one scale the parent packets meeting the fixed observation region are at most Cr^{C_0} . Hence, after discarding packets with $\|\widetilde{f}_T\|_6 < r^{-K}$, choosing the decay order larger than $K + 200$ gives for every retained packet

$$\|\widetilde{f}_T\|_{L^6(\mathbb{R}^2 \setminus 2.1T)} \leq r^{-200} \|\widetilde{f}_T\|_6, \quad (10.36ag)$$

and the retained norms split into $O(\log r)$ dyadic classes.

Proof. Before multiplication by η_1 , the Fourier transform is supported on the radius- r circle inside the angular parent associated with T . Multiplication by η_1 convolves that measure with the Schwartz function $\hat{\eta}_1$, so $\hat{f}_T$ is rapidly decreasing away from a $r^{\delta/2}$ -neighbourhood of the slightly enlarged parent arc. Choose $\chi_T = 1$ on that neighbourhood and supported in the $r^{3\delta/4}$ -neighbourhood of $\theta(T)$, with the natural anisotropic derivative bounds. Hausdorff–Young gives the first error in (10.36ae). The inverse transform $\check{\chi}_T$ has uniformly bounded L^1 norm and anisotropic rapid decay. Convolution with the already localized f_T therefore gives (10.36af); outside $2.1T$ there is a fixed fraction of the transverse packet width between the point and $2T$. Moreover $\|M_T \mu_1\|_1 \lesssim 1$ by the uniform L^1 multiplier-kernel bound, so the normalized-circle extension formula and the compact cut-off give $\|f_T\|_2 \lesssim 1$; Young’s inequality then gives $\|\tilde{f}_T\|_2 \lesssim 1$. The absolute cutoff converts the absolute rapid tail into (10.36ag). Because the number of parent packets at one scale is only polynomial in r , the discarded family is negligible once the desired rapid-decay order is fixed first and the absolute cutoff exponent K is chosen afterwards. $\square$

Lemma (corrected activity incidence and local constancy). *Let $\mathbb{W}_\lambda$ be one dyadic L^6 class of retained good parent packets at scale r , let $W = |\mathbb{W}_\lambda|$, and put $s = r^{-1/2}$. On the grid of s -squares set*

$$m(q) = \#\{T \in \mathbb{W}_\lambda : 100q \cap 3T \neq \emptyset\}, \quad Y_M = \bigcup_{M \leq m(q) < 2M} q.$$

Then

$$\mu_2(\mathcal{N}_{5s}(Y_M)) \lesssim \frac{W}{M} r^{-1/2+100\delta}. \quad (10.36ah)$$

Moreover, if H has Fourier support in $B(0, Cr)$ and Y is a union of s -squares, then for every K ,

$$\int_Y |H|^2 d\mu_2 \lesssim \int_{Y^+} |H(z)|^2 (\mu_2 * \Phi_r)(z) dz + r^{-K} \|H\|_2^2, \quad Y^+ = \mathcal{N}_{r^{-3/4}}(Y), \quad (10.36ai)$$

where $\Phi_r(z) = C_N r^2 (1 + r|z|)^{-N}$,

$$\|\mu_2 * \Phi_r\|_\infty \lesssim r^{2-\alpha}, \quad \int_{Y^+} \mu_2 * \Phi_r \lesssim \mu_2(\mathcal{N}_{2s}(Y)) + O_K(r^{-K}). \quad (10.36aj)$$

If $Y = Y_M$ and Y_M^{cov} is the union of all s -squares meeting Y_M^+ , then every $q' \subset Y_M^{\text{cov}}$ is contained in $4q$ for some $q \subset Y_M$ and $60q' \subset 100q$. Hence

$$\#\{T \in \mathbb{W}_\lambda : 3T \cap 60q' \neq \emptyset\} < 2M, \quad (10.36ak)$$

so (10.36w) applies on Y_M^{cov} .

Proof. For the incidence estimate define

$$J = \sum_{q \subset Y_M} \sum_{T : 100q \cap 3T \neq \emptyset} \mu_2(200q).$$

Every $q \subset Y_M$ has at least M incidences, and $200q$ contains the $5s$ -neighbourhood of q , so $J \geq M\mu_2(\mathcal{N}_{5s}(Y_M))$. If $100q \cap 3T \neq \emptyset$ and $y \in 200q$, then $\text{dist}(y, \ell_T) \leq 3\rho_j + 213s < A_{\text{good}}\rho_j$

once r is large. Thus $200q \subset T^{\text{good}}$. The family $\{200q\}$ has bounded overlap, and goodness gives

$$\sum_{q: 100q \cap 3T \neq \emptyset} \mu_2(200q) \lesssim \mu_2(T^{\text{good}}) \lesssim r^{-1/2+100\delta}.$$

Summing over T proves (10.36ah).

For local constancy choose a Schwartz φ with $\hat{\varphi} = 1$ on $B(0, 2C)$ and put $\varphi_r(z) = r^2\varphi(rz)$. Then $H = H * \varphi_r$. Cauchy–Schwarz followed by Fubini gives

$$\int_Y |H|^2 d\mu_2 \lesssim \int |H(z)|^2 ((\mu_2|_Y) * |\varphi_r|)(z) dz.$$

Outside Y^+ the kernel contributes $O_K(r^{-K})$, while on Y^+ it is bounded by a constant multiple of $\mu_2 * \Phi_r$. Dyadic annuli and the Frostman estimate give the L^∞ bound in (10.36aj); Fubini and the rapid tail of Φ_r give its integral bound. Since $r^{-3/4} = o(s)$, any grid square meeting Y_M^+ lies in a fixed enlargement $4q$ of some $q \subset Y_M$, and $60q' \subset 100q$ after fixing the harmless grid constants. This proves (10.36ak). $\square$

Lemma (local Liu identity). *Let σ_t be normalised arclength on the circle of radius t . For Schwartz $f : \mathbb{R}^2 \rightarrow \mathbb{C}$ and every $x \in \mathbb{R}^2$,*

$$\int_0^\infty |f * \sigma_t(x)|^2 t dt = \int_0^\infty |f * \hat{\sigma}_r(x)|^2 r dr. \quad (10.36al)$$

Proof. Apply the radialisation–Plancherel identity (10.16) to $g_x(z) = f(x - z)$. The radialisation of g_x at radius t is $f * \sigma_t(x)$. Since $\hat{g}_x(\xi) = e^{-2\pi i x \cdot \xi} \hat{f}(-\xi)$, its radialisation at radius r is $f * \hat{\sigma}_r(x)$. In dimension two the polar Jacobian is the same factor $2\pi t$ (or $2\pi r$) on the two sides, so it cancels and gives (10.36al). Approximation extends the identity to the smooth packet sums used below. $\square$

Lemma (good-part averaged L^2 estimate). *Under the hypotheses and notation of the train-track lemma, if $\alpha > 5/4$ and $\delta > 0$ is chosen sufficiently small, then*

$$\int_{E_2} \|d_*^x \mu_{1,g}\|_{L^2(\mathbb{R})}^2 d\mu_2(x) < \infty. \quad (10.36am)$$

Proof. The packet kernels show that $\mu_{1,g}$ is rapidly decreasing away from E_1 ; more precisely, outside a neighbourhood of radius $R_0^{-1/2+\delta}$ the tail is $O_N(R_0^{-N})$. Since E_1, E_2 are separated, the factor t^2 in the radial coarea formula for $\|d_*^x \mu_{1,g}\|_2^2$ is therefore comparable with t , up to the same rapidly decreasing error. The local Liu identity reduces (10.36am) to estimating

$$\int_1^\infty \int_{E_2} |\mu_{1,g} * \hat{\sigma}_r(x)|^2 d\mu_2(x) r dr. \quad (10.36an)$$

We prove the scale estimate

$$\int_{E_2} |\mu_{1,g} * \hat{\sigma}_r(x)|^2 d\mu_2(x) \lesssim_{R_0, \varepsilon} r^{-(\alpha+1)/3+\varepsilon} r^{-1} \int_{\mathbb{R}^2} |\hat{\mu}_1(\xi)|^2 \Psi_r(\xi) d\xi + O_N(r^{-N}), \quad (10.36ao)$$

where $\Psi_r(\xi) \lesssim_N (1 + \|\xi\| - r)^{-N}$ and $\Psi_r \asymp 1$ on the unit annulus about $\|\xi\| = r$.

Assume first $r > 10R_0$. Only $O(1)$ packet scales $R_j \asymp r$ survive convolution with $\hat{\sigma}_r$. After the fixed observation cutoff to $B(0, 3/2)$ put $f_T = \eta_1(M_T \mu_1 * \hat{\sigma}_r)$ and apply the exact-preparation lemma. Replacing f_T by $\tilde{f}_T$ changes the required estimate by $O_N(r^{-N})$ after choosing the truncation order larger than the polynomial packet count. Discard also the packets with $\|\tilde{f}_T\|_6 < r^{-K}$. Indeed, if H_{sm} is their sum, then $\|H_{\text{sm}}\|_6 \lesssim r^{C_0-K}$ by Minkowski, while its L^2 norm is polynomial in r . Applying (10.36ai) on a fixed square cover of $B(0, 2)$ and using (10.36aj) makes $\int |H_{\text{sm}}|^2 d\mu_2 = O_N(r^{-N})$ once K and the local-constancy decay order are chosen large enough. Thus every remaining parent packet has exact frequency support, the relative $2.1T$ tail (10.36ag), and belongs to one of $O(\log r)$ dyadic L^6 classes.

Fix one scale and one norm class W_λ , put $W = |W_\lambda|$, and use the multiplicity decomposition from the corrected activity-incidence lemma. For one nonzero class Y_M let

$$A_M := \mu_2(\mathcal{N}_{5r^{-1/2}}(Y_M)).$$

Apply (10.36ai) to $H = \sum_{T \in W_\lambda} \tilde{f}_T$. Hölder with exponents 3 and $3/2$, then (10.36aj), gives

$$\int_{Y_M} |H|^2 d\mu_2 \lesssim \|H\|_{L^6(Y_M^{\text{cov}})}^2 r^{(2-\alpha)/3} (A_M + r^{-K})^{2/3} + O_N(r^{-N}). \quad (10.36ap)$$

The neighbourhood Y_M^{cov} is essential: local constancy does not preserve the original set Y_M . By (10.36ak), however, every covering square meets at most $2M$ counted activity tubes. Therefore the enlargement-stable refined packet estimate (10.36w) applies and yields

$$\|H\|_{L^6(Y_M^{\text{cov}})}^2 \lesssim r^\varepsilon \left(\frac{M}{W}\right)^{2/3} \sum_{T \in W_\lambda} \|\tilde{f}_T\|_6^2. \quad (10.36aq)$$

On the other hand (10.36ah) gives

$$\left(\frac{M}{W}\right)^{2/3} A_M^{2/3} \lesssim r^{-1/3+(200/3)\delta}. \quad (10.36ar)$$

It remains to estimate one retained parent packet. By (10.36af), $\tilde{f}_T$ is rapidly decreasing off $2.1T$, whose area in the observation ball is $O(r^{-1/2+\delta})$, and $\|\tilde{\chi}_T\|_1 \lesssim 1$. Hence

$$\|\tilde{f}_T\|_6 \lesssim r^{-1/12+\delta/6} \|f_T\|_\infty + O_N(r^{-N}).$$

The circle extension formula and Cauchy–Schwarz on the $O(r^{-1/2})$ fraction of normalized circle measure meeting the parent angular arc give

$$\|f_T\|_\infty \lesssim r^{-1/4} \|\widehat{M_T \mu_1}\|_{L^2(d\sigma_r)} + O_N(r^{-N}).$$

Thus

$$\|\tilde{f}_T\|_6^2 \lesssim r^{-2/3+\delta/3} \|\widehat{M_T \mu_1}\|_{L^2(d\sigma_r)}^2 + O_N(r^{-N}). \quad (10.36as)$$

Combining (10.36ap)–(10.36as) gives

$$\int_{Y_M} |H|^2 d\mu_2 \lesssim r^{-(\alpha+1)/3+\varepsilon+C\delta} \sum_{T \in W_\lambda} \|\widehat{M_T \mu_1}\|_{L^2(d\sigma_r)}^2 + O_N(r^{-N}). \quad (10.36at)$$

There are only $O(\log r)$ norm classes and $O(\log r)$ nonzero multiplicity classes. On the zero-multiplicity squares, every $100q$ misses $3T$ for every retained parent, so the relative $2.1T$ tail and the fixed geometric margin give a rapidly decreasing contribution. Cauchy–Schwarz over the finitely many dyadic classes and absorption of logarithms into r^ε therefore reduce the whole scale to the packet square sum in (10.36at).

Finally the packets are square-summable on the thickened circle. More precisely,

$$\sum_{R_j \asymp r} \sum_{\tau} \sum_T \|\widehat{M_T \mu_1}\|_{L^2(d\sigma_r)}^2 \lesssim r^{-1} \int |\hat{\mu}_1(\xi)|^2 \Psi_r(\xi) d\xi. \quad (10.36au)$$

To see this, thicken $d\sigma_r$ to a unit annulus, producing the factor r^{-1} ; write $\widehat{M_T \mu_1} = \hat{\eta}_T * (\psi_{j,\tau} \hat{\mu}_1)$; enlarge the corresponding frequency cutoff once; then Plancherel and bounded overlap of the physical η_T give the square sum for fixed (j, τ) . Summing the finitely overlapping τ and the $O(1)$ relevant j proves (10.36au). Combining this with (10.36at), choosing δ so that $C\delta < \varepsilon$, proves (10.36ao) for $r > 10R_0$. The bounded range $r \leq 10R_0$ follows by Cauchy–Schwarz on the circle and (10.36au), with the finite loss absorbed into $C(R_0)$.

Insert (10.36ao) in (10.36an) and use Tonelli. The r -weight cancels the explicit r^{-1} , and rapid decay of Ψ_r gives

$$\int_{E_2} \|d_*^x \mu_{1,g}\|_2^2 d\mu_2(x) \lesssim_{R_0} \int_{\mathbb{R}^2} |\hat{\mu}_1(\xi)|^2 (1 + |\xi|)^{-(\alpha+1)/3+2\varepsilon} d\xi + 1.$$

For large frequencies this is the Fourier form of $I_\beta(\mu_1)$ with $\beta = 2 - (\alpha + 1)/3 + 2\varepsilon$. Since $\alpha > 5/4$, choose $\varepsilon > 0$ so small that $\beta < \alpha$; then the Frostman bound (10.36p) gives $I_\beta(\mu_1) < \infty$.

For complete limiting bookkeeping, first truncate the packet decomposition to finitely many frequency scales. Every estimate above is uniform in that truncation. Hence the truncated radial push-forwards are bounded in $L^2(E_2 \times \mathbb{R})$ and admit weakly convergent subsequences. The train-track lemma gives absolute L^1 summability of the discarded bad push-forwards, while the good distribution is defined as the complementary packet sum; therefore distributional convergence together with the L^1 identity identifies every weak L^2 limit with $d_*^x \mu_{1,g}$ for μ_2 -almost every x . Weak lower semicontinuity then yields (10.36am). Thus the proof never treats the infinite good packet sum as an a priori finite measure, and no smoothing-dependent reclassification of packets is used. $\square$

Theorem (local planar $5/4$ pinned positive-measure theorem). *If $A \subset \mathbb{R}^2$ is Borel and $\dim_{\text{H}} A > 5/4$, then there exists $x \in A$ such that $|\Delta^x(A)| > 0$.*

Proof. Choose $5/4 < \alpha < \dim_{\text{H}} A$. Capacitability and Frostman’s lemma give a compact subset of A carrying an α -Frostman probability μ . Choose distinct $z_1, z_2 \in \text{supp } \mu$ and then disjoint sufficiently small balls $B_i = B(z_i, \rho_*)$ of positive μ -mass. After restricting and renormalising, $\mu_i = \mu|_{B_i} / \mu(B_i)$ are still α -Frostman, their compact supports E_i are separated, and taking $\rho_* \ll |z_1 - z_2|$ puts every direction $(x - y)/|x - y|$, $y \in E_1$, $x \in E_2$, in one arbitrarily short arc. A fixed similarity places the pair in the normalised geometry of the train-track lemma. Apply that lemma and the good-part L^2 lemma. Since E'_2 has positive μ_2 -mass and the average in (10.36am) is finite, choose $x \in E'_2$ for which $d_*^x \mu_{1,g} \in L^2$ and (10.36r) holds.

The probability $d_*^x \mu_1$ is supported on $\Delta^x(A)$. Therefore

$$\int_{\Delta^x(A)} |d_*^x \mu_{1,g}| \geq 1 - 2 \cdot 10^{-3}.$$

Cauchy–Schwarz gives

$$1 - 2 \cdot 10^{-3} \leq |\Delta^x(A)|^{1/2} \|d_*^x \mu_{1,g}\|_2,$$

so $|\Delta^x(A)| > 0$. □

The local proof uses the corrected enlargement-stable form of the packet estimate rather than the invalid literal arbitrary-packet refined-decoupling formulation: the reader now uses an enlargement-stable activity estimate, exact frequency preparation, an absolute small-packet cutoff, and the enlarged local-constancy incidence set. These ingredients are proved from the Bourgain–Demeter endpoint estimate (10.36o); the Singh–Singh Parmar paper identifies the correct formulation. The higher-dimensional Peres–Schlag branch is also proved locally.

Lemma (local Peres–Schlag pinned positive-measure escape). *Let $n \geq 2$ and let $A \subset \mathbb{R}^n$ be Borel. If*

$$\dim_H A > \frac{n+1}{2}, \tag{10.37a}$$

then some $y \in A$ satisfies $|\Delta^y(A)| > 0$.

Proof. Choose

$$\frac{n+1}{2} < \alpha < \dim_H A.$$

By the energy/capacitability package of Volume I, there is a compact $K \subset A$ carrying a probability ω with $I_\alpha(\omega) < \infty$. In particular ω is non-atomic. Choose two disjoint closed balls B_0, B_1 with positive ω -mass and positive mutual distance, and let λ, τ be the normalized restrictions of ω to $K \cap B_0$ and $K \cap B_1$. Both restrictions still have finite α -energy, so

$$\dim_H \text{supp } \lambda, \dim_H \text{supp } \tau \geq \alpha. \tag{10.37b}$$

Apply the locally proved Peres–Schlag generalized-projection theorem from Chapter III.4 to the separated distance family

$$\Pi_y(x) = |x - y|, \quad y \in Q, \quad x \in \text{supp } \tau,$$

where Q is a bounded open neighbourhood of $\text{supp } \lambda$ whose closure stays a positive distance from $\text{supp } \tau$. We verify its $\beta = 0$ transversality directly. For distinct $x, z \in \text{supp } \tau$, put

$$a = |x - y|, \quad b = |z - y|, \quad e_x = \frac{y - x}{a}, \quad e_z = \frac{y - z}{b}.$$

The exact cosine-law identity

$$|x - z|^2 = (a - b)^2 + ab|e_x - e_z|^2 \tag{10.37c}$$

shows that if $|\Pi_y(x) - \Pi_y(z)| \leq \frac{1}{2}|x - z|$, then

$$\left| D_y \frac{\Pi_y(x) - \Pi_y(z)}{|x - z|} \right| = \frac{|e_x - e_z|}{|x - z|} \geq c_Q > 0. \quad (10.37d)$$

Here c_Q is uniform because a, b are uniformly bounded above on the compact parameter-source configuration. All parameter derivatives satisfy the required regularity bounds: away from the diagonal $x = y$, every derivative of $y \mapsto |x - y|$ is smooth, and the difference of the corresponding derivatives at x and z is $O(|x - z|)$ uniformly on $\bar{Q} \times \text{supp } \tau$. Thus division by $|x - z|$ leaves bounded derivatives of the normalised secant map, with arbitrary finite regularity order.

In the notation of the Chapter III.4 theorem, the parameter dimension is n , the target dimension is $m = 1$, and $\beta = 0$. Taking its regularity order larger than α and using the exceptional-set estimate at $\sigma = 1$ gives

$$\dim_H \{y \in Q : \dim_S(\Delta^y \tau) \leq 1\} \leq n + 1 - \alpha < \alpha. \quad (10.37e)$$

By (10.37b), $\text{supp } \lambda$ cannot be contained in this exceptional set. Choose $y \in \text{supp } \lambda$ outside it. Then $\dim_S(\Delta^y \tau) > 1$; in particular $\Delta^y \tau$ has an $L^2(\mathbb{R})$ density f . Since it is a probability, Cauchy-Schwarz gives

$$1 = \int f \leq |\text{supp}(\Delta^y \tau)|^{1/2} \|f\|_2,$$

so $|\text{supp}(\Delta^y \tau)| > 0$. The source support is compact and contained in A , hence $\text{supp}(\Delta^y \tau) \subset \Delta^y(A)$, proving the claim. $\square$

The n -dimensional Peres-Schlag clause is proved locally above. The stronger planar 5/4 theorem is proved locally relative to the harmonic-analysis decoupling estimate stated above.

Lemma III.10.17 (average energy of slices). *Let $1 \leq q \leq d - 1$ and $q < u < d$. If $I_u(\mu) < \infty$, then*

$$\int_{G(d, d-q)} \int_{V^\perp} I_{u-q}(\mu_{V,a}) d\mathcal{H}^q(a) d\gamma(V) \lesssim_{d,u,q} I_u(\mu). \quad (10.38)$$

In particular almost every nonzero affine $(d - q)$ -slice has finite $(u - q)$ -energy.

Proof. Approximate the conditional measure on $V + a$ by

$$d\mu_{V,a}^{(\rho)}(x) = c_q \rho^{-q} \mathbf{1}_{\{|P_{V^\perp} x - a| < \rho\}} d\mu(x).$$

Tonelli reduces the averaged slice energy to

$$\iint |x - z|^{-(u-q)} K_\rho(P_{V^\perp}(x - z)) d\mu(x) d\mu(z),$$

where $K_\rho(w) \lesssim \rho^{-q} \mathbf{1}_{|w| < 2\rho}$. Averaging over the Grassmannian gives

$$\int K_\rho(P_{V^\perp} w) d\gamma(V) \lesssim |w|^{-q},$$

because after scaling $|w| = 1$ the set of V for which $|P_{V^\perp} w| < \rho$ has measure $O(\rho^q)$. Hence the average is $O(I_u(\mu))$. Pass to weak slice limits; lower semicontinuity of energy and Fatou give (10.38). $\square$

Lemma III.10.18 (generic upper-box slicing). *If $E \subset \mathbb{R}^d$ is bounded and $1 \leq q \leq d-1$, then for almost every affine $(d-q)$ -plane W ,*

$$\overline{\dim}_B(E \cap W) \leq \max\{\overline{\dim}_B E - q, 0\}. \quad (10.39)$$

Proof. Fix $\beta > \overline{\dim}_B E$. At every sufficiently small dyadic scale δ , cover E by $O(\delta^{-\beta})$ dyadic cubes. A fixed cube meets fibres of a chosen orientation whose translation parameters occupy q -dimensional measure $O(\delta^q)$. Hence

$$\int N_\delta(E \cap (V + a)) da \lesssim \delta^{q-\beta}.$$

Markov and Borel–Cantelli imply that, for every $\epsilon > 0$, almost every fibre satisfies $N_\delta \lesssim \delta^{-(\beta-q+\epsilon)_+}$ at all fine dyadic scales. Let $\epsilon \downarrow 0$ and $\beta \downarrow \overline{\dim}_B E$. $\square$

Proposition III.10.19 (cylindrical recombination). *Let $2 \leq k \leq d$ and $k-1 < s < k$. Suppose μ, ν have finite s -energy and their projections to a generic k -plane have separated supports. For almost every such plane and μ -almost every centre x , the cylindrical slice recombination $\nu^{V,x}$ is mutually absolutely continuous with ν after positive-mass compact restriction, and for some $p > 1$ its radial projection has an L^p density. The radial density equals the mass of the corresponding cylindrical slice almost everywhere.*

Proof. By Lemma III.10.17 and projection monotonicity, the projected measures have finite s -energy for almost every V . Apply the locally proved radial L^p slicing proposition above in $V \cong \mathbb{R}^k$. Orthogonal disintegration followed by polar disintegration gives

$$\int f d\nu = \int_{S_V^{k-1}} \int f d\nu_{\Pi_{V,x}^{-1}(\theta)} w_{V,x}(\theta) d\sigma_V(\theta), \quad (10.40)$$

where $w_{V,x}$ is precisely the radial density. Truncate to a positive-mass region $j^{-1} \leq w_{V,x} \leq j$. On this region the unweighted recombination and ν have bounded Radon–Nikodym densities in both directions. The L^p assertion and identification of slice mass with $w_{V,x}$ follow from the same disintegration formula. $\square$

Lemma (Borel selection of thin-tube witnesses). *Fix a closed ball B_0 , $t, K > 0$, $c \in (0, 1)$ and $\delta \geq 0$. The set of pairs $(x, \rho) \in B_0 \times \mathcal{P}(B_0)$ having (t, K, c, δ) -thin tubes is compact. Moreover, one may choose the witnessing submeasure $\rho' \leq \rho$, $|\rho'| \geq c$, as a Borel function of (x, ρ) on this compact set. The same conclusion holds for nonzero finite measures after normalisation; on a region where their total masses lie in a fixed compact subinterval of $(0, \infty)$, the selected witnesses may be rescaled Borel-measurably and retain a fixed positive fraction of the original mass.*

Proof. Consider triples (x, ρ, ρ') with $\rho' \leq \rho$, $|\rho'| \geq c$, and

$$\rho'(T) \leq Kr^t \quad \text{for every open } r\text{-tube } T \text{ through } x, r > \delta.$$

The domination relation $\rho' \leq \rho$ is weakly closed. If a convergent sequence of such triples has limit (x, ρ, ρ') , then an open r -tube through x contains the corresponding centres x_n for

all large n ; enlarging the tube by an arbitrarily small amount and using the Portmanteau theorem gives the same tube bound for ρ' . Thus the set of triples is compact, and so is its projection to (x, ρ) . A standard Borel-selection theorem applied to the nonempty compact fibres supplies the witness ρ' measurably. This is the compactness argument used in Shmerkin–Wang [33, Lemma 5.7]. $\square$

Theorem III.10.20 (lifting thin tubes from slices). *Let μ, ν have finite s -energy in $\mathbb{R}^d$, where $s \in (k, k+1]$ and $1 \leq k \leq d-2$. If almost every affine $(d-k)$ -slice pair has t -thin tubes, then (μ, ν) has $(k+t)$ -thin tubes.*

Proof. After a change of coordinates, rescaling and positive-mass restriction, place the two supports in small separated balls so that projection to a fixed $(k+1)$ -plane V has separated images. A small generic perturbation of V preserves this separation and ensures the energy and slicing conclusions used below. Proposition III.10.19 then gives, for μ -almost every x ,

$$\nu^x = \int_{\mathbb{S}^k} v_{\theta,x} d\sigma^k(\theta) \sim \nu \quad \text{and} \quad \int_{\mathbb{S}^k} |v_{\theta,x}|^p d\sigma^k(\theta) < \infty \quad (10.40a)$$

for some $p > 1$. Here the fibres are affine $(d-k)$ -planes through x .

By hypothesis, for almost every (θ, x) the sliced pair has t -thin tubes. The preceding Borel-selection lemma makes the set of such pairs Borel and supplies the witness submeasures $\nu'_{\theta,x} \leq v_{\theta,x}$ measurably. Pigeonhole the thin-tube constant and retained fibre fraction, and then pass to a compact positive-incidence subset on which they are uniform. The L^p estimate (10.40a) implies

$$\iint_{\{|v_{\theta,x}| > M\}} |v_{\theta,x}| d\sigma^k(\theta) d\mu(x) \lesssim M^{1-p}.$$

Choose M so large that deleting this set loses less than half of the already retained incidence mass. A second pigeonholing and Fubini step yields a compact X of positive μ -mass and, for each $x \in X$, a compact set $\Theta_x \subset \mathbb{S}^k$ of uniformly positive spherical measure such that

$$M^{-1} \leq |v_{\theta,x}| \leq M, \quad |v'_{\theta,x}| \geq c|v_{\theta,x}|, \quad \nu'_{\theta,x}(T \cap W_{\theta,x}) \leq Kr^t |v_{\theta,x}| \quad (10.40b)$$

for $\theta \in \Theta_x$ and every width- r tube T through x . This is exactly what results from applying the selector to the normalised finite slice and then rescaling. In particular $|v'_{\theta,x}| \geq c/M$ on the retained incidence set.

Since $\nu^x \sim_C \nu$ after the fixed restriction, define

$$\tilde{\nu}_x = C^{-1} \int_{\Theta_x} \nu'_{\theta,x} d\sigma^k(\theta) \leq \nu.$$

Its total mass is uniformly positive. If T is a width- r tube through x , the separation of the projected supports implies that only directions in an angular $O(r)$ ball of $\mathbb{S}^k$ can have $W_{\theta,x} \cap T \cap \text{supp } \nu \neq \emptyset$. Therefore

$$\tilde{\nu}_x(T) \lesssim r^t \int_{B(\theta_T, Cr) \cap \Theta_x} |v_{\theta,x}| d\sigma^k(\theta) \lesssim Mr^{k+t}.$$

Thus every $x \in X$ has a uniformly positive witness with the $(k + t)$ tube bound, proving the claim. This is the measurable recombination mechanism of Shmerkin–Wang [33, Theorem 6.7], written here in the present notation. $\square$

Proposition III.10.21 (higher-rank radial dimension growth). *Let $d \geq 2$ and $d - 2 < s < d$. There is $\eta(d, s) > 0$ such that if μ, ν are separated probabilities of finite s -energy and neither charges an affine hyperplane, then for μ -almost every x and every Borel Y with $\nu(Y) > 0$,*

$$\dim_{\mathbb{H}} \pi_x(Y) \geq \frac{d-1}{d}s + \eta(d, s). \quad (10.41)$$

Proof. Trim the measures on sets of arbitrarily large mass to obtain s -Frostman bounds. The spherical-direction decay theorem gives, after another arbitrarily small loss, radial direction laws with positive power decay near every hyperplane. On finitely many non-singular radial charts this is exactly hypothesis (10.36l) of the locally proved co-rank-one all-scales nonlinear engine above, with $\alpha = s$. It yields robust radial images with exponent $(d - 1)s/d + \eta$. The robustness-to-dimension lemma III.6.3, followed by letting the discarded mass tend to zero, gives (10.41) for every positive-mass restriction Y . $\square$

Proposition III.10.22 (generic two-slices do not share a common line). *Let $d \geq 3$. Suppose μ, ν have disjoint supports and finite s -energy with*

$$d - 1 - \frac{1}{d-1} - \eta_0 < s \leq d - 1 \quad (10.42)$$

for sufficiently small $\eta_0(d) > 0$. Assume neither measure charges a hyperplane and $\dim_{\mathbb{H}} \text{supp } \nu < s + \eta_0$. After positive-mass restriction, almost every nonzero affine 2-slice pair is not simultaneously supported on one affine line.

Proof. Assume contrariwise that a positive family of affine 2-slices is bad. Let $\mathcal{B}$ be the set of affine 2-planes W for which the two nonzero sliced measures are supported on one common line ℓ_W . The slice maps are Borel. Using the compactness of the line space and Lusin's theorem, pass to a positive subfamily on which $\mathcal{B}$ is compact and $W \mapsto \ell_W$ is continuous; this is exactly the regularisation used in the proof of Shmerkin–Wang [33, Proposition 6.8].

Lemma III.10.17 gives finite $(s - d + 2)$ -energy on almost every bad slice, hence

$$\dim_{\mathbb{H}}(\text{supp } \nu \cap \ell_W) \geq s - d + 2. \quad (10.42a)$$

Parameterise 2-planes by first choosing a hyperplane H and then $\theta \in \mathbb{S}_H^{d-2}$, with

$$V_{\theta, x} = H^\perp + \langle \theta \rangle + x.$$

Orthogonal invariance shows that this has the same Grassmannian law as the original 2-plane slicing. Fubini and the positive bad-slice incidence therefore give a positive-measure set of pairs (H, x) for which

$$\sigma_H^{d-2} \{\theta : V_{\theta, x} \in \mathcal{B}\} > 0. \quad (10.42b)$$

Choose a density point (H_0, x_0) of this set. Since $s > d - 2$, the measure ν cannot be supported on the line $x_0 + H_0^\perp$; choose a density point y_0 of ν with $P_{H_0}y_0 \neq P_{H_0}x_0$. Restrict μ, ν to sufficiently small balls around x_0, y_0 . Then, for every H in a small neighbourhood of H_0 , the two projected supports in H remain separated, while (10.42b) still holds for a positive family of such H . Fix one H for which these conclusions and the standard projection/slicing identities all hold.

For this fixed H , suppress it from the notation. For almost every θ , define the averaged slice measure

$$\mu_\theta := \int \mu_{\theta,x} d\mu(x).$$

Because the projection of μ to $\theta^\perp \cap H$ is absolutely continuous for almost every θ (the projection dimension is $d - 2 < s$), the disintegration formula gives $\mu_\theta \ll \mu$. Now set

$$L_\theta = \bigcup \{\ell_{\theta,x} : V_{\theta,x} \in \mathcal{B}\}.$$

After the compact regularisation of $\mathcal{B}$, the sets L_θ are compact. The positive incidence (10.42b) implies, by Fubini, that $\mu_\theta(L_\theta) > 0$ for θ in a positive- σ_H^{d-2} set. Since $\mu_\theta \ll \mu$, also $\mu(L_\theta) > 0$ for those θ .

If $x \in L_\theta$, then $x \in \ell_{\theta,x'}$ for some x' with $V_{\theta,x'} \in \mathcal{B}$. But $x \in V_{\theta,x'}$ and the planes $V_{\theta,z}$ are the parallel fibres of the same cylindrical projection, so $V_{\theta,x} = V_{\theta,x'}$ and hence

$$x \in \ell_{\theta,x} = \ell_{\theta,x'}. \quad (10.42c)$$

A final Fubini/inner-regularity step therefore gives a compact set X with $\mu(X) > 0$ and compact sets $\Theta_x \subset \mathbb{S}_H^{d-2}$ of positive spherical measure such that (10.42c) holds for every $x \in X, \theta \in \Theta_x$. Proposition III.10.19, after shrinking X once more, also gives

$$\nu^x := \int \nu_{\theta,x} d\sigma_H^{d-2}(\theta) \sim \nu \quad (x \in X). \quad (10.42d)$$

Set $Z_x = \bigcup_{\theta \in \Theta_x} \ell_{\theta,x}$ and let S_x be its set of line directions. By (10.42d), $\nu(Z_x) > 0$ for every $x \in X$, while $\pi_x(Z_x) = S_x$.

Choose $\eta_1 > 0$ smaller than the gain in Proposition III.10.21; the admissible range (10.42) ensures

$$\frac{d-1}{d}s + \eta(d, s) > d - 2 + \eta_1. \quad (10.42e)$$

If $\dim_H S_x \leq d - 2 + \eta_1$ for every $x \in X$, then the positive- ν set Z_x has radial image of dimension at most $d - 2 + \eta_1$ for a positive set of centres, contradicting Proposition III.10.21.

Hence for some $x \in X$, $\dim_H S_x > d - 2 + \eta_1$. On the separated support the radial map is Lipschitz, and every fibre over S_x contains a subset of $\text{supp } \nu$ of dimension at least $s - d + 2$ by (10.42a). The standard Hausdorff dimension inequality for a Lipschitz map and its fibres (Volume I) gives

$$\dim_H \text{supp } \nu \geq (s - d + 2) + (d - 2 + \eta_1) = s + \eta_1,$$

contrary to $\dim_H \text{supp } \nu < s + \eta_0$ after choosing the source slack below η_1 . Therefore the bad slice family is null. $\square$

Theorem III.10.23 (roughly equal-dimensional higher-dimensional thin tubes). *Fix $d \geq 3$ and $k \in \{2, \dots, d-1\}$. Let μ, ν be disjointly supported probabilities with finite s -energy, where*

$$s \in (k - 1/k - \eta_d, k], \quad (10.43)$$

and assume no affine k -plane has positive mass for both measures. If

$$\overline{\dim}_B \operatorname{supp} \mu, \overline{\dim}_B \operatorname{supp} \nu \leq s + \epsilon,$$

then

$$(\mu, \nu) \text{ has } (s - \zeta_d(\epsilon))\text{-thin tubes}, \quad \zeta_d(\epsilon) \rightarrow 0. \quad (10.44)$$

Proof. We follow the reduction in the primary proof of Shmerkin–Wang [33, Theorem 6.14], spelling out the dimension bookkeeping.

Step 1: reduction to ambient dimension $k+1$. Choose an affine plane W of minimal dimension with $\mu(W)\nu(W) > 0$; such a plane exists since $W = \mathbb{R}^d$ is admissible. The hypothesis excluding common k -planes forces $\dim W \geq k+1$. Restrict and renormalise the measures on W . By minimality, no affine hyperplane of W has positive mass for both restricted measures. Now project generically from W to a $(k+1)$ -plane V . Finite s -energy survives for almost every V , upper box dimension cannot increase, and if a relative hyperplane $H \subset V$ had positive mass for both projected measures then $P_V^{-1}(H) \cap W$ would be a common hyperplane in W , a contradiction. Thin tubes for the projected pair lift to the original pair because the orthogonal projection of an r -tube through x is contained in an r -tube through $P_V x$. Hence we may assume from now on that

$$d = k + 1. \quad (10.43a)$$

Step 2: typical 2-slices. Put

$$u = s - d + 2 = s - k + 1. \quad (10.43b)$$

The range (10.43) becomes $u > 1 - 1/k - \eta_d$, exactly the range needed in Proposition III.10.22 after $d = k + 1$. By Lemmas III.10.17 and III.10.18, for almost every affine 2-slice L the nonzero slice measures satisfy

$$I_u(\mu_L), I_u(\nu_L) < \infty, \quad \overline{\dim}_B \operatorname{supp} \mu_L, \overline{\dim}_B \operatorname{supp} \nu_L \leq u + \epsilon. \quad (10.43c)$$

Because $\overline{\dim}_B \operatorname{supp} \nu \leq s + \epsilon$, taking ϵ below the slack in Proposition III.10.22 also gives the required $\dim_H \operatorname{supp} \nu < s + \eta_0$. That proposition therefore shows that almost no nonzero slice pair is supported on a common line.

Step 3: planar rough-regular bootstrap on each slice. Fix a slice satisfying (10.43c). Choose $0 < \delta \ll \epsilon$. Finite u -energy and the Frostman extraction lemma give positive-mass compact restrictions for which every sufficiently small dyadic square $Q \in \mathcal{D}_m$ satisfies

$$\mu_L(Q), \nu_L(Q) \leq 2^{-m(u-\delta)}. \quad (10.43c1)$$

On the other hand, the upper-box bounds in (10.43c) imply, after increasing the starting scale, that each support meets at most $2^{m(u+2\epsilon)}$ squares in $\mathcal{D}_m$. Therefore the total mass contained in occupied squares with $\mu_L(Q) < 2^{-m(u+3\epsilon)}$ is at most

$$2^{m(u+2\epsilon)} 2^{-m(u+3\epsilon)} = 2^{-\epsilon m},$$

and the same estimate holds for ν_L . Summing in m and discarding the resulting null tail, then using inner regularity, leaves positive-mass compact restrictions satisfying for all large m and every occupied $Q \in \mathcal{D}_m$

$$2^{-m(u+3\epsilon)} \leq \mu_L(Q), \nu_L(Q) \leq 2^{-m(u-\delta)}. \quad (10.43c2)$$

This is precisely the rough regularity required by the planar bootstrap, and is the quantitative cube deletion used in the proof of Shmerkin–Wang [33, Theorem 6.14]. Since the two restrictions are not supported on a common line, the intermediate planar statement proved in Proposition III.10.14 yields

$$(\mu_L, \nu_L) \text{ has } (u - \omega(\epsilon))\text{-thin tubes}, \quad \omega(\epsilon) \rightarrow 0. \quad (10.43d)$$

No uniform choice of the slice constants is needed at this stage: the measurable-selection and pigeonholing required to recombine the witnesses are already part of the proof of Theorem III.10.20.

Step 4: recombination. Apply Theorem III.10.20 with lifting parameter $d - 2 = k - 1$. Combining (10.43b) and (10.43d) gives

$$(d - 2) + (u - \omega(\epsilon)) = s - \omega(\epsilon).$$

Taking $\zeta_d(\epsilon)$ to dominate the finitely many trimming and planar losses proves (10.44), with $\zeta_d(\epsilon) \rightarrow 0$. $\square$

Corollary III.10.24 (endpoint slicing dichotomy). *Let $d \geq 3$ and let $X \subset \mathbb{R}^d$ satisfy $\dim_{\text{H}} X = \dim_{\text{p}} X = d/2$. For every sufficiently small $\epsilon > 0$, choose separated regular pieces as in Lemma III.10.4, with finite $(d/2 - \epsilon)$ -energy and upper box dimension at most $d/2 + \epsilon$, by applying that lemma with the smaller parameter $\epsilon/2$. Then either*

- (i) $\Delta(X)$ has positive Lebesgue measure; or
- (ii) after positive-mass restriction the two measures have

$$(d/2 - \zeta_d(\epsilon))\text{-thin tubes}, \quad \zeta_d(\epsilon) \rightarrow 0. \quad (10.45)$$

Proof. Write

$$s = \frac{d}{2} - \epsilon, \quad k_* = \begin{cases} d/2 - 1, & d \text{ even,} \\ (d - 1)/2, & d \text{ odd.} \end{cases} \quad (10.45a)$$

This is the index used in the proof of Shmerkin–Wang [33, Corollary 6.15 and §7.1].

Plane-concentration branch. Suppose first that some affine $(k_* + 1)$ -plane W has positive mass for both regular pieces. After restriction, the support inside W has Hausdorff dimension at least s . If $d = 3$, then W is two-dimensional and $s = 3/2 - \epsilon > 5/4$ for sufficiently small ϵ ; apply the local planar $5/4$ theorem attached to III.10.5. If $d = 4$, then again W is two-dimensional but now $s = 2 - \epsilon > 3/2$, so the local Peres–Schlag pinned-positive-measure lemma above gives the positive-measure pinned distance set.

For $d \geq 5$, put $n = k_* + 1$. The elementary inequalities

$$\frac{d}{2} - \epsilon > \frac{n + 1}{2} \quad (10.45b)$$

hold for all sufficiently small ϵ (for even d , the gap tends to $d/4 - 1/2$; for odd d , to $(d-3)/4$). The local Peres–Schlag pinned-positive-measure lemma above, applied inside $W \cong \mathbb{R}^n$, therefore gives a pinned distance set of positive Lebesgue measure. In every case $|\Delta(X)| > 0$, which is alternative (i).

Assume henceforth that no such $(k_* + 1)$ -plane concentration occurs.

Dimensions 3 and 4. For $d = 3$, $s = 3/2 - \epsilon$ lies in the range (10.43) for $k = 2$, so Theorem III.10.23 gives alternative (ii). For $d = 4$, first perform the generic projection reduction from the proof of that theorem to a 3-space; the projected energy exponent remains $s = 2 - \epsilon$ and the box bounds do not increase. Apply the $k = 2$ case there and lift the resulting thin-tube estimate back by orthogonal projection.

Dimensions $d \geq 5$. Slice by generic affine $(d - k_* + 1)$ -planes L . By Lemmas III.10.17 and III.10.18, almost every nonzero sliced pair has

$$I_{s-k_*+1}(\mu_L), I_{s-k_*+1}(\nu_L) < \infty, \quad \overline{\dim}_B \text{supp } \mu_L, \overline{\dim}_B \text{supp } \nu_L \leq s - k_* + 1 + \epsilon. \quad (10.45c)$$

The slice exponent is

$$s - k_* + 1 = \begin{cases} 2 - \epsilon, & d \text{ even,} \\ 3/2 - \epsilon, & d \text{ odd.} \end{cases} \quad (10.45d)$$

If on a positive family of such slices there is a common affine 2-plane of positive mass, then use (10.45d). When d is even, the planar restriction has dimension at least $2 - \epsilon > 3/2$, so the local Peres–Schlag pinned-positive-measure lemma above gives alternative (i). When d is odd, the available exponent is $3/2 - \epsilon$; for small ϵ it is still $> 5/4$, so the local planar theorem attached to III.10.5 applies. Thus alternative (i) again holds.

Otherwise almost every relevant sliced pair has no common 2-plane. Apply Theorem III.10.23 inside the slice, with internal parameter $k = 2$ and source exponent $s - k_* + 1$. The values in (10.45d) lie in its admissible range for small ϵ , so the slices have

$$(s - k_* + 1 - \omega_d(\epsilon))\text{-thin tubes,} \quad \omega_d(\epsilon) \rightarrow 0. \quad (10.45e)$$

Finally apply Theorem III.10.20 with lifting parameter $k_* - 1$. The exponent recombines exactly as

$$(k_* - 1) + (s - k_* + 1 - \omega_d(\epsilon)) = s - \omega_d(\epsilon) = \frac{d}{2} - \epsilon - \omega_d(\epsilon).$$

Absorb the explicit ϵ and the finitely many trimming/slicing losses into $\zeta_d(\epsilon) = \epsilon + \omega_d(\epsilon) + o(1)$. Then $\zeta_d(\epsilon) \rightarrow 0$ and (10.45) follows. $\square$

Theorem III.10.25 (higher-dimensional slicing/thin-tube block). *Under the hypotheses of Theorem III.10.30 with $d \geq 3$, every sequence $\epsilon_j \downarrow 0$ admits regular separated pieces for which either the distance set already has positive Lebesgue measure, or there are positive-mass restrictions with strong $(d/2 - o(1))$ -thin tubes.*

Proof. Apply Corollary III.10.24. In the second branch use Lemma III.10.7; the exponent is unchanged and only the constants and retained masses change. This is the only strong-fibre upgrade needed later. $\square$

III.10.6 Adapted projections and the endpoint distance engine

We use the [quantitative adaptedness definition](#) above; the loss, exceptional exponent, and uniformity range cannot be omitted when this property is an actual proof premise. The lemma below supplies those parameters from a directional Frostman estimate.

Lemma III.10.26 (finite-scale Falconer adaptedness). *Fix $d \geq 2$ and $0 < q \leq d - 1$. Let ρ be a probability on $\mathbb{S}^{d-1}$ with*

$$\rho(B(\theta, r)) \leq Kr^q, \quad \delta \leq r \leq 1, \quad (10.46)$$

and let μ be a probability on $[0, 1]^d$ with

$$\mu(B(x, r)) \leq \delta^{-\epsilon} r^t, \quad \delta \leq r \leq 1. \quad (10.47)$$

Put $a = d - q$. Outside a set of ρ -mass at most $\delta^{c\epsilon}$, the line projection $P_\theta \mu$ is

$$\begin{cases} (\delta, 1 - C\epsilon, \delta^{c\epsilon})\text{-robust}, & t > a, \\ (\delta, q + t + 1 - d - C\epsilon, \delta^{c\epsilon})\text{-robust}, & t < a. \end{cases} \quad (10.48)$$

The constants are uniform on compact parameter ranges away from $t = a$.

Proof. Smooth both measures at scale δ . The averaged Falconer–Sobolev projection estimate from Chapter III.2 gives

$$\int \|P_\theta \mu\|_2^2 d\rho(\theta) \leq KI_{a-q}^{(\delta)}(\mu). \quad (10.49)$$

Dyadic annuli and (10.47) give

$$I_a^{(\delta)}(\mu) \lesssim \begin{cases} \delta^{-2\epsilon}, & t > a, \\ \delta^{t-a-2\epsilon}, & t < a. \end{cases}$$

Markov removes at most $\delta^{c\epsilon}$ direction mass and gives the corresponding L^2 bounds. If a probability density λ on the line satisfies $\|\lambda\|_2^2 \leq \delta^{-u}$ and $\lambda(A) \geq \delta^\eta$, then Cauchy–Schwarz gives $|A^{(\delta)}| \gtrsim \delta^{2\eta+u}$, so $N_\delta(A) \gtrsim \delta^{-(1-u-2\eta)}$. Choosing η as a small multiple of ϵ yields (10.48). $\square$

Parameter convention. *The pointwise lower regularity and global upper regularity losses are one quarter of the adaptedness loss. The proof below establishes both the local conditional Frostman bound and the simultaneous all-scales entropy assembly from the stated adaptedness premise; it uses no additional research-source interface beyond the earlier proved entropy lemmas.*

Proposition III.10.27 (pointwise-regular scalar nonlinear images). *Let $d \geq 2$, $0 < t \leq d$, $0 < u \leq 1$, $0 < \epsilon < \min\{t, u, 1/16\}$, and let μ be a probability with compact support in $(0, 1)^d \subset U$ satisfying $\mu(Q) \leq C_{\text{reg}} 2^{-k(t-\epsilon/4)}$ for every occupied dyadic k -cube Q with $k \geq k_0$.*

Let $E \subset \text{supp } \mu$ be compact with $\mu(E) > 0$ and suppose $\mu(Q_k(x)) \geq C_{\text{reg}}^{-1} 2^{-k(t+\epsilon/4)}$ for every $x \in E$ and $k \geq k_0$. Let $F_y : U \rightarrow \mathbb{R}$ be a compact nonsingular C^2 family indexed by a compact parameter space carrying a Borel probability ν . Suppose $G \subset E \times \text{supp } \nu$ is compact with $(\mu \times \nu)(G) > 0$. Let $E_{\text{good}} \subset E$ be a Borel set with $\mu(E \setminus E_{\text{good}}) = 0$. For every $x \in E_{\text{good}}$ with $\nu(G_x) > 0$, where $G_x = \{y : (x, y) \in G\}$, assume the derivative direction law $\rho_x = (\text{dir } \nabla_x F_y(x))_{\#} \nu_{G_x}$ is uniformly $(2^{-k}, t - \epsilon \rightarrow u; \epsilon, \gamma, K)$ -adapted at all $k \geq k_0$, for fixed $C_{\text{reg}}, K \geq 1$ and $\gamma > 0$, where $K \geq C_d C_{\text{reg}}^2$ for the dimensional ball-cube overlap constant C_d . There is a dimensional loss constant $C_* > 0$ with the following property. If $u > C_* \sqrt{\epsilon}$, then there are a Borel parameter set Y with $\nu(Y) > 0$, a number $b > 0$, compact sets $Z_y \subset G^y$ with $\mu(Z_y) \geq b$ for every $y \in Y$, and $\eta > 0$ such that, uniformly in $y \in Y$, for all sufficiently large m ,

$$F_y(\mu_{Z_y}) \text{ is } (2^{-m}, u - C_* \sqrt{\epsilon}, 2^{-\eta m})\text{-robust.} \quad (10.50)$$

Consequently $\dim_{\text{H}} F_y(\text{supp } \mu) \geq u - C_* \sqrt{\epsilon}$ for every $y \in Y$. If $u \leq C_* \sqrt{\epsilon}$, the corresponding dimension lower bound is only the trivial nonnegative bound; no robustness with a nonpositive exponent is asserted.

Proof. Assume $u > C_* \sqrt{\epsilon}$, as required for the robustness claim. Put $\tau = \sqrt{\epsilon}$. We follow the simultaneous-scale selection mechanism in Shmerkin–Wang [33, Proposition 4.2], but spell it out in the present scalar notation. By a fibre-mass level selection, choose $c_0 > 0$ so that $\{x \in E_{\text{good}} : \nu(G_x) \geq c_0\}$ has positive μ -mass. The section mass $x \mapsto \nu(G_x)$ is Borel because G is compact. By inner regularity choose a compact positive-mass $X' \subset \{x \in E_{\text{good}} : \nu(G_x) \geq c_0\}$. Retain the *full* original fibres $Y_x = G_x$ for $x \in X'$; the graph $G \cap (X' \times \text{supp } \nu)$ is compact. Thus we may write it as

$$G = \{(x, y) : x \in X', y \in Y_x\},$$

where $X' \subset E$, $\mu(X') \geq c_0$ and $\nu(Y_x) \geq c_0$ uniformly in $x \in X'$, and all adaptedness constants are uniform after decreasing c_0 if needed. Since no fibre is thinned, its assumed adaptedness is preserved. We keep the *original* μ for every dyadic cube conditional: restricting μ to X' need not preserve the lower half of rough regularity, even up to a fixed constant.

Fix an output level M for the moment. Set $J = \lfloor \log_2(1/\tau) \rfloor$ and $m_j = \lceil M/2^{J-j} \rceil$ for $0 \leq j \leq J$; in particular $m_J = M$. Write $m_* = m_0$ for this block construction (so $m_* \geq k_0$ once M is large) and $L_j = m_{j+1} - m_j$. For $0 \leq j < J$ and all sufficiently large M ,

$$m_j/2 \leq L_j \leq m_j, \quad L_j \geq \tau M/2, \quad m_{j+1} \leq 2m_j. \quad (10.50a0)$$

Thus every block, including the terminal block, satisfies the endpoint-ratio hypothesis of Lemma III.6.22; the uncovered initial range has length $m_* = O(\tau M)$. If $x \in E \cap Q \in \mathcal{D}_{m_j}$, the pointwise lower bound for the parent and the global upper bound for its children give, after rescaling Q to unit size,

$$\mu^Q(B(z, r)) \leq K_0 2^{\epsilon L_j} r^{t-\epsilon}, \quad 2^{-L_j} \leq r \leq 1, \quad (10.50a)$$

for M sufficiently large, with K_0 depending only on d, C_{reg} . Indeed, a ball of radius $r \asymp 2^{-l}$ meets $O_d(1)$ dyadic $(m_j + l)$ -cubes, and the two pointwise/global bounds at loss $\sigma = \epsilon/4$ give, after division by the mass of Q ,

$$\mu^Q(B(z, r)) \leq C_d C_{\text{reg}}^2 \cdot 2^{2\sigma m_j} r^{t-\sigma} \leq K_0 \cdot 2^{\epsilon L_j} r^{t-\epsilon}. \quad (10.50a1)$$

The last inequality uses $2\sigma m_j \leq \epsilon L_j$ and $r^{t-\sigma} \leq r^{t-\epsilon}$ for $r \leq 1$. Thus the rescaled conditional source really is admissible for the stated $(2^{-L_j}, t - \epsilon \rightarrow u; \epsilon, \gamma, K)$ premise, because $K \geq K_0$ was stipulated in the hypothesis. For each j and $x \in X'$, let $\text{Bad}_M(x, j) \subset Y_x$ be the *canonical* set of parameters for which the corresponding linear projection of $\mu^{Q(x, m_j)}$ fails the displayed robustness test. Adaptedness gives

$$\nu_{Y_x}(\text{Bad}_M(x, j)) \leq 2^{-\gamma L_j}, \quad (10.50b)$$

and, whenever $y \in Y_x \setminus \text{Bad}_M(x, j)$, the linearized projection of the conditional source is

$$(2^{-L_j}, u - \epsilon, 2^{-\gamma L_j})\text{-robust}. \quad (10.50c)$$

These bad relations are Borel. Indeed, at a fixed dyadic scale a projection of a measure supported in a fixed bounded interval is not (δ, a, p) -robust exactly when some union of at most $\lfloor \delta^{-a} \rfloor$ terminal dyadic intervals has mass greater than p . There are finitely many relevant unions, and each push-forward mass is a Borel function of (x, y) by the measurability of the kernel and continuity of the derivative family. Thus the bad set can be used in Fubini directly; no compact-superset assertion is needed.

Let $\text{Bad}_M(x) = \bigcup_{j < J} \text{Bad}_M(x, j)$. Since $L_j \geq m_*$ and the number of blocks depends only on τ ,

$$\nu_{Y_x}(\text{Bad}_M(x)) \leq C_\tau 2^{-\gamma m_*} \leq C_\tau 2^{-\gamma \tau M/2}. \quad (10.50d)$$

Now take the union over all $M \geq M_0$:

$$\text{Bad}(x) = \bigcup_{M \geq M_0} \text{Bad}_M(x).$$

Choosing M_0 large enough gives $\nu_{Y_x}(\text{Bad}(x)) \leq c_0/2$ uniformly in x . Hence

$$G_* = \{(x, y) \in G : y \notin \text{Bad}(x)\}$$

has positive $(\mu \times \nu)$ -mass. By inner regularity of the product measure choose a compact $H \subset G_*$ of positive product mass. Fubini and fibre-mass pigeonholing give $b > 0$ and a Borel $Y \subset \text{supp } \nu$ of positive ν -mass with $\mu(H^y) \geq b$ for every $y \in Y$. Put $Z_y = H^y$; these fibres are compact. Every pair (y, Z_y) is good for *every* output level $M \geq M_0$; this is the simultaneous-scale selection, uniform over a positive-measure parameter set.

Fix $y \in Y$ and such an M , and let $Z' \subset Z_y$ satisfy $\mu_{Z_y}(Z') \geq 2^{-\eta M}$, where $\eta > 0$ is chosen with

$$0 < \eta < \min\{\gamma\tau/16, \tau^2/16\}. \quad (10.50e)$$

Put $c_y = \mu(Z_y) \geq b$. Since $L_j \geq \tau M/2$, for all large M and every block we have the explicit threshold comparison

$$\frac{C_\tau M}{\mu(Z_y)\mu_{Z_y}(Z')} \leq C_\tau M b^{-1} 2^{\eta M} \leq 2^{\gamma L_j/4}. \quad (10.50f)$$

Set $\lambda = (\mu_{Z_y})_{Z'} = \mu|_{Z'}/\mu(Z')$. Then $\lambda \leq \Delta\mu$ with $\Delta = 1/\mu(Z')$. Apply the smooth-image entropy linearisation III.6.22 directly to μ, λ on the blocks $[m_j, m_{j+1}]$, choosing its linearisation point x_Q inside $Z' \cap Q$ whenever $\lambda(Q) > 0$. This is permitted because the lemma

allows any point of Q , and $Z' \subset Z_y$. The robust input in that lemma is $H_{L_j}^{M\Delta}(P_{V(x_Q)}\mu^Q)$. By (10.50f), $M\Delta \leq 2^{Y_{L_j}/4} \leq 2^{Y_{L_j}/2}$ for all large M . Since $y \notin \text{Bad}_M(x_Q, j)$, (10.50c) and the robust-entropy Lemma III.6.4 yield

$$H_{L_j}^{M\Delta}(P_{V(x_Q)}\mu^Q) \geq (u - \epsilon - o(1))L_j$$

uniformly in all parents carrying λ -mass. Consequently

$$H_M((F_y)_*\lambda) \geq (u - C\tau) \sum_{j < J} (m_{j+1} - m_j) - O(\tau M) - O(\tau^{-1}) \geq (u - C'\sqrt{\epsilon})M$$

for all sufficiently large M . Entropy is bounded above by the logarithm of the number of occupied dyadic intervals, so

$$N_{2^{-M}}(F_y Z') \geq 2^{(u - C'\sqrt{\epsilon})M}.$$

This is (10.50), after renaming the constant. Finally apply the robustness-to-dimension lemma III.6.3. $\square$

Proposition III.10.28 (strong thin tubes give adapted distance directions). *Let μ, ν have separated compact supports and strong (q, K, c) -thin tubes at all sufficiently small scales. For $F_y(x) = |x - y|$, the derivative direction law on the strong incidence graph is q -Frostman. Fix $0 < t < d - q$ and a sufficiently small $\beta > 0$. For every fixed source constant $K_* \geq 1$ there are $A = A_d > 0$, $\gamma = \gamma(d, q, t, \beta) > 0$, and a scale $\delta_0 > 0$ such that, whenever $\delta \leq \delta_0$ is dyadic and $u = q + t + 1 - d - A\beta$ belongs to $(\beta, 1]$, these laws are uniformly*

$$(\delta, t \rightarrow u; \beta, \gamma, K_*)\text{-adapted.} \quad (10.51)$$

Proof. The derivative direction is $\text{dir}(x - y)$. Since the supports are separated, the inverse image of a spherical cap of radius r is contained in a tube through x of width $O(r)$. Strong thin tubes therefore give $\rho_x(B(\theta, r)) \leq K_1 r^q$ with K_1 uniform over the retained centres. Take any test source ω allowed by the [quantitative adaptedness definition](#); it obeys $\omega(B(z, r)) \leq K_* \delta^{-\beta} r^t$. For sufficiently small δ the fixed factor K_* is at most $\delta^{-\beta}$, so the source obeys the hypothesis of Lemma III.10.26 with 2β in place of its ϵ . That lemma gives, outside ρ_x -mass at most $\delta^{2c_1\beta}$, robustness with exponent $q + t + 1 - d - 2C_1\beta$ and mass threshold $\delta^{2c_1\beta}$. Set $A = 2C_1 - 1$ and $\gamma = c_1\beta$. Then the exponent is $u - \beta$, and both the exceptional mass and the robustness threshold are at most δ^γ . These choices are uniform over the compact retained graph. The exponent-loss coefficient C_1 can be chosen dimensionally: in the dyadic-annulus proof of Lemma III.10.26, a factor depending on the positive gap $d - q - t$ is absorbed into an additional $\delta^{-\beta}$ once δ is sufficiently small. The starting scale may depend on that gap, but the exponent-loss coefficient does not. $\square$

Theorem III.10.29 (endpoint adapted-distance engine). *Let $d \geq 3$, $\epsilon > 0$, $\zeta \geq 0$, and $t_0 = d/2$. Suppose X contains separated compact positive-mass pieces carrying probability measures μ, ν such that μ is $(t_0 - \epsilon)$ -Frostman, $\dim_B \text{supp } \mu \leq t_0 + \epsilon$, and, after positive-mass restriction, (μ, ν) has strong*

$$(t_0 - \zeta, K, c)\text{-thin tubes.} \quad (10.52)$$

Then some $y \in \text{supp } \nu$ satisfies

$$\dim_{\text{H}} \Delta^y(X) \geq 1 - \zeta - C_d \sqrt{\epsilon}. \quad (10.53)$$

Proof. Apply one fixed similarity to both separated compact pieces so that the source support lies compactly inside $(0, 1)^d$. This changes only Frostman and thin-tube constants, preserves all exponents and separation, and leaves pinned-distance Hausdorff dimension unchanged after rescaling the distance variable. Let A_d and C_* be the loss constants in Propositions III.10.28 and III.10.27, respectively, and choose the dimensional constant $C_d \geq 2C_*\sqrt{10}$ in (10.53) large enough to handle the following split. If $1 - \zeta - C_d\sqrt{\epsilon} \leq 0$, the conclusion is trivial. For ϵ above a fixed dimensional threshold, enlarge C_d so that this trivial case applies. We may therefore assume $0 \leq \zeta < 1$, $\lambda := 10\epsilon < 1/16$, and $(A_d + 2)\lambda \leq C_d\sqrt{\epsilon}/2$; in particular the target $u = 1 - \zeta - (A_d + 1)\lambda$ below satisfies $u > \max\{\lambda, C_*\sqrt{\lambda}\}$ whenever the desired lower bound is nontrivial. Apply Lemma III.10.8 with parameter λ to the *original* μ . Choose its loss parameter q_0 smaller than half the mass of the strong incidence graph, and retain a compact positive-mass subgraph over the resulting regular core E . Thus the global upper cube bound and the lower bound along E hold with centre t_0 and loss $\lambda/4$.

Put $q = t_0 - \zeta$ and $t' = t_0 - \lambda$. Then $t' < d - q = t_0 + \zeta$. Proposition III.10.28, with $\beta = \lambda$, supplies the quantified law

$$(2^{-k}, t' \rightarrow u; \lambda, \gamma, K_*)\text{-adapted}, \quad u = q + t' + 1 - d - A_d\lambda = 1 - \zeta - (A_d + 1)\lambda$$

for all sufficiently large k , where the fixed K_* is chosen to meet the source constant in Proposition III.10.27. Apply that proposition with regularity centre t_0 and core E to the distance family. The resulting pinned distance measure has Hausdorff dimension at least $1 - \zeta - (A_d + 1)\lambda - C_*\sqrt{\lambda}$, which is (10.53) after absorbing the linear error. $\square$

The preceding three mechanisms now meet at the critical dimension.

Proof status. *The planar adaptedness gain and the planar endpoint decoupling estimate used below are both proved locally.*

Theorem III.10.30 (Shmerkin–Wang $d/2$ distance theorem). *Let $d \geq 2$, and let $X \subset \mathbb{R}^d$ be Borel. If*

$$\dim_{\text{H}} X = \dim_{\text{P}} X = \frac{d}{2}, \quad (10.11)$$

then

$$\dim_{\text{H}} \Delta(X) = 1. \quad (10.12)$$

Proof. If $d = 2$, Theorem III.10.16 with $t = 1$ gives, for every $\eta > 0$, a centre $y \in X$ with $\dim_{\text{H}} \Delta^y(X) \geq 1 - \eta$. Lemma III.10.5 finishes the proof.

Assume $d \geq 3$ and choose $\epsilon_j \downarrow 0$. Apply Lemma III.10.4 with parameter $\epsilon_j/2$. The resulting measures are $(d/2 - \epsilon_j/2)$ -Frostman, so the dyadic energy sum gives finite $(d/2 - \epsilon_j)$ -energy; their upper box dimensions are at most $d/2 + \epsilon_j/2$. Apply Theorem III.10.25. If the positive-measure branch occurs for some j , then $|\Delta(X)| > 0$ and hence $\dim_{\text{H}} \Delta(X) = 1$.

Otherwise, after positive-mass restriction, the measures have strong $(d/2 - \zeta_j)$ -thin tubes with $\zeta_j \rightarrow 0$. Theorem III.10.29 gives centres $y_j \in X$ such that

$$\dim_{\mathrm{H}} \Delta^{y_j}(X) \geq 1 - \zeta_j - C_d \sqrt{\epsilon_j}. \quad (10.54)$$

Since every pinned distance set is contained in $\Delta(X)$, the right side tends to one and (10.12) follows. $\square$

III.10.7 Scope and consequences

The conclusion of Theorem III.10.30 is a dimension-one statement. It does not assert positive Lebesgue measure at the critical endpoint. It also requires equal Hausdorff and packing dimension, not merely $\dim_{\mathrm{H}} X \geq d/2$, and nowhere is this equality replaced by Ahlfors regularity.

Theorem 1.4 of Shmerkin–Wang [33] proves the stronger source statement that for every $\eta > 0$ the set of centres

$$\{y \in X : \dim_{\mathrm{H}} \Delta^y(X) < 1 - \eta\}$$

has Hausdorff dimension strictly smaller than $\dim_{\mathrm{H}} X$; when $\mathcal{H}^{d/2}(X) > 0$, the full pinned supremum is realised at almost every centre. These stronger conclusions require additional arguments beyond the unpinned theorem proved here.

Three mechanisms drive the endpoint deduction: the planar thin-tube bootstrap III.10.4, the higher-dimensional slicing block III.10.5, and the adapted-distance block III.10.6. Together with the locally proved endpoint decoupling estimate, they yield Theorem III.10.30; the classical Mattila criterion III.10.2 remains a separate harmonic-analytic baseline rather than a substitute for the endpoint proof.

Chapter III.11

Tubes, Heavy Lines, and Visible Parts

Three questions that at first look rather different are governed by the same geometric currency: how a set distributes across families of affine lines. The first asks whether dynamically defined sets can be covered cheaply by thin tubes. The second asks how much of a planar set can lie on lines whose intersections with the set are larger than the generic Marstrand slice. The third asks how large the part of a compact planar set visible from a given direction can be.

The chapter develops all three themes to sharp reader-level statements. For closed proper $\times N$ -invariant sets we obtain a finite rational-direction decomposition and the resulting tube-nullity theorem of Pyörälä–Shmerkin–Suomala–Wu [42]. For heavy lines we prove the Dąbrowski–Orponen–Wang upper theorem from the sharp planar Furstenberg estimate proved in the preceding chapters and reconstruct the matching all-directions examples after proving the finite double-scale block of their Appendix A locally from the explicit sheared-grid/Farey construction [43]. Finally, we prove the sharp almost-everywhere visible-part bound and reconstruct the matching all-directions $3/2$ example after proving the Orponen–Rutar basic visibility block locally from the same sheared-grid/Farey geometry. In the upper branch, we prove the complete phase-space ray-incidence engine locally: the ray high–low theorem, uniformisation, branching functions, the effective-triple criterion, and the branching-function theorem. Thus the Orponen–Rutar discretised visibility theorem is proved here in a normalisation calibrated against [45, 44].

A recurring principle is that a finite-scale research input should remain visible at exactly the point where it is used. The continuous heavy-line and visibility constructions are therefore proved here rather than imported as headline theorems.

Prerequisites and source boundary. The chapter uses the affine-line metric and point–line duality from Chapter III.7, the locally proved low-parameter Furstenberg branch and the locally proved high-parameter Fu–Ren branch, and the standard Frostman, slicing, elementary Shannon-entropy, and symbolic results from the foundation volumes. The almost-AD branch of Ren–Wang Theorem 6.1 is proved below from the Chapter III.9 regular-set Furstenberg estimate (9.46), together with a local almost-AD-to-regular conversion and a finite compactness argument in the intermediate exponent. Thus the middle branch $s < u < 2 - s$ uses the locally proved regular-set theorem; the low branch $s + u$ is proved in Chapter 6 and the high branch $s + 1$ is proved in Chapter 7. The Orponen–Rutar discretised visibility theo-

rem is proved here from simultaneous uniformisation, the branching-function compactness calculus, the ray initial estimate, the ray high–low inequality, the effective-triple criterion, the branching-function theorem, the polynomial-constant upgrade, and the visibility reduction are all proved locally below. The Dąbrowski–Orponen–Wang finite double-scale block and the Orponen–Rutar basic visibility block are now proved locally from explicit sheared-grid/Farey constructions. The WSC entropy package and the Bowen empirical-accumulation estimate used in the tube-null branch are proved locally below. The continuous sharpness constructions and all parameter conversions are proved here. The Assouad-visible theorem of Järvenpää–Järvenpää–Suomala–Wu is included only for comparison and is not used below.

III.11.1 Tubes, covers, and invariant sets

Definition (tube and width). A tube in $\mathbb{R}^d$ is the r -neighbourhood of an affine line. Its width is r . Changing the width by a fixed multiplicative constant does not affect any nullity or dimension statement below.

Definition (tube-nullity). A set $E \subset \mathbb{R}^d$ is tube-null if, for every $\epsilon > 0$, there are tubes T_i such that

$$E \subset \bigcup_i T_i, \quad \sum_i w(T_i)^{d-1} < \epsilon. \quad (11.1)$$

The exponent $d - 1$ is the codimension-one volume scaling of a unit-length tube.

Definition (quantitative hyperplane-cover exponent). For $0 < c < 1$, a bounded set $E \subset \mathbb{R}^d$ admits a c -hyperplane cover if for every $\epsilon > 0$ there are affine hyperplanes H_i and $r_i > 0$ with

$$E \subset \bigcup_i H_i^{(r_i)}, \quad \sum_i r_i^c < \epsilon. \quad (11.2)$$

The hyperplane estimate is stronger than tube-nullity: a bounded r -neighbourhood of a hyperplane is coverable by $O_d(r^{2-d})$ line tubes of width $O(r)$.

Definition (T_N -invariance). On the torus $[0, 1]^d$ write

$$T_N(x) = Nx \pmod{1}. \quad (11.3)$$

A closed set K is T_N -invariant if $T_N(K) \subset K$.

III.11.2 Invariant sets, low-dimensional projections, and tube-nullity

For a unit vector v , let P_v denote orthogonal projection onto the line spanned by v . The next four results are formulated for a proper homogeneous N -adic attractor generated by a finite digit alphabet Γ . This is not a restriction on the final theorem: its proof first enlarges an arbitrary proper closed T_N -invariant set to such an attractor and then restricts the resulting cover back to the original set. In this homogeneous setting the natural coding $\pi : \Gamma^{\mathbb{N}} \rightarrow K$ is a factor of the full shift.

Lemma III.11.1 (finite rational direction detection). *Let $K \subseteq [0, 1]^d$ be a proper homogeneous N -adic attractor. There is a finite set V of rational directions such that every T_N -invariant probability measure μ supported on K has some $v \in V$ for which $P_v\mu$ is not absolutely continuous.*

Proof. Let $\mathcal{M}_K$ be the compact set of T_N -invariant probabilities supported on K . Since K is a proper closed subset, no element of $\mathcal{M}_K$ is Lebesgue measure on the torus. Hence for each $\mu \in \mathcal{M}_K$ there is a nonzero $k(\mu) \in \mathbb{Z}^d$ with $\hat{\mu}(k(\mu)) \neq 0$. The maps $v \mapsto \hat{\mu}(k)$ are weak- $*$ continuous, so compactness gives finitely many nonzero integer vectors $k_1, \dots, k_M$ such that every $\mu \in \mathcal{M}_K$ has $\hat{\mu}(k_j) \neq 0$ for at least one j . Let V be the set of directions spanned by these vectors.

Fix such a k_j . Invariance gives

$$\hat{\mu}(N^n k_j) = \hat{\mu}(k_j), \quad n \geq 0, \quad (11.10)$$

because the integer correction in T_N has trivial integer Fourier phase. If $P_{k_j}\mu$ were absolutely continuous, Riemann–Lebesgue would force its Fourier coefficients at frequencies $N^n|k_j|$ to tend to zero, contradicting (11.10). $\square$

Lemma III.11.2 (rational projected WSC). *For every rational direction v arising in Lemma III.11.1, the projected homogeneous N -adic iterated-function system satisfies the weak separation condition.*

Proof. Scale v so that it has primitive integer coordinates. There is a fixed denominator Q_v such that every level- n projected cylinder translation lies in $N^{-n}Q_v^{-1}\mathbb{Z}$. Consequently two level- n cylinder maps either coincide exactly or their translations differ by at least $Q_v^{-1}N^{-n}$. This is the weak separation condition for a homogeneous system of ratio N^{-1} . $\square$

Proposition (WSC entropy package). *Let $\mathcal{F} = \{f_i(x) = N^{-1}x + t_i\}_{i \in \Gamma}$ be a homogeneous one-dimensional IFS satisfying WSC, let π be its coding map, and let ν be shift-invariant. For a compactly supported probability λ on $\mathbb{R}$, write*

$$\underline{\dim}_e \lambda = \liminf_{n \rightarrow \infty} \frac{H(\lambda, \mathcal{D}_{N^{-n}})}{n \log N},$$

where $\mathcal{D}_{N^{-n}}$ is the grid partition into intervals of length N^{-n} . Then

$$\pi\nu \ll \mathcal{L}^1 \iff \underline{\dim}_e(\pi\nu) = 1,$$

and the map $\nu \mapsto \underline{\dim}_e(\pi\nu)$ is upper semicontinuous on the compact space of shift-invariant probabilities.

Proof. We give the finite-partition argument because it is exactly what is needed later. For each n , let A_n be the set of distinct level- n translations $f_w(0)$, and partition the symbolic space by

$$\mathcal{P}_n = \left\{ \bigcup_{f_w(0)=a} [w] : a \in A_n \right\}.$$

Put $a_n(v) = H(v, \mathcal{P}_n)$. WSC says that distinct points of A_n are separated by at least cN^{-n} . Since a level- n cylinder image has diameter $O(N^{-n})$, there is a constant M , independent of n and v , such that each atom of $\mathcal{P}_n$ meets at most M atoms of $\pi^{-1}\mathcal{D}_{N^{-n}}$, and conversely. The elementary bounded-incidence entropy comparison therefore gives

$$|H(\pi v, \mathcal{D}_{N^{-n}}) - a_n(v)| \leq \log M. \quad (11.11a)$$

Moreover $\mathcal{P}_n \vee \sigma^{-n}\mathcal{P}_m$ refines $\mathcal{P}_{n+m}$: the two translations belonging to the first n symbols and the following m symbols determine the translation of the concatenated word. Shift invariance hence gives

$$a_{n+m}(v) \leq a_n(v) + a_m(v). \quad (11.11b)$$

Fekete's lemma and (11.11a) yield

$$\underline{\dim}_e(\pi v) = \frac{1}{\log N} \inf_{n \geq 1} \frac{a_n(v)}{n}. \quad (11.11c)$$

Each $\mathcal{P}_n$ is a finite clopen partition, so a_n is weak- $*$ continuous. Formula (11.11c), as an infimum of continuous functions, proves upper semicontinuity.

Suppose next that $\underline{\dim}_e(\pi v) = 1$. The right side of (11.11c) then forces $a_n(v) \geq n \log N$ for every n , and hence

$$H(\pi v, \mathcal{D}_{N^{-n}}) \geq n \log N - O(1). \quad (11.11d)$$

If πv were not absolutely continuous, there would be a compact Lebesgue-null set E with $q = (\pi v)(E) > 0$. Given $\eta > 0$, for some arbitrarily fine scale n the set E is contained in a union F of at most ηN^n grid intervals. Splitting the measure into its normalized restrictions to F and F^c , the entropy-of-a-mixture inequality gives

$$H(\pi v, \mathcal{D}_{N^{-n}}) \leq n \log N + (\pi v)(F) \log \eta + O(1).$$

Since $(\pi v)(F) \geq q$ and η is arbitrarily small, this contradicts (11.11d). Thus full entropy dimension implies absolute continuity.

Conversely, assume $\lambda = \pi v \ll \mathcal{L}^1$. Given $\varepsilon > 0$, choose a Borel set G with $\lambda(G) > 1 - \varepsilon$ on which the density of λ is bounded by some constant C . The normalized restriction λ_G assigns at most $C'N^{-n}$ to every level- n grid interval, and therefore $H(\lambda_G, \mathcal{D}_{N^{-n}}) \geq n \log N - O(1)$. Concavity of entropy under the mixture $\lambda = \lambda(G)\lambda_G + (1 - \lambda(G))\lambda_{G^c}$ gives $\underline{\dim}_e \lambda \geq 1 - \varepsilon$. Letting $\varepsilon \downarrow 0$, and using the trivial upper bound $\underline{\dim}_e \lambda \leq 1$ for compactly supported measures on $\mathbb{R}$, proves the converse. $\square$

Proposition III.11.3 (uniform projected entropy-dimension gap). *There exists $\delta_0 = \delta_0(K) > 0$ such that for every $m \geq 1$ and every T_N^m -invariant probability μ supported on K , some $v \in V$ satisfies*

$$\underline{\dim}_e P_v \mu \leq 1 - \delta_0. \quad (11.11)$$

Proof. First take $m = 1$. If there were no uniform gap, compactness of $\mathcal{M}_K$ would give a sequence $\mu_j \rightarrow \mu$ weak- $*$ such that $\underline{\dim}_e P_v \mu_j \rightarrow 1$ for every $v \in V$. Every invariant geometric measure on this full-shift attractor has a shift-invariant symbolic lift: start from any lift and average its first n shift iterates; any weak- $*$ accumulation point is invariant

and still projects to the original measure. Apply this to the convergent sequence and, after passing to a subsequence in the compact symbolic measure space, obtain invariant lifts converging to a lift of μ . The local WSC entropy package and Lemma III.11.2 then imply $\underline{\dim}_e P_v \mu = 1$ for every $v \in V$, hence every $P_v \mu$ is absolutely continuous. This contradicts Lemma III.11.1. Thus a common $\delta_0 > 0$ exists for T_N -invariant measures.

Now let μ be T_N^m -invariant and put

$$\bar{\mu} = \frac{1}{m} \sum_{j=0}^{m-1} (T_N^j)_\# \mu. \quad (11.12)$$

Then $\bar{\mu}$ is T_N -invariant. Fix v . On each of the finitely many N^{-j} -adic cells on which T_N^j is affine, $P_v \circ T_N^j$ is a translate of $N^j P_v$. If $P_v \mu$ is decomposed according to these cells, the entropy of the mixture differs by at most the entropy of the finite cell index from the weighted sum of the component entropies. Translation changes a fixed grid entropy by $O(1)$, and multiplication by N^j only shifts the grid level by j . Consequently

$$\underline{\dim}_e P_v((T_N^j)_\# \mu) = \underline{\dim}_e P_v \mu. \quad (11.13)$$

Concavity of partition entropy under the finite mixture in (11.12) then gives

$$\underline{\dim}_e P_v \bar{\mu} \geq \frac{1}{m} \sum_{j=0}^{m-1} \underline{\dim}_e P_v((T_N^j)_\# \mu) = \underline{\dim}_e P_v \mu. \quad (11.14)$$

Choose $v \in V$ for $\bar{\mu}$ from the $m = 1$ case. The preceding inequality proves (11.11). $\square$

Lemma III.11.4 (high-level entropy transfer). *There exists m_0 such that for every $m \geq m_0$, every $v \in V$, and every shift-invariant probability θ on the level- m symbolic alphabet,*

$$\underline{\dim}_e P_v(\pi\theta) \leq 1 - \delta_0 \implies \frac{h(\Pi_v \theta)}{m \log N} \leq 1 - \frac{\delta_0}{2}. \quad (11.15)$$

Here Π_v identifies exact projected overlaps at level m .

Proof. After quotienting exact level- m overlaps, distinct first-level translations of the projected IFS are separated by $\geq_v N^{-m}$, while every first-level image of the convex hull has length $O_v(N^{-m})$. Hence there is an integer M_v , independent of m , such that each first-level image meets at most M_v other first-level images. By concatenation, a level- q image meets at most M_v^q level- q images.

Let $\rho = \Pi_v \theta$. Compare the length- q cylinder partition with a geometric grid of mesh comparable with N^{-mq} . Each cylinder image meets $O_v(1)$ grid intervals and each grid interval meets $O_v(M_v^q)$ cylinder images. The bounded-incidence entropy comparison therefore gives

$$H(P_v(\pi\theta), \mathcal{D}_{N^{-mq}}) \geq H(\rho, \mathcal{C}_q) - q \log M_v - O_v(1), \quad (11.16)$$

where $\mathcal{C}_q$ is the length- q cylinder partition. Divide by $qm \log N$ and let $q \rightarrow \infty$. Since $q^{-1} H(\rho, \mathcal{C}_q) \rightarrow h(\rho)$,

$$\underline{\dim}_e P_v(\pi\theta) \geq \frac{h(\Pi_v \theta)}{m \log N} - \frac{\log M_v}{m \log N}.$$

The set V is finite. Choose m_0 so that $\max_{v \in V} \log M_v / (m \log N) \leq \delta_0/2$ for $m \geq m_0$, and rearrange. $\square$

Proposition (empirical-accumulation entropy bound). *Let $\Sigma = \Gamma^{\mathbb{N}}$ be a full shift over a finite alphabet, and let $V(i)$ be the set of weak- * accumulation points of the empirical measures $n^{-1} \sum_{j=0}^{n-1} \delta_{\sigma^j i}$. For every $a \geq 0$,*

$$h_{\text{top}}\{i : \exists \mu \in V(i) \text{ with } h(\mu, \sigma) \leq a\} \leq a. \quad (11.17)$$

Proof. Let $\mathcal{C}_k$ be the partition into k -cylinders and fix $b > a$. For $n, k \geq 1$ set

$$F_{n,k}(b) = \left\{ i : \frac{1}{k} H\left(\frac{1}{n} \sum_{j=0}^{n-1} \delta_{\sigma^j i}, \mathcal{C}_k\right) \leq b \right\}.$$

We first count prefixes meeting $F_{n,k}(b)$. The empirical distribution of the k -blocks starting at positions $0, \dots, n-1$ is the weighted average of the k empirical block distributions obtained by fixing the starting position modulo k . Entropy is concave, so if the first distribution has entropy at most kb , at least one residue-class distribution does also. For a fixed residue class there are $m = n/k + O(1)$ disjoint k -blocks. The number of possible empirical types is polynomial in n , and a type p contributes at most $\exp(mH(p))$ block strings. The $O(k)$ boundary symbols contribute only a constant depending on k . Thus the number of length- $n+k$ cylinders meeting $F_{n,k}(b)$ is at most

$$C_k(n+1)^{|\Gamma|^k} \exp(bn + O_k(1)). \quad (11.17a)$$

For every $s > b$, the sum of e^{-sn} over these covers is therefore summable in n . The cylinder Carathéodory definition of Bowen entropy gives

$$h_{\text{top}}\left(\limsup_{n \rightarrow \infty} F_{n,k}(b)\right) \leq b. \quad (11.17b)$$

Now suppose i has an empirical accumulation measure μ with $h(\mu, \sigma) \leq a$. For the full shift, $h(\mu, \sigma) = \inf_k k^{-1} H(\mu, \mathcal{C}_k)$, so some k has $k^{-1} H(\mu, \mathcal{C}_k) < b$. Along an empirical subsequence converging to μ , the masses of the finitely many clopen k -cylinders, and hence their entropy, converge. Therefore $i \in \limsup_n F_{n,k}(b)$. The target set in (11.17) is contained in the countable union of these limsup sets over k . Bowen entropy is countably stable, so it is at most b . Let $b \downarrow a$. $\square$

Theorem III.11.5 (finite-direction projection decomposition). *Let $K \subsetneq [0, 1]^d$ be closed and T_N -invariant. There are a finite set V of rational directions, a number $c < 1$, and sets $E_v \subset K$ such that*

$$K = \bigcup_{v \in V} E_v, \quad \dim_{\text{H}} P_v(E_v) < c \quad (v \in V). \quad (11.18)$$

Proof. We first make a finite-type enlargement. Since K is proper and closed, some N -adic cube of some generation q misses K . Let $\mathcal{A}$ be the set of length- q words that occur as initial blocks of codings of points of K , and let K' be the homogeneous N^q -adic attractor obtained by allowing the blocks in $\mathcal{A}$ independently. Invariance ensures that every successive q -block of a coding of a point of K lies in $\mathcal{A}$, hence

$$K \subset K' \subseteq [0, 1]^d. \quad (11.19)$$

A decomposition of K' restricts to one of K , so we replace (N, K) by (N^q, K') .

Choose $m \geq m_0$ from Lemma III.11.4, and let Σ be the full symbolic space for the level- m alphabet. For $i \in \Sigma$, write $V(i)$ for the nonempty compact set of empirical accumulation measures. Every $\theta \in V(i)$ is shift-invariant and geometrically codes a T_N^m -invariant measure.

For $v \in V$ put

$$B_v = \{\theta : \underline{\dim}_e P_v(\pi\theta) \leq 1 - \delta_0\}, \quad D_v = \{i : V(i) \cap B_v \neq \emptyset\}. \quad (11.20)$$

Proposition III.11.3 implies that the B_v cover the invariant-measure simplex, hence the D_v cover Σ .

If $i \in D_v$, choose $\theta \in V(i) \cap B_v$. The factor map Π_v is continuous and commutes with the shift, so $\Pi_v\theta$ is an empirical accumulation measure of $\Pi_v(i)$. Lemma III.11.4 gives

$$h(\Pi_v\theta, \sigma) \leq (1 - \delta_0/2)m \log N. \quad (11.21)$$

The locally proved empirical-accumulation estimate (11.17) therefore yields

$$h_{\text{top}}(\Pi_v D_v) \leq (1 - \delta_0/2)m \log N. \quad (11.22)$$

The level- m projected system has contraction ratio N^{-m} . For completeness, a cylinder of length q has image diameter $O(N^{-mq})$; hence any Bowen cylinder cover with cost $\sum_w e^{-s|w|}$ pushes forward to a Hausdorff cover with cost comparable with $\sum_w (N^{-m|w|})^s / (m \log N)$. Thus the coding entropy–dimension inequality is local and gives

$$\dim_H P_v(\pi D_v) \leq 1 - \delta_0/2. \quad (11.23)$$

Set $E_v = K \cap \pi(D_v)$ and $c = 1 - \delta_0/4$. The E_v cover K and (11.23) gives the strict inequality in (11.18). $\square$

Corollary III.11.6 (quantitative hyperplane cover and tube-nullity). *Assume $d \geq 2$. Under the hypotheses of Theorem III.11.5, for the same $c < 1$ and every $\epsilon > 0$ there are affine hyperplanes H_i and radii $r_i > 0$ such that*

$$K \subset \bigcup_i H_i^{(r_i)}, \quad \sum_i r_i^c < \epsilon. \quad (11.24)$$

Consequently K is tube-null.

Proof. For each $v \in V$, cover $P_v(E_v)$ by intervals $I_{v,j}$ of radii $r_{v,j} < 1$ with $\sum_j r_{v,j}^c < \epsilon/|V|$. The inverse image of $I_{v,j}$ is a slab about the affine hyperplane perpendicular to v , of thickness comparable with $r_{v,j}$. This proves (11.24) after an inessential change of constants.

Inside a fixed bounded region, an r -slab is coverable by $O_d(r^{2-d})$ line tubes of width $O(r)$. Its contribution to the tube-null cost is therefore

$$O_d(r^{2-d}r^{d-1}) = O_d(r).$$

Since $0 < r < 1$ and $c < 1$, $r \leq r^c$. Summing the tube costs and first choosing the slab content arbitrarily small proves tube-nullity. $\square$

III.11.3 Heavy lines: the sharp generic upper bound

The heavy-line problem is a direct application of the sharp planar Furstenberg theorem once the dimension of the family of heavy lines is related to the dimension of the heavy part. We isolate that relation because it is the only line-space bookkeeping needed in the upper theorem.

Definition (heavy lines and heavy parts). *Let $A \subset \mathbb{R}^2$ with $\dim_H A = t \in [1, 2]$, let $e \in \mathbb{S}^1$, and let $s > t - 1$. A line $\ell \parallel e$ is s -heavy for A if $\dim_H(A \cap \ell) \geq s$. The corresponding heavy part is*

$$H(A, e, s) = A \cap \bigcup_{\substack{\ell \parallel e \\ \dim_H(A \cap \ell) \geq s}} \ell, \quad (11.4)$$

and its generic profile is

$$\mathfrak{h}(A, s) = \operatorname{ess\,sup}_{e \in \mathbb{S}^1} \dim_H H(A, e, s). \quad (11.5)$$

The full heavy part is $H(A, e) = \bigcup_{s > t-1} H(A, e, s)$, with $\mathfrak{h}(A) = \operatorname{ess\,sup}_e \dim_H H(A, e)$.

The parameter space of lines will be used as a geometric object in its own right. In a bounded nonvertical chart we identify $y = ax + b$ with (a, b) . Point-line duality sends (x, y) to

$$D(x, y) = \{(a, b) : b = y - ax\}. \quad (11.9)$$

These identifications are bi-Lipschitz, with uniform constants on bounded charts; this is the line-space metric fixed in Chapter III.7.

Lemma III.11.7 (from a heavy part to a line family). *Let $A \subset \mathbb{R}^2$, fix $s \in [0, 1]$, and let $E \subset \mathbb{S}^1$ be a Borel set of directions. Suppose that*

$$\dim_H H(A, e, s) > h' \quad (e \in E), \quad (11.25)$$

with $h' > 1$. Then the family

$$\mathcal{L}_{E,s}(A) = \{\ell : \ell \parallel e \text{ for some } e \in E, \dim_H(A \cap \ell) \geq s\}$$

satisfies

$$\dim_H \mathcal{L}_{E,s}(A) \geq \dim_H E + h' - 1. \quad (11.26)$$

In particular, if $\dim_H E = 1$, then $\dim_H \mathcal{L}_{E,s}(A) \geq h'$.

Proof. Work in finitely many bounded nonvertical line charts. For a fixed direction e , projection onto the orthogonal intercept coordinate maps $H(A, e, s)$ onto the set B_e of intercepts of s -heavy lines. Every fibre of this projection is contained in one affine line and therefore has Hausdorff dimension at most 1. The standard Hausdorff fibre inequality gives

$$\dim_H B_e \geq \dim_H H(A, e, s) - 1 > h' - 1. \quad (11.27)$$

In line coordinates the family $\mathcal{L}_{E,s}(A)$ has direction projection containing E and its fibre over e contains B_e . Applying the same fibre inequality in the reverse direction, or equivalently the standard slicing lower bound for analytic subsets of a product chart, gives

$$\dim_H \mathcal{L}_{E,s}(A) \geq \dim_H E + \inf_{e \in E} \dim_H B_e,$$

which is (11.26). Passing to compact/analytic subsets if necessary uses the usual capacity interface from Volume I. $\square$

Imported theorem (almost-AD Furstenberg kernel). *The following finite-scale form is deduced below from the regular-set Furstenberg theorem in Chapter III.9. Fix $0 < s < 1$ and $\varepsilon > 0$. There is $\eta = \eta(s, \varepsilon) > 0$, independent of $q \in [s, 2 - s]$, such that for each such q and all sufficiently small dyadic δ the following holds for a $(\delta, s, \delta^{-\eta}, M)$ -nice configuration $(P, \mathcal{T})$ with $|P| \asymp \delta^{-q}$. If P is (q, η) -almost Ahlfors–David regular in the sense that*

$$\delta^\eta r^q |P| \leq |P \cap B(p, r)| \leq \delta^{-\eta} r^q |P| \quad (p \in P, \delta \leq r \leq 1), \quad (11.28a)$$

then

$$\frac{|\mathcal{T}|}{M} \gtrsim_{s, \varepsilon} \delta^{-(s+q)/2+\varepsilon}. \quad (11.28b)$$

This is the almost-AD branch of Ren–Wang Theorem 6.1 [32]. Ren–Wang reduce this case to the Orponen–Shmerkin regular-set Furstenberg theorem, with two additional points: the change of regularity convention and uniformity of the small exponent in q . The regular-set theorem is stated in (9.46); the next lemma and proposition supply those two reductions.

Lemma III.11.7.1 (almost-AD implies Orponen–Shmerkin regularity). *Let $q \in (0, 2]$, $0 < \eta < 1/10$, and let $P \subset \mathcal{D}_\delta^2$ satisfy $|P| \asymp \delta^{-q}$ and (11.28a). For all sufficiently small $\delta = \delta(q, \eta)$, P is a $(\delta, q, \delta^{-3\eta})$ -regular set in the sense of Chapter III.9. Concretely, besides the $(\delta, q, \delta^{-3\eta})$ set estimate, one has*

$$|P \cap Q|_r \lesssim \delta^{-2\eta} \left(\frac{R}{r}\right)^q \leq \delta^{-3\eta} \left(\frac{R}{r}\right)^q$$

for every dyadic $\delta \leq r \leq R \leq 1$ and every dyadic R -square Q .

Proof. The upper half of (11.28a), with a fixed enlargement of a ball, gives

$$|P \cap B(x, r)| \lesssim \delta^{-\eta} r^q |P| \quad (x \in \mathbb{R}^2, \delta \leq r \leq 1),$$

so P is a $(\delta, q, C\delta^{-\eta})$ -set for an absolute C .

Identify each occupied δ -square with a chosen point in it; this changes all ball estimates only by fixed enlargements. Fix a dyadic R -square Q and $\delta < r \leq R$. Partition its occupied dyadic r -squares into a bounded number of residue classes so that squares in each class

are separated by at least $3r$. Choose $p_{Q'}$ in the union of P inside each occupied r -square Q' . Within one residue class the balls $B(p_{Q'}, r/2)$ are disjoint, while their union lies in a fixed $O(R)$ -enlargement of Q . If R is bounded below by an absolute constant, the lower half of (11.28a) on these disjoint balls and the trivial total bound $|P|$ give

$$|P \cap Q|_r \delta^\eta (r/2)^q |P| \lesssim |P|,$$

hence $|P \cap Q|_r \lesssim \delta^{-\eta} r^{-q} \lesssim \delta^{-2\eta} (R/r)^q$ once δ is small. If R is smaller than that fixed constant, the upper half of (11.28a) on the fixed $O(R)$ -enlargement, together with the same lower bounds on the disjoint balls, gives

$$|P \cap Q|_r \delta^\eta (r/2)^q |P| \lesssim \delta^{-\eta} R^q |P|.$$

Boundedly many residue classes and fixed ball enlargements are absorbed in the implicit constants, so again $|P \cap Q|_r \lesssim \delta^{-2\eta} (R/r)^q$. For $r = \delta$, the same estimate follows directly from the upper bound and $|P| \asymp \delta^{-q}$. Finally, once δ is sufficiently small, every fixed implicit constant is at most $\delta^{-\eta}$, giving the stated $\delta^{-3\eta}$ regularity constant. $\square$

Proposition III.11.7.2 (almost-AD kernel from the regular-set theorem). *The estimate (11.28b) follows from (9.46); in particular, its parameter $\eta = \eta(s, \varepsilon)$ can be chosen independently of $q \in [s, 2 - s]$.*

Proof. If $\varepsilon \geq s$, the assertion is immediate from $|\mathcal{T}| \geq M$, so assume $0 < \varepsilon < s$. Fix temporarily $q_0 \in [s, 2 - s]$ and put

$$u_0 := \frac{s + q_0}{2} - \frac{\varepsilon}{4}.$$

Then $0 < u_0 < \min\{(s + q_0)/2, 1\}$. Apply the regular-set theorem (9.46) with parameters (s, q_0, u_0) , and denote by $\theta(q_0) > 0$ and $\delta_0(q_0) > 0$ its regularity budget and scale threshold. Set

$$\kappa(q_0) := \min\{\varepsilon/4, \theta(q_0)/4\}.$$

The intervals $B(q_0, \kappa(q_0))$, $q_0 \in [s, 2 - s]$, cover the compact interval $[s, 2 - s]$. Choose a finite subcover with centres $q_1, \dots, q_N$, and define

$$\eta := \min_{1 \leq j \leq N} \frac{\theta(q_j)}{12} > 0.$$

Decrease the common scale threshold below the finitely many $\delta_0(q_j)$. Decrease it also below the thresholds in Lemma III.11.7.1.

Now take $q \in [s, 2 - s]$ and choose q_j with $|q - q_j| < \kappa(q_j)$. By Lemma III.11.7.1, P is $(\delta, q, \delta^{-3\eta})$ -regular. Changing the exponent from q to q_j costs at most $\delta^{-|q - q_j|}$ in each of the two defining regularity inequalities: indeed $r^q \leq \delta^{-|q - q_j|} r^{q_j}$ for $\delta \leq r \leq 1$, and $(R/r)^q \leq \delta^{-|q - q_j|} (R/r)^{q_j}$ for $\delta \leq r \leq R \leq 1$. Thus P is $(\delta, q_j, \delta^{-(3\eta + \kappa(q_j))})$ -regular. Since $3\eta + \kappa(q_j) \leq \theta(q_j)/2$, and also $\eta \leq \theta(q_j)$, the hypotheses of (9.46) hold with its larger allowed constant $\delta^{-\theta(q_j)}$. Consequently

$$\frac{|\mathcal{T}|}{M} \geq \delta^{-u_0}.$$

Finally,

$$u_0 - \left(\frac{s+q}{2} - \varepsilon \right) = \frac{q_j - q}{2} + \frac{3\varepsilon}{4} \geq -\frac{\kappa(q_j)}{2} + \frac{3\varepsilon}{4} > 0.$$

Hence $\delta^{-u_0} \geq \delta^{-(s+q)/2+\varepsilon}$, proving the claimed estimate uniformly in q . This is the regularity-conversion and compactness refinement isolated in the source proof of Ren–Wang Theorem 6.1; its only upstream research ingredient is the regular-set theorem proved in Chapter III.9. $\square$

The semi-well-spaced kernel. The source proof of the semi-well-spaced kernel consists of three finite ingredients: a richness-pruning energy lemma, a high–low induction for rich tubes, and an all-scale refinement in the endpoint regime $s+t \geq 2$. We record all three because the high–low inequality and the two-scale nice-configuration refinement already have local proofs in Chapters III.7 and III.6. Throughout this paragraph, fixed enlargements of dyadic tubes and squares are harmless and are suppressed from the notation.

Lemma III.11.7.3 (local fibre-pruning energy lemma). *Let $0 < \delta < \Delta \leq 1$, $0 < s \leq 1$, and let P be a finite family of distinct dyadic δ -squares contained in a square of side Δ . Suppose*

$$|P \cap B(x, w)| \leq K(w/\delta)^s \quad (\delta \leq w \leq \Delta). \quad (11.28c)$$

For each $p \in P$, let $\mathcal{T}(p)$ be a (δ, s, K) -family of $\asymp M$ incident tubes. Then, after discarding at most half the points and at most half the fibre of every surviving point, every surviving tube is incident to at most

$$CK^2 \log(1/\delta) \quad (11.28d)$$

surviving points, where C is absolute.

Proof. Call a tube bad if its incidence degree is at least $C_1 K^2 \log(1/\delta)$. If the conclusion fails, there is a set $P' \subset P$ with $|P'| \geq |P|/2$ such that every $p \in P'$ retains at least $M/2$ bad tubes. Put $\mathcal{T}' = \bigcup_{p \in P'} \mathcal{T}'(p)$ and let J count triples (p, q, T) with $T \in \mathcal{T}'(p) \cap \mathcal{T}'(q)$. For fixed p , split q into dyadic distance classes w . By (11.28c) there are $O(K(w/\delta)^s)$ possible q in a class. The s -nonconcentration of the slope fibre at p shows that at most $O(KM(\delta/w)^s)$ tubes through p can also meet such a q . Hence

$$J \lesssim M(K^2 \log(1/\delta) + 1)|P|.$$

On the other hand, Cauchy–Schwarz in the tube variable gives

$$(M|P|)^2 \lesssim |\mathcal{T}'|J,$$

so $|\mathcal{T}'| \gtrsim (K^2 \log(1/\delta))^{-1} M|P|$. But every member of $\mathcal{T}'$ is bad, and double counting incidences gives $|\mathcal{T}'| \leq M|P|/(C_1 K^2 \log(1/\delta))$. Choosing C_1 larger than the implicit constant is a contradiction. $\square$

Proposition III.11.7.4 (local endpoint all-scale tube-occupancy refinement). *Fix a sufficiently small dyadic Δ , an integer $n \geq 1$, and put $\delta = \Delta^n$. Let $(P_0, \mathcal{T}_0)$ be a (δ, s, C_0, M_0) -nice configuration, and suppose P_0 is a (δ, t, C_0) -set with $s+t \geq 2$. For every $0 < \varepsilon \leq 1/n$ there*

is a refinement $(P, \mathcal{T})$ preserving a polylogarithmic fraction of the restricted incidences such that, for $w = \Delta^k$, $1 \leq k \leq n$, and every $T \in \mathcal{T}$,

$$\#\{Q \in \mathcal{D}_w(P) : Q \subset 6T^w\} \lesssim_{n,\epsilon,\Delta_0} C_0^2 \Delta^{-1} w^{1-\epsilon} |\mathcal{D}_w(P)|. \quad (11.28e)$$

Here the implicit loss is at most a fixed power of $\log(1/\delta)$, and T^w denotes the dyadic w -ancestor of T .

Proof. We induct on n . For $n = 1$, a width- Δ tube meets $O(\Delta^{-1})$ dyadic Δ -squares, which is stronger than (11.28e).

Assume the statement at depth $n - 1$. First uniformize P_0 on the finite grid $1, \Delta, \dots, \Delta^n$, losing only a polylogarithmic factor. Apply the locally proved two-scale refinement III.6.5 at scale Δ^{n-1} . It produces a fine refinement $(P_1, \mathcal{T}_1)$ and a coarse $(\Delta^{n-1}, s, C_{n-1}, M_{n-1})$ -nice configuration. Apply the induction hypothesis to the coarse configuration and lift the resulting coarse refinement back to the fine scale by the covering clause in III.6.5. Denote the lift by $(P_2, \mathcal{T}_2)$. For $w = \Delta^k$, $k < n$, the induction hypothesis and uniformity give (11.28e).

It remains to control the bottom scale $w = \delta$. Pigeonhole the fine tubes by the number $r \asymp |6T \cap P_2|$ of points met, retaining a polylogarithmic fraction of the restricted incidences. Apply the local weighted Fourier thickening estimate III.7.23 with unit point weights and thickening factor $S = \Delta^{-1} = \delta^{-1/n}$. In the present normalized chart it gives

$$I(P_2, \mathcal{T}) \lesssim \Delta^{-1/2} |P_2|^{1/2} (|\mathcal{T}| \delta^{-1})^{1/2} + \Delta^{1-\epsilon} I(P_2, \mathcal{T}^{\Delta^{n-1}}), \quad (11.28f)$$

after decreasing the harmless Fourier loss parameter.

If the first term dominates, then both $I \gtrsim |P_2| M_0$ and $I \gtrsim r |\mathcal{T}|$ hold. Multiplying these inequalities and comparing with the square of the first term yields

$$r \lesssim \Delta^{-1} \delta^{-1} M_0^{-1}.$$

The relative s -regularity of every fibre gives $M_0 \gtrsim C_0^{-1} \delta^{-s}$, while the relative t -regularity of P_0 gives $|P_2| \gtrsim C_0^{-1} \delta^{-t}$ up to the already budgeted pruning loss. Since $s + t \geq 2$, this implies $r \lesssim C_0^2 \Delta^{-1} |P_2| \delta^{1-\epsilon}$ after absorbing the polylogarithm.

If the second term in (11.28f) dominates, group the points by their Δ^{n-1} -parents. The induction hypothesis at $w = \Delta^{n-1}$ bounds the number of such parents inside every thickened coarse tube, while uniformity identifies the number of fine points per parent. Hence

$$r |\mathcal{T}| \lesssim C_0^2 \Delta^{-1} |P_2| |\mathcal{T}| \delta^{1-\epsilon},$$

and the same estimate for r follows. This is exactly (11.28e) at $w = \delta$. The refinement preserves a polylogarithmic fraction of the incidences at each of the finitely many steps, completing the induction. $\square$

For a multiset P of dyadic δ -squares and an integer $b \geq 1$, write $N_{\Delta,b}(T)$ for the number of occupied dyadic Δ -squares Q for which $|4T \cap Q \cap P| \geq b$, counting the multiplicity of P .

Lemma III.11.7.5 (coarse-square energy bound). *Assume distinct occupied Δ -squares are separated by at least Δ and that*

$$|P \cap R| \leq Kr |P| \quad (R \in \mathcal{D}_r, \Delta \leq r \leq 1). \quad (11.28g)$$

Then

$$\sum_{T: N_{\Delta,b}(T) \geq 2} N_{\Delta,b}(T)^2 \lesssim K \log(1/\Delta) \frac{|P|^2}{b^2}. \quad (11.28h)$$

Consequently, for a family $\mathcal{T}_{a,b}$ of essentially distinct tubes with $N_{\Delta,b}(T) \geq a \geq 2$,

$$|\mathcal{T}_{a,b}| \lesssim K \log(1/\Delta) \frac{|P|^2}{(ab)^2}. \quad (11.28i)$$

Proof. Count triples (p, q, T) with p, q in different occupied Δ -squares and both meeting $6T$. A tube with $N_{\Delta,b}(T) = m \geq 2$ contributes at least a constant multiple of $m^2 b^2$. For fixed p and a dyadic distance $r \in [\Delta, 1]$, (11.28g) gives $O(Kr|P|)$ choices for q , while two δ -squares at distance $\asymp r$ lie in only $O(r^{-1})$ essentially distinct δ -tubes. Summing over $O(\log(1/\Delta))$ distance classes gives $O(K \log(1/\Delta) |P|^2)$ triples. Divide by b^2 . The second assertion follows from $N_{\Delta,b}(T)^2 \geq a^2$ on $\mathcal{T}_{a,b}$. $\square$

Lemma III.11.7.6 (local high-low rich-tube induction). *Fix $0 < \epsilon < 1/10$ with $\epsilon^{-1} \in \mathbb{N}$. Let $\delta < \Delta^{32}$, and suppose the multiset P is uniform on the dyadic scales nearest $\Delta^\epsilon, \Delta^{2\epsilon}, \dots, \Delta$ and has pairwise Δ -separated occupied Δ -parents. Let $a \geq 2$ and $ab > \delta^{1-2\epsilon}|P|$. Suppose every $T \in \mathcal{T}_{a,b}$ is essentially distinct, satisfies $N_{\Delta,b}(T) \geq a$, and for $w \in \{\Delta^\epsilon, \dots, \Delta\}$ obeys*

$$\#\{Q \in \mathcal{D}_w(P) : Q \subset 6T^w\} \leq \Delta^{-\epsilon} w |\mathcal{D}_w(P)|. \quad (11.28j)$$

Then

$$|\mathcal{T}_{a,b}| \lesssim_\epsilon \delta^{-5\epsilon} \frac{|P|^2}{a^3 b^2}. \quad (11.28k)$$

Proof. The proof is an induction on the dyadic ratio Δ/δ , with a quantitative contraction. We give the full stopping-scale calculation, which is the decisive Fourier step.

First, (11.28j) and uniformity imply that P is a $(\Delta, 1, O(\Delta^{-2\epsilon}))$ -set above scale Δ . Therefore Lemma III.11.7.5 gives the weak estimate

$$|\mathcal{T}_{a,b}| \lesssim_\epsilon \Delta^{-3\epsilon} \frac{|P|^2}{(ab)^2}. \quad (11.28l)$$

This settles all bounded Δ , and also the case $a \leq 2\Delta^{-2\epsilon}$ after enlarging the constant. If $\delta > \Delta^{1+\epsilon}$, the richness condition $ab > \delta^{1-2\epsilon}|P|$, uniformity at scale Δ , and (11.28j) at $w = \Delta$ are incompatible for sufficiently small Δ . These are the base cases.

Assume now $\delta < \Delta^{1+\epsilon}$ and $a > 2\Delta^{-2\epsilon}$. Choose the largest $w = \Delta^{k\epsilon}$ for which

$$a > 2\Delta^{-\epsilon} w |\mathcal{D}_w(P)|. \quad (11.28m)$$

Maximality and uniformity give

$$2\Delta^{-\epsilon} w |\mathcal{D}_w(P)| < a \leq \Delta^{-2\epsilon} w |\mathcal{D}_w(P)|, \quad (11.28n)$$

and, after rescaling any occupied w -square Q to the unit square, the descendant multiset P_Q is a $(\Delta/w, 1, O(\Delta^{-\epsilon}))$ -set at all scales between Δ/w and 1.

Inside each w -square choose a maximal essentially distinct family of $\delta \times w$ tubelets induced by tubes of $\mathcal{T}_{a,b}$. Give a tubelet u the weight $N(u) = N_{\Delta,b}(u)$. Bounded geometric overlap of the tubelets and (11.28m) show, after discarding at most half the incidences, that all surviving tubelets have $N(u) \geq 2$. Cover line space by $O(1)$ bounded charts and, in each chart, rescale w to one. Apply the local weighted Fourier thickening estimate III.7.23 with

$$S = \Delta^{-\epsilon/100}.$$

Summing over the w -squares/charts gives

$$a|\mathcal{T}_{a,b}| \lesssim (Sw^{-1})^{1/2} \sum_Q |\mathcal{T}_Q|^{1/2} \left(\sum_{u \subset Q} N(u)^2 \right)^{1/2} + S^{-1+\epsilon} \mathcal{J}_{\text{low}}. \quad (11.28o)$$

For the high-frequency term, Cauchy–Schwarz in Q , the bounded multiplicity of the tube charts, and Lemma III.11.7.5 applied to each rescaled P_Q give

$$\sum_Q |\mathcal{T}_Q|^{1/2} \left(\sum_{u \subset Q} N(u)^2 \right)^{1/2} \lesssim |\mathcal{T}_{a,b}|^{1/2} \left[|\mathcal{D}_w(P)| \Delta^{-2\epsilon} \left(\frac{|P|}{b|\mathcal{D}_w(P)|} \right)^2 \right]^{1/2}.$$

If this term dominates (11.28o), then (11.28n) yields directly (11.28k), with room to absorb the factor S .

Suppose instead that the low-frequency term dominates. For each fine tube T let $F(T)$ be the sum of the weights of tubelets contained in its $S\delta$ -thickening. A half-sized subfamily has $F(T) \gtrsim S^{1-\epsilon}a$. If $g_T(Q)$ denotes the number of contributing tubelets in the Δ -square Q , bounded tubelet overlap gives

$$|4T^{2S\delta} \cap Q \cap P^{2S\delta}| \gtrsim b g_T(Q), \quad \sum_Q g_T(Q) \gtrsim S^{1-\epsilon}a, \quad g_T(Q) \lesssim S.$$

Pigeonhole $g_T(Q)$ dyadically, first for each T and then uniformly over a subfamily. Thus for some $1 \leq k \lesssim S$ there are

$$a' \gtrsim \frac{S^{1-\epsilon}a}{k \log S} \quad \text{distinct } \Delta\text{-parents, each containing at least } b' = kb$$

points of the thickened multiset along $T^{2S\delta}$. Delete duplicate $2S\delta$ -ancestors; at most $O(S^2)$ fine tubes have the same ancestor, so the ancestor family has size $\gtrsim S^{-2}(\log S)^{-1}|\mathcal{T}_{a,b}|$.

The condition (11.28j) is inherited at the coarser scale, and $a'b' > (2S\delta)^{1-2\epsilon}|P|$ once Δ is sufficiently small. The induction hypothesis at scale $2S\delta$ therefore gives

$$|\mathcal{T}_{a,b}| \lesssim_\epsilon S^2 \log S (2S\delta)^{-5\epsilon} \frac{|P|^2}{(a')^3(b')^2} \lesssim S^{-\epsilon} \delta^{-5\epsilon} \frac{|P|^2}{a^3 b^2}.$$

Because $S = \Delta^{-\epsilon/100} > 1$, the factor $S^{-\epsilon}$ is a strict contraction. Choosing the base threshold so that the contraction absorbs the implicit constant closes the induction and proves (11.28k). $\square$

Proposition III.11.7.7 (local semi-well-spaced transition estimate). *Let $0 < s \leq 1$, $\epsilon > 0$, and $\delta \leq \Delta$ be dyadic. Suppose $(P, \mathcal{T})$ is a $(\delta, s, \delta^{-\epsilon^2}, M)$ -nice configuration satisfying*

$$|P \cap B(x, r)| \leq \delta^{-\epsilon^2} r^{2-s} |P| \quad (\Delta \leq r \leq 1), \quad |P \cap B(x, r)| \leq \delta^{-\epsilon^2} (r/\delta)^s \quad (\delta \leq r \leq \Delta). \quad (11.28p)$$

Then, after decreasing $\delta_0(s, \epsilon)$ if necessary,

$$|\mathcal{T}| \gtrsim_{\epsilon} \delta^{C\epsilon} \min\{M|P|, M^{3/2}|P|^{1/2}, \delta^{-1}M\}, \quad (11.28q)$$

where $C > 0$ is absolute.

Proof. It is enough to treat ϵ below a fixed absolute threshold: for larger ϵ , enlarge C in (11.28q) and use $|\mathcal{T}| \geq M$ together with $|P| \leq O(\delta^{-2})$. Replacing ϵ by a comparable larger reciprocal-integer parameter changes only the absolute constant C , so all auxiliary grids below may be taken dyadic.

We first dispose of the two easy regimes exactly as in the source proof. If $\delta^{8\epsilon} < \Delta$, then for $\rho < \Delta$ the second inequality in (11.28p) applies, while for $\rho \geq \Delta$ a packing by $O((\rho/\Delta)^2)$ many Δ -balls gives

$$|P \cap B(x, \rho)| \lesssim \delta^{-17\epsilon} (\rho/\delta)^s$$

after decreasing the small-scale threshold. Hence P is a global $(\delta, s, \delta^{-17\epsilon})$ Katz–Tao set. Lemma III.11.7.3 then yields a refinement in which each tube contains at most $\delta^{-O(\epsilon)}$ selected incidences, and therefore $|\mathcal{T}| \gtrsim \delta^{O(\epsilon)} M|P|$.

At the other extreme, suppose $\delta > \Delta^{100}$. The high-scale estimate in (11.28p), together with the low-scale one and the relation $\delta > \Delta^{100}$, makes P a global $(\delta, 2-s, \delta^{-O(\epsilon)})$ -set. Apply Proposition III.11.7.4 with a fixed depth $O(\epsilon^{-1})$ (rounding the geometric grid to adjacent dyadic scales). Its bottom-scale occupancy estimate bounds the selected incidence degree of every tube by $\delta^{1-O(\epsilon)}|P|$. Comparing this with the surviving $\gtrsim |P|M$ incidences gives $|\mathcal{T}| \gtrsim \delta^{-1+O(\epsilon)}M$. This is the third term in (11.28q).

We may therefore assume

$$\delta^{8\epsilon} \geq \Delta, \quad \delta \leq \Delta^{100}. \quad (11.28r_0)$$

In particular $\delta < \Delta^{32}$ once the harmless endpoint equality cases are absorbed into the previous regime.

Uniformize P on the dyadic scales nearest $\Delta^\epsilon, \Delta^{2\epsilon}, \dots, \Delta$, and apply the local two-scale refinement III.6.5 at Δ . For the resulting coarse Δ -scale nice configuration, take $n \asymp 4/\epsilon$ and a dyadic base $\Lambda \asymp \Delta^{1/n}$, so $\Lambda^n \asymp \Delta$. Apply Proposition III.11.7.4 with its internal loss parameter $\epsilon/4$ and lift the refinement back to the fine scale using III.6.5. The factor $C_0^2 \Lambda^{-1}$ in (11.28e), the first inequality in (11.28r₀), and $w \geq \Delta$ imply, after absorbing polylogarithms,

$$\#\{Q \in \mathcal{D}_w(P) : Q \subset 6T^w\} \leq \Delta^{-\epsilon} w |\mathcal{D}_w(P)| \quad (11.28r_1)$$

for every w in the coarser grid $\{\Delta^\epsilon, \Delta^{2\epsilon}, \dots, \Delta\}$. Thus the precise all-scale hypothesis needed by Lemma III.11.7.6 is available.

Inside each surviving Δ -square apply Lemma III.11.7.3. Retain one of four parity subgrids so distinct occupied Δ -parents are Δ -separated, and uniformize once more. All these operations cost only $\delta^{-O(\epsilon)}$.

For each surviving fine tube T , pigeonhole the number $a(T)$ of occupied Δ -parents that carry a selected incidence and then the number $b(T)$ of points contributed by a typical such parent. A second pigeonhole produces common dyadic values (a, b) and a subfamily $\mathcal{T}'$ preserving a $\delta^{O(\epsilon)}$ fraction of all restricted incidences. The pruning lemma, applied separately in the occupied parents, gives

$$\#\{p \in P : T \in \mathcal{T}(p)\} \lesssim \delta^{-O(\epsilon)} a,$$

and hence

$$|\mathcal{T}'| \gtrsim \delta^{O(\epsilon)} \frac{|P|M}{a}. \quad (11.28r)$$

Moreover (11.28r₁) is exactly the spacing hypothesis of Lemma III.11.7.6.

If $a \lesssim \delta^{1-2\epsilon}|P|$, then (11.28r) gives $|\mathcal{T}'| \gtrsim \delta^{-1+O(\epsilon)}M$. Otherwise $ab > \delta^{1-2\epsilon}|P|$ (since $b \geq 1$, after absorbing the pigeonholing loss), so Lemma III.11.7.6 gives

$$|\mathcal{T}'| \lesssim \delta^{-O(\epsilon)} \frac{|P|^2}{a^3 b^2}.$$

Comparison with (11.28r) yields $a^2 b^2 \lesssim \delta^{-O(\epsilon)}|P|/M$. Since $b \geq 1$, $a \lesssim \delta^{-O(\epsilon)}(|P|/M)^{1/2}$, and substitution back into (11.28r) gives $|\mathcal{T}'| \gtrsim \delta^{O(\epsilon)}|P|^{1/2}M^{3/2}$. In each of the three regimes we have proved at least one member of the minimum in (11.28q), completing the proof. $\square$

Proposition III.11.7.8 (semi-well-spaced Ren–Wang kernel, local). *Fix $0 < s < 1$ and $\epsilon > 0$. There is $\eta = \eta(s, \epsilon) > 0$ such that, uniformly for $q \in [s, 2-s]$, the following holds for all sufficiently small dyadic δ . Let $(P, \mathcal{T})$ be a $(\delta, s, \delta^{-\eta}, M)$ -nice configuration with $|P| \asymp \delta^{-q}$ and suppose that, for every dyadic $\rho \in [\delta, 1]$ and every dyadic ρ -square Q ,*

$$|P \cap Q| \lesssim \delta^{-\eta} \max \left\{ \rho^{2-s}|P|, \left(\frac{\rho}{\delta} \right)^s \right\}. \quad (11.28s)$$

Then

$$\frac{|\mathcal{T}|}{M} \gtrsim_{s, \epsilon} \delta^{-(s+q)/2+\epsilon}. \quad (11.28t)$$

Proof. Choose $\eta \ll_s \epsilon^2$, and let Δ be the dyadic scale nearest

$$(|P|\delta^s)^{-1/(2-2s)}.$$

Because $q \in [s, 2-s]$, one has $\delta \lesssim \Delta \lesssim 1$. The two terms in the maximum of (11.28s) balance at Δ . Choose an auxiliary reciprocal-integer parameter $\epsilon_1 \asymp_s \sqrt{\eta}$ so small that the eventual $O(\epsilon_1)$ loss is below $\epsilon/4$. After changing Δ by at most one dyadic scale, the hypotheses of Proposition III.11.7.7 hold with ϵ_1 in place of ϵ .

A relative $(\delta, s, \delta^{-\eta})$ fibre containing M tubes necessarily satisfies $M \gtrsim \delta^{-s+\eta}$ by applying its nonconcentration inequality at radius δ . Therefore each term in the minimum of (11.28q), after division by M , is at least the target power up to $\delta^{O(\eta)}$:

$$|P| \gtrsim \delta^{-q}, \quad (M|P|)^{1/2} \gtrsim \delta^{-(s+q)/2+O(\eta)}, \quad \delta^{-1} \gtrsim \delta^{-(s+q)/2}$$

because $q \geq s$ and $q \leq 2-s$. Taking the transition loss and η sufficiently small in terms of ϵ proves (11.28t). $\square$

Local middle-regime assembly. We now prove the rest of the Ren–Wang Section 6 mechanism. After Proposition III.11.7.2, both finite kernels used by the assembly follow from the Chapter III.9 regular-set theorem and the arguments above.

Lemma III.11.7.9 (upper-capped almost-linear decomposition). *Fix $0 < t < v < 2$. There is a constant $C = C(t, v)$ with the following property. Let $0 < \eta < 1/10$ and $0 < \rho < (v - t)/100$, and let $f : [0, m] \rightarrow \mathbb{R}$ be continuous, piecewise affine and 2-Lipschitz, with $f(0) = 0$ and*

$$f(x) \geq tx - \rho m \quad (0 \leq x \leq m), \quad f(m) \leq (t + \rho)m. \quad (11.28u_0)$$

Then there are pairwise disjoint intervals $[a_i, b_i] \subset [0, m]$ such that

$$b_i - a_i \geq \tau m, \quad \mathcal{L}\left([0, m] \setminus \bigcup_i [a_i, b_i]\right) \leq C\left(\eta^2 + \frac{\rho}{v - t}\right)m, \quad (11.28u_1)$$

where $\tau = \tau(t, v, \eta) > 0$, and every retained interval has one of the following two forms:

- (U1) (f, a_i, b_i) is $C\eta^2$ -linear and its chord slope is at most v ;
- (U2) (f, a_i, b_i) is $C\eta^2$ -superlinear and its chord slope is exactly v .

Proof. This is the elementary upper stopping-time which underlies the first half of the Ren–Wang profile decomposition. Apply the local almost-linear covering lemma Lemma III.6.15 to $f/2$ with accuracy $c\eta^4$, where $c > 0$ is a sufficiently small absolute constant, and rescale. We obtain disjoint base intervals $[a_i, b_i]$, in increasing order, each of length at least $\tau_0 m$, whose complement has length $O(\eta^4 m)$ and on which f is $O(\eta^4)$ -linear.

If all base chord slopes are at most v , there is nothing to change. Otherwise let k be the first base interval whose chord slope is larger than v . If

$$a_k \geq m - \frac{3\rho m}{v - t},$$

we discard the remaining tail. This costs $O_{t,v}(\rho m)$ and ends the construction. In the opposite case, (11.28u₀) gives

$$s_f(a_k, m) := \frac{f(m) - f(a_k)}{m - a_k} \leq t + \frac{2\rho m}{m - a_k} < v.$$

Since $s_f(a_k, b_k) > v$ and f is piecewise affine, there is a first point $b' > b_k$ with $s_f(a_k, b') = v$. For $b_k < x < b'$ the minimality of b' gives $s_f(a_k, x) > v$, whereas on $[a_k, b_k]$ the almost-linearity and the inequality $s_f(a_k, b_k) > v$ give the same lower affine barrier up to $O(\eta^4)(b' - a_k)$. Hence $[a_k, b']$ is $O(\eta^2)$ -superlinear with endpoint slope v .

Keep the intervals preceding a_k and replace the run crossed by $[a_k, b']$ by this single bridge. If b' falls in a gap of the base cover, continue with the first base interval to its right. If $b' \in [a_\ell, b_\ell]$ and $b_\ell - b' \leq \eta^2(b_\ell - a_\ell)$, discard that short tail and continue to the right. Finally, if $b_\ell - b' > \eta^2(b_\ell - a_\ell)$, replace $[a_\ell, b_\ell]$ by $[b', b_\ell]$ and continue. The last interval remains $O(\eta^2)$ -linear: restricting an $O(\eta^4)$ -linear interval to a subinterval of relative length at least η^2 enlarges the relative affine error by at most a constant multiple of η^{-2} .

Every pass consumes at least one original base interval, so the procedure terminates after at most $O(\tau_0^{-1})$ steps. The newly created bridges have length at least $\tau_0 m$; a shortened

tail is retained only when its length is at least $\eta^2 \tau_0 m$. The extra discarded pieces have total length $O(\eta^2 m)$, while the possible terminal tail has length $O_{t,v}(\rho m/(v-t))$. Decreasing τ and enlarging C proves the lemma. $\square$

Lemma III.11.7.10 (capped branching-profile decomposition). *Fix $0 < s < t < u \leq 2$. There is $\epsilon_0 = \epsilon_0(s, t, u) > 0$ such that for $0 < \epsilon < \epsilon_0$ there is $\tau = \tau(s, t, u, \epsilon) > 0$ with the following property. Let $f : [0, m] \rightarrow \mathbb{R}$ be continuous, piecewise affine and 2-Lipschitz, with $f(0) = 0$ and*

$$f(x) \geq tx - \epsilon^2 m \quad (0 \leq x \leq m), \quad f(m) \leq (t + \epsilon^2)m. \quad (11.28v)$$

Then there are pairwise disjoint intervals $[c_j, d_j] \subset [0, m]$, each of length at least τm , covering all but $O_{s,t,u}(\epsilon)m$ of $[0, m]$, such that, writing $q_j = (f(d_j) - f(c_j))/(d_j - c_j)$, one of the following holds:

(a) $q_j \in [s, u]$ and

$$|f(x) - f(c_j) - q_j(x - c_j)| \leq C\epsilon(d_j - c_j) \quad (c_j \leq x \leq d_j); \quad (11.28w)$$

(b) $q_j \in [s, u]$ and

$$f(x) \geq \min\{f(c_j) + u(x - c_j), f(d_j) - s(d_j - x)\} - C\epsilon(d_j - c_j). \quad (11.28x)$$

Here C depends only on the fixed parameters.

Proof. Choose an auxiliary number $\eta = c_{s,t,u}\epsilon$, with $c_{s,t,u} > 0$ so small that

$$s_* := s + 4\eta < t < u_* := u - 4\eta, \quad u_* - s_* > 10\eta. \quad (11.28x_0)$$

Apply Lemma III.11.7.9 with $v = u_*$, $\rho = \epsilon^2$, and auxiliary accuracy η . After absorbing its $O(\epsilon^2 m)$ exceptional set, we have a bounded family of long intervals, ordered from left to right, of two kinds:

- (I1) $O(\eta^2)$ -linear intervals with chord slope at most u_* ;
- (I2) $O(\eta^2)$ -superlinear intervals with chord slope exactly u_* .

We now impose the lower slope s_* by processing this family from right to left. Type (I2) intervals are already retained. A type (I1) interval whose chord slope is at least s_* is also retained. It remains to treat a rightmost unprocessed type (I1) interval $[c_k, d_k]$ with chord slope $q_k < s_*$. If

$$d_k \leq \frac{2\epsilon^2 m}{t - s_*},$$

we discard all remaining intervals to its left and stop; this costs only $O_{s,t}(\epsilon^2 m)$. Otherwise the floor in (11.28v) gives $s_f(0, d_k) > s_*$, whereas $s_f(c_k, d_k) = q_k < s_*$. Hence there is a largest point $c' \in (0, c_k)$ such that

$$s_f(c', d_k) = s_*. \quad (11.28x_1)$$

For $c' < x < c_k$, maximality gives $s_f(x, d_k) < s_*$. Since the chord slope on $[c', d_k]$ equals s_* , the convex-combination identity for secant slopes implies $s_f(c', x) \geq s_*$. On $[c_k, d_k]$, the $O(\eta^2)$ -linearity and $q_k < s_*$ give the same lower barrier from the right. Thus

$$f(x) \geq f(d_k) - s_*(d_k - x) - C\eta^2(d_k - c') \quad (c' \leq x \leq d_k). \quad (11.28y)$$

In particular, $[c', d_k]$ is a legitimate right-hand branch of a future type-(b) interval.

There are four landing cases, and keeping them separate is what controls both the error and the lost length.

Gap landing. If c' lies outside all unprocessed intervals, retain the bridge $[c', d_k]$ and resume with the last interval lying strictly to its left. All intervals swallowed by the bridge remain covered.

Near-left landing. If $c' \in [c_\ell, d_\ell]$ and $c' - c_\ell \leq 2\eta(d_\ell - c_\ell)$, discard the short piece $[c_\ell, c']$, retain the bridge, and resume with index $\ell - 1$. The discarded length is at most a 2η fraction of that interval.

Landing in a linear interval. Suppose $c' \in [c_\ell, d_\ell]$, $c' - c_\ell > 2\eta(d_\ell - c_\ell)$, and $[c_\ell, d_\ell]$ is of type (I1). Replace it by $[c_\ell, c']$, retain the bridge, and resume with this shortened interval. Restricting an $O(\eta^2)$ -linear interval to a subinterval of relative length at least 2η makes it $O(\eta)$ -linear, and changes its chord slope by $O(\eta)$. Because the original chord slope is at most $u_* = u - 4\eta$, the shortened slope remains $< u$ after $c_{s,t,u}$ is chosen small enough.

Landing in an upper-superlinear interval. Finally suppose the same size condition holds and $[c_\ell, d_\ell]$ is of type (I2). Merge $[c_\ell, d_\ell]$ with the bridge into $[c_\ell, d_k]$. The upper endpoint-slope bound follows because $s_f(c_\ell, d_\ell) = u_*$ while, by maximality of c' , $s_f(d_\ell, d_k) < s_*$. For the lower bound, type (I2) and $c' - c_\ell > 2\eta(d_\ell - c_\ell)$ give

$$s_f(c_\ell, c') \geq u_* - C\eta > s_*;$$

combining this with (11.28x₁) shows $s_f(c_\ell, d_k) \geq s_*$. Thus the merged chord slope lies in $[s_*, u_*] \subset [s, u]$. Moreover, the type-(I2) lower barrier controls the left part and (11.28y) controls the right part. Replacing u_* by u and s_* by s costs at most $4\eta(d_k - c_\ell)$ on either side, so the merged interval satisfies (11.28x).

The bridge retained in the first three landing cases has chord slope $s_* \in [s, u]$ and satisfies the right-hand barrier in (11.28y); since the minimum in (11.28x) is no larger than that right-hand barrier, it is itself a type-(b) interval after the same $O(\eta)$ error enlargement. Likewise an untouched type-(I2) interval is type (b), using its left-hand u_* -barrier. An accepted type-(I1) interval is type (a); after a possible shortening its chord slope lies in $[s, u]$ and its affine error is $O(\eta)$.

Every operation either consumes a whole interval or shortens one once before moving strictly to the left. Hence the number of operations is bounded in terms of the upper-cap decomposition. The near-left losses sum to $O(\eta m)$, the initial-tail loss is $O(\epsilon^2 m)$, and the upper-cap exceptional set is $O(\epsilon^2 m)$. Thus the total uncovered length is $O_{s,t,u}(\epsilon)m$. Retained or merged intervals have length bounded below by a positive parameter-dependent fraction of m ; decreasing τ if necessary completes the proof. $\square$

Lemma III.11.7.11 (profile pieces give the two geometric kernels). *Let $P \subset [0, 1]^2$ be uniform on the scales Δ^k , $0 \leq k \leq m$, with branching profile $f(k) = \log |P|_{\Delta^k} / \log(1/\Delta)$, extended affinely. Let $[c, d]$ be an integer-endpoint interval from Lemma III.11.7.10, put $\delta_0 = \Delta^{d-c}$, and let $P_{c,d}$ be the rescaled descendants inside any occupied Δ^c -square. Write $q = (f(d) - f(c))/(d - c)$.*

If (11.28w) holds, then $P_{c,d}$ is $(q, C\epsilon)$ -almost AD-regular at scale δ_0 . If (11.28x) holds with

$u = 2 - s$, then $|P_{c,d}| \asymp \delta_0^{-q}$ and, for every dyadic $r \in [\delta_0, 1]$ and every dyadic r -square Q ,

$$|P_{c,d} \cap Q| \lesssim \delta_0^{-C\epsilon} \max \left\{ r^{2-s} |P_{c,d}|, \left(\frac{r}{\delta_0} \right)^s \right\}. \quad (11.28z)$$

Proof. Uniformity gives an exact count formula. If $0 \leq k \leq d - c$ and Q is an occupied Δ^k -square in the rescaled picture, then

$$|P_{c,d} \cap Q|_{\delta_0} = \Delta^{f(c+k)-f(d)}, \quad |P_{c,d}|_{\delta_0} = \Delta^{f(c)-f(d)}. \quad (11.28aa)$$

In the linear case, subtracting the affine approximation at c from (11.28aa) gives matching upper and lower bounds $\delta_0^{C\epsilon} (\Delta^k)^q |P_{c,d}| \lesssim |P_{c,d} \cap Q| \lesssim \delta_0^{-C\epsilon} (\Delta^k)^q |P_{c,d}|$. A ball is covered by $O_\Delta(1)$ comparable grid squares and contains a comparable smaller one, which is the asserted almost-AD regularity after enlarging C .

In the wedge case, the left barrier in (11.28x) gives $f(c+k) - f(c) \geq (2-s)k - C\epsilon(d-c)$, while the right barrier gives $f(d) - f(c+k) \leq s(d-c-k) + C\epsilon(d-c)$. Insert these inequalities in (11.28aa). The first yields the term $r^{2-s} |P_{c,d}|$ and the second yields $(r/\delta_0)^s$, with the common factor $\delta_0^{-C\epsilon}$. Rounding a general dyadic r to the nearest Δ -scale changes only the implicit constant. $\square$

Lemma III.11.7.12 (iterated local nice-configuration product). *Let $(P_0, \mathcal{T}_0)$ be a (δ, s, C, M) -nice configuration and let $1 = \Delta_0 > \Delta_1 > \dots > \Delta_N = \delta$ be a fixed finite scale chain. After restrictions costing only a fixed power of $\log(1/\delta)$ at each step, one may choose, for every $0 \leq j < N$, a rescaled nice configuration $(P_j, \mathcal{T}_j)$ at scale $r_j = \Delta_{j+1}/\Delta_j$ with fibre size M_j such that*

$$\frac{|\mathcal{T}_0|}{M} \gtrsim_{\log, \delta} \prod_{j=0}^{N-1} \frac{|\mathcal{T}_j|}{M_j}. \quad (11.28ab)$$

The same conclusion holds after deleting scale intervals whose total logarithmic length is $O(\epsilon \log(1/\delta))$, at the cost of a factor $\delta^{O(\epsilon)}$.

Proof. Apply the locally proved two-scale refinement III.6.5 at Δ_1 . Equation (6.42) gives the first factor and a rescaled fine configuration. Repeat inside the retained fine configuration at Δ_2/Δ_1 , and continue. Multiplication telescopes the fibre sizes and yields (11.28ab). A deleted logarithmic interval can be bounded by the trivial estimate $|\mathcal{T}_j|/M_j \geq 1$; the product of its scale ratios is at most $\delta^{-O(\epsilon)}$, which is absorbed by the displayed loss. This is the finite iteration of the two-scale argument already proved in Chapter III.6, not an additional external multiscale theorem. $\square$

Proposition III.11.7.13 (local middle-regime discrete theorem). *Fix $0 < s < 1$, $s < t < 2 - s$, and $\epsilon > 0$. Relative to the Chapter III.9 regular-set theorem (9.46), there exists $\eta = \eta(s, t, \epsilon) > 0$ such that every sufficiently small dyadic δ and every $(\delta, s, \delta^{-\eta}, M)$ -nice configuration $(P, \mathcal{T})$ with P a $(\delta, t, \delta^{-\eta})$ -set satisfy*

$$\frac{|\mathcal{T}|}{M} \gtrsim_{s, t, \epsilon} \delta^{-(s+t)/2+\epsilon}. \quad (11.28ac)$$

Proof. By thinning P and dyadically pigeonholing, we may assume $|P| = \delta^{-t+O(\eta)}$. Choose a large dyadic base $\Delta = 2^{-T}$ depending only on the accuracy parameters. Repeated popularity pruning on the grid $1, \Delta, \dots, \Delta^m \asymp \delta$ makes P uniform at these scales. There are m stages and only $O(T)$ branching classes per stage, so the loss is $\exp(O(m \log T)) = \delta^{-O((\log T)/T)}$; choosing T first makes this smaller than any prescribed power $\delta^{-\epsilon_0}$. Endpoint scales of length $< T$ cost only an absolute factor.

Let f be the resulting branching profile. The $(\delta, t, \delta^{-\eta})$ condition and the cardinality normalization give

$$f(x) \geq tx - O(\eta)m, \quad f(m) = tm + O(\eta)m. \quad (11.28ad)$$

Choose a profile tolerance $\epsilon_0 \ll_{s,t} \epsilon$ and then $\eta \ll \epsilon_0^2$. Apply Lemma III.11.7.10 with $u = 2 - s$. Round each interval endpoint to the nearest integer scale, shrinking the interval if necessary. Since the profile family contains only $O_{s,t}(1)$ intervals, this alters the total logarithmic length by $O_{s,t}(1)$; for m sufficiently large it is absorbed by the $O(\epsilon_0 m)$ exceptional length. The 2-Lipschitz bound for f changes every endpoint increment by $O_{s,t}(1)$ as well, hence changes the corresponding chord slope and the linear/wedge error by $O_{s,t}(1/m)$, again absorbed after decreasing δ . Thus we obtain structured integer-scale intervals of length at least $\tau m/2$, separated by bad gaps whose total length is $O(\epsilon_0 m)$. Write $R_j = \Delta_j/\Delta_{j+1} > 1$ for their scale ratios and t_j for their endpoint slopes. Lemma III.11.7.11 says that every structured piece is either almost AD-regular of exponent $t_j \in [s, 2 - s]$ or satisfies the semi-well-spaced condition (11.28z). Since every structured ratio obeys $R_j \geq \delta^{-\tau'}$ for a fixed $\tau' > 0$, all logarithmic and inherited Frostman losses are absorbed by the kernel parameter after the choices are made in the order $\epsilon, \epsilon_0, \eta, \delta$.

Apply Lemma III.11.7.12. On each structured interval, use the local Proposition III.11.7.2 for almost-AD linear pieces and Proposition III.11.7.8 for wedge/semi-well-spaced pieces. In either case

$$\frac{|\mathcal{T}_j|}{M_j} \gtrsim R_j^{(s+t_j)/2-\epsilon_0}.$$

The bad gaps contribute only $\delta^{O(\epsilon_0)}$. Since the structured intervals cover all but $O(\epsilon_0)m$ logarithmic scales,

$$\prod_j R_j \geq \delta^{-1+O(\epsilon_0)}, \quad \prod_j R_j^{t_j} \geq \delta^{-t+O(\epsilon_0)}; \quad (11.28ae)$$

the second inequality is the endpoint telescoping identity for f , with the bad gaps bounded by the 2-Lipschitz norm. Therefore

$$\frac{|\mathcal{T}|}{M} \gtrsim \left(\prod_j R_j \right)^{s/2-\epsilon_0} \left(\prod_j R_j^{t_j} \right)^{1/2} \delta^{O(\epsilon_0)} \geq \delta^{-(s+t)/2+O_{s,t}(\epsilon_0)}.$$

Taking ϵ_0 sufficiently small proves (11.28ac). $\square$

Theorem III.11.7.14 (local continuum middle branch). *Relative only to the Chapter III.9 regular-set theorem (9.46), if $0 < s < 1$, $s < t < 2 - s$, and $F \subset \mathbb{R}^2$ is an (s, t) -Furstenberg set, then*

$$\dim_{\text{H}} F \geq \frac{3s+t}{2}. \quad (11.28af)$$

Proof. Fix $q < (3s + t)/2$. Choose $a < s$ and $b < t$ so close to (s, t) that $a < b < 2 - a$ and $(3a + b)/2 > q$. Suppose for contradiction that $\dim_{\text{H}} F < q$. Then F admits arbitrarily fine ball covers of arbitrarily small q -cost. Apply the common-scale Furstenberg discretisation Lemma III.9.2 with exponents (a, b) and with a loss smaller than the parameter in Proposition III.11.7.13. At the resulting scale δ we obtain a $(\delta, b, \delta^{-o(1)})$ family of line-tubes, each carrying a $(\delta, a, \delta^{-o(1)})$ family of incident squares from one scale layer of the cover.

Apply the local bounded-chart point–line duality Lemma III.6.29.1. After a dyadic pigeonhole of the fibre cardinalities, the dual configuration is a $(\delta, a, \delta^{-\eta}, M)$ -nice configuration whose point set is a $(\delta, b, \delta^{-\eta})$ -set and

$$M \geq \delta^{-a+O(\eta)}.$$

Proposition III.11.7.13 gives

$$|\mathcal{T}_{\text{dual}}| \geq \delta^{-a-(a+b)/2+O(\eta)+\varepsilon} = \delta^{-(3a+b)/2+O(\eta)+\varepsilon}. \quad (11.28\text{ag})$$

Duality has bounded multiplicity, so the corresponding scale layer of the original cover contains at least this many balls up to an absolute factor. Its q -cost is therefore at least $\delta^{q-(3a+b)/2+O(\eta)+\varepsilon}$, which tends to infinity after choosing η, ε sufficiently small because $q < (3a + b)/2$. This contradicts the chosen covers. Letting $q \uparrow (3s + t)/2$ proves the theorem. $\square$

Reconstruction of the sharp minimum from the preceding chapters. If $F \subset \mathbb{R}^2$ is an (s, u) -Furstenberg set with $0 < s \leq 1$ and $0 < u \leq 2$, then

$$\dim_{\text{H}} F \geq \min \left\{ s + u, \frac{3s + u}{2}, s + 1 \right\}. \quad (11.28)$$

The intermediate branch follows from the Chapter III.9 regular-set Furstenberg theorem together with the almost-AD conversion and semi-well-spaced kernel proved above.

If $u \leq s$, the local low-parameter estimate (6.84) gives $\dim_{\text{H}} F \geq s + u$. If $u \geq 2 - s$, the Fu–Ren high branch (7.371) gives $\dim_{\text{H}} F \geq s + 1$. In the remaining range $s < u < 2 - s$, Theorem III.11.7.14 gives $(3s + u)/2$. These are exactly the three branches of the displayed minimum; the boundary values agree by continuity.

Proof status. The heavy-line theorem below applies the full sharp Furstenberg minimum (11.28). Its middle branch uses the Chapter 9 regular-set Furstenberg proof, whose high-multiplicity input X-III.9.ABC49 is proved there; its high branch is proved in Chapter 7. Thus all three branches used by the displayed deduction have local proof owners.

Theorem III.11.8 (generic heavy-line upper theorem). Let $A \subset \mathbb{R}^2$ with $\dim_{\text{H}} A = t \in [1, 2]$. If

$$s > \frac{2t - 1}{3}, \quad (11.29)$$

then s -heavy lines are absent for almost every direction. If

$$t - 1 < s \leq \frac{2t - 1}{3}, \quad (11.30)$$

then

$$\mathfrak{h}(A, s) \leq \min\{2t - 3s, t\}. \quad (11.31)$$

Consequently

$$\mathfrak{h}(A) \leq \min\{3 - t, t\}. \quad (11.32)$$

Proof from the preceding Furstenberg theorems. Assume first that (11.29) holds and that a positive-measure set of directions carries an s -heavy line. Choosing one such line in each direction gives a line family of Hausdorff dimension at least 1, so A is an $(s, 1)$ -Furstenberg set. The interface (11.28) gives

$$t \geq \min\left\{s + 1, \frac{3s + 1}{2}, s + 1\right\} = \frac{3s + 1}{2} > t,$$

a contradiction.

Now suppose $t - 1 < s < (2t - 1)/3$ and write $h = \mathfrak{h}(A, s)$. The trivial bound is $h \leq t$. Suppose, toward a contradiction, that $h > 2t - 3s$. Since $s < (2t - 1)/3$, the latter number is strictly larger than 1, so choose

$$1 < h' < h \quad \text{with} \quad h' > 2t - 3s. \quad (11.33)$$

By the definition of essential supremum there is a positive-measure set E of directions for which $\dim_{\mathbb{H}} H(A, e, s) > h'$. In particular $\dim_{\mathbb{H}} E = 1$, and Lemma III.11.7 gives a family of s -heavy lines of dimension at least h' . Hence A is an (s, h') -Furstenberg set. The sharp lower bound yields

$$t \geq \min\left\{s + h', \frac{3s + h'}{2}, s + 1\right\}. \quad (11.34)$$

Each candidate on the right is $> t$. Indeed,

$$\frac{3s + h'}{2} > t, \quad s + 1 > t,$$

and

$$s + h' > s + 2t - 3s = 2t - 2s \geq t,$$

because $s \leq (2t - 1)/3 \leq t/2$ for $t \leq 2$. This contradiction proves $h \leq 2t - 3s$ in the open range.

At $s = (2t - 1)/3$, take $s_j \uparrow s$. Since $H(A, e, s) \subset H(A, e, s_j)$, the preceding bounds pass to the endpoint. Finally,

$$H(A, e) = \bigcup_{q \in \mathbb{Q}, q > t-1} H(A, e, q),$$

so countable stability of Hausdorff dimension and the limit $s \downarrow t - 1$ give (11.32). $\square$

III.11.4 Heavy lines: a matching all-directions construction

Fix

$$t \in (1, 2], \quad s \in \left[t - 1, \frac{2t - 1}{3}\right], \quad \tau = \min\{2t - 3s, t\}. \quad (11.35)$$

Then

$$1 \leq \tau \leq 2, \quad 0 \leq s \leq 2 - \tau. \quad (11.36)$$

The construction alternates a two-dimensional free branching scale with a τ -dimensional structured scale. The heavy-line construction proves the structured finite block locally. The Ren–Wang boundary reduction also removes the former independent Ren–Wang authority from the upper-bound branch: its middle-regime ingredient is now local relative to the earlier Chapter III.9 regular-set Furstenberg G interface.

Proposition III.11.9 (local Dąbrowski–Orponen–Wang finite block). *With t, s, τ as in (11.35)–(11.36), for every sufficiently small dyadic Δ there are a finite union $\mathcal{P}_\Delta$ of dyadic Δ -squares in $[0, 1]^2$ and a family $\mathcal{E}_\Delta$ of Δ -arcs in a fixed bounded direction chart such that, with a constant depending only on (s, τ) :*

(P1) *for every vertical line $\{x\} \times \mathbb{R}$, the corresponding vertical slice of $\mathcal{P}_\Delta$ contains a nonempty $(\Delta, \tau - 1, C)$ set;*

(P2) *$\mathcal{E}_\Delta$ is $c\Delta^s$ -separated, $|\mathcal{E}_\Delta| \asymp \Delta^{-s}$, and for every $e \in \bigcup \mathcal{E}_\Delta$,*

$$N_\Delta(P_e(\mathcal{P}_\Delta)) \lesssim \Delta^{-(s+\tau)/2}; \quad (11.37)$$

(P3) *the relative scales may be chosen recursively, with $\bar{\Delta}_n < \bar{\Delta}_{n-1}^n$ and arbitrarily stronger finite separation conditions, so that*

$$E = \bigcap_n \bigcup \mathcal{E}_{\bar{\Delta}_n} \quad \text{satisfies} \quad \dim_H E = s. \quad (11.38)$$

Equivalently, after a finite retained family at level $n - 1$ is fixed, $\bar{\Delta}_n$ can be chosen so small that every retained parent arc contains a relative $(\bar{\Delta}_n/\bar{\Delta}_{n-1}, s - 1/n, C)$ family of level- n descendants.

Proof. This is the explicit sheared-grid/Farey construction from Appendix A of Dąbrowski et al. [43], with the integer rounding and the endpoint $\tau = 1$ written out because both are useful in the present self-contained proof. In the parameter range (11.35) one has $s > 0$ and $1 \leq \tau < 2$.

Step 1: the sheared lattice. Choose

$$N = \lfloor c_0 \Delta^{\tau/2-1} \rfloor, \quad h = N\Delta, \quad \theta = \frac{\Delta}{h} = \frac{1}{N}, \quad M = \left\lceil \frac{2}{h} \right\rceil, \quad (11.38a)$$

where $c_0 \in (0, 1/100)$ is fixed and Δ is so small that $N \geq 2$. Then

$$h \asymp \Delta^{\tau/2}, \quad \theta \asymp \Delta^{1-\tau/2}, \quad M \asymp \Delta^{-\tau/2}. \quad (11.38b)$$

Put

$$P' = \{(hk, h\ell) : 0 \leq k, \ell \leq M\}, \quad R_\theta(x, y) = (x + \theta y, y), \quad P = R_\theta(P'). \quad (11.38c)$$

Since $h = N\Delta$ and $\theta h = \Delta$,

$$R_\theta(hk, h\ell) = (\Delta(Nk + \ell), h\ell) \in (\Delta\mathbb{Z})^2. \quad (11.38d)$$

Let $\tilde{P} = P \cup (P - (1/2, 0))$, and let $\mathcal{P}_\Delta$ be the dyadic Δ -squares in $[0, 1]^2$ whose lower-left corner belongs to $\tilde{P}$. Clipping to $[0, 1]^2$ and taking this fixed translate can only improve projection counts.

We verify (P1). First let x lie in a dyadic Δ -column with left endpoint $x_0 = j\Delta \in [1/2, 1)$. Euclidean division gives

$$j = Nk + \ell, \quad 0 \leq \ell < N. \quad (11.38e)$$

Since $x_0 < 1$, $k < 1/(N\Delta) = 1/h$, so $0 \leq k \leq M$; also $\ell < N \leq M$ for small Δ . Thus

$$z_0 = (x_0, h\ell) = R_\theta(hk, h\ell) \in P. \quad (11.38f)$$

If $\tau = 1$, this single point already gives the required nonempty $(\Delta, 0, C)$ slice. Suppose $\tau > 1$. Set

$$v = (0, hN) = (0, N^2\Delta), \quad L = \left\lfloor \frac{1}{8hN} \right\rfloor. \quad (11.38g)$$

Then $hN \asymp \Delta^{\tau-1}$ and $L \asymp \Delta^{1-\tau}$. Moreover

$$R_\theta(h(k-m), h(\ell+mN)) = z_0 + mv. \quad (11.38h)$$

For $0 \leq m \leq L$, the indices on the left remain in $[0, M]$ once Δ is small: from $x_0 \geq 1/2$ we have $k \geq (3h)^{-1}$, whereas $L \leq (8hN)^{-1}$, and $\ell + LN \leq N + (8h)^{-1} < M$. Also the resulting heights stay in $[0, 1]$. Thus the vertical slice contains an arithmetic progression of $\asymp \Delta^{1-\tau}$ points with spacing $\asymp \Delta^{\tau-1}$. For $\Delta \leq r \leq 1$, the number of these points in an interval of length $O(r)$ is

$$\lesssim 1 + r\Delta^{1-\tau} \lesssim (r/\Delta)^{\tau-1}, \quad (11.38i)$$

because $0 \leq \tau-1 \leq 1$ and $r \leq 1$. Hence the selected slice is a $(\Delta, \tau-1, C)$ set. The translated copy $P - (1/2, 0)$ gives the same conclusion in columns with $x < 1/2$, proving (P1) on all of $[0, 1]$.

Step 2: rational directions and small projections. Write

$$\pi_\lambda(x, y) = x + \lambda y.$$

The shear obeys the exact identity

$$\pi_e(R_\theta(x, y)) = \pi_{e+\theta}(x, y). \quad (11.38j)$$

Choose a small fixed $c_1 > 0$ and put

$$Q = \left\lfloor c_1 \Delta^{-s/2} \right\rfloor. \quad (11.38k)$$

Let $\mathcal{F}_Q$ be the reduced fractions $p/q \in [1/4, 3/4]$ with $1 \leq q \leq Q$, and for $\lambda = p/q \in \mathcal{F}_Q$ take the slope $e = \lambda - \theta$. If $(hk, h\ell) \in P'$, then

$$qh^{-1}\pi_\lambda(hk, h\ell) = qk + p\ell.$$

The integer on the right ranges over an interval of length $O(QM)$. Therefore

$$N_\Delta(\pi_e(P)) \lesssim QM \lesssim \Delta^{-(s+\tau)/2}. \quad (11.38l)$$

The same estimate holds for the translated copy. In the fixed bounded slope chart the orthogonal coordinate P_e differs from π_e by multiplication by $(1 + e^2)^{-1/2}$, bounded above and below by absolute constants. Replacing an exact slope e by a slope within $O(\Delta)$ moves the projection of every point in the fixed bounded square by $O(\Delta)$, so either normalization changes the covering number only by a fixed factor.

Distinct reduced fractions with denominators at most Q are separated by at least Q^{-2} . Taking c_1 small makes the corresponding slope centres $c\Delta^s$ -separated even when $s = 1$. We now record the elementary local Farey count needed both for cardinality and for (P3): if I is an interval in a fixed compact subinterval of $(0, 1)$ and

$$|I| \geq CQ^{-1} \log Q, \quad (11.38m)$$

then

$$\#\{p/q \in \mathcal{F}_Q : p/q - \theta \in I\} \asymp |I|Q^2. \quad (11.38n)$$

The upper bound follows from Q^{-2} -separation. For the lower bound, apply Dirichlet's pigeonhole theorem to every $x \in I + \theta$: there are $1 \leq q \leq Q$ and $p \in \mathbb{Z}$ with

$$|x - p/q| \leq (qQ)^{-1}. \quad (11.38o)$$

The union of the approximation intervals with $q \leq cQ$ has total length

$$\leq c|I| + Q^{-1} \log Q.$$

Choose c small and the constant in (11.38m) large. A fixed positive proportion of $I + \theta$ is then covered using denominators $cQ \leq q \leq Q$. For such denominators the intervals in (11.38o) have radius $O(Q^{-2})$, so covering that positive proportion requires $\gtrsim |I|Q^2$ distinct rational centres. Reduction of fractions can only decrease denominators, and the same separation estimate applies, proving (11.38n).

Apply (11.38n) to a fixed interior interval to obtain $|\mathcal{F}_Q| \asymp Q^2 \asymp \Delta^{-s}$. Let $\mathcal{E}_\Delta$ be the closed Δ -arcs around the corresponding slopes $e = \lambda - \theta$. Then the separation, cardinality and projection claims in (P2) follow from (11.38l).

Step 3: the quasi-nested direction limit. We choose the relative scales recursively. Begin with one level-1 direction arc contained strictly inside the fixed slope chart. Once the finite retained parent family at level $n - 1$ has been chosen, take $\bar{\Delta}_n$ so small that, in addition to $\bar{\Delta}_n < \bar{\Delta}_{n-1}^n$,

$$\bar{\Delta}_{n-1} \geq C\bar{\Delta}_n^{s/2} \log(1/\bar{\Delta}_n) \quad (11.38q)$$

and all finitely many parent arcs stay inside the interior chart after the common shift $\theta(\bar{\Delta}_n)$. Then (11.38n), applied in every retained $\bar{\Delta}_{n-1}$ -arc, supplies

$$\asymp \bar{\Delta}_{n-1} \bar{\Delta}_n^{-s} \quad (11.38r)$$

level- n arcs whose centres are $\gtrsim \bar{\Delta}_n^s$ -separated. Discard the $O(1)$ boundary children and retain all the others.

Put $u_n = s - 1/n$. For $n > 1/s$, superlacunarity and (11.38r) imply that every parent has at least $(\bar{\Delta}_{n-1}/\bar{\Delta}_n)^{u_n}$ children. If $\bar{\Delta}_n \leq r \leq \bar{\Delta}_{n-1}$, separation gives

$$\#\{\text{children meeting an interval of length } r\} \lesssim 1 + r\bar{\Delta}_n^{-s}. \quad (11.38s)$$

The second term is at most $(r/\bar{\Delta}_{n-1})^{u_n}$ times the total child count because $u_n \leq 1$. For the first term it is enough to take $r = \bar{\Delta}_n$; then

$$(r/\bar{\Delta}_{n-1})^{u_n} \bar{\Delta}_{n-1} \bar{\Delta}_n^{-s} = \bar{\Delta}_n^{-1/n} \bar{\Delta}_{n-1}^{1-s+1/n} \geq 1 \quad (11.38t)$$

for all large n , since $\bar{\Delta}_n < \bar{\Delta}_{n-1}^n$. Hence, inside every parent, the retained children form a relative family with parameters

$$(\bar{\Delta}_n/\bar{\Delta}_{n-1}, u_n, C).$$

Let F be the resulting nested Cantor set. By construction $F \subset \bigcap_n U_n$, where $U_n = \bigcup \mathcal{E}_{\bar{\Delta}_n}$. For every $\epsilon > 0$, all sufficiently late generations are relative $(s - \epsilon)$ -sets, so the mass-distribution argument of Lemma III.11.11 gives $\dim_H F \geq s - \epsilon$. Thus $\dim_H F \geq s$. Conversely, U_n is covered by $O(\bar{\Delta}_n^{-s})$ intervals of length $O(\bar{\Delta}_n)$, and therefore

$$\dim_H \bigcap_n U_n \leq s. \quad (11.38p)$$

This proves (P3). □

At the endpoint $\tau = 1$, property (P1) asks only for one retained point in every vertical slice, supplied by (11.38f); the remaining construction is unchanged because $s > 0$ in the heavy-line parameter range. This is the only endpoint adjustment to the sheared-lattice and Farey construction.

Choose dyadic scales

$$1 = \Delta_0 \gg \delta_0 \gg \Delta_1 \gg \delta_1 \gg \dots \quad (11.39)$$

so rapidly that

$$\delta_n < \Delta_n^{2n}, \quad \Delta_{n+1} < \delta_n^{2(n+1)}. \quad (11.40)$$

Starting with $R_{\Delta_0} = [0, 1]^2$, let R_{δ_n} be the union of all dyadic δ_n -squares contained in R_{Δ_n} . Inside each such parent square place an affine copy of the block $\mathcal{P}_{\Delta_{n+1}/\delta_n}$ and let their union be $R_{\Delta_{n+1}}$. Put

$$R = \bigcap_n R_{\Delta_n}. \quad (11.41)$$

Thus the passage $\Delta_n \rightarrow \delta_n$ has full two-dimensional branching, while $\delta_n \rightarrow \Delta_{n+1}$ has the structured τ -dimensional branching supplied by the block.

Write $\bar{\Delta}_n = \Delta_n/\delta_{n-1}$. Choose these relative scales recursively by Proposition III.11.9, imposing $\bar{\Delta}_n < \bar{\Delta}_{n-1}^n$ together with the finite descendant conditions from Step 3 of its proof, and define E by (11.38). Then $\dim_H E = s$. For $e \in E$ let

$$\mathcal{L}_F(e) = \{P_e^{-1}\{P_e(z)\} : z \in R\}, \quad \mathcal{L}_F = \bigcup_{e \in E} \mathcal{L}_F(e). \quad (11.42)$$

Lemma III.11.10 (small projections along the structured scales). *For $e \in \bigcup \mathcal{E}_{\bar{\Delta}_n}$,*

$$N_{\Delta_n}(P_e R_{\Delta_n}) \lesssim \Delta_n^{-(s+\tau)/2-1/n}. \quad (11.43)$$

Consequently

$$\dim_{\mathrm{H}} \mathcal{L}_F \leq \frac{3s + \tau}{2}. \quad (11.44)$$

Moreover, for every $z \in R$,

$$\dim_{\mathrm{H}} \{\ell \in \mathcal{L}_F : z \in \ell\} \geq s. \quad (11.45)$$

Proof. At level Δ_n there are at most δ_{n-1}^{-2} parent squares. In each parent, (11.37) at relative scale $\bar{\Delta}_n$ gives at most $\bar{\Delta}_n^{-(s+\tau)/2}$ projected cells. Hence

$$N_{\Delta_n}(P_e R_{\Delta_n}) \lesssim \delta_{n-1}^{-2} \bar{\Delta}_n^{-(s+\tau)/2}.$$

The superlacunarity in (11.40) absorbs every fixed power of δ_{n-1}^{-1} into $\Delta_n^{-1/n}$, proving (11.43).

At scale Δ_n , E is coverable by $\lesssim \Delta_n^{-s-1/n}$ arcs, again by the superlacunary choice. For each such arc the intercept coordinate is covered by $\lesssim \Delta_n^{-(s+\tau)/2-1/n}$ intervals. Bounded line coordinates therefore give

$$N_{\Delta_n}(\mathcal{L}_F) \lesssim \Delta_n^{-(3s+\tau)/2-O(1/n)}.$$

Taking a liminf along the scales proves (11.44). Finally, for fixed $z \in R$ the map $e \mapsto P_e^{-1}\{P_e(z)\}$ is bi-Lipschitz on bounded direction charts, and one such line belongs to $\mathcal{L}_F$ for every $e \in E$. Since $\dim_{\mathrm{H}} E = s$, this proves (11.45). $\square$

We now build a complementary line family directly in line-parameter space. At scale Δ_0 take finitely many bounded parameter squares covering all lines meeting $[0, 1]^2$. If $\mathbf{T}$ is a retained $4\Delta_n$ -tube, choose inside it a maximally separated family of $4\delta_n$ -tubes with cardinality

$$\asymp (\Delta_n/\delta_n)^\tau. \quad (11.46)$$

As in Dąbrowski–Orponen–Wang, we choose this family moderately well distributed in the rescaled two-dimensional line-parameter square: after dilation by $(4\Delta_n)^{-1}$, the nonempty restriction to every fixed side-1/100 parameter square is a $(\delta_n/\Delta_n, \tau, C)$ -set, with C independent of n and of the parent tube. Such a family may be obtained from a fixed Ahlfors- τ model in the parameter square and a maximal δ_n/Δ_n discretisation. Inside each retained $4\delta_n$ -tube keep all dyadic $4\Delta_{n+1}$ subtubes. The limit line set is

$$\mathcal{L}_G = \bigcap_n \bigcup \mathcal{T}_{\delta_n} = \bigcap_n \bigcup \mathcal{T}_{\Delta_n}. \quad (11.47)$$

The scale count gives

$$\dim_{\mathrm{H}} \mathcal{L}_G \leq \tau. \quad (11.48)$$

Indeed, at scale δ_n there are $\lesssim \Delta_n^{-2} \delta_n^{-\tau} \leq \delta_n^{-\tau-1/n}$ retained parameter cells.

Lemma III.11.11 (nested relative-set dimension lemma). *Let F be a nested Cantor set in a bounded Euclidean parameter space, built at super-geometrically decreasing scales ρ_n . Suppose that inside every retained parent the next-generation children form a relative $(\rho_{n+1}/\rho_n, u, C)$ set. Then*

$$\dim_{\mathrm{H}} F \geq u. \quad (11.49)$$

Proof. Distribute mass equally among the children of each parent. Relative u -regularity gives at least $C^{-1}(\rho_n/\rho_{n+1})^u$ children, so a level- n cell has mass at most $C^n \rho_n^u$ up to a fixed normalisation. If $\rho_n \leq r \leq \rho_{n-1}$, relative regularity inside the level- $(n-1)$ parent gives

$$\mu(B(x, r)) \leq C^{n+1} r^u.$$

Super-geometric decay implies $C^n \leq \rho_{n-1}^{-\eta}$ for every fixed $\eta > 0$ and all large n . Thus $\mu(B(x, r)) \lesssim_\eta r^{u-\eta}$. Frostman's lemma gives $\dim_{\text{H}} F \geq u - \eta$, and $\eta \downarrow 0$ proves the claim. $\square$

For $x \in [0, 1]$, let $L_x = \{x\} \times \mathbb{R}$. The free branching from Δ_n to δ_n retains every dyadic interval in each vertical parent, while property (P1) of the imported block gives, from δ_n to Δ_{n+1} , a relative $(\Delta_{n+1}/\delta_n, \tau - 1, C)$ family of vertical descendants. We use these intervals to select a nested subfamily of $\mathcal{L}_G$ whose tubes meet the corresponding vertical fibre.

Lemma III.11.12 (two relative- τ half-steps). *Fix $x \in [0, 1]$. Inside every retained Δ_n -tube meeting a retained vertical Δ_n -interval on L_x , the δ_n -subtubes that continue to meet a retained vertical interval form a relative $(\delta_n/\Delta_n, \tau, C)$ set. Inside every such δ_n -tube, the Δ_{n+1} -subtubes meeting one of the structured vertical descendants form a relative $(\Delta_{n+1}/\delta_n, \tau, C)$ set.*

Proof. This is the local parameter-space argument in Claims 5.10 and 5.12 of Dąbrowski–Orponen–Wang [43], written here because it is the geometric point on which the τ -dimensional fibre conclusion rests.

For the first half-step, let $\mathbf{T}$ be a retained $4\Delta_n$ -tube and let $\mathbf{I} \subset L_x \cap \mathbf{T}$ be the retained vertical Δ_n -interval attached to it. All $4\delta_n$ -children of $\mathbf{T}$ meet L_x at an angle bounded away from zero, so the map from line parameters to the centre of the crossing interval on L_x is uniformly Lipschitz. Restrict to the children whose entire crossing segment on L_x lies in the middle half of $\mathbf{I}$. In the rescaled line-parameter square this condition contains a fixed side-1/100 parameter square and is contained in a bounded union of such squares. By the local regularity built into the choice preceding (11.46), the retained children form a relative $(\delta_n/\Delta_n, \tau, C)$ set. Every one of these children contains a dyadic δ_n -interval of $R_{\delta_n} \cap L_x$, since the free step $\Delta_n \rightarrow \delta_n$ retains all vertical descendants. This proves the first assertion.

For the second half-step, fix one of those $4\delta_n$ -tubes T and its retained vertical δ_n -interval $I \subset T \cap L_x$. Use the uniformly bi-Lipschitz line coordinates

$$(a, b) \mapsto (\text{height of intersection with } L_x, \text{slope}).$$

Property (P1) provides inside I a relative $(\Delta_{n+1}/\delta_n, \tau - 1, C)$ family of permitted heights for the structured vertical children. At each permitted height the slope coordinate ranges over a full one-dimensional interval of $4\Delta_{n+1}$ -subtubes contained in T . Thus the permitted child parameters are, up to bounded multiplicity and a uniform bi-Lipschitz change of coordinates, the product of a relative $(\tau - 1)$ -set with a one-dimensional interval. The product counting inequality gives a relative $(\Delta_{n+1}/\delta_n, \tau, C')$ set, with C' independent of n, x, T . This proves the second assertion. $\square$

Proof status. *The finite double-scale construction and its fibre-counting assertions are proved locally. The equality $\dim_{\text{H}} \mathcal{L} = t$ below uses Theorem III.11.8 for its lower bound; that theorem's Chapter 9 high-multiplicity input X-III.9.ABC49 is locally proved. The heavy-line sharpness theorem therefore has the same local dependency chain.*

Proposition III.11.13 (double-scale line-family assembly). *For every $x \in [0, 1]$,*

$$\dim_H\{\ell \in \mathcal{L}_G : \ell \cap (R \cap L_x) \neq \emptyset\} = \tau. \quad (11.50)$$

If $\mathcal{L} = \mathcal{L}_F \cup \mathcal{L}_G$, then

$$\dim_H \mathcal{L} = t, \quad (11.51)$$

and for every $x \in [0, 1]$ the set

$$R_x = \{z \in R \cap L_x : \dim_H\{\ell \in \mathcal{L} : z \in \ell\} \geq s\} \quad (11.52)$$

satisfies

$$\dim_H\{\ell \in \mathcal{L} : \ell \cap R_x \neq \emptyset\} \geq \tau. \quad (11.53)$$

Proof using the preceding Furstenberg bound for (11.51). Fix x . Starting with a parent tube that crosses the initial vertical interval, retain only those children described in Lemma III.11.12. A nested sequence of retained tubes converges to a line in $\mathcal{L}_G$, and the associated nested vertical intervals converge to a point of $R \cap L_x$. Both half-steps have relative exponent τ , so Lemma III.11.11 gives a τ -dimensional subfamily of lines meeting $R \cap L_x$. The upper bound follows from (11.48); this proves (11.50).

By (11.45), every $z \in R$ lies on an s -dimensional subfamily of $\mathcal{L}_F$, hence $R \cap L_x \subset R_x$. Therefore (11.50) implies (11.53).

For the dimension of $\mathcal{L}$, Lemma III.11.10 and (11.48) give

$$\dim_H \mathcal{L} \leq \max\left\{\frac{3s + \tau}{2}, \tau\right\} = t, \quad (11.54)$$

where the last identity follows from $\tau = \min\{2t - 3s, t\}$. Suppose $u = \dim_H \mathcal{L} < t$. Under point-line duality $\mathcal{L}$ becomes a planar set of dimension u . For every x , (11.53) says that in the dual direction the s -heavy lines cover a set of dimension at least τ . Since $s \geq t - 1 > u - 1$, Theorem III.11.8 applies to the dual set. If $s > (2u - 1)/3$, it says there are no such heavy lines in almost every direction, contradicting the conclusion for every x . Otherwise it gives

$$\dim_H H(\mathcal{L}^*, e, s) \leq \min\{2u - 3s, u\} < \min\{2t - 3s, t\} = \tau$$

for almost every direction, again a contradiction. Hence $u \geq t$, and (11.54) proves (11.51). $\square$

Theorem III.11.14 (heavy-line sharpness in every direction). *For every*

$$t \in (1, 2], \quad s \in \left[t - 1, \frac{2t - 1}{3}\right],$$

there is a compact $K \subset \mathbb{R}^2$ such that

$$\dim_H K = t, \quad \dim_H H(K, e, s) \geq \min\{2t - 3s, t\} \quad (e \in \mathbb{S}^1). \quad (11.55)$$

Thus Theorem III.11.8 is sharp, even with the lower bound required in every direction.

Proof from the preceding Furstenberg theorems. Represent the lines of $\mathcal{L}$ from Proposition III.11.13 in a bounded nonvertical chart as $\ell_{a,b} : y = ax + b$, and let

$$K_0 = \{(a, b) : \ell_{a,b} \in \mathcal{L}\}. \quad (11.56)$$

Bounded point–line duality is bi-Lipschitz, so (11.51) gives $\dim_{\mathbb{H}} K_0 = t$.

Fix $x \in [0, 1]$. For every $z = (x, y) \in R_x$, the dual line $D(-x, y)$ satisfies

$$\dim_{\mathbb{H}}(K_0 \cap D(-x, y)) = \dim_{\mathbb{H}}\{\ell \in \mathcal{L} : z \in \ell\} \geq s. \quad (11.57)$$

Moreover, by bi-Lipschitz duality and (11.53), the union of these heavy slices inside K_0 has Hausdorff dimension at least τ . All lines $D(-x, y)$ have the same direction. As x ranges over a bounded interval, this produces a nontrivial direction arc for which the s -heavy part has dimension at least τ .

Finally take finitely many rotated copies of K_0 so that the corresponding direction arcs cover $\mathbb{S}^1$. A finite union preserves Hausdorff dimension t , and in each direction one copy contributes a τ -dimensional heavy part. Since $\tau = \min\{2t - 3s, t\}$, this is (11.55). $\square$

III.11.5 Visible parts: the sharp almost-everywhere upper bound

We work first in the bounded slope chart $[-1, 1]$. The passage to the full direction circle is a finite chart decomposition. The continuum theorem is reduced to one finite point–ray statement.

Definition (visible and bi-visible parts). *For a compact set $K \subset \mathbb{R}^2$, an oriented slope or direction σ , and $z \in \mathbb{R}^2$, let $\ell_{z,\sigma}^+$ be the forward ray from z and $\ell_{z,\sigma}$ the full line. Put*

$$\text{Vis}_{\sigma}(K) = \{z \in K : K \cap \ell_{z,\sigma}^+ = \{z\}\}, \quad (11.6)$$

$$\text{Vis}_{\sigma}^2(K) = \{z \in K : K \cap \ell_{z,\sigma} = \{z\}\}. \quad (11.7)$$

Thus $\text{Vis}_{\sigma}^2(K) \subset \text{Vis}_{\sigma}(K)$.

Put $\Omega = [-1, 1]^3$, write $\omega = (x, y, \sigma)$ and $z_{\omega} = (x, y)$, and let ℓ_{ω}^+ denote the forward ray from z_{ω} of slope σ . For $u, w \in (0, 1]$ with $uw \geq \delta$, set

$$\mathbf{R}_{u \times uw \times w}(x, y, \sigma) = \{(x + r_1, y + \sigma r_1 + r_2, \sigma + r_3) : |r_1| \leq u, |r_2| \leq uw, |r_3| \leq w\}. \quad (11.58a)$$

A finite nonempty $\mathbf{X} \subset \Omega$ is phase-space Frostman (δ, a, b, C) if

$$|\mathbf{X} \cap \mathbf{R}_{u \times uw \times w}(\omega)| \leq Cu^a w^b |\mathbf{X}| \quad (11.58b)$$

for every admissible u, w, ω . We write

$$\mathcal{J}_{\delta}^+(\mathbf{X}) = |\{(\omega_1, \omega_2) \in \mathbf{X}^2 : \text{dist}(z_{\omega_2}, \ell_{\omega_1}^+) \leq \delta\}|. \quad (11.58c)$$

Imported theorem (discretised ray-incidence principle). *The finite δ^ε ray-incidence statement needed for visibility is proved locally below. Its proof is the ray analogue of the Cohen–Pohoata–Zakharov phase-space incidence scheme [44]: finite-scale uniformisation, branching functions, the local ray initial estimate, the local ray high–low theorem, the effective-triple criterion, and the branching-function theorem. This route is independent of the Furstenberg estimate used in the heavy-line branch.*

For the next lemmas, if $\mathbf{A}, \mathbf{B} \subset \Omega$ are finite, write

$$\mathcal{J}_w^+(\mathbf{A}, \mathbf{B}) = |\{(\omega_1, \omega_2) \in \mathbf{A} \times \mathbf{B} : \text{dist}(z_{\omega_2}, \ell_{\omega_1}^+) \leq w\}|. \quad (11.58e)$$

Choose once and for all a nonnegative radial $\chi \in C_c^\infty(B(0, 1))$ with $\chi \geq 1$ on $B(0, 1/2)$, put $\chi_w = w^{-2}\chi(\cdot/w)$, and define

$$\begin{aligned} f_w &= w^{-1} \sum_{\omega \in \mathbf{A}} \mathbf{1}_{[\ell_\omega^+]_{w/2}}, & g &= \sum_{\omega \in \mathbf{B}} \delta_{z_\omega}, \\ \mathcal{J}_w^+(\mathbf{A}, \mathbf{B}) &= w \langle \chi_w * \chi_{w/2} * f_w, g \rangle. \end{aligned} \quad (11.58f)$$

Finally let

$$\begin{aligned} M_{w \times w}(\mathbf{X}) &= \sup_z |\{\omega \in \mathbf{X} : z_\omega \in B(z, w)\}|, \\ M_{1 \times w}(\mathbf{X}) &= \sup_{\ell_0} |\{\omega \in \mathbf{X} : d_{\mathcal{A}}(\ell_\omega, \ell_0) \leq w\}|, \end{aligned} \quad (11.58g)$$

where on the bounded line chart $d_{\mathcal{A}}$ is any fixed metric bi-Lipschitz equivalent to intercept-plus-slope distance.

Lemma III.11.15 (smoothed ray incidences versus raw incidences). *For $0 < w \leq 1$,*

$$\mathcal{J}_w^+(\mathbf{A}, \mathbf{B}) \lesssim \mathcal{J}_w^+(\mathbf{A}, \mathbf{B}) \lesssim \mathcal{J}_{2w}^+(\mathbf{A}, \mathbf{B}). \quad (11.58h)$$

Proof. The lower estimate follows from $\chi \geq 1$ on $B(0, 1/2)$: after the two convolutions, every point within distance w of a ray receives a fixed positive mass. For the upper estimate, the support of $\chi_w * \chi_{w/2} * \mathbf{1}_{[\ell^+]_{w/2}}$ lies in a fixed enlargement of $[\ell^+]_w$, and its size is $O(w)$. Summing over the two finite families gives the result. $\square$

Lemma III.11.16 (line cancellation kernel). *This is the full-line cancellation estimate used in Appendix A of Orponen–Rutar [45]; we include the proof because it is the only line-theoretic input needed for the local ray high–low theorem. For a full line ℓ , define*

$$\Phi_\ell = \chi_w * \frac{\mathbf{1}_{[\ell]_{w/2}}}{w} - \chi_{w/4} * \frac{\mathbf{1}_{[\ell]_{w/4}}}{w/2}.$$

If $\psi \in C_c^\infty([-3, 3]^2)$, then for lines meeting a fixed bounded enlargement of the square,

$$\left| \int \Phi_{\ell_1} \Phi_{\ell_2} \psi \right| \lesssim \min\{w^{-1}, w^2 d_{\mathcal{A}}(\ell_1, \ell_2)^{-3}\}. \quad (11.58i)$$

Proof. The w^{-1} bound is Cauchy–Schwarz, since each Φ_ℓ has size $O(w^{-1})$ on an $O(w)$ -neighbourhood of a unit-length line segment inside the fixed square. Assume now that $d = d_{\mathcal{A}}(\ell_1, \ell_2) \gg w$ and that the enlarged strips meet. Then their angle ϑ satisfies $\vartheta \gtrsim d$. Their overlap is contained in a parallelogram R of dimensions $O(w)$ by $O(w/\vartheta)$, centred at the intersection point x_R .

Because χ is radial, Φ_{ℓ_j} is even in the signed normal coordinate to ℓ_j . Moreover its normal integral is zero: the two normalized strip profiles have the same total mass. In the affine coordinates given by the signed distances to ℓ_1 and ℓ_2 , this implies that the integral of $\Phi_{\ell_1} \Phi_{\ell_2}$ against every affine function is zero. Expand $\psi(x_R + y) = \psi(x_R) + \nabla \psi(x_R) \cdot y + O(|y|^2)$. Only the quadratic remainder remains. Since $|R| \lesssim w^2 \vartheta^{-1}$, $\text{diam } R \lesssim w \vartheta^{-1}$ and $|\Phi_{\ell_1} \Phi_{\ell_2}| \lesssim w^{-2}$, the integral is $O(w^2 \vartheta^{-3}) = O(w^2 d^{-3})$. $\square$

Theorem III.11.17 (local high–low inequality for rays). *For finite $\mathbf{A}, \mathbf{B} \subset \Omega$ and $0 < w \leq 1/2$,*

$$\left| \frac{\mathcal{J}_w^+(\mathbf{A}, \mathbf{B})}{w|\mathbf{A}||\mathbf{B}|} - \frac{\mathcal{J}_{w/2}^+(\mathbf{A}, \mathbf{B})}{(w/2)|\mathbf{A}||\mathbf{B}|} \right| \lesssim \left(\frac{M_{1 \times w}(\mathbf{A}) M_{w \times w}(\mathbf{B})}{|\mathbf{A}||\mathbf{B}|} w^{-3} \log w^{-1} \right)^{1/2}. \quad (11.58j)$$

Proof. Set $B(w) = \mathcal{J}_w^+ / (w|\mathbf{A}||\mathbf{B}|)$. Factoring the common smoothing at scale $w/2$ and applying Cauchy–Schwarz gives

$$|B(w) - B(w/2)| \lesssim \frac{\|F_w\|_{L^2([-2,2]^2)} \|\chi_{w/2} * g\|_2}{|\mathbf{A}||\mathbf{B}|},$$

where

$$F_w = \sum_{\omega \in \mathbf{A}} \Phi_\omega^+, \quad \Phi_\omega^+ = \chi_w * \frac{\mathbf{1}_{[\ell_\omega^+]_{w/2}}}{w} - \chi_{w/4} * \frac{\mathbf{1}_{[\ell_\omega^+]_{w/4}}}{w/2}.$$

The atomic factor satisfies

$$\|\chi_{w/2} * g\|_2^2 \lesssim w^{-2} |\mathbf{B}| M_{w \times w}(\mathbf{B}). \quad (11.58k)$$

It remains to show

$$\|F_w\|_{L^2([-2,2]^2)}^2 \lesssim w^{-1} \log(w^{-1}) |\mathbf{A}| M_{1 \times w}(\mathbf{A}). \quad (11.58l)$$

Choose ψ equal to one on $[-2, 2]^2$ and supported in $[-3, 3]^2$, and expand the square. Call a pair (ω_1, ω_2) bad if $\text{dist}(z_{\omega_1}, \ell_{\omega_2}) \leq 20w$ or $\text{dist}(z_{\omega_2}, \ell_{\omega_1}) \leq 20w$.

For a non-bad pair, either the two ray profiles are disjoint or, throughout their common support, they coincide with the corresponding full-line profiles. Indeed, if two forward rays fail to meet, common support within $O(w)$ would force one initial point to lie within $20w$ of the other line; if they meet, a point where a ray profile differs from its line profile again lies within $O(w)$ of an initial point and gives the same contradiction. Thus [Theorem III.11.16](#) applies. Dyadically summing in the line-space distance, and using that an r -ball in the two-dimensional bounded line chart is coverable by $O((r/w)^2)$ w -balls, gives total contribution

$$\lesssim w^{-1} |\mathbf{A}| M_{1 \times w}(\mathbf{A}).$$

For bad pairs use only the non-cancellative estimate

$$\left| \int \Phi_{\omega_1}^+ \Phi_{\omega_2}^+ \psi \right| \lesssim \min\{w^{-1}, d_{\mathcal{A}}(\ell_{\omega_1}, \ell_{\omega_2})^{-1}\}. \quad (11.58m)$$

Fixing ω_1 , the set of lines within $20w$ of z_{ω_1} is contained, in line parameter space, in an $O(w)$ -neighbourhood of a line. Its annular piece at line-space distance $r \in [w, 1]$ from ℓ_{ω_1} is coverable by $O(r/w)$ line-space w -balls. Hence each dyadic annulus contributes $O(w^{-1}M_{1 \times w}(\mathbf{A}))$, and there are $O(\log w^{-1})$ annuli. The symmetric bad alternative is identical. This proves (11.58l); combining it with (11.58k) proves (11.58j). $\square$

III.11.5.1 The local phase-space branching engine

We now prove the finite ray-incidence theorem. The argument follows the phase-space architecture of Cohen–Pohoata–Zakharov [44] and the ray adaptation of Orponen–Rutar [45], but all ingredients used below are stated and proved here. The only analytic input specific to rays is the high–low theorem [Theorem III.11.17](#), already proved locally above.

Lemma III.11.18 (finite simultaneous uniformisation). *Let $m, T \in \mathbb{N}$, put $\delta = 2^{-mT}$ and $\Delta_j = 2^{-jT}$. There is an absolute $A \geq 1$ such that every finite nonempty $\mathbf{X} \subset \Omega$ contains a subset $\mathbf{X}'$ with*

$$|\mathbf{X}'| \geq K^{-1}|\mathbf{X}|, \quad K = (Am^3 \log(1/\delta))^{m^3+2}, \quad (11.58r1)$$

which is K -uniform simultaneously for every rectangle scale $\Delta_i \times \Delta_j \times \Delta_k$ with $0 \leq i, j, k \leq m$ and $\Delta_j \geq \Delta_i \Delta_k$. Here K -uniform means that every occupied rectangle at the specified scale contains at least K^{-1} times the maximum occupancy at that scale. For fixed m , $K = \delta^{-o_T \rightarrow \infty(1)}$.

Proof. There are at most $(m+1)^3$ admissible scale triples. Fix one such triple and partition a bounded enlargement of Ω into one of finitely many translated bounded-overlap grids of corresponding phase rectangles. The number of grid cells meeting Ω is at most $C\delta^{-3}$. Discard all cells containing fewer than $\delta^4|\mathbf{X}|$ points; their union contains at most $C\delta|\mathbf{X}|$ points. On the remaining cells every occupancy lies between $\delta^4|\mathbf{X}|$ and $|\mathbf{X}|$, so there are only $O(\log(1/\delta))$ dyadic occupancy classes. One class therefore carries at least $c(\log(1/\delta))^{-1}$ of the points left after the discard. Retain precisely the cells in this class. Their occupancies differ by at most a factor two, hence the retained set is uniformly distributed at the chosen scale up to an absolute factor.

Repeat the operation over the finite list of admissible scales, always applying it to the set retained by the previous stage. Since δ is exponentially small in T , the losses $1 - C\delta$ from the low-occupancy discard may be bounded below by $1/2$ at every stage. The product of the remaining logarithmic losses is at least $(Am^3 \log(1/\delta))^{-(m^3+2)}$ after enlarging A . Finally pass, at each scale, to one of a fixed finite family of translated grids so that every retained δ -cube lies away from grid boundaries; this only changes A . This proves (11.58r1). For fixed m , $\log K = O_m(\log \log(1/\delta)) = o_T(\log(1/\delta))$. $\square$

For an (m, T) -uniform set $\mathbf{X}$ define

$$D_m = \{(x, y, z) \in m^{-1}\mathbb{Z}^3 : 0 \leq x, y \leq 1, 0 \leq z \leq \min\{1, x + y\}\}$$

and its branching function by

$$|\mathbf{X}|_{\delta^x \times \delta^z \times \delta^y} = \delta^{-f_{\mathbf{X}}(x, y, z)}. \quad (11.58r2)$$

The covering number is computed with bounded-overlap phase rectangles of the indicated side lengths. Write $f_{\mathbf{X}}(x, y) = f_{\mathbf{X}}(x, y, x + y)$ and, whenever defined,

$$f_{\mathbf{X}}(u, v, w; x, y) = f_{\mathbf{X}}(x + u, y + v, x + y + w) - f_{\mathbf{X}}(x, y, x + y). \quad (11.58r3)$$

Lemma III.11.19 (branching compactness and algebra). *Along any sequence $m_j \rightarrow \infty$ and then $T_{j,k} \rightarrow \infty$, branching functions of $(m_j, T_{j,k})$ -uniform sets have uniformly convergent subsequences on*

$$D = \{(x, y, z) : 0 \leq x, y \leq 1, 0 \leq z \leq \min\{1, x + y\}\}.$$

Every limit f is nonnegative, coordinatewise monotone, 1-Lipschitz in the ℓ^1 metric, and submodular under taking coordinatewise joins/meets of admissible boxes. Moreover the class of limits is closed under uniform limits and under the blowups

$$Q_w^{x_0, y_0} f(u, v, r) = w^{-1} f(wu, wv, wr; x_0, y_0). \quad (11.58r4)$$

If the original sets are phase-space Frostman $(\delta, a, b, \delta^{-\eta})$ with $\eta \rightarrow 0$, then every limiting branching function satisfies

$$f(x, y) \geq ax + by. \quad (11.58r5)$$

Proof. Nested covering numbers change by at most the reciprocal volume factor when one side length is refined. Taking $\log_{\delta^{-1}}$ gives monotonicity and the 1-Lipschitz estimate. The standard two-cover refinement inequality $N(R \wedge S)N(R \vee S) \leq C_K N(R)N(S)$ for a K -uniform set gives submodularity after division by $\log(1/\delta)$; the $\log C_K$ error is $o_T(1)$ for fixed m . Arzelà–Ascoli gives the compactness. Restricting to a heavily occupied phase rectangle and applying the affine rescaling from (11.58a) changes covering numbers only by powers of K , which vanish in the limit and yields (11.58r4). Finally the Frostman inequality says that each $\delta^x \times \delta^{x+y} \times \delta^y$ rectangle contains at most $\delta^{ax+by-\eta} |\mathbf{X}|$ points, so at least $\delta^{-ax-by+\eta}$ such rectangles are needed; passage to the limit gives (11.58r5). $\square$

For a limiting branching function set

$$\begin{aligned} d(t; x, y) &= f(0, t, 0; x, y), & d^\vee(t; x, y) &= f(t, 0, 0; x, y), \\ b(t; x, y) &= f(t, t, t; x, y) - (f(t, 0, t; x, y) + f(0, t, t; x, y) - t), \\ e(s; x, y) &= \frac{1}{2} (f(s, 0, s; x, y) + f(0, s, s; x, y) - 3s). \end{aligned} \quad (11.58r6)$$

Lemma III.11.20 (direction-number calculus). *For every limiting branching function and every admissible choice of parameters,*

$$d(t + s; x, y) \geq d(t; x, y) + d(s; x, y + t), \quad d(t; x, y) \geq f(0, t, t; x, y) - t, \quad (11.58r6a)$$

and

$$0 \leq d(t + s; x, y) - d(t; x, y) \leq s, \quad 0 \leq d(t; x, y) - d(t; x + x', y) \leq 2x', \quad (11.58r6b)$$

$$|d(t; x, y + y') - d(t; x, y)| \leq 2y'. \quad (11.58r6c)$$

The dual direction number $d^\vee$ satisfies the same statements with the roles of x and y interchanged. In particular,

$$b(t; x, y) \geq d(t; x + t, y) - d(t; x, y), \quad b(t; x, y) \geq d^\vee(t; x, y + t) - d^\vee(t; x, y). \quad (11.58r6d)$$

Proof. All identities are for the renormalised increments in (11.58r3). Submodularity applied to the two phase boxes with increment triples $(0, t + s, t)$ and $(0, t, 0)$ gives

$$f(0, t + s, t; x, y) + f(0, t, 0; x, y) \leq f(0, t + s, 0; x, y) + f(0, t, t; x, y).$$

After subtracting the common increment $f(0, t, t; x, y)$, the left difference is $d(s; x, y + t)$ and the right difference is $d(t + s; x, y) - d(t; x, y)$, proving the first inequality in (11.58r6a). Since the third coordinate changes by t between $(0, t, 0)$ and $(0, t, t)$, monotonicity and the 1-Lipschitz property give $d(t; x, y) = f(0, t, 0; x, y) \geq f(0, t, t; x, y) - t$.

Monotonicity and the 1-Lipschitz bound in the second coordinate give the first pair of bounds in (11.58r6b) concerning the t variable. For the x variable, submodularity gives

$$f(0, t, 0; x, y) \geq f(x', t, x'; x, y) - f(x', 0, x'; x, y) = d(t; x + x', y),$$

and the two renormalised terms change by at most x' each, proving the $2x'$ upper bound. Changing y by y' changes the two terms defining d by at most y' each, which proves (11.58r6c). The dual statements follow by interchanging point and line coordinates.

Finally, by the definitions,

$$b(t; x, y) = d(t; x + t, y) - (f(0, t, t; x, y) - t).$$

The second inequality in (11.58r6a) therefore gives the first inequality in (11.58r6d); the dual identity gives the second. $\square$

A triple $(t; x, y)$ is (c_1, c_2) -effective if $\max\{t, x, y\} \leq c_2 \leq 1/3$ and

$$b(t; x, y) + e(s; x, y) \geq c_1 \quad (t \leq s \leq 1 - x - y). \quad (11.58r7)$$

Call f good if for every $c_2 > 0$ it has a (c_1, c_2) -effective triple for some $c_1 > 0$.

Lemma III.11.21 (local ray initial estimate). *Let $0 < \eta \leq 1/10$ and suppose $\mathbf{X}$ is K -uniform at the three scales $1 \times \eta \times \eta$, $\eta \times \eta \times 1$, and $\eta \times \eta \times \eta$. Then*

$$\frac{\mathcal{J}_{10\eta}^+(\mathbf{X})}{\eta|\mathbf{X}|^2} \gtrsim K^{-5} \frac{|\mathbf{X}|_{\eta \times \eta \times \eta}}{\eta|\mathbf{X}|_{\eta \times \eta \times 1}|\mathbf{X}|_{1 \times \eta \times \eta}}. \quad (11.58r8)$$

Proof. Let $A = |\mathbf{X}|_{\eta \times \eta \times \eta}$ and cover $\mathbf{X}$ by a minimal family of $2 \times \eta \times \eta$ phase rectangles R . Inside each R , the base points lie in an $O(\eta)$ -tube around one line. Choose a maximal η -separated subset P_R of their base points. Uniformity at the three scales gives

$$|P_R| \gtrsim K^{-1}A, \quad |\mathbf{X} \cap R| \gtrsim K^{-1}|\mathbf{X}|/|\mathbf{X}|_{1 \times \eta \times \eta}.$$

If $|P_R| \lesssim 1$, the same inequalities already imply (11.58r8) from the diagonal incidences in R . Otherwise order P_R along the central line of R and split it into left and right subfamilies of comparable size. Every point on the right lies within $O(\eta)$ of the forward ray based at every left point whose associated slope belongs to R . Uniformity at scale $\eta \times \eta \times 1$ converts each selected base point into $\gtrsim K^{-1}|\mathbf{X}|/|\mathbf{X}|_{\eta \times \eta \times 1}$ phase points. Hence R contributes

$$\gtrsim K^{-4}|P_R|^2 \frac{|\mathbf{X}|^2}{|\mathbf{X}|_{\eta \times \eta \times 1} |\mathbf{X}|_{1 \times \eta \times \eta}}$$

ordered ray incidences. Summing over the bounded-overlap family and using $\sum_R |P_R| \gtrsim K^{-1}A|\mathbf{X}|/|\mathbf{X}|_{1 \times \eta \times \eta}$ gives (11.58r8); the displayed power K^{-5} absorbs all bounded-overlap losses. $\square$

Proposition III.11.22 (effective triples yield ray incidences). *For every $c_1, c_2 > 0$ there is $m_0 = m_0(c_1)$ such that, for $m \geq m_0$ and then T sufficiently large, if $\delta = 2^{-m^T}$, $\mathbf{X}$ is (m, T) -uniform, and its branching function has a (c_1, c_2) -effective triple, then*

$$\mathcal{J}_{10\delta}^+(\mathbf{X}) \geq \delta^{1+40c_2} |\mathbf{X}|^2. \quad (11.58r9)$$

Proof. First suppose the effective triple is $(t; 0, 0)$ and put $w = \delta^t$. By uniformity, the right side of (11.58r8) is $K^{-O(1)}\delta^{-b(t;0,0)}$. The local ray high–low theorem Theorem III.11.17, summed dyadically from w down to δ , shows that the normalized incidence density can decrease by at most

$$K^{O(1)} \sum_{s \in [t, 1] \cap m^{-1}\mathbb{Z}} \delta^{e(s;0,0)+O(1/m)}.$$

Effectivity gives $b(t; 0, 0) + e(s; 0, 0) \geq c_1$. Take m so large that the $O(1/m)$ loss is below $c_1/4$, then T so large that every $K^{O(1)}$ is at most $\delta^{-c_1/4}$. The total high–low loss is at most half the initial normalized density. Since $|b(t; 0, 0)| \leq O(t) \leq O(c_2)$ by the Lipschitz bound, the surviving incidence density is at least δ^{40c_2} after harmlessly enlarging the numerical constant.

For a general $(t; x, y)$ fix an occupied $\delta^x \times \delta^{x+y} \times \delta^y$ rectangle R , rescale it by the affine map from (11.58a), and extract an (m', T) -uniform subset $\mathbf{X}_R$ with $m' = m(1 - x - y)$. The covering-number comparison furnished by uniformity is

$$|\mathbf{X}_R|_{\delta^u \times \delta^r \times \delta^v} = K^{O(1)} \delta^{-f_{\mathbf{X}}(u,v,r;x,y)}. \quad (11.58r9a)$$

Apply Theorem III.11.21 to $\mathbf{X}_R$ at $w = \delta^t$, and telescope Theorem III.11.17 over the discrete scales from δ^t down to $\bar{\delta} = \delta^{1-x-y}$. By (11.58r9a), the initial normalized incidence density is $K^{-O(1)}\delta^{-b(t;x,y)}$, while the high–low error at scale δ^s is $K^{O(1)}\delta^{e(s;x,y)-O(1/m)}$. Effectivity therefore gives, after choosing m and then T exactly as in the case $(x, y) = (0, 0)$,

$$\mathcal{J}_{2a\bar{\delta}}^+(\mathbf{X}_R) \gtrsim a\delta^{1+24c_2} |\mathbf{X}|^2 \quad (11.58r9b)$$

for a sufficiently small absolute $a > 0$. Here we used

$$|\mathbf{X}_R| \geq K^{-O(1)}\delta^{3(x+y)}|\mathbf{X}|, \quad \max\{x, y\} \leq c_2,$$

to convert the local cardinality to the global one.

There is one genuinely ray-specific transfer issue: an incidence in the rescaled box transfers to a forward incidence only when the initial first coordinate does not exceed the terminal one. The pairs violating this order and lying within $3a\bar{\delta}$ of a forward ray have base points at distance $O(a\bar{\delta})$. Their number is at most

$$|\mathbf{X}_R| M_{\bar{\delta} \times \bar{\delta}}(\mathbf{X}_R) \leq K^{O(1)} \delta^{1+25c_2} |\mathbf{X}|^2, \quad (11.58r9c)$$

where (11.58r9a), the elementary monotonicity of covering numbers, and $\max\{x, y\} \leq c_2$ are used in the last step. For T large this is smaller than half of (11.58r9b). Hence at least half the incidences in (11.58r9b) have the correct order. The inverse affine rescaling then sends them to genuine forward-ray incidences of width $O(\delta)$ in the original coordinates. After decreasing a once more, their width is at most 10δ . This proves (11.58r9). $\square$

The remaining point is purely about the geometry of limiting branching functions. We give the argument because it is precisely where the condition $a + b > 3$ enters.

Lemma III.11.23 (zooming effective triples). *Let f be a limiting branching function, let $(x_0, y_0) \in [0, 1/5]^2$, and let $0 < w \leq 1 - x_0 - y_0$. Suppose that*

$$e(s; x_0, y_0) \geq rw \quad (w/2 \leq s \leq 1 - x_0 - y_0) \quad (11.58r9d)$$

for some $r > 0$. If the blowup $Q_w^{x_0, y_0} f$ has a (c_1, c_2) -effective triple with $c_2 < r/20$, then f has a

$$(\min\{wc_1, rw/2\}, \max\{x_0, y_0\} + wc_2)$$

-effective triple.

Proof. Let $(t; x, y)$ be effective for the blowup. On scales $wt \leq s \leq w(1 - x - y)$ the scaling identities for b and e give

$$b(wt; x_0 + wx, y_0 + wy) + e(s; x_0 + wx, y_0 + wy) \geq wc_1.$$

For the remaining scales, (11.58r6d) and (11.58r6b) give the crude bound $b(wt; x_0 + wx, y_0 + wy) \geq -2wt \geq -2wc_2$. Moving the base point by at most wc_2 in each coordinate changes each renormalised increment entering e by $O(wc_2)$; the 1-Lipschitz bound gives explicitly

$$e(s; x_0 + wx, y_0 + wy) \geq e(s; x_0, y_0) - 4wc_2 \geq rw - 4wc_2.$$

Thus $b + e \geq w(r - 6c_2) \geq rw/2$ on all larger admissible scales. The pulled-back triple has size at most $\max\{x_0, y_0\} + wc_2$, proving the claim. $\square$

Lemma III.11.24 (maximal-sublevel good-point curve). *Let F be C -Lipschitz on a closed planar domain containing $[0, L/5]^2$, and suppose*

$$F(x, y) \geq \alpha x + \beta y \quad (\alpha, \beta > 0). \quad (11.58r9e)$$

There are a 1-Lipschitz function $p : [0, L/10] \rightarrow [-L/10, L/10]$ and a closed set $E \subset [0, L/10]$ of positive Lebesgue measure such that every $(t + p(t), t - p(t))$, $t \in E$, is 0-good for F . Moreover

$$|E| \gtrsim \frac{L \min\{\alpha, \beta\}}{C}, \quad (11.58r9f)$$

with an absolute implied constant.

Proof. Put $v = L \min\{\alpha, \beta\}/10$. For a dyadic integer $M = 2^k$ set $h = v/M$. Construct points (x_j, y_j) , $0 \leq j < M$, recursively, starting from $(0, 0)$ and choosing $(x_j, y_j) \geq (x_{j-1}, y_{j-1})$ maximal in the coordinatewise order subject to $F(x_j, y_j) \leq jh$. Such a maximal point exists by compactness, and maximality makes it 0-good. The lower bound (11.58r9e) gives $\max\{x_j, y_j\} \leq L/10$. Since $F(L/5, L/5) \geq (\alpha + \beta)L/5 > jh$, continuity along the segment from (x_j, y_j) to $(L/5, L/5)$ shows that maximality forces

$$F(x_j, y_j) = jh. \quad (11.58r9g)$$

Let $t_j = (x_j + y_j)/2$ and $p_M(t_j) = (x_j - y_j)/2$. The monotonicity of the coordinates and (11.58r9g) imply

$$h \leq C((x_{j+1} - x_j) + (y_{j+1} - y_j)) = 2C(t_{j+1} - t_j),$$

so the t_j are $\geq h/C$ -separated. Also $|p_M(t_i) - p_M(t_j)| \leq |t_i - t_j|$. Extend p_M linearly between the t_j and constantly after the last point; it remains 1-Lipschitz. By Arzelà–Ascoli, along a subsequence $p_M \rightarrow p$ uniformly. Let

$$E = \bigcap_{N \geq 1} \overline{\bigcup_{M \geq N} \{t_0, \dots, t_{M-1}\}}.$$

The set of 0-good points is closed, so every $(t + p(t), t - p(t))$ with $t \in E$ is 0-good. Finally, for every fixed $r > 0$ and all sufficiently large M with $h < r$, the separation estimate gives

$$N_r(\{t_j\}) \geq \frac{v}{Cr}, \quad |N_r(\{t_j\})| \geq \frac{v}{C}.$$

Passing first to the limsup set and then letting $r \downarrow 0$ gives $|E| \geq v/C$, which is (11.58r9f). $\square$

Lemma III.11.25 (good-line extraction from a bad branching function). *Suppose f is a limiting branching function with $f(x, y) \geq ax + by$, where $a, b > 0$. Fix $a' < a$ and $b' < b$ with $a' + b' > 3$. If f is not good, then there is a limiting branching function $\tilde{f}$ which is still not good and some $q \in [-1, 1]$ such that every point*

$$(t + qt, t - qt), \quad 0 \leq t \leq 1/10, \quad (11.58r10a)$$

is (a', b') -good for $\tilde{f}$, meaning every admissible positive increment satisfies

$$\tilde{f}(x + u, y + v) - \tilde{f}(x, y) \geq a'u + b'v. \quad (11.58r10b)$$

Proof. Since f is not good, there is a fixed $c_* > 0$ such that f has no (c_1, c_*) -effective triple for any $c_1 > 0$. Set

$$F(x, y) = f(x, y) - a'x - b'y, \quad \alpha = a - a' > 0, \quad \beta = b - b' > 0.$$

Apply Theorem III.11.24 with $L = c_*/5$. We obtain a 1-Lipschitz function p and a positive-measure closed set E such that every $(t + p(t), t - p(t))$, $t \in E$, is 0-good for F , hence (a', b') -good for f . Choose a Lebesgue density point t_0 of E at which p is differentiable, put $q = p'(t_0) \in [-1, 1]$, and write

$$(x_0, y_0) = (t_0 + p(t_0), t_0 - p(t_0)).$$

The construction in [Theorem III.11.24](#) keeps $x_0, y_0 \leq c_*/50$.

For $w \downarrow 0$ consider

$$f_w(u, v, r) = w^{-1} f(wu, wv, wr; x_0, y_0).$$

By branching compactness choose $w_j \downarrow 0$ such that $f_{w_j} \rightarrow \tilde{f}$ uniformly. For each fixed $t \in [0, 1/10]$, density of E at t_0 gives $t_j \rightarrow t$ with $t_0 + w_j t_j \in E$, and differentiability gives

$$\left(t_j + \frac{p(t_0 + w_j t_j) - p(t_0)}{w_j}, t_j - \frac{p(t_0 + w_j t_j) - p(t_0)}{w_j} \right) \rightarrow (t + qt, t - qt).$$

The points on the left are (a', b') -good for f_{w_j} , and the good-point inequality is closed under uniform convergence. Hence every point on the segment [\(11.58r10a\)](#) is (a', b') -good for $\tilde{f}$.

It remains to show that the blowup has not become good. Put $c'_0 = a' + b' - 3 > 0$. Goodness of (x_0, y_0) gives

$$e(s; x_0, y_0) \geq c'_0 s/2.$$

Thus for every small w the hypothesis of [Theorem III.11.23](#) holds with $r = c'_0/4$ on $s \geq w/2$. Let

$$c_b = \min\{c'_0/100, c_*/100\}.$$

If f_w had a (c_1, c_b) -effective triple, the zooming lemma would give an effective triple for f with size parameter at most $c_*/50 + wc_b < c_*$. Such a triple is also (c'_1, c_*) -effective, contradicting the choice of c_* . Therefore every sufficiently small f_w has no effective triple with size parameter c_b . This property is closed under uniform limits: if $\tilde{f}$ had a (c_1, c_b) -effective triple, uniform convergence of b and e would give a $(c_1/2, c_b)$ -effective triple for f_{w_j} for all large j . Hence $\tilde{f}$ is not good. $\square$

Theorem III.11.26 (local branching-function theorem). *Let $a, b \in (1, 2]$ with $a + b > 3$. If a limiting branching function f satisfies $f(x, y) \geq ax + by$ for all admissible (x, y) , then f is good.*

Proof. Put $c_0 = a + b - 3 > 0$. Suppose for contradiction that f is not good. Choose $a' < a$ and $b' < b$ so close to a, b that $a' + b' - 3 > 0$. By [Theorem III.11.25](#), after replacing f by a bad microscopic blowup we may assume that every point on a segment

$$(t + qt, t - qt), \quad 0 \leq t \leq 1/10, \quad (11.58r10)$$

is (a', b') -good. Rename a', b' as a, b and again write $c_0 = a + b - 3 > 0$. We show that for every prescribed $c_2 \in (0, 1/3)$ there is a (c_1, c_2) -effective triple, contradicting badness.

Horizontal and vertical good lines. Suppose first that $q = 1$, so every $(x, 0)$ with $0 \leq x \leq 1/10$ is good. Fix $\eta = c_0/4$ and choose $0 < t < \eta c_2/2$. By [Theorem III.11.20](#),

$$D_t(x) := d(t; x, 0) - d(t; x + t, 0) \geq 0.$$

Since $0 \leq d(t; x, 0) \leq t$,

$$\frac{1}{t} \int_0^{c_2} D_t(x) dx = \frac{1}{t} \left(\int_0^t d(t; x, 0) dx - \int_{c_2}^{c_2+t} d(t; x, 0) dx \right) \leq t. \quad (11.58r10c)$$

Hence the set where $D_t(x) > \eta t$ has length at most $t/\eta < c_2$, and there is some $x \in [0, c_2]$ with $D_t(x) \leq \eta t$. At this point (11.58r6d) gives $b(t; x, 0) \geq -\eta t$, while goodness gives, for every $s \geq t$ for which the inputs are admissible,

$$e(s; x, 0) \geq \frac{1}{2}(a + b - 3)s = \frac{1}{2}c_0 s.$$

Thus $b(t; x, 0) + e(s; x, 0) \geq c_0 t/4$ for all such s , so $(t; x, 0)$ is effective. The case $q = -1$ is identical, using $d^\vee$ and the second inequality in (11.58r6d).

Diagonal good lines. It remains to treat $-1 < q < 1$. By interchanging the two coordinates if necessary, write the good line as (x, mx) with $0 < m \leq 1$. Let $c'_2 = c_2/20$ and parameterize candidate triples by $(x, t) \in B = [0, c'_2]^2$ through

$$(t; x, mx + mt).$$

For $\rho > 0$ put

$$E_{ds}^\rho = \{(x, t) \in B : d(t; x, mx + mt) - d(t; x + t, mx + mt) \leq \rho t\},$$

$$E_{htr}^\rho = \{(x, t) \in B : e(s; x, mx + mt) \geq 2\rho t \text{ for every } s \geq t \text{ in the domain}\}. \quad (11.58r11)$$

If (x, t) lies in both sets, then (11.58r6d) gives $b(t; x, mx + mt) \geq -\rho t$ and hence $b + e \geq \rho t$ for every $s \geq t$; since $x, t, mx + mt < c_2$, this is a $(\rho t, c_2)$ -effective triple. We therefore show that the two sets intersect.

Equip B with $d\mu = dx dt/t$ and let μ_ϵ be its normalization on $t \in [\epsilon, c'_2]$. We first prove that directionally stable pairs have asymptotically full μ_ϵ -measure. Write

$$d(t; x) = d(t; x, mx), \quad \bar{d}(t) = \frac{1}{c'_2} \int_0^{c'_2} d(t; x) dx.$$

The chain inequality (11.58r6a) and monotonicity in the x variable give

$$0 \leq d(t; x, mx + mt) - d(t; x + t, mx + mt) \leq d((m+1)t; x) - d(mt; x) - d(t; x + t). \quad (11.58r11a)$$

Set the right-hand side divided by t equal to $F(x, t) \geq 0$. For fixed t ,

$$\frac{1}{c'_2} \int_0^{c'_2} F(x, t) dx = \frac{\bar{d}((m+1)t) - \bar{d}(mt) - \bar{d}(t)}{t} + O(t/c'_2), \quad (11.58r11b)$$

because shifting the last integral by t creates only two boundary intervals and $0 \leq d(t; x) \leq t$. For every fixed $\lambda > 0$,

$$\frac{1}{\log(c'_2/\epsilon)} \int_\epsilon^{c'_2} \bar{d}(\lambda t) \frac{dt}{t^2} = \frac{\lambda}{\log(c'_2/\epsilon)} \int_\epsilon^{c'_2} \bar{d}(t) \frac{dt}{t^2} + O_\lambda\left(\frac{1}{\log(1/\epsilon)}\right), \quad (11.58r11c)$$

since $0 \leq \bar{d}(t) \leq t$ and the change of variables leaves only the two endpoint intervals. Applying this with $\lambda = m, m+1$ in (11.58r11b), the main terms cancel and yield

$$\int F d\mu_\epsilon \leq \frac{C(c_2, m)}{\log(1/\epsilon)}. \quad (11.58r11d)$$

Outside E_{ds}^ρ , (11.58r11a) gives $F \geq \rho$, so

$$\mu_\epsilon(E_{ds}^\rho) \geq 1 - \frac{C(c_2, m)}{\rho \log(1/\epsilon)}. \quad (11.58r12)$$

We next obtain a positive logarithmic density of high–low regular pairs. Define

$$s_x(u) = u^{-1}[f(x, mx + u) - f(x, mx)].$$

Goodness and the Lipschitz bound give $b \leq s_x(u) \leq 2$, and monotonicity together with Lipschitz regularity gives, for $r \geq 1$,

$$r^{-1}s_x(u) \leq s_x(ru) \leq r^{-1}s_x(u) + 2(r-1)/r. \quad (11.58r13)$$

Fix $\gamma = c_0/4$ and $H = 10/c_0$. Call (x, t) (γ, H) -slope-minimal when $t \leq c'_2/(H+1)$ and

$$s_x(mt + t') \geq s_x(mt) - \gamma \quad (t \leq t' \leq Ht). \quad (11.58r13a)$$

For such a pair and $t \leq s \leq Ht$,

$$f(0, s; x, mx + mt) \geq s s_x(mt) - 2\gamma s,$$

while goodness at (x, mx) gives

$$f(s, 0; x, mx + mt) \geq as + bmt - mt s_x(mt).$$

Since $m \leq 1$, $s \geq t$, and $s_x(mt) \geq b$, these inequalities imply

$$e(s; x, mx + mt) \geq (c_0/2 - \gamma)s \geq c_0 t/4.$$

For $s \geq Ht$, goodness and the 2-Lipschitz bound instead give

$$e(s; x, mx + mt) \geq c_0 s/2 - 2mt \geq (c_0 H/2 - 2)t \geq c_0 t/4.$$

Hence every (γ, H) -slope-minimal pair belongs to $E_{hlr}^{c_0/8}$.

The regularity (11.58r13) also has two elementary consequences. First, if (x, t) is $(\gamma/2, H)$ -slope-minimal and $t \leq c'_2/(2H+1)$, then every $\tilde{t} \in [t, (1+\gamma/100)t]$ is (γ, H) -slope-minimal: apply the lower bound in (11.58r13) at $(m+u)t$ and the upper bound at mt , for $1 \leq u \leq H$. Second, slope-minimal scales occur in every sufficiently long multiplicative interval. Indeed, put

$$k = \lceil 4/\gamma \rceil, \quad L = (m+H)/m, \quad R = L^k.$$

If no $t \in [t_0, Rt_0]$ were $(\gamma/2, H)$ -slope-minimal, start with $t^{(0)} = t_0$. Failure at $t^{(j)}$ gives $u_j \in [t^{(j)}, Ht^{(j)}]$ with $s_x(mt^{(j)} + u_j) < s_x(mt^{(j)}) - \gamma/2$, and set $t^{(j+1)} = (mt^{(j)} + u_j)/m$. Then $t^{(j+1)} \leq Lt^{(j)}$ and

$$s_x(mt^{(j+1)}) < s_x(mt^{(j)}) - \gamma/2.$$

After k steps this contradicts $b \leq s_x \leq 2$. Thus such a scale exists whenever t_0 is small enough that $Rt_0 \leq c'_2/(2H+1)$.

Apply this conclusion on the disjoint logarithmic intervals

$$\left[R^{-2j-1} \frac{c'_2}{3H}, R^{-2j} \frac{c'_2}{3H} \right].$$

For each fixed x choose a $(\gamma/2, H)$ -slope-minimal t_j in each interval. The preceding persistence statement gives a disjoint interval $[t_j, (1 + \gamma/100)t_j]$ of (γ, H) -slope-minimal scales. Fubini with dt/t therefore yields constants $c, C > 0$, depending only on c_0, c_2, m , such that

$$\mu_\epsilon(E_{hlr}^{c_0/8}) \geq c - \frac{C}{\log(1/\epsilon)}. \quad (11.58r14)$$

Choose $\rho = c_0/8$. For sufficiently small ϵ , (11.58r12) and (11.58r14) have sum larger than one, so the two sets intersect. Any point of the intersection gives a $(\rho t, c_2)$ -effective triple, the desired contradiction. The remaining diagonal case, corresponding to slope $m \geq 1$, follows by interchanging the point and line coordinates and using $d^\vee$ in place of d . Hence f is good. $\square$

Theorem III.11.27 (local δ^ϵ ray-incidence theorem). *Let $a, b \in (1, 2]$ satisfy $a + b > 3$, and let $\epsilon > 0$. There are $\zeta = \zeta(a, b, \epsilon) > 0$ and $\delta_0 > 0$ such that every phase-space Frostman $(\delta, a, b, \delta^{-\zeta})$ set $\mathbf{X} \subset \Omega$ with dyadic $0 < \delta \leq \delta_0$ satisfies*

$$\mathcal{J}_\delta^+(\mathbf{X}) \geq \delta^{1+\epsilon} |\mathbf{X}|^2. \quad (11.58d)$$

Proof. Assume the conclusion fails for some a, b, ϵ . Then there are $\eta_k \downarrow 0$, $\delta_k \downarrow 0$ and phase-space Frostman $(\delta_k, a, b, \delta_k^{-\eta_k})$ sets $\mathbf{X}_k$ with $\mathcal{J}_{\delta_k}^+(\mathbf{X}_k) < \delta_k^{1+\epsilon} |\mathbf{X}_k|^2$. Fix $m \geq 5$. For each k choose an integer T_k so that the working dyadic scale $\bar{\delta}_k = 2^{-mT_k}$ satisfies

$$2^{-2m}\delta_k < \bar{\delta}_k \leq 2^{-m}\delta_k. \quad (11.58r15)$$

Thus $10\bar{\delta}_k \leq \delta_k$. Apply Theorem III.11.18 at the scale $\bar{\delta}_k$ and pass to a subsequence, then diagonalize as $m \rightarrow \infty$. Although the last few working boxes may have product below the original cutoff δ_k , the ratio $\bar{\delta}_k/\delta_k$ is bounded above and below by constants depending only on m . Enlarging such a box to the smallest admissible δ_k -box changes its normalized logarithmic covering exponent by $O_m(1/\log(1/\delta_k))$. Hence the original Frostman inequality still yields (11.58r5) in the limit. Therefore Theorem III.11.19 produces a limiting branching function f with $f(x, y) \geq ax + by$, and Theorem III.11.26 makes f good.

Choose $c_2 < \epsilon/100$ and an effective triple of f . Uniform convergence transfers that triple, with half its c_1 , to all sufficiently large uniformized subsets $\mathbf{X}'_k$. Proposition III.11.22 gives

$$\mathcal{J}_{10\bar{\delta}_k}^+(\mathbf{X}'_k) \geq \bar{\delta}_k^{1+40c_2} |\mathbf{X}'_k|^2.$$

The uniformisation loss is $|\mathbf{X}'_k| \geq \bar{\delta}_k^{o(1)} |\mathbf{X}_k|$, and (11.58r15) shows that $\bar{\delta}_k^{1+40c_2}$ differs from $\delta_k^{1+40c_2}$ only by a fixed m -dependent factor. Since $10\bar{\delta}_k \leq \delta_k$ and $40c_2 < \epsilon/2$, the last display is, for k large, strictly larger than $\delta_k^{1+\epsilon} |\mathbf{X}_k|^2$. This contradicts the assumed upper bound for $\mathcal{J}_{\delta_k}^+(\mathbf{X}_k)$ and proves the theorem. $\square$

Lemma III.11.28 (intermediate-scale spatial bipartisation). *Fix $a, b \in (1, 2]$ with $a + b > 3$. Let $\varepsilon > 0$. If $\zeta > 0$ is chosen sufficiently small and $0 < \delta \leq \Delta \ll 1$, then every phase-space Frostman $(\delta, a, b, \Delta^{-\zeta})$ set $\mathbf{X}$ contains spatially Δ -separated subsets $\mathbf{A}, \mathbf{B}$ with*

$$\mathcal{J}_{\Delta}^{+}(\mathbf{A}, \mathbf{B}) \gtrsim \Delta^{1+\varepsilon} |\mathbf{X}|^2, \quad (11.58n)$$

provided the core interface (11.58d) is available at scale Δ .

Proof. Replace ε , if necessary, by a smaller positive number; a bound with the smaller exponent is stronger for $0 < \Delta < 1$. Thus we may assume $\varepsilon < (a-1)/2$. Let ζ_0 be the core exponent associated to $\varepsilon/2$, and choose $0 < \zeta < \zeta_0/2$ also so small that $a - \zeta > 1 + \varepsilon/2$.

Colour the dyadic Δ -squares in the base-point plane by four colours so distinct squares of one colour are Δ -separated, and retain a colour containing a fixed positive fraction of $\mathbf{X}$. Restricting to a fixed positive fraction changes the Frostman constant only by an absolute factor. Moreover the same finite set may be viewed at scale Δ : the phase-space Frostman definition has no separation requirement, and the admissible tests at scale Δ form a subset of those at scale δ . Hence, for Δ sufficiently small, the retained set is Frostman $(\Delta, a, b, \Delta^{-\zeta_0})$. Apply (11.58d) at scale Δ with exponent $\varepsilon/2$.

Write m_{Q_1, Q_2} for the number of resulting Δ -incidences with initial point in Q_1 and terminal point in Q_2 . The diagonal sum is at most $|\mathbf{X}| \max_Q |\{\omega : z_\omega \in Q\}| \lesssim \Delta^{a-\zeta} |\mathbf{X}|^2 = o(\Delta^{1+\varepsilon/2} |\mathbf{X}|^2)$. Apply the elementary random bipartition lemma to the directed graph of distinct Δ -squares weighted by m_{Q_1, Q_2} . One bipartition retains at least one quarter of the off-diagonal weight, and the corresponding phase subsets are spatially Δ -separated. Since $\Delta^{1+\varepsilon/2} \gg \Delta^{1+\varepsilon}$, fixed constants are absorbed for small Δ , proving (11.58n). $\square$

Proposition III.11.29 (polynomial-constant separated ray incidence). *Let $a, b \in (1, 2]$ with $a + b > 3$. There is $K = K(a, b) \geq 1$ such that for every $C \geq 1$ and all sufficiently small dyadic $\delta \leq K^{-1}C^{-K}$, every phase-space Frostman (δ, a, b, C) set $\mathbf{X}$ contains $\mathbf{A}, \mathbf{B} \subset \mathbf{X}$ with*

$$\text{dist}(P[\mathbf{A}], P[\mathbf{B}]) \gtrsim C^{-K}, \quad \mathcal{J}_{\delta}^{+}(\mathbf{A}, \mathbf{B}) \gtrsim C^{-K} \delta |\mathbf{X}|^2. \quad (11.58o)$$

Proof. Put $\varepsilon_0 = (a+b-3)/6 > 0$ and take the core exponent ζ corresponding to ε_0 . Choose a dyadic intermediate scale

$$\Delta \asymp \min\{C^{-1/\zeta}, c_{\varepsilon_0} C^{-1/\varepsilon_0}, \Delta_0\}. \quad (11.58p)$$

Then Theorem III.11.28 gives spatially Δ -separated $\mathbf{A}, \mathbf{B}$ with $\mathcal{J}_{\Delta}^{+}(\mathbf{A}, \mathbf{B}) \gtrsim \Delta^{1+\varepsilon_0} |\mathbf{X}|^2$ by Theorem III.11.15. The phase Frostman condition gives, for dyadic $w \in [\delta/2, \Delta]$,

$$M_{w \times w}(\mathbf{B}) \leq C w^a |\mathbf{X}|, \quad M_{1 \times w}(\mathbf{A}) \leq C w^b |\mathbf{X}|. \quad (11.58q)$$

We telescope with the normalization $|\mathbf{X}|^2$, not with $|\mathbf{A}||\mathbf{B}|$. Indeed, Theorem III.11.17 gives

$$\begin{aligned} & \left| \frac{\mathcal{J}_w^{+}(\mathbf{A}, \mathbf{B})}{w |\mathbf{X}|^2} - \frac{\mathcal{J}_{w/2}^{+}(\mathbf{A}, \mathbf{B})}{(w/2) |\mathbf{X}|^2} \right| \\ & \leq \left(\frac{|\mathbf{A}||\mathbf{B}|}{|\mathbf{X}|^2} \right)^{1/2} \left(\frac{M_{1 \times w}(\mathbf{A}) M_{w \times w}(\mathbf{B})}{|\mathbf{X}|^2} w^{-3} \log w^{-1} \right)^{1/2} \\ & \lesssim C w^{(a+b-3)/2} (\log w^{-1})^{1/2}, \end{aligned}$$

since $|\mathbf{A}|, |\mathbf{B}| \leq |\mathbf{X}|$. Summing over the dyadic scales between Δ and $\delta/2$, the total normalized loss is

$$\lesssim C \sum_{w \leq \Delta} w^{(a+b-3)/2} (\log w^{-1})^{1/2} \leq \frac{1}{2} \Delta^{\varepsilon_0}$$

by the choice of Δ (note $(a+b-3)/2 = 3\varepsilon_0$). Hence $\mathcal{F}_{\delta/2}^+(\mathbf{A}, \mathbf{B}) \gtrsim \Delta^{\varepsilon_0} \delta |\mathbf{X}|^2$, and the upper comparison in [Theorem III.11.15](#) gives the same lower bound for $\mathcal{F}_{\delta}^+(\mathbf{A}, \mathbf{B})$. Since Δ is a fixed negative power of C , this is [\(11.58o\)](#) after enlarging K . $\square$

Proposition III.11.30 (local Orponen–Rutar discretised visibility statement). *Fix $\alpha \in (0, 1]$, $q > 2 - \alpha/2$, and $C \geq 1$. There exist $\eta, \delta_0 > 0$ such that for every dyadic $0 < \delta \leq \delta_0$ the following holds. Let Σ be a δ -separated Frostman (δ, α, C) slope set, and for each $\sigma \in \Sigma$ let P_{σ} be a δ -separated Frostman (δ, q, C) planar point set, with the relative Frostman inequality at the respective exponent (α for Σ and q for P_{σ}). Put $P = \bigcup_{\sigma} P_{\sigma}$. Then for some $\sigma \in \Sigma$ and $z \in P_{\sigma}$,*

$$\text{diam}(P \cap [\ell_{z,\sigma}^+]_{\delta}) \geq \eta. \quad (11.58)$$

Proof. For sufficiently small δ , the case $q > 2$ is vacuous by planar packing: a nonempty δ -separated planar set cannot have cardinality $\asymp \delta^{-q}$ with $q > 2$. Hence we may assume $q \leq 2$. Since $q > 2 - \alpha/2$ and $\alpha \in (0, 1]$, we also have $1 < q + \alpha - 1 \leq 2$, so both phase-space exponents below lie in the range required by the ray-incidence core.

By the standard finite-scale Frostman extraction lemma from Volume I, we may replace Σ and each P_{σ} by subsets, without weakening the desired conclusion, so that they satisfy the absolute Katz–Tao bounds

$$|\Sigma \cap I| \lesssim_C (|I|/\delta)^{\alpha}, \quad |P_{\sigma} \cap B(z, r)| \lesssim_C (r/\delta)^q,$$

and $|\Sigma| \asymp_C \delta^{-\alpha}$, $|P_{\sigma}| \asymp_C \delta^{-q}$. Define

$$\mathbf{X} = \{(x, y, \sigma) : (x, y) \in P_{\sigma}, \sigma \in \Sigma\} \subset \Omega.$$

Then $|\mathbf{X}| \asymp_C \delta^{-q-\alpha}$.

We claim that $\mathbf{X}$ is phase-space Frostman $(\delta, q, q + \alpha - 1, C')$ for some $C' = O_{\alpha,q}(C^{O(1)})$. A rectangle $\mathbf{R}_{u \times uw \times w}$ is coverable by $O(w^{-1})$ rectangles of the form $\mathbf{R}_{uw \times uw \times w}$. The base-point projection of each latter rectangle lies in a ball of radius $O(uw)$ and its slope projection lies in an interval of length $O(w)$. Hence

$$|\mathbf{X} \cap \mathbf{R}_{uw \times uw \times w}| \lesssim_C (w/\delta)^{\alpha} (uw/\delta)^q \lesssim_C w^{\alpha} (uw)^q |\mathbf{X}|.$$

Multiplying by the $O(w^{-1})$ covering multiplicity gives $|\mathbf{X} \cap \mathbf{R}_{u \times uw \times w}| \lesssim_C u^q w^{q+\alpha-1} |\mathbf{X}|$, as required.

Since

$$q + (q + \alpha - 1) > 3$$

is equivalent to $q > 2 - \alpha/2$, [Theorem III.11.29](#) applies. It gives $(z_1, \sigma_1), (z_2, \sigma_2) \in \mathbf{X}$ with $|z_1 - z_2| \geq \eta \asymp_{\alpha,q,C} 1$ and $\text{dist}(z_2, \ell_{z_1, \sigma_1}^+) \leq \delta$. Both z_1, z_2 belong to P , while $z_1 \in P_{\sigma_1}$, so they both lie in $P \cap [\ell_{z_1, \sigma_1}^+]_{\delta}$ and the displayed diameter is at least η . $\square$

For $\eta > 0$ define the truncated visible part

$$\text{Vis}_\sigma^\eta(K) = \{x \in K : K \cap \ell_{x,\sigma}^+ \subset B(x, \eta/8)\}, \quad (11.59)$$

and for $\epsilon > 0$ the robust truncation

$$\text{Vis}_\sigma^{\eta,\epsilon}(K) = \{x \in K : K \cap [\ell_{x,\sigma}^+]_\epsilon \subset B(x, \eta/8)\}. \quad (11.60)$$

Because K is compact,

$$\text{Vis}_\sigma^\eta(K) = \bigcup_{m \geq 1} \text{Vis}_\sigma^{\eta, 2^{-m}}(K). \quad (11.61)$$

Indeed, if the exact ray meets $K \setminus B(x, \eta/8)$ nowhere, compactness gives a positive distance between these two closed sets.

Proposition III.11.31 (continuum-to-discrete visibility contradiction). *Fix $\alpha \in (0, 1]$ and $q > 2 - \alpha/2$. There do not exist $\kappa > 0$ and a compact $K \subset [-1, 1]^2$ such that*

$$\mathcal{H}_\infty^\alpha \{\sigma : \mathcal{H}_\infty^q(\text{Vis}_\sigma(K)) > \kappa\} > \kappa. \quad (11.62)$$

Proof. Assume (11.62). Let η be supplied by the local Proposition III.11.30, with its Frostman constant enlarged by a fixed multiple of κ^{-1} . Since $\text{Vis}_\sigma(K) \subset \text{Vis}_\sigma^\eta(K)$, the bad slope set still has positive α -content for the truncated visible part. By (11.61) and Davies' increasing-sets lemma for Hausdorff content [41], for every bad slope σ the q -content of $\text{Vis}_\sigma^{\eta,\epsilon}(K)$ increases to that of $\text{Vis}_\sigma^\eta(K)$ as $\epsilon \searrow 0$. Hence the slope sets

$$E_\epsilon = \{\sigma : \mathcal{H}_\infty^q(\text{Vis}_\sigma^{\eta,\epsilon}(K)) > \kappa\} \quad (11.63)$$

increase to the truncated bad-slope set. A second application of the same increasing-sets lemma therefore gives one $\epsilon > 0$ for which $\mathcal{H}_\infty^\alpha(E_\epsilon) > \kappa$.

Choose dyadic

$$0 < \delta \ll \min\{\epsilon, \eta, \delta_0\}. \quad (11.64)$$

The finite-scale Frostman extraction lemma from Volume I gives a nonempty $(\delta, \alpha, C/\kappa)$ slope set Σ whose elements are $O(\delta)$ -close to E_ϵ . For each $\sigma \in \Sigma$, choose a nearby $\bar{\sigma} \in E_\epsilon$ and extract a Frostman $(\delta, q, C/\kappa)$ set P_σ of representatives of dyadic δ -squares meeting $\text{Vis}_{\bar{\sigma}}^{\eta,\epsilon}(K)$. Put $P = \bigcup_\sigma P_\sigma$.

We claim that

$$\text{diam}(P \cap [\ell_{z,\sigma}^+]_\delta) \leq \eta/2 \quad (z \in P_\sigma). \quad (11.65)$$

Choose $x \in \text{Vis}_{\bar{\sigma}}^{\eta,\epsilon}(K)$ in the dyadic square represented by z . Then $|x - z| = O(\delta)$ and $|\sigma - \bar{\sigma}| = O(\delta)$. Every point of P is within $O(\delta)$ of K . On the fixed bounded square, an $O(\delta)$ perturbation of the ray $\ell_{z,\sigma}^+$ lies inside the ϵ -neighbourhood of $\ell_{x,\bar{\sigma}}^+$ once (11.64) is strong enough. Definition (11.60) therefore confines all such points to $B(x, \eta/4)$, and (11.65) follows after enlarging constants.

The local discretised visibility Proposition III.11.30 produces some σ, z for which the same diameter is at least η , a contradiction. $\square$

Theorem III.11.32 (sharp parameterised visible-part upper bound). *Let $K \subset \mathbb{R}^2$ be compact and $\alpha \in [0, 1]$. Then*

$$\dim_{\text{H}} \text{Vis}_{\sigma}(K) \leq 2 - \frac{\alpha}{2} \quad \text{for } \mathcal{H}^{\alpha}\text{-almost every direction } \sigma. \quad (11.66)$$

In particular,

$$\dim_{\text{H}} \text{Vis}_{\sigma}(K) \leq \frac{3}{2} \quad \text{for } \mathcal{H}^1\text{-almost every } \sigma. \quad (11.67)$$

Proof. For $\alpha = 0$, the bound is the trivial inequality $\dim_{\text{H}} \text{Vis}_{\sigma}(K) \leq 2$. Assume $\alpha > 0$ and suppose (11.66) fails. By countable exhaustion in the dimension threshold, there are $q > 2 - \alpha/2$ and a positive- $\mathcal{H}^{\alpha}$ set of directions on which $\dim_{\text{H}} \text{Vis}_{\sigma}(K) > q$. By the definition of Hausdorff dimension and a second countable exhaustion, there is $\kappa > 0$ such that the set in (11.62) has positive α -content. This contradicts Proposition III.11.31. A finite slope-chart decomposition gives the statement on the full circle. $\square$

III.11.6 Visible parts: a matching all-directions construction

The lower construction is finite-scale in origin. We first prove the basic avoiding block locally, and then reconstruct the multiscale theorem and the compact limit.

Proposition III.11.33 (basic visibility block). *Fix dyadic $\epsilon, \Delta > 0$. For all sufficiently small dyadic $\delta \leq \Delta/2$ there is a finite family $\mathfrak{P} \subset \mathcal{D}_{\delta}([0, 1]^2)$ with the following property. For every $t \in [1 + \epsilon, 3/2]$ and every $\mathcal{Q} \subset \mathcal{D}_{\Delta}([0, 1]^2)$:*

(B1) *content survives in the basic set,*

$$\mathcal{H}_{\infty}^{t-\epsilon}(\mathcal{Q} \cap \mathfrak{P}) \geq \mathcal{H}_{\infty}^t(\mathcal{Q}); \quad (11.68)$$

(B2) *for every dyadic slope interval $\theta \in \mathcal{D}_{\delta}([0, 1])$ there is a family $\mathcal{T}_{\theta}$ of pairwise 10δ -separated δ -tubes with slopes in θ , and one $(\delta \times \Delta)$ rectangle $R_T \subset T$ for every $T \in \mathcal{T}_{\theta}$, such that*

$$\bigcup \mathfrak{P} \cap \bigcup_{T \in \mathcal{T}_{\theta}} 10T = \emptyset; \quad (11.69)$$

(B3) *with $\mathcal{R}_{\theta} = \{R_T : T \in \mathcal{T}_{\theta}\}$,*

$$\mathcal{H}_{\infty}^{t-\epsilon}(\mathcal{Q} \cap \mathcal{D}_{\delta}(\bigcup \mathcal{R}_{\theta})) \geq \mathcal{H}_{\infty}^t(\mathcal{Q}). \quad (11.70)$$

The normalisation agrees with Orponen–Rutar [45], Proposition 5.1 and Appendix B.

Proof. If $\epsilon > 1/2$, the content interval $[1 + \epsilon, 3/2]$ is empty; take $\mathfrak{P} = \emptyset$ and, for each slope interval, any single tube and rectangle. Thus we assume $0 < \epsilon \leq 1/2$.

We reconstruct the dual-grid argument, using only the sheared-grid/Farey calculation already proved in Proposition III.11.9. The point worth recording is that Steps 1–2 of that proof do not use the heavy-line relation (11.35): for arbitrary

$$1 < \tau < 2, \quad 0 < u \leq 2 - \tau, \quad (11.70a)$$

the same rounded sheared lattice and Farey family give the two finite-scale conclusions used below. We spell them out in the special parameters needed here.

Put

$$\beta = \frac{1-\epsilon}{2}, \quad \tau = \frac{3}{2}. \quad (11.70b)$$

Choose an auxiliary dyadic scale $\underline{\Delta} \leq \Delta/100$ so small that

$$\underline{\Delta}^{-\epsilon/2} \geq C_0 \max\{1, \Delta^{-1}\}, \quad (11.70c)$$

where C_0 is a sufficiently large absolute constant. We shall then take δ dyadic and so small, depending on $(\epsilon, \Delta, \underline{\Delta})$, that every inequality below holds; in particular

$$C_0 \delta^{\epsilon/4} \leq \underline{\Delta}/100, \quad C_0 \delta^{\epsilon/2} \leq \underline{\Delta}^4/100, \quad C_0 \delta^{\beta/2} \log(1/\delta) \leq \underline{\Delta}. \quad (11.70d)$$

The local grid input. Repeat the construction in (11.38a)–(11.38o) with $\tau = 3/2$ and with the Farey exponent s there replaced by β . For this auxiliary lemma use the same lattice on a fixed enlarged box $[-3, 3]^2$ rather than clipping it to the unit square; all index calculations are unchanged up to absolute constants. Concretely, with $N \asymp \delta^{-1/4}$ and $h = N\delta \asymp \delta^{3/4}$, take indices $|k|, |\ell| \leq C/h$ in the sheared lattice $R_{1/N}(hk, h\ell)$. If $\sigma = j\delta \in [0, 1)$, Euclidean division $j = Nk_0 + \ell_0$, $0 \leq \ell_0 < N$, gives a point in the vertical σ -slice. The identity

$$R_{1/N}(h(k_0 - m), h(\ell_0 + mN)) = R_{1/N}(hk_0, h\ell_0) + (0, mN^2\delta)$$

then supplies an arithmetic progression of intercepts with spacing $N^2\delta \asymp \delta^{1/2}$. Because the lattice was taken on the fixed enlarged box, one may let $|m| \lesssim \delta^{-1/2}$, so this progression covers $[-2, 2]$ up to a fixed endpoint margin. Hence, for every $\sigma \in \delta\mathbb{Z} \cap [0, 1)$, there is a set of intercepts $Y_\sigma \subset [-2, 2]$ which is $c\delta^{1/2}$ -separated and satisfies

$$|Y_\sigma \cap I| \asymp |I|\delta^{-1/2} \quad \text{whenever } |I| \geq \underline{\Delta}. \quad (11.70e)$$

The proof of the Farey count (11.38n) is uniform on bounded slope intervals: the same Dirichlet argument applies to reduced fractions in $[0, 2]$. Use all such fractions $\lambda = p/q$ with $q \leq Q \asymp \delta^{-\beta/2}$ for which $e = \lambda - \vartheta \in [0, 1]$. Then every interval $I \subset [0, 1]$ of length at least $CQ^{-1} \log Q$ contains $\asymp |I|Q^2$ such e 's, and distinct ones are $\gtrsim Q^{-2} \asymp \delta^\beta$ apart. Together with the last inequality in (11.70d), this gives the required local count on the whole interval $[0, 1]$. Snap every centre to the nearest point of $\delta\mathbb{Z} \cap [0, 1]$. Since $\delta^\beta \gg \delta$, separation and all local counts are preserved up to constants; on the fixed box the corresponding projections move by only $O(\delta)$, so their δ -covering numbers change by at most a fixed factor. We therefore obtain a set $\Sigma \subset \delta\mathbb{Z} \cap [0, 1]$ satisfying

$$|\Sigma \cap I| \asymp |I|\delta^{-\beta} \quad \text{whenever } I \subset [0, 1], \quad |I| \geq \underline{\Delta}. \quad (11.70f)$$

For the same sheared grid $\mathcal{P} \subset \mathcal{D}_\delta([-3, 3]^2)$, the rational-projection calculation (11.38l) yields

$$N_\delta(\pi_\zeta(\mathcal{P})) \lesssim \delta^{-(\beta+3/2)/2} = \delta^{-1+\epsilon/4}, \quad \zeta \in \Sigma, \quad (11.70g)$$

where $\pi_\zeta(a, b) = \zeta a + b$. This is the only place where the numerical exponent matters; its positive gain $\epsilon/4$ is more than enough for the deletion argument below.

Dual tubes and well-spread rectangles. Fix $\sigma \in \delta\mathbb{Z} \cap [0, 1)$ and enumerate $Y_\sigma = \{b_1 < \dots < b_n\}$. Let

$$\ell_{\sigma,b} = \{(x, y) : y = \sigma x + b\}, \quad T_{\sigma,b} = [\ell_{\sigma,b}]_{\delta/2}, \quad \mathcal{T}_\sigma = \{T_{\sigma,b} : b \in Y_\sigma\}. \quad (11.70h)$$

By (11.70e), distinct intercepts are separated by $\gtrsim \delta^{1/2} \gg \delta$, so, after decreasing δ , the tubes are pairwise 10δ -separated.

Let $\bar{\pi}_\sigma$ denote orthogonal projection onto the line parallel to slope σ . Enumerate the dyadic intervals of length $\Delta/2$ in $[-2, 2]$ as $I_1, \dots, I_m$, with $m \asymp \Delta^{-1}$. Since $n \asymp \delta^{-1/2} \gg m$, define

$$R_{\sigma, b_j} = T_{\sigma, b_j} \cap \bar{\pi}_\sigma^{-1}(2I_i), \quad j \equiv i \pmod{m}. \quad (11.70i)$$

Up to harmless fixed endpoint trimming, these are $(\delta \times \Delta)$ rectangles. Put $\mathcal{R}_\sigma = \{R_{\sigma, b} : b \in Y_\sigma\}$.

Let $\mathcal{T} = \bigcup_{\sigma \in \delta\mathbb{Z} \cap [0, 1)} \mathcal{T}_\sigma$. For $\zeta \in \Sigma$, let $(10\mathcal{T})_\zeta$ be the family of dyadic δ -squares in the vertical ζ -column which meet some tenfold tube $10T$, $T \in \mathcal{T}$. Point-line duality says that the vertical coordinates appearing in this slice lie in an $O(\delta)$ -neighbourhood of $\pi_\zeta(\mathcal{P})$. Hence

$$|(10\mathcal{T})_\zeta| \lesssim N_\delta(\pi_\zeta(\mathcal{P})) \lesssim \delta^{-1+\epsilon/4}. \quad (11.70j)$$

Define

$$\mathfrak{P} = \bigcup_{\zeta \in \Sigma} \{[\zeta, \zeta + \delta) \times [b, b + \delta) \subset [0, 1]^2 : [\zeta, \zeta + \delta) \times [b, b + \delta) \notin (10\mathcal{T})_\zeta\}. \quad (11.70k)$$

By construction, $\bigcup \mathfrak{P}$ misses every $10T$, proving (B2) for the slope σ . If $\theta \in \mathcal{D}_\delta([0, 1))$, take σ to be its left endpoint and write $\mathcal{T}_\theta = \mathcal{T}_\sigma$, $\mathcal{R}_\theta = \mathcal{R}_\sigma$. Thus all tube slopes belong to θ .

Content retained by $\mathfrak{P}$. It suffices to prove (B1) with the coarse family subdivided to scale $\underline{\Delta}$: dyadic subdivision does not change the union represented by $\mathcal{Q}$, and the admissible covering class in the definition of dyadic Hausdorff content is unchanged. Thus assume $\mathcal{Q} \subset \mathcal{D}_{\underline{\Delta}}([0, 1]^2)$. Let $\mathcal{W}$ be an arbitrary dyadic cover of $\mathcal{Q} \cap \mathfrak{P}$. Remove nonmaximal members. We may assume that the members of $\mathcal{W}$ are disjoint and meet the set being covered. We may also assume $\ell(W) \geq \delta$: if a subfamily of smaller dyadic squares covers one retained δ -square, then, since $t - \epsilon < 2$, its total $(t - \epsilon)$ -cost is at least the cost $\delta^{t-\epsilon}$ of that δ -square (use area and $\ell(W)^{t-\epsilon} \geq \delta^{t-\epsilon-2}\ell(W)^2$). Let $\mathcal{Q}_0$ consist of those $Q \in \mathcal{Q}$ not strictly contained in an element of $\mathcal{W}$, and let $\mathcal{W}''$ consist of those $W \in \mathcal{W}$ which are not contained in any $Q \in \mathcal{Q}$.

Fix $Q = I \times J \in \mathcal{Q}_0$. Separation and (11.70f) give, for every dyadic W with $\delta \leq \ell(W) \leq \underline{\Delta}$,

$$|\mathfrak{P} \cap W| \lesssim \begin{cases} \ell(W)^2 \delta^{-1-\beta}, & \delta^\beta \leq \ell(W) \leq \underline{\Delta}, \\ \ell(W) \delta^{-1}, & \delta \leq \ell(W) \leq \delta^\beta. \end{cases} \quad (11.70l)$$

On the other hand, every $\zeta \in \Sigma \cap I$ contributes $\underline{\Delta}/\delta + O(1)$ potential squares in the vertical portion of Q , and at most $C\delta^{-1+\epsilon/4}$ of them were deleted by (11.70j). By (11.70f) and (11.70d),

$$|\mathfrak{P} \cap Q| \gtrsim \underline{\Delta}^2 \delta^{-1-\beta}. \quad (11.70m)$$

Because $Q \in \mathcal{Q}_0$, the set $\mathfrak{P} \cap Q$ is covered by those $W \in \mathcal{W}$ with $W \subset Q$. At least half of its squares are covered by W in one of the two scale ranges in (11.70l).

If the large range $\delta^\beta \leq \ell(W) \leq \underline{\Delta}$ carries at least half, then $t - \epsilon - 2 \leq 0$ and

$$\sum_{\substack{W \in \mathcal{W} \\ W \subset Q}} \ell(W)^{t-\epsilon} \gtrsim \underline{\Delta}^{t-\epsilon-2} \delta^{1+\beta} |\mathfrak{P} \cap Q| \gtrsim \underline{\Delta}^{t-\epsilon} \geq \underline{\Delta}^t. \quad (11.70n)$$

If the small range $\delta \leq \ell(W) \leq \delta^\beta$ carries at least half, then $t - \epsilon \geq 1$ and

$$\sum_{\substack{W \in \mathcal{W} \\ W \subset Q}} \ell(W)^{t-\epsilon} \gtrsim \delta^{t-\epsilon} |\mathfrak{P} \cap Q| \gtrsim \underline{\Delta}^2 \delta^{t-\epsilon-1-\beta}. \quad (11.70o)$$

Here the first factorisation is the counting estimate $\ell(W)^{t-\epsilon} \geq \delta^{t-\epsilon} \ell(W) \delta^{-1}$ together with (11.70l); equivalently one may sum the number of covered δ -squares. Since $\beta = (1 - \epsilon)/2$ and $t \leq 3/2$, $t - \epsilon - 1 - \beta \leq -\epsilon/2$. The second condition in (11.70d) therefore makes the last expression at least $\underline{\Delta}^t$.

Thus every $Q \in \mathcal{Q}_0$ costs at least $\ell(Q)^t$ in the $(t - \epsilon)$ -cover. For completeness, the global replacement is exact. The family $\mathcal{Q}_0 \cup \mathcal{W}''$ covers $\mathcal{Q}$: if $Q \notin \mathcal{Q}_0$, then $Q \subsetneq W$ for some $W \in \mathcal{W}$, and the dyadic nesting forces $W \in \mathcal{W}''$. Conversely, every $W \in \mathcal{W}$ is either in $\mathcal{W}''$ or is contained in some $Q \in \mathcal{Q}_0$; otherwise the disjointness of the maximal cover squares would be violated. Therefore

$$\begin{aligned} \sum_{W \in \mathcal{W}} \ell(W)^{t-\epsilon} &= \sum_{W \in \mathcal{W}''} \ell(W)^{t-\epsilon} + \sum_{Q \in \mathcal{Q}_0} \sum_{\substack{W \in \mathcal{W} \\ W \subset Q}} \ell(W)^{t-\epsilon} \\ &\geq \sum_{W \in \mathcal{W}''} \ell(W)^t + \sum_{Q \in \mathcal{Q}_0} \ell(Q)^t \geq \mathcal{H}_\infty^t(\mathcal{Q}), \end{aligned}$$

where in the first inequality we also used $\ell(W) \leq 1$ and $t - \epsilon \leq t$. Taking the infimum over $\mathcal{W}$ proves (B1).

Content retained by the rectangles. Fix σ and abbreviate

$$\mathfrak{R}_\sigma = \mathcal{D}_\delta \left(\bigcup \mathcal{R}_\sigma \right).$$

Again assume $\mathcal{Q} \subset \mathcal{D}_{\underline{\Delta}}$ and let $Q \in \mathcal{Q}_0$ be as above for an arbitrary dyadic cover of $\mathcal{Q} \cap \mathfrak{R}_\sigma$. The $\delta^{1/2}$ separation of the tubes gives

$$|\mathfrak{R}_\sigma \cap W| \lesssim \begin{cases} \ell(W)^2 \delta^{-3/2}, & \delta^{1/2} \leq \ell(W) \leq \underline{\Delta}, \\ \ell(W) \delta^{-1}, & \delta \leq \ell(W) \leq \delta^{1/2}. \end{cases} \quad (11.70p)$$

We also need a lower bound inside Q . There is an interval $I_Q \subset [-2, 2]$ of length $\asymp \underline{\Delta}$ such that every $T_{\sigma, b}$ with $b \in I_Q$ crosses Q in $\asymp \underline{\Delta}/\delta$ dyadic δ -squares. By (11.70e), $|Y_\sigma \cap I_Q| \asymp \underline{\Delta} \delta^{-1/2}$. Moreover $Y_\sigma \cap I_Q$ is a consecutive block in the increasing enumeration $b_1 < \dots < b_n$. Since $\ell(Q) = \underline{\Delta} \leq \Delta/100$, one of the intervals I_i in (11.70i) satisfies $\tilde{\pi}_\sigma(Q) \subset 2I_i$. Among any $m \asymp \Delta^{-1}$ consecutive indices one has the required residue class modulo m . Hence at least

$$\gtrsim \underline{\Delta} \delta^{-1/2} \geq c \underline{\Delta}^{1+\epsilon/2} \delta^{-1/2} \quad (11.70q)$$

of the selected rectangles cross Q for length $\asymp \underline{\Delta}$. The last inequality is exactly (11.70c). Therefore

$$|\mathfrak{R}_\sigma \cap Q| \gtrsim \underline{\Delta}^{2+\epsilon/2} \delta^{-3/2}. \quad (11.70r)$$

Now repeat the two-scale covering argument. If at least half of $\mathfrak{R}_\sigma \cap Q$ is covered by squares with $\delta^{1/2} \leq \ell(W) \leq \underline{\Delta}$, then

$$\sum_{\substack{W \in \mathcal{W} \\ W \subset Q}} \ell(W)^{t-\epsilon} \gtrsim \underline{\Delta}^{t-\epsilon-2} \delta^{3/2} |\mathfrak{R}_\sigma \cap Q| \gtrsim \underline{\Delta}^{t-\epsilon/2} \geq \underline{\Delta}^t. \quad (11.70s)$$

If instead the range $\delta \leq \ell(W) \leq \delta^{1/2}$ carries at least half, then

$$\sum_{\substack{W \in \mathcal{W} \\ W \subset Q}} \ell(W)^{t-\epsilon} \gtrsim \underline{\Delta}^{2+\epsilon/2} \delta^{t-\epsilon-3/2} \geq \underline{\Delta}^{2+\epsilon/2} \delta^{-\epsilon/2} \geq \underline{\Delta}^t, \quad (11.70t)$$

where we used $t \leq 3/2$ and (11.70d). Replacing local cover pieces by the corresponding coarse squares, exactly as for (B1), proves (B3). This completes the local proof of the basic block. $\square$

For a dyadic slope interval θ and square families $\mathcal{P}, \mathcal{F} \subset \mathcal{D}_\delta$, say that they are θ -disconnected if no width- $\delta/2$ tube with slope in θ meets both $\bigcup \mathcal{P}$ and $\bigcup \mathcal{F}$. Also put

$$\text{diam}_\theta(\mathcal{P}) = \sup_{\sigma(\ell) \in \theta} \text{diam} \left(\bigcup \mathcal{P} \cap [\ell]_{\delta/2} \right). \quad (11.71)$$

Lemma (finite obstacle-refinement lemma). *Fix dyadic $\epsilon, \Delta > 0$ and $M \in \mathbb{N}$. There is a dyadic $\delta \leq \Delta/2$ with the following property. Suppose that for every $\theta \in \mathcal{D}_\Delta([0, 1])$ a family $\mathcal{P}_\theta \subset \mathcal{D}_\Delta([0, 1]^2)$ is given, and let $\mathcal{M}$ be a family of at most M obstacle families $\mathcal{F} \subset \mathcal{D}_\Delta([0, 1]^2)$. Then one can find descendants $\mathcal{P}_\theta \subset \mathcal{D}_\delta$ for every $\theta \in \mathcal{D}_\delta([0, 1])$ and descendants $\mathcal{F}' \subset \mathcal{D}_\delta(\mathcal{F})$ for every $\mathcal{F} \in \mathcal{M}$ such that:*

- (a) *if $\theta \subset \theta$, then $\mathcal{P}_\theta \subset \mathcal{D}_\delta(\mathcal{P}_\theta)$ and $\text{diam}_\theta(\mathcal{P}_\theta) \leq 2\Delta$;*
- (b) *for every $t \in [1 + \epsilon, 3/2]$,*

$$\mathcal{H}_\infty^{t-\epsilon}(\mathcal{P}_\theta) \geq \mathcal{H}_\infty^t(\mathcal{P}_\theta), \quad \mathcal{H}_\infty^{t-\epsilon}(\mathcal{F}') \geq \mathcal{H}_\infty^t(\mathcal{F}); \quad (11.71a)$$

- (c) *every pair $(\mathcal{P}_\theta, \mathcal{F}')$ is θ -disconnected.*

Proof. This is the mechanism of Orponen–Rutar, Lemma 5.6 [45]; we reconstruct it from the basic block because it is the finite induction used below.

First suppose $|\mathcal{M}| = 1$, say $\mathcal{M} = \{\mathcal{F}\}$. Enumerate the coarse slope intervals as

$$\mathcal{D}_\Delta([0, 1]) = \{\theta_1, \dots, \theta_m\}, \quad m = \Delta^{-1}.$$

Set $\rho_0 = \Delta$ and $\mathcal{F}_0 = \mathcal{F}$. At stage k , apply the basic block with ambient scale ρ_{k-1} and content loss ϵ/m . It gives a finer dyadic scale $\rho_k \leq \rho_{k-1}/2$ and a basic family $\mathfrak{P}_k$. Define

$$\mathcal{F}_k = \mathcal{D}_{\rho_k}(\mathcal{F}_{k-1}) \cap \mathfrak{P}_k.$$

Repeated use of (B1) gives, for $1 \leq k \leq m$,

$$\mathcal{H}_\infty^{t-k\epsilon/m}(\mathcal{F}_k) \geq \mathcal{H}_\infty^t(\mathcal{F}), \quad t \in [1 + \epsilon, 3/2]. \quad (11.71b)$$

For every fine slope interval $\theta \subset \theta_k$, property (B3), applied to the full ρ_{k-1} -refinement of $\mathcal{P}_\theta$, provides descendants supported in the rectangles R_T of (B2)–(B3), with

$$\mathcal{H}_\infty^{t-\epsilon/m}(\mathcal{P}_\theta) \geq \mathcal{H}_\infty^t(\mathcal{P}_\theta). \quad (11.71c)$$

The $10\rho_k$ -avoidance in (B2) implies that $(\mathcal{P}_\theta, \mathcal{F}_k)$ is θ -disconnected. Indeed, if a $\rho_k/2$ -neighbourhood of a line with slope in θ meets one of the selected rectangles, then within

the bounded unit square it lies in the corresponding tenfold tube, which is disjoint from $\mathcal{F}_k$. Also each selected rectangle has long side at most ρ_{k-1} , so $\text{diam}_\theta(\mathcal{P}_\theta) \leq 2\rho_{k-1} \leq 2\Delta$.

The obstacle families are nested: $\bigcup \mathcal{F}_k \subset \bigcup \mathcal{F}_{k-1}$. Hence every disconnection created at an earlier stage persists. After the last stage set $\delta = \rho_m$ and $\mathcal{F}' = \mathcal{F}_m$. Descendants constructed at the intermediate scales are refined trivially to scale δ . Refinement preserves nestedness, Hausdorff-content lower bounds, the diameter estimate, and disconnection because $\delta \leq \rho_k$. Equations (11.71b)–(11.71c) therefore prove the lemma when $|\mathcal{M}| = 1$.

For a family of at most M obstacles, iterate the one-obstacle construction at most M times, each time with content loss ϵ/M . At a later stage all previously constructed families are replaced only by descendants. Consequently every old disconnection persists, while the losses add to at most ϵ . This proves the finite-obstacle statement. $\square$

Theorem III.11.34 (finite multiscale separation theorem). *For every dyadic $\epsilon, \Delta > 0$ there is a dyadic $\delta \leq \Delta/2$ with the following property. Suppose that for each $\theta \in \mathcal{D}_\Delta([0, 1])$ a nonempty family $\mathcal{P}_\theta \subset \mathcal{D}_\Delta([0, 1]^2)$ is given. Then for every $\theta \in \mathcal{D}_\delta([0, 1])$ there is a family $\mathcal{P}_\theta \subset \mathcal{D}_\delta([0, 1]^2)$ such that:*

- (i) if $\theta \subset \theta$, then $\mathcal{P}_\theta \subset \mathcal{D}_\delta(\mathcal{P}_\theta)$;
- (ii) $\text{diam}_\theta(\mathcal{P}_\theta) \leq 2\Delta$;
- (iii) for every $t \in [1 + \epsilon, 3/2]$,

$$\mathcal{H}_\infty^{t-\epsilon}(\mathcal{P}_\theta) \geq \mathcal{H}_\infty^t(\mathcal{P}_\theta); \quad (11.72)$$

- (iv) if θ, θ' have different parents at scale Δ , then $\mathcal{P}_\theta$ and $\mathcal{P}_{\theta'}$ are both θ - and θ' -disconnected.

Proof. We follow the invariant-driven proof of Orponen–Rutar, Theorem 5.5 [45]. The point of spelling out the induction is that bilateral disconnection is not preserved merely by processing unordered pairs: at each stage one must know which coarse slope class has already been turned into an obstacle and which direction of disconnection has been established.

The case $\Delta = 1$ is immediate, so assume $m = \Delta^{-1} \geq 2$ and enumerate

$$\mathcal{D}_\Delta([0, 1]) = \{\theta_1, \dots, \theta_m\}.$$

Put $\delta_0 = \Delta$. We construct decreasing dyadic scales $\delta_0 > \delta_1 > \dots > \delta_m$ and, at generation k , families $\mathcal{P}_\theta \subset \mathcal{D}_{\delta_k}$ indexed by $\theta \in \mathcal{D}_{\delta_k}([0, 1])$. The following invariants are maintained.

- (I1) If $0 \leq i \leq j \leq k$, $\theta_i \in \mathcal{D}_{\delta_i}$ and $\theta_j \subset \theta_i$, then

$$\mathcal{P}_{\theta_j} \subset \mathcal{D}_{\delta_j}(\mathcal{P}_{\theta_i}). \quad (11.72a)$$

- (I2) If $\theta \subset \theta$ at generation k , then, for $t \in [1 + \epsilon, 3/2]$,

$$\mathcal{H}_\infty^{t-k\epsilon/m}(\mathcal{P}_\theta) \geq \mathcal{H}_\infty^t(\mathcal{P}_\theta). \quad (11.72b)$$

- (I3) If $j \leq k$, $i \neq j$, and $\theta_i \subset \theta_i$, $\theta_j \subset \theta_j$ are generation- k descendants, then $(\mathcal{P}_{\theta_i}, \mathcal{P}_{\theta_j})$ is θ_i -disconnected.
- (I4) If $k \geq 1$ and $j \neq k$, then every generation- k descendant $\theta \subset \theta_j$ satisfies $\text{diam}_\theta(\mathcal{P}_\theta) \leq \Delta$.

The first two properties hold at $k = 0$, and the last two are vacuous.

Assume the generation- k objects have been constructed with $k < m$. Let

$$\mathcal{M}_k = \{\mathcal{P}_\vartheta : \vartheta \in \mathcal{D}_{\delta_k}(\theta_{k+1})\}.$$

This is a finite family. Apply the finite obstacle-refinement lemma at ambient scale δ_k , with content loss ϵ/m , to all generation- k families and to $\mathcal{M}_k$. We obtain a finer scale δ_{k+1} , source descendants $\overline{\mathcal{P}}_\theta$ for every fine slope θ , and, for every $\mathcal{P}_\vartheta \in \mathcal{M}_k$, an obstacle descendant $\mathcal{P}'_\vartheta$ which is θ -disconnected from every $\overline{\mathcal{P}}_\theta$.

Define the generation- $k + 1$ families by

$$\mathcal{P}_\theta = \begin{cases} \overline{\mathcal{P}}_\theta, & \theta \not\subset \theta_{k+1}, \\ \mathcal{D}_{\delta_{k+1}}(\mathcal{P}'_\vartheta), & \theta \subset \vartheta \subset \theta_{k+1}, \quad \vartheta \in \mathcal{D}_{\delta_k}. \end{cases} \quad (11.72c)$$

The refinement lemma and the inductive hypothesis immediately give (11.72a). Their content estimates compose to give (11.72b) with $k + 1$ in place of k .

We verify the disconnection invariant. If the processed class is an old one $j \leq k$, take the generation- k parents of the two fine slope intervals. By induction the parent families are already disconnected in the required direction, and both new families are subsets of those parents; hence the finer pair remains disconnected. If the newly processed class is $j = k + 1$, then one member of the pair is an obstacle descendant $\mathcal{P}'_\vartheta$ and the other is a source descendant $\overline{\mathcal{P}}_\theta$, so disconnection is exactly the conclusion of the finite obstacle-refinement lemma. This proves (I3). Finally, outside the newly processed coarse class the new family is a source descendant, so the same lemma gives $\text{diam}_\theta \leq \delta_k \leq \Delta$, proving (I4).

After m stages put $\delta = \delta_m$. Property (I1) gives (i), and (I2) with $k = m$ gives (iii). For (iv), take descendants of two distinct coarse parents. Applying (I3) with the first parent as i and the second as the processed j gives disconnection in the first fine direction; exchanging the two parents gives disconnection in the second. For (ii), (I4) directly handles every coarse class except the last one. If $\theta \subset \theta_m$, let $\bar{\theta}$ be its generation- $(m - 1)$ parent. At stage $m - 1$ this class was not the newly processed class, so (I4) gives $\text{diam}_{\bar{\theta}}(\mathcal{P}_{\bar{\theta}}) \leq \Delta$; nestedness and $\theta \subset \bar{\theta}$ then give the same bound for $\mathcal{P}_\theta$. This is stronger than the stated 2Δ bound and completes the proof. $\square$

Proposition III.11.35 (nested compact visibility construction and endpoint diagonalisation). *There exists a compact $K_0 \subset [0, 1]^2$ such that for every $\sigma \in [0, 1]$ there is a compact set $K_0(\sigma) \subset K_0$ satisfying*

$$K_0(\sigma) \subset \text{Vis}_\sigma^2(K_0), \quad \dim_H K_0(\sigma) \geq \frac{3}{2}. \quad (11.73)$$

Proof. We first construct the nested geometry with a flexible sequence of content losses. Start at scale $\delta_0 = 1$ with the unit square. Having constructed nonempty families $\mathcal{P}_\theta$ at slope scale δ_k , apply Theorem III.11.34 with $\Delta = \delta_k$ and a loss parameter $\epsilon_{k+1} > 0$, chosen later. This yields a finer scale $\delta_{k+1} \leq \delta_k/2$ and new families with nestedness, the line-section estimate

$$\text{diam}_\theta(\mathcal{P}_\theta) \leq 2\delta_{k-1} \quad (11.74)$$

for $k \geq 1$, and cross-slope disconnection whenever two fine intervals have different parents. Put

$$K_k = \bigcup_{\theta \in \mathcal{D}_{\delta_k}([0,1))} \overline{\mathcal{P}_\theta}, \quad K_0 = \bigcap_{k=0}^{\infty} K_k. \quad (11.75)$$

For a fixed slope σ , let $\theta_k(\sigma)$ be its dyadic interval and define

$$K_k(\sigma) = \overline{\mathcal{P}_{\theta_k(\sigma)}}, \quad K_0(\sigma) = \bigcap_k K_k(\sigma). \quad (11.76)$$

We first prove bi-visibility, independently of the precise losses. Fix $z \in K_0(\sigma)$ and let ℓ be the full line of slope σ through z . At level k , all families whose slope parents differ from the parent of $\theta_k(\sigma)$ are excluded from ℓ by the bilateral disconnection property. Therefore

$$K_k \cap \ell \subset K_{k-1}(\sigma), \quad \text{diam}(K_k \cap \ell) \leq 2\delta_{k-2}. \quad (11.77)$$

Letting $k \rightarrow \infty$ gives $K_0 \cap \ell = \{z\}$, so $K_0(\sigma) \subset \text{Vis}_\sigma^2(K_0)$.

It remains to choose the losses so that every $K_0(\sigma)$ has dimension at least $3/2$. Let $q_j \uparrow 3/2$ with $1 < q_j < 3/2$. We choose increasing activation stages k_j and the future losses inductively. At stage k_j , every one of the finitely many nonempty families has positive r_j -Hausdorff content for some $q_j < r_j < 3/2$; let $c_j > 0$ be the minimum of these finitely many positive contents. From that stage onward choose the losses so small that

$$\sum_{k > k_j} \epsilon_k < r_j - q_j, \quad \epsilon_k < \frac{q_j - 1}{2} \quad (k > k_j). \quad (11.78)$$

This can be done simultaneously for all previously activated j by choosing the tail losses successively smaller. The second inequality is not cosmetic: it guarantees that every intermediate exponent used to propagate the j th lower bound remains in the admissible range $t \in [1 + \epsilon_k, 3/2]$ of Theorem III.11.34.

The content inequality (11.72), applied at each subsequent generation with the current exponent as its parameter t , first preserves a positive content exponent strictly larger than q_j . Since all relevant diameters are at most 1, $p > q_j$ implies $\mathcal{H}_\infty^{q_j}(E) \geq \mathcal{H}_\infty^p(E)$. Consequently, for all sufficiently large k and every slope family,

$$\mathcal{H}_\infty^{q_j}(\mathcal{P}_{\theta_k(\sigma)}) \geq c_j. \quad (11.79)$$

If $\{U_i\}$ is an open cover of the compact limit $K_0(\sigma)$, then for all large k the compact set $K_k(\sigma)$ is contained in $\bigcup_i U_i$. Estimate (11.79) therefore implies $\sum_i \text{diam}(U_i)^{q_j} \gtrsim c_j$. Thus $\mathcal{H}^{q_j}(K_0(\sigma)) > 0$ for every j , uniformly in the chosen slope. Letting $q_j \uparrow 3/2$ proves (11.73). $\square$

Theorem III.11.36 (matching all-directions $3/2$ visibility construction). *There exists a compact $K \subset \mathbb{R}^2$ such that*

$$\dim_H \text{Vis}_\sigma(K) \geq \frac{3}{2} \quad \text{for every } \sigma \in \mathbb{S}^1. \quad (11.80)$$

In fact

$$\dim_H \text{Vis}_\sigma^2(K) \geq \frac{3}{2} \quad \text{for every } \sigma \in \mathbb{S}^1. \quad (11.81)$$

Proof. We use the four-chart gluing mechanism from Orponen–Rutar, Remark 5.8 [45], but we make the geometric separation explicit. After a similarity we may assume that the one-chart set K_0 from Proposition III.11.35 lies in $B(0, 1/4)$. Let R_φ denote rotation through angle φ , and put

$$w_0 = (0, 0), \quad w_1 = (-5, 2), \quad w_2 = (5, -2), \quad w_3 = (-2, -5). \quad (11.82)$$

Set

$$K = \bigcup_{j=0}^3 (R_{j\pi/4} K_0 + w_j). \quad (11.83)$$

We first record the separation geometry. Write $I_j = [j\pi/4, (j+1)\pi/4)$, with angles understood modulo π since we are considering full lines. For every $i \neq j$, the direction of the vector $w_i - w_j$ stays at angular distance at least

$$\omega_0 = \arctan(2/5) > 0$$

from I_j . This is an elementary check from (11.82): the six unoriented connecting directions are

$$\pi - \arctan(2/5), \quad \arctan(5/2), \quad \pi - \arctan(7/3), \quad \arctan(3/7),$$

with repetitions among the six pairs. None of these directions lies in a sector assigned to either endpoint. Moreover $|w_i - w_j| \geq \sqrt{29}$. Hence, if a line ℓ of direction in I_j meets $B(w_j, 1/4)$, then the distance from w_i to ℓ is at least

$$|w_i - w_j| \sin \omega_0 - \frac{1}{4} \geq \sqrt{29} \frac{2}{\sqrt{29}} - \frac{1}{4} = \frac{7}{4} > \frac{1}{4}. \quad (11.84)$$

Thus such a line misses every $B(w_i, 1/4)$ with $i \neq j$.

Now fix an arbitrary direction $\sigma \in \mathbb{S}^1$. Choose j so that its unoriented angle lies in I_j . Rotating Proposition III.11.35 by $j\pi/4$ gives a compact set

$$K_j(\sigma) \subset R_{j\pi/4} K_0 + w_j, \quad \dim_{\text{H}} K_j(\sigma) \geq \frac{3}{2},$$

such that the full σ -line through every $z \in K_j(\sigma)$ meets the j th component only at z . By (11.84) the same line misses the other three components altogether. Therefore

$$K_j(\sigma) \subset \text{Vis}_\sigma^2(K),$$

and (11.81) follows. Since bi-visible points are visible, (11.80) follows as well. $\square$

III.11.7 Assouad-visible comparison and conclusions

The comparison theorem uses a stronger local covering dimension than the Hausdorff dimension in the preceding visibility results.

Definition (Assouad dimension). *For bounded E , $\dim_{\text{A}} E$ is the infimum of a for which there is C such that*

$$N_r(E \cap B(x, R)) \leq C(R/r)^a \quad (11.8)$$

for all $x \in E$ and $0 < r < R \leq 1$.

External result (Järvenpää–Järvenpää–Suomala–Wu comparison). *For planar self-similar sets generated by an iterated-function system with finite rotation group and satisfying the weak separation condition, the Assouad dimension of a visible part is bounded in terms of the penetrable part as in [46, Theorem 2.4]. In particular, if the relevant orthogonal projection is a finite or countable union of intervals, the visible part is one-dimensional [46, Corollary 2.6].*

This comparison result concerns a structured self-similar setting and an Assouad-dimension conclusion. It is not an input to Theorem III.11.32 or Theorem III.11.36. Terminology from another source should be identified with (11.8) only when the defining covering laws agree.

Theorem III.11.5 and Corollary III.11.6 are fully local relative to the standard entropy/Fourier foundations: the WSC entropy package, the high-level overlap estimate, the empirical-accumulation entropy bound, and the coding Hausdorff estimate are all proved in this chapter. The heavy-line upper theorem uses only the sharp Furstenberg inequality, while the sharpness theorem reconstructs the continuous example from the finite DOW block. The visible-part upper theorem uses only the discretised point-ray proposition, and the all-directions lower example reconstructs the compact limit from the basic visibility block. The comparison theorem above is logically independent of these six conclusions.

Chapter III.12

Fourier Analysis on Curves and Frontiers

The final chapter returns to a theme already present in the distance-set arguments: Fourier information can be converted into geometric information, but only when the quantifiers and the mode of control are kept exact. For Frostman measures on the parabola, the theorem of Orponen–Puliatti–Pyörälä gives averaged L^p control over large frequency balls. From this research-scale input we derive finite energy for suitable convolution powers and a finite-scale approximate quadratic Vinogradov bound. A separate argument uses the sharp planar Furstenberg theorem to prove the explicit growth law for repeated sumsets on the parabola. Its low Furstenberg branch was proved in Chapter 6; the high branch developed in Chapter 7 is proved by the weighted Fu–Ren induction, while the middle branch is supplied by Chapter 11 from the regular-set theorem of Chapter 9.

The chapter ends at the September 2026 frontier. The transversal-family theorem of Orponen–Pyörälä–Yi extends the sharp Furstenberg numerology, and hence the OPP-type averaged Fourier theorem, to general nonzero-curvature graphs. More recent two-ends work gives new L^6 progress below the $s = 2/3$ threshold, but does not close the sharp low-dimensional problem. These frontier statements are recorded for orientation only and are not used in the proofs below.

Prerequisites and source boundary. Fourier-energy and Frostman results come from the foundation volume. The low and high Furstenberg branches are proved in Chapters 6 and 7; Chapter 11 derives the middle branch from the Chapter 9 regular-set theorem (9.46). Below we prove the general-range parabola Fourier estimate by quantitative Furstenberg reduction, capped-energy flattening, and iteration, using Chapter 9’s local robust-projection theorem. The sharp $p = 6$ clause uses local fibrewise Katz–Tao incidence, weighted smoothing, and pair-convolution wave packets. On-parabola smoothing gives the finite-scale estimate; the convolution-energy, sumset, and counting deductions are local. The OPY and two-ends statements at the end are terminal previews.

III.12.1 Parabola measures and Fourier conventions

Definition (parabola and curved graph). *Write*

$$\mathbb{P} = \{(x, x^2) : x \in [-1, 1]\}. \quad (12.1)$$

For $g \in C^3(\mathbb{R})$, write

$$\Gamma_g^1 = \{(x, g(x)) : x \in [-1, 1]\}. \quad (12.2)$$

Definition (Fourier normalisation). For a finite Borel measure σ on $\mathbb{R}^2$,

$$\hat{\sigma}(\xi) = \int e^{-2\pi i x \cdot \xi} d\sigma(x). \quad (12.3)$$

This is the same normalisation used in Chapter III.10.

Definition (curve Frostman measure). An s -Frostman measure on a curve is a finite Borel measure σ supported on that curve and satisfying

$$\sigma(B(x, r)) \leq Cr^s, \quad x \in \mathbb{R}^2, r > 0. \quad (12.4)$$

When a theorem is stated with $C = 1$, a fixed Frostman constant is absorbed into the implicit constant after normalisation.

III.12.2 The parabola Fourier theorem and convolution energy

The first theorem controls Fourier mass on a whole frequency ball. To keep that distinction visible, call an inequality

$$\|\hat{\sigma}\|_{L^p(B(R))} \lesssim R^{(2-t)/p} \quad (12.5)$$

an *averaged Fourier L^p estimate*; it is not a pointwise decay statement.

Its geometric consequence is expressed through convolution powers. For an integer $n \geq 1$ set

$$\sigma^{*n} = \underbrace{\sigma * \cdots * \sigma}_{n \text{ factors}}, \quad nK = \underbrace{K + \cdots + K}_{n \text{ summands}}. \quad (12.6)$$

If $K = \text{supp } \sigma$, then $\text{supp}(\sigma^{*n}) \subset nK$.

For $0 < \tau < 2$, the Riesz energy of a finite measure ν is

$$I_\tau(\nu) = c_\tau \int_{\mathbb{R}^2} |\hat{\nu}(\xi)|^2 |\xi|^{\tau-2} d\xi. \quad (12.7)$$

This Fourier-energy identity is the bridge between the averaged estimate and dimension.

The Fourier argument below needs a finite-scale, not merely a Hausdorff-dimensional, Furstenberg bound. We record its exact form before turning to convolution. Its three parameter branches come from the incidence arguments already developed in Chapters 6, 7, and 11.

Lemma III.12.1 (finite Katz–Tao thinning). Fix $0 \leq d \leq 2$. Let F be a relative (δ, d, C) -set of δ -cells in a fixed bounded two-dimensional chart, with $C \geq 1$. There is a subset $F_0 \subset F$ such that

$$|F_0| \geq cC^{-1}\delta^{-d}, \quad |F_0 \cap Q| \leq C'C(1 + \log(1/\delta))(r/\delta)^d \quad (\delta \leq r \leq 1), \quad (12.7a)$$

for every dyadic r -cell Q . The constants depend only on the fixed chart. The same assertion applies to dyadic tube families through their two-dimensional tube-parameter cells. Every parameter or spatial ball of radius $r \geq \delta$ meets only $O(1)$ adjacent dyadic cells at a comparable scale, so the same family is an absolute Katz–Tao family in the ball formulation after changing C' by a fixed factor.

Proof. The relative bound on one occupied δ -cell gives $|F| \geq C^{-1}\delta^{-d}$. Choose uniformly without replacement $N = \max\{1, \lfloor cC^{-1}\delta^{-d} \rfloor\}$ members of F , with $c > 0$ small enough that $N \leq |F|$. For an r -cell Q , the selected occupancy X_Q has mean at most $NCr^d \leq C(r/\delta)^d$. The elementary factorial-moment bound for sampling without replacement gives

$$\Pr\{X_Q \geq k\} \leq \left(\frac{e \mathbb{E} X_Q}{k} \right)^k.$$

Take k to be a sufficiently large multiple of $C(1 + \log(1/\delta))(r/\delta)^d$. The resulting probability is at most δ^{10} . There are only $O(\delta^{-2}(1 + \log(1/\delta)))$ cells at all dyadic scales in the fixed chart. A union bound gives a choice for which every displayed occupancy bound holds. The case $N = 1$ is immediate. $\square$

Theorem III.12.2 (quantitative planar Furstenberg bound). *Fix $0 < s \leq 1$, $0 \leq t \leq 2$, and $\kappa > 0$, and put*

$$\gamma(s, t) = \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}. \quad (12.7b)$$

There are $\eta > 0$ and $\delta_0 > 0$ such that the following holds for $0 < \delta \leq \delta_0$. Let $\mathcal{L}$ be a nonempty relative $(\delta, t, \delta^{-\eta})$ -family of δ -tubes in a fixed bounded line chart. For every $T \in \mathcal{L}$, let $P(T)$ be a relative $(\delta, s, \delta^{-\eta})$ -family of δ -squares in one fixed bounded spatial region, each meeting a fixed enlargement of T , with $|P(T)| \geq \delta^{-s+\eta}$. Then

$$\left| \bigcup_{T \in \mathcal{L}} P(T) \right| \geq \delta^{-\gamma(s, t) + \kappa}. \quad (12.7c)$$

Proof. We first thin $\mathcal{L}$ by Lemma III.12.1, then thin each fibre independently by the same lemma and trim their sizes to a common integer M . After decreasing η , this leaves

$$|\mathcal{L}'| \geq \delta^{-t+2\eta}, \quad M \geq \delta^{-s+2\eta}, \quad (12.7d)$$

while both the tube family and every fibre are absolute Katz–Tao families with constants at most $\delta^{-3\eta}$. All retained cells and incidences were in the original configuration, so it suffices to bound their union. To apply the line-incidence estimates, replace each square by its centre and use Lemma III.7.4. A square meeting a fixed enlargement of a tube gives an incidence with a further fixed enlargement; finitely many colour classes restore the separated-tube convention. The proofs of Lemma III.6.30.3 and Proposition III.7.28 use only this $O(\delta)$ -incidence relation, so their estimates hold with constants depending on the fixed enlargement. All such constants are absorbed by the strict exponent slack.

If $t = 0$, one fibre already gives $|\bigcup P(T)| \geq M \geq \delta^{-s+2\eta}$, and $\gamma(s, 0) = s$. If $0 < t \leq s$, apply the proved overlap estimate Lemma III.6.30.3 with its parameter $A \leq \delta^{-O(\eta)}$. It yields

$$\left| \bigcup_{T \in \mathcal{L}'} P(T) \right| \gtrsim \delta^{-s-t+O(\eta)}. \quad (12.7e)$$

Here $s + t = \gamma(s, t)$ in this range.

If $s < t < 2 - s$, the bounded-chart point–line duality Lemma III.6.29.1 sends the configuration to a $(\delta, s, \delta^{-O(\eta)}, M)$ -nice configuration whose point family is a $(\delta, t, \delta^{-O(\eta)})$ -set. The middle-regime discrete theorem Proposition III.11.7.13, with its loss chosen below $\kappa/4$, gives

$$\left| \bigcup_{T \in \mathcal{L}'} P(T) \right| \gtrsim M \delta^{-(s+t)/2 + \kappa/4} \geq \delta^{-(3s+t)/2 + \kappa/3} \quad (12.7f)$$

after choosing η small. The first equality in (12.7b) is not the active branch here; $\gamma(s, t) = (3s + t)/2$.

Finally suppose $t \geq 2 - s$. The case $s = t = 1$ is included. Apply the fibrewise Fu–Ren incidence estimate (7.37g) with $\alpha = t$, $\beta = s$, and an arbitrarily small incidence loss $\xi > 0$. Since $s + t \geq 2$, its parameter is $c = 1/(s + t) \leq 1/2$. Writing $P' = \bigcup_{T \in \mathcal{L}'} P(T)$, it gives

$$M|\mathcal{L}'| \lesssim \delta^{-c-\xi-O(\eta)} |P'|^{1-c} |\mathcal{L}'|^{1-c}.$$

Use (12.7d) and

$$\frac{s + tc - c}{1 - c} = s + 1 \quad (c = (s + t)^{-1})$$

to obtain $|P'| \geq \delta^{-s-1+O(\eta+\xi)}$. Since $\gamma(s, t) = s + 1$ here, choosing η, ξ sufficiently small in terms of s, t, κ proves the stronger bound with $\kappa/2$ in place of κ at dyadic scales. The same stronger bound holds in the low and middle cases by reducing their input losses.

For an arbitrary δ , let δ' be the next larger dyadic scale, so $\delta \leq \delta' < 2\delta$. Replace each δ -cell and tube-parameter cell by the dyadic δ' -cell containing its centre; the original cell lies in a fixed enlargement of the chosen cell. Choose one representative from each occupied δ' -cell and trim the fibres to a common degree. The bounded ratio $\delta'/\delta < 2$, together with δ -separation, means that each chosen cell represents only $O(1)$ original cells; the point and tube cardinalities, relative nonconcentration constants, and every fibre degree therefore change by at most absolute factors. The centre/enlargement conversion above preserves incidence. Choose η initially smaller so that these fixed factors fit the dyadic theorem's input threshold. The dyadic estimate at loss $\kappa/2$ then implies (12.7c), because $\delta' \asymp \delta$ and the remaining $\kappa/2$ absorbs all fixed factors for small δ . $\square$

The low and high branches of Theorem III.12.2 are local consequences of Chapters 6 and 7. Its middle branch uses the Chapter 11 discrete theorem, which is proved from Theorem III.9.38; the latter, including its robust-projection input X-III.9.ABC49, is proved in Chapter 9. No independent quantitative Furstenberg research theorem is being imported here.

For the finite-scale convolution argument, fix the ball-average kernel $\psi = \pi^{-1} \mathbf{1}_{B(1)}$ and $\psi_r(x) = r^{-2} \psi(x/r)$. For a finite measure ν and $0 \leq u \leq 2$, define the *capped u -energy at scale r* by

$$I_u^r(\nu) = \iint \max\{|x - y|, r\}^{-u} d\nu(x) d\nu(y). \quad (12.7g)$$

Thus $I_0^r(\nu) = \nu(\mathbb{R}^2)^2$. For $0 < u < 2$, $I_u^r(\nu) \asymp_u I_u(\nu * \psi_r)$, up to a fixed change of scale in the mollifier. We do not identify the capped endpoint $u = 2$ with ordinary Riesz energy; the latter has a logarithmic singularity.

Lemma III.12.3 (capped energy under convolution and across scales). *For probabilities ν, τ on $\mathbb{R}^2$ and $0 < u \leq 2$, there is C_u such that*

$$I_u^r(\nu * \tau) \leq C_u I_u^r(\nu). \quad (12.7h)$$

If $2^{-m-1} < r \leq 2^{-m}$, then for some fixed $c_\psi > 1$,

$$I_u^r(\nu) \lesssim_u 1 + \sum_{j=0}^m 2^{-j(2-u)} \|\nu * \psi_{c_\psi 2^{-j}}\|_2^2. \quad (12.7i)$$

For $u = 0$, the corresponding assertion is the identity $I_0^r = 1$ on probabilities.

Proof. For $u > 0$, put

$$K_{u,r}(z) = u \int_r^\infty \rho^{-u-1} e^{-|z|^2/\rho^2} d\rho.$$

A change of variables gives $K_{u,r}(z) \asymp_u \max\{|z|, r\}^{-u}$, uniformly in z, r . Each Gaussian in this integral is positive definite. Hence Plancherel and $|\hat{\tau}| \leq 1$ show that the energy with kernel $K_{u,r}$ cannot increase under convolution by τ . Comparing kernels before and after convolution proves (12.7h); no false positive-definiteness claim is made for the sharply capped kernel itself.

The layer-cake formula gives

$$I_u^r(\nu) \lesssim_u 1 + \int_r^1 \rho^{-u-1} (\nu \times \nu) \{|x - y| \leq \rho\} d\rho.$$

The autocorrelation of $\psi_{c_\psi \rho}$ is bounded below by $c\rho^{-2}$ when $|x - y| \leq \rho$. Consequently the pair mass in braces is at most $C\rho^2 \|\nu * \psi_{c_\psi \rho}\|_2^2$. Dyadically summing the last integral proves (12.7i), including $u = 2$. $\square$

Lemma III.12.4 (translated-parabola incidence growth). *Fix $0 < s \leq 1$, $0 \leq t \leq 2$, $\kappa > 0$, and a bounded region $B(R_0)$. There are $\eta > 0$ and $\delta_0 > 0$ such that the following holds for $0 < \delta \leq \delta_0$. Let A be a nonempty $(\delta, t, \delta^{-\eta})$ -family of δ -squares in $B(R_0)$, and choose a centre $c(a)$ for each $a \in A$. For every a , let $F(a)$ be a $(\delta, s, \delta^{-\eta})$ -family of δ -squares in a second fixed bounded region, with $|F(a)| \geq \delta^{-s+\eta}$, each meeting the $O(\delta)$ -neighbourhood of $c(a) + \mathbb{P}$. Then*

$$\left| \bigcup_{a \in A} F(a) \right| \geq \delta^{-\gamma(s,t)+\kappa}. \quad (12.7j)$$

Proof. On the fixed bounded region the involution $\Psi(x, y) = (x, x^2 - y)$ and its inverse are Lipschitz. For $c(a) = (b, c)$, the image of $c(a) + \mathbb{P}$ lies on the line

$$y = 2bx - b^2 - c.$$

The parameter map $(b, c) \mapsto (2b, -b^2 - c)$ and its inverse are Lipschitz on bounded sets. Thus Ψ changes the parameter and fibre regularity constants only by fixed factors, after replacing image cells by a bounded number of adjacent grid cells. Every image fibre cell meets a fixed enlargement of its corresponding line. Apply Theorem III.12.2 with an input exponent small enough to absorb these constants and an output loss $\kappa/2$. The inverse Lipschitz map changes covering numbers by only a fixed factor, absorbed by the remaining loss. $\square$

An energy bound is an integral assertion, whereas the incidence theorem asks for finite families with uniform local counts. The next lemma is the exact conversion used below. Its small-scale threshold is *uniform* in the energy bound C , which will itself be a power of δ^{-1} in the application.

Lemma III.12.5 (energy-to-regular-set extraction). *Fix $0 < u < 2$, $0 < \zeta < 1$, and a bounded region $B(R_0)$. There are $\delta_0 > 0$ and C_0 , depending only on u, ζ, R_0 , such that for every $0 < \delta \leq \delta_0$, every $C \geq 1$, every probability μ with $I_u^\delta(\mu) \leq C$, and every Borel $K \subset B(R_0)$ with $\mu(K) \geq \delta^\zeta$, there is a nonempty family $\mathcal{P}$ of δ -grid squares meeting $[K]_{C_0\delta}$ such that*

$$|\mathcal{P}| \geq cC^{-1}\delta^{-u+3\zeta}, \quad |\mathcal{P} \cap B(x, r)| \leq C_0C\delta^{-3\zeta}r^u|\mathcal{P}| \quad (\delta \leq r \leq 1). \quad (12.7k)$$

In particular $\mathcal{P}$ is a $(\delta, u, C_0C\delta^{-3\zeta})$ -set.

Proof. Let $\mu_\delta = \mu * \psi_\delta$, and let K_δ be the union of grid squares meeting $[K]_{2\delta}$. Positivity of the ball kernel gives $\mu_\delta(K_\delta) \geq \mu(K) \geq \delta^\zeta$. Put $U(x) = \int |x - y|^{-u} d\mu_\delta(y)$. The kernel comparison for $0 < u < 2$ gives

$$\int U d\mu_\delta = I_u(\mu_\delta) \leq C_u I_u^\delta(\mu) \leq C_u C.$$

Markov's inequality leaves a subset of K_δ of μ_δ -mass at least $\delta^\zeta/2$ on which $U(x) \leq 4C_u C\delta^{-\zeta}$. At every such point,

$$\mu_\delta(B(x, r)) \leq C'_u C\delta^{-\zeta}r^u \quad (r > 0). \quad (12.7l)$$

At most $O_{R_0}(\delta^{-2})$ grid squares meet K_δ . Consider only the squares that meet the marked subset just selected; their total μ_δ -mass is at least $\delta^\zeta/2$. Discard those whose μ_δ -mass is below $\delta^{2+2\zeta}$; their total mass is $O_{R_0}(\delta^{2\zeta})$, less than half the retained mass for small δ , independently of C . There are $O_\zeta(\log(1/\delta))$ dyadic mass levels among the remaining squares. One level $\mathcal{P}$ carries mass at least $c\delta^{2\zeta}$, after absorbing its logarithm into the spare δ^ζ . If all squares in this level have mass comparable to a , then $a|\mathcal{P}| \gtrsim \delta^{2\zeta}$. For any ball $B(x, r)$ meeting selected squares, choose a marked point in one selected square; all selected squares meeting the ball lie in a ball of radius $O(r)$ about that point. Equation (12.7l) gives $a|\mathcal{P} \cap B(x, r)| \lesssim C\delta^{-\zeta}r^u$. Dividing by the lower mass bound proves the relative estimate in (12.7k). Finally, each selected square has mass at most $C'_u C\delta^{u-\zeta}$ by (12.7l) at radius $O(\delta)$; the level's total mass therefore implies $|\mathcal{P}| \gtrsim C^{-1}\delta^{-u+3\zeta}$. Every selected square meets the specified neighbourhood of K by its construction. $\square$

Proposition III.12.6 (large sets for parabola convolution powers). *Fix $0 < s \leq 1$, $0 \leq t \leq 2$, and $\kappa > 0$. There are $\varepsilon > 0$ and $\delta_0 > 0$ such that the following holds. Suppose μ, σ are probabilities with*

$$\text{spt } \mu \subset B(1), \quad \text{spt } \sigma \subset \mathbb{P}, \quad I_t^\delta(\mu), I_s^\delta(\sigma) \leq \delta^{-\varepsilon}, \quad 0 < \delta \leq \delta_0, \quad (12.7m)$$

and put $\Pi = \mu * \sigma$. If a Borel set $E \subset \mathbb{R}^2$ satisfies $\Pi^{*k}(E) \geq \delta^\varepsilon$ for some integer $k \geq 1$, then

$$|E|_\delta \geq \delta^{-\gamma(s,t)+\kappa}. \quad (12.7n)$$

The covering number may be taken with balls or dyadic squares; fixed enlargements change only the implicit constant absorbed by κ .

Proof. For $k > 1$, Fubini gives a translate $E - z$ with $\Pi(E - z) \geq \delta^\varepsilon$. Since covering numbers are translation invariant up to fixed grid constants, it suffices to treat $k = 1$.

If $t = 0$, choose z with $\sigma(E - z) \geq \delta^\varepsilon$. Cover $E - z$ by N squares of side δ . The within-square contribution to capped energy and Cauchy–Schwarz give

$$I_s^\delta(\sigma) \gtrsim_s \delta^{-s} \frac{\sigma(E - z)^2}{N}.$$

Thus $N \gtrsim_s \delta^{-s+3\varepsilon}$, which proves the claim when $3\varepsilon < \kappa$, since $\gamma(s, 0) = s$. If $t = 2$, choose $t' \in (2 - s, 2)$. Then $\gamma(s, t') = s + 1 = \gamma(s, 2)$, and bounded support gives $I_{t'}^\delta(\mu) \lesssim_s I_2^\delta(\mu)$. The interior result below, with a smaller input exponent, therefore implies the endpoint. We assume $0 < t < 2$ for the remainder.

Let $\varepsilon_{\text{inc}} > 0$ be the input exponent in Lemma III.12.4 with output loss $\kappa/2$. Choose the present ε so small that $10\varepsilon < \varepsilon_{\text{inc}}$, as well as the endpoint requirement. Put

$$K = \{z \in \text{spt } \mu : \sigma(E - z) \geq \delta^{2\varepsilon}\}.$$

The inequality $\delta^\varepsilon \leq \int \sigma(E - z) d\mu(z)$ implies $\alpha := \mu(K) \geq \delta^\varepsilon/2$ for small δ . Apply Lemma III.12.5 to the probability $\mu_K = \alpha^{-1}\mu|_K$, its full-mass set K , and $\zeta = \varepsilon$. Since

$$I_t^\delta(\mu_K) \leq \alpha^{-2} I_t^\delta(\mu) \leq 4\delta^{-3\varepsilon},$$

we obtain a $(\delta, t, O(\delta^{-6\varepsilon}))$ family A of parameter squares, with $|A| \gtrsim \delta^{-t+6\varepsilon}$, meeting $[K]_{O(\delta)}$.

For each $a \in A$, choose $z_a \in K$ within $O(\delta)$ of its centre. Write $\beta_a = \sigma(E - z_a) \geq \delta^{2\varepsilon}$ and apply the same extraction lemma to $\sigma_a = \beta_a^{-1}\sigma|_{E-z_a}$ and its full-mass set $K_a = (E - z_a) \cap \mathbb{P}$, now with $u = s$ and $\zeta = \varepsilon$. Its capped energy is at most $\delta^{-5\varepsilon}$. We obtain a $(\delta, s, O(\delta^{-8\varepsilon}))$ family $F(a)$ of at least $c\delta^{-s+8\varepsilon}$ squares. Translate these squares by z_a and replace them by a fixed number of grid squares. They lie in $[E]_{O(\delta)}$ and meet the $O(\delta)$ -neighbourhood of $c(a) + \mathbb{P}$. All parameter and fibre squares lie in fixed bounded regions because both original measures have fixed bounded supports. Our choice $10\varepsilon < \varepsilon_{\text{inc}}$ ensures the hypotheses of Lemma III.12.4 for small δ . That lemma gives at least $\delta^{-\gamma(s,t)+\kappa/2}$ squares in the union. A fixed enlargement changes the covering number only by an absolute factor, absorbed by the remaining $\kappa/2$. This proves (12.7n). $\square$

Proposition III.12.7 (finite-scale parabola L^2 -flattening). *Fix $0 < s \leq 1$, $0 \leq t \leq 2$, $0 < \kappa \leq 1$, and $R_0 > 0$. There are $\varepsilon > 0$, $k_0 \in \mathbb{N}$, and $\delta_0 > 0$ such that, if $0 < \delta \leq \delta_0$, μ, σ are probabilities with*

$$\text{spt } \mu \subset B(R_0), \quad \text{spt } \sigma \subset \mathbb{P}, \quad I_t^\delta(\mu), I_s^\delta(\sigma) \leq \delta^{-\varepsilon},$$

and $\Pi = \mu * \sigma$, then

$$I_{\gamma(s,t)}^\delta(\Pi^{*k}) \leq \delta^{-\kappa}, \quad k \geq k_0. \quad (12.7o)$$

Proof. Write $v_j = \Pi^{*2^j}$ and $J_r(j) = \|v_j * \psi_r\|_2$. Young's inequality gives $J_r(j+1) \leq J_r(j)$. We first show that there is an integer $J_* = J_*(s, t, \kappa)$ such that for every $\delta \leq r \leq 1$ one can find $j(r) \leq J_*$ with

$$J_r(j(r)) \leq \delta^{-\kappa/4} r^{\gamma(s,t)-2)/2}. \quad (12.7p)$$

Monotonicity then gives the same estimate at the *single* index J_* , simultaneously for all r . Lemma III.12.3 yields

$$I_{\gamma(s,t)}^\delta(v_{J_*}) \lesssim (1 + \log(1/\delta))\delta^{-\kappa/2} \leq \frac{1}{2}\delta^{-\kappa}$$

for sufficiently small δ ; if $\gamma = 0$ the assertion is the identity $I_0^\delta = 1$. For $k \geq 2^{J_*}$, convolve v_{J_*} with the remaining Π -factors and use (12.7h). The fixed comparison factor is absorbed by the same positive power margin. Thus (12.7p) implies (12.7o) with $k_0 = 2^{J_*}$.

It remains to prove (12.7p). Let e_0 be the input exponent of Proposition III.12.6 with output loss $\kappa/8$. Choose

$$0 < \varepsilon < \kappa e_0/16, \quad 0 < e < \min\{e_0/4, \kappa/48\}, \quad J_* = \lceil e^{-1} \rceil + 1. \quad (12.7q)$$

If $r \geq \delta^{\kappa/8}$, the crude density bound $J_r(0) \lesssim_{R_0} r^{-1}$, together with $\gamma(s, t) \leq 2$, proves (12.7p) after shrinking δ_0 . Assume $\delta \leq r < \delta^{\kappa/8}$. If $J_r(j+1) < r^{2e} J_r(j)$ for every $0 \leq j < J_*$, iteration from $J_r(0) \lesssim r^{-1}$ gives $J_r(J_*) \lesssim 1$, which again implies (12.7p). Otherwise, for some $j < J_*$,

$$J_r(j) \geq J_r(j+1) \geq r^{2e} J_r(j). \quad (12.7r)$$

Put $v = v_j$, $f = v * \psi_r$, and $g = v * \psi_{8r}$. The support of f lies in a fixed ball depending only on R_0, J_* , and $\|f\|_2 = J_r(j) \geq c_{R_0, J_*} > 0$. For each square Q of a fixed r -grid, let $a_Q = \sup_Q f$. Discard squares with $a_Q \leq r^{10}$; their union contributes a function f_{l_0} of L^1 -norm $O_{R_0, J_*}(r^{10})$. Group the other squares into sets A_ℓ according to $2^{\ell-1} < a_Q \leq 2^\ell$. There are $O(\log(1/r))$ nonempty levels, and

$$f \leq f_{l_0} + \sum_\ell 2^\ell \mathbf{1}_{A_\ell}, \quad 2^\ell \mathbf{1}_{A_\ell} \lesssim g. \quad (12.7s)$$

For the second inequality, if $x, x' \in Q$ and $\psi_r(x' - y) > 0$, then $|x - y| < (1 + \sqrt{2})r < 8r$, so the ball-average kernels give $\psi_r(x' - y) \leq 64\psi_{8r}(x - y)$. Taking a supremum in x' proves the claim. The overlap of ball-average kernels also gives $\psi_r \lesssim \psi_r * \psi_r$ and $\psi_{8r} \lesssim \psi_{8r} * \psi_r$. Consequently

$$J_r(j+1) \lesssim \|f * f\|_2, \quad \|g\|_2 \lesssim J_r(j).$$

The low part contributes at most $\|f_{l_0} * f\|_2 \lesssim r^{10} J_r(j)$, negligible compared with (12.7r). Pigeonholing the remaining levels produces one $A = A_\ell$ with

$$J_r(j+1) \lesssim (\log(1/r))2^\ell \|\mathbf{1}_A * f\|_2.$$

Plancherel and Cauchy–Schwarz give

$$\|\mathbf{1}_A * f\|_2^2 \leq \|\mathbf{1}_A * \mathbf{1}_A\|_2 J_r(j+1), \quad \|\mathbf{1}_A * \mathbf{1}_A\|_2 \leq |A|^{3/2}.$$

Use (12.7r), absorb $\log^2(1/r)$ into r^{-e} , and enlarge fixed constants to obtain

$$J_r(j) \lesssim r^{-3e} 2^{2\ell} |A|^{3/2}. \quad (12.7t)$$

Since $2^\ell \mathbf{1}_A \lesssim g$, $2^\ell |A| \lesssim 1$ and $2^\ell |A|^{1/2} \lesssim \|g\|_2 \lesssim J_r(j)$. Combining these with (12.7t) gives

$$J_r(j) \lesssim r^{-1-3e} |A|^{-1/2}, \quad \int_A g \gtrsim r^{3e}, \quad (12.7u)$$

because A is a union of r -squares and $|A| \asymp r^2|A|_r$.

The support of ψ_{8r} now implies $\nu([A]_{8r}) \geq r^{3e}$. Since $I_u^r \leq I_u^\delta$ for $r \geq \delta$, the energy hypotheses at δ remain valid at r , with upper bound $\delta^{-\varepsilon}$. The inequality $r < \delta^{\kappa/8}$ and the choice of ε give $\delta^{-\varepsilon} \leq r^{-e_0/2}$ for sufficiently small δ , hence the r^{-e_0} energy threshold of Proposition III.12.6. Also $3e < e_0$, so $r^{3e} \geq r^{e_0}$. If $R_0 > 1$, first apply the fixed parabolic contraction $(x_1, x_2) \mapsto (S^{-1}x_1, S^{-2}x_2)$, $S = 2 \max\{1, R_0\}$. It preserves the truncated parabola, brings the translation support into $B(1)$, and changes energies and scales by constants depending only on R_0 ; the strict exponent margins absorb them. Apply the heavy-set proposition at a scale comparable to r to $[A]_{8r}$, or to its image under the fixed parabolic contraction. A fixed enlargement of a union of r -squares has covering number at most a constant multiple of $|A|_r$, so the proposition gives

$$|A|_r \gtrsim r^{-\gamma(s,t)+\kappa/8}.$$

Insert this into (12.7u):

$$J_r(j) \lesssim r^{(\gamma(s,t)-2)/2-\kappa/16-3e} \leq \delta^{-\kappa/4} r^{(\gamma(s,t)-2)/2},$$

since $r \geq \delta$ and $e \leq \kappa/48$. This is (12.7p) and finishes the proof. $\square$

Proposition III.12.8 (iteration of parabola energy gain). *Fix $0 < s \leq 1$, $0 < t < \min\{3s, s+1\}$, and $\kappa > 0$. There are $\varepsilon > 0$, $m \in \mathbb{N}$, and $\delta_0 > 0$ such that every probability σ supported on $\mathbb{P}$ with $I_s^\delta(\sigma) \leq \delta^{-\varepsilon}$ satisfies*

$$I_t^\delta(\sigma^{*k}) \leq \delta^{-\kappa} \quad (k \geq m, 0 < \delta \leq \delta_0). \quad (12.7v)$$

Proof. It suffices to prove the assertion with $\kappa_0 = \min\{\kappa, 1\}$ in place of κ : for $0 < \delta < 1$, a bound by $\delta^{-\kappa_0}$ implies the stated bound. Put $t_0 = s$ and

$$t_{j+1} = \gamma(s, t_j) = \min \left\{ \frac{3s + t_j}{2}, s + 1 \right\}.$$

Since $t_j \geq s$, the term $s + t_j$ is never smaller than $(3s + t_j)/2$. The sequence increases to $\min\{3s, s+1\}$. Choose j_0 with $t_{j_0} > t$.

Here is the quantifier order. Set $\beta_{j_0} = \kappa_0/2$. For $j = j_0 - 1, \dots, 0$, working backwards, let ε_j be an admissible input exponent in Proposition III.12.7 with output loss $\beta_{j+1}/2$; decrease ε_j to at most 1, and set $\beta_j = \varepsilon_j/2$. Choose the exponent in the present proposition so that

$$0 < \varepsilon_* < \frac{1}{2} \min\{\beta_0, \varepsilon_0, \dots, \varepsilon_{j_0-1}\}. \quad (12.7va)$$

If $j_0 = 0$, take the minimum here to be β_0 . The hypothesis also gives $I_s^\delta(\sigma) \leq \delta^{-\varepsilon_j}$ for every later stage. By the capped convolution comparison (12.7h), for every $k \geq 1$ we have

$$I_{t_0}^\delta(\sigma^{*k}) \leq C_s I_s^\delta(\sigma) \leq C_s \delta^{-\varepsilon_*} \leq \delta^{-\beta_0}$$

after decreasing δ_0 . Thus the induction already holds for all powers at its base stage.

Set $k_0 = 1$. Suppose that all powers σ^{*k} , $k \geq k_j$, have $I_{t_j}^\delta \leq \delta^{-\beta_j}$. Apply Proposition III.12.7 to $\mu = \sigma^{*k_j}$ and the original σ , with output loss $\beta_{j+1}/2$. The support of μ

lies in the fixed ball $B(3k_j)$, which changes only the smallness threshold δ_0 . If K_j is the convolution threshold returned by flattening, set $k_{j+1} = (k_j + 1)K_j$. The proposition gives

$$I_{t_{j+1}}^\delta(\sigma^{*k_{j+1}}) \leq \delta^{-\beta_{j+1}/2}.$$

For every $k \geq k_{j+1}$, convolve with the remaining σ -factors and apply (12.7h). Its fixed factor is at most $\delta^{-\beta_{j+1}/2}$ for small δ , so the inductive assertion holds at stage $j+1$. Iterate to j_0 . Finally, for every probability measure the pointwise kernel inequality gives $I_t^\delta \leq I_{t_0}^\delta + 1$; since $\beta_{j_0} = \kappa_0/2$, another decrease of δ_0 proves (12.7v). $\square$

Lemma III.12.9 (critical Frostman energy budget). *If $0 < s \leq 1$ and a finite measure σ on $\mathbb{P}$ satisfies $\sigma(B(x, r)) \leq r^s$ for all x, r , then*

$$I_s^\delta(\sigma) \lesssim_s 1 + \log(1/\delta), \quad 0 < \delta < 1/2. \quad (12.7w)$$

Proof. For each x , divide distances from x into dyadic annuli between δ and the fixed diameter of $\mathbb{P}$. Frostman decay bounds the integral of the capped kernel on each annulus by $O_s(1)$, and there are $O(1 + \log(1/\delta))$ annuli. Integrate in x ; the total mass of σ is bounded by an absolute constant from the same Frostman condition at a fixed radius. $\square$

Theorem III.12.10 (general-range averaged parabola Fourier estimate). *Let $0 < s \leq 1$ and $0 \leq t < \min\{3s, s+1\}$. There is an even integer $p = p(s, t) \geq 2$ such that every finite Borel measure σ supported on $\mathbb{P}$ and satisfying $\sigma(B(x, r)) \leq r^s$ obeys*

$$\int_{B(R)} |\hat{\sigma}(\xi)|^p d\xi \leq C_{s,t} R^{2-t}, \quad R \geq 1. \quad (12.7x)$$

Proof. For $t = 0$, take $p = 2$ and use $|\hat{\sigma}| \leq \sigma(\mathbb{R}^2) \lesssim_s 1$. Assume $t > 0$, and choose $t < t_1 < \min\{3s, s+1\}$ and $0 < \lambda < t_1 - t$. Let $\varepsilon > 0$ and m be supplied by Proposition III.12.8 for t_1, λ .

First suppose σ is a probability satisfying $I_s^\delta(\sigma) \leq \delta^{-\varepsilon}$. The iteration gives $I_{t_1}^\delta(\sigma^{*m}) \leq \delta^{-\lambda}$. For $|\xi| \leq c_0\delta^{-1}$, the Fourier transform of the fixed ball-average kernel obeys $|\hat{\psi}(\delta\xi)| \geq c > 0$. The Fourier–Riesz identity for $0 < t_1 < 2$, combined with the capped-energy comparison, gives

$$\int_{B(c_0\delta^{-1})} |\hat{\sigma}(\xi)|^{2m} d\xi \lesssim_{s,t} \delta^{-(2-t_1)} I_{t_1}^\delta(\sigma^{*m}) \leq \delta^{-(2-t_1+\lambda)} \leq \delta^{-(2-t)}. \quad (12.7y)$$

To make the constant uniform for an arbitrary Frostman measure, put $M = \sigma(\mathbb{R}^2) \lesssim_s 1$, and ignore the zero measure. Choose one even integer $p \geq 2m$ so large that $2t/p < \varepsilon/2$. Set $\delta \asymp R^{-1}$. If $M \leq R^{-t/p}$, the trivial bound $|\hat{\sigma}| \leq M$ gives $\int_{B(R)} |\hat{\sigma}|^p \lesssim M^p R^2 \leq R^{2-t}$. Otherwise normalize $\tilde{\sigma} = \sigma/M$. Lemma III.12.9 gives

$$I_s^\delta(\tilde{\sigma}) = M^{-2} I_s^\delta(\sigma) \lesssim R^{2t/p} (1 + \log R) \leq \delta^{-\varepsilon}$$

for large R . Apply (12.7y) to $\tilde{\sigma}$, enlarging the fixed frequency ball by a constant through the choice of δ . Since $|\hat{\tilde{\sigma}}| \leq 1$ and $p \geq 2m$,

$$\int_{B(R)} |\hat{\sigma}|^p \leq M^p \int_{B(R)} |\hat{\tilde{\sigma}}|^{2m} \lesssim_{s,t} R^{2-t}.$$

The bounded range of R is absorbed in the constant. This proves (12.7x). $\square$

OPP general flattening block. The general-range proof of the next theorem is given locally above: the quantitative Furstenberg input, parabola-energy extraction, finite-scale L^2 flattening, multiscale energy iteration, and the passage to the averaged Fourier estimate. Its geometric middle branch uses the Chapter 9 regular-set Furstenberg theorem, whose high-multiplicity input $\text{int}:X\text{--III.9.ABC49}$ is proved locally. The argument follows the research-scale route of Orponen–Puliatti–Pyörälä [47]; the sharp $p = 6$ smoothing branch below is not claimed here.

III.12.2.1 A local smoothing step and the remaining pair-energy input

The square-root Katz–Tao incidence estimate is now local to this book: Corollary III.7.29 includes the full range $s + t \leq 2$. We spell out how its weighted form improves the energy of a measure convolved with a curved Frostman measure. This separates the smoothing argument from the sharp pair-energy theorem, which we prove next and then combine with smoothing for the sixth moment.

Definition (discrete Frostman measures). *Fix a bounded region and its dyadic δ -squares. A discrete (δ, u, C) measure is a nonnegative mass $\lambda(q)$ on these squares with total mass at most one and*

$$\sum_{q: q \cap B(x, r) \neq \emptyset} \lambda(q) \leq Cr^u, \quad \delta \leq r \leq 1. \quad (12.12a)$$

The mass of the measure is $\|\lambda\| = \sum_q \lambda(q)$. For a family of square masses $\lambda_p(q)$, the condition is required uniformly in p .

Lemma III.12.11 (weighted translated-parabola smoothing). *Let $0 < s \leq 1$, $0 < t \leq 2$, and $s + t \leq 2$. Let $\mu(p)$ be a discrete (δ, t, C_μ) measure. For every p let $\sigma_p(q)$ be a discrete (δ, s, C_σ) measure supported on the $O(\delta)$ -neighbourhood of $c(p) + \mathbb{P}$, where $c(p)$ is the centre of p . All squares lie in fixed bounded regions, and $C_\mu, C_\sigma \geq 1$. Define*

$$f_\delta(x) = \sum_p \mu(p) \sum_q \sigma_p(q) \delta^{-2} \mathbf{1}_q(x).$$

For every $\varepsilon > 0$,

$$\|f_\delta\|_2^2 \lesssim_{s,t,\varepsilon} C_\mu C_\sigma^2 \|\mu\| \delta^{t+2s-3-\varepsilon} + 1. \quad (12.12b)$$

Proof. There are $O(\delta^{-2})$ possible p and $O(\delta^{-2})$ possible q in the bounded regions. Discard pairs (p, q) for which $\mu(p)\sigma_p(q) < \delta^6$. At every q their total mass is at most $O(\delta^4)$, so their density is $O(\delta^2)$ and their contribution to the L^2 norm is $O(1)$. For the remaining pairs, dyadically pigeonhole $\mu(p) \asymp a$ and $\sigma_p(q) \asymp b$. Because $a, b \leq 1$ and $ab \geq \delta^6$, there are $O(\log^2(2/\delta))$ pairs of levels.

Fix one pair of levels. Write $\mathcal{P}$ for its nonempty base squares, $F(p)$ for its fibre squares, and $m(q) = \#\{p \in \mathcal{P} : q \in F(p)\}$. The discrete Frostman conditions imply that the unweighted families are Katz–Tao with constants

$$C_1 \lesssim C_\mu a^{-1} \delta^t, \quad C_2 \lesssim C_\sigma b^{-1} \delta^s. \quad (12.12c)$$

The bounded bi-Lipschitz parabola-to-line change of variables in the proof of Lemma III.12.4 converts these fibres to line fibres. Finitely many adjacent image squares and colours only

change C_1, C_2 by fixed factors. Thus Corollary III.7.29 gives, on every dyadic multiplicity level $\mathcal{F}_M = \{q : M \leq m(q) < 2M\}$,

$$M|\mathcal{F}_M| \lesssim_\varepsilon \delta^{-\varepsilon/4} (\delta^{-1} C_1 C_2 |\mathcal{F}_M| |\mathcal{P}|)^{1/2}.$$

After squaring and using $m(q) < 2M$,

$$\sum_{q \in \mathcal{F}_M} m(q)^2 \lesssim_\varepsilon \delta^{-1-\varepsilon/2} C_1 C_2 |\mathcal{P}|. \quad (12.12d)$$

There are only $O(\log(2/\delta))$ multiplicity levels. The density of the fixed (a, b) level at q is at most $4ab\delta^{-2}m(q)$, so (12.12c) and (12.12d) imply

$$\begin{aligned} \|f_{a,b}\|_2^2 &\lesssim_\varepsilon \delta^{-3-\varepsilon/2} a^2 b^2 C_1 C_2 |\mathcal{P}| \log(2/\delta) \\ &\lesssim_\varepsilon C_\mu C_\sigma \delta^{t+s-3-\varepsilon/2} ab |\mathcal{P}| \log(2/\delta). \end{aligned}$$

The one-cell case of the fibre Frostman condition gives $b \leq C_\sigma \delta^s$, while $a|\mathcal{P}| \leq \|\mu\|$. Hence each level contributes at most $C_\mu C_\sigma^2 \|\mu\| \delta^{t+2s-3-\varepsilon/2}$ times one logarithm. Apply the triangle inequality over the $O(\log^2(2/\delta))$ weight levels and absorb every resulting logarithm into the remaining $\delta^{-\varepsilon/2}$. The discarded part contributes $O(1)$, proving (12.12b). $\square$

Proposition III.12.12 (one-step Sobolev smoothing). *Let $1/2 < s \leq 1$, $0 < t < 2$, and let μ be a probability measure supported in a fixed ball with $I_t(\mu) \leq M$. Let σ be an s -Frostman probability on $\mathbb{P}$ with Frostman constant C_σ . Then*

$$I_\zeta(\mu * \sigma) \lesssim_{s,t,\zeta,M,C_\sigma} 1 \quad \text{whenever} \quad 0 < \zeta < \min\{t + 2s - 1, s + 1\}. \quad (12.12e)$$

Proof. First suppose $s+t \leq 2$. Put the mass of μ in dyadic δ -squares p and define its capped discrete potential by

$$U_t(p) = \sum_q \frac{\mu(q)}{\max\{|c(p) - c(q)|, \delta\}^t}. \quad (12.12f)$$

For $x \in p$ and $y \in q$, $|x-y| \lesssim \max\{|c(p)-c(q)|, \delta\}$; the self term is included. Integration over all pairs gives $\sum_p \mu(p) U_t(p) \lesssim I_t(\mu) \leq M$. Partition the occupied p into the $O(\log(2/\delta))$ dyadic potential levels $U_t(p) \asymp 2^j$ and write μ_j for the restriction to one level. Then

$$2^j \|\mu_j\| \lesssim M. \quad (12.12g)$$

Also μ_j is a discrete $(\delta, t, C2^j)$ measure. Indeed if a ball $B(x, r)$ meets an occupied level- j square p , every square q meeting the same ball has $|c(p) - c(q)| \lesssim r$; its mass is bounded by $Cr^t U_t(p) \lesssim 2^j r^t$.

For each p with $\mu(p) > 0$, distribute the conditional convolution of $\mu|_p / \mu(p)$ with σ over its destination δ -squares q , and call these masses $\sigma_p(q)$. They are probability measures supported in an $O(\delta)$ -neighbourhood of $c(p) + \mathbb{P}$. Averaging translates of σ preserves the uniform (δ, s, C_s) Frostman condition. Therefore Lemma III.12.11, (12.12g), and the triangle inequality over the potential levels give

$$\|(\mu * \sigma) * \varphi_\delta\|_2^2 \lesssim_{s,t,\varepsilon,M} \delta^{t+2s-3-\varepsilon}, \quad (12.12h)$$

for every fixed nonnegative smooth unit-mass mollifier φ_δ supported in $B(0, O(\delta))$. Here the square-mass density dominates the mollified measure after replacing each square by $O(1)$ neighbours; the corresponding L^2 operator norm is absolute. We used $t + 2s - 3 \leq 0$ to absorb the constant term and spent half of ε on the logarithmic number of levels.

If $s + t > 2$, choose $t' < 2 - s$ arbitrarily close to $2 - s$. Compact support and $I_t(\mu) < \infty$ imply $I_{t'}(\mu) \lesssim_{t,t'} 1 + M$. Applying the preceding case with t' proves (12.12h) with any exponent smaller than $s - 1$. Finally, the dyadic Fourier representation of Riesz energy gives

$$I_\zeta(\nu) \lesssim 1 + \sum_{k \geq 0} 2^{k(\zeta-2)} \|\nu * \varphi_{2^{-k}}\|_2^2, \quad \nu = \mu * \sigma.$$

Choose the available exponent in (12.12h) strictly larger than $\zeta - 2$; the series converges and proves (12.12e). $\square$

The following wave-packet argument follows the architecture of Orponen [48, Section 3]. We include the non-concentration step because this is where the sharp exponent $2s$ enters; the packet frame itself is an elementary Fourier-series construction.

Lemma III.12.13 (parabolic wave-packet frame). *Let $R \geq 2$, $\delta = R^{-1}$, and let Θ be a finitely overlapping cover of the δ -neighbourhood of $\mathbb{P}$ by caps θ of tangential length $R^{-1/2}$ and normal width R^{-1} . For each θ there is a tiling $\mathcal{T}_\theta$ of the frequency plane by rectangles T of tangential width $R^{1/2}$ and normal length R , with Schwartz functions V_T, W_T such that every $F \in L^2$ supported in θ has the expansion*

$$\widehat{F} = \sum_{T \in \mathcal{T}_\theta} \langle \widehat{F}, V_T \rangle W_T, \quad \sum_T |\langle \widehat{F}, V_T \rangle|^2 \asymp \|F\|_2^2. \quad (12.12i)$$

The Fourier transforms of V_T, W_T are supported, up to the fixed Fourier-sign reflection, in a fixed enlargement of θ ; also $\|V_T\|_2 + \|W_T\|_2 \lesssim 1$, and, for every integer $N \geq 1$,

$$|V_T(x)| + |W_T(x)| \lesssim_N |T|^{-1/2} \left(1 + \frac{|(x - x_T) \cdot e_T|}{R} + \frac{|(x - x_T) \cdot e_T^\perp|}{R^{1/2}} \right)^{-N}, \quad (12.12j)$$

where e_T is the long-axis direction, x_T is the centre, and $|T| \asymp R^{3/2}$.

Proof. Choose a rectangle B_θ of dimensions comparable to θ and containing it in its interior, and a smooth cutoff χ_θ equal to one on θ and supported strictly inside B_θ . The normalized exponentials on B_θ form an orthonormal Fourier-series basis. Expand F in that basis, then multiply each basis exponential by χ_θ ; because $\chi_\theta = 1$ on the support of F , the synthesis series still equals F . Parseval gives the coefficient identity in (12.12i). The exponentials are indexed by the dual lattice of B_θ ; their inverse transforms are translated smooth bumps on the dual rectangles T . Repeated integration by parts in the two rescaled coordinates gives (12.12j) for the analysis and synthesis functions, including the normalization $|T|^{-1/2}$. Taking the cutoff and a second cutoff for the analysis functions inside one fixed enlargement of B_θ gives the stated Fourier supports. This construction also proves the claim for rotated caps with uniform constants. $\square$

Lemma III.12.14 (packet non-concentration). *Let $0 < s \leq 1$, and let σ be an s -Frostman probability on $\mathbb{P}$ with constant C_σ . Smooth it at scale $\delta = R^{-1}$, write $\sigma_\delta = \sum_{\theta \in \Theta} \sigma_\theta$ by a nonnegative smooth cap partition, and put $a_T = \langle \widehat{\sigma}_\theta, V_T \rangle$. In estimates involving $R^{C\rho}$, C denotes an absolute constant that may increase from line to line. Fix $0 < \rho < 1/20$. If S is any rectangle of dimensions $\Delta \times R^{1+\rho}$ with $R^{1/2} \leq \Delta \leq R^{1+\rho}$, then*

$$\sum_{\theta} \sum_{\substack{T \in \mathcal{T}_\theta \\ T \subset S}} |a_T|^2 \lesssim_{s,\rho,C_\sigma} \Delta R^{1-s+C\rho}, \quad (12.12k)$$

Proof. The long axes of packets contained in S make angle $O(\Delta/R)$ with the long axis of S . Curvature implies that their caps lie in an arc J of $\mathbb{P}$ of length $O(\delta\Delta)$; the assertion is trivial if $\delta\Delta \gtrsim 1$. Thus

$$\sum_{\theta \subset J} \sigma(2\theta) \lesssim \sigma(B(J, C\delta\Delta)) \lesssim (\delta\Delta)^s. \quad (12.12l)$$

Fix such a cap θ . Colour its dual-lattice tiling into $O(R^{2\rho})$ classes by the two lattice coordinates, so two distinct rectangles of one class are at least R^ρ lattice steps apart in one of the two anisotropically rescaled coordinates. Let S^+ be the concentric R^ρ enlargement of S , and choose a smooth cutoff η_{S^+} equal to one on S . The anisotropic decay (12.12j) makes the coefficient error from replacing $\widehat{\sigma}_\theta$ by $\widehat{\sigma}_\theta \eta_{S^+}$ smaller than R^{-N} for any prescribed N . On one separated class the Gram matrix of the normalized V_T has diagonal $O(1)$ and off-diagonal row sums $O(R^{-N})$, again by (12.12j). The elementary Bessel inequality for this Gram matrix and Plancherel therefore give

$$\sum_{\substack{T \subset S \\ T \text{ in one class}}} |a_T|^2 \lesssim \|\sigma_\theta * \widehat{\eta}_{S^+}\|_2^2 + R^{-N}. \quad (12.12m)$$

For clarity, duality gives $\sum_T |\langle g, V_T \rangle|^2 = \sup_{\|c\|_{\ell^2}=1} |\langle g, \sum_T c_T V_T \rangle|^2$. The diagonal Gram entries are $O(1)$ and their off-diagonal row sums are $O(R^{-N})$, so $\|\sum_T c_T V_T\|_2^2 \lesssim \|c\|_{\ell^2}^2$. This proves the displayed Bessel bound without assuming an orthonormal packet family.

Write $w_r(x) = (1 + |x|/r)^{-N_0}$ with $N_0 > 4$. The dual rectangle of S^+ is contained in $B(C/\Delta)$, and integration by parts gives

$$|\widehat{\eta}_{S^+}(x)| \lesssim_\rho \Delta R^{1+C\rho} w_{C/\Delta}(x), \quad \|\widehat{\eta}_{S^+}\|_1 \lesssim 1. \quad (12.12n)$$

The Frostman condition, summed over dyadic annuli, yields $\|\sigma_\theta * w_{C/\Delta}\|_\infty \lesssim \Delta^{-s}$. This also holds when $\Delta^{-1} < \delta$: the smoothed measure has density at most $C\delta^{s-2}$, so convolution with the unnormalised weight w_r is at most $C\delta^{s-2}r^2 \leq Cr^s$ for $r \leq \delta$. Since $\|\sigma_\theta\|_1 \lesssim \sigma(2\theta)$, Young's inequality and (12.12n) give

$$\|\sigma_\theta * \widehat{\eta}_{S^+}\|_2^2 \leq \|\sigma_\theta * \widehat{\eta}_{S^+}\|_\infty \|\sigma_\theta * \widehat{\eta}_{S^+}\|_1 \lesssim \Delta^{1-s} R^{1+C\rho} \sigma(2\theta).$$

Sum (12.12m) over the $O(R^{C\rho})$ classes and over caps in J , use (12.12l), and choose N to absorb all polynomially many errors. The result is $\Delta^{1-s} R^{1+C\rho} (\Delta/R)^s = \Delta R^{1-s+C\rho}$. $\square$

Theorem III.12.15 (sharp pair-convolution energy). *Let $0 < s \leq 1$ and let σ be an s -Frostman probability supported on $\mathbb{P}$. For every $0 < \eta < 2s$,*

$$I_{2s-\eta}(\sigma * \sigma) < \infty. \quad (12.12o)$$

The bound depends only on s, η and the Frostman constant.

Proof. We first prove, for every $\varepsilon > 0$,

$$\int_{B(R)} |\hat{\sigma}(\xi)|^4 d\xi \lesssim_{s,\varepsilon,C_\sigma} R^{2-2s+\varepsilon}. \quad (12.12p)$$

Choose σ_δ and the packet expansion above, with $\hat{\sigma}_\delta = \sum_{\theta,T} a_T W_T$. Parseval in (12.12i) and the bounded overlap of caps yield

$$\sum_{\theta,T} |a_T|^2 \lesssim \|\sigma_\delta\|_2^2 \leq \|\sigma_\delta\|_\infty \|\sigma_\delta\|_1 \lesssim_{C_\sigma} R^{2-s}. \quad (12.12q)$$

We record the only square-function fact used here, with its proof. If f_θ has Fourier support in 2θ , then

$$\left\| \sum_\theta f_\theta \right\|_4^4 \lesssim \log(2R) \int \left(\sum_\theta |f_\theta|^2 \right)^2. \quad (12.12r)$$

To see this, use Plancherel on the square and group pairs (θ, θ') by dyadic separation $\alpha \in [R^{-1/2}, 1]$. At a fixed α , the frequency sumsets $2\theta + 2\theta'$ have bounded overlap: writing the two parabola parameters as u, v , their sum and quadratic sum determine the unordered pair $\{u, v\}$, and the inverse Jacobian on $|u - v| \asymp \alpha$ is $O(\alpha^{-1})$. For a fixed output frequency, choose actual curve parameters u, v from any represented cap pair. The normal errors in the two coordinates are $O(R^{-1})$; comparing two representations and inverting the sum map determines each unordered parameter to $O(R^{-1}/\alpha) \leq O(R^{-1/2})$. The cap tangential width is $R^{-1/2}$, so there are only $O(1)$ cap pairs when $\alpha \geq CR^{-1/2}$. At the lowest separation level $\alpha \asymp R^{-1/2}$, the sum coordinate fixes $u + v$ up to $O(R^{-1})$ and $|u - v| = O(R^{-1/2})$; hence both parameters lie in $O(1)$ caps near $(u + v)/2$. Thus bounded overlap also holds at this level. Cauchy–Schwarz over the $O(\log R)$ separation levels proves (12.12r).

For packets in one cap, colour the dual lattice into $O(R^{2\rho})$ classes by both anisotropic lattice coordinates, as in the proof of Lemma III.12.14. Equation (12.12j), followed by the inequality $|\sum_{j=1}^m z_j|^2 \leq m \sum_j |z_j|^2$, gives

$$\int \left(\sum_\theta \left| \sum_{T \in \mathcal{T}_\theta} a_T W_T \right|^2 \right)^2 \lesssim_\rho R^{C\rho} \sum_{T,T'} |a_T|^2 |a_{T'}|^2 \int |W_T W_{T'}|^2 + O(R^{-N}). \quad (12.12s)$$

The error is rapid because σ_δ is smooth and compactly supported, so coefficients outside a sufficiently large polynomial ball are rapidly decreasing; inside it there are only polynomially many packets. This justifies every finite-sum approximation in this argument.

Fix T and group T' by dyadic angle $R^{-1/2} \leq \alpha \leq 1$. Enlarge T, T' by R^ρ in both directions. If the enlarged rectangles are disjoint, (12.12j) contributes $O(R^{-N})$. Otherwise their intersection has area $O(R^{1+2\rho}/\alpha)$, so

$$\int |W_T W_{T'}|^2 \lesssim_\rho R^{-2+2\rho} \alpha^{-1}. \quad (12.12t)$$

Every such T' lies in a rectangle S about T of dimensions $\Delta \times R^{1+\rho}$, where $\Delta \asymp \alpha R^{1+\rho}$ and $R^{1/2} \leq \Delta \leq R^{1+\rho}$. Lemma III.12.14 now gives

$$\sum_{\substack{T': \angle(T, T') \asymp \alpha \\ T^+ \cap T'^+ \neq \emptyset}} |a_{T'}|^2 \int |W_T W_{T'}|^2 \lesssim R^{-2+2\rho} \alpha^{-1} \Delta R^{1-s+C\rho} \lesssim R^{-s+C\rho}.$$

Sum the $O(\log R)$ angles and then T using (12.12q). Equations (12.12r) and (12.12s) give $\|\widehat{\sigma}_\delta\|_4^4 \lesssim R^{2-2s+O(\rho)}$. Choose ρ sufficiently small compared with ε . Since the mollifier has Fourier transform bounded below on $B(cR)$, a fixed change of R proves (12.12p).

By the Riesz-energy Fourier identity,

$$I_{2s-\eta}(\sigma * \sigma) \asymp \int_{\mathbb{R}^2} |\hat{\sigma}(\xi)|^4 |\xi|^{2s-\eta-2} d\xi.$$

Choose $\varepsilon < \eta$ and apply (12.12p) on dyadic annuli. The resulting terms are $O(2^{-k(\eta-\varepsilon)})$, so their sum is finite. $\square$

Corollary III.12.16 (sharp sixth moment). *For $2/3 \leq s \leq 1$ and $0 \leq t < s+1$, every s -Frostman probability σ on $\mathbb{P}$ with Frostman constant C_σ satisfies*

$$\int_{B(R)} |\hat{\sigma}(\xi)|^6 d\xi \lesssim_{s,t,C_\sigma} R^{2-t}, \quad R \geq 1. \quad (12.12u)$$

Proof. For $t = 0$ the estimate follows from $|\hat{\sigma}| \leq 1$. For $t > 0$ choose $\eta > 0$ so small that $u_0 = 2s - \eta > 0$ and $t < \min\{u_0 + 2s - 1, s + 1\}$. This is possible because $s \geq 2/3$ implies $4s - 1 \geq s + 1$. Theorem III.12.15 gives $I_{u_0}(\sigma * \sigma) < \infty$. Proposition III.12.12, applied to $\mu = \sigma * \sigma$, gives $I_u(\sigma^{*3}) < \infty$ for some $t < u < s + 1$. Fourier energy then yields

$$\int_{B(R)} |\hat{\sigma}|^6 \leq R^{2-u} \int_{\mathbb{R}^2} |\widehat{\sigma^{*3}}(\xi)|^2 |\xi|^{u-2} d\xi \lesssim R^{2-u} I_u(\sigma^{*3}) \lesssim R^{2-t}.$$

$\square$

Theorem III.12.17 (Orponen–Puliatti–Pyörälä). *Let $0 \leq s \leq 1$ and*

$$0 \leq t < \min\{3s, s + 1\}. \quad (12.13)$$

There exists $p = p(s, t) \geq 1$ such that every Borel measure σ supported on $\mathbb{P}$ and satisfying $\sigma(B(x, r)) \leq r^s$ obeys

$$\|\hat{\sigma}\|_{L^p(B(R))} \leq C_{s,t} R^{(2-t)/p}, \quad R \geq 1. \quad (12.14)$$

If $s \geq 2/3$, one may take $p = 6$ for every $0 \leq t < s + 1$.

Proof. The general-range assertion is (12.7x); that proof uses the locally proved Chapter 9 input int :X-III.9.ABC49. We verify the uniformity needed to apply Corollary III.12.16 to the sharp clause. The Frostman hypothesis and bounded support give $\sigma(\mathbb{P}) \leq 2$. Let ν_0 be normalized arclength on $\mathbb{P}$, so $\nu_0(B(x, r)) \leq Cr^s$ for $0 < r \leq 1$. If $t = 0$, the bound $|\hat{\sigma}| \leq 2$ gives the asserted R^2 estimate directly. Assume henceforth $t > 0$. The probability

$$\lambda = \frac{1}{4}\sigma + \left(1 - \frac{1}{4}\sigma(\mathbb{P})\right)\nu_0$$

has an s -Frostman constant bounded independently of σ , and $\sigma \leq 4\lambda$ as positive measures. For any $t < s + 1$, the proof of Corollary III.12.16 gives a fixed $u \in (t, s + 1)$ with $I_u(\lambda^{*3}) \lesssim_{s,t} 1$.

Positivity of the Riesz kernel and $\sigma^{*3} \leq 4^3 \lambda^{*3}$ imply $I_u(\sigma^{*3}) \leq 4^6 I_u(\lambda^{*3})$. The Fourier-energy identity then yields

$$\int_{B(R)} |\hat{\sigma}|^6 \lesssim R^{2-u} I_u(\sigma^{*3}) \lesssim_{s,t} R^{2-t}.$$

This is (12.14) with $p = 6$. The theorem is due to Orponen–Puliatti–Pyörälä [47]; the proof here separates its general-range geometric input from the local sharp smoothing chain. $\square$

Lemma III.12.18 (even-exponent upgrade). *Assume (12.14) for some p . If $2n \geq p$ is even, then*

$$\int_{B(R)} |\hat{\sigma}(\xi)|^{2n} d\xi \lesssim_{\sigma,s,t,n} R^{2-t}, \quad R \geq 1. \quad (12.15)$$

Proof. Since σ is finite, $\|\hat{\sigma}\|_\infty \leq \sigma(\mathbb{R}^2) =: M$. Hence

$$\int_{B(R)} |\hat{\sigma}|^{2n} \leq M^{2n-p} \int_{B(R)} |\hat{\sigma}|^p \lesssim R^{2-t}.$$

$\square$

Proposition III.12.19 (finite energy of a convolution power). *Let σ satisfy the hypotheses of Theorem III.12.17. For every*

$$0 < \tau < \min\{3s, s+1\} \quad (12.16)$$

there is an integer $n = n(s, \tau)$ such that

$$I_\tau(\sigma^{*n}) < \infty. \quad (12.17)$$

If $s \geq 2/3$, then $n = 3$ is valid for every $0 < \tau < s+1$.

Proof. Choose

$$\tau < t < \min\{3s, s+1\}$$

and let $p = p(s, t)$ be supplied by Theorem III.12.17. Choose n with $2n \geq p$. Since

$$\widehat{\sigma^{*n}} = \hat{\sigma}^n,$$

Lemma III.12.18 gives

$$\int_{B(R)} |\widehat{\sigma^{*n}}(\xi)|^2 d\xi \lesssim R^{2-t}. \quad (12.18)$$

Split $\{\xi : |\xi| \geq 1\}$ into dyadic annuli. By the Fourier-energy identity,

$$\begin{aligned} I_\tau(\sigma^{*n}) &\lesssim 1 + \sum_{j \geq 0} 2^{j(\tau-2)} \int_{B(2^{j+1})} |\widehat{\sigma^{*n}}(\xi)|^2 d\xi \\ &\lesssim 1 + \sum_{j \geq 0} 2^{j(\tau-t)} < \infty. \end{aligned}$$

If $s \geq 2/3$, choose t with $\tau < t < s+1$ and use $p = 6$ in Theorem III.12.17; then $2n = 6$, so $n = 3$. $\square$

Corollary III.12.20 (support growth from energy). *Under the hypotheses of Proposition III.12.19,*

$$\dim_{\mathbb{H}}(n \operatorname{supp} \sigma) \geq \tau. \quad (12.19)$$

Proof. Finite τ -energy forces $\dim_{\mathbb{H}} \operatorname{supp}(\sigma^{*n}) \geq \tau$, while $\operatorname{supp}(\sigma^{*n}) \subset n \operatorname{supp} \sigma$. $\square$

The dependence of n here is implicit. The next section gives a separate, explicit recurrence for the support dimension.

III.12.3 Parabola sumsets from the sharp Furstenberg theorem

Imported theorem (sharp Furstenberg input). *The almost-AD estimate (11.28b) follows from the regular-set Furstenberg theorem of Chapter III.9, and the semi-well-spaced kernel is Proposition III.11.7.8. Chapter III.11 combines them to prove the middle continuum bound (11.28af). The sumset argument also uses the locally proved Chapter III.7 high branch (7.37l). Thus the wrapper below remains relative to the middle research interface only.*

Sharp Furstenberg wrapper relative to its three branches. If F is an (s, t) -Furstenberg set with $0 < s \leq 1$ and $0 < t \leq 2$, then

$$\dim_{\mathbb{H}} F \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}. \quad (12.20)$$

For $t \leq s$, use the local low-parameter bound (6.84); for $t \geq 2 - s$, use the proved Furstenberg high branch (7.37l); and for $s < t < 2 - s$, use the locally assembled middle theorem (11.28af).

Lemma III.12.21 (parabola-to-line diffeomorphism). *Define*

$$\Psi(x, y) = (x, x^2 - y). \quad (12.21)$$

On every bounded set, Ψ is bi-Lipschitz. If $z = (a, b)$ and $z + \mathbb{P}$ is a translate of the parabola, then $\Psi(z + \mathbb{P})$ lies on the line

$$L_z : \quad y = 2ax - a^2 - b. \quad (12.22)$$

Moreover, on bounded sets the parameter map

$$z = (a, b) \mapsto (2a, -a^2 - b) \quad (12.23)$$

is bi-Lipschitz from the translation parameter to the usual slope-intercept chart of line space.

Proof. The Jacobian of Ψ is

$$D\Psi(x, y) = \begin{pmatrix} 1 & 0 \\ 2x & -1 \end{pmatrix}, \quad \det D\Psi = -1,$$

and both $D\Psi$ and $D\Psi^{-1}$ are bounded on bounded sets. Thus Ψ is bi-Lipschitz there. A point of $z + \mathbb{P}$ has the form $(a + u, b + u^2)$, and

$$\Psi(a + u, b + u^2) = (a + u, 2au + a^2 - b).$$

Writing $x = a + u$ gives (12.22). Finally, the inverse of $(a, b) \mapsto (m, c) = (2a, -a^2 - b)$ is

$$a = m/2, \quad b = -m^2/4 - c,$$

so the parameter map and its inverse have bounded derivatives on bounded sets. $\square$

Proof status. The recurrence below applies the full Furstenberg wrapper. The active $s+1$ branch is proved in Chapter 7; the middle branch uses the Chapter 9 theorem, including its locally proved X-III.9.ABC49 input. The parabola-to-line calculation, recurrence, and explicit-growth descendant therefore have local proof chains relative only to the declared foundational inputs.

Lemma III.12.22 (sumset recurrence). *Let $K \subset \mathbb{P}$ be Borel with $\dim_{\text{H}} K \geq s$, and put $d_n = \dim_{\text{H}}(nK)$. For every $n \geq 2$,*

$$d_n \geq \min \left\{ \frac{3s + d_{n-1}}{2}, s + 1 \right\}. \quad (12.24)$$

Proof from the preceding Furstenberg branches. Put $Z = (n-1)K$ and $t = d_{n-1}$. If $t = 2$, choose any $k \in K$; then $Z + k \subset nK$, so $d_n = 2$ and the conclusion is immediate. Assume $t < 2$.

By Lemma III.12.21, the line family $\mathcal{L}_Z = \{L_z : z \in Z\}$ has Hausdorff dimension t . For every $z \in Z$,

$$\Psi(nK) \cap L_z \supset \Psi(z + K),$$

and the set on the right has Hausdorff dimension at least s . Hence $\Psi(nK)$ is an (s, t) -Furstenberg set. Applying (12.20), and using bi-Lipschitz invariance of Hausdorff dimension, gives

$$d_n \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}.$$

Since $t = d_{n-1} \geq s$ (a translate of K is contained in $(n-1)K$), we have $s + t \geq (3s + t)/2$, which yields (12.24). $\square$

The explicit formula is the elementary iteration of the preceding locally proved recurrence and uses its Furstenberg inputs.

Theorem III.12.23 (explicit parabola sumset growth). *Let $K \subset \mathbb{P}$ be Borel with $\dim_{\text{H}} K \geq s$, $0 \leq s \leq 1$. Then, for every integer $n \geq 1$,*

$$\dim_{\text{H}}(nK) \geq \min \left\{ 3s - s \cdot 2^{-(n-2)}, s + 1 \right\}. \quad (12.25)$$

In particular,

$$\dim_{\text{H}}(K + K + K) \geq \min \left\{ \frac{5s}{2}, s + 1 \right\}. \quad (12.26)$$

Proof. The case $s = 0$ is immediate. Define

$$b_n = \min\{3s - s2^{-(n-2)}, s + 1\}.$$

For $n = 1$, the first entry is $3s - 2s = s$, so $d_1 \geq b_1$. If $d_{n-1} \geq b_{n-1}$, Lemma III.12.22 and monotonicity give

$$d_n \geq \min\left\{\frac{3s + b_{n-1}}{2}, s + 1\right\} = b_n.$$

Induction proves (12.25); the case $n = 3$ is (12.26). $\square$

Theorem III.12.23 and Theorem III.12.17 are related but logically different. The additive theorem controls the dimension of the support of repeated convolutions. The Fourier theorem gives quantitative information about how convolution mass is distributed, which is why it yields finite-energy conclusions unavailable from support dimension alone.

III.12.4 Approximate quadratic Vinogradov counts

Fix a scale $0 < \delta < 1$. A finite δ -separated set $P \subset \mathbb{P}$ is a (δ, s, C) set if its normalised counting measure

$$\sigma_P = |P|^{-1} \sum_{p \in P} \delta_p \quad (12.8)$$

satisfies

$$\sigma_P(B(x, r)) \leq Cr^s, \quad \delta \leq r \leq 1. \quad (12.9)$$

For such P , the n -fold approximate quadratic Vinogradov count is

$$\#\left\{(p_1, \dots, p_n, q_1, \dots, q_n) \in P^{2n} : \left|\sum_{i=1}^n p_i - \sum_{i=1}^n q_i\right| \leq \delta\right\}. \quad (12.10)$$

Because $p = (x, x^2)$, the two coordinates encode approximate equality of the first and second power sums.

Proposition III.12.24 (finite-scale OPP Fourier estimate from the continuous theorem). *Fix $0 \leq s \leq 1$ and $0 \leq t < \min\{3s, s + 1\}$. There is an even integer $p = p(s, t)$ such that, for every (δ, s, C) set $P \subset \mathbb{P}$ with normalised counting measure μ , one has at $R \asymp \delta^{-1}$*

$$\int_{B(C_0\delta^{-1})} |\hat{\mu}(\xi)|^p d\xi \lesssim_{s,t,C} \delta^{t-2}. \quad (12.27)$$

The constants may change with the fixed enlargement C_0 . This is the exact finite-scale form needed for the quadratic Vinogradov consequence of [47].

Proof. The admissible range is empty when $s = 0$, so assume $s > 0$. Let $p_0 = p_0(s, t)$ be supplied by Theorem III.12.17 and choose an even integer $p = 2n \geq p_0$. Fix a small constant $\lambda < 1/(10n)$. For $\delta \leq h \leq h_0$, replace each atom $p_u = (u, u^2) \in P$ by normalized arclength on the part of $\mathbb{P}$ with parameter $v \in [-1, 1]$ and $|v - u| \leq \lambda h$; call the resulting

probability σ_h . The boundary intervals are one-sided but still have length comparable to h .

We first check a uniform Frostman bound, including scales below h . For $r \geq h$, a ball meeting one of the replacement arcs has its original atom in a ball of radius $r + O(h)$, so $\sigma_h(B(x, r)) \lesssim_C r^s$. If $0 < r < h$, only atoms within $O(h)$ of x contribute; their total μ -mass is $O_C(h^s)$ by (12.9). Each associated normalized arc puts at most $O(r/h)$ of its mass in $B(x, r)$, since the parabola is a Lipschitz graph with arclength comparable to parameter length. Hence

$$\sigma_h(B(x, r)) \lesssim_C (r/h)h^s \leq C_* r^s, \quad 0 < r < h, \quad (12.27a)$$

where C_* is independent of h and P . Apply Theorem III.12.17 to $C_*^{-1}\sigma_h$. Since $\|\hat{\sigma}_h\|_\infty \leq 1$ and $2n \geq p_0$, it gives

$$\int_{B(R)} |\hat{\sigma}_h(\xi)|^{2n} d\xi \lesssim_{s,t,C} R^{2-t}, \quad R \geq 1, \quad (12.27b)$$

uniformly in h .

Put $v_h = \sigma_h^{*n} * \tilde{\sigma}_h^{*n}$. Choose a nonnegative Schwartz function $\phi \geq 1$ on $B(0, 2)$ with compactly supported Fourier transform. Fourier inversion and (12.27b) give

$$v_h(B(0, 2h)) \leq \int \phi(z/h) dv_h(z) \lesssim h^2 \int_{B(C_1/h)} |\hat{\sigma}_h(\xi)|^{2n} d\xi \lesssim h^t. \quad (12.27c)$$

Every parameter displacement $|v - u| \leq \lambda h$ moves its parabola point by at most $\sqrt{5}\lambda h$. Thus every selection of $2n$ replacement points lies within $2n\sqrt{5}\lambda h < h$ in total sum-difference distance of the corresponding original tuple. Consequently, with $v = \mu^{*n} * \tilde{\mu}^{*n}$,

$$v(B(0, h)) \leq v_h(B(0, 2h)) \lesssim h^t, \quad \delta \leq h \leq h_0. \quad (12.27d)$$

For $h > h_0$ the same estimate follows from $v(\mathbb{R}^2) = 1$ with a constant depending on h_0 .

Finally choose a nonnegative Schwartz $\psi \geq 1$ on $B(0, C_0)$. Since $\hat{v} = |\hat{\mu}|^{2n}$, Fourier inversion and the rapid decay of $\hat{\psi}$ yield, for any $N > t + 2$,

$$\begin{aligned} \int_{B(C_0\delta^{-1})} |\hat{\mu}(\xi)|^{2n} d\xi &\leq \int \psi(\delta\xi) |\hat{\mu}(\xi)|^{2n} d\xi \\ &\leq C_N \delta^{-2} \sum_{k \geq 0} 2^{-Nk} v(B(0, 2^{k+1}\delta)) \\ &\lesssim \delta^{-2} \sum_{k \geq 0} 2^{-Nk} (2^{k+1}\delta)^t \lesssim \delta^{t-2}. \end{aligned} \quad (12.27e)$$

The term with $k = 0$ also covers $|z| \leq 2\delta$. All constants are uniform in P and δ , as claimed. $\square$

Lemma III.12.25 (Fourier-to-tuple-count conversion). *Assume a probability measure μ supported on a finite set satisfies, for some integer n ,*

$$\int_{B(C_0\delta^{-1})} |\hat{\mu}(\xi)|^{2n} d\xi \lesssim \delta^{t-2}. \quad (12.28)$$

Then, with

$$\nu = \mu^{*n} * \tilde{\mu}^{*n}, \quad \tilde{\mu}(A) = \mu(-A),$$

one has

$$\nu(B(0, \delta)) \lesssim \delta^t. \quad (12.29)$$

Proof. Choose a nonnegative Schwartz function ϕ such that $\phi \geq 1$ on $B(1)$ and $\text{supp } \hat{\phi} \subset B(C_0)$. One concrete construction is to take a smooth compactly supported frequency bump whose inverse transform is nonzero near the origin, square its modulus, and rescale. Since

$$\hat{\nu} = |\hat{\mu}|^{2n},$$

Fourier inversion gives

$$\begin{aligned} \nu(B(0, \delta)) &\leq \int \phi(x/\delta) d\nu(x) \\ &= \delta^2 \int \hat{\phi}(\delta\xi) |\hat{\mu}(\xi)|^{2n} d\xi \\ &\lesssim \delta^2 \delta^{t-2} = \delta^t. \end{aligned}$$

□

Proposition III.12.26 (approximate quadratic Vinogradov bound). *Fix $0 \leq s \leq 1$ and $0 \leq t < \min\{3s, s+1\}$. There is an integer $n = n(s, t)$ such that, for every sufficiently small δ and every (δ, s, C) set $P \subset \mathbb{P}$,*

$$\#\left\{(p_1, \dots, p_n, q_1, \dots, q_n) \in P^{2n} : \left| \sum_{i=1}^n p_i - \sum_{i=1}^n q_i \right| \leq \delta\right\} \lesssim_{s,t,C} \delta^t |P|^{2n}. \quad (12.30)$$

Proof. Let $\mu = |P|^{-1} \sum_{p \in P} \delta_p$. The finite-scale OPP interface gives (12.27). Since $|\hat{\mu}| \leq 1$, choosing n with $2n \geq p$ yields (12.28). Lemma III.12.25 therefore gives $\nu(B(0, \delta)) \lesssim \delta^t$.

By definition of ν , this probability is exactly $|P|^{-2n}$ times the number of tuples in (12.30). Multiplying by $|P|^{2n}$ proves the claim. □

III.12.5 Average control is not pointwise decay

A pointwise power-decay assertion would have the form

$$|\hat{\sigma}(\xi)| \lesssim (1 + |\xi|)^{-\beta/2}. \quad (12.12)$$

The averaged estimates above do not by themselves imply such a bound.

Lemma III.12.27 (large-frequency-set estimate). *Under Theorem III.12.17, for every $\lambda > 0$,*

$$\{|\xi| \in B(R) : |\hat{\sigma}(\xi)| > \lambda\} \lesssim \lambda^{-p} R^{2-t}. \quad (12.31)$$

Proof. Chebyshev's inequality and (12.14) give

$$\lambda^p \{|\xi| \in B(R) : |\hat{\sigma}(\xi)| > \lambda\} \leq \int_{B(R)} |\hat{\sigma}|^p \lesssim R^{2-t}.$$

□

The conclusion is a measure estimate for the exceptional frequency set. It does not imply any bound of the form $\sup_{|\xi| \sim R} |\hat{\sigma}(\xi)| \lesssim R^{-\beta}$. Sparse frequency sets can carry large values without violating (12.31). This average/pointwise firewall is an essential limitation of the conclusion.

III.12.6 Transversal curves: the 2026 frontier

Let I be compact and $\mathcal{F} \subset C^2(I)$. We call $\mathcal{F}$ *transversal with constant T* when, for all distinct $f, g \in \mathcal{F}$,

$$\inf_{x \in I} (|f(x) - g(x)| + |f'(x) - g'(x)|) \geq T^{-1} \|f - g\|_{C^2(I)}. \quad (12.11)$$

The main OPP theorem was proved first for the parabola. The subsequent paper of Orponen–Pyörälä–Yi extends the geometric mechanism to transversal graph families. The cited arXiv preprint is the manuscript dated 27 August 2025 [50].

Theorem (OPY transversal Furstenberg theorem – frontier preview). *Let $s \in (0, 1]$, $t \in [0, 2]$, and let $\mathcal{F} \subset C^2(I)$ be a transversal family with $\dim_{\text{H}} \mathcal{F} \geq t$. If a planar set F satisfies*

$$\dim_{\text{H}}(F \cap \Gamma_f) \geq s \quad (f \in \mathcal{F}),$$

then

$$\dim_{\text{H}} F \geq \min \left\{ s + t, \frac{3s + t}{2}, s + 1 \right\}. \quad (12.32)$$

This is Theorem 1.9 of [50]. It is recorded only as a frontier theorem; no argument in Volume III depends on it.

Theorem (OPY curved-graph Fourier theorem – frontier preview). *Let $g \in C^3(\mathbb{R})$ and assume g'' never vanishes. If $0 \leq s \leq 1$ and $0 \leq t < \min\{3s, s + 1\}$, then there exists $p = p(g, s, t) \geq 1$ such that every s -Frostman measure σ on Γ_g^1 satisfies*

$$\|\hat{\sigma}\|_{L^p(B(R))}^p \lesssim_{g,s,t} R^{2-t}, \quad R \geq 1. \quad (12.33)$$

This is Theorem 1.16 of [50]. The same source stresses that the effective p is not expected to be sharp and records the $p = 6$ theorem for $s \geq 2/3$ from Yi [49]. This preview is not used in the preceding arguments.

A low-dimensional L^6 refinement

O’Regan–Wu–Yi, Theorem 1.8, prove that if $0 < s \leq 2/3$ and σ is s -Frostman on a uniformly nonzero-curvature C^3 graph, then for every $\epsilon > 0$,

$$\|\hat{\sigma}\|_{L^6(B(R))}^6 \lesssim_{\epsilon} R^{2-5s/2+\epsilon}. \quad (12.34)$$

Their manuscript explicitly contrasts this with the conjecturally sharp exponent $2 - \min\{3s, s + 1\}$ and notes that the low- s sharp problem remains open [52]. Thus (12.34) is genuine progress on the first question below, but it does not close it. It is included only to describe the current boundary of the method.

III.12.7 Terminal questions and synthesis

Q-III.12.01: sharp L^6 below $2/3$. For $0 < s < 2/3$, does the OPP/OPY estimate hold with $p = 6$ throughout the full sharp range $t < \min\{3s, s + 1\}$? The 2026 bound (12.34) does not reach this range.

Q-III.12.02: sharp triple sumset growth. For every Borel $K \subset \mathbb{P}$ with $\dim_{\mathbb{H}} K = s$, is

$$\dim_{\mathbb{H}}(K + K + K) \geq \min\{3s, s + 1\} \quad (12.35)$$

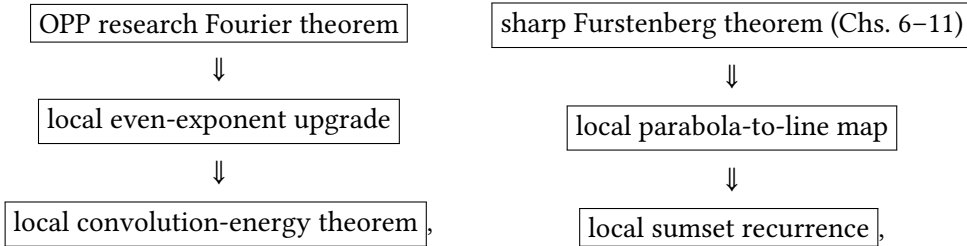
This is the natural additive counterpart of the sharp L^6 problem; the local theorem (12.26) gives $\min\{5s/2, s + 1\}$.

Q-III.12.03: tangential and cinematic families. How far can the sharp transversal-family theory be extended to curve families that permit tangencies, for example cinematic families? OPY explicitly distinguishes transversality from the cinematic condition and identifies this as an interesting further direction [50].

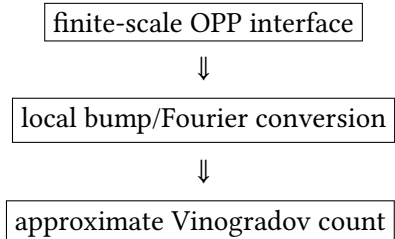
All three questions are terminal. They are included to mark the frontier and are not hypotheses for any result in this volume.

How the conclusions fit together

The Fourier theorem has two immediate routes to geometric information:



and, separately,



The transversal-family results of OPY and the two-ends refinement of O'Regan–Wu–Yi indicate where these arguments may next be sharpened, but neither is used in the preceding deductions.

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